

STM32 microcontroller system memory boot mode

Introduction

This document applies to the products listed in [Table 1](#), referred to as STM32 throughout the document.

It describes the supported peripherals and hardware requirements to consider when using the bootloader, stored in the internal boot ROM (system memory) of STM32 devices, and programmed during production.

Its main task is to download the application program to the internal flash memory through one of the available serial peripherals (such as USART, CAN, USB, I²C, I³C, SPI, FDCAN). A communication protocol is defined for each interface, with a compatible command set and sequence.

Table 1. Applicable products

Type	Part number or product series
Microcontrollers	STM32C0 series: STM32C011xx, STM32C031xx, STM32C051xx, STM32C071xx, STM32C091xx, STM32C092xx
	STM32F0 series: STM32F03xxx, STM32F04xxx, STM32F05xxx, STM32F07xxx, STM32F09xxx
	STM32F1 series
	STM32F2 series
	STM32F3 series: STM32F301xx, STM32F302xx, STM32F303xx, STM32F318xx, STM32F328xx, STM32F334xx, STM32F358xx, STM32F373xx, STM32F378xx, STM32F398xx
	STM32F4 series: STM32F401xx, STM32F405xx, STM32F407xx, STM32F410xx, STM32F411xx, STM32F412xx, STM32F413xx, STM32F415xx, STM32F417xx, STM32F423xx, STM32F427xx, STM32F429xx, STM32F437xx, STM32F439xx, STM32F446xx, STM32F469xx, STM32F479xx
	STM32F7 series: STM32F722xx, STM32F723xx, STM32F732xx, STM32F733xx, STM32F745xx, STM32F746xx, STM32F756xx, STM32F765xx, STM32F767xx, STM32F769xx, STM32F777xx, STM32F779xx
	STM32G0 series: STM32G030xx, STM32G031xx, STM32G041xx, STM32G07xxx, STM32G08xxx, STM32G0B0xx, STM32G0B1xx, STM32G0C1xx, STM32G050xx, STM32G051xx, STM32G061xx
	STM32G4 series: STM32G431xx, STM32G441xx, STM32G47xxx, STM32G48xxx, STM32G491xx, STM32G4A1xx
	STM32H5 series: STM32H503xx, STM32H562xx, STM32H563xx, STM32H573xx, STM32H523xx, STM32H533xx
	STM32H7 series: STM32H72xxx, STM32H73xxx, STM32H74xxx, STM32H75xxx, STM32H7A3xx, STM32H7B0xx, STM32H7B3xx, STM32H7R3xx, STM32H7R7xx, STM32H7S3xx, STM32H7S7xx
	STM32L0 series
	STM32L1 series: STM32L100xx, STM32L151xx, STM32L152xx, STM32L162xx
	STM32L4 series: STM32L431xx, STM32L432xx, STM32L433xx, STM32L442xx, STM32L443xx, STM32L451xx, STM32L452xx, STM32L462xx, STM32L471xx, STM32L475xx, STM32L476xx, STM32L486xx, STM32L496xx, STM32L4A6xx, STM32L4R5xx, STM32L4R7xx, STM32L4R9xx, STM32L4S5xx, STM32L4S7xx, STM32L4S9xx, STM32L412xx, STM32L422xx, STM32L423xx, STM32L442xx, STM32L443xx, STM32L451xx, STM32L452xx, STM32L462xx, STM32L471xx, STM32L475xx, STM32L476xx, STM32L486xx, STM32L496xx, STM32L4A6xx, STM32L4R5xx, STM32L4R7xx, STM32L4R9xx, STM32L4S5xx, STM32L4S7xx, STM32L4S9xx, STM32L412xx, STM32L422xx, STM32L4P5xx, STM32L4Q5xx
	STM32L5 series: STM32L552xx, STM32L562xx
	STM32U0 series: STM32U031xx, STM32U073xx, STM32U083xx
	STM32U3 series: STM32U375xx, STM32U385xx
	STM32U5 series: STM32U535xx, STM32U545xx, STM32U575xx, STM32U585xx, STM32U595xx, STM32U599xx, STM32U5A5xx, STM32U5A9xx, STM32U5F7xx, STM32U5F9xx, STM32U5G7xx, STM32U5G9xx
	STM32WB series: STM32WB10xx, STM32WB15xx, STM32WB30xx, STM32WB35xx, STM32WB50xx, STM32WB55xx
	STM32WBA series: STM32WBA52xx, STM32WBA54xx, STM32WBA55xx, STM32WBA62xx, STM32WBA63xx, STM32WBA64xx, STM32WBA65xx
	STM32WB0 series: STM32WB05xx, STM32WB06xx, STM32WB07xx, STM32WB09xx
	STM32WL series: STM32WLE5xx STM32WL55xx

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1 General information

This document applies to Arm^{®(a)}-based devices.

2 Related documents

For each supported product refer to the following documents, available on www.st.com:

- Datasheet or databrief
- Reference manual
- Application notes
 - AN3154: *CAN protocol used in the STM32 bootloader*
 - AN3155: *USART protocol used in the STM32 bootloader*
 - AN3156: *USB DFU protocol used in the STM32 bootloader*
 - AN4221: *I2C protocol used in the STM32 bootloader*
 - AN4286: *SPI protocol used in the STM32 bootloader*
 - AN5405: *FDCAN protocol used in the STM32 bootloader*
 - AN5927: *I3C protocol used in the STM32 bootloader*

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3 Glossary

C0 series:

STM32C011xx indicates STM32C011xx devices

STM32C031xx indicates STM32C031xx devices

STM32C051xx indicates STM32C051xx devices

STM32C071xx indicates STM32C071xx devices

STM32C091xx/92xx indicates STM32C091xx and STM32C092xx devices

F0 series:

STM32F03xxx indicates STM32F030x4, STM32F030x6, STM32F038x6, STM32F030xC, STM32F031x4, and STM32F031x6 devices

STM32F04xxx indicates STM32F042x4 and STM32F042x6 devices

STM32F05xxx and STM32F030x8 devices indicates STM32F051x4, STM32F051x6, STM32F051x8, STM32F058x8, and STM32F030x8 devices

STM32F07xxx indicates STM32F070x6, STM32F070xB, STM32F071xB, STM32F072x8, and STM32F072xB devices

STM32F09xxx indicates STM32F091xx and STM32F098xx devices

F1 series:

STM32F10xxx indicates Low-density, Medium-density, High-density, Low-density value line, Medium-density value line, and High-density value line devices:

Low-density devices are STM32F101xx, STM32F102xx, and STM32F103xx microcontrollers, where the flash memory density ranges between 16 and 32 Kbytes.

Medium-density devices are STM32F101xx, STM32F102xx, and STM32F103xx microcontrollers, where the flash memory density ranges between 64 and 128 Kbytes.

High-density devices are STM32F101xx and STM32F103xx microcontrollers, where the flash memory density ranges between 256 and 512 Kbytes.

Low-density value line devices are STM32F100xx microcontrollers, where the flash memory density ranges between 16 and 32 Kbytes.

Medium-density value line devices are STM32F100xx microcontrollers, where the flash memory density ranges between 64 and 128 Kbytes.

High-density value line devices are STM32F100xx microcontrollers, where the flash memory density ranges between 256 and 512 Kbytes.

STM32F105xx/107xx indicates STM32F105xx and STM32F107xx devices

STM32F10xxx XL-density indicates STM32F101xx and STM32F103xx devices, where the flash memory density ranges between 768 Kbytes and 1 Mbyte.

F2 series:

STM32F2xxxx indicates STM32F215xx, STM32F205xx, STM32F207xx, and STM32F217xx devices

F3 series:

STM32F301xx/302x4(6/8) indicates STM32F301x4, STM32F301x6, STM32F301x8, STM32F302x4, STM32F302x6, and STM32F302x8 devices

STM32F302xB(C)/303xB(C) indicates STM32F302xB, STM32F302xC, STM32F303xB and STM32F303xC devices

STM32F302xD(E)/303xD(E) indicates STM32F302xD, STM32F302xE, STM32F303xD, and STM32F303xE devices

STM32F303x4(6/8)/334xx/328xx indicates STM32F303x4, STM32F303x6, STM32F303x8, STM32F334x4, STM32F334x6, STM32F334x8, and STM32F328x8 devices

STM32F318xx indicates STM32F318x8 devices

STM32F358xx indicates STM32F358xC devices

STM32F373xx indicates STM32F373x8, STM32F373xB and STM32F373xC devices

STM32F378xx indicates STM32F378xC devices

STM32F398xx indicates STM32F398xE devices

F4 series:

STM32F40xxx/41xxx indicates STM32F405xx, STM32F407xx, STM32F415xx, and STM32F417xx devices

STM32F401xB(C) indicates STM32F401xB and STM32F401xC devices

STM32F401xD(E) indicates STM32F401xD and STM32F401xE devices

STM32F410xx indicates STM32F410x8 and STM32F410xB devices

STM32F411xx indicates STM32F411xD and STM32F411xE devices

STM32F412xx indicates STM32F412Cx, STM32F412Rx, STM32F412Vx and STM32F412Zx devices

STM32F413xx/423xx indicates STM32F413xG, STM32F413xH and STM32F423xH devices

STM32F42xxx/43xxx indicates STM32F427xx, STM32F429xx, STM32F437xx, and STM32F439xx devices

STM32F446xx indicates STM32F446xE and STM32F446xC devices

STM32F469xx/479xx indicates STM32F469xE, STM32F469xG, STM32F469xI, STM32F479xG, and STM32F479xI devices

F7 series:

STM32F72xxx/73xxx indicates STM32F722xx, STM32F723xx, STM32F732xx, and STM32F733xx devices

STM32F74xxx/75xxx indicates STM32F745xx, STM32F746xx, and STM32F756xx devices

STM32F76xxx/77xxx indicates STM32F765xx, STM32F767xx, STM32F769xx, STM32F777xx, and STM32F779xx devices

G0 series:

STM32G03xxx/04xxx indicates STM32G03xxx, and STM32G04xxx devices

STM32G07xxx/08xxx indicates STM32G07xxx, and STM32G08xxx devices

STM32G0B0xx indicates STM32G0B0xx devices

STM32G0B1xx/C1xx indicates STM32G0B1xx, and STM32G0C1xxx devices

STM32G05xxx/061xx indicates STM32G050xx, STM32G051xx, and STM32G061xx devices

G4 series:

STM32G431xx/441xx indicates STM32G431xx and STM32G441xx devices

STM32G47xxx/48xxx indicates STM32G471xx, STM32G473xx, STM32G474xx, STM32G483xx, and STM32G484xx devices

STM32G491xx/A1xx indicates STM32G491xx and STM32G4A1xx devices

H5 series:

STM32H503xx indicates STM32H503xx devices

STM32H562/63xx/73xx indicates STM32H562xx, STM32H563xx, and STM32H573xx devices

STM32H523xx/33xx indicates STM32H523xx and STM32H533xx devices

H7 series:

STM32H72xxx/73xxx indicates STM32H72xxx and STM32H73xxx devices

STM32H74xxx/75xxx indicates STM32H74xxx and STM32H75xxx devices

STM32H7A3xx/7B3xx/7B0xx indicates STM32H7A3xx, STM32H7B3xx, and STM32H7B0xx devices

STM32H7Rxxx/7Sxxx indicates STM32H7R3xx, STM32H7R7xx, STM32H7S3xx and STM32H7S7xx devices

L0 series:

STM32L01xxx/02xxx indicates STM32L011xx and STM32L021xx devices

STM32L031xx/041xx indicates STM32L031xx and STM32L041xx devices

STM32L05xxx/06xxx indicates STM32L051xx, STM32L052xx, STM32L053xx, STM32L062xx, and STM32L063xx ultra-low power devices

STM32L07xxx/08xxx indicates STM32L071xx, STM32L072xx, STM32L073xx, STM32L081xx, STM32L082xx, and STM32L083xx devices

L1 series:

STM32L1xxx6(8/B) indicates STM32L1xxV6T6, STM32L1xxV6H6, STM32L1xxR6T6, STM32L1xxR6H6, STM32L1xxC6T6, STM32L1xxC6H6, STM32L1xxV8T6, STM32L1xxV8H6, STM32L1xxR8T6, STM32L1xxR8H6, STM32L1xxC8T6, STM32L1xxC8H6, STM32L1xxVBT6, STM32L1xxVBH6, STM32L1xxRBT6, STM32L1xxRBH6, STM32L1xxCBT6, and STM32L1xxCBH6 ultra-low power devices

STM32L1xxx6(8/B)A indicates STM32L1xxV6T6-A, STM32L1xxV6H6-A, STM32L1xxR6T6-A, STM32L1xxR6H6-A, STM32L1xxC6T6-A, STM32L1xxC6H6-A, STM32L1xxV8T6-A, STM32L1xxV8H6-A, STM32L1xxR8T6-A, STM32L1xxR8H6-A, STM32L1xxC8T6-A, STM32L1xxC8H6-A, STM32L1xxVBT6-A, STM32L1xxVBH6-A, STM32L1xxRBT6-A, STM32L1xxRBH6-A, STM32L1xxCBT6-A, and STM32L1xxCBH6-A ultra-low power devices

STM32L1xxxC indicates STM32L1xxVCT6, STM32L1xxVCH6, STM32L1xxRCT6, STM32L1xxUCY6, STM32L1xxCCT6, and STM32L1xxCCU6 ultra-low power devices

STM32L1xxxD indicates STM32L1xxZDT6, STM32L1xxQDH6, STM32L1xxVDT6, STM32L1xxRDY6, STM32L1xxRDT6, STM32L1xxZCT6, STM32L1xxQCH6, STM32L1xxRCY6, STM32L1xxVCT6-A, and STM32L1xxRCT6-A ultra-low power devices

STM32L1xxxE indicates STM32L1xxZET6, STM32L1xxQEH6, STM32L1xxVET6, STM32L1xxVEY6, and STM32L1xxRET6 ultra-low power devices

L4 series:

STM32L412xx/422xx indicates STM32L412xB, STM32L412x8, and STM32L422xB devices

STM32L43xxx/44xxx indicates STM32L431xx, STM32L432xx, STM32L433xx and STM32L442xx, and STM32L443xx devices

STM32L45xxx/46xxx indicates STM32L451xx, STM32L452xx, and STM32L462xx devices

STM32L47xxx/48xxx indicates STM32L471xx, STM32L475xx, STM32L476xx, and STM32L486xx devices

STM32L496xx/4A6xx indicates STM32L496xE, STM32L496xG, and STM32L4A6xG devices

STM32L4Rxxx/4Sxxx indicates STM32L4R5xx, STM32L4R7xx, STM32L4R9xx, STM32L4S5xx, STM32L4S7xx, and STM32L4S9xx devices

STM32L4P5xx/4Q5xx indicates STM32L4P5xx/STM32L4Q5xx devices

L5 series:

STM32L552xx/62xx indicates STM32L552xx and STM32L562xx devices

U0 series:

STM32U031xx indicates STM32U031xx devices

STM32U073xx/83xx indicates STM32U073xx and STM32U083xx devices

U3 series:

STM32U375xx/85xx indicates STM32U375xx and STM32U385xx devices

U5 series:

STM32U535xx/45xx indicates STM32U535xx and STM32U545xx devices

STM32U575xx/85xx indicates STM32U575xx and STM32U585xx devices

STM32U595xx/99xx/A5xx/A9xx indicates STM32U595xx, STM32U599xx, STM32U5A5xx, and STM32U5A9xx devices

STM32U5F7xx/F9xx/G7xx/G9xx indicates STM32U5F7xx, STM32U5F9xx, STM32U5G7xx, and STM32U5G9xx devices

WB series:

STM32WB10xx/15xx indicates STM32WB10xx and STM32WB15xx devices

STM32WB30xx/35xx/50xx/55xx indicates STM32WB30xx, STM32WB35xx, STM32WB50xx, and STM32WB55xx devices

WBA series:

STM32WBA52xx indicates STM32WBA52xx devices

STM32WBA54xx/55xx indicates STM32WBA54xx and STM32WBA55xx devices

STM32WBA62xx/63xx/64xx/65xx indicates STM32WBA54xx, STM32WBA55xx, STM32WBA64xx, and STM32WBA65xx devices

WB0 series:

STM32WB0xx indicates STM32WB05xx, STM32WB06xx, STM32WB07xx, and STM32WB09xx devices

WL series:

STM32WLE5xx/55xx indicates STM32WLE5xx and STM32WL55xx devices

Note:

- BL_USART_Loop refers to the USART bootloader execution loop.*
- BL_CAN_Loop refers to the CAN bootloader execution loop.*
- BL_FDCAN_Loop refers to the FDCAN execution loop.*
- BL_I2C_Loop refers to the I2C bootloader execution loop.*
- BL_I3C_Loop refers to the I3C bootloader execution loop.*
- BL_SPI_Loop refers to the SPI bootloader execution loop.*

4 General bootloader description

4.1 Bootloader activation

The bootloader is activated by applying one of the patterns described in [Table 2](#).

If boot from Bank2 option is activated (for products supporting this feature), the bootloader executes Dual Boot mechanism as described in figures “Dual bank boot implementation for STM32xxxx” (example: [Figure 48](#)), otherwise bootloader selection protocol is executed as described in figures “Bootloader VY.x selection for STM32xxxx” (example: [Figure 29](#)), where STM32xxxx is the relative STM32 product.

When readout protection Level2 is activated, the MCU does not boot on system memory, and bootloader cannot be executed (unless jumping to it from flash user code, all commands are not accessible except Get, GetID, and GetVersion).

Table 2. Bootloader activation patterns

Pattern	Condition
Pattern 1	Boot0(pin) = 1 and Boot1(pin) = 0
Pattern 2	Boot0(pin) = 1 and nBoot1(bit) = 1
Pattern 3	Boot0(pin) = 1, Boot1(pin) = 0 and BFB2(bit) = 1
	Boot0(pin) = 0, BFB2(bit) = 0 and both banks do not contain valid code
	Boot0(pin) = 1, Boot1(pin) = 0, BFB2(bit) = 0 and both banks do not contain valid code
Pattern 4	Boot0(pin) = 1, Boot1(pin) = 0 and BFB2(bit) = 1
	Boot0(pin) = 0, BFB2(bit) = 0 and both banks do not contain valid code
	Boot0(pin) = 1, Boot1(pin) = 0 and BFB2(bit) = 0
Pattern 5	Boot0(pin) = 1, Boot1(pin) = 0 and BFB2(bit) = 0
	Boot0(pin) = 0, BFB2(bit) = 1 and both banks do not contain valid code
	Boot0(pin) = 1, Boot1(pin) = 0 and BFB2 (bit) = 1
Pattern 6	Boot0(pin) = 1, nBoot1(bit) = 1 and nBoot0_SW(bit) = 1
	nBoot0(bit) = 0, nBoot1(bit) = 1 and nBoot0_SW(bit) = 0
	Boot0(pin) = 0, nBoot0_SW(bit) = 1 and main flash memory empty
	nBoot0(bit) = 1, nBoot0_SW(bit)=0 and main flash memory empty
Pattern 7	Boot0(pin) = 1, nBoot1(bit) = 1 and BFB2(bit) = 0
	Boot0(pin) = 0, BFB2(bit) = 1 and both banks do not contain valid code
	Boot0(pin) = 1, nBoot1(bit) = 1 and BFB2(bit) = 1
Pattern 8	Boot(pin) = 0 and BOOT_ADD0(optionbyte) = 0x0040
	Boot(pin) = 1 and BOOT_ADD1(optionbyte) = 0x0040

Table 2. Bootloader activation patterns (continued)

Pattern	Condition
Pattern 9	nDBANK(bit) = 1, Boot(pin) = 0 and BOOT_ADD0(optionbyte) = 0x0040
	nDBANK(bit) = 1, Boot(pin) = 1 and BOOT_ADD1(optionbyte) = 0x0040
	nDBANK(bit) = 0, nDBOOT(bit) = 1, Boot(pin) = 0 and BOOT_ADD0(optionbyte) = 0x0040
	nDBANK(bit) = 0, nDBOOT(bit) = 1, Boot(pin) = 1 and BOOT_ADD1(optionbyte) = 0x0040
	nDBANK(bit) = 0, nDBOOT(bit) = 0, BOOT_ADDx(optionbyte) out of memory range or in ICP memory range
	nDBANK(bit) = 0, nDBOOT(bit) = 0, BOOT_ADDx(optionbyte) in flash memory range and both banks do not contain valid code
Pattern 10	Boot(pin) = 0 and BOOT_ADD0(optionbyte) = 0x1FF0
	Boot(pin) = 1 and BOOT_ADD1(optionbyte) = 0x1FF0
Pattern 11	BOOT_LOCK(bit) = 0, nBoot1(bit) = 1, nBOOT0_SEL(bit) = 1 and nBoot0(bit) = 0
	BOOT_LOCK(bit) = 0, nBoot1(bit) = 1, Boot0(pin) = 1 and nBOOT0_SEL(bit) = 1
	BOOT_LOCK(bit) = 0, nBOOT0_SEL(bit) = 1, nBoot0(bit) = 1 and main flash empty
	BOOT_LOCK(bit) = 0, Boot0(pin) = 0, nBOOT0_SEL(bit) = 0 and main flash empty
Pattern 12	TZEN = 1 = 0, Boot0(pin) = 0, nSWBoot0(bit) = 1 and NSBOOTADD0 [24:0] = Address ⁽¹⁾
	TZEN = 1 = 0, Boot0(pin) = 1, nSWBoot0(bit) = 1 and NSBOOTADD1 [24:0] = Address ⁽¹⁾
	TZEN = 1 = 0, nBoot0(bit) = 0, nSWBoot0(bit) = 0 and NSBOOTADD1 [24:0] = Address ⁽¹⁾
	TZEN = 0, nBoot0(bit) = 1, nSWBoot0(bit) = 0 and NSBOOTADD0 [24:0] = Address ⁽¹⁾
	TZEN = 1, Boot0(pin) = 0, nSWBoot0(bit) = 1 and SECBOOTADD0 [24:0] = Address ⁽¹⁾ and RSSCMD = 0
	TZEN = 1, Boot0(pin) = 1, nSWBoot0(bit) = 1 and RSSCMD = 0, BOOT_LOCK = 0 or (BOOT_LOCK = 1 and SECBOOTADD0 [24:0] = Address ⁽¹⁾)
	TZEN = 1, nBoot0(bit) = 1, nSWBoot0(bit) = 0 and SECBOOTADD0 [24:0] = Address ⁽¹⁾ and RSSCMD = 0, BOOT_LOCK = 0 or (BOOT_LOCK = 1 and SECBOOTADD0 [24:0] = Address ⁽¹⁾)
	TZEN = 1, nBoot0(bit) = 0, nSWBoot0(bit) = 0 and RSSCMD = 0, BOOT_LOCK = 0 or BOOT_LOCK = 1 and SECBOOTADD1 [24:0] = Address ⁽¹⁾
	TZEN = 1, RSSCMD = 0x1C0, BOOT_LOCK=0 or (BOOT_LOCK = 1 and SECBOOTADD0 [24:0] = Address ⁽¹⁾)
Pattern 13	nBoot0(bit) = 0, nBoot1(bit) = 1 and nSWBoot0(bit) = 0
	nBoot0(bit) = 1, nBoot1(bit) = 1, nSWBoot0(bit) = 0 and user flash empty
	nBoot1(bit) = 1, nSWBoot0(bit) = 1 and Boot0(pin) = 1
	nBoot1(bit) = 1, nSWBoot0(bit) = 1, Boot0(pin) = 0 and user flash empty
Pattern 14	BOOT_LOCK(bit) = 0, nBoot1(bit) = 1, Boot0(pin) = 1 and nSWBoot0(bit) = 1
	BOOT_LOCK(bit) = 0, nBoot1(bit) = 1, nBoot0(bit) = 0 and nSWBoot0(bit) = 0
	BOOT_LOCK(bit) = 0, Boot0(pin) = 0, nSWBoot0(bit) = 1, BFB2(bit) = 1 and both banks do not contain valid code
	BOOT_LOCK(bit) = 0, nBoot0(bit) = 1, nSWBoot0(bit) = 0, BFB2(bit) = 1 and both banks do not contain valid code

Table 2. Bootloader activation patterns (continued)

Pattern	Condition
Pattern 15	BOOT_LOCK(bit)=0, Boot0(pin) = 1, nBoot1(bit) = 1 and nBoot0_SW(bit) = 1
	BOOT_LOCK(bit)=0, nBoot0(bit) = 0, nBoot1(bit) = 1 and nBoot0_SW(bit) = 0
Pattern 16	Boot0(pin) = 1, nBoot1(bit) = 1 and nBoot0_SW(bit) = 1
	nBoot0(bit) = 0, nBoot1(bit) = 1 and nBoot0_SW(bit) = 0
	Boot0(pin) = 0, nBoot0_SW(bit) = 1 and main flash memory empty
Pattern 17	PRODUCT_STATE = Open and Boot0(pin) = 1
	PRODUCT_STATE = Provisioning
Pattern 18	Force PA10 high during HW reset

1. Device dependent: 0x17F200 for STM32L5, STM32U5, and STM32WBA6, 0x17F1E00 for STM32U3, 0x17F1000 for STM32WBA5.

Note: *nBoot0_SW means either nSWBoot0 or nBOOT0_SEL, depending upon the product.*

Note: *BOOT_LOCK implementation is product dependent. See the reference manual for more details.*

In addition to the patterns described above, the user can execute bootloader by performing a jump to system memory from user code. Before jumping to bootloader:

- Disable all peripheral clocks
- Disable used PLL
- Disable interrupts
- Clear pending interrupts

System memory boot mode can be exited by getting out from bootloader activation condition and generating hardware reset or using Go command to execute user code.

Note: *When executing the Go command, the peripheral registers used by the bootloader are not initialized to their default reset values before jumping to the user application. They must be reconfigured in the user application if they are used. So, if the application uses the IWDG, the IWDG prescaler value must be adapted to meet requirements (since the prescaler was set to its maximum value). For some products, not all reset values are set. For more information, refer to the known limitations detailed for each product bootloader version.*

Note: *On devices with dual bank boot, to jump to system memory from user code the user must first remap the system memory bootloader at address 0x00000000 using SYSCFG register (except for STM32F7 series), then jump to bootloader. For the STM32F7 series, the user must disable nDBOOT and/or nDBANK features (in option bytes), then jump to bootloader. For STM32L0 series, the jump to system memory from user code is not possible.*

Note: *For STM32 devices embedding bootloader using the DFU/CAN interface in which the external clock source (HSE) is required for DFU/CAN operations, the detection of the HSE value is done dynamically by the bootloader firmware and is based on the internal oscillator clock (HSI, MSI). When (because of temperature variations or other conditions) the internal oscillator precision is altered above the tolerance band (1% around the theoretical value), the bootloader can calculate a wrong HSE frequency value. In this case, the bootloader DFU/CAN interfaces can malfunction, or not work at all.*

4.2 Bootloader identification

Depending upon the device, the bootloader can support one or more embedded serial peripherals used to download the code to the internal flash memory. The bootloader identifier (ID) provides information about the supported serial peripherals.

For a given STM32 device, the bootloader is identified by means of the:

1. **Bootloader (protocol) version:** version of the serial peripheral (e.g. USART, CAN, USB) communication protocol used in the bootloader. This version can be retrieved using the bootloader Get Version command.
2. **Bootloader identifier (ID):** version of the STM32 device bootloader, coded on one byte in the **0xXY** format, where:
 - **X** specifies the embedded serial peripheral(s) used by the device bootloader:
 - X = 1: one USART is used
 - X = 2: two USARTs are used
 - X = 3: USART, CAN, and DFU are used
 - X = 4: USART and DFU are used
 - X = 5: USART and I²C are used
 - X = 6: I²C is used
 - X = 7: USART, CAN, DFU, and I²C are used
 - X = 8: I2C and SPI are used
 - X = 9: USART, CAN (or FDCAN), DFU, I²C, and SPI are used
 - X = 10: USART, I2C, and DFU are used
 - X = 11: USART, I2C, and SPI are used
 - X = 12: USART and SPI are used
 - X = 13: USART, DFU, I2C, and SPI are used
 - X = 14: USART, DFU, I2C, I3C, FDCAN, and SPI are used
 - X = 15: USART, USB-DFU, I2C, and I3C are used
 - X = 16: USART, USB-DFU, FDCAN, and SPI are used
 - X = 17: USART, SPI, and FDCAN are used
 - X = 18: USART, SPI, FDCAN, and I2C are used
 - **Y** specifies the device bootloader version
 For example, if the bootloader ID is 0x10, this is the first version, which uses only one USART.
 The bootloader ID is programmed in the last byte address - 1 of the device system memory and can be read by using the "Read memory" command or by direct access to the system memory via JTAG/SWD.

Note: The bootloader ID format is applied to all STM32 products, except the STM32F1xx devices. The bootloader version for the STM32F1xx applies only to the embedded device bootloader version and not to its supported protocols.

[Table 3](#) provides identification information of the bootloaders embedded in STM32 devices.

Table 3. Embedded bootloaders

Series	Device		Supported serial peripherals	Bootloader ID		Bootloader (protocol) version
				ID	Memory location	
C0	STM32C011xx		USART1 I2C1	0x51	0x1FFF17FE	USART (V3.1) I2C1(V1.1)
	STM32C031xx		USART1 I2C1	0x52	0x1FFF17FE	USART (V3.1) I2C1(V1.1)
	STM32C051xx		USART1/USART2 I2C1/I2C2 SPI1/SPI2	0xB0	0x1FFF2FFE	USART (V4.0) I2C (V2.0) SPI (V2.0)
	STM32C071xx		USART1/USART2 I2C1/I2C2 SPI1/SPI2 USB DFU	0xD1	0x1FFF67FE	USART (V3.1) I2C1(V1.2) SPI (V1.1) USB (V2.2)
	STM32C091xx/92xx		USART1/USART2/USART3 I2C1/I2C2 SPI1/SPI2 FDCAN1	0x121	0x1FFF3FFE	USART (V4.0) I2C (V2.0) SPI (V2.0) FDCAN (V2.2)
F0	STM32F05xxx/STM32F030x8		USART1/USART2	0x21	0x1FFFF7A6	USART (V3.1)
	STM32F03xx4/6		USART1	0x10	0x1FFFF7A6	USART (V3.1)
	STM32F030xC		USART1 I2C1	0x52	0x1FFFF796	USART (V3.1) I2C1(V1.0)
	STM32F04xxx		USART1/USART2 DFU (USB device FS) I2C1	0xA1	0x1FFFF6A6	USART (V3.1) DFU (V2.2) I2C (V1.0)
	STM32F071xx/072xx		USART1/USART2 DFU (USB device FS) I2C1	0xA1	0x1FFFF6A6	USART (V3.1) DFU (V2.2) I2C (V1.0)
	STM32F070x6		USART1/USART2 DFU (USB device FS) I2C1	0xA2	0x1FFFF6A6	USART (V3.1) DFU (V2.2) I2C (V1.0)
	STM32F070xB		USART1/USART2 DFU (USB device FS) I2C1	0xA3	0x1FFFF6A6	USART (V3.1) DFU (V2.2) I2C (V1.0)
	STM32F09xxx		USART1/USART2 I2C1	0x50	0x1FFFF796	USART (V3.1) I2C (V1.0)
F1	STM32F10xxx	Low-density	USART1	NA	NA	USART (V2.2)
		Medium-density	USART1	NA	NA	USART (V2.2)
		High-density	USART1	NA	NA	USART (V2.2)
		Medium-density value line	USART1	0x10	0x1FFFF7D6	USART (V2.2)
		High-density value line	USART1	0x10	0x1FFFF7D6	USART (V2.2)
	STM32F105xx/107xx		USART1/USART2 (remapped) CAN2 (remapped) DFU (USB device)	NA	NA	USART (V2.2 ⁽¹⁾) CAN (V2.0) DFU(V2.2)
	STM32F10xxx XL-density		USART1/USART2 (remapped)	0x21	0x1FFFF7D6	USART (V3.0)

Table 3. Embedded bootloaders (continued)

Series	Device	Supported serial peripherals	Bootloader ID		Bootloader (protocol) version
			ID	Memory location	
F2	STM32F2xxxx	USART1/USART3	0x20	0x1FFF77DE	USART (V3.0)
		USART1/USART3 CAN2 DFU (USB device FS)	0x33	0x1FFF77DE	USART (V3.1) CAN (V2.0) DFU (V2.2)
F3	STM32F373xx	USART1/USART2 DFU (USB device FS)	0x41	0x1FFFF7A6	USART (V3.1) DFU (V2.2)
	STM32F378xx	USART1/USART2 I2C1	0x50	0x1FFFF7A6	USART (V3.1) I2C (V1.0)
	STM32F302xB(C)/303xB(C)	USART1/USART2 DFU (USB device FS)	0x41	0x1FFFF796	USART (V3.1) DFU (V2.2)
	STM32F358xx	USART1/USART2 I2C1	0x50	0x1FFFF796	USART (V3.1) I2C (V1.0)
	STM32F301xx/302x4(6/8)	USART1/USART2 DFU (USB device FS)	0x40	0x1FFFF796	USART (V3.1) DFU (V2.2)
	STM32F318xx	USART1/USART2 I2C1/ I2C3	0x50	0x1FFFF796	USART (V3.1) I2C (V1.0)
	STM32F302xD(E)/303xD(E)	USART1/USART2 DFU (USB device FS)	0x40	0x1FFFF796	USART (V3.1) DFU (V2.2)
	STM32F303x4(6/8)/334xx/328xx	USART1/USART2 I2C1	0x50	0x1FFFF796	USART (V3.1) I2C (V1.0)
	STM32F398xx	USART1/USART2 I2C1/I2C3	0x50	0x1FFFF796	USART (V3.1) I2C (V1.0)

Table 3. Embedded bootloaders (continued)

Series	Device	Supported serial peripherals	Bootloader ID		Bootloader (protocol) version
			ID	Memory location	
F4	STM32F40xxx/41xxx	USART1/USART3 CAN2 DFU (USB device FS)	0x31	0x1FFF77DE	USART (V3.1) CAN (V2.0) DFU (V2.2)
		USART1/USART3 CAN2 DFU (USB device FS) I2C1/I2C2/I2C3 SPI1/SPI2	0x91	0x1FFF77DE	USART (V3.1) CAN (V2.0) DFU (V2.2) SPI(V1.1) I2C (V1.0)
	STM32F42xxx/43xxx	USART1/USART3 CAN2 DFU (USB device FS) I2C1	0x70	0x1FFF76DE	USART (V3.1) CAN (V2.0) DFU (V2.2) I2C (V1.0)
		USART1/USART3 CAN2 DFU (USB device FS) SPI1/ SPI2/ SPI4 I2C1/I2C2/I2C3	0x91	0x1FFF76DE	USART (V3.1) CAN (V2.0) DFU (V2.2) SPI(V1.1) I2C (V1.0)
	STM32F401xB(C)	USART1/USART2 DFU (USB device FS) SPI1/SPI2/ SPI3 I2C1/I2C2/I2C3	0xD1	0x1FFF76DE	USART (V3.1) DFU (V2.2) SPI(V1.1) I2C (V1.0)
	STM32F401xD(E)	USART1/USART2 DFU (USB device FS) SPI1/SPI2/SPI3 I2C1/I2C2/I2C3	0xD1	0x1FFF76DE	USART (V3.1) DFU (V2.2) SPI(V1.1) I2C (V1.1)
	STM32F410xx	USART1/USART2 I2C1/I2C2/I2C4 SPI1/SPI2	0xB1	0x1FFF76DE	USART (V3.1) I2C (V1.2) SPI (V1.1)
	STM32F411xx	USART1/USART2 DFU (USB device FS) SPI1/SPI2/ SPI3 I2C1/I2C2/I2C3	0xD0	0x1FFF76DE	USART (V3.1) DFU (V2.2) SPI(V1.1) I2C (V1.1)
	STM32F412xx	USART1/USART2 USART3/CAN2 DFU (USB device FS) SPI1/SPI3/SPI4 I2C1/I2C2/I2C3/I2C4	0x91	0x1FFF76DE	USART (V3.1) CAN (V2.0) DFU (V2.2) SPI (V1.1) I2C (V1.2)
	STM32F413xx/423xx	USART1/USART2 USART3/CAN2 DFU (USB device FS) I2C1/I2C2/I2C3/I2C4 SPI1/SPI3/SPI4	0x90	0x1FFF76DE	USART (V3.1) CAN (V2.0) DFU (V2.2) I2C (V1.2) SPI (V1.1)
	STM32F446xx	USART1/USART3 CAN2 DFU (USB device FS) SPI1/ SPI2/SPI4 I2C1/I2C2/I2C3	0x90	0x1FFF76DE	USART (V3.1) CAN (V2.0) DFU (V2.2) SPI(V1.1) I2C (V1.2)
	STM32F469xx/479xx	USART1/USART3 I2C1/I2C2/I2C3 CAN2 DFU (USB device FS) SPI1/SPI2/ SPI4	0x90	0x1FFF76DE	USART (V3.1) I2C (V1.2) CAN (V2.0) DFU (V2.2) SPI (V1.1)

Table 3. Embedded bootloaders (continued)

Series	Device	Supported serial peripherals	Bootloader ID		Bootloader (protocol) version
			ID	Memory location	
F7	STM32F72xxx/73xxx	USART1/USART3 CAN1 DFU (USB device FS) I2C1/I2C2/I2C3 SPI1/SPI2/SPI4	0x90	0x1FF0EDBE	USART (V3.1) CAN (V2.0) DFU (V2.2) I2C (V1.2) SPI (V1.2)
	STM32F74xxx/75xxx	USART1/USART3 I2C1/I2C2/I2C3 CAN2 DFU (USB device FS)	0x70	0x1FF0EDBE	USART (V3.1) I2C (V1.2) CAN (V2.0) DFU (V2.2)
		USART1/USART3 I2C1/I2C2/I2C3 CAN2 DFU (USB device FS) SPI1/SPI2/SPI4	0x90	0x1FF0EDBE	USART (V3.1) I2C (V1.2) CAN (V2.0) DFU (V2.2) SPI (V1.2)
	STM32F76xxx/77xxx	USART1/USART3 CAN2 DFU (USB device FS) I2C1/I2C2/I2C3 SPI1/SPI2/SPI4	0x93	0x1FF0EDBE	USART (V3.1) CAN (V2.0) DFU (V2.2) I2C (V1.2) SPI (V1.2)
G0	STM32G07xxx/08xxx	USART1/USART2/USART3 I2C1/I2C2 SPI1/SPI2	0xB4	0x1FFF6FFE	USART (V3.1) I2C (V1.2) SPI (V1.1)
	STM32G03xxx/04xxx	USART1/USART2 I2C1/I2C2	0x53	0x1FFF1FFE	USART (V3.1) I2C (V1.2)
	STM32G0B0xx	USART1/USART2/USART3 I2C1/I2C2 SPI1/SPI2 DFU (USB device FS)	0xD0	0x1FFF9FFE	USART (V3.1) I2C (V1.2) SPI (V1.1) DFU (V2.2)
	STM32G0B1xx/0C1xx	USART1/USART2/USART3 I2C1/I2C2 SPI1/SPI2 DFU (USB device FS) FDCAN	0x92	0x1FFF9FFE	USART (V3.1) I2C (V1.2) SPI (V1.1) DFU (V2.2) FDCAN (V1.0)
	STM32G05xxx/061xx	USART1/USART2 I2C1/I2C2	0x51	0x1FFF1FFE	USART (V3.1) I2C (V1.2)
G4	STM32G431xx/441xx	USART1/USART2/USART3 I2C2/I2C3 SPI1/SPI2 DFU (USB device FS)	0xD4	0x1FFF6FFE	USART (V3.1) I2C (V1.2) SPI (V1.1) DFU (V2.2)
	STM32G47xxx/48xxx	USART1/USART2/USART3 I2C2/I2C3/I2C4 SPI1/SPI2 DFU (USB device FS)	0xD5	0x1FFF6FFE	USART (V3.1) I2C (V1.2) SPI (V1.1) DFU (V2.2)
	STM32G491xx/4A1xx	USART1/USART2/USART3 I2C2/I2C3 SPI1/SPI2 DFU (USB device FS)	0xD2	0x1FFF6FFE	USART (V3.1) I2C (V1.2) SPI (V1.1) DFU (V2.2)

Table 3. Embedded bootloaders (continued)

Series	Device	Supported serial peripherals	Bootloader ID		Bootloader (protocol) version
			ID	Memory location	
H5	STM32H503xx	USART1/USART2/USART3 I2C2 I3C1 SPI1/SPI2/SPI3 USB DFU FDCAN1	0xE2	0x0BF8FFFE	USART (V4.0) I2C (V2.0) I3C (V1.0) SPI (V2.0) USB (V3.0) FDCAN (V2.0)
	STM32H562xx/63xx/73xx	USART1/USART2/USART3 I2C3/I2C4 I3C1 SPI1/SPI2/SPI3 USB DFU FDCAN2	0xE4	0x0BF9FAFE	USART (V4.0) I2C (V2.0) I3C (V1.0) SPI (V2.0) USB (V3.0) FDCAN (V2.0)
	STM32H523xx/33xx	USART1/USART2/USART3 I2C1/I2C3 I3C1 SPI1/SPI2/SPI3 USB DFU FDCAN2	0xE2	0x0BF8FFFE	USART (V4.0) I2C (V2.0) I3C (V1.0) SPI (V2.0) USB (V3.0) FDCAN (V2.0)
H7	STM32H72xxx/73xxx	USART1/USART2/USART3 I2C1/I2C2/I2C3 DFU (USB device FS) SPI1/SPI2/SPI3/SPI4 FDCAN1	0x93	0x1FF1E7FE	USART (V3.1) I2C (V1.2) DFU (V2.2) SPI (V1.1) FDCAN (V1.1)
	STM32H74xxx/75xxx	USART1/USART2/USART3 I2C1/I2C2/I2C3 DFU (USB device FS) SPI1/SPI2/SPI3/SPI4 FDCAN1	0x92	0x1FF1E7FE	USART (V3.1) I2C (V1.1) DFU (V2.2) SPI (V1.1) FDCAN (V1.1)
	STM32H7A3xx/7B3xx/7B0xx	USART1/USART2/USART3 I2C1/I2C2/I2C3 DFU (USB device FS) SPI1/SPI2/SPI3 FDCAN1	0x92	0x1FF13FFE	USART (V3.1) I2C (V1.2) DFU (V2.2) SPI (V1.2) FDCAN (V1.1)
	STM32H7Rxxx/7Sxxx	USART1/USART2/USART3 UART4 I2C1/I2C2/I2C3 DFU (USB device FS) SPI1/SPI2/SPI3 FDCAN2 I3C1	0xE3	0x1FF1FCFE	USART (V4.0) I2C (V2.0) DFU (V2.2) SPI (V2.0) FDCAN (V2.1) I3C (V1.0)

Table 3. Embedded bootloaders (continued)

Series	Device	Supported serial peripherals	Bootloader ID		Bootloader (protocol) version
			ID	Memory location	
L0	STM32L01xxx/02xxx	USART2 SPI1	0xC3	0x1FF00FFE	USART (V3.1) SPI (V1.1)
	STM32L031xx/041xx	USART2 SPI1	0xC0	0x1FF00FFE	USART (V3.1) SPI (V1.1)
	STM32L05xxx/06xxx	USART1/USART2 SPI1/ SPI2	0xC0	0x1FF00FFE	USART (V3.1) SPI (V1.1)
	STM32L07xxx/08xxx	USART1/USART2 DFU (USB device FS)	0x41	0x1FF01FFE	USART (V3.1) DFU (V2.2)
		USART1/USART2 SPI1/SPI2 I2C1/I2C2	0xB2	0x1FF01FFE	USART (V3.1) SPI (V1.1) I2C (V1.2)
L1	STM32L1xxx6(8/B)	USART1/USART2	0x20	0x1FF00FFE	USART (V3.0)
	STM32L1xxx6(8/B)A	USART1/USART2	0x20	0x1FF00FFE	USART (V3.1)
	STM32L1xxxC	USART1/USART2 DFU (USB device FS)	0x40	0x1FF01FFE	USART (V3.1) DFU (V2.2)
	STM32L1xxxD	USART1/USART2 DFU (USB device FS)	0x45	0x1FF01FFE	USART (V3.1) DFU (V2.2)
	STM32L1xxxE	USART1/USART2 DFU (USB device FS)	0x40	0x1FF01FFE	USART (V3.1) DFU (V2.2)

Table 3. Embedded bootloaders (continued)

Series	Device	Supported serial peripherals	Bootloader ID		Bootloader (protocol) version
			ID	Memory location	
L4	STM32L412xx/422xx	USART1/USART2/USART3 I2C1/I2C2/I2C3 DFU (USB device FS) SPI1/SPI2	0xD1	0x1FFF6FFE	USART (V3.1) I2C (V1.2) DFU (V2.2) SPI (V1.1)
	STM32L43xxx/44xxx	USART1/USART2/USART3 I2C1/I2C2/I2C3 CAN1 DFU (USB device FS) SPI1/SPI2	0x91	0x1FFF6FFE	USART (V3.1) I2C (V1.2) CAN (V2.0) DFU (V2.2) SPI (V1.1)
	STM32L45xxx/46xxx	USART1/USART2/USART3 I2C1/I2C2/I2C3 CAN1 DFU (USB device FS) SPI1/SPI2	0x92	0x1FFF6FFE	USART (V3.1) I2C (V1.2) CAN (V2.0) DFU (V2.2) SPI (V1.1)
	STM32L47xxx/48xxx	USART1/USART2/ USART3 I2C1/I2C2/I2C3 DFU (USB device FS)	0xA3	0x1FFF6FFE	USART (V3.1) I2C (V1.2) DFU (V2.2)
		USART1/USART2/ USART3 I2C/I2C2/I2C3 SPI1/SPI2 CAN1 DFU (USB device FS)	0x92	0x1FFF6FFE	USART (V3.1) I2C (V1.2) SPI (V1.1) CAN(V2.0) DFU(V2.2)
	STM32L496xx/4A6xx	USART1/USART2/USART3 I2C1/I2C2/I2C3 CAN1 DFU (USB device FS) SPI1/SPI2	0x93	0x1FFF6FFE	USART (V3.1) I2C (V1.2) CAN (V2.0) DFU (V2.2) SPI (V1.1)
	STM32L4Rxxx/STM32L4Sxxx	USART1/USART2/USART3 I2C1/I2C2/I2C3 CAN1 DFU (USB device FS) SPI1/SPI2	0x95	0x1FFF6FFE	USART (V3.1) I2C (V1.2) CAN (V2.0) DFU (V2.2) SPI (V1.1)
L5	STM32L4P5xx/Q5xx	USART1/USART2/USART3 I2C1/I2C2/I2C3 CAN1 DFU (USB device FS) SPI1/SPI2	0x90	0x1FFF6FFE	USART (V3.1) I2C (V1.2) CAN (V2.0) DFU (V2.2) SPI (V1.1)
		USART1/USART2/USART3 I2C1/I2C2/I2C3 SPI1/SPI2/SPI3 DFU (USB device FS) FDCAN1	0x92	0x0BF97FFE	USART (V3.1) I2C (V1.2) SPI (V1.1) DFU (V2.2) FDCAN (V1.0)
U0	STM32U031xx	USART1/USART2/USART3 I2C1/I2C2/I2C3 SPI1/SPI2	0xB0	0x1FFF37FE	USART (V3.1) I2C (V1.2) SPI (V1.1)
	STM32U073xx/83xx	USART1/USART2/USART3 I2C1/I2C2/I2C3 SPI1/SPI2 DFU (USB device FS)	0xD0	0x1FFF67FE	USART (V3.1) I2C (V1.2) SPI (V1.1) DFU (V2.2)

Table 3. Embedded bootloaders (continued)

Series	Device	Supported serial peripherals	Bootloader ID		Bootloader (protocol) version
			ID	Memory location	
U3	STM32U375xx/385xx	USART1/USART3 I2C1/I2C2/I2C3 I3C1 SPI1/SPI2/SPI3 DFU (USB device FS) FDCAN1	0xE2	0x0BF98FFE	USART(V4.0) I2C (V2.0) I3C(V1.0) SPI (V2.0) DFU (V3.0) FDCAN (V2.2)
U5	STM32U535xx/545xx	USART1/USART3 I2C1/I2C2/I2C3 SPI1/SPI2/SPI3 DFU (USB device FS) FDCAN1	0x91	0x0BF99EFE	USART (V3.1) I2C (V1.2) SPI (V1.1) DFU (V2.2) FDCAN (V1.1)
	STM32U575xx/STM32U585xx	USART1/USART2/USART3 I2C1/I2C2/I2C3 SPI1/SPI2/SPI3 DFU (USB device FS) FDCAN1	0x92	0x0BF99EFE	USART (V3.1) I2C (V1.2) SPI (V1.1) DFU (V2.2) FDCAN (V1.1)
	STM32U595xx/599xx/ STM32U5A5xx/5A9xx	USART1/USART2/USART3 I2C1/I2C2/I2C3 SPI1/SPI2/SPI3 DFU (USB device HS) FDCAN1	0x92	0x0BF99EFE	USART (V3.1) I2C (V1.2) SPI (V1.1) DFU (V2.2) FDCAN (V1.1)
	STM32U5F7xx/5F9xx/ STM32U5G7xx/5G9xx	USART1/USART2/USART3 I2C1/I2C2/I2C3 SPI1/SPI2/SPI3 DFU (USB device HS) FDCAN1	0x90	0x0BF99EFE	USART (V3.1) I2C (V1.2) SPI (V1.1) DFU (V2.2) FDCAN (V1.1)
WB	STM32WB10xx/15xx	USART1 I2C1 SPI1	0xB1	0x1FFF6FFE	USART (V3.1) I2C (V1.2) SPI (V1.1)
	STM32WB30xx/35xx/50xx/55xx	USART1 I2C1/I2C3 SPI1/SPI2 DFU (USB device FS)	0xD5	0x1FFF6FFE	USART (V3.2) I2C (V1.2) SPI (V1.1) DFU (V2.2)
WBA	STM32WBA52xx	USART1/USART2 I2C1/I2C3 SPI3	0xB0	0x0BF8FEFE	USART (V3.1) I2C (V1.2) SPI (V1.1)
	STM32WBA54xx/55xx	USART1/USART2 I2C1/I2C3 SPI3	0xB1	0x0BF8FEFE	USART (V3.1) I2C (V1.2) SPI (V1.1)
	STM32WBA62xx/63xx/64xx/65xx	USART1/USART2 I2C1/I2C3 SPI2/SPI3 DFU (USB device FS)	0xD2	0x0BF97EFE	USART (V4.0) I2C (V2.0) SPI (V2.0) DFU (V3.0)
WB0	STM32WB05xx	USART1	NA	NA	USART (V2.0)
	STM32WB06xx/7xx				USART (V4.0)
	STM32WB09xx				USART (V1.0)

Table 3. Embedded bootloaders (continued)

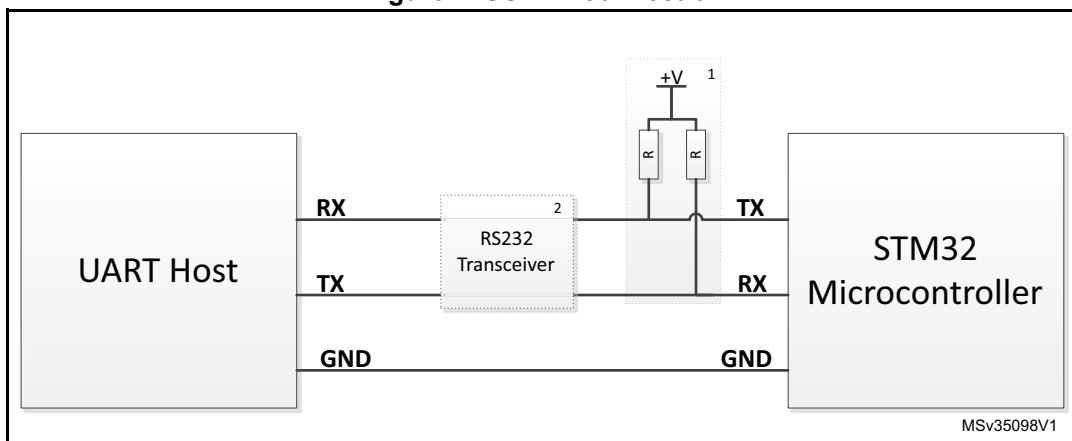
Series	Device	Supported serial peripherals	Bootloader ID		Bootloader (protocol) version
			ID	Memory location	
WL	STM32WLE5xx/55xx	USART1/USART2 SPI1/SPI2	0xC4	0x1FFF3EFE	USART (V3.1) SPI (V1.1)

1. For connectivity line devices, the USART bootloader returns V2.0 instead of V2.2 for the protocol version. For more details refer to the "STM32F105xx and STM32F107xx revision Z" errata sheet available from www.st.com.

4.3 Hardware connection requirements

To use the USART bootloader, the host must be connected to the RX and TX pins of the desired USARTx interface via a serial cable.

Figure 1. USART connection

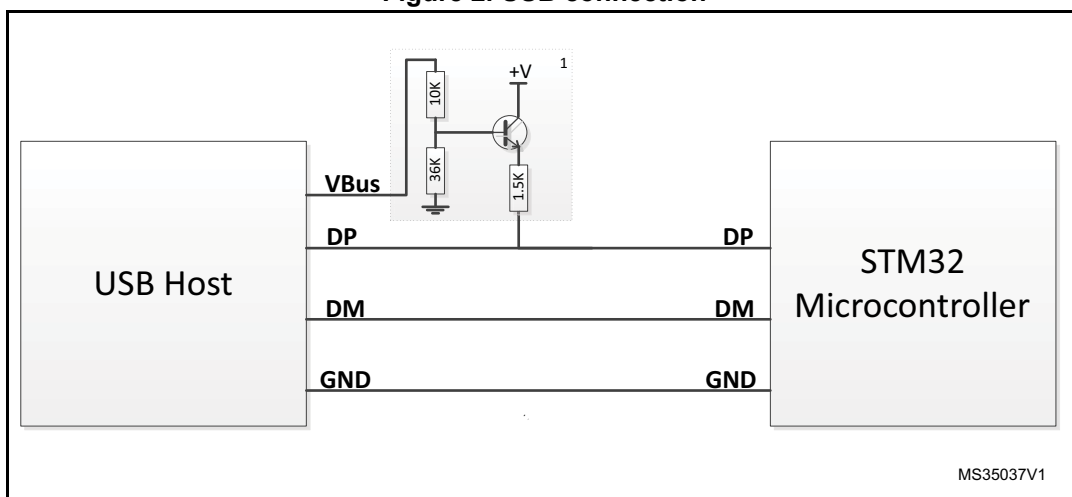


1. A pull-up resistor must be added, if they are not connected on host side.
2. An RS232 transceiver must be connected to adapt the voltage level (3.3 to 12 V) between the STM32 device and the host.

Note: Typically V is 3.3 V, and R is 100 k Ω . These values depend upon the application and the used hardware.

To use the DFU, connect the microcontroller USB interface to a USB host (such as a PC).

Figure 2. USB connection

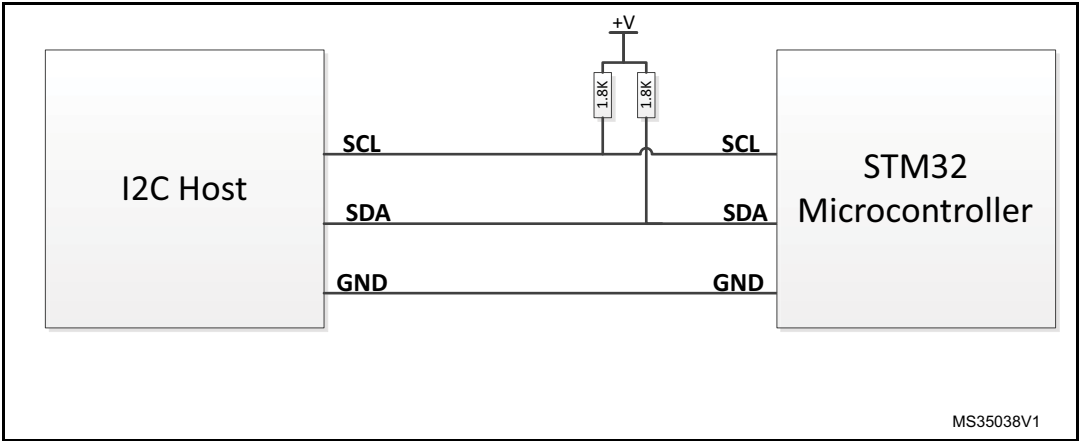


1. This additional circuit permits to connect a pull-up resistor to DP pin using VBus when needed. Refer to product section (table describing STM32 configuration in system memory boot mode) to know if an external pull-up resistor must be connected to DP pin.

Note: V typically is 3.3 V. This value depends upon the application and the used hardware.

To use the I2C bootloader, connect the host (controller) and the desired I2Cx interface (target) together via the data (SDA) and clock (SCL) pins. A 1.8 K Ω pull-up resistor must be connected to both SDA and SCL lines.

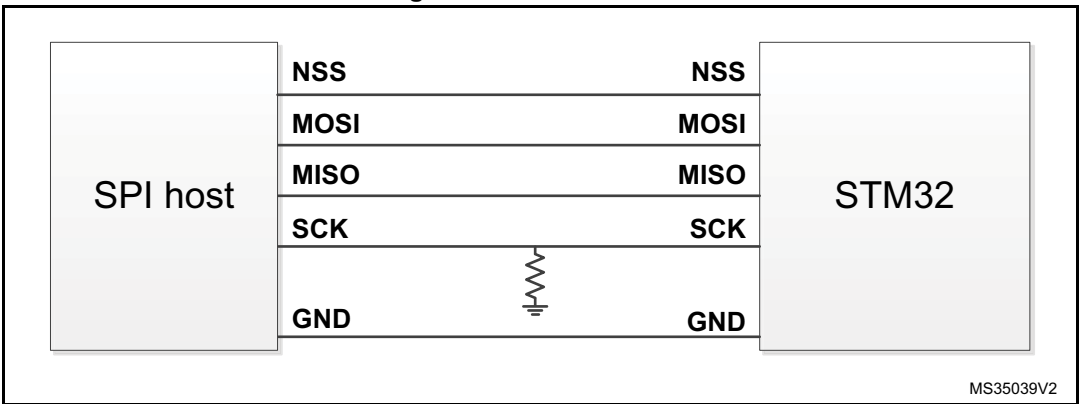
Figure 3. I2C connection



Note: V is typically 3.3 V. This value depends upon the application and the used hardware.

To use the SPI bootloader, connect the host (master) and the desired SPIx interface (slave) together via the MOSI, MISO, SCK and NSS pins. The NSS pin can be connected to GND. A pull-down resistor must be connected to the SCK line.

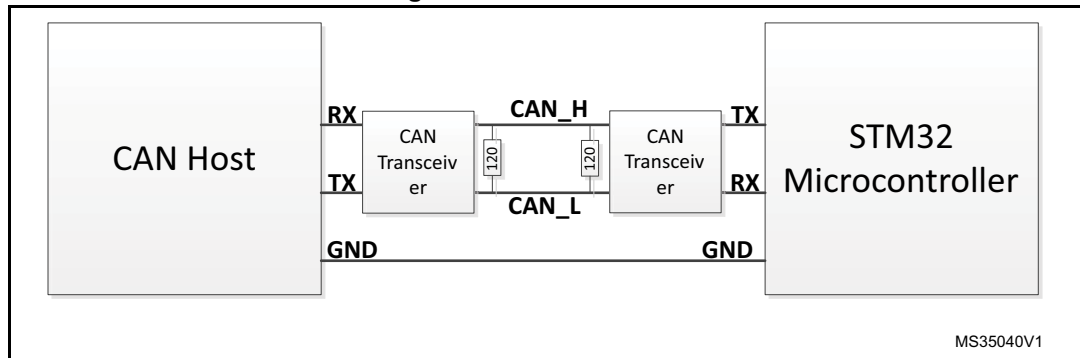
Figure 4. SPI connection



Note: The resistor is typically 10 K Ω , its value depends upon the application and the used hardware.

To use the CAN interface, the host must be connected to the RX and TX pins of the desired CANx interface via CAN transceiver and a serial cable. A 120 Ω resistor must be added as terminating resistor.

Figure 5. CAN connection



Note: When a bootloader firmware supports DFU, it is mandatory that no USB host is connected to the USB peripheral during the selection phase of the other interfaces. After selection phase, the user can plug a USB cable without impacting the selected bootloader execution, except for commands generating a system reset.

It is recommended to keep the RX pins of unused bootloader interfaces (USART_RX, SPI_MOSI, and CAN_RX, if present) at a known (low or high) level and keep the USB D+/D- lines, if present, on the same level (low/high) at the startup of the bootloader (detection phase). Leaving these pins floating during the detection phase can result in activation of unused interfaces.

4.4 Bootloader memory management

All write operations using bootloader commands must be word-aligned (the address must be a multiple of 4). The number of data to write must be a multiple of 4 as well (non-aligned half page write addresses are accepted).

Some products embed a bootloader with specific features:

- On products that do not support mass erase operation, to perform this operation using the bootloader, two options are available:
 - Erase all sectors one by one using the Erase command
 - Set protection level to Level 1. Then, set it to Level 0 (using the Read protect and then the Read unprotect command). This operation results in a mass erase of the internal flash memory.
- Bootloader firmware of STM32 L1 and L0 series supports Data memory in addition to standard memories (internal flash, internal SRAM, option bytes and System memory). The start address and the size of this area depends on product, refer to the reference manual for more information. Data memory can be read and written but cannot be erased using the Erase command. When writing in a Data memory location, the bootloader firmware manages the erase operation of this location before any write. A write to Data memory must be word-aligned (address to be written must be a multiple of 4) and the number of data must also be a multiple of 4. To erase a Data memory location, write 0s at this location.
- Bootloader firmware of the F2, F4, F7, and L4 series supports OTP memory in addition to standard memories (internal flash, internal SRAM, option bytes and System memory). The start address and the size of this area depends on product, refer to the reference manual for more information. OTP memory can be read and written but

cannot be erased using Erase command. When writing in an OTP memory location, make sure that the relative protection bit is not reset.

- For STM32 F2, F4, and F7 series the internal flash memory write operation format depends on voltage range. By default write operations are allowed by one byte format (half-word, word, and double-word operations are not allowed). To increase the speed of write operation, the user must apply the adequate voltage range that allows write operation by half-word, word or double-word and update this configuration on the fly by the bootloader software through a virtual memory location. This memory location is not physical but can be read and written using usual bootloader read/write operations according to the protocol in use. This memory location contains four bytes, described in [Table 4](#). It can be accessed by 1, 2, 3, or 4 bytes. However, reserved bytes must remain at their default values (0xFF), otherwise the request is NACK-ed.

Table 4. STM32 F2, F4, and F7 voltage range configuration using bootloader

Address	Size	Description
0xFFFF0000	1 byte	This byte controls the current value of the voltage range. – 0x00: voltage range [1.8 V, 2.1 V] – 0x01: voltage range [2.1 V, 2.4 V] – 0x02: voltage range [2.4 V, 2.7 V] – 0x03: voltage range [2.7 V, 3.6 V] – 0x04: voltage range [2.7 V, 3.6 V] and double word write/erase operation is used. In this case it is mandatory to supply 9 V through the VPP pin (refer to the product reference manual for more details about the double-word write procedure). – Others: all other values are not supported and are NACK-ed.
0xFFFF0001	1 byte	Reserved. 0xFF is the default value, all other values are not supported and are NACK-ed.
0xFFFF0002	1 byte	
0xFFFF0003	1 byte	

[Table 5](#) lists the valid memory areas, depending upon the bootloader commands.

Table 5. Supported memory area by Write, Read, Erase, and Go commands

Memory area	Write command	Read command	Erase command	Go command
Flash	Supported	Supported	Supported	Supported
RAM	Supported	Supported	Not supported	Supported
System memory	Not supported	Supported	Not supported	Not supported
Data memory	Supported	Supported	Not supported	Not supported
OTP memory	Supported	Supported	Not supported	Not supported

4.5 Bootloader UART baudrate detection

For the UART interface baudrate detection, there are two implemented mechanisms:

- Software baudrate detection using internal HSI and timer (use GPIO as input, detect falling edge and rising edge as explained in AN3155).

The devices using this mechanism are subject to software jitter (variable error of baudrate calculation) that can reach up to $\pm 5\%$. In this case, the host connecting to the STM32 bootloader UART interface must support a $\pm 5\%$ deviation in baudrate.

The software jitter value is variable and different at each retry, so it is possible to use multiple retry connections to overcome it. Connect and check for correct bootloader answer, if the answer is not correct, reset the device and retry connection until the correct answer is received. At this point, the rest of the communication is not impacted. It is also possible to reduce the software jitter by reducing baudrate value (as an example, use 56000 instead of 115200 bps).

[Table 6](#) provides the maximum software jitter value for the 115200 bps baudrate. The lower the baudrate, the lower the software jitter.

- Baudrate detection using UART auto-baudrate feature. Devices using this mechanism do not present any software jitter.

Table 6. Jitter software calculation on bootloader USART detection

Series or product	Detection method	Maximum software jitter
STM32C011xx	Auto-baudrate	Not applicable
STM32C031xx		
STM32C051xx USART1		
STM32C051xx USART2	Software baudrate detection	-2%
STM32C071xx USART1	Auto-baudrate	Not applicable
STM32C071xx USART2	Software baudrate detection	-2%
STM32C091xx/92xx USART1	Auto-baudrate	Not applicable
STM32C091xx/92xx USART2/3	Software baudrate detection	-2%
STM32F0	Software baudrate detection	-1%
STM32F1		-3%
STM32F2		-5%
STM32F3		-2%
STM32F4		-6%
STM32F7		-6%
STM32G05xxx/061xx USART1	Auto-baudrate	Not applicable
STM32G05xxx/061xx USART2	Software baudrate detection	-2%
STM32G07/8x USART3 STM32G03/4x USART2		-4%
STM32G07/8x USART1/USART2 STM32G03/4x USART1	Auto-baudrate	Not applicable
STM32G0B/Cxxx		
STM32G4		
STM32H5		
STM32H7		

Table 6. Jitter software calculation on bootloader USART detection (continued)

Series or product	Detection method	Maximum software jitter
STM32L0	Software baudrate detection	-2%
STM32L1		-3%
STM32L4		-5%
STM32L5	Auto-baudrate	Not applicable
STM32U031xx USART1/2		
STM32U073/83xx USART1/2		
STM32U073/83xx USART3	Software baudrate detection	-2%
STM32U3	Auto-baudrate	Not applicable
STM32U5		
STM32WB		
STM32WBA		
STM32WB0		
STM32WL		

4.6 Programming constraints

When using the bootloader interface to write in the flash memory, respect the alignment on the programmed address detailed in [Table 7](#). If the address is not aligned, the operation fails, and all following program operations fail as well.

Table 7. Flash memory alignment constraints

Series	Alignment
STM32C0	8 bytes
STM32F0	4 bytes
STM32F1	4 bytes
STM32F2	4 bytes
STM32F3	4 bytes
STM32F4	4 bytes
STM32F7	8 bytes
STM32G0	4 bytes
STM32G4	8 bytes
STM32H5	16 bytes
STM32H7	16 bytes (H7Rxx/H7Sxx devices)
	8 bytes (all other devices)
STM32L0	8 bytes
STM32L1	8 bytes

Table 7. Flash memory alignment constraints (continued)

Series	Alignment
STM32L4	8 bytes
STM32L5	16 bytes
STM32U0	8 bytes
STM32U3	8 bytes
STM32U5	16 bytes
STM32WB	8 bytes
STM32WBA	16 bytes
STM32WB0	4 bytes
STM32WL	8 bytes

Examples of alignment:

- 4 bytes: 0x0800 0014 is aligned and passes, 0x0800 0012 is not aligned and fails
- 8 bytes: 0x0800 0010 is aligned and passes, 0x0800 0014 is not aligned and fails

Note: On STM32F4 and STM32F7 it is possible to change the alignment constraint by writing in the device feature space.

4.7 ExitSecureMemory feature

The securable memory area is used to isolate secure boot code/data, which handles sensitive information (secrets), from application code. The secure boot code access is controlled by HW (FLASH registers and option bytes, depending on the product). The code is executed once at boot, then locked by HW until the next reset.

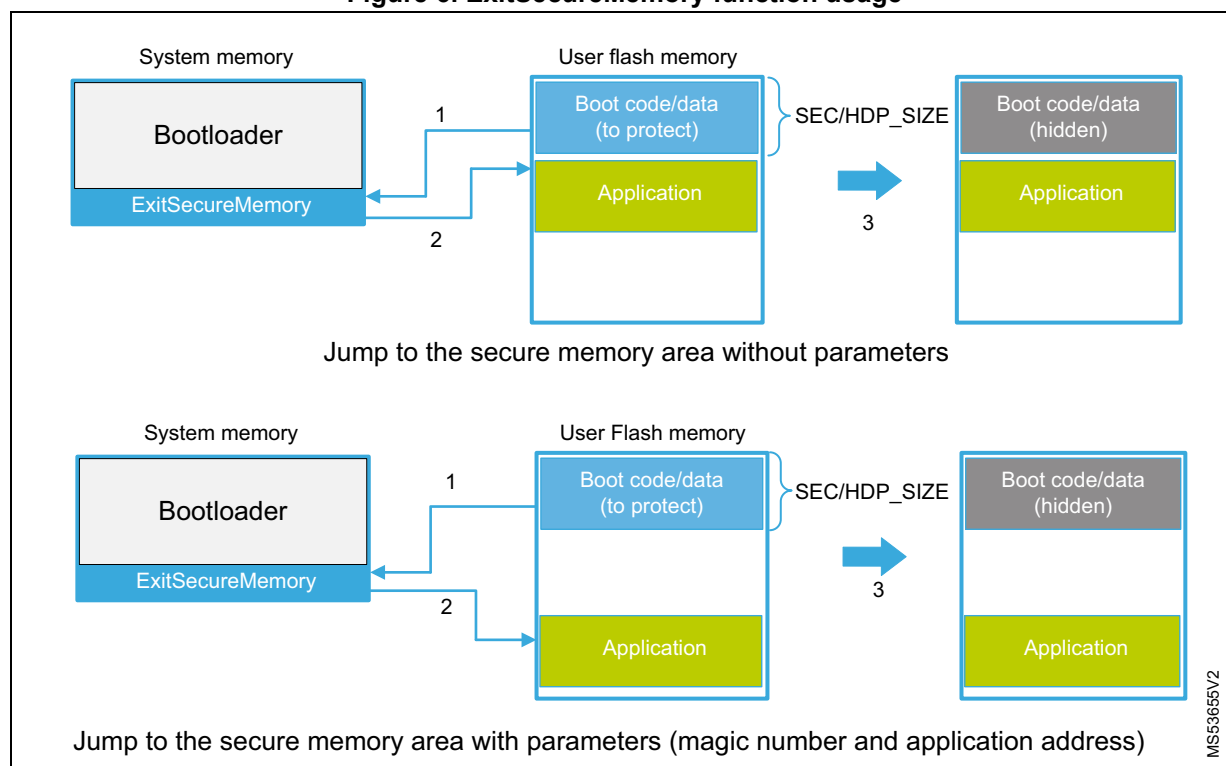
The ExitSecureMemory is a software hosted on the system memory. When the user boot code jumps to it, it is possible to enable the hide protection area, and then jump to the application code. The size of the Hide Protection Area (HDP) must be set by the user to the needed value before jumping to the ExitSecureMemory function.

4.7.1 ExitSecureMemory v1.0

As shown in [Figure 6](#), two methods can be used.

1. Jump to the secure memory function without parameters: the application must be loaded just after the defined secure memory area (HDP and size).
2. Jump to the secure memory function using two parameters:
 - a) Magic number: 0x08192A3C is used to secure boot code/data in flash/Bank1 and jump in case of a single/dual bank product, 0x08192A3D is used to secure boot code/data and jump to application in Bank2 in case of a dual bank product
 - b) User address = Application address: the application can be loaded to any desired address

Figure 6. ExitSecureMemory function usage

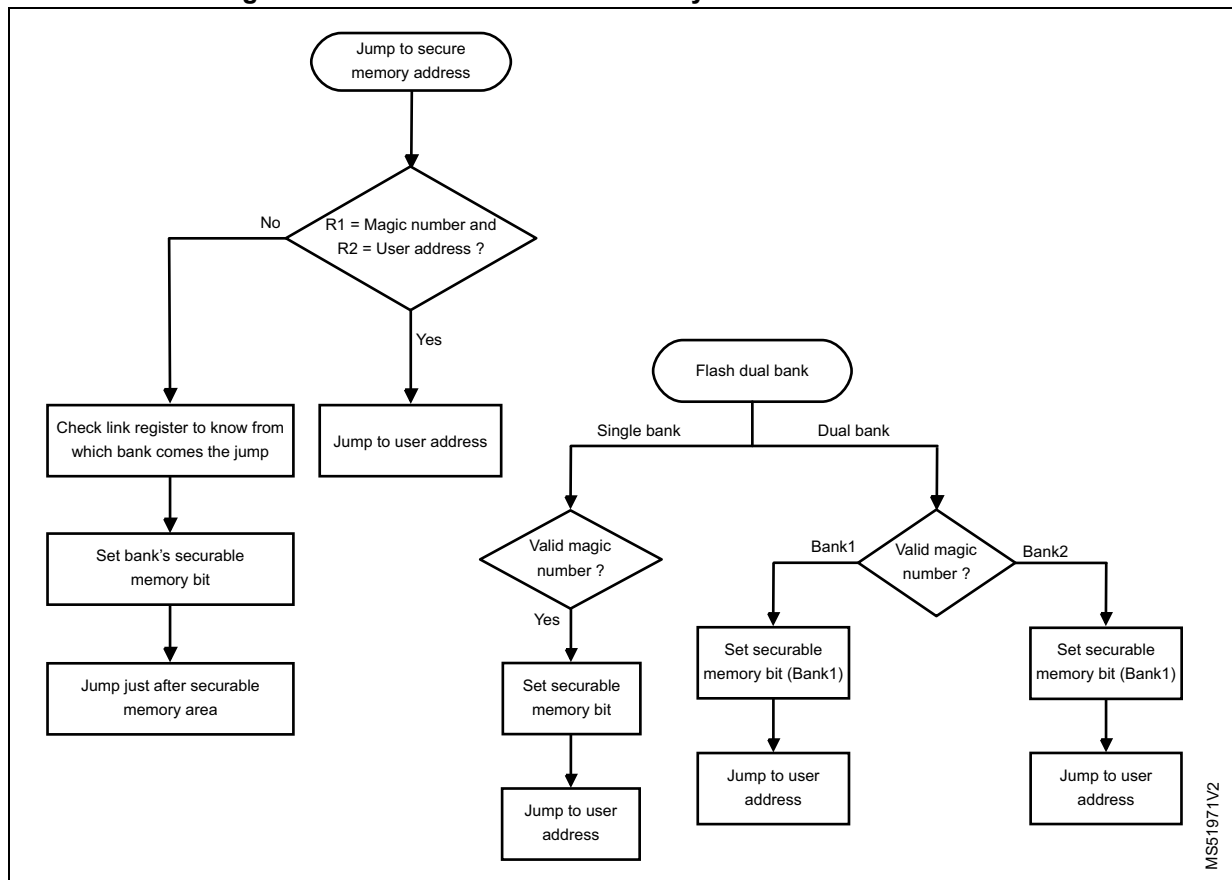


For more information regarding the option bytes configuration, see the reference manual.

For examples of functions that can be used to call ExitSecureMemory, see [Appendix A](#).

For more details refer to [Figure 7](#).

Figure 7. Access to securable memory area from the bootloader



1. The bootloader does not check the integrity of the user address, it is up to the user to ensure the validity of the address to jump to.

4.7.2 ExitSecureMemory v1.1

Compared to the ExitSecureMemory v1.0, the user can define an MPU region. This is done using the CPU R3 register, before jumping to the software, that runs as shown in [Figure 8](#).

Figure 8. Defining an MPU region

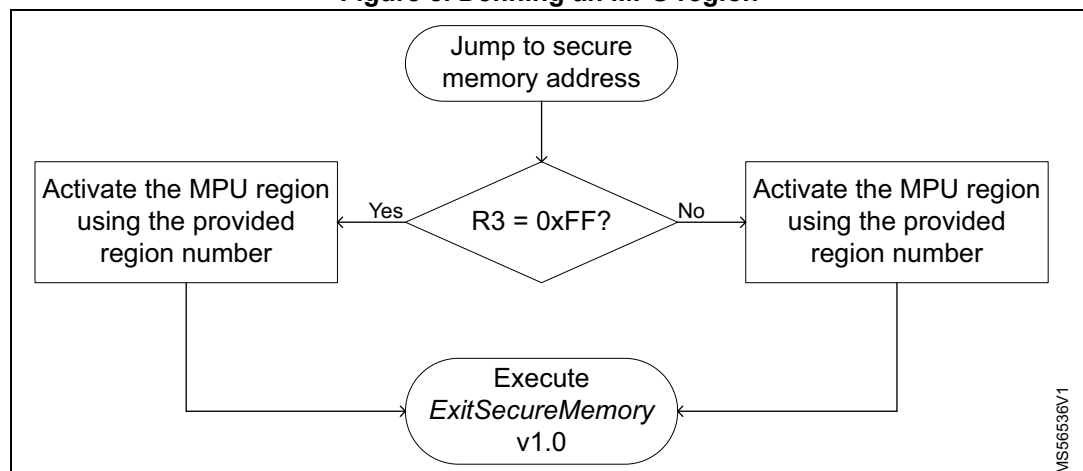


Table 8. ExitSecureMemory entry address

MCU		ExitSecureMemory address	Version address	Version
STM32G0	STM32G07xxx/08xxx	0x1FFF6800	Not available	V1.0
	STM32G03xxx/04xxx	0x1FFF1E00		
	STM32G0Bxxx/0Cxxx	0x1FFF6800		
	STM32G05xxx/061xx	0x1FFF6800		
STM32G4	STM32G47xxx/48xxx	0x1FFF6800		
	STM32G431xx/441xx	0x1FFF6800		
	STM32G491xx/4A1xx	0x1FFF6800		
STM32C0	STM32C011xx	0x1FFF1600	0x1FFF2F8C	V1.1 (0x11)
	STM32C031xx	0x1FFF1600		
	STM32C051xx	0x1FFF2E00		
	STM32C071xx	0x1FFF1600		
	STM32C091xx/92xx	0x1FFF3E00		
STM32U0	STM32U031xx	0x1FFF3500	0x1FFF368C	V1.1 (0x11)
	STM32U073xx/83xx	0x1FFF6000	0x1FFF618C	

4.8 IWDG usage

The bootloader does not enable IWDG, it only tries to update the prescaler value if the IWDG was enabled by HW (through option bytes) or by SW in case of an application that jumps to bootloader.

If the IWDG was not enabled before the boot on bootloader (using HW boot or by a jump from an application), the watchdog prescaler value update bit (PVU) is set to 1 when the bootloader tries to change the prescaler value.

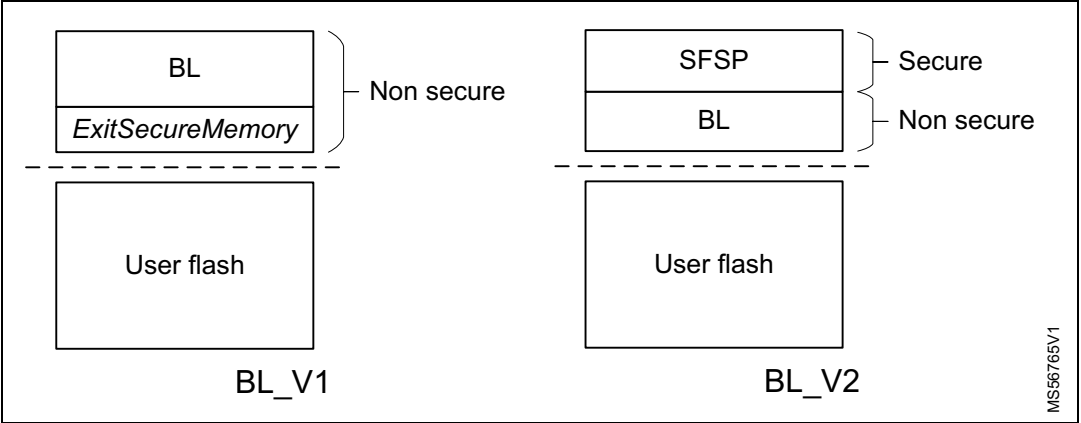
This value does not change, it remains at 0x1 as the prescaler update never happens (IWDG is not enabled), even after the jump. When using the bootloader to jump to the application, and when there is the need to enable the IWDG, consider that the PVU bit in the IWDG_SR register is set to 1.

4.9 Bootloader models

To address the evolution of STM32 devices on security, the bootloader (BL) is now available on different models.

1. Legacy model - BL_V1 (see left side of [Figure 9](#)):
The system memory is non secure and allocated to the bootloader. In some projects a SW (functionally independent from the bootloader) called ExitSecureMemory is implemented on the same zone as the BL.
2. New BL model - BL_V2 (see right side of [Figure 9](#)):
The system memory is split between secure and nonsecure areas. The secure area contains the System Flash Secure Package (SFSP), the nonsecure contains the BL.

Figure 9. BL_V1 (left) and BL_V2 (right) models



4.10 Boot constraints on BL

Boot depends upon the MCU (see [Table 9](#)), and adds new constraints to the BL.

- Legacy products - Boot_V1

When correctly set, the boot is directly on the BL (left side of [Figure 10](#))

- Products with security but no TrustZone isolation - Boot_V2 (right side of [Figure 10](#))

When correctly set and boot on the BL is possible, the boot starts on the SFSP, then it jumps to the bootloader. This adds a constraint to the boot timing.

- Products with security using TrustZone isolation - Boot_V3

On this category there are two possibilities:

- a) Boot depending on TrustZone value - Boot_V3_1 (see [Figure 11](#))

When TZEN is enabled, some constraints are added to the BL functionalities:

- > Boot timing is different from the TZEN disabled use case
- > Before jumping to the BL, the SFSP maps all the needed resources by the BL to the nonsecure domain. A jump from the BL ("Go" command) to an application using other resources does not work.
- > Some secure option bytes are not be accessible as the bootloader is nonsecure. Some APIs are added on the SFSP to guarantee that the BL can modify them on open products states (Open, RDP L0).
- > Some parts of the SRAM are used by the SFSP and remain on Secure domain when jumping to the BL, so are not accessible by the customer through BL

- b) Boot not depending on TrustZone value - Boot_V3_2 (see [Figure 12](#))

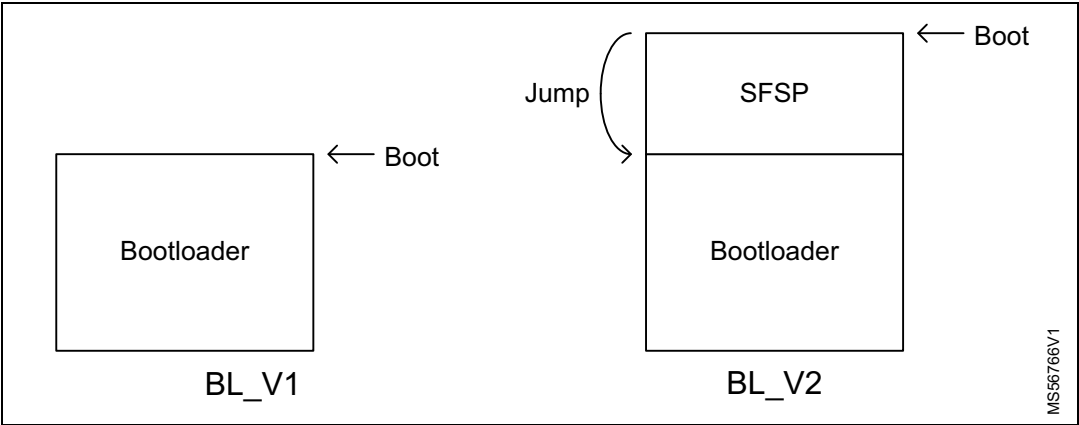
Boot goes through SFSP first, then jumps to the BL. In this model, some constraints are added to the BL functionalities on both TZEN states:

- > Boot timing includes SFSP timing.
- > Before jumping to the BL, the SFSP map all the needed resources by the BL to the nonsecure domain. A jump from the BL ("Go" command) to an application using other resources does not work.
- > Some secure option bytes are not be accessible by the Bootloader as it is nonsecure. Some APIs are added on the SFSP to guarantee that the BL can modify them on open products states (Open).
- > Some parts of the SRAM are used by the SFSP and remain on Secure domain when jumping to the BL, consequently they are not accessible by the customer through BL.

Table 9. BL and boot by product series

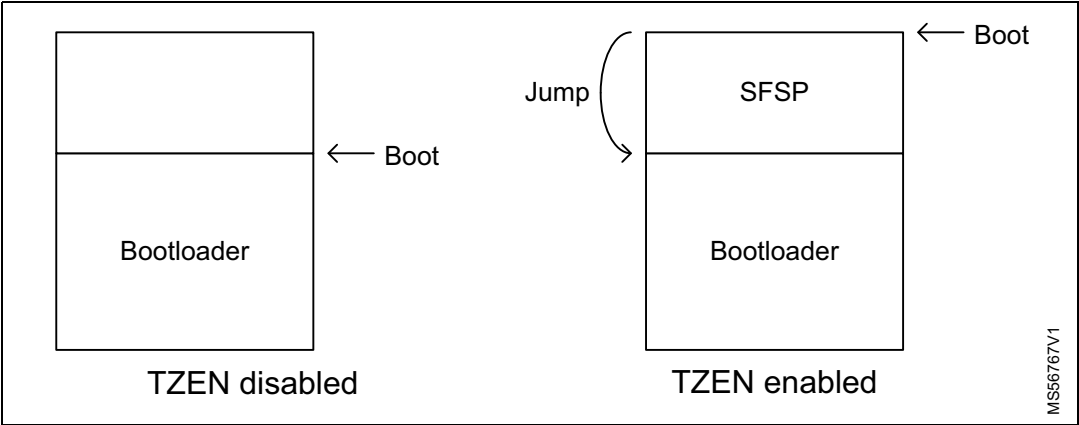
Series	BL model	Boot
STM32H7	BL_V2	Boot_V2 (see Figure 10)
STM32L5, STM32U3, STM32U5, STM32WBA		Boot_V3_1 (see Figure 11)
STM32H5		Boot_V3_2 (see Figure 12)
Others	BL_V1	Boot_V1 (see Figure 10)

Figure 10. Boot_V1 (left) and Boot_V2 (right)



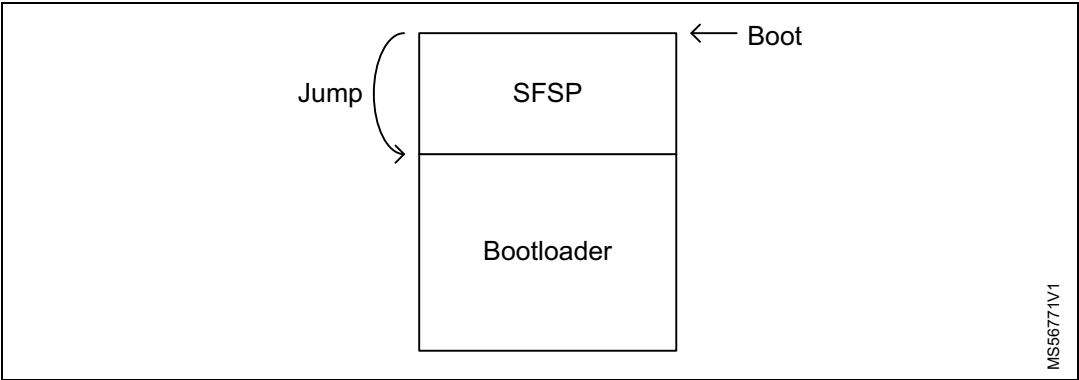
MS5676V1

Figure 11. Boot_V3_1



MS5676V1

Figure 12. Boot_V3_2



MS5677V1

5 STM32C011xx devices

5.1 Bootloader configuration

The STM32C011xx bootloader is activated by applying Pattern 11 (see [Table 2](#)). [Table 10](#) shows the hardware resources used by this bootloader.

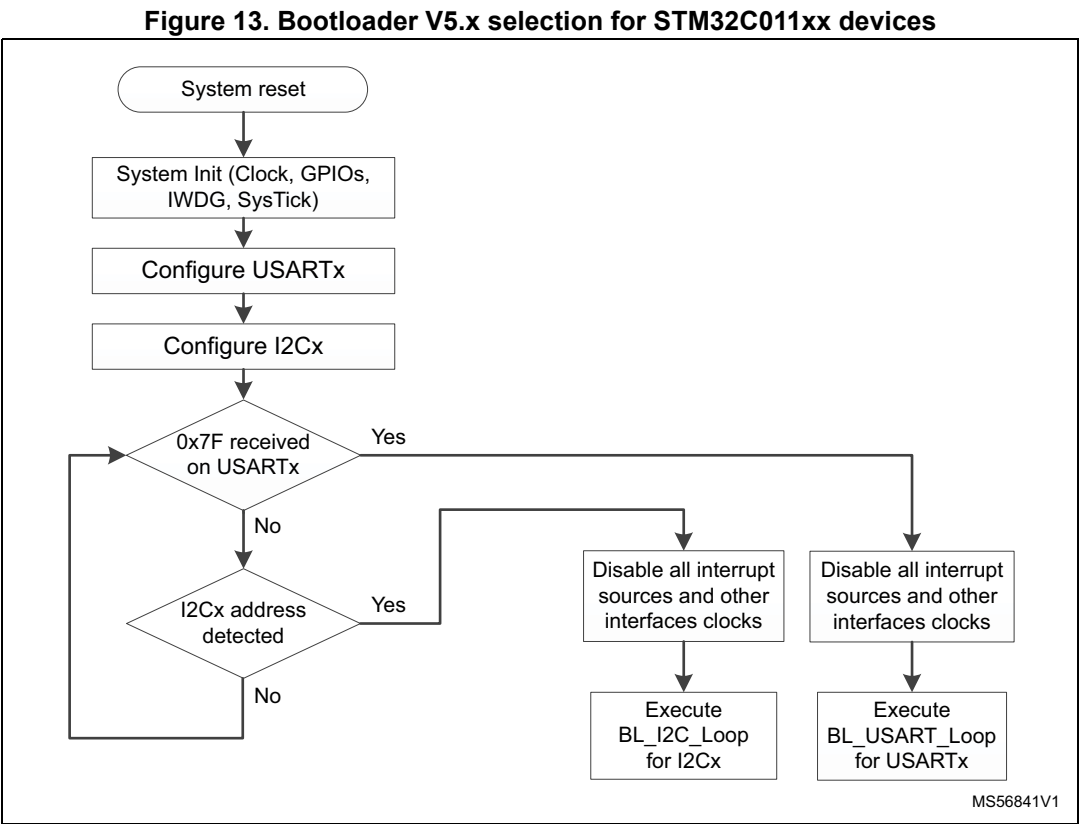
Table 10. STM32C011xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 24 MHz (no PLL)
	RAM	-	3.5 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	6 Kbytes, starting from address 0x1FFF0000 contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset if the hardware IWDG option was previously enabled by the user.
Securable memory area	-	-	The address to jump to for the securable memory area is 0x1FFF1600
USART1	USART1	Enabled	Once initialized, the USART1 configuration is 8 bits, even parity, and one stop bit.
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1100100x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain pull-up mode
	I2C1_SDA pin		PB7 pin: data line is used in open-drain pull-up mode

Note: On WLCSP12, SO8N, TSSOP20, and UFQFN20 packages USART1 PA9/PA10 IOs are remapped on PA11/PA12.

5.2 Bootloader selection

Figure 13 shows the bootloader selection mechanism.



5.3 Bootloader version

Table 11 lists the STM32C011xx devices bootloader versions.

Table 11. STM32C011xx bootloader versions

Version number	Description	Known limitations
V5.1	Initial bootloader version	None

6 STM32C031xx devices

6.1 Bootloader configuration

The STM32C031xx bootloader is activated by applying Pattern 11 (see [Table 2](#)). [Table 12](#) shows the hardware resources used by this bootloader.

Table 12. STM32C031xx configuration in system memory boot mode

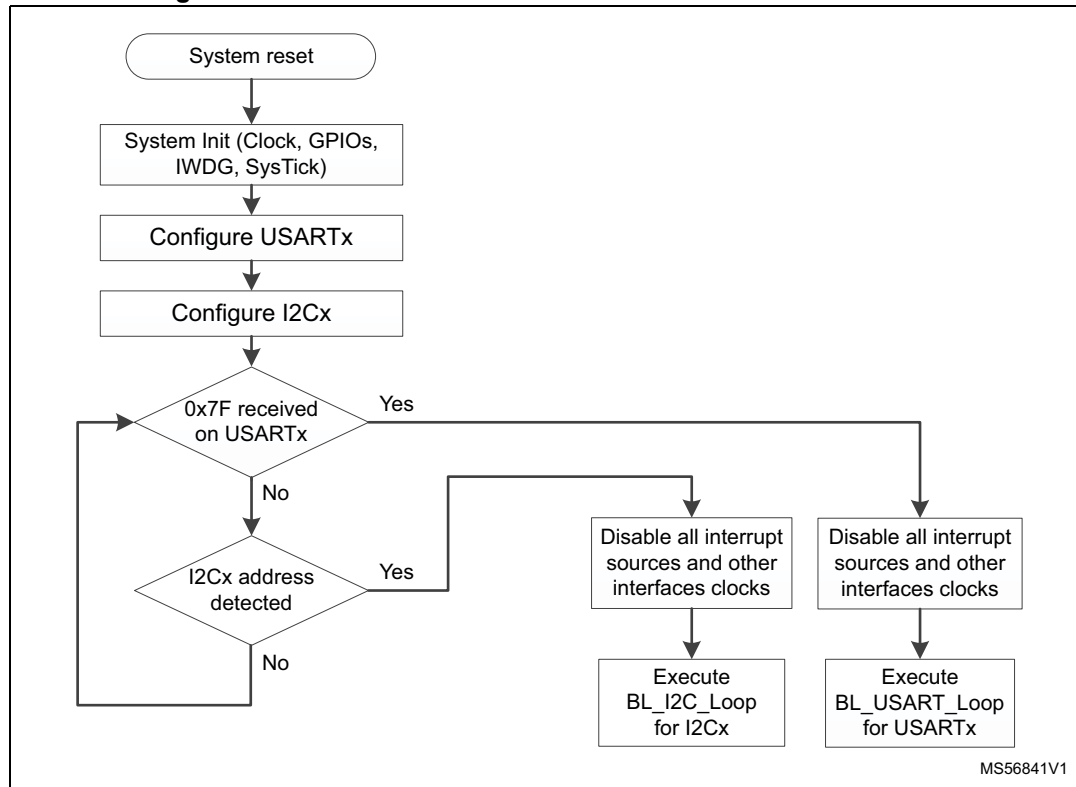
Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 24 MHz (no PLL)
	RAM	-	3.5 Kbytes, starting from address 0x20000000, are used by the bootloader firmware.
	System memory	-	6 Kbytes, starting from address 0x1FFF0000, contain the bootloader firmware.
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset if the hardware IWDG option was previously enabled by the user.
Securable memory area	-	-	The address to jump to for the securable memory area is 0x1FFF1600
USART1	USART1	Enabled	Once initialized, the USART1 configuration is 8 bits, even parity, and one stop bit.
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1100011x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain no pull mode.

Note: On TSSOP20 and UFQFN28 packages USART1 PA9/PA10 IOs are remapped on PA11/PA12.

6.2 Bootloader selection

Figure 14 shows the bootloader selection mechanism.

Figure 14. Bootloader V5.x selection for STM32C031xx devices



6.3 Bootloader version

Table 13 lists the STM32C031xx devices bootloader versions.

Table 13. STM32C031xx bootloader versions

Version number	Description	Known limitations
V5.2	Initial bootloader version	None

7 STM32C051xx devices

7.1 Bootloader configuration

The STM32C051xx bootloader is activated by applying Pattern 11 (see [Table 2](#)). [Table 14](#) shows the hardware resources used by this bootloader.

Table 14. STM32C051xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 24 MHz, derived directly from the HSI
	RAM	-	5 Kbytes, starting from address 0x2000 0000, are used by the bootloader firmware.
	System memory	-	12 Kbytes, starting from address 0x1FFF 0000, contain the bootloader firmware.
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset if the hardware IWDG option was previously enabled by the user.
Securable memory area	-	-	The address to jump to for the securable memory area is 0x1FFF 2E00
USART1	USART1	Enabled	Once initialized, the USART1 configuration is 8 bits, even parity, and one stop bit.
	USART1_RX pin	Input	PA10 pin: USART in reception mode. PA12 pin: as PA10 pin does not exist on WLCSP15, TSSOP20, and UFQFN28, PA12 is remapped to PA10. Used in alternate function with pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART in transmission mode. PA11 pin: as PA9 pin does not exist on WLCSP15, TSSOP20, and UFQFN28, PA11 is to PA9. Kept in reset configuration until 0x7F detected on USART_RX.
USART2	USART2	Enabled	Once initialized, the USART2 configuration is 8 bits, even parity, and one stop bit.
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate function, pull-up mode
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Kept in reset configuration until 0x7F detected on USART_RX
	EXTI line 11	Input	Used for USART detection. Baudrate calculation is based on this line interrupt.

Table 14. STM32C051xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1110110x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain pull-up mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain pull-up mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1110110x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PB10 pin: clock line is used in open-drain pull-up mode.
	I2C2_SDA pin		PB11 pin: data line is used in open-drain pull-up mode.
SPI1	SPI1	Enabled	The SPI1 configuration is: – Target mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: target data Input line, used in alternate function, pull down mode.
	SPI1_MISO pin ⁽¹⁾	Output	PA6 pin: target data output line, used in alternate function, pull down mode
	SPI1_SCK pin	Input	PA5 pin: target clock line, used in alternate function, pull down mode.
	SPI1_NSS pin		PA4 pin: slave chip select pin used in alternate function, pull down mode.

Table 14. STM32C051xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PB15 pin: Slave data Input line, used in alternate function, pull down mode.
	SPI2_MISO pin ⁽¹⁾	Output	PB14 pin: Slave data output line, used in alternate function, pull down mode
	SPI2_SCK pin	Input	PB13 pin: Slave clock line, used in alternate function, pull down mode.
	SPI2_NSS pin		PB12 pin: slave chip select pin used in alternate function, pull down mode.

1. SPI Tx (MISO) is handled by DMA. On the bootloader start-up after SPI initialization as soon as the bit DMATx enable on SPI CR2 register is set to 0x1, the MISO line is set to 3.3 V.

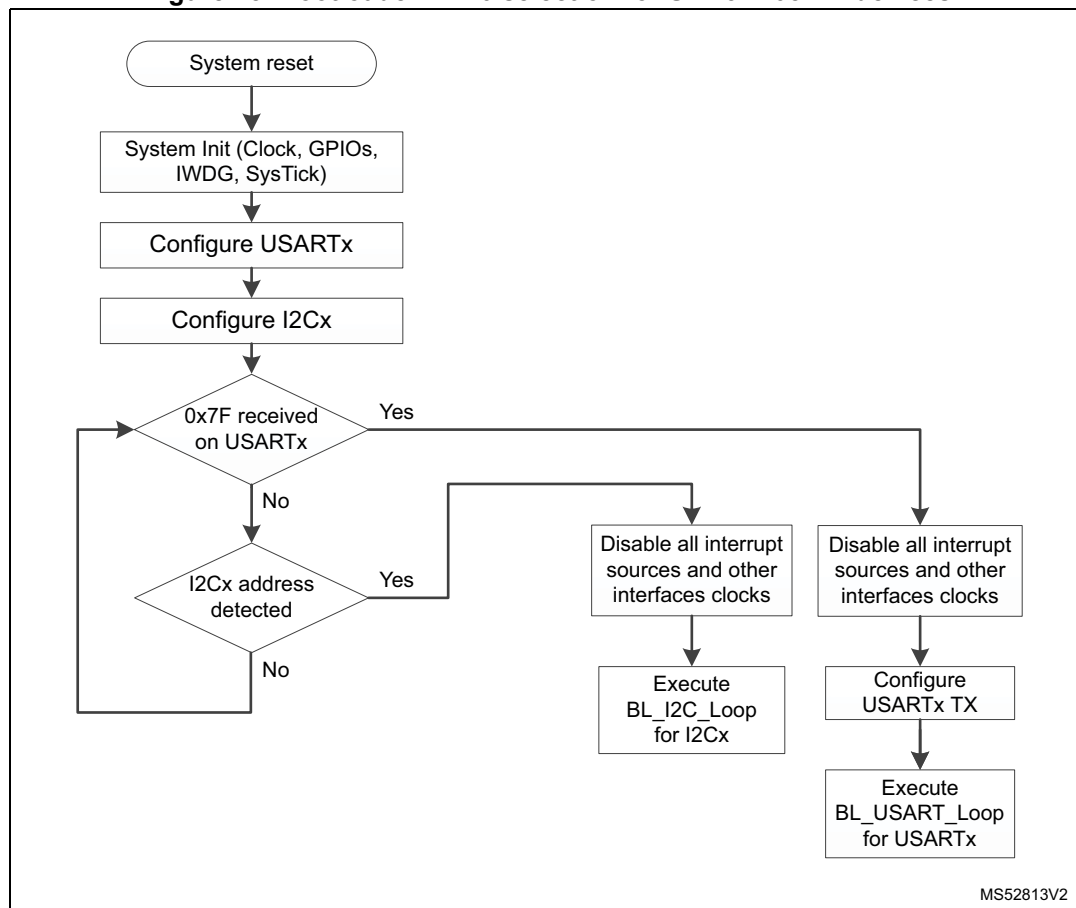
7.2 Boot model

The bootloader follows boot model V2 (see [Section 4.10](#)), there are no specific constraints.

7.3 Bootloader selection

Figure 15 shows the bootloader selection mechanism.

Figure 15. Bootloader V11.0 selection for STM32C051xx devices



7.4 Bootloader version

Table 15 lists the STM32C051xx devices bootloader versions.

Table 15. STM32C051xx bootloader versions

Version number	Description	Known limitations
V11.0	Initial bootloader version	Empty check flag cleared by error on the bootloader startup phase – Root cause: on the startup phase the bootloader SW performs a system deinitialization, leading to write the default value on the FLASH_ACR register, which overrides the Empty check bit with 0 – Behavior: when Empty check boot mode is used and the flash memory is empty, the MCU boots on the bootloader but the flag is cleared by the SW. If a reset is triggered, the system tries to boot on the empty flash memory, and crashes. – Caution: Avoid using reset on this case. If the system crashes, an option byte change or POR is needed to reboot.

8 STM32C071xx devices

8.1 Bootloader configuration

The STM32C071xx bootloader is activated by applying Pattern 11 (see [Table 2](#)). [Table 14](#) shows the hardware resources used by this bootloader.

Table 16. STM32C071xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 24 MHz (no PLL)
		HSI48 enabled	The clock recovery system (CRS) is enabled for the DFU bootloader so that USB can be clocked by HSI48 48 MHz
	RAM	-	8.75 Kbytes, starting from address 0x20000000, are used by the bootloader firmware.
	System memory	-	28 Kbytes, starting from address 0x1FFF0000, contain the bootloader firmware.
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset if the hardware IWDG option was previously enabled by the user.
Securable memory area	-	-	The address to jump to for the securable memory area is 0x1FFF1600
USART1	USART1	Enabled	Once initialized, the USART1 configuration is 8 bits, even parity, and one stop bit.
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Set as input until USART1 is detected.
USART2	USART2	Enabled	Once initialized, the USART2 configuration is 8 bits, even parity, and one stop bit.
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Set as input until USART2 is detected.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1110001x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain pull-up mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain pull-up mode.

Table 16. STM32C071xx configuration in system memory boot mode (continued)

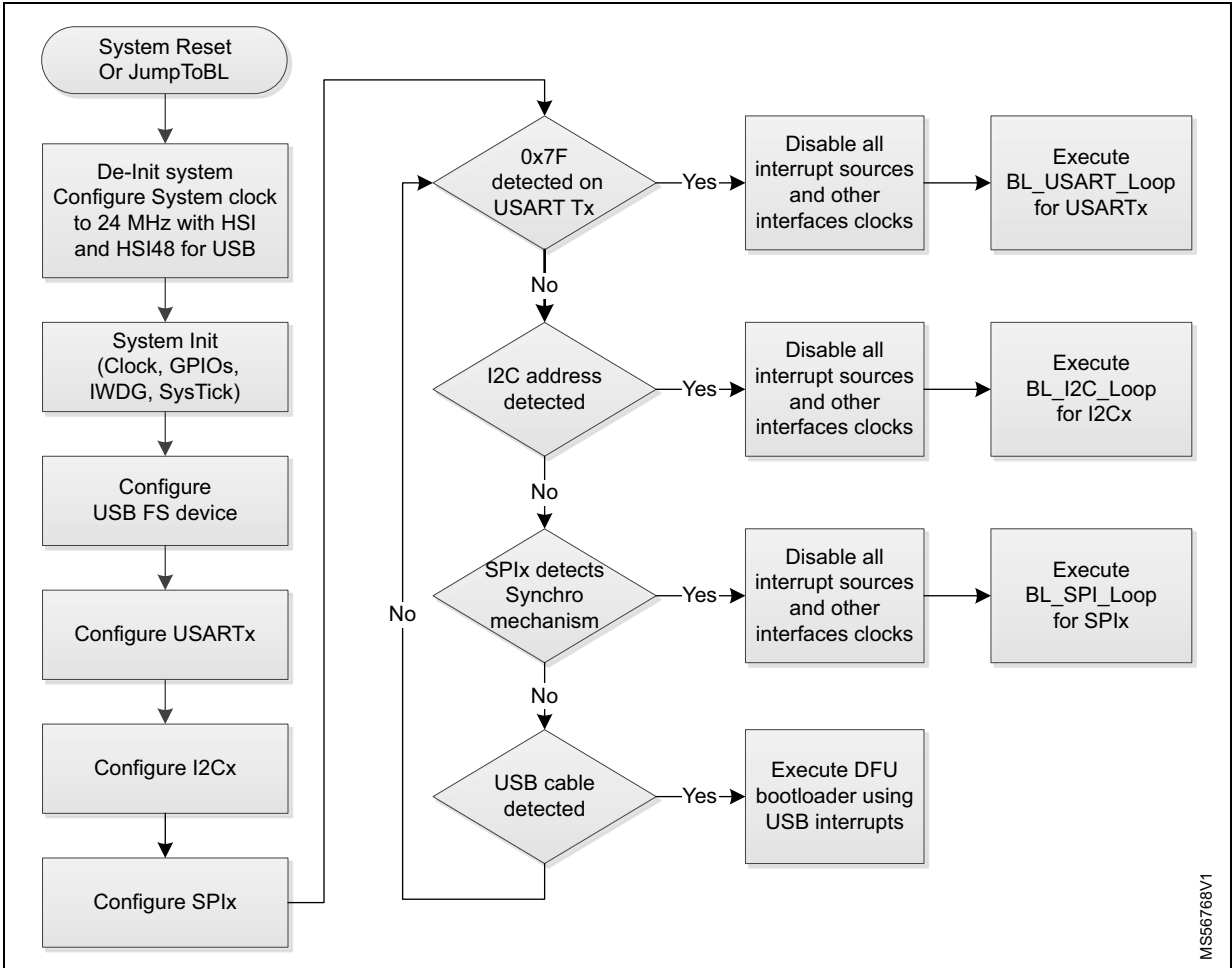
Bootloader	Feature/Peripheral	State	Comment
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1110001x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PB10 pin: clock line is used in open-drain pull-up mode.
	I2C2_SDA pin		PB11 pin: data line is used in open-drain pull-up mode.
SPI1	SPI1	Enabled	The SPI1 configuration is: – Target mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: target data Input line, used in push-pull, pull down mode.
	SPI1_MISO pin ⁽¹⁾	Output	PA6 pin: target data output line, used in push-pull, pull down mode
	SPI1_SCK pin	Input	PA5 pin: target clock line, used in push-pull, pull down mode.
	SPI1_NSS pin		PA4 pin: slave chip select pin used in push-pull, pull down mode.
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PB15 pin: Slave data Input line, used in push-pull, pull down mode.
	SPI2_MISO pin ⁽¹⁾	Output	PB14 pin: Slave data output line, used in push-pull, pull down mode
	SPI2_SCK pin	Input	PB13 pin: Slave clock line, used in push-pull, pull down mode.
	SPI2_NSS pin		PB12 pin: slave chip select pin used in push-pull, pull down mode.
DFU	USB	Enabled	USB FS configured in forced device mode. USB FS interrupt vector is enabled and used for USB DFU communications.
	USB_DM pin	Input/output	PA11: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12: USB DP line. Used in alternate push-pull, no pull mode. No external pull-up resistor is required.

- 1. SPI Tx (MISO) is handled by DMA. On the bootloader start-up after SPI initialization as soon as the bit DMATx enable on SPI CR2 register is set to 0x1, the MISO line is set to 3.3 V.

8.2 Bootloader selection

Figure 15 shows the bootloader selection mechanism.

Figure 16. Bootloader V13.1 selection for STM32C071xx devices



8.3 Bootloader version

Table 15 lists the STM32C071xx devices bootloader versions.

Table 17. STM32C071xx bootloader versions

Version number	Description	Known limitations
V13.1	Initial bootloader version	None

9 STM32C091xx/92xx devices

9.1 Bootloader configuration

The STM32C091xx/92xx bootloader is activated by applying Pattern 11 (see [Table 2](#)). [Table 14](#) shows the hardware resources used by this bootloader.

Table 18. STM32C091xx/92xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 24 MHz, derived directly from the HSI
	RAM	-	5 Kbytes, starting from address 0x2000 0000, are used by the bootloader firmware.
	System memory	-	12 Kbytes, starting from address 0x1FFF 0000, contain the bootloader firmware.
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset if the hardware IWDG option was previously enabled by the user.
Securable memory area	-	-	The address to jump to for the securable memory area is 0x1FFF 3E00
USART1	USART1	Enabled	Once initialized, the USART1 configuration is 8 bits, even parity, and one stop bit.
	USART1_RX pin	Input	PA10 pin: USART in reception mode. PA12 pin: as PA10 pin does not exist on TSSOP20, WLCSP24, and UFQFN28, PA12 is remapped to PA10. Used in alternate function with pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART in transmission mode. PA11 pin: as PA9 pin does not exist on TSSOP20, WLCSP24, and UFQFN28, PA11 is remapped to PA9. Kept in reset configuration until 0x7F detected on USART_RX.
USART2	USART2	Enabled	Once initialized, the USART2 configuration is 8 bits, even parity, and one stop bit.
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate function, pull-up mode
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Kept in reset configuration until 0x7F detected on USART_RX
	EXTI line 3	Input	Used for USART detection. Baudrate calculation is based on this line interrupt.

Table 18. STM32C091xx/92xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
USART3	USART3	Enabled	Once initialized the USART3 configuration is: 8- bits, even parity and 1 Stop bit
	USART3_RX pin	Input	PC11 pin: USART in reception mode. Used in alternate function with pull-up mode
	USART3_TX pin	Output	PC10 pin: USART in transmission mode. Kept in reset configuration until 0x7F detected on USART_RX
	EXTI line 11	Input	Used for USART detection. Baudrate calculation is based on this line interrupt
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1110110x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain pull-up mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain pull-up mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1110110x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PB10 pin: clock line is used in open-drain pull-up mode.
	I2C2_SDA pin		PB11 pin: data line is used in open-drain pull-up mode.
SPI1	SPI1	Enabled	The SPI1 configuration is: – Target mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: target data Input line, used in alternate function, pull down mode.
	SPI1_MISO pin ⁽¹⁾	Output	PA6 pin: target data output line, used in alternate function, pull down mode
	SPI1_SCK pin	Input	PA5 pin: target clock line, used in alternate function, pull down mode.
	SPI1_NSS pin		PA4 pin: slave chip select pin used in alternate function, pull down mode.

Table 18. STM32C091xx/92xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PB15 pin: Slave data Input line, used in alternate function, pull down mode.
	SPI2_MISO pin ⁽¹⁾	Output	PB14 pin: Slave data output line, used in alternate function, pull down mode
	SPI2_SCK pin	Input	PB13 pin: Slave clock line, used in alternate function, pull down mode.
	SPI2_NSS pin		PB12 pin: slave chip select pin used in alternate function, pull down mode.
FDCAN1	FDCAN	Enabled ⁽²⁾	Once initialized the FDCAN configuration is: – Connection bit rate 250 kbit/s – Data bit rate 1000 kbit/s – FrameFormat = FDCAN_FRAME_FD_BRS – Mode = FDCAN_MODE_NORMAL – AutoRetransmission = ENABLE – TransmitPause = DISABLE – ProtocolException = ENABLE
	FDCAN_Rx pin	Input	FDCAN in reception mode. Used in alternate function with pull up mode. – PB0 for WLCSP24, UFQFN28, UFQFN32, and LQFP32 – PD0 for UFQFN48, LQFP48, UFBGA64, and LQFP644
	FDCAN_Tx pin	Output	FDCAN in transmission mode. Used in alternate function with pull up mode – PB1 for WLCSP24, UFQFN28, UFQFN32, and LQFP32 – PD1 for UFQFN48, LQFP48, UFBGA64, and LQFP64

1. SPI Tx (MISO) is handled by DMA. On the bootloader start-up after SPI initialization, as soon as the bit DMATx enable on SPI CR2 register is set to 0x1, the MISO line is set to 3.3 V.

2. Only when enabled by engineering bytes. Not supported on TSSOP20 package, even if enabled by engineering bytes.

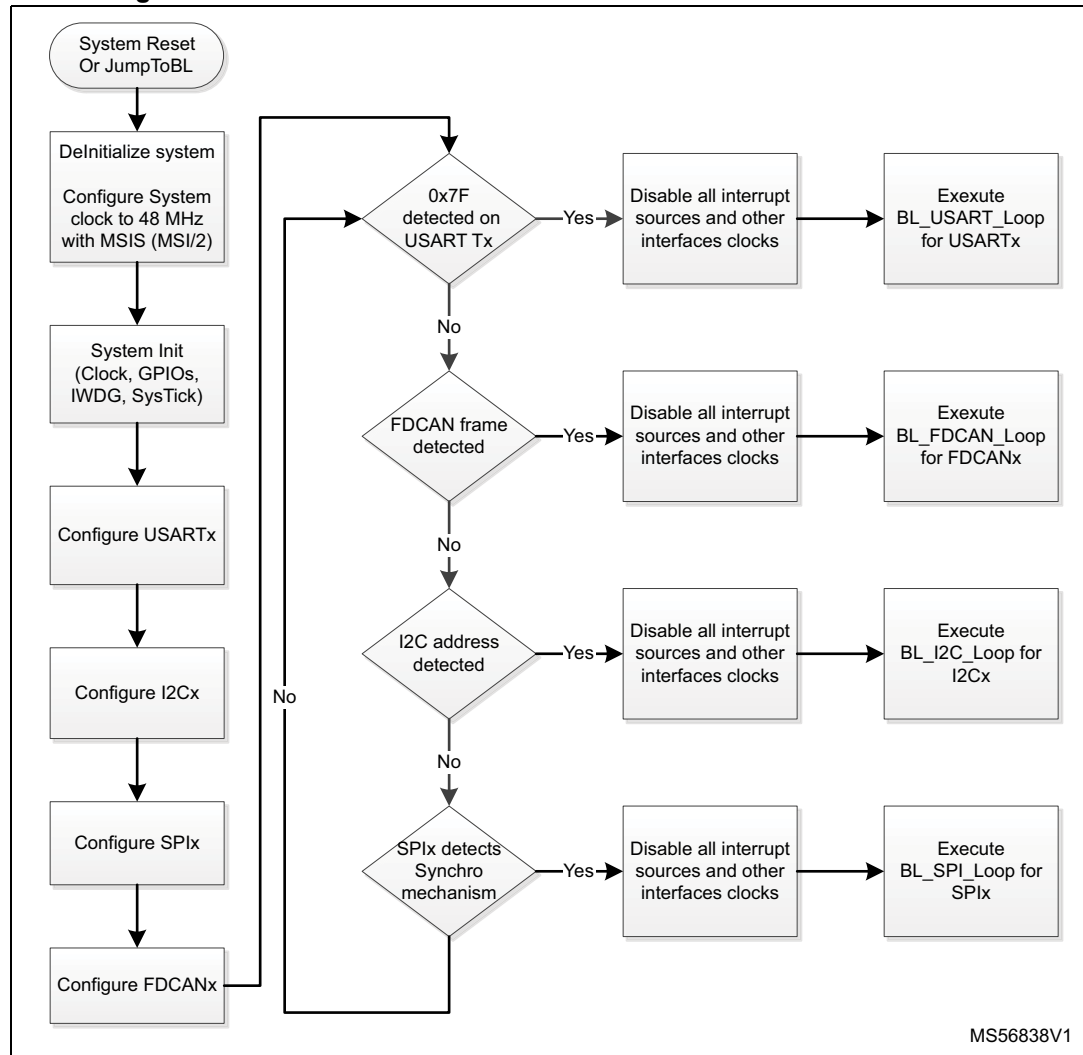
9.2 Boot model

The bootloader follows boot model V2 (see [Section 4.10](#)), there are no specific constraints.

9.3 Bootloader selection

Figure 15 shows the bootloader selection mechanism.

Figure 17. Bootloader V18.1 selection for STM32C091xx/92xx devices



9.4 Bootloader version

[Table 15](#) lists the STM32C091xx/92xx devices bootloader versions.

Table 19. STM32C091xx/92xx bootloader versions

Version number	Description	Known limitations
V18.1	Initial bootloader version	Empty check flag cleared by error on the bootloader startup phase – Root cause: on the startup phase the bootloader SW performs a system deinitialization, leading to write the default value on the FLASH_ACR register, which overrides the Empty check bit with 0 – Behavior: when Empty check boot mode is used and the flash memory is empty, the MCU boots on the bootloader but the flag is cleared by the SW. If a reset is triggered, the system tries to boot on the empty flash memory, and crashes. – Caution: Avoid using reset on this case. if the system crashes, an option byte change or POR is needed to reboot.

10 STM32F03xx4/6 devices

10.1 Bootloader configuration

The STM32F03xx4/6 bootloader is activated by applying Pattern 2 (see [Table 2](#)). [Table 20](#) shows the hardware resources used by this bootloader.

Table 20. STM32F03xx4/6 configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 24 MHz (using PLL clocked by HSI). 1 flash Wait State.
	RAM	-	2 Kbytes, starting from address 0x20000000, are used by the bootloader firmware.
	System memory	-	3 Kbytes, starting from address 0x1FFFE000 contain the bootloader firmware.
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset if the hardware IWDG option was previously enabled by the user.
USART1 (on PA10/PA9)	USART1	Enabled	Once initialized, the USART1 configuration is 8 bits, even parity, and one stop bit.
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
USART1 (on PA14/PA15)	USART1	Enabled	Once initialized, the USART1 configuration is 8 bits, even parity, and one stop bit.
	USART1_RX pin	Input	PA15 pin: USART1 in reception mode. Used in input pull-up mode.
	USART1_TX pin	Output	PA14 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
USART1s	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host.

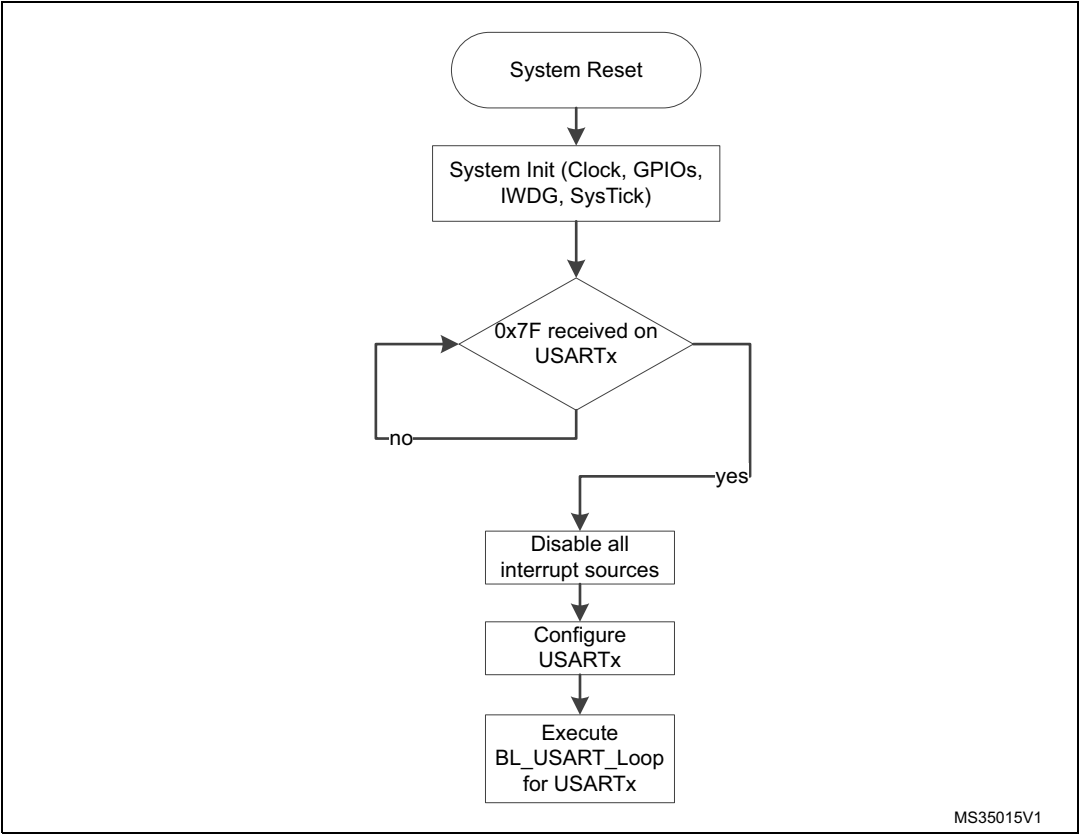
The system clock is derived from the embedded internal high-speed RC. No external quartz is required for the bootloader execution.

Note: *After the STM32F03xx4/6 device has booted in bootloader mode, serial wire debug (SWD) communication is no longer possible until the system is reset. This is because the SWD uses the PA14 pin (SWCLK), already used by the bootloader (USART1_TX).*

10.2 Bootloader selection

Figure 18 shows the bootloader selection mechanism.

Figure 18. Bootloader selection for STM32F03xx4/6 devices



10.3 Bootloader version

Table 21 lists the STM32F03xx4/6 devices bootloader versions.

Table 21. STM32F03xx4/6 bootloader versions

Version number	Description	Known limitations
V1.0	Initial bootloader version	For the USART interface, two consecutive NACKs instead of one are sent when a Read Memory or Write Memory command is sent and the RDP level is active.

11 STM32F030xC devices

11.1 Bootloader configuration

The STM32F030xC bootloader is activated by applying Pattern 2 (see [Table 2](#)). [Table 22](#) shows the hardware resources used by this bootloader.

Table 22. STM32F030xC configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 48 MHz with HSI 8 MHz as clock source.
	RAM	-	6 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	8 Kbytes, starting from address 0x1FFFD800, contain the bootloader firmware.
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
USART2 (on PA2/PA3)	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in input pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in input pull-up mode.
USART2 (on PA14/PA15)	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA15 pin: USART2 in reception mode. Used in input pull-up mode.
	USART2_TX pin	Output	PA14 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON Target 7-bit address: 0b1000001x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain no pull mode.

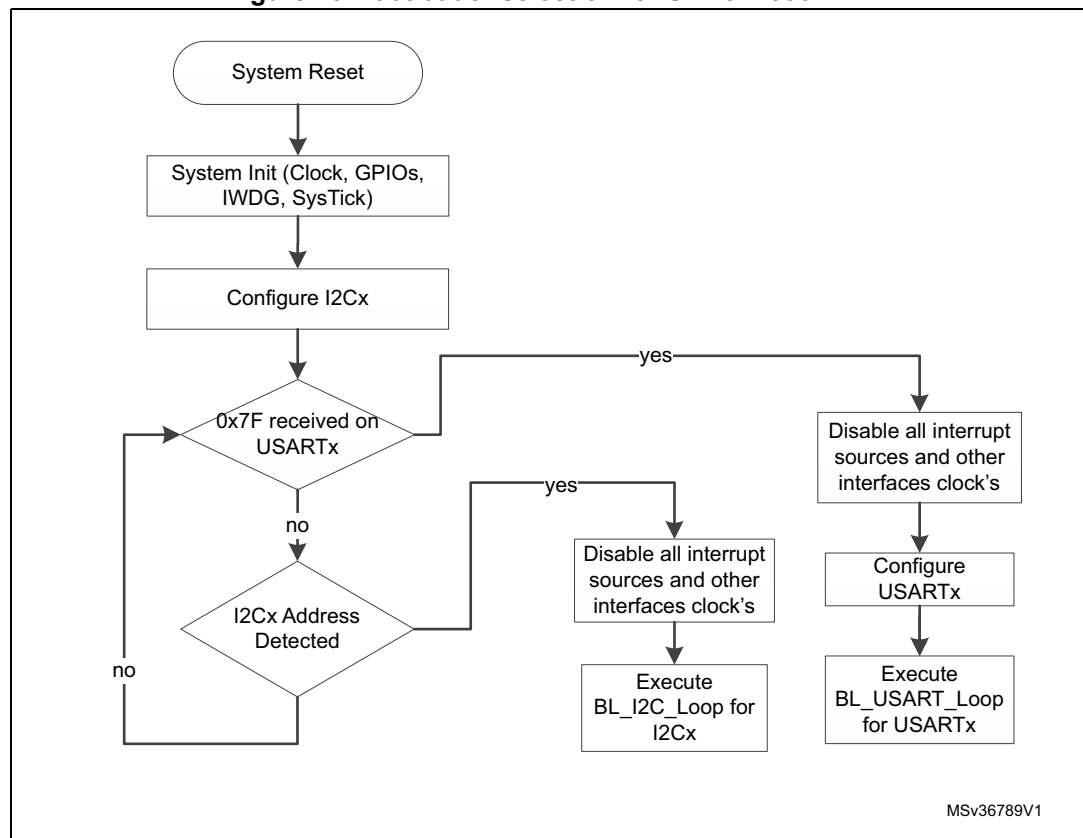
Note: After the devices have booted in bootloader mode using USART2, the serial wire debug (SWD) communication is no more possible until the system is reset, because SWD uses PA14 pin (SWCLK), already used by the bootloader (USART2_RX).

The system clock is derived from the embedded internal high-speed RC. No external quartz is required for the bootloader execution.

11.2 Bootloader selection

Figure 19 shows the bootloader selection mechanism.

Figure 19. Bootloader selection for STM32F030xC



11.3 Bootloader version

Table 23 lists the STM32F030xC devices bootloader versions.

Table 23. STM32F030xC bootloader versions

Version number	Description	Known limitations
V5.2	Initial bootloader version	PA13 is set in input pull-up mode even if not used by the bootloader

12 STM32F05xxx and STM32F030x8 devices

12.1 Bootloader configuration

The STM32F05xxx and STM32F030x8 devices bootloader is activated by applying Pattern 2 (described in [Table 2](#)). [Table 24](#) shows the hardware resources used by this bootloader.

Table 24. STM32F05xxx and STM32F030x8 devices configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 24 MHz (using PLL clocked by HSI). 1 flash Wait State.
	RAM	-	2 Kbytes, starting from address 0x20000000, are used by the bootloader firmware.
	System memory	-	3 Kbytes, starting from address 0x1FFFE000, contain the bootloader firmware.
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset if the hardware IWDG option was previously enabled by the user.
USART1	USART1	Enabled	Once initialized, the USART1 configuration is 8 bits, even parity, and one stop bit.
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
USART2	USART2	Enabled	Once initialized, the USART2 configuration is 8 bits, even parity, and one stop bit.
	USART2_RX pin	Input	PA15 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA14 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host.

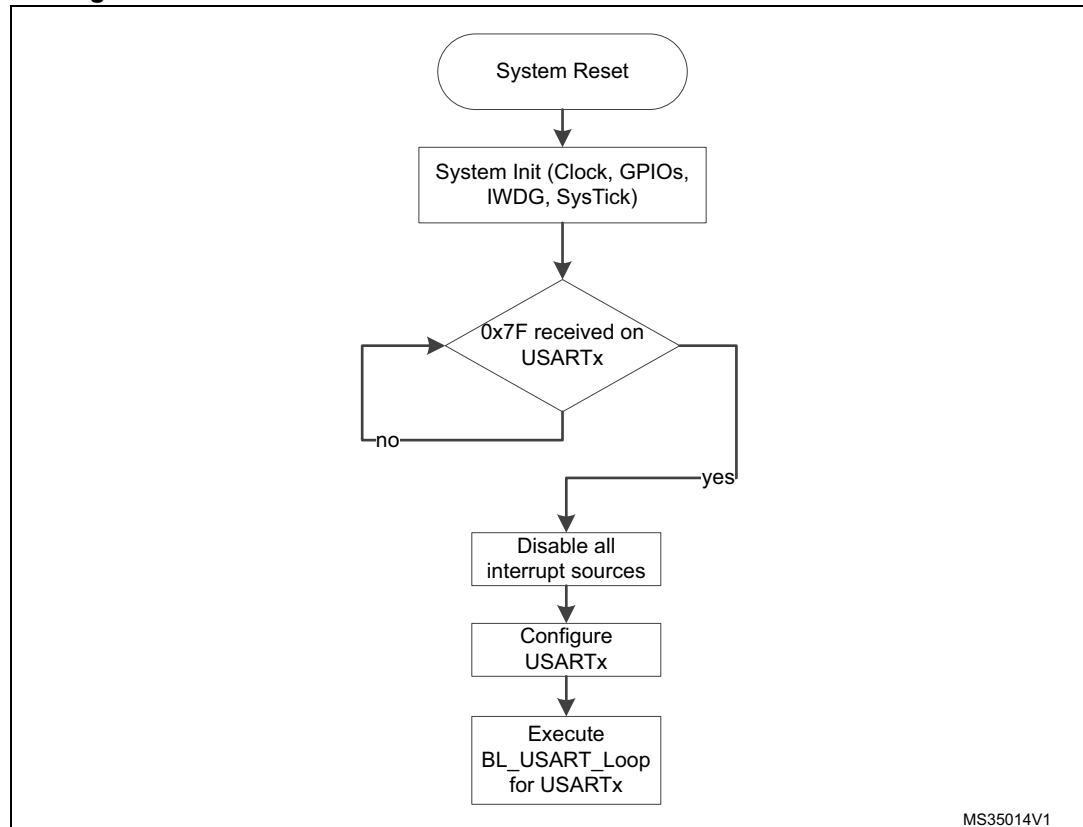
The system clock is derived from the embedded internal high-speed RC. No external quartz is required for the bootloader execution.

Note: *After the STM32F05xxx and STM32F030x8 devices have booted in bootloader mode, the serial wire debug (SWD) communication is no more possible until the system is reset, because SWD uses PA14 pin (SWCLK), already used by the bootloader (USART2_TX).*

12.2 Bootloader selection

Figure 20 shows the bootloader selection mechanism.

Figure 20. Bootloader selection for STM32F05xxx and STM32F030x8 devices



12.3 Bootloader version

Table 25 lists the STM32F05xxx and STM32F030x8 devices bootloader versions.

Table 25. STM32F05xxx and STM32F030x8 devices bootloader versions

Version number	Description	Known limitations
V2.1	Initial bootloader version	– At bootloader startup, the HSITRIM value is set to 0 (in HSITRIM bits on RCC_CR register) instead of default value (16), as a consequence a deviation is generated in crystal measurement. For better results, use the smallest supported crystal value (i.e. 4 MHz). – For the USART interface, two consecutive NACKs instead of 1 NACK are sent when a Read Memory or Write Memory command is sent and the RDP level is active. – PA13 is set in input pull-up mode even if not used by the Bootloader.

13 STM32F04xxx devices

13.1 Bootloader configuration

The STM32F04xxx bootloader is activated by applying Pattern 6 (described in [Table 2](#)). [Table 26](#) shows the hardware resources used by this bootloader.

Table 26. STM32F04xxx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 48 MHz with HSI48 48 MHz as clock source.
		-	The clock recovery system (CRS) is enabled for the DFU to allow USB to be clocked by HSI48 48 MHz.
	RAM	-	6 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	13 Kbytes, starting from address 0x1FFFC400, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA15 pin: USART2 in reception mode. Used in input pull-up mode.
	USART2_TX pin	Output	PA14 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b0111110x (x = 0 for write and x = 1 for read).
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin	Input/output	PB7 pin: data line is used in open-drain no pull mode.

Table 26. STM32F04xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
DFU	USB	Enabled	USB used in FS mode
	USB_DM pin	Input/output	PA11: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12: USB DP line No external pull-up resistor is required. Used in alternate push-pull, no pull mode.

Note: After the devices have booted in bootloader mode using USART2, the serial wire debug (SWD) communication is no more possible until the system is reset, because SWD uses PA14 pin (SWCLK), already used by the bootloader (USART2_RX).

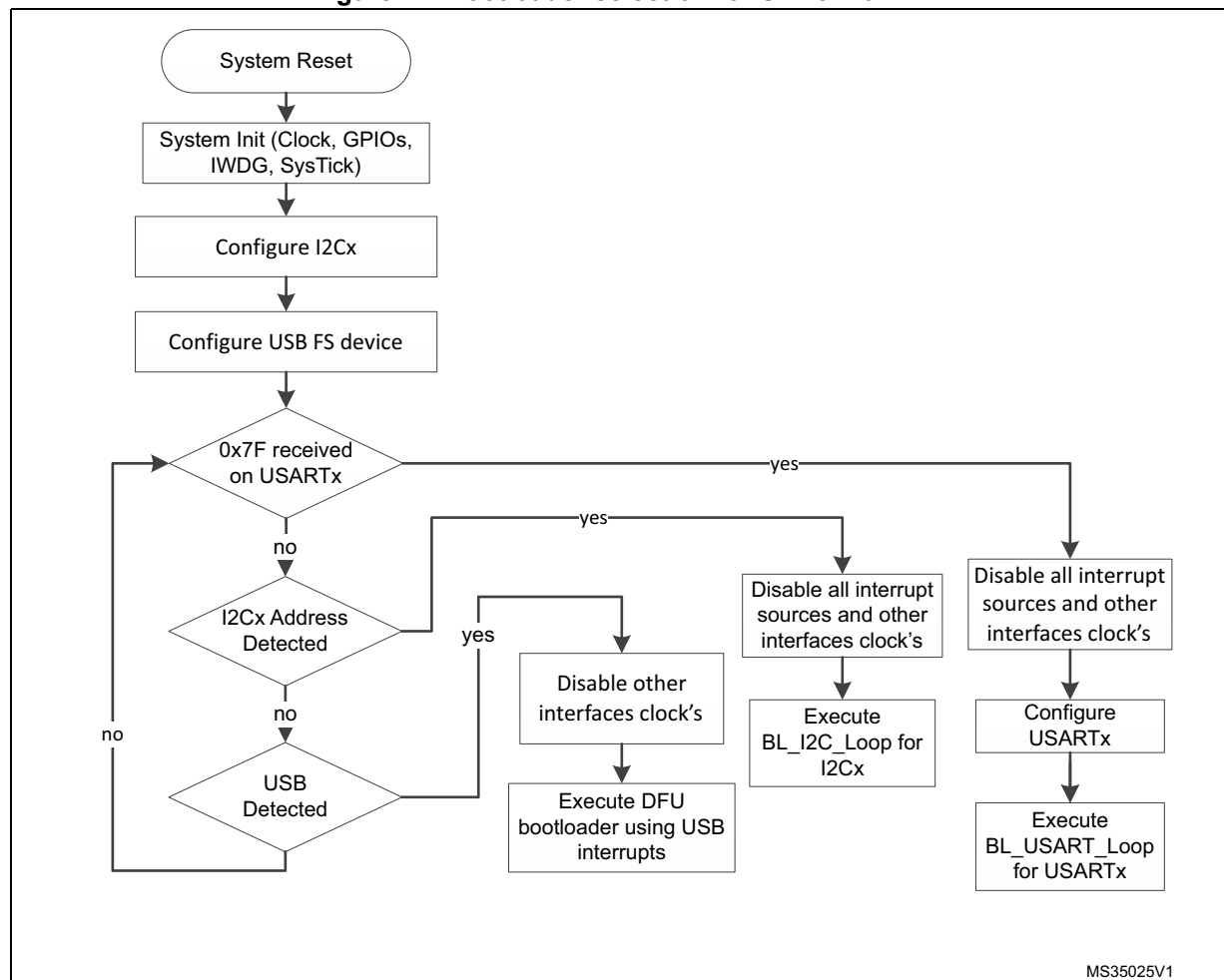
The system clock is derived from the embedded internal high-speed RC. No external quartz is required for the bootloader execution.

Note: Due to empty check mechanism present on these products, it is not possible to jump from user code to system bootloader. Such jump results in a jump back to user flash memory space. If the first four bytes of User flash (at 0x0800 0000) are empty at the moment of jump (i.e. erase first sector before jump or execute code from SRAM while flash is empty), then system bootloader is executed when jumped to.

13.2 Bootloader selection

Figure 21 shows the bootloader selection mechanism.

Figure 21. Bootloader selection for STM32F04xxx



13.3 Bootloader version

Table 27. STM32F04xxx bootloader versions

Version number	Description	Known limitations
V10.0	Initial bootloader version	At bootloader startup, the HSITRIM value is set to 0 (in HSITRIM bits on RCC_CR register) instead of default value (16), as a consequence a deviation is generated in crystal measurement. For better results, use the smallest supported crystal value (4 MHz). PA13 is set in input pull-up mode even if not used by the bootloader.
V10.1	Add dynamic support of USART/USB interfaces on PA11/12 IOs for small packages.	

14 STM32F070x6 devices

14.1 Bootloader configuration

The STM32F070x6 bootloader is activated by applying Pattern 6 (described in [Table 2](#)). [Table 28](#) shows the hardware resources used by this bootloader.

Table 28. STM32F070x6 configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	At startup, the system clock frequency is configured to 48 MHz using the HSI. If an external clock (HSE) is not present, the system is kept clocked from the HSI.
		HSE enabled	The external clock can be used for all bootloader interfaces and must have one of the following values: 24, 18, 16, 12, 8, 6, 4 MHz. The PLL is used to generate 48 MHz for USB and system clock.
		-	The CSS interrupt is enabled for HSE. Any failure (or removal) of the external clock generates system reset.
	RAM	-	6 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	13 Kbytes, starting from address 0x1FFFC400, contain the bootloader firmware.
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA15 pin: USART2 in reception mode
	USART2_TX pin	Output	PA14 pin: USART2 in transmission mode
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz – 7-bit address, Target mode – Analog filter ON – Target 7-bit address: 0b0111110x x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain mode.
DFU	USB	Enabled	USB FS configured in forced device mode. USB FS interrupt vector is enabled and used for USB DFU communications.
	USB_DM pin	Input/output	PA11 pin: USB FS DM line
	USB_DP pin		PA12 pin: USB FS DP line. No external pull-up resistor is required.

Note: If HSI deviation exceeds 1% the bootloader might not function correctly.

Note: After the STM32F070x6 devices have booted in bootloader mode using USART2, the serial wire debug (SWD) communication is no more possible until the system is reset, because SWD uses PA14 pin (SWCLK), already used by the bootloader (USART2_RX).

The bootloader has two cases of operation depending on the presence of the external clock (HSE) at bootloader startup:

- If HSE is present and has a value of 24, 18, 16, 12, 8, 6, or 4 MHz, the system clock is configured to 48 MHz with HSE as clock source. The DFU interface, USART1, USART2, and I2C1 are functional and can be used to communicate with the bootloader device.
- If HSE is not present, the HSI is kept as default clock source and only USART1, USART2, and I2C1 are functional.

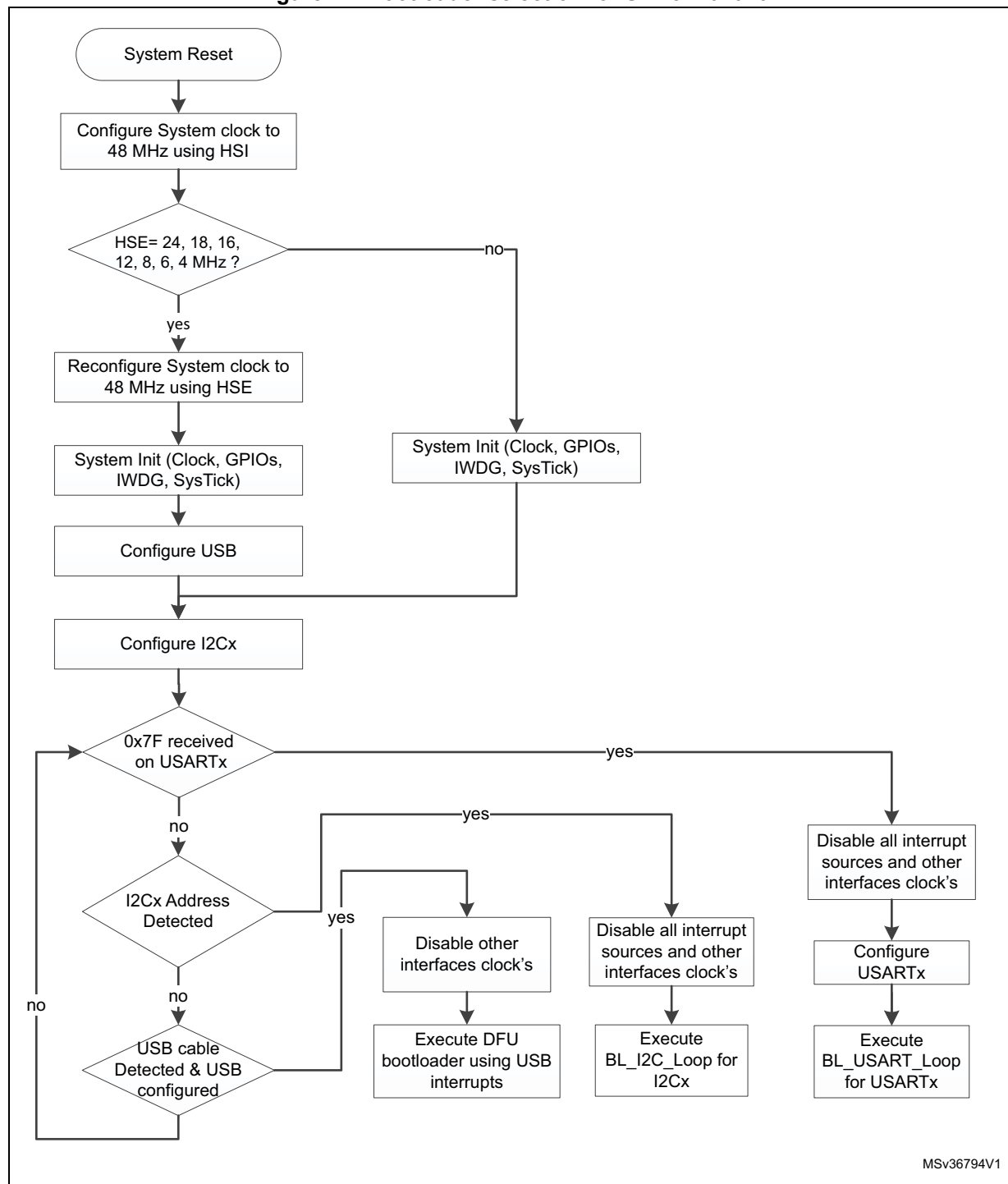
The external clock (HSE) must be kept if it is connected at bootloader startup, because it is used as system clock source.

Note: Due to empty check mechanism present on this product, it is not possible to jump from user code to system bootloader. Such jump results in a jump back to user flash space, but if the first four bytes of user flash (at 0x0800 0000) are empty at the moment of jump (i.e. erase first sector before jump or execute code from SRAM while flash is empty), then system bootloader is executed when jumped to.

14.2 Bootloader selection

Figure 22 shows the bootloader selection mechanism.

Figure 22. Bootloader selection for STM32F070x6



14.3 Bootloader version

[Table 29](#) lists the STM32F070x6 devices bootloader versions.

Table 29. STM32F070x6 bootloader versions

Version number	Description	Known limitations
V10.2	Initial bootloader version	At bootloader startup, the HSITRIM value is set to 0 (in HSITRIM bits on RCC_CR register) instead of default value (16), as a consequence a deviation is generated in crystal measurement. For better results, use the smallest supported crystal value (4 MHz).
V10.3	Clock configuration fixed to HSI 8 MHz	

15 STM32F070xB devices

15.1 Bootloader configuration

The STM32F070xB bootloader is activated by applying Pattern 2 (described in [Table 2](#)). [Table 30](#) shows the hardware resources used by this bootloader.

Table 30. STM32F070xB configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	At startup, the system clock frequency is configured to 48 MHz using the HSI. If an external clock (HSE) is not present, the system is kept clocked from the HSI.
		HSE enabled	The external clock can be used for all bootloader interfaces and must have one of the following values: 24, 18, 16, 12, 8, 6, 4 MHz. The PLL is used to generate 48 MHz for USB and system clock.
		-	The clock security system (CSS) interrupt is enabled for HSE. Any failure (or removal) of the external clock generates a system reset.
	RAM	-	6 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	12 Kbytes, starting from address 0x1FFFC800, contain the bootloader firmware.
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in input pull-up mode
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA15 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA14 pin: USART2 in transmission mode. Used in input pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b0111011x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain no pull mode.

Table 30. STM32F070xB configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
DFU	USB	Enabled	USB FS configured in forced device mode. USB FS interrupt vector is enabled and used for USB DFU communications.
	USB_DM pin	Input/output	PA11 pin: USB FS DM line used in alternate push-pull, no pull mode.
	USB_DP pin		PA12 pin: USB FS DP line used in alternate push-pull, no pull mode. No external pull-up resistor is required.

Note: *If HSI deviation exceeds 1% the bootloader might not function correctly.*

Note: *After the devices have booted in bootloader mode using USART2, the serial wire debug (SWD) communication is no more possible until the system is reset, because SWD uses PA14 pin (SWCLK), already used by the bootloader (USART2_RX).*

The bootloader has two cases of operation depending on the presence of the external clock (HSE) at bootloader startup:

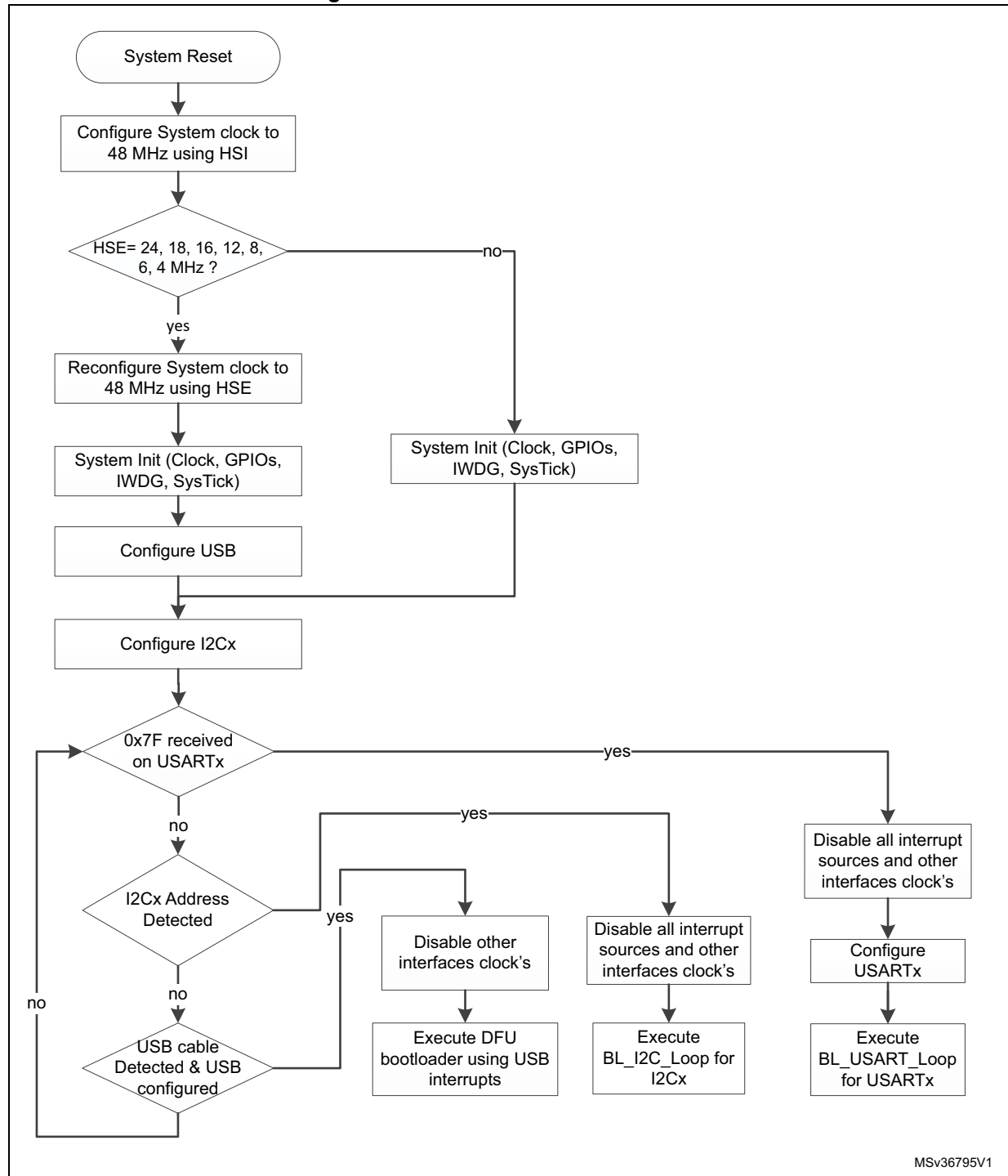
- If HSE is present and has a value of 24, 18, 16, 12, 8, 6, or 4 MHz, the system clock is configured to 48 MHz with HSE as clock source. The DFU interface, USART1, USART2, and I2C1 are functional and can be used to communicate with the bootloader device.
- If HSE is not present, the HSI is kept as default clock source and only USART1, USART2, and I2C1 are functional.

The external clock (HSE) must be kept if it is connected at bootloader startup, because it is used as system clock source.

15.2 Bootloader selection

Figure 23 shows the bootloader selection mechanism.

Figure 23. Bootloader selection for STM32F070xB



15.3 Bootloader version

[Table 31](#) lists the STM32F070xB devices bootloader versions.

Table 31. STM32F070xB bootloader versions

Version number	Description	Known limitations
V10.2	Initial bootloader version	At bootloader startup, the HSITRIM value is set to 0 (in HSITRIM bits on RCC_CR register) instead of default value (16), as a consequence a deviation is generated in crystal measurement. For better results, use the smallest supported crystal value (4 MHz). PA13 is set in alternate push-pull mode even if not used by the bootloader.
V10.3	Clock configuration fixed to HSI 8 MHz	

16 STM32F071xx/072xx devices

16.1 Bootloader configuration

The STM32F071xx/072xx bootloader is activated by applying Pattern 2 (described in [Table 2](#)). [Table 32](#) shows the hardware resources used by this bootloader.

Table 32. STM32F071xx/072xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 48 MHz with HSI48 48 MHz as clock source.
		-	CRS is enabled for the DFU to allow USB to be clocked by HSI48 48 MHz.
	RAM	-	6 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	12 Kbytes, starting from address 0x1FFFC800, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA15 pin: USART2 in reception mode. Used in input pull-up mode
	USART2_TX pin	Output	PA14 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz – 7-bit address, Target mode – Analog filter ON – Target 7-bit address: 0b0111011x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain pull-up mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain pull-up mode.

Table 32. STM32F071xx/072xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
DFU	USB	Enabled	USB used in FS mode
	USB_DM pin	Input/output	PA11: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12: USB DP line No external pull-up resistor is required. Used in alternate push-pull, no pull mode.

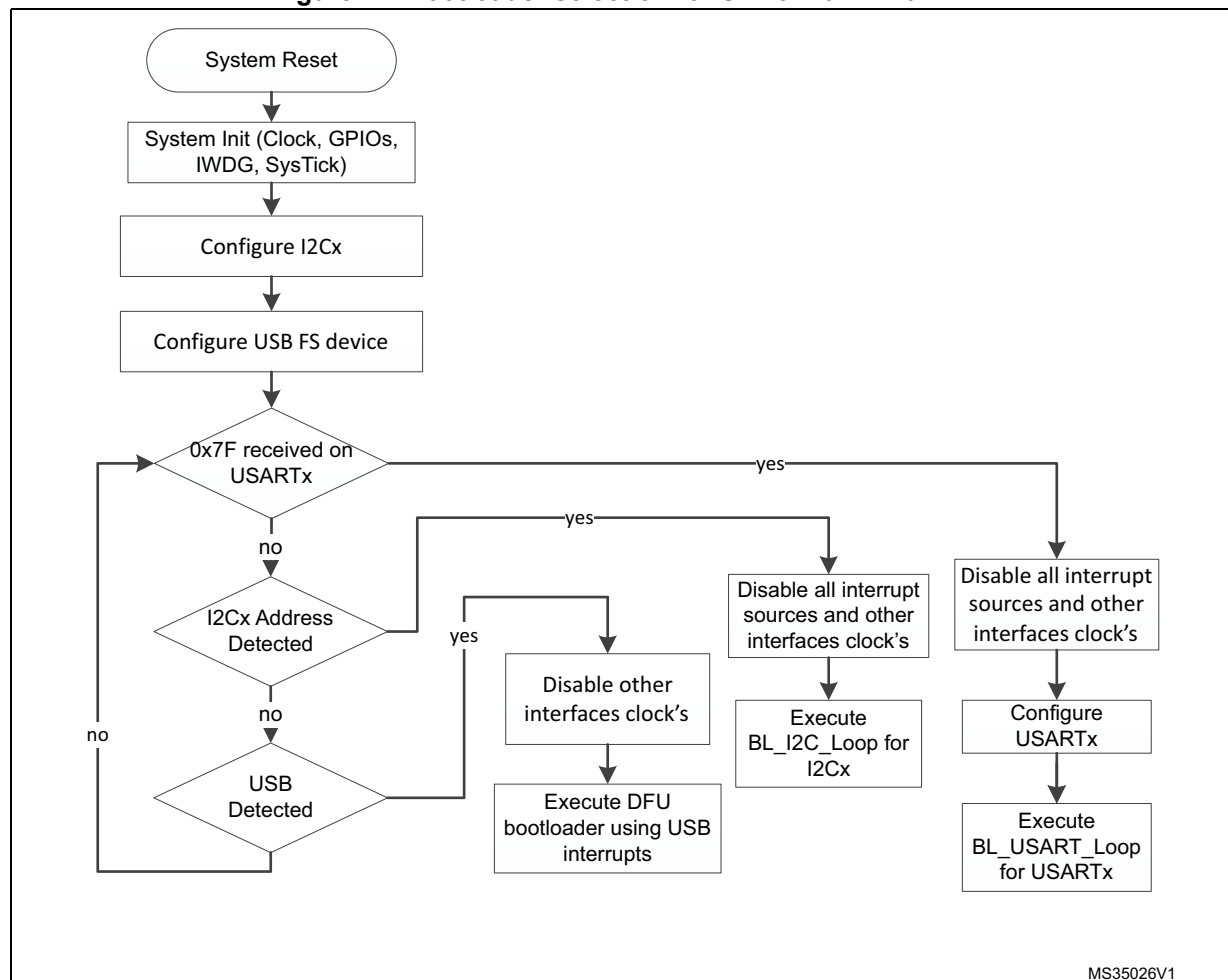
Note: *After the devices have booted in bootloader mode using USART2, the serial wire debug (SWD) communication is no more possible until the system is reset, because SWD uses PA14 pin (SWCLK), already used by the bootloader (USART2_RX).*

The system clock is derived from the embedded internal high-speed RC. No external quartz is required for the bootloader execution.

16.2 Bootloader selection

Figure 24 shows the bootloader selection mechanism.

Figure 24. Bootloader selection for STM32F071xx/072xx



16.3 Bootloader version

Table 33 lists the STM32F071xx/072xx devices bootloader versions.

Table 33. STM32F071xx/072xx bootloader versions

Version number	Description	Known limitations
V10.1	Initial bootloader version	At bootloader startup, the HSITRIM value is set to (0) (in HSITRIM bits on RCC_CR register) instead of default value (16), as a consequence a deviation is generated in crystal measurement. For better results, use the smallest supported crystal value (4 MHz). PA13 set in alternate push-pull, pull-up mode even if not used by bootloader.

17 STM32F09xxx devices

17.1 Bootloader configuration

The STM32F09xxx bootloader is activated by applying Pattern 6 (described in [Table 2](#)). [Table 34](#) shows the hardware resources used by this bootloader.

Table 34. STM32F09xxx configuration in system memory boot mode

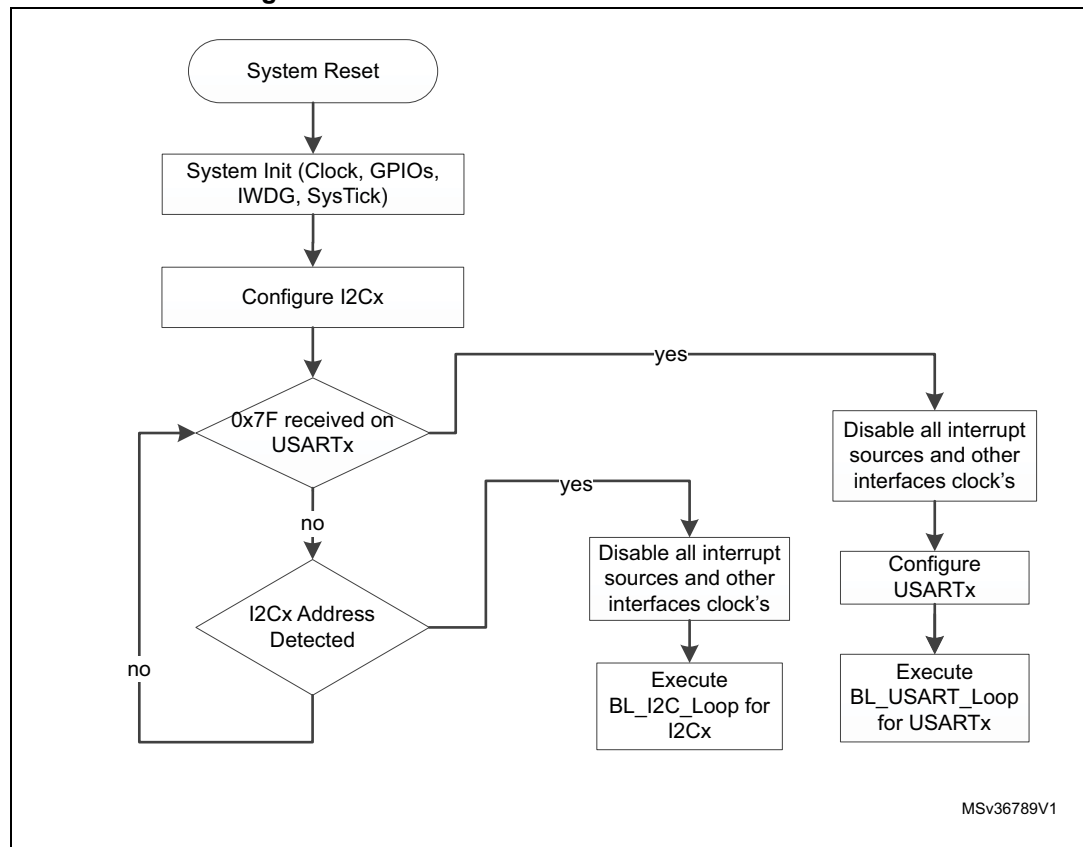
Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 48 MHz with HSI48 48 MHz as clock source.
	RAM	-	6 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	8 Kbytes, starting from address 0x1FFFD800, contain the bootloader firmware.
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input pull-up mode
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in input pull-up mode.
			PA15 pin: USART2 in reception mode. Used in input pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
			PA14 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON Target 7-bit address: 0b1000001x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain no pull mode.

Note: After the devices have booted in bootloader mode using USART2, the serial wire debug (SWD) communication is no longer possible until the system is reset, because SWD uses PA14 pin (SWCLK), already used by the bootloader (USART2_RX).

The system clock is derived from the embedded internal high-speed RC. No external quartz is required for the bootloader execution.

17.2 Bootloader selection

Figure 25. Bootloader selection for STM32F09xxx



17.3 Bootloader version

[Table 35](#) lists the STM32F09xxx devices bootloader versions.

Table 35. STM32F09xxx bootloader versions

Version number	Description	Known limitations
V5.0	Initial bootloader version	At bootloader startup, the HSITRIM value is set to 0 in HSITRIM bits on RCC_CR register instead of default value (16). As a consequence, a deviation is generated in crystal measurement. For better results, use the smallest supported crystal value (4 MHz). PA13 set in input pull-up mode even if not used by the bootloader.

18 STM32F10xxx devices

18.1 Bootloader configuration

The STM32F10xxx bootloader is activated by applying Pattern 1 (described in [Table 2](#)). [Table 36](#) shows the hardware resources used by this bootloader.

Table 36. STM32F10xxx configuration in system memory boot mode

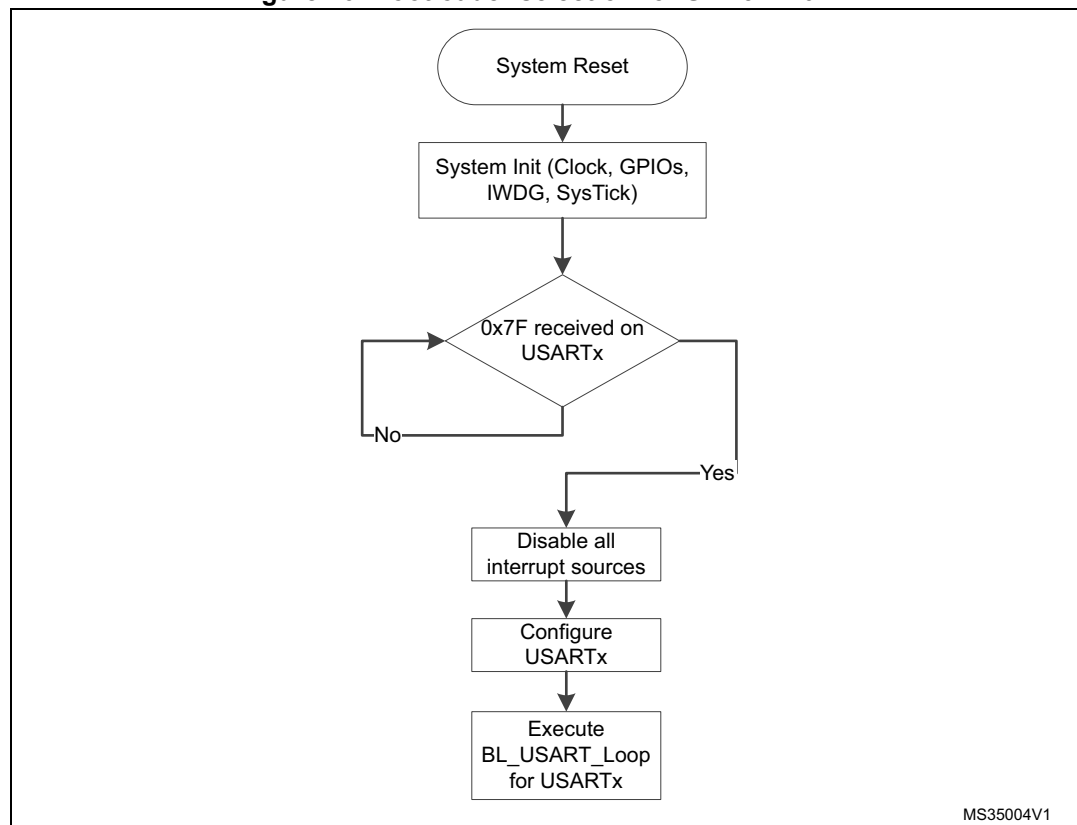
Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 24 MHz using the PLL.
	RAM	-	512 byte starting from address 0x20000000, are used by the bootloader firmware.
	System memory	-	2 Kbytes, starting from address 0x1FFFF000 contain the bootloader firmware.
	IWDG	-	The IWDG prescaler is configured to its maximum value and is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	USART1	Enabled	Once initialized, the USART1 configuration is 8 bits, even parity, and one stop bit.
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input no pull mode.
	USART1_TX pin	Output push-pull	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host.

The system clock is derived from the embedded internal high-speed RC. No external quartz is required for the bootloader execution.

18.2 Bootloader selection

Figure 26 shows the bootloader selection mechanism.

Figure 26. Bootloader selection for STM32F10xxx



18.3 Bootloader version

Table 37 lists the STM32F10xxx devices bootloader versions:

Table 37. STM32F10xxx bootloader versions

Version number	Description
V2.0	Initial bootloader version
V2.1	– Updated Go Command to initialize the main stack pointer – Updated Go command to return NACK when jump address is in the Option byte area or System memory area – Updated Get ID command to return the device ID on two bytes – Update the bootloader version to V2.1
V2.2	– Updated Read Memory, Write Memory and Go commands to deny access with a NACK response to the first 0x200 bytes of RAM used by the bootloader – Updated Readout Unprotect command to initialize the whole RAM content to 0x0 before ROP disable operation

Note: The bootloader ID format is applied to all STM32 devices except the STM32F1xx devices. The bootloader version for the STM32F1xx applies only to the embedded device's bootloader version and not to its supported protocols.

19 STM32F105xx/107xx devices

19.1 Bootloader configuration

The STM32F105xx/107xx bootloader is activated by applying Pattern 1 (described in [Table 2](#)). [Table 38](#) shows the hardware resources used by this bootloader.

Table 38. STM32F105xx/107xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 24 MHz using the PLL. This is used only for USARTx and during CAN2, USB detection for CAN and DFUs (once CAN or DFU is selected, the clock source is derived from the external crystal).
		HSE enabled	The external clock is mandatory only for DFU and CAN bootloaders and it must provide one of the following frequencies: 8 MHz, 14.7456 MHz or 25 MHz. For CAN bootloader, the PLL is used only to generate 48 MHz when 14.7456 MHz is used as HSE. For DFU, the PLL is used to generate a 48 MHz system clock from all supported external clock frequencies.
		-	The CSS interrupt is enabled for the CAN and DFUs. Any failure (or removal) of the external clock will generate system reset.
	RAM	-	4 Kbytes, starting from address 0x20000000, are used by the bootloader firmware.
	System memory	-	18 Kbytes, starting from address 0x1FFFB000 contain the bootloader firmware.
	IWDG	-	The IWDG prescaler is configured to its maximum value and is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	USART1	Enabled	Once initialized, the USART1 configuration is 8 bits, even parity, and one stop bit.
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input no pull mode.
	USART1_TX pin	Output push-pull	PA9 pin: USART1 in transmission mode. Used in input no pull mode.
USART2	USART2	Enabled	Once initialized, the USART2 configuration is 8 bits, even parity, and one stop bit. The USART2 uses its remapped pins.
	USART2_RX pin	Input	PD6 pin: USART2 receive (remapped pin)
	USART2_TX pin	Output push-pull	PD5 pin: USART2 transmit (remapped pin)
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx bootloader.

Table 38. STM32F105xx/107xx configuration in system memory boot mode (continued)

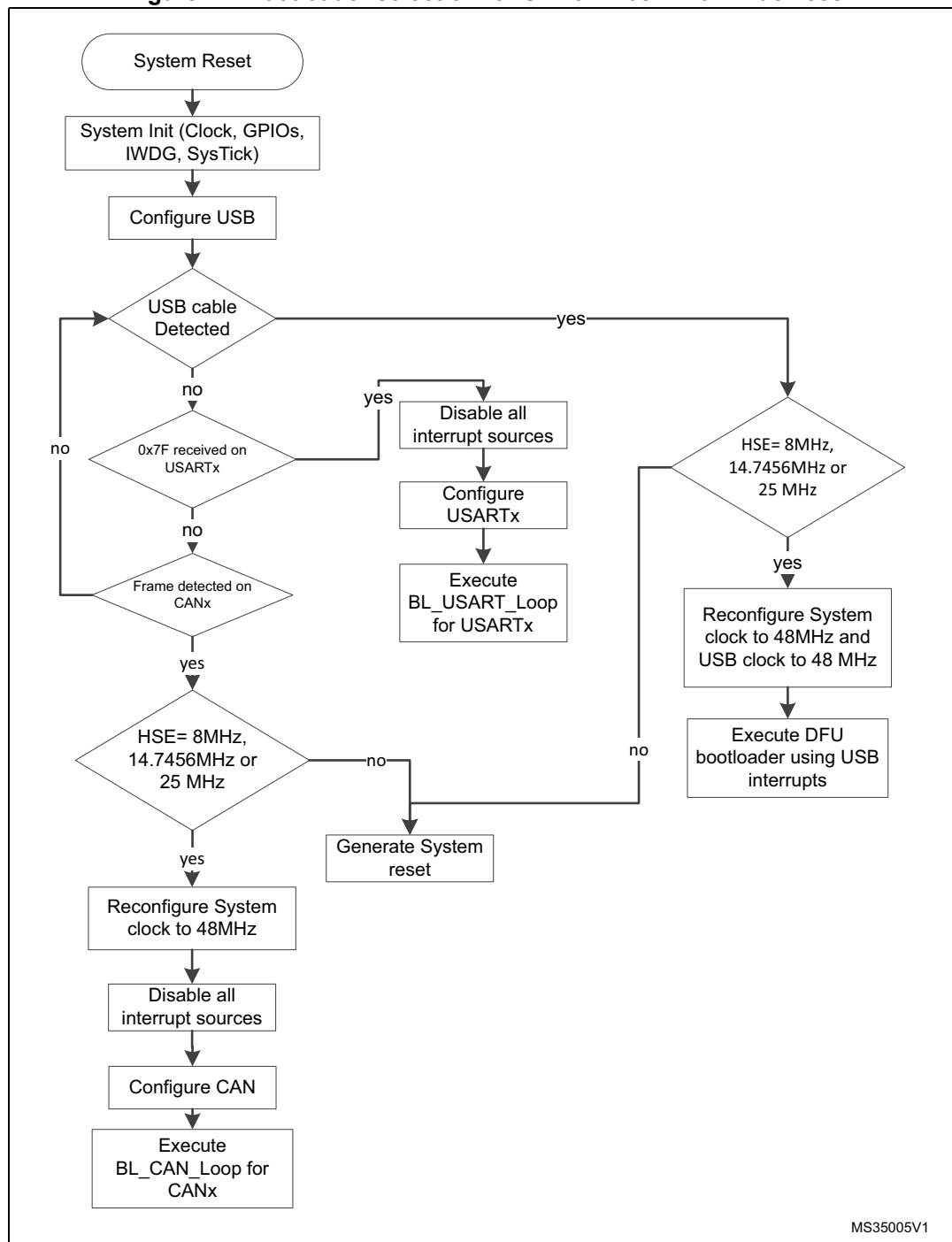
Bootloader	Feature/Peripheral	State	Comment
CAN2	CAN2	Enabled	Once initialized, the CAN2 configuration is baudrate 125 kbps, 11-bit identifier. Note: CAN1 is clocked during the CAN bootloader execution because CAN1 manages the communication between CAN2 and SRAM.
	CAN2_RX pin	Input	PB5 pin: CAN2 receives (remapped pin). Used in alternate push-pull, pull-up mode.
	CAN2_TX pin	Output push-pull	PB6 pin: CAN2 transmits (remapped pin). Used in input no pull mode.
DFU	USB	Enabled	USB OTG FS configured in forced device mode
	USB_VBUS pin	Input	PA9: Power supply voltage line
	USB_DM pin	Input/output	PA11 pin: USB_DM line
	USB_DP pin		PA12 pin: USB_DP line. No external pull-up resistor is required

The system clock is derived from the embedded internal high-speed RC for USARTx bootloader. This internal clock is used also for DFU and CAN bootloaders, but only for the selection phase. An external clock (8, 14.7456, or 25 MHz) is required for DFU and CAN bootloader execution after the selection phase.

19.2 Bootloader selection

Figure 27 shows the bootloader selection mechanism.

Figure 27. Bootloader selection for STM32F105xx/107xx devices



19.3 Bootloader version

[Table 39](#) lists the STM32F105xx/107xx devices bootloader versions:

Table 39. STM32F105xx/107xx bootloader versions

Version number	Description
V1.0	Initial bootloader version
V2.0	– Bootloader detection mechanism updated to fix the issue when GPIOs of unused peripherals in this bootloader are connected to low level or left floating during the detection phase. For more details refer to Section 19.3.2. – Vector table set to 0x1FFFB000 instead of 0x00000000 – Go command updated (for all bootloaders): USART1, USART2, CAN2, GPIOA, GPIOB, GPIOD and SysTick peripheral registers are set to their default reset values – DFU: USB pending interrupt cleared before executing the Leave DFU command – DFU subprotocol version changed from V1.0 to V1.2 – Bootloader version updated to V2.0
V2.1	– Fixed PA9 excessive consumption described in Section 19.3.4. – Get-Version command (defined in AN3155) corrected. It returns 0x22 instead of 0x20 in bootloader V2.0. Refer to Section 19.3.3 for more details. – Bootloader version updated to V2.1
V2.2	– Fixed DFU option bytes descriptor (set to 'e' instead of 'g' because it is read/write and not erasable). – Fixed DFU polling timings for flash Read/Write/Erase operations. – Robustness enhancements for DFU interface. – Updated bootloader version to V2.2.

Note: *The bootloader ID format is applied to all STM32 devices except the STM32F1xx products. The version for STM32F1xx applies only to the embedded device's bootloader version and not to its supported protocols.*

19.3.1 How to identify STM32F105xx/107xx bootloader versions

Bootloader V1.0 is implemented on devices whose date code is lower than 937. Bootloader V2.0 and V2.1 are implemented on devices with a date code higher than or equal to 937. Bootloader V2.2 is implemented on devices with a date code higher than or equal to 227.

Refer to the datasheets to find the date code on the device marking)

There are two ways to distinguish between bootloader versions:

- When using the USART bootloader, the Get-Version command defined in AN3155 has been corrected in V2.1 version. It returns 0x22 instead of 0x20 as in bootloader V2.0.
- The values of the vector table at the beginning of the bootloader code are different. The user software (or via JTAG/SWD) reads 0x1FFFE945 at address 0x1FFFB004 for bootloader V2.0 0x1FFFE9A1 for bootloader V2.1, and 0x1FFFE9C1 for bootloader V2.2.

The DFU version can be read through the `bcdDevice` field of the DFU Device Descriptor:

- V2.1 in bootloader V2.1
- V2.2 in bootloader V2.2.

19.3.2 Bootloader unavailability on STM32F105xx/STM32F107xx devices with date code lower than 937

Description

The bootloader cannot be used if the USART1_RX (PA10), USART2_RX (PD6, remapped), CAN2_Rx (PB5, remapped), OTG_FS_DM (PA11), and/or OTG_FS_DP (PA12) pin(s) are held low or left floating during the bootloader activation phase.

The bootloader cannot be connected through CAN2 (remapped), DFU (OTG FS in Device mode), USART1 or USART2 (remapped).

On 64-pin packages, the USART2_RX signal remapped PD6 pin is not available and it is internally grounded. In this case, the bootloader cannot be used at all.

Workaround

- For 64-pin packages
None. The bootloader cannot be used.
- For 100-pin packages
Depending on the used peripheral, the pins for the unused peripherals must be kept at a high level during the bootloader activation phase as described below:
 - If USART1 is used to connect to the bootloader, PD6 and PB5 must be kept at a high level.
 - If USART2 is used to connect to the bootloader, PA10, PB5, PA11, and PA12 must be kept at a high level.
 - If CAN2 is used to connect to the bootloader, PA10, PD6, PA11, and PA12 must be kept at a high level.
 - If DFU is used to connect to the bootloader, PA10, PB5, and PD6 must be kept at a high level.

Note: This limitation applies only to STM32F105xx and STM32F107xx devices with a date code lower than 937. STM32F105xx and STM32F107xx devices with a date code higher or equal to 937 are not impacted. See STM32F105xx and STM32F107xx datasheets for where to find the date code on the device marking.

19.3.3 USART bootloader Get-Version command returns 0x20 instead of 0x22

Description

In USART mode, the Get-Version command (defined in AN3155) returns 0x20 instead of 0x22. This limitation is present on bootloader versions V1.0 and V2.0, while it is fixed in bootloader version 2.1.

Workaround

None.

19.3.4 PA9 excessive power consumption when USB cable is plugged in bootloader V2.0

Description

When connecting a USB cable after booting from System-Memory mode, PA9 pin (connected to $V_{BUS} = 5\text{ V}$) is also shared with USART TX pin, configured as alternate push-pull and forced to 0 since the USART peripheral is not yet clocked. As a consequence, a current higher than 25 mA is drained by PA9 I/O and may affect the I/O pad reliability.

This limitation is fixed in bootloader version 2.1 by configuring PA9 as alternate function push-pull when a correct 0x7F is received on RX pin and the USART is clocked. Otherwise, PA9 is configured as alternate input floating.

Workaround

None.

20 STM32F10xxx XL-density devices

20.1 Bootloader configuration

The STM32F10xxx XL-density bootloader is activated by applying Pattern 3 (described in [Table 2](#)). [Table 40](#) shows the hardware resources used by this bootloader.

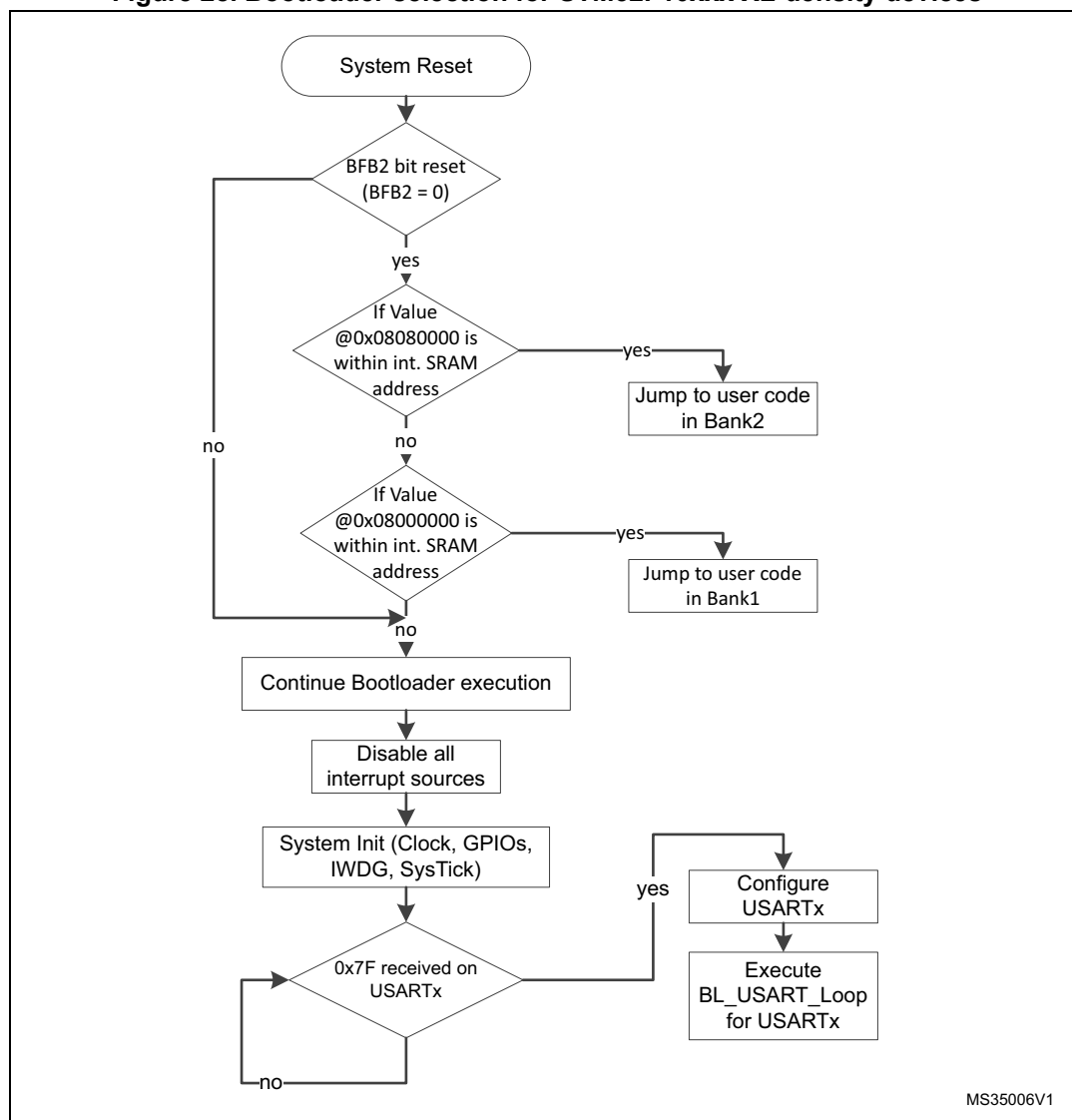
Table 40. STM32F10xxx XL-density configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 24 MHz using the PLL.
	RAM	-	2 Kbytes, starting from address 0x20000000, are used by the bootloader firmware.
	System memory	-	6 Kbytes, starting from address 0x1FFFE000 contain the bootloader firmware.
	IWDG	-	The IWDG prescaler is configured to its maximum value and is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	USART1	Enabled	Once initialized, the USART1 configuration is 8 bits, even parity, and one stop bit.
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input pull-up mode.
	USART1_TX pin	Output push-pull	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
USART2	USART2	Enabled	Once initialized, the USART2 configuration is 8 bits, even parity, and one stop bit.
	USART2_RX pin	Input	PD6 pin: USART2 receives (remapped pins). Used in input pull-up mode.
	USART2_TX pin	Output push-pull	PD5 pin: USART2 transmits (remapped pins). Used in alternate push-pull, pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host.

The system clock is derived from the embedded internal high-speed RC. No external quartz is required for the bootloader execution.

20.2 Bootloader selection

Figure 28. Bootloader selection for STM32F10xxx XL-density devices



20.3 Bootloader version

Table 41. STM32F10xxx XL-density bootloader versions

Version number	Description
V2.1	Initial bootloader version

Note: The bootloader ID format is applied to all STM32 devices families except the STM32F1xx family. The bootloader version for the STM32F1xx applies only to the embedded device bootloader version and not to its supported protocols.

21 STM32F2xxxx devices

Two bootloader versions are available on STM32F2xxxx devices:

- V2.x supporting USART1 and USART3
This version is embedded in revisions A, Z, and B
- V3.x supporting USART1, USART3, CAN2, and DFU (USB FS device)
This version is embedded in all other revisions (Y, X, W, 1, V, 2, 3, and 4)

21.1 Bootloader V2.x

21.1.1 Bootloader configuration

The STM32F2xxxx bootloader is activated by applying Pattern 1 (described in [Table 2](#)). [Table 42](#) shows the hardware resources used by this bootloader.

Table 42. STM32F2xxxx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 24 MHz.
	RAM	-	8 Kbytes, starting from address 0x20000000.
	System memory	-	29 Kbytes, starting from address 0x1FFF0000, contain the bootloader firmware.
	IWDG	-	The IWDG prescaler is configured to its maximum value and is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
	Power	-	Voltage range is set to [1.62 V, 2.1 V]. In this range internal flash write operations are allowed only in byte format (half-word, word, and double-word operations are not allowed). The voltage range can be configured in run time using bootloader commands.
USART1	USART1	Enabled	Once initialized, the USART1 configuration is 8 bits, even parity, and one stop bit.
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode
USART3 (on PC10/PC11)	USART3	Enabled	Once initialized, the USART3 configuration is 8 bits, even parity, and one stop bit.
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode

Table 42. STM32F2xxxx configuration in system memory boot mode (continued)

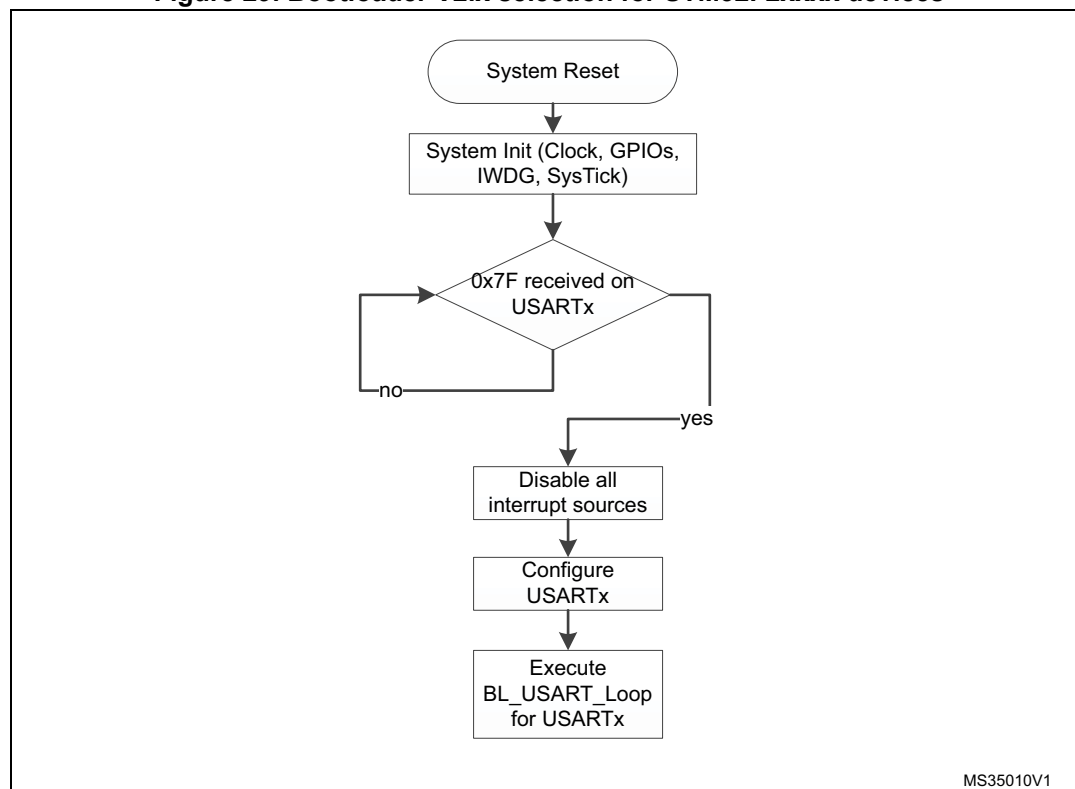
Bootloader	Feature/Peripheral	State	Comment
USART3 (on PB10/PB11)	USART3	Enabled	Once initialized, the USART3 configuration is 8 bits, even parity, and one stop bit
	USART3_RX pin	Input	PB11 pin: USART3 in reception mode
	USART3_TX pin	Output	PB10 pin: USART3 in transmission mode
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host.

The system clock is derived from the embedded internal high-speed RC. No external quartz is required for the bootloader code.

21.1.2 Bootloader selection

Figure 29 shows the bootloader selection mechanism.

Figure 29. Bootloader V2.x selection for STM32F2xxxx devices



21.1.3 Bootloader version

[Table 43](#) lists the STM32F2xxxx devices V2.x bootloader versions:

Table 43. STM32F2xxxx bootloader V2.x versions

Version number	Description	Known limitations
V2.0	Initial bootloader version	When a Read Memory command or Write Memory command is issued with an unsupported memory address and a correct address checksum (i.e. address 0x6000 0000), the command is aborted by the bootloader device, but the NACK (0x1F) is not sent to the host. As a result, the next two bytes (the number of bytes to be read/written and its checksum) are considered as a new command and its checksum. For the CAN interface, the Write Unprotect command is not functional. Use Write Memory command and write directly to the option bytes in order to disable the write protection.⁽¹⁾

1. If the "number of data - 1" (N-1) to be read/written is not equal to a valid command code (0x00, 0x01, 0x02, 0x11, 0x21, 0x31, 0x43, 0x44, 0x63, 0x73, 0x82 or 0x92), the limitation is not perceived from the host, as the command is NACK-ed anyway (as an unsupported new command).

21.2 Bootloader V3.x

21.2.1 Bootloader configuration

The STM32F2xxx bootloader is activated by applying Pattern 1 (described in [Table 2](#)). [Table 44](#) shows the hardware resources used by this bootloader.

Table 44. STM32F2xxx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 24 MHz using the PLL. The HSI clock source is used at startup (interface detection phase) and when USARTx interfaces are selected (once CAN or DFU is selected, the clock source is derived from the external crystal).
		HSE enabled	The system clock frequency is 60 MHz. The HSE clock source is used only when the CAN or the DFU (USB FS device) interfaces are selected. The external clock must provide a frequency multiple of 1 MHz and ranging from 4 MHz to 26 MHz.
		-	The CSS interrupt is enabled for the CAN and DFUs. Any failure (or removal) of the external clock generates system reset.
	RAM	-	8 Kbytes, starting from address 0x20000000, are used by the bootloader firmware.
	System memory	-	29 Kbytes, starting from address 0x1FF00000 contain the bootloader firmware.
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
	Power	-	Voltage range is set to 1.62 V, 2.1 V. In this range internal flash write operations are allowed only in byte format (half-word, word, and double-word operations are not allowed). The voltage range can be configured in run time using bootloader commands.
USART1	USART1	Enabled	Once initialized, the USART1 configuration is 8 bits, even parity, and one stop bit.
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input no pull mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in no pull mode.
USART3 (on PB10/PB11)	USART3	Enabled	Once initialized, the USART3 configuration is 8 bits, even parity, and one stop bit.
	USART3_RX pin	Input	PB11 pin: USART3 in reception mode. Used in pull-up mode
	USART3_TX pin	Output	PB10 pin: USART3 in transmission mode. Used in pull-up mode

Table 44. STM32F2xxxx configuration in system memory boot mode (continued)

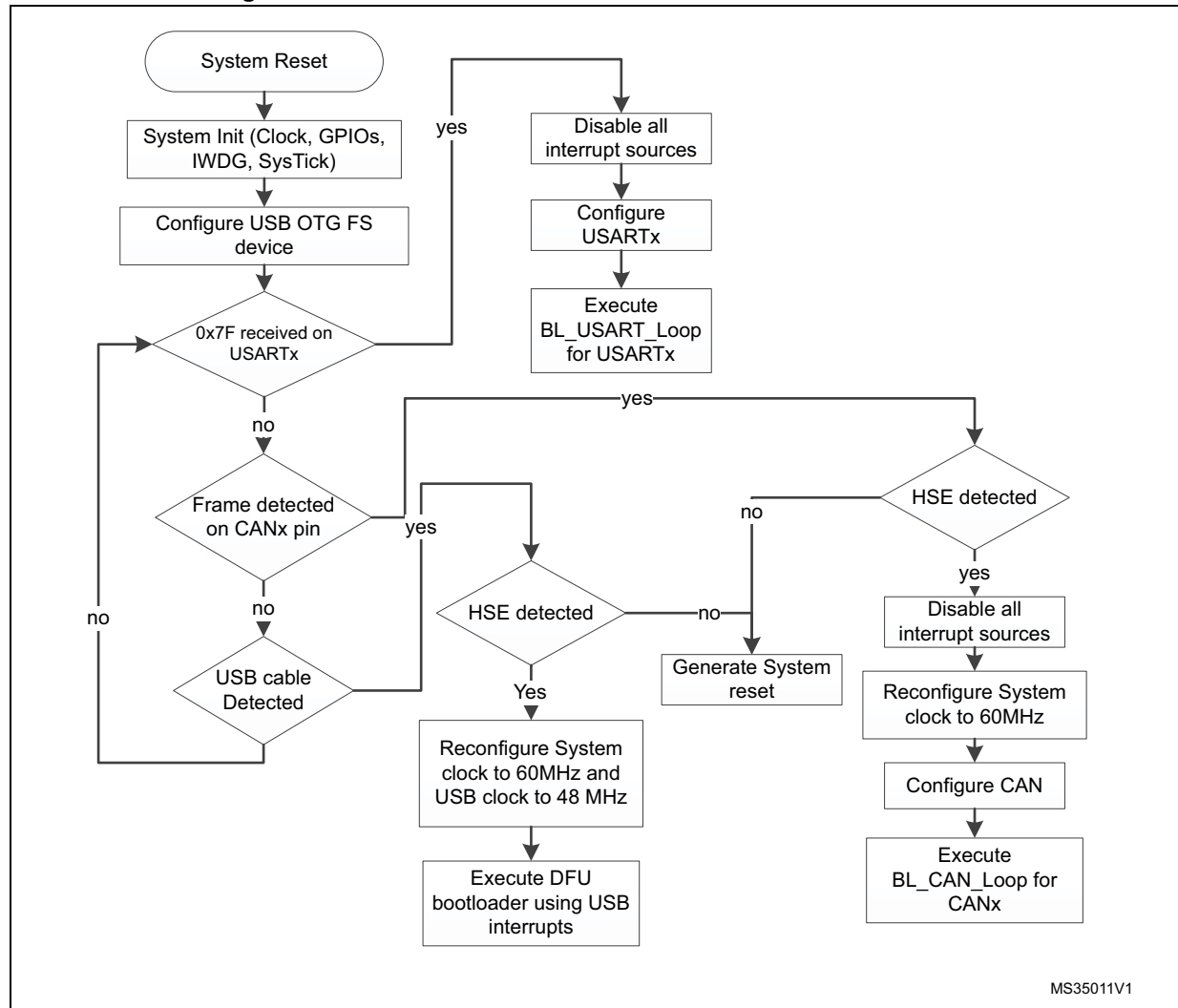
Bootloader	Feature/Peripheral	State	Comment
USART3 (on PC10/PC11)	USART3	Enabled	Once initialized, the USART3 configuration is 8 bits, even parity, and one stop bit.
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. Used in pull-up mode.
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode. Used in pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
CAN2	CAN2	Enabled	Once initialized, the CAN2 configuration is: Baudrate 125 kbps, 11-bit identifier. Note: CAN1 is clocked during CAN2 bootloader execution because CAN1 manages the communication between CAN2 and SRAM.
	CAN2_RX pin	Input	PB5 pin: CAN2 in reception mode. Used in alternate push-pull, pull-up mode.
	CAN2_TX pin	Output	PB13 pin: CAN2 in transmission mode. Used in alternate push-pull, pull-up mode.
DFU	USB	Enabled	USB OTG FS configured in forced device mode
	USB_DM pin	Input/output	PA11: USB DM line. Used in input no pull mode.
	USB_DP pin		PA12: USB DP line. Used in inpt no pull mode. No external pull-up resistor is required
CAN2 and DFUs	TIM11	Enabled	This timer is used to determine the value of the HSE. Once the HSE frequency is determined, the system clock is configured to 60 MHz using PLL and HSE.

The system clock is derived from the embedded internal high-speed RC for USARTx. This internal clock is also used for CAN and DFU (USB FS device), but only for the selection phase. An external clock multiple of 1 MHz (between 4 and 26 MHz) is required for CAN and DFU execution after the selection phase.

21.2.2 Bootloader selection

Figure 30 shows the bootloader selection mechanism.

Figure 30. Bootloader V3.x selection for STM32F2xxx devices



21.2.3 Bootloader version

[Table 45](#) lists the STM32F2xxxx devices V3.x bootloader versions:

Table 45. STM32F2xxxx bootloader V3.x versions

Version number	Description	Known limitations
V3.2	Initial bootloader version	– When a Read Memory command or Write Memory command is issued with an unsupported memory address and a correct address checksum (i.e. address 0x6000 0000), the command is aborted by the bootloader device, but the NACK (0x1F) is not sent to the host. As a result, the next two bytes (which are the number of bytes to be read/written and its checksum) are considered as a new command and its checksum⁽¹⁾. – Option bytes, OTP and Device Feature descriptors (in DFU interface) are set to “g” instead of “e” (not erasable memory areas).
V3.3	Fix V3.2 limitations. DFU interface robustness enhancement	– For the USART interface, two consecutive NACKs (instead of 1 NACK) are sent when a Read Memory or Write Memory command is sent and the RDP level is active. – For the CAN interface, the Write Unprotect command is not functional. Use Write Memory command and write directly to the option bytes in order to disable the write protection.

1. If the “number of data - 1” (N-1) to be read/written is not equal to a valid command code (0x00, 0x01, 0x02, 0x11, 0x21, 0x31, 0x43, 0x44, 0x63, 0x73, 0x82 or 0x92), the limitation is not perceived from the host, as the command is NACK-ed anyway (as an unsupported new command).

22 STM32F301xx/302x4(6/8) devices

22.1 Bootloader configuration

The STM32F301xx/302x4(6/8) bootloader is activated by applying Pattern 2 (described in [Table 2](#)). [Table 46](#) shows the hardware resources used by this bootloader.

Table 46. STM32F301xx/302x4(6/8) configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 48 MHz with HSI48 48 MHz as clock source.
		HSE enabled	The external clock can be used for all bootloader interfaces and must have one the following values: 24,18,16,12,9,8,6,4,3 MHz. The PLL is used to generate the USB48 MHz clock and the 48 MHz clock for the system clock.
		-	The CSS interrupt is enabled for the DFU. Any failure (or removal) of the external clock generates system reset.
	RAM	-	6 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	8 Kbytes, starting from address 0x1FFFD800, contain the bootloader firmware
USART1	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
USART2	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
USARTx	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.

Table 46. STM32F301xx/302x4(6/8) configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
DFU	USB	Enabled	USB used in FS mode
	USB_DM pin	Input/output	PA11: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12: USB DP line. Used in alternate push-pull, no pull mode. An external pull-up resistor 1.5 K Ω must be connected to USB_DP pin.

The bootloader has two cases of operation, depending upon the presence of the external clock (HSE) at bootloader startup:

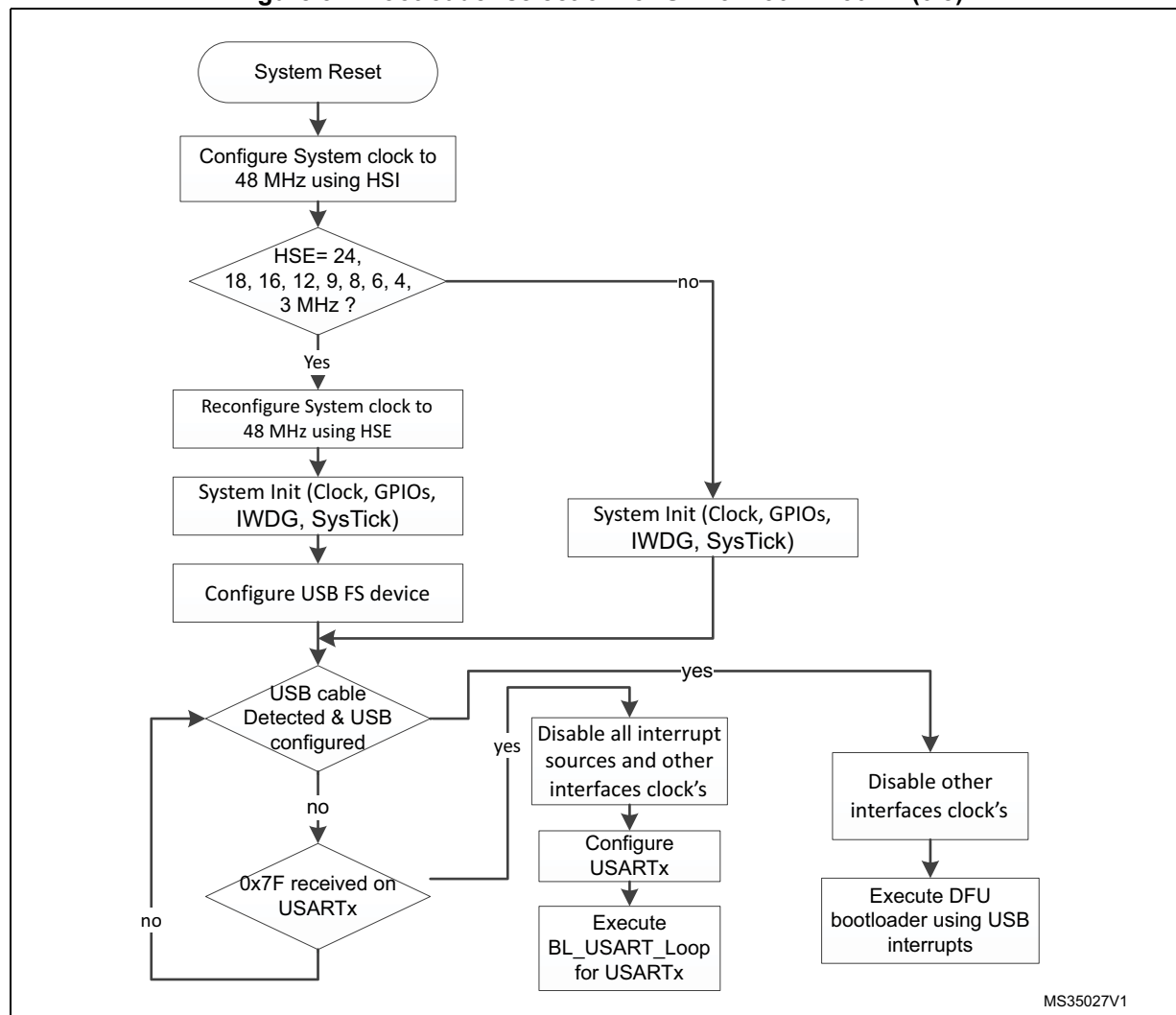
- If HSE is present and has a value of 24, 18, 16, 12, 9, 8, 6, 4, or 3 MHz, the system clock is configured to 48 MHz with HSE as clock source. The DFU interface, USART1 and USART2 are functional and can be used to communicate with the bootloader device.
- If HSE is not present, the HSI is kept as default clock source, and only USART1 and USART2 are functional.

The external clock (HSE) must be kept if it is connected at bootloader startup, because it is used as system clock source.

22.2 Bootloader selection

Figure 31 shows the bootloader selection mechanism.

Figure 31. Bootloader selection for STM32F301xx/302x4(6/8)



22.3 Bootloader version

Table 47 lists the STM32F301xx/302x4(6/8) devices bootloader versions:

Table 47. STM32F301xx/302x4(6/8) bootloader versions

Version number	Description	Known limitations
V4.0	Initial bootloader version	None

23 STM32F302xB(C)/303xB(C) devices

23.1 Bootloader configuration

The STM32F302xB(C)/303xB(C) bootloader is activated by applying Pattern 2 (described in [Table 2](#)). [Table 48](#) shows the hardware resources used by this bootloader.

Table 48. STM32F302xB(C)/303xB(C) configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	At startup, the system clock frequency is configured to 48 MHz using the HSI. If an external clock (HSE) is not present, the system is kept clocked from the HSI.
		HSE enabled	The external clock can be used for all bootloader interfaces and must have one the following values: 24, 18, 16, 12, 9, 8, 6, 4, 3 MHz. The PLL is used to generate the USB 48 MHz clock and the 48 MHz clock for the system clock.
		-	The CSS interrupt is enabled for the DFU. Any failure (or removal) of the external clock generates system reset.
	RAM	-	5 Kbytes, starting from address 0x20000000, are used by the bootloader firmware.
	System memory	-	8 Kbytes, starting from address 0x1FFFD800, contains the bootloader firmware.
USART1	IWDG	-	The IWDG prescaler is configured to its maximum value and is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
	USART1	Enabled	Once initialized, the USART1 configuration is 8 bits, even parity, and one stop bit.
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
USART2	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
	USART2	Enabled	Once initialized, the USART2 configuration is 8 bits, even parity, and one stop bit. The USART2 uses its remapped pins.
	USART2_RX pin	Input	PD6 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
USARTx	USART2_TX pin	Output	PD5 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx bootloader.

Table 48. STM32F302xB(C)/303xB(C) configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
DFU	USB	Enabled	USB used in FS mode
	USB_DM pin	Input/output	PA11: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12: USB DP line. Used in alternate push-pull, no pull mode. An external pull-up resistor 1.5 K Ω must be connected to USB_DP pin.

The bootloader has two cases of operation depending on the presence of the external clock (HSE) at bootloader startup:

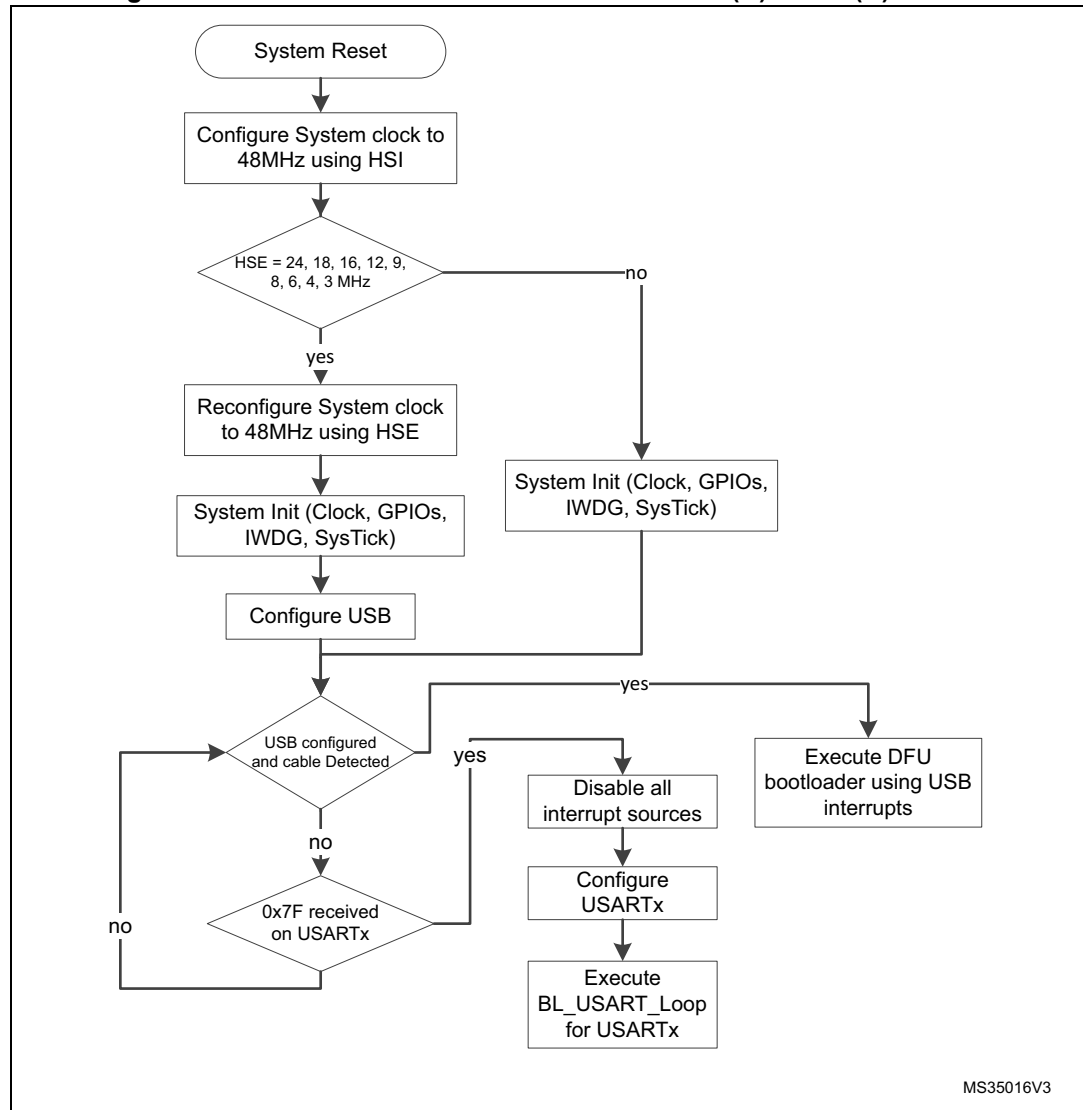
- If HSE is present and has a value of 24, 18, 16, 12, 9, 8, 6, 4 or 3 MHz, the system clock is configured to 48 MHz with HSE as clock source. The DFU interface, USART1 and USART2 are functional and can be used to communicate with the bootloader device.
- If HSE is not present, the HSI is kept as default clock source, and only USART1 and USART2 are functional.

The external clock (HSE) must be kept if it is connected at bootloader startup, because it is used as system clock source.

23.2 Bootloader selection

Figure 32 shows the bootloader selection mechanism.

Figure 32. Bootloader selection for STM32F302xB(C)/303xB(C) devices



23.3 Bootloader version

Table 49 lists the STM32F302xB(C)/303xB(C) devices bootloader versions.

Table 49. STM32F302xB(C)/303xB(C) bootloader versions

Version number	Description	Known limitations
V4.1	Initial bootloader version	None

24 STM32F302xD(E)/303xD(E) devices

24.1 Bootloader configuration

The STM32F302xD(E)/303xD(E) bootloader is activated by applying Pattern 2 (described in [Table 2](#)). [Table 50](#) shows the hardware resources used by this bootloader.

Table 50. STM32F302xD(E)/303xD(E) configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 48 MHz with HSI48 48 MHz as clock source.
		HSE enabled	The external clock can be used for all bootloader interfaces and must have one the following values: 24, 18, 16, 12, 9, 8, 6, 4, 3 MHz. The PLL is used to generate the USB 48 MHz clock and the 48 MHz clock for the system clock.
		-	The CSS interrupt is enabled for the DFU. Any failure (or removal) of the external clock generates system reset.
	RAM	-	6 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	8 Kbytes, starting from address 0x1FFFD800, contain the bootloader firmware
USART1	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
USART2	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
USART2	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
DFU	USB	Enabled	USB FS configured in forced device mode. USB FS interrupt vector is enabled and used for USB DFU communications.
	USB_DM pin	Input/output	PA11 pin: USB FS DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12 pin: USB FS DP line. Used in alternate push-pull, no pull mode. An external pull-up resistor 1.5 KΩ must be connected to USB_DP pin.

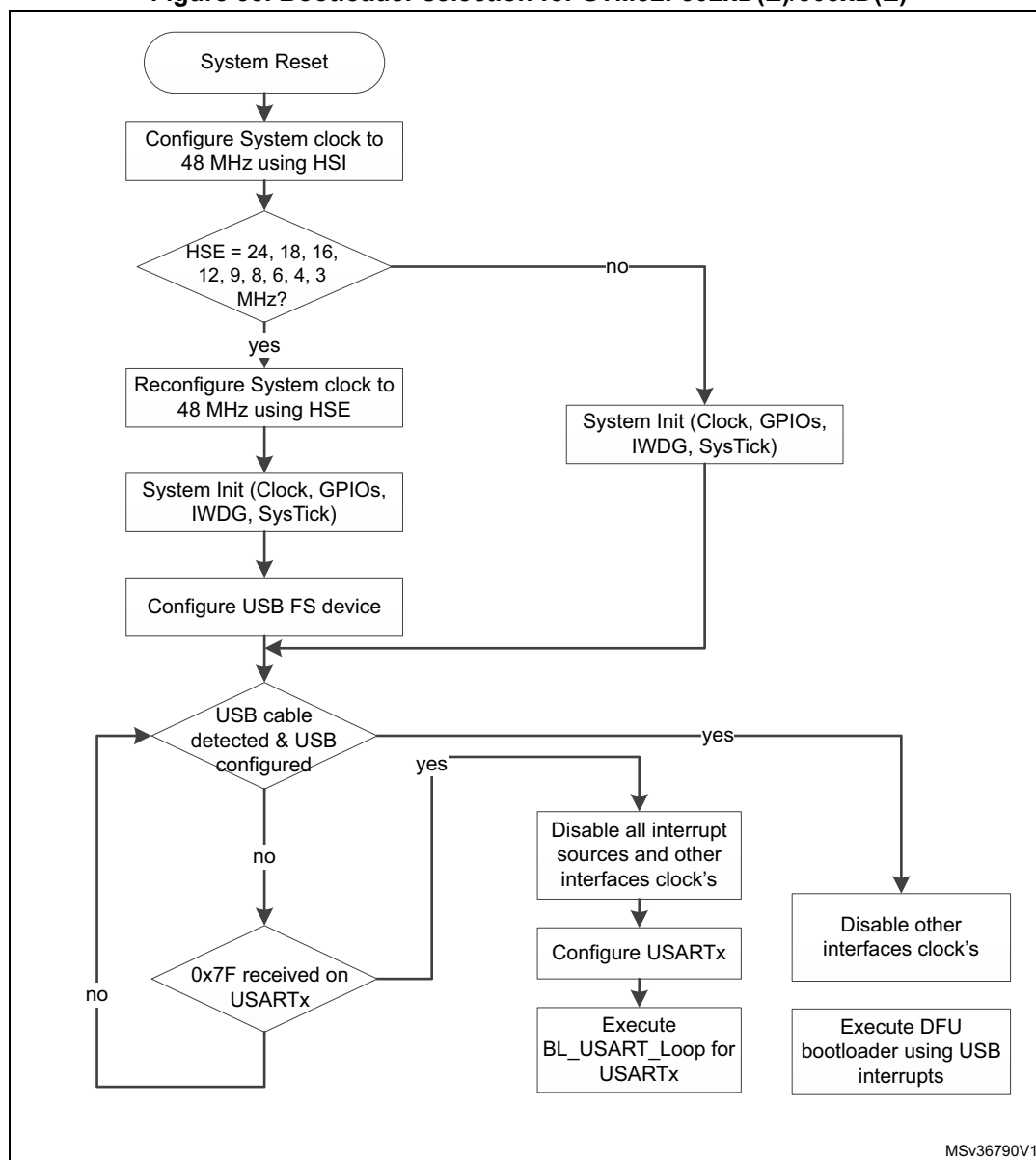
The bootloader has two cases of operation depending on the presence of the external clock (HSE) at bootloader startup:

- If HSE is present and has a value of 24, 18, 16, 12, 9, 8, 6, 4 or 3 MHz, the system clock is configured to 48 MHz with HSE as clock source. The DFU interface, USART1 and USART2 are functional and can be used to communicate with the bootloader device.
- If HSE is not present, the HSI is kept as default clock source, and only USART1 and USART2 are functional.

The external clock (HSE) must be kept if it is connected at bootloader startup, because it is used as system clock source.

24.2 Bootloader selection

Figure 33. Bootloader selection for STM32F302xD(E)/303xD(E)



24.3 Bootloader version

Table 51. STM32F302xD(E)/303xD(E) bootloader versions

Version number	Description	Known limitations
V4.0	Initial bootloader version	None

25 STM32F303x4(6/8)/334xx/328xx devices

25.1 Bootloader configuration

The STM32F303x4(6/8)/334xx/328xx bootloader is activated by applying Pattern 2 (described in [Table 2](#)). [Table 52](#) shows the hardware resources used by this bootloader.

Table 52. STM32F303x4(6/8)/334xx/328xx configuration in system memory boot mode

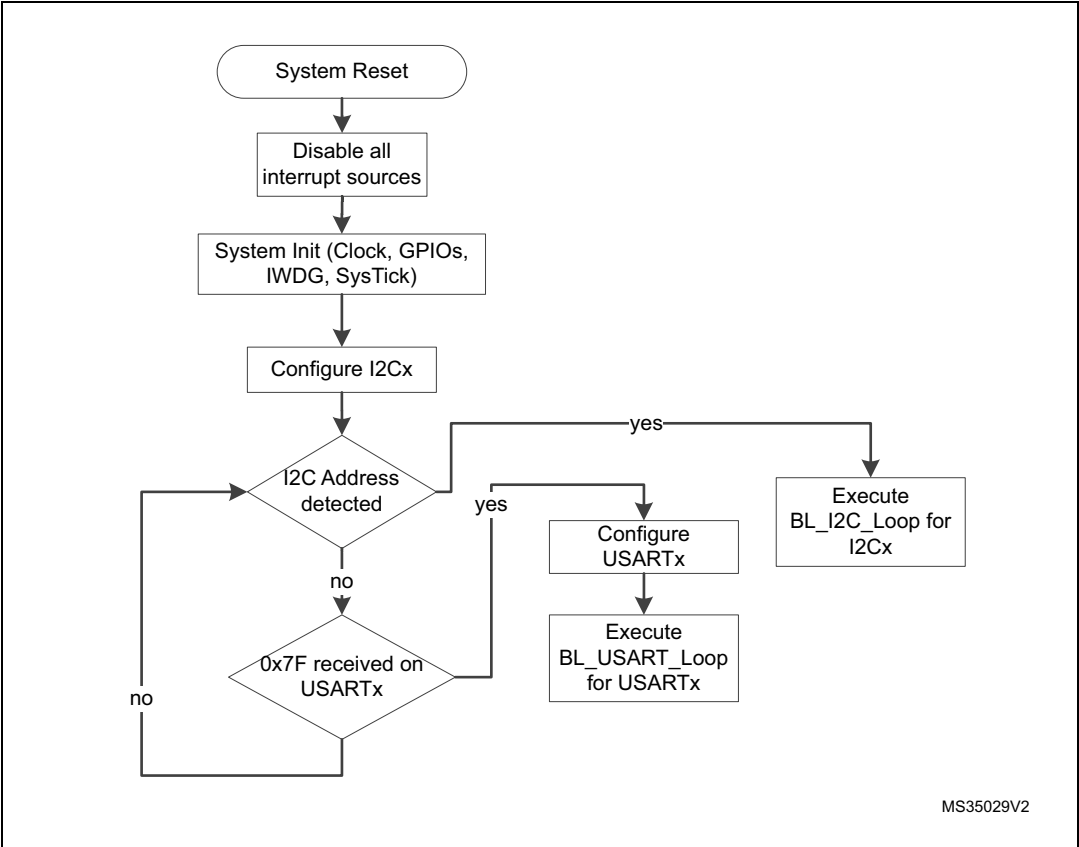
Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 60 MHz with HSI 8 MHz as clock source.
	RAM	-	6 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	8 Kbytes, starting from address 0x1FFFD800, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
I2C1	I2C1	Enabled	The I2C1 configuration is: I2C speed: up to 400 kHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b0111111x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin	Input/output	PB7 pin: data line is used in open-drain no pull mode.

The system clock is derived from the embedded internal high-speed RC. No external quartz is required for the bootloader execution.

25.2 Bootloader selection

Figure 34 shows the bootloader selection mechanism.

Figure 34. Bootloader selection for STM32F303x4(6/8)/334xx/328xx



25.3 Bootloader version

Table 53 lists the STM32F303x4(6/8)/334xx/328xx devices bootloader versions:

Table 53. STM32F303x4(6/8)/334xx/328xx bootloader versions

Version number	Description	Known limitations
V5.0	Initial bootloader version	None

26 STM32F318xx devices

26.1 Bootloader configuration

The STM32F318xx bootloader is activated by applying Pattern 2 (described in [Table 2](#)). [Table 54](#) shows the hardware resources used by this bootloader.

Table 54. STM32F318xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 60 MHz with HSI 8 MHz as clock source.
	RAM	-	6 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	8 Kbytes, starting from address 0x1FFFD800, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value, and periodically refreshed to prevent a reset (if the hardware IWDG option was previously enabled by the user).
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b0111101x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain no pull mode.

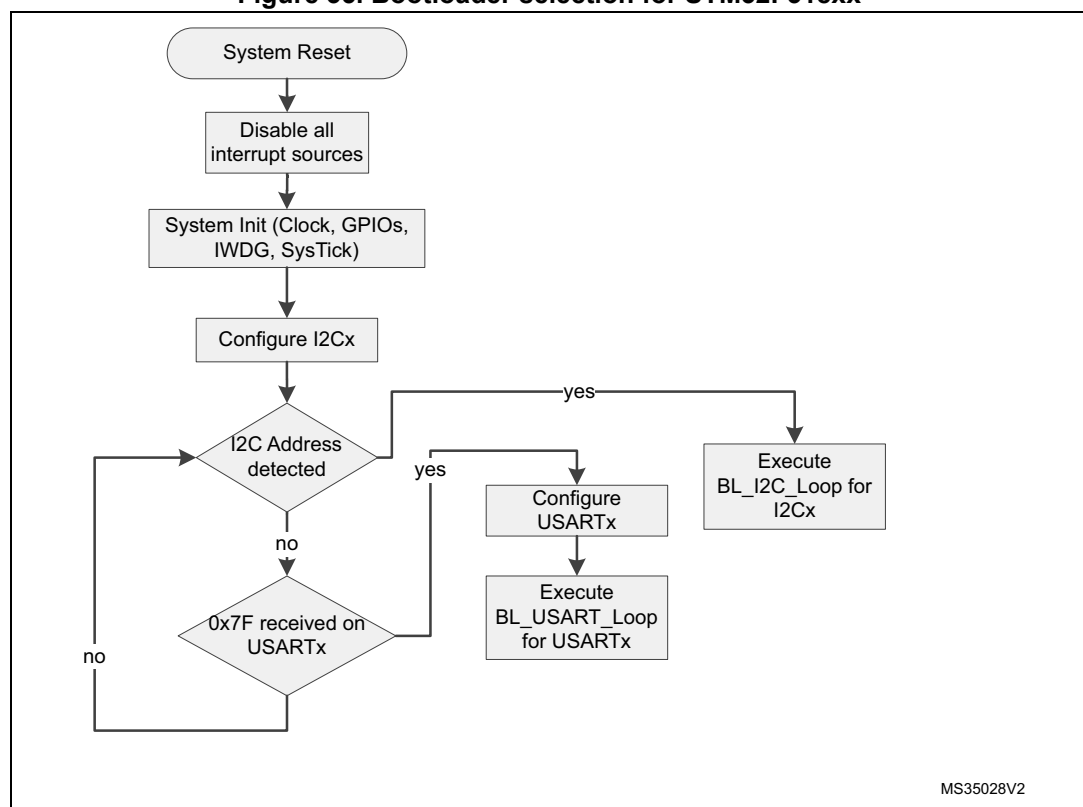
Table 54. STM32F318xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b0111101x (x = 0 for write and x = 1 for read) and digital filter disabled.
	I2C3_SCL pin	Input/output	PA8 pin: clock line is used in open-drain no pull mode.
	I2C3_SDA pin		PB5 pin: data line is used in open-drain no pull mode.

The system clock is derived from the embedded internal high-speed RC. No external quartz is required for the bootloader execution.

26.2 Bootloader selection

Figure 35. Bootloader selection for STM32F318xx



26.3 Bootloader version

Table 55. STM32F318xx bootloader versions

Version number	Description	Known limitations
V5.0	Initial bootloader version	None

27 STM32F358xx devices

27.1 Bootloader configuration

The STM32F358xx bootloader is activated by applying Pattern 2 (described in [Table 2](#)). [Table 56](#) shows the hardware resources used by this bootloader.

Table 56. STM32F358xx configuration in system memory boot mode

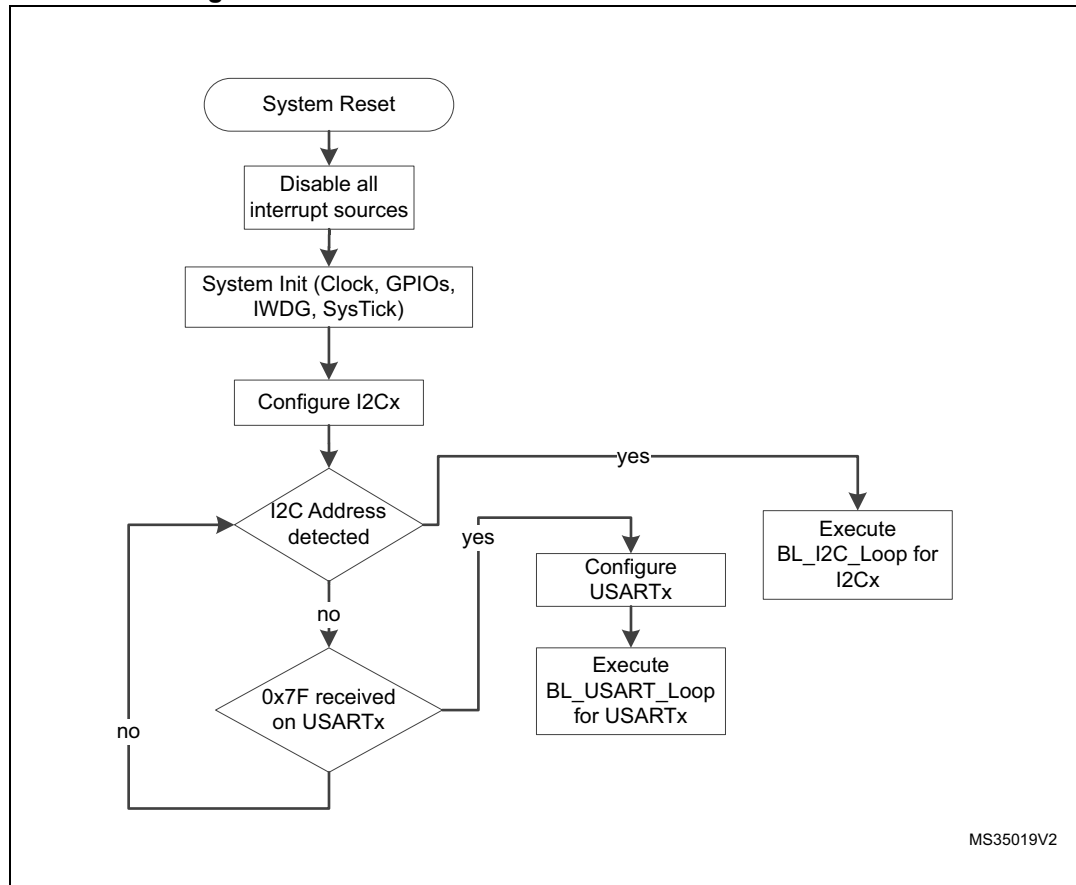
Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 8 MHz using the HSI.
	RAM	-	5 Kbytes, starting from address 0x20000000, are used by the bootloader firmware.
	System memory	-	8 Kbytes, starting from address 0x1FFFD800, contains the bootloader firmware.
	IWDG	-	The IWDG prescaler is configured to its maximum value and is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user). Window feature is disabled.
USART1	USART1	Enabled	Once initialized, the USART1 configuration is 8 bits, even parity, and one stop bit.
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
USART2	USART2	Enabled	Once initialized, the USART2 configuration is 8 bits, even parity, and one stop bit. The USART2 uses its remapped pins.
	USART2_RX pin	Input	PD6 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PD5 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx bootloader.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b0110111x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain no pull mode.

The system clock is derived from the embedded internal high-speed RC. No external quartz is required for the bootloader execution.

27.2 Bootloader selection

Figure 36 shows the bootloader selection mechanism.

Figure 36. Bootloader selection for STM32F358xx devices



27.3 Bootloader version

Table 57 lists the STM32F358xx devices bootloader versions.

Table 57. STM32F358xx bootloader versions

Version number	Description	Known limitations
V5.0	Initial bootloader version	For USART1 and USART2 interfaces, the maximum baudrate supported by the bootloader is 57600 baud.

28 STM32F373xx devices

28.1 Bootloader configuration

The STM32F373xx bootloader is activated by applying Pattern 2 (described in [Table 2](#)). [Table 58](#) shows the hardware resources used by this bootloader.

Table 58. STM32F373xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	At startup, the system clock frequency is configured to 48 MHz using the HSI. If an external clock (HSE) is not present, the system is kept clocked from the HSI.
		HSE enabled	The external clock can be used for all bootloader interfaces and must have one the following values: 24, 18, 16, 12, 9, 8, 6, 4, 3 MHz. The PLL is used to generate the USB 48 MHz clock and the 48 MHz clock for the system clock.
		-	The CSS interrupt is enabled for the DFU. Any failure (or removal) of the external clock generates system reset.
	RAM	-	5 Kbytes, starting from address 0x20000000, are used by the bootloader firmware.
	System memory	-	8 Kbytes, starting from address 0x1FFFD800, contains the bootloader firmware
USART1	IWDG	-	The IWDG prescaler is configured to its maximum value and is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
	USART1	Enabled	Once initialized, the USART1 configuration is 8 bits, even parity, and one stop bit.
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
USART2	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
	USART2	Enabled	Once initialized, the USART2 configuration is 8 bits, even parity, and one stop bit. The USART2 uses its remapped pins.
	USART2_RX pin	Input	PD6 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
USARTx	USART2_TX pin	Output	PD5 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx bootloader.

Table 58. STM32F373xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
DFU	USB	Enabled	USB used in FS mode
	USB_DM pin	Input/output	PA11: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12: USB DP line. Used in alternate push-pull, no pull mode. An external pull-up resistor 1.5 K Ω must be connected to USB_DP pin.

There are two operation modes, depending upon the presence of the external clock (HSE) at bootloader startup:

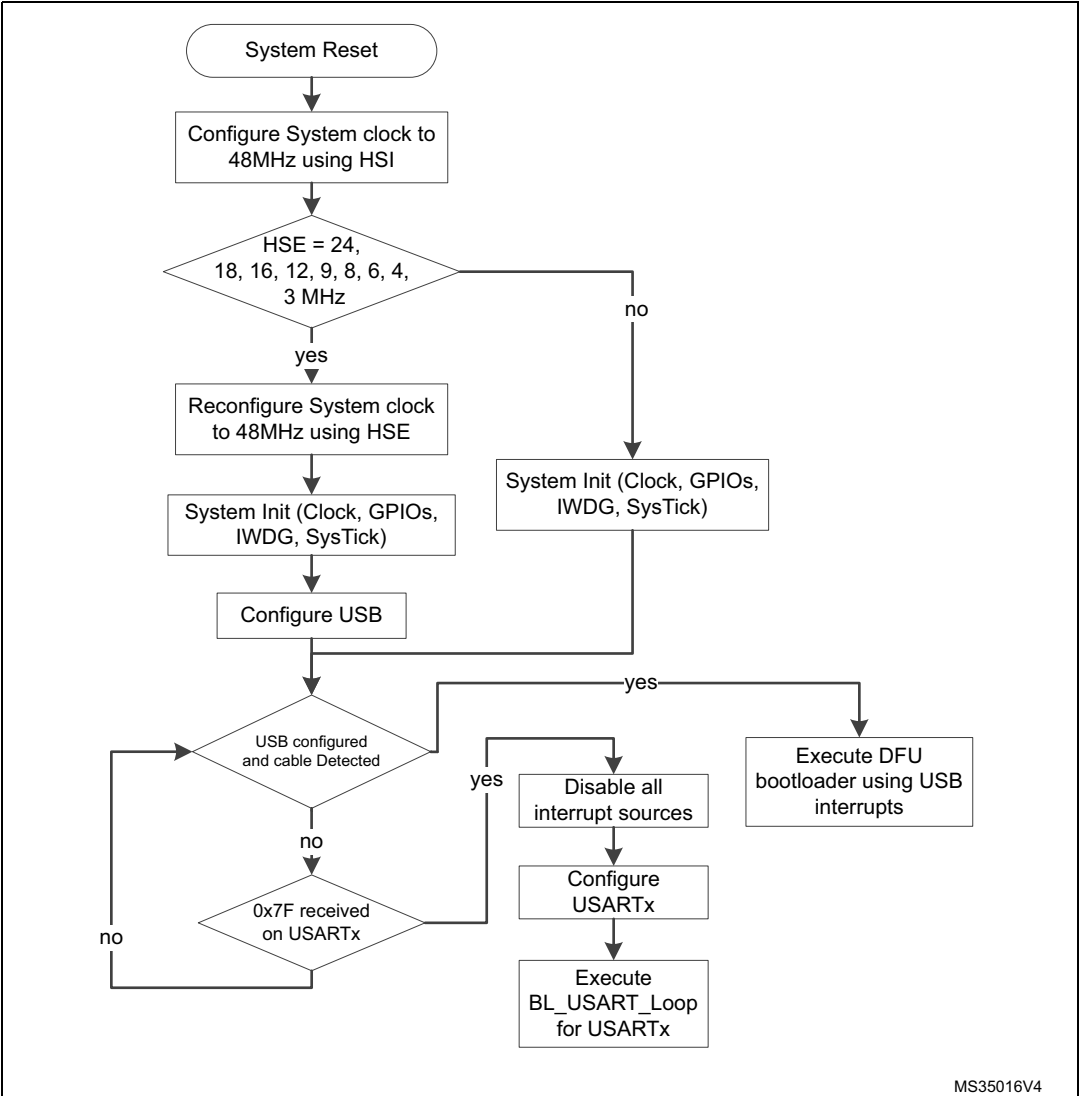
- If HSE is present and has a value of 24, 18, 16, 12, 9, 8, 6, 4, or 3 MHz, the system clock is configured to 48 MHz with HSE as clock source. The DFU interface, USART1 and USART2 are functional and can be used to communicate with the bootloader.
- If HSE is not present, the HSI is kept as default clock source, and only USART1 and USART2 are functional.

Note: The external clock (HSE) must be kept if it is connected at bootloader startup, because it is used as system clock source.

28.2 Bootloader selection

Figure 37 shows the bootloader selection mechanism.

Figure 37. Bootloader selection for STM32F373xx devices



MS35016V4

28.3 Bootloader version

Table 59 lists the STM32F373xx devices bootloader versions.

Table 59. STM32F373xx bootloader versions

Version number	Description	Known limitations
V4.1	Initial bootloader version	None

29 STM32F378xx devices

29.1 Bootloader configuration

The STM32F378xx bootloader is activated by applying Pattern 2 (described in [Table 2](#)). [Table 60](#) shows the hardware resources used by this bootloader.

Table 60. STM32F378xx configuration in system memory boot mode

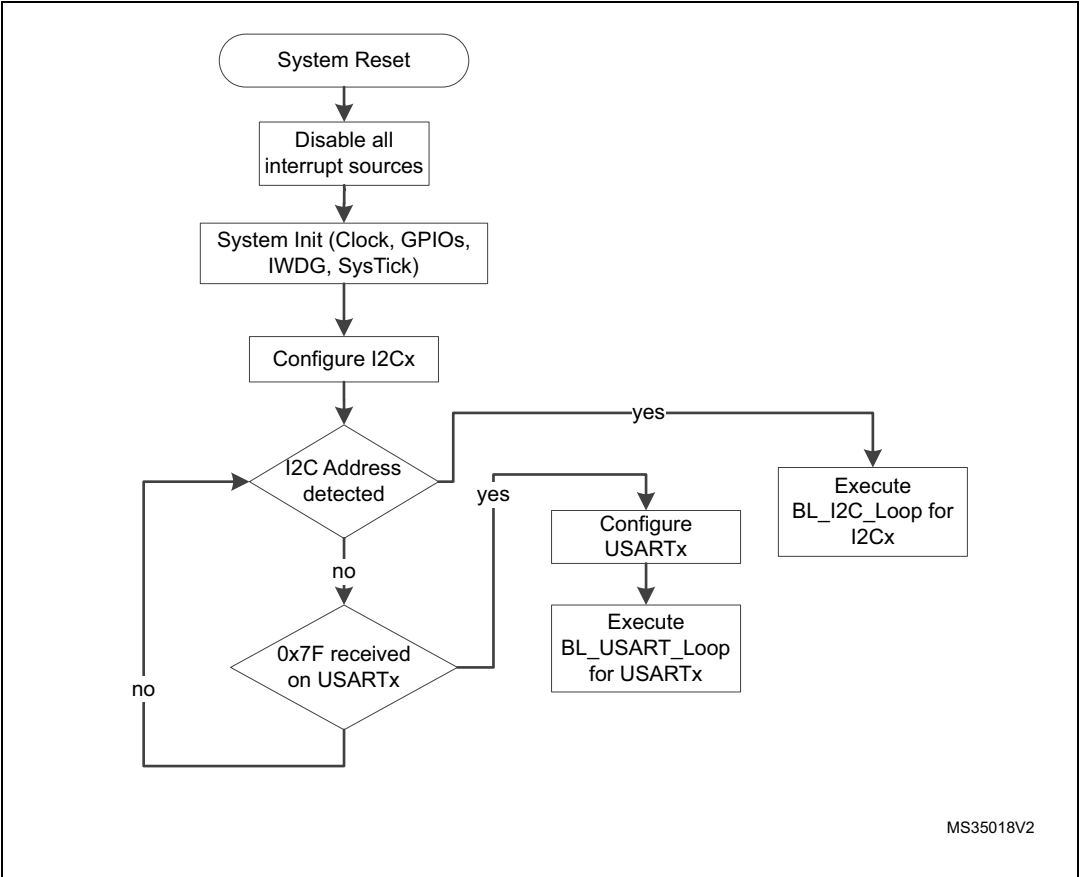
Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 8 MHz using the HSI.
	RAM	-	4 Kbytes, starting from address 0x20000000, are used by the bootloader firmware.
	System memory	-	8 Kbytes, starting from address 0x1FFFD800, contains the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value and is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user). Window feature is disabled.
USART1	USART1	Enabled	Once initialized, the USART1 configuration is 8 bits, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
USART2	USART2	Enabled	Once initialized, the USART2 configuration is 8 bits, even parity, and one stop bit. The USART2 uses its remapped pins.
	USART2_RX pin	Input	PD6 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PD5 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx bootloader.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b0110111x (x = 0 for write and x = 1 for read).
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain no pull mode.

The system clock is derived from the embedded internal high-speed RC. No external quartz is required for the bootloader execution.

29.2 Bootloader selection

Figure 38 shows the bootloader selection mechanism.

Figure 38. Bootloader selection for STM32F378xx devices



29.3 Bootloader version

Table 61 lists the STM32F378xx devices bootloader versions.

Table 61. STM32F378xx bootloader versions

Version number	Description	Known limitations
V5.0	Initial bootloader version	For USART1 and USART2 interfaces, the maximum baudrate supported by the bootloader is 57600 baud.

30 STM32F398xx devices

30.1 Bootloader configuration

The STM32F398xx bootloader is activated by applying Pattern 2 (described in [Table 2](#)). [Table 62](#) shows the hardware resources used by this bootloader.

Table 62. STM32F398xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 60 MHz with HSI 8 MHz as clock source.
	RAM	-	6 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	7 Kbytes, starting from address 0x1FFFD800, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1000000x (x = 0 for write and x = 1 for read).
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain no pull mode.

Table 62. STM32F398xx configuration in system memory boot mode (continued)

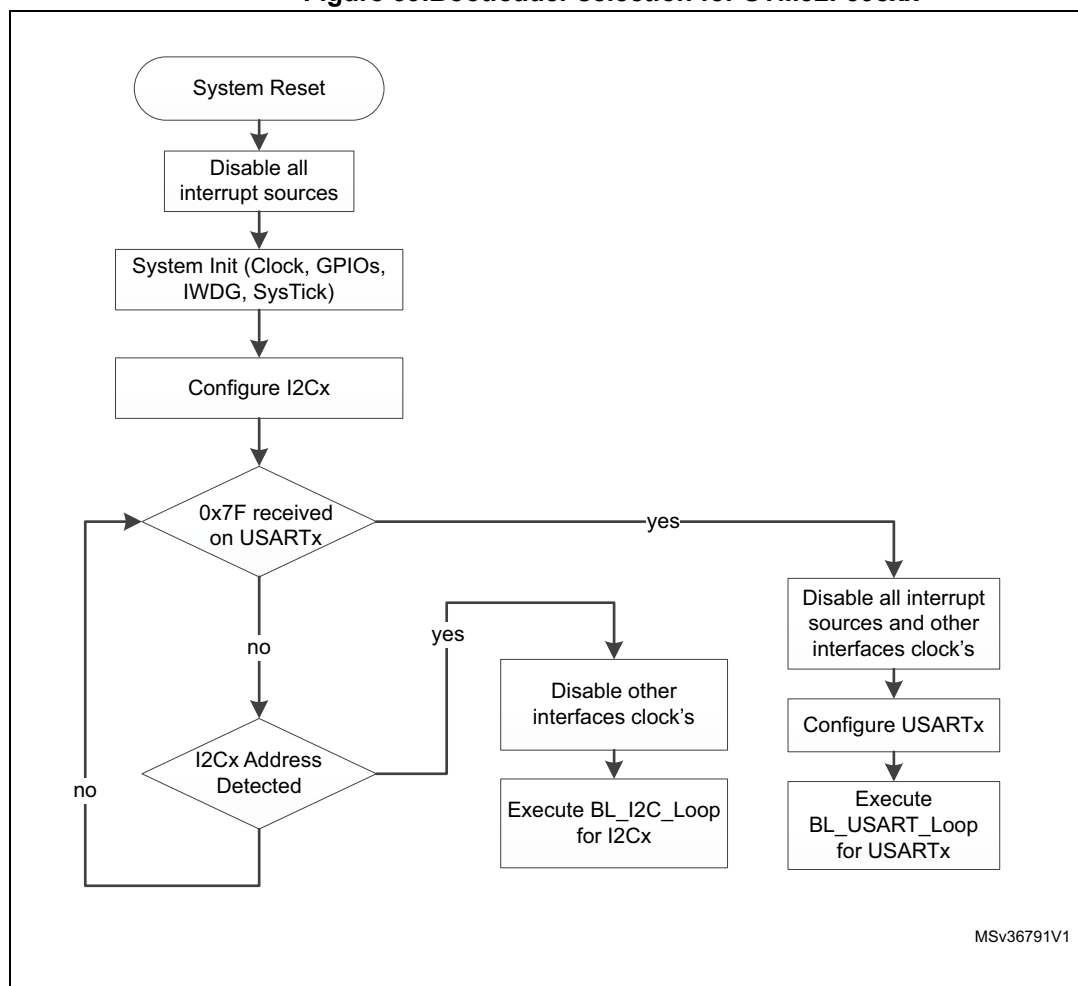
Bootloader	Feature/Peripheral	State	Comment
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 400 kHz – 7-bit address, Target mode – Analog filter ON – Target 7-bit address: 0b1000000x (x = 0 for write and x = 1 for read).
	I2C3_SCL pin	Input/output	PA8 pin: clock line is used in open-drain no pull mode.
	I2C3_SDA pin		PB5 pin: data line is used in open-drain no pull mode.

The system clock is derived from the embedded internal high-speed RC for all bootloader interfaces. No external quartz is required for bootloader operations.

30.2 Bootloader selection

Figure 39 shows the bootloader selection mechanism.

Figure 39. Bootloader selection for STM32F398xx



30.3 Bootloader version

[Table 63](#) lists the STM32F398xx devices bootloader versions.

Table 63. STM32F398xx bootloader versions

Version number	Description	Known limitations
V5.0	Initial bootloader version	None

31 STM32F40xxx/41xxx devices

31.1 Bootloader V3.x

31.1.1 Bootloader configuration

The STM32F40xxx/41xxx bootloader is activated by applying Pattern 1 (described in [Table 2](#)). [Table 64](#) shows the hardware resources used by this bootloader.

Table 64. STM32F40xxx/41xxx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 24 MHz using the PLL. The HSI clock source is used at startup (interface detection phase) and when USARTx interfaces are selected (once CAN or DFU is selected, the clock source is derived from the external crystal).
		HSE enabled	The system clock frequency is 60 MHz. The HSE clock source is used only when the CAN or the DFU (USB FS device) interfaces are selected. The external clock must provide a frequency multiple of 1 MHz and ranging from 4 to 26 MHz.
		-	The CSS interrupt is enabled for the CAN and DFUs. Any failure (or removal) of the external clock generates system reset.
	RAM	-	8 Kbytes, starting from address 0x20000000, are used by the bootloader firmware.
	System memory	-	29 Kbytes, starting from address 0x1FFF 0000 contain the bootloader firmware.
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
	Power	-	Voltage range is set to [1.62 V, 2.1 V]. In this range internal flash write operations are allowed only in byte format (half-word, word, and double-word operations are not allowed). The voltage range can be configured in run time using bootloader commands.
USART1	USART1	Enabled	Once initialized, the USART1 configuration is 8 bits, even parity, and one stop bit.
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input no pull mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in input no pull mode

Table 64. STM32F40xxx/41xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
USART3 (on PB10/PB11)	USART3	Enabled	Once initialized, the USART3 configuration is 8 bits, even parity, and one stop bit.
	USART3_RX pin	Input	PB11 pin: USART3 in reception mode. Used in input pull-up mode.
	USART3_TX pin	Output	PB10 pin: USART3 in transmission mode. Used in input pull-up mode.
USART3 (on PC10/PC11)	USART3	Enabled	Once initialized, the USART3 configuration is 8 bits, even parity, and one stop bit.
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. used in input pull-up mode.
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode. used in input pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
CAN2	CAN2	Enabled	Once initialized, the CAN2 configuration is: Baudrate 125 kbps, 11-bit identifier. Note: CAN1 is clocked during CAN2 bootloader execution because CAN1 manages the communication between CAN2 and SRAM.
	CAN2_RX pin	Input	PB5 pin: CAN2 in reception mode. Used in alternate push-pull, pull-up mode.
	CAN2_TX pin	Output	PB13 pin: CAN2 in transmission mode. Used in alternate push-pull, pull-up mode.
DFU	USB	Enabled	USB OTG FS configured in forced device mode
	USB_DM pin	Input/output	PA11: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12: USB DP line. Used in alternate push-pull, no pull mode. No external pull-up resistor is required
CAN2 and DFUs	TIM11	Enabled	This timer is used to determine the value of the HSE. Once the HSE frequency is determined, the system clock is configured to 60 MHz using PLL and HSE.

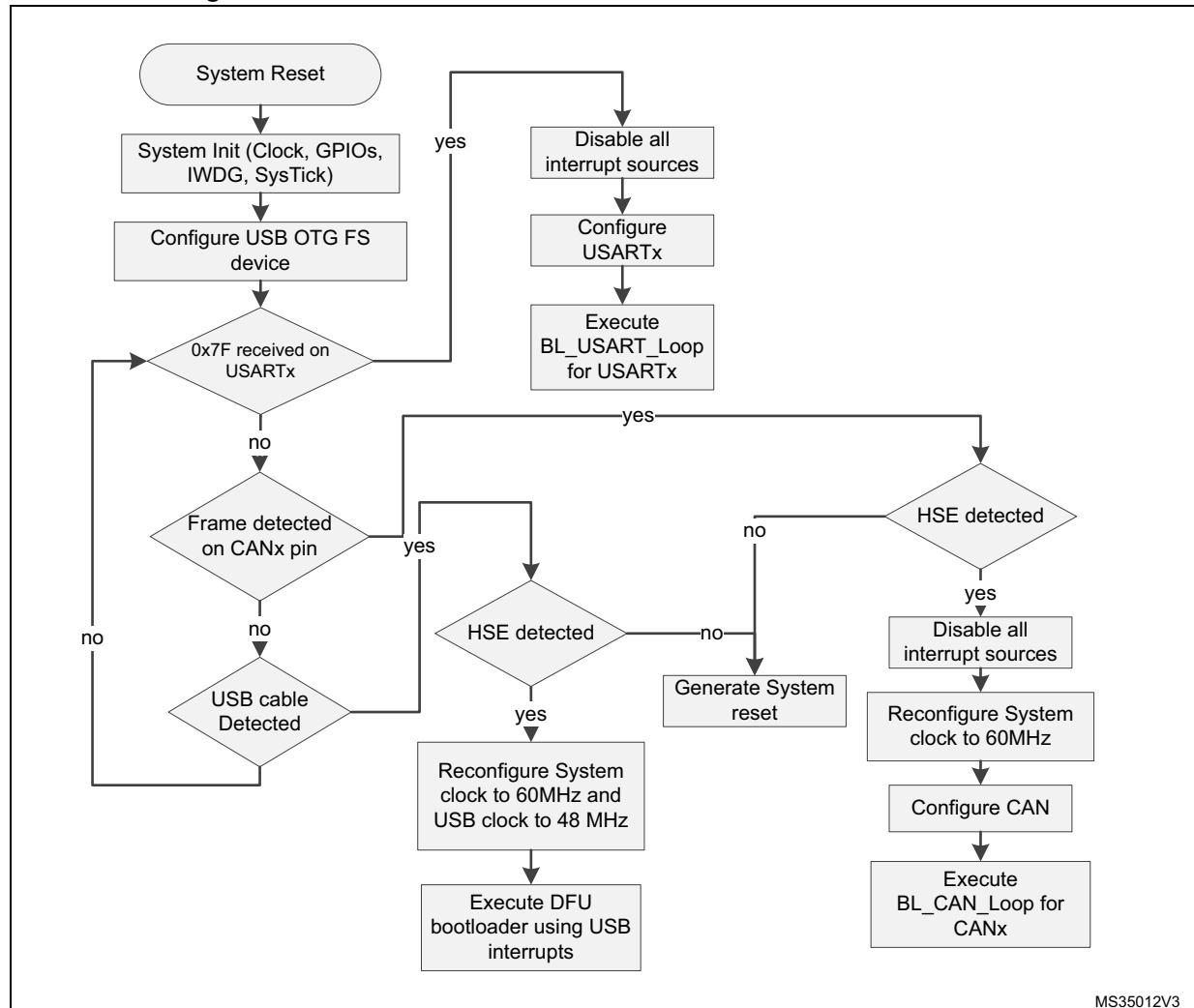
The system clock is derived from the embedded internal high-speed RC for USARTx. This internal clock is also used for CAN and DFU (USB FS device), but only for the selection phase. An external clock multiple of 1 MHz (between 4 and 26 MHz) is required for CAN and DFU execution after the selection phase.

Note: Due to HSI deviation and since HSI is used to detect HSE value, use low rather than high frequency HSE crystals (low frequency values are better detected due to larger error margin). For example, it is better to use 8 MHz instead of 25 MHz.

31.1.2 Bootloader selection

Figure 40 shows the bootloader selection mechanism.

Figure 40. Bootloader V3.x selection for STM32F40xxx/41xxx devices



MS35012V3

31.1.3 Bootloader version

[Table 65](#) lists the STM32F40xxx/41xxx devices V3.x bootloader versions:

Table 65. STM32F40xxx/41xxx bootloader V3.x versions

Version number	Description	Known limitations
V3.0	Initial bootloader version	- When a Read Memory command or Write Memory command is issued with an unsupported memory address and a correct address checksum (i.e. address 0x6000 0000), the command is aborted by the bootloader device, but the NACK (0x1F) is not sent to the host. As a result, the next two bytes (which are the number of bytes to be read/written and its checksum) are considered as a new command and its checksum⁽¹⁾. - Option bytes, OTP and Device Feature descriptors (in DFU interface) are set to "g" instead of "e" (not erasable memory areas). After executing Go command (jump to user code) the bootloader resets AHB1ENR value to 0x0000 0000 and thus CCM RAM, when present, is not active (must be re-enabled by user code at startup)
V3.1	- Fix V3.0 limitations - DFU interface robustness enhancement	- For the USART interface, two consecutive NACKs (instead of 1 NACK) are sent when a Read Memory or Write Memory command is sent and the RDP level is active. - For the CAN interface, the Write Unprotect command is not functional. Use Write Memory command and write directly to the option bytes in order to disable the write protection. After executing Go command (jump to user code) the bootloader resets AHB1ENR value to 0x0000 0000 and thus CCM RAM, when present, is not active (must be re-enabled by user code at startup)

1. If the "number of data - 1" (N-1) to be read/written is not equal to a valid command code (0x00, 0x01, 0x02, 0x11, 0x21, 0x31, 0x43, 0x44, 0x63, 0x73, 0x82 or 0x92), the limitation is not perceived from the host, as the command is NACK-ed anyway (as an unsupported new command).

31.2 Bootloader V9.x

31.2.1 Bootloader configuration

The STM32F40xxx/41xxx bootloader is activated by applying Pattern 1 (described in [Table 2](#)). [Table 66](#) shows the hardware resources used by this bootloader.

Note: *The bootloader version V9.0 is embedded only in STM32F405xx/415xx devices in WLCSP90 package.*

Version V9.1 is populated in all packages of the product.

Table 66. STM32F40xxx/41xxx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 60 MHz using the PLL. The HSI clock source is used at startup (interface detection phase) and when USART or SPI or I2C interfaces are selected (once CAN or DFU is selected, the clock source is derived from the external crystal).
		HSE enabled	The system clock frequency is 60 MHz. The HSE clock source is used only when the CAN or the DFU (USB FS device) interfaces are selected. The external clock must provide a frequency multiple of 1 MHz and ranging from 4 MHz to 26 MHz.
		-	The CSS interrupt is enabled for the CAN and DFUs. Any failure (or removal) of the external clock generates system reset.
	RAM	-	12 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	29 Kbytes, starting from address 0x1FFF0000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	Power	-	Voltage range is set to [1.62 V, 2.1 V]. In this range internal flash write operations are allowed only in byte format (half-word, word, and double-word operations are not allowed). The voltage range can be configured in run time using bootloader commands.
	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input no pull mode.
USART3 (on PB10/PB11)	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in input no pull mode.
	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PB11 pin: USART3 in reception mode. Used in input pull-up mode.
USART3 (on PC10/PC11)	USART3_TX pin	Output	PB10 pin: USART3 in transmission mode
	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. Used in input pull-up mode.
USARTx	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode. Used in input pull-up mode.
	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.

Table 66. STM32F40xxx/41xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
CAN2	CAN2	Enabled	Once initialized the CAN2 configuration is: Baudrate 125 kbps, 11-bit identifier. Note: CAN1 is clocked during CAN2 bootloader execution because CAN1 manages the communication between CAN2 and SRAM.
	CAN2_RX pin	Input	PB5 pin: CAN2 in reception mode. Used in alternate push-pull, pull-up mode.
	CAN2_TX pin	Output	PB13 pin: CAN2 in transmission mode. Used in alternate push-pull, pull-up mode.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b0111010x (x = 0 for write and x = 1 for read).
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain no pull mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b0111010x (x = 0 for write and x = 1 for read).
	I2C2_SCL pin	Input/output	PF1 pin: clock line is used in open-drain no pull mode.
	I2C2_SDA pin		PF0 pin: data line is used in open-drain no pull mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b0111010x (x = 0 for write and x = 1 for read).
	I2C3_SCL pin	Input/output	PA8 pin: clock line is used in open-drain no pull mode.
	I2C3_SDA pin		PC9 pin: data line is used in open-drain no pull mode.

Table 66. STM32F40xxx/41xxx configuration in system memory boot mode (continued)

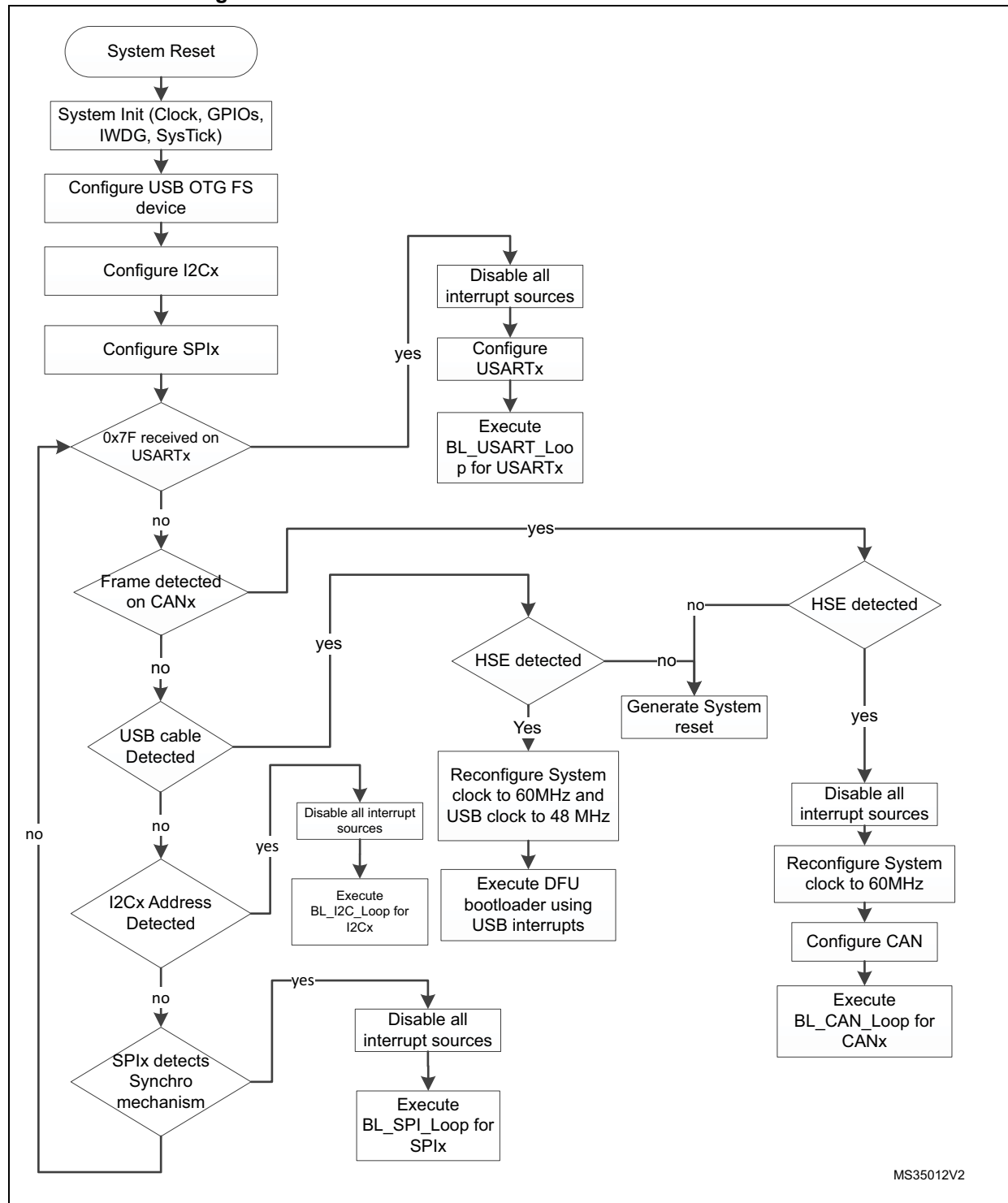
Bootloader	Feature/Peripheral	State	Comment
SPI1	SPI1	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz, Polarity: CPOL low, CPHA low, – NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode
	SPI1_NSS pin		PA4 pin: slave chip select pin used in push-pull, pull-down mode.
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz, Polarity: CPOL low, CPHA low, – NSS hardware.
	SPI2_MOSI pin	Input	PI3 pin: slave data input line, used in push-pull, pull-down mode
	SPI2_MISO pin	Output	PI2 pin: slave data output line, used in push-pull, pull-down mode
	SPI2_SCK pin	Input	PI1 pin: slave clock line, used in push-pull, pull-down mode
	SPI2_NSS pin	Input	PI0 pin: slave chip select pin used in push-pull, pull-down mode.
DFU	USB	Enabled	USB OTG FS configured in forced device mode
	USB_DM pin	Input/output	PA11: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12: USB DP line. Used in alternate push-pull, no pull mode. No external pull-up resistor is required
CAN2 and DFUs	TIM11	Enabled	This timer is used to determine the value of the HSE. Once the HSE frequency is determined, the system clock is configured to 60 MHz using PLL and HSE.

The system clock is derived from the embedded internal high-speed RC for USARTx, I2Cx, and SPIx bootloaders. This internal clock is also used for CAN and DFU (USB FS device), but only for the selection phase. An external clock multiple of 1 MHz (between 4 and 26 MHz) is required for CAN and DFU execution after the selection phase.

Note: *Due to HSI deviation and since HSI is used to detect HSE value, use low rather than high frequency HSE crystal values (low frequency values are better detected due to larger error margin). For example, it is better to use 8 MHz instead of 25 MHz.*

31.2.2 Bootloader selection

Figure 41. Bootloader V9.x selection for STM32F40xxx/41xxx



31.2.3 Bootloader version

[Table 67](#) lists the STM32F40xxx/41xxx devices V9.x bootloader versions.

Table 67. STM32F40xxx/41xxx bootloader V9.x versions

Version number	Description	Known limitations
V9.0	This bootloader is an updated version of bootloader v3.1. This new version of bootloader supports I2C1, I2C2, I2C3, SPI1 and SPI2 interfaces. The RAM used by this bootloader is increased from 8 to 12 Kb. The ID of this bootloader is 0x90. The connection time is increased.	– For the USART interface, two consecutive NACKs (instead of 1 NACK) are sent when a Read Memory or Write Memory command is sent and the RDP level is active. – For the CAN interface, the Write Unprotect command is not functional. Use Write Memory command and write directly to the option bytes in order to disable the write protection. After executing Go command (jump to user code) the bootloader resets AHB1ENR value to 0x0000 0000 and thus CCM RAM, when present, is not active (must be re-enabled by user code at startup)
V9.1	This bootloader is an updated version of bootloader v9.0 that will be populated in all packages even the one embedding the V3.1 bootloader version. It contains fixes of the known limitations of the V9.0	None

32 STM32F401xB(C) devices

32.1 Bootloader configuration

The STM32F401xB(C) bootloader is activated by applying Pattern 1 (described in [Table 2](#)). [Table 68](#) shows the hardware resources used by this bootloader.

Table 68. STM32F401xB(C) configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 60 MHz using the PLL. The HSI clock source is used at startup (interface detection phase) and when USART or SPI or I2C interface is selected (once DFU is selected, the clock source is derived from the external crystal).
		HSE enabled	The system clock frequency is 60 MHz. The HSE clock source is used only when the DFU (USB FS device) interface is selected. The external clock must provide a frequency multiple of 1 MHz and ranging from 4 to 26 MHz.
		-	The CSS interrupt is enabled for the CAN and DFUs. Any failure (or removal) of the external clock generates system reset.
	RAM	-	12 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	29 Kbytes, starting from address 0x1FFF0000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	Power	-	Voltage range is set to [1.62 V, 2.1 V]. In this range internal flash write operations are allowed only in byte format (half-word, word, and double-word operations are not allowed). The voltage range can be configured in run time using bootloader commands.
	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input no pull mode.
USART2	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in input no pull mode.
	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PD6 pin: USART2 in reception mode. Used in input pull-up mode.
USART2	USART2_TX pin	Output	PD5 pin: USART2 in transmission mode. Used in input pull-up mode.

Table 68. STM32F401xB(C) configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b0111010x (x = 0 for write and x = 1 for read).
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain no pull mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b0111010x (x = 0 for write and x = 1 for read).
	I2C2_SCL pin	Input/output	PB10 pin: clock line is used in open-drain no pull mode.
	I2C2_SDA pin		PB3 pin: data line is used in open-drain no pull mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b0111010x (x = 0 for write and x = 1 for read).
	I2C3_SCL pin	Input/output	PA8 pin: clock line is used in open-drain no pull mode.
	I2C3_SDA pin		PB4 pin: data line is used in open-drain no pull mode.
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode
	SPI1_NSS pin		PA4 pin: slave chip select pin used in push-pull, pull-down mode.

Table 68. STM32F401xB(C) configuration in system memory boot mode (continued)

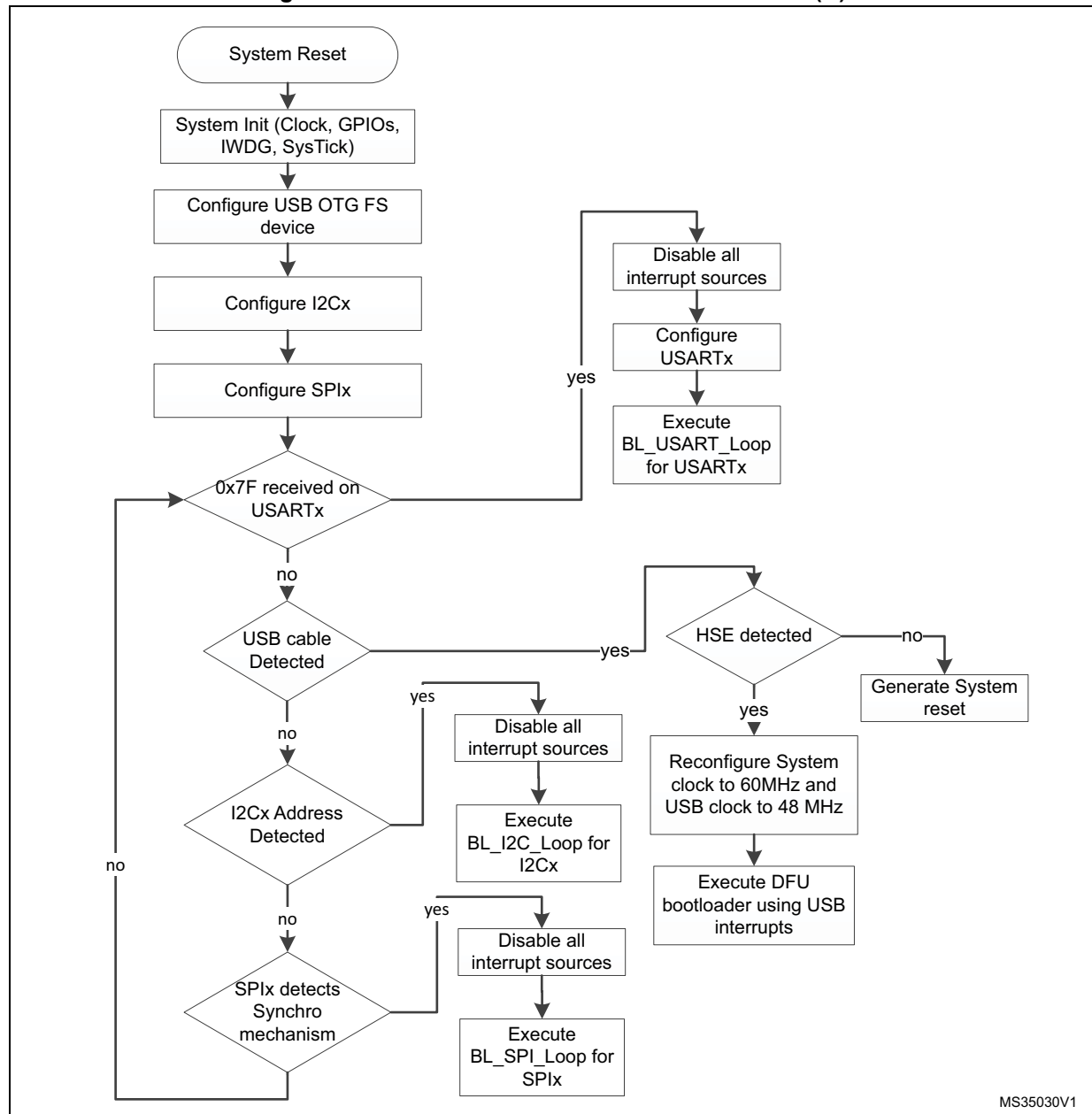
Bootloader	Feature/Peripheral	State	Comment
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PB15 pin: slave data input line, used in push-pull, pull-down mode
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, pull-down mode
	SPI2_SCK pin	Input	PB13 pin: slave clock line, used in push-pull, pull-down mode
	SPI2_NSS pin		PB12 pin: slave chip select pin used in push-pull, pull-down mode.
SPI3	SPI3	Enabled	The SPI3 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI3_MOSI pin	Input	PC12 pin: slave data input line, used in push-pull, pull-down mode
	SPI3_MISO pin	Output	PC11 pin: slave data output line, used in push-pull, pull-down mode
	SPI3_SCK pin	Input	PC10 pin: slave clock line, used in push-pull, pull-down mode
	SPI3_NSS pin		PA15 pin: slave chip select pin used in push-pull, pull-down mode.
DFU	USB	Enabled	USB OTG FS configured in forced device mode
	USB_DM pin	Input/output	PA11: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12: USB DP line. Used in alternate push-pull, no pull mode. No external pull-up resistor is required
	TIM11	Enabled	This timer is used to determine the value of the HSE. Once the HSE frequency is determined, the system clock is configured to 60 MHz using PLL and HSE.

The system clock is derived from the embedded internal high-speed RC for USARTx, I2Cx, and SPIx bootloaders. This internal clock is also used for CAN and DFU (USB FS device), but only for the selection phase. An external clock, multiple of 1 MHz (between 4 and 26 MHz), is required for CAN and DFU execution after the selection phase.

Note: Due to HSI deviation and since HSI is used to detect HSE value, use low rather than high frequency HSE crystal values (low frequency values are better detected due to larger error margin). For example, it is better to use 8 MHz instead of 25 MHz.

32.2 Bootloader selection

Figure 42. Bootloader selection for STM32F401xB(C)



32.3 Bootloader version

Table 69. STM32F401xB(C) bootloader versions

Version number	Description	Known limitations
V13.0	Initial bootloader version	– After executing Go command (jump to user code) the bootloader resets AHB1ENR value to 0x0000 0000 and thus CCM RAM, when present, is not active (must be re-enabled by user code at startup) – The bootloader does not support the reset of SPRMOD bit during RDP regression

33 STM32F401xD(E) devices

33.1 Bootloader configuration

The STM32F401xD(E) bootloader is activated by applying Pattern 1 (described in [Table 2](#)). [Table 70](#) shows the hardware resources used by this bootloader.

Table 70. STM32F401xD(E) configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 60 MHz using the PLL. The HSI clock source is used at startup (interface detection phase) and when USART or SPI or I2C interface is selected (once DFU is selected, the clock source is derived from the external crystal).
		HSE enabled	The system clock frequency is 60 MHz. The HSE clock source is used only when the DFU (USB FS device) interface is selected. The external clock must provide a frequency multiple of 1 MHz, ranging from 4 to 26 MHz.
		-	The CSS interrupt is enabled for the DFU. Any failure (or removal) of the external clock generates system reset.
	RAM	-	12 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	29 Kbytes, starting from address 0x1FFF0000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	Power	-	Voltage range is set to 1.62 V, 2.1 V. In this range internal flash write operations are allowed only in byte format (half-word, word, and double-word operations are not allowed). The voltage range can be configured in run time using bootloader commands.
	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in no pull mode.
USART2	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in no pull mode.
	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PD6 pin: USART2 in reception mode. Used in pull-up mode.
USARTx	USART2_TX pin	Output	PD5 pin: USART2 in transmission mode. Used in pull-up mode.
	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.

Table 70. STM32F401xD(E) configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b0111001x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain no pull mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b0111001x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PB10 pin: clock line is used in open-drain no pull mode.
	I2C2_SDA pin		PB3 pin: data line is used in open-drain no pull mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b0111001x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PA8 pin: clock line is used in open-drain no pull mode.
	I2C3_SDA pin		PB4 pin: data line is used in open-drain no pull mode.
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode.

Table 70. STM32F401xD(E) configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PB15 pin: slave data input line, used in push-pull, pull-down mode
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, pull-down mode
	SPI2_SCK pin	Input	PB13 pin: slave clock line, used in push-pull, pull-down mode
	SPI2_NSS pin	Input	PB12 pin: slave chip select pin used in push-pull, pull-down mode.
SPI3	SPI3	Enabled	The SPI3 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI3_MOSI pin	Input	PC12 pin: slave data input line, used in push-pull, pull-down mode
	SPI3_MISO pin	Output	PC11 pin: slave data output line, used in push-pull, pull-down mode
	SPI3_SCK pin	Input	PC10 pin: slave clock line, used in push-pull, pull-down mode
	SPI3_NSS pin	Input	PA15 pin: slave chip select pin used in push-pull, pull-down mode.
DFU	USB	Enabled	USB OTG FS configured in forced device mode
	USB_DM pin	Input/output	PA11: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12: USB DP line. Used in alternate push-pull, no pull mode. No external pull-up resistor is required
	TIM11	Enabled	This timer is used to determine the value of the HSE. Once the HSE frequency is determined, the system clock is configured to 60 MHz using PLL and HSE.

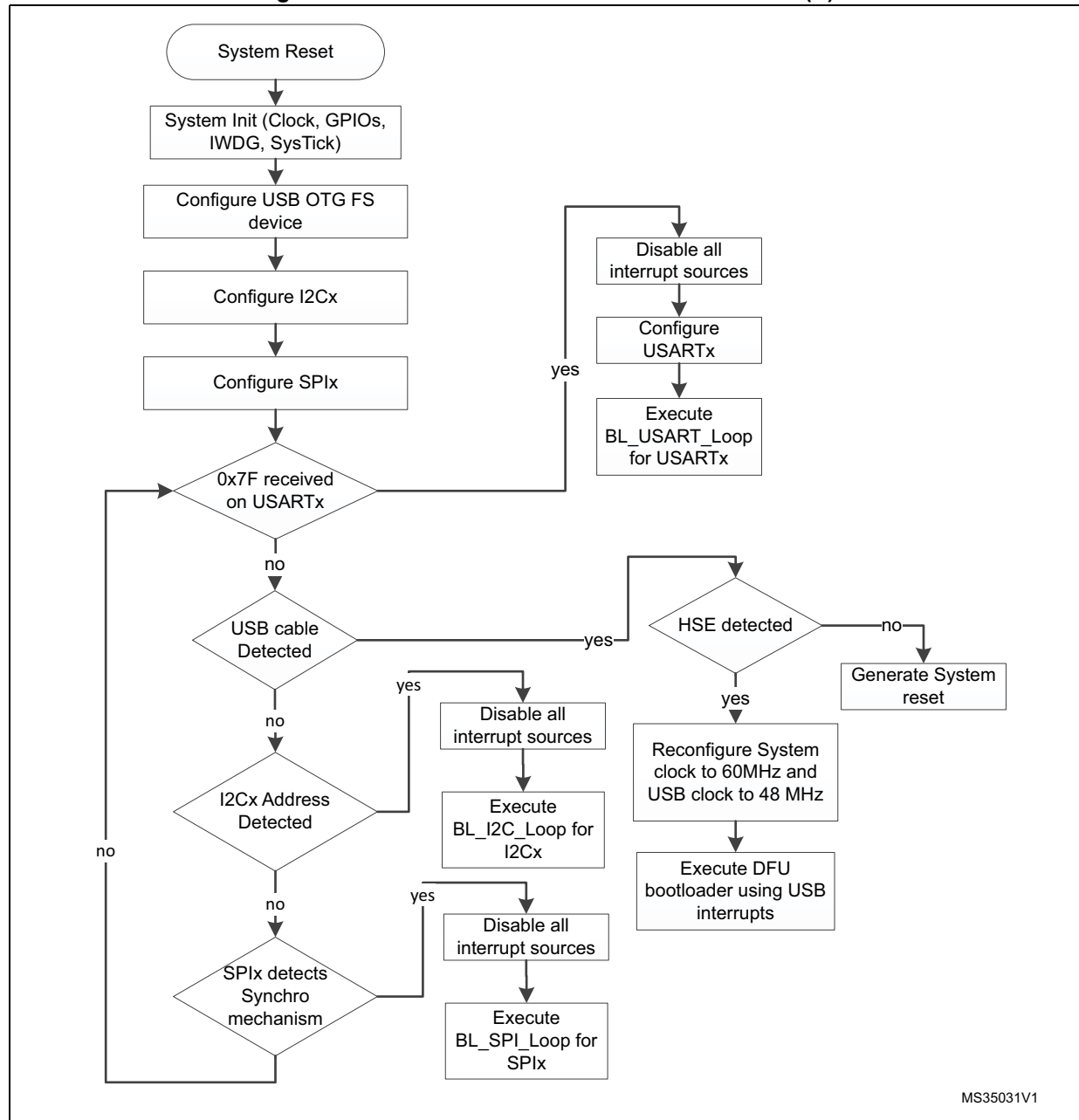
The system clock is derived from the embedded internal high-speed RC for USARTx, I2Cx, and SPIx bootloaders. This internal clock is also used for DFU (USB FS device), but only for the selection phase. An external clock multiple of 1 MHz (between 4 and 26 MHz) is required for DFU execution after the selection phase.

Note: Due to HSI deviation and since HSI is used to detect HSE value, use low rather than high frequency HSE crystal values (low frequency values are better detected due to larger error margin). For example, it is better to use 8 MHz instead of 25 MHz.

33.2 Bootloader selection

Figure 43 shows the bootloader selection mechanism.

Figure 43. Bootloader selection for STM32F401xD(E)



33.3 Bootloader version

[Table 71](#) lists the STM32F401xD(E) devices bootloader version.

Table 71. STM32F401xD(E) bootloader versions

Version number	Description	Known limitations
V13.1	Initial bootloader version	After executing Go command (jump to user code) the bootloader resets AHB1ENR value to 0x0000 0000 and thus CCM RAM, when present, is not active (must be re-enabled by user code at startup)

34 STM32F410xx devices

34.1 Bootloader configuration

The STM32F410xx bootloader is activated by applying Pattern 1 (described in [Table 2](#)). [Table 72](#) shows the hardware resources used by this bootloader.

Table 72. STM32F410xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The HSI is used at startup as clock source for system clock configured to 60 MHz and for USART and I2C bootloader operation.
	RAM	-	5 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	29 Kbytes, starting from address 0x1FFF0000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
	Power	-	The voltage range is [1.8V, 3.6V]. In this range: – Flash wait states: 3. – System clock frequency 60 MHz. – ART Accelerator enabled. – Flash write operation by byte (refer to bootloader memory management section for more information).
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.

Table 72. STM32F410xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1000111x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain no pull mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1000111x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PB10 pin: clock line is used in open-drain no pull mode.
	I2C2_SDA pin		PB11 pin: data line is used in open-drain no pull mode.
I2C4	I2C4	Enabled	The I2C4 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1000111x (x = 0 for write and x = 1 for read)
	I2C4_SCL pin	Input/output	PB15 pin: clock line is used in open-drain no pull mode for STM32F410Cx/Rx devices. PB10 pin: clock line is used in open-drain no pull mode for STM32F410Tx devices.
	I2C4_SDA pin	Input/output	PB14 pin: data line is used in open-drain no pull mode for STM32F410Cx/Rx devices. PB3 pin: data line is used in open-drain no pull mode for STM32F410Tx devices.

Table 72. STM32F410xx configuration in system memory boot mode (continued)

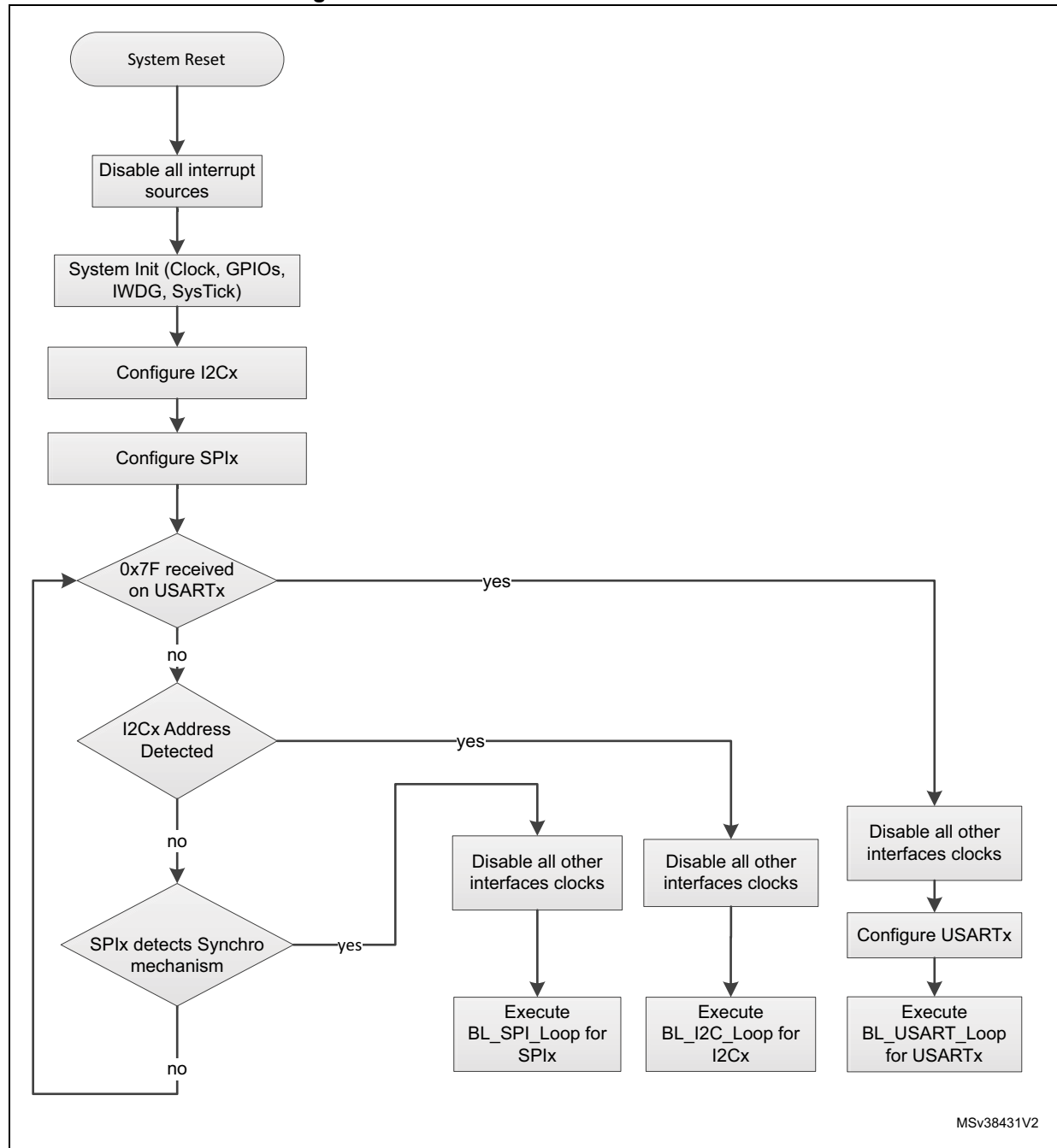
Bootloader	Feature/Peripheral	State	Comment
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode for STM32F410Cx/Rx devices. PB5 pin: slave data input line, used in push-pull, pull-down mode for STM32F410Tx devices.
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode for STM32F410Cx/Rx devices. PB4 pin: slave data output line, used in push-pull, pull-down mode for STM32F410Tx devices.
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode.
	SPI1_NSS pin		PA4 pin: slave chip select pin used in push-pull, pull-up mode for STM32F410Cx/Rx devices. PA15 pin: slave chip select pin used in push-pull, pull-down mode for STM32F410Tx devices.
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PC3 pin: slave data input line, used in push-pull, pull-down mode
	SPI2_MISO pin	Output	PC2 pin: slave data output line, used in push-pull, pull-down mode
	SPI2_SCK pin	Input	PB13 pin: slave clock line, used in push-pull, pull-down mode
	SPI2_NSS pin	Input	PB12 pin: slave chip select pin used in push-pull, pull-down mode.

The system clock is derived from the embedded internal high-speed RC for all bootloader interfaces. No external quartz is required for bootloader operations.

34.2 Bootloader selection

Figure 44 shows the bootloader selection mechanism.

Figure 44. Bootloader V11.x selection for STM32F410xx



34.3 Bootloader version

[Table 73](#) lists the STM32F410xx devices bootloader V11.x versions.

Table 73. STM32F410xx bootloader V11.x versions

Version number	Description	Known limitations
V11.0	Initial bootloader version	After executing Go command (jump to user code) the bootloader resets AHB1ENR value to 0x0000 0000 and thus CCM RAM, when present, is not active (must be re-enabled by user code at startup)
V11.1	Support I2C4 and SPI1 for STM32F410Tx devices	After executing Go command (jump to user code) the bootloader resets AHB1ENR value to 0x0000 0000 and thus CCM RAM, when present, is not active (must be re-enabled by user code at startup)

35 STM32F411xx devices

35.1 Bootloader configuration

The STM32F411xx bootloader is activated by applying Pattern 1 (described in [Table 2](#)). [Table 74](#) shows the hardware resources used by this bootloader.

Table 74. STM32F411xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 60 MHz using the PLL. The HSI clock source is used at startup (interface detection phase) and when USART or SPI or I2C interface is selected (once DFU is selected, the clock source is derived from the external crystal).
		HSE enabled	The system clock frequency is 60 MHz. The HSE clock source is used only when the DFU (USB FS device) interface is selected. The external clock must provide a frequency multiple of 1 MHz and ranging from 4 MHz to 26 MHz.
		-	The CSS interrupt is enabled for the DFU. Any failure (or removal) of the external clock generates system reset.
	RAM	-	12 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	29 Kbytes, starting from address 0x1FFF0000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	Power	-	Voltage range is set to [1.62 V, 2.1 V]. In this range internal flash write operations are allowed only in byte format (half-word, word, and double-word operations are not allowed). The voltage range can be configured in run time using bootloader commands.
	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input no pull mode.
USART2	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in input no pull mode.
	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PD6 pin: USART2 in reception mode. Used in input pull-up mode.
USART2	USART2_TX pin	Output	PD5 pin: USART2 in transmission mode. Used in input pull-up mode.

Table 74. STM32F411xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b0111001x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain no pull mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b0111001x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PB10 pin: clock line is used in open-drain no pull mode.
	I2C2_SDA pin		PB3 pin: data line is used in open-drain no pull mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b0111001x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PA8 pin: clock line is used in open-drain no pull mode.
	I2C3_SDA pin		PB4 pin: data line is used in open-drain no pull mode.
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode.

Table 74. STM32F411xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware
	SPI2_MOSI pin	Input	PB15 pin: slave data input line, used in push-pull, pull-down mode
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, pull-down mode
	SPI2_SCK pin	Input	PB13 pin: slave clock line, used in push-pull, pull-down mode
	SPI2_NSS pin	Input	PB12 pin: slave chip select pin used in push-pull, pull-down mode.
SPI3	SPI3	Enabled	The SPI3 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware
	SPI3_MOSI pin	Input	PC12 pin: slave data input line, used in push-pull, pull-down mode
	SPI3_MISO pin	Output	PC11 pin: slave data output line, used in push-pull, pull-down mode
	SPI3_SCK pin	Input	PC10 pin: slave clock line, used in push-pull, pull-down mode
	SPI3_NSS pin	Input	PA15 pin: slave chip select pin used in push-pull, pull-down mode.
DFU	USB	Enabled	USB OTG FS configured in forced device mode
	USB_DM pin	Input/output	PA11: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12: USB DP line. Used in alternate push-pull, no pull mode. No external pull-up resistor is required
	TIM11	Enabled	This timer is used to determine the value of the HSE. Once the HSE frequency is determined, the system clock is configured to 60 MHz using PLL and HSE.

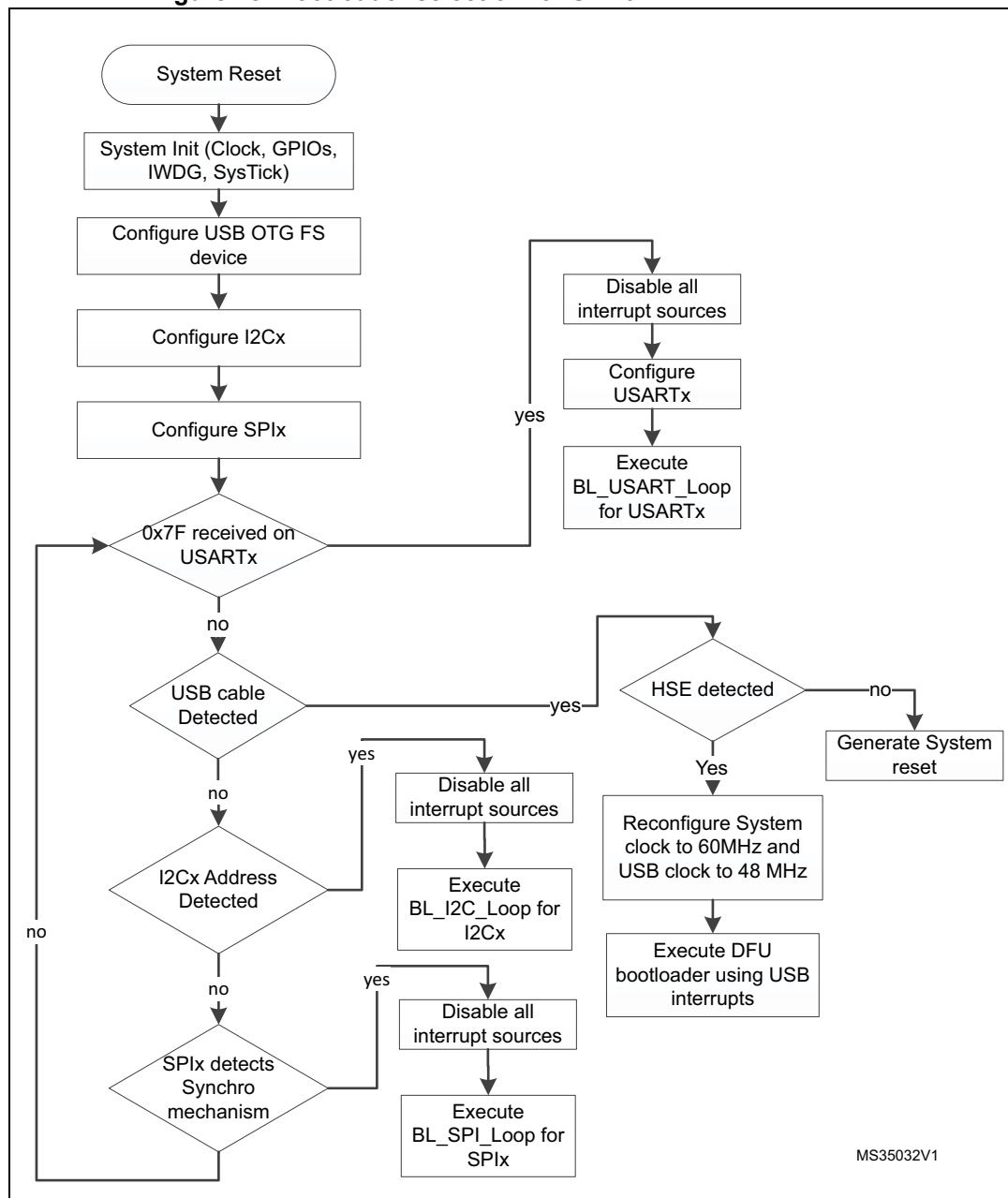
The system clock is derived from the embedded internal high-speed RC for USARTx, I2Cx, I2Cx, and SPIx bootloaders. This internal clock is also used for DFU (USB FS device), but only for the selection phase. An external clock multiple of 1 MHz (between 4 and 26 MHz) is required for DFU execution after the selection phase.

Note: *Due to HSI deviation and since HSI is used to detect HSE value, use low rather than high frequency HSE crystal values (low frequency values are better detected due to larger error margin). For example, it is better to use 8 MHz instead of 25 MHz.*

35.2 Bootloader selection

Figure 45 shows the bootloader selection mechanism.

Figure 45. Bootloader selection for STM32F411xx



35.3 Bootloader version

The following table lists the STM32F411xx devices bootloader version.

Table 75. STM32F411xx bootloader versions

Version number	Description	Known limitations
V13.0	Initial bootloader version	After executing Go command (jump to user code) the bootloader resets AHB1ENR value to 0x0000 0000 and thus CCM RAM, when present, is not active (must be re-enabled by user code at startup)

36 STM32F412xx devices

36.1 Bootloader configuration

The STM32F412xx bootloader is activated by applying Pattern 1 (described in [Table 2](#)). [Table 76](#) shows the hardware resources used by this bootloader.

Table 76. STM32F412xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The HSI is used at startup as clock source for system clock configured to 60 MHz and for USART and I2C bootloader operation.
		HSE enabled	The HSE is used only when the CAN or the DFU (USB FS device) interfaces are selected. In this case the system clock is configured to 60 MHz with HSE as clock source. The HSE frequency must be a multiple of 1 MHz, ranging from 4 to 26 MHz.
		-	The CSS interrupt is enabled for the CAN and DFUs. Any failure (or removal) of the external clock generates a system reset.
	RAM	-	16 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	29 Kbytes, starting from address 0x1FFF0000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
	Power	-	The voltage range is [1.8V, 3.6V]. In this range: - Flash wait states: 3. - System clock frequency 60 MHz. - ART Accelerator enabled. - Flash write operation by byte (refer to bootloader memory management section for more information).
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input no pull mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in input no pull mode.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PD6 pin: USART2 in reception mode. Used in input pull-up mode.
	USART2_TX pin	Output	PD5 pin: USART2 in transmission mode. Used in input pull-up mode.

Table 76. STM32F412xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
USART3	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PB11 pin: USART3 in reception mode. Used in input pull-up mode.
	USART3_TX pin	Output	PB10 pin: USART3 in transmission mode. Used in input pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
CAN2	CAN2	Enabled	Once initialized the CAN2 configuration is: Baudrate 125 kbps, 11-bit identifier. Note: CAN1 is clocked during CAN2 bootloader execution because CAN1 manages the communication between CAN2 and SRAM.
	CAN2_RX pin	Input	PB5 pin: CAN2 in reception mode. Used in alternate push-pull, pull-up mode.
	CAN2_TX pin	Output	PB13 pin: CAN2 in transmission mode. Used in alternate push-pull, pull-up mode.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1000110x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain no pull mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1000110x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PF1 pin: clock line is used in open-drain no pull mode.
	I2C2_SDA pin		PF0 pin: data line is used in open-drain no pull mode.

Table 76. STM32F412xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1000110x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PA8 pin: clock line is used in open-drain no pull mode.
	I2C3_SDA pin		PB4 pin: data line is used in open-drain no pull mode.
I2C4	I2C4	Enabled	The I2C4 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1000110x (x = 0 for write and x = 1 for read)
	I2C4_SCL pin	Input/output	PB15 pin: clock line is used in open-drain no pull mode.
	I2C4_SDA pin		PB14 pin: data line is used in open-drain no pull mode.
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode
	SPI1_NSS pin		PA4 pin: slave chip select pin used in push-pull, pull-up mode.

Table 76. STM32F412xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI3	SPI3	Enabled	The SPI3 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI3_MOSI pin	Input	PC12 pin: slave data input line, used in push-pull, pull-down mode
	SPI3_MISO pin	Output	PC11 pin: slave data output line, used in push-pull, pull-down mode
	SPI3_SCK pin	Input	PC10 pin: slave clock line, used in push-pull, pull-down mode
	SPI3_NSS pin		PA15 pin: slave chip select pin used in push-pull, pull-up mode.
SPI4	SPI4	Enabled	The SPI4 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI4_MOSI pin	Input	PE14 pin: slave data input line, used in push-pull, pull-down mode
	SPI4_MISO pin	Output	PE13 pin: slave data output line, used in push-pull, pull-down mode
	SPI4_SCK pin	Input	PE12 pin: slave clock line, used in push-pull, pull-down mode
	SPI4_NSS pin		PE11 pin: slave chip select pin used in push-pull, pull-up mode.
DFU	USB	Enabled	USB OTG FS configured in forced device mode
	USB_DM pin	Input/output	PA11 pin: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12 pin: USB DP line. Used in alternate push-pull, no pull mode. No external pull-up resistor is required.
CAN2 and DFUs	TIM11	Enabled	This timer is used to determine the value of the HSE. Once HSE frequency is determined, the system clock is configured to 60 MHz using PLL and HSE.

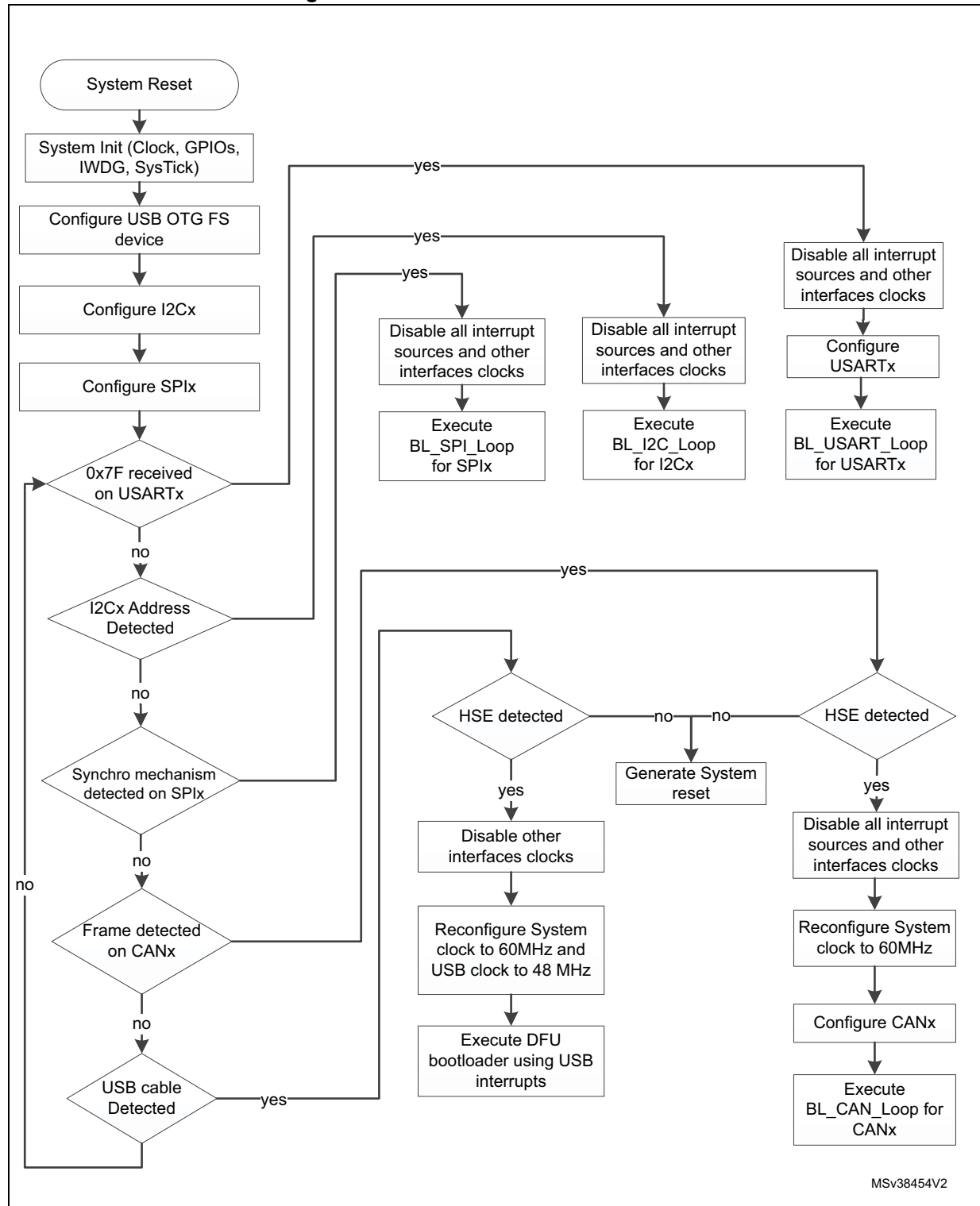
The system clock is derived from the embedded internal high-speed RC for USARTx and I2Cx bootloaders. This internal clock is also used for CAN and DFU (USB FS device), but only for the selection phase. An external clock multiple of 1 MHz (between 4 and 26 MHz) is required for CAN and DFU execution after the selection phase.

Note: Due to HSI deviation and since HSI is used to detect HSE value, use low rather than high frequency HSE crystal values (low frequency values are better detected due to larger error margin). For example, it is better to use 8 MHz instead of 25 MHz.

36.2 Bootloader selection

Figure 46 shows the bootloader selection mechanism.

Figure 46. Bootloader V9.x selection for STM32F412xx



36.3 Bootloader version

The following table lists the STM32F412xx devices bootloader V9.x versions.

Table 77. STM32F412xx bootloader V9.x versions

Version number	Description	Known limitations
V9.0	Initial bootloader version	After executing Go command (jump to user code) the bootloader resets AHB1ENR value to 0x0000 0000 and thus CCM RAM, when present, is not active (must be re-enabled by user code at startup)
V9.1	Fix USART3 interface pinout	After executing Go command (jump to user code) the bootloader resets AHB1ENR value to 0x0000 0000 and thus CCM RAM, when present, is not active (must be re-enabled by user code at startup)

37 STM32F413xx/423xx devices

37.1 Bootloader configuration

The STM32F413xx/423xx bootloader is activated by applying Pattern 1 (described in [Table 2h](#)). The following table shows the hardware resources used by this bootloader.

Table 78. STM32F413xx/423xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The HSI is used at startup as clock source for system clock configured to 60 MHz and for USART and I2C bootloader operation.
		HSE enabled	The HSE is used only when the CAN or the DFU (USB FS device) interfaces are selected. In this case the system clock configured to 60 MHz with HSE as clock source. The HSE frequency must be multiple of 1 MHz and ranging from 4 MHz to 26 MHz.
		-	The CSS interrupt is enabled for the CAN and DFUs. Any failure (or removal) of the external clock generates system reset.
	RAM	-	16 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	60 Kbytes, starting from address 0x1FF00000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
	Power	-	The voltage range is [1.8V, 3.6V] In this range: – Flash wait states 4. – System clock frequency 60 MHz. – ART Accelerator enabled. – Flash write operation by byte (refer to Bootloader memory management for more information).
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input no pull mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in input no pull mode.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PD6 pin: USART2 in reception mode. Used in input pull-up mode.
	USART2_TX pin	Output	PD5 pin: USART2 in transmission mode. Used in input pull-up mode.

Table 78. STM32F413xx/423xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
USART3	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PB11 pin: USART3 in reception mode. Used in input pull-up mode.
	USART3_TX pin	Output	PB10 pin: USART3 in transmission mode. Used in input pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
CAN2	CAN2	Enabled	Once initialized the CAN2 configuration is: Baudrate 125 kbps, 11-bit identifier. Note: CAN1 is clocked during CAN2 bootloader execution because CAN1 manages the communication between CAN2 and SRAM.
	CAN2_RX pin	Input	PB5 pin: CAN2 in reception mode. Used in alternate push-pull, pull-up mode.
	CAN2_TX pin	Output	PB13 pin: CAN2 in transmission mode. Used in alternate push-pull, pull-up mode.
I2C1	I2C1	Enabled	The I2C1 configuration is: I2C speed: up to 400 kHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b1001011x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin	Input/output	PB7 pin: data line is used in open-drain no pull mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: I2C speed: up to 400 kHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b1001011x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PF1 pin: clock line is used in open-drain no pull mode.
	I2C2_SDA pin	Input/output	PF0 pin: data line is used in open-drain no pull mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: I2C speed: up to 400 kHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b1001011x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PA8 pin: clock line is used in open-drain no pull mode.
	I2C3_SDA pin	Input/output	PB4 pin: data line is used in open-drain no pull mode.

Table 78. STM32F413xx/423xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I2C4	I2C4	Enabled	The I2C4 configuration is: I2C speed: up to 1 MHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b1001011x (x = 0 for write and x = 1 for read)
	I2C4_SCL pin	Input/output	PB15 pin: clock line is used in open-drain no pull mode.
	I2C4_SDA pin	Input/output	PB14 pin: data line is used in open-drain no pull mode.
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB, speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode.
SPI3	SPI3	Enabled	The SPI3 configuration is: – Slave mode – Full Duplex – 8-bit MSB, speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI3_MOSI pin	Input	PC12 pin: slave data input line, used in push-pull, pull-down mode
	SPI3_MISO pin	Output	PC11 pin: slave data output line, used in push-pull, pull-down mode
	SPI3_SCK pin	Input	PC10 pin: slave clock line, used in push-pull, pull-down mode
	SPI3_NSS pin	Input	PA15 pin: slave chip select pin used in push-pull, pull-down mode.

Table 78. STM32F413xx/423xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI4	SPI4	Enabled	The SPI4 configuration is: – Slave mode – Full Duplex – 8-bit MSB, speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI4_MOSI pin	Input	PE14 pin: slave data input line, used in push-pull, pull-down mode
	SPI4_MISO pin	Output	PE13 pin: slave data output line, used in push-pull, pull-down mode
	SP4_SCK pin	Input	PE12 pin: slave clock line, used in push-pull, pull-down mode
	SPI4_NSS pin	Input	PE11 pin: slave chip select pin used in push-pull, pull-down mode.
DFU	USB	Enabled	USB OTG FS configured in forced device mode
	USB_DM pin	Input/output	PA11 pin: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12 pin: USB DP line. Used in alternate push-pull, no pull mode. No external pull-up resistor is required.
CAN2 and DFUs	TIM11	Enabled	This timer is used to determine the value of the HSE. Once HSE frequency is determined, the system clock is configured to 60 MHz using PLL and HSE.

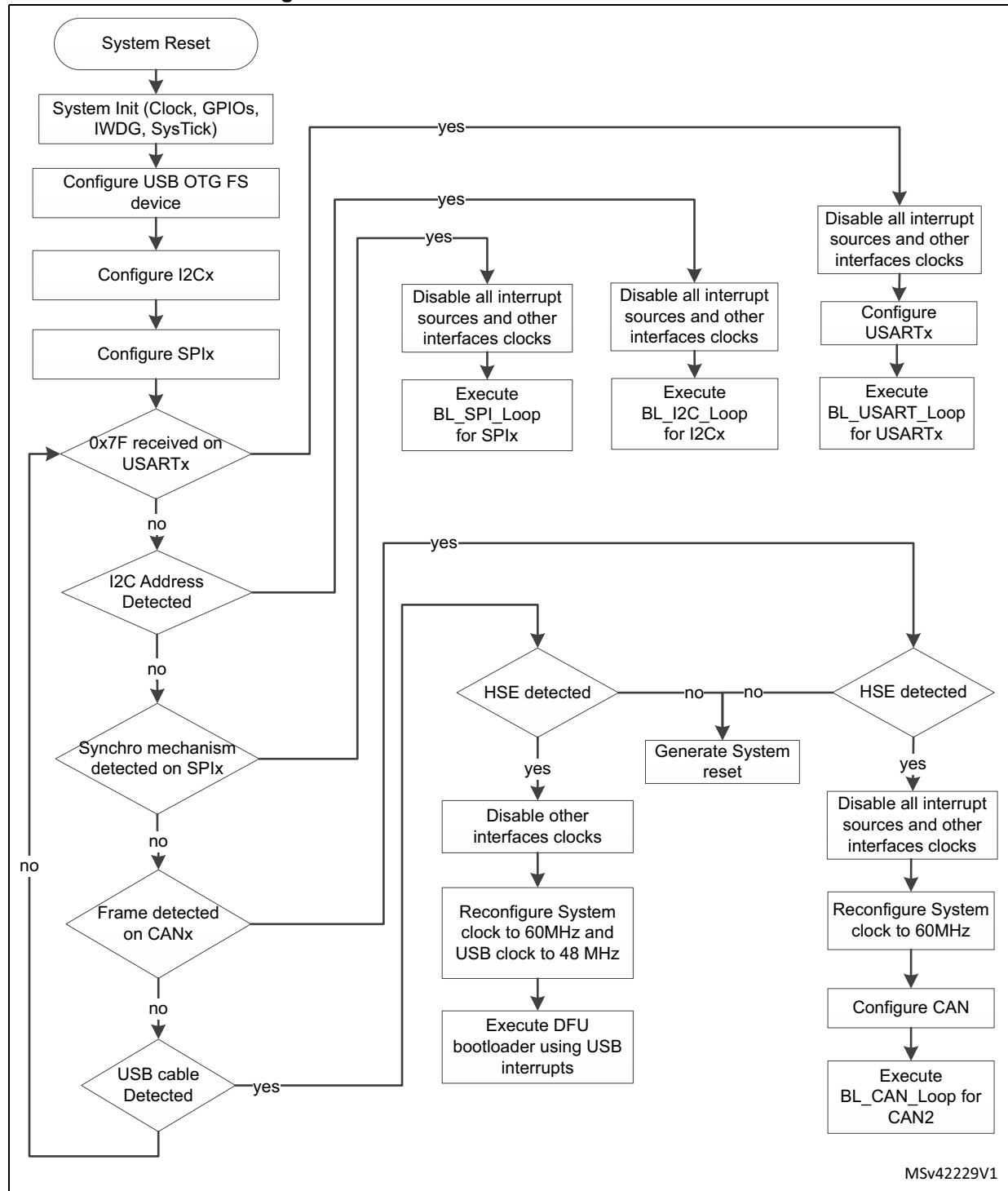
The system clock is derived from the embedded internal high-speed RC for USARTx and I2Cx bootloaders. This internal clock is also used for CAN and DFU (USB FS device), but only for the selection phase. An external clock multiple of 1 MHz (between 4 and 26 MHz) is required for CAN and DFU execution after the selection phase.

Note: *Due to HSI deviation and since HSI is used to detect HSE value, use low rather than high frequency HSE crystal values (low frequency values are better detected due to larger error margin). For example, it is better to use 8 MHz instead of 25 MHz.*

37.2 Bootloader selection

Figure 47 shows the bootloader selection mechanism.

Figure 47. Bootloader V9.x selection for STM32F413xx/423xx



37.3 Bootloader version

The following table lists the STM32F413xx/423xx devices bootloader V9.x versions.

Table 79. STM32F413xx/423xx bootloader V9.x versions

Version number	Description	Known limitations
V9.0	Initial bootloader version	After executing Go command (jump to user code) the bootloader resets AHB1ENR value to 0x0000 0000 and thus CCM RAM, when present, is not active (must be re-enabled by user code at startup)

38 STM32F42xxx/43xxx devices

38.1 Bootloader V7.x

38.1.1 Bootloader configuration

The STM32F42xxx/43xxx bootloader is activated by applying Pattern 5 (described in [Table 2](#)). The following table shows the hardware resources used by this bootloader.

Table 80. STM32F42xxx/43xxx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 24 MHz using the PLL. The HSI clock source is used at startup (interface detection phase) and when USART or I2C interfaces are selected (once CAN or DFU is selected, the clock source is derived from the external crystal).
		HSE enabled	The system clock frequency is 60 MHz. The HSE clock source is used only when the CAN or the DFU (USB FS device) interfaces are selected. The external clock must provide a frequency multiple of 1 MHz and ranging from 4 MHz to 26 MHz.
		-	The CSS interrupt is enabled for the CAN and DFUs. Any failure (or removal) of the external clock /generates system reset.
	RAM	-	8 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	29 Kbytes, starting from address 0x1FFF0000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
	Power	-	Voltage range is set to [1.62 V, 2.1 V]. In this range internal flash write operations are allowed only in byte format (half-word, word, and double-word operations are not allowed). The voltage range can be configured in run time using bootloader commands.

Table 80. STM32F42xxx/43xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
USART1	USART1	Enabled	Once initialized, the configuration is 8 bits, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input no pull mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in input no pull mode.
USART3 (on PB10/PB11)	USART3	Enabled	Once initialized, the configuration is 8 bits, even parity, and one stop bit
	USART3_RX pin	Input	PB11 pin: USART3 in reception mode. Used in input pull-up mode.
	USART3_TX pin	Output	PB10 pin: USART3 in transmission mode. Used in input pull-up mode.
USART3 (on PC10/PC11)	USART3	Enabled	Once initialized, the configuration is 8 bits, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. Used in input pull-up mode.
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode. Used in input pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
CAN2	CAN2	Enabled	Once initialized the CAN2 configuration is: Baudrate 125 kbps, 11-bit identifier. Note: CAN1 is clocked during CAN2 bootloader execution because CAN1 manages the communication between CAN2 and SRAM.
	CAN2_RX pin	Input	PB5 pin: CAN2 in reception mode. Used in alternate push pull, pull-up mode.
	CAN2_TX pin	Output	PB13 pin: CAN2 in transmission mode. Used in alternate push pull, pull-up mode.
I2C1	I2C1	Enabled	The I2C1 configuration is: I2C speed: up to 400 kHz, 7-bit address, Slave mode, Analog filter ON Target 7-bit address: 0b0111000x (x = 0 for write and x = 1 for read).
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin	Input/output	PB9 pin: data line is used in open-drain no pull mode.

Table 80. STM32F42xxx/43xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
DFU	USB	Enabled	USB OTG FS configured in forced device mode
	USB_DM pin	Input/output	PA11: USB DM line. Used in alternate push pull no pull mode.
	USB_DP pin		PA12: USB DP line. Used in alternate push pull no pull mode. No external pull-up resistor is required
CAN2 and DFUs	TIM11	Enabled	This timer is used to determine the value of the HSE. Once the HSE frequency is determined, the system clock is configured to 60 MHz using PLL and HSE.

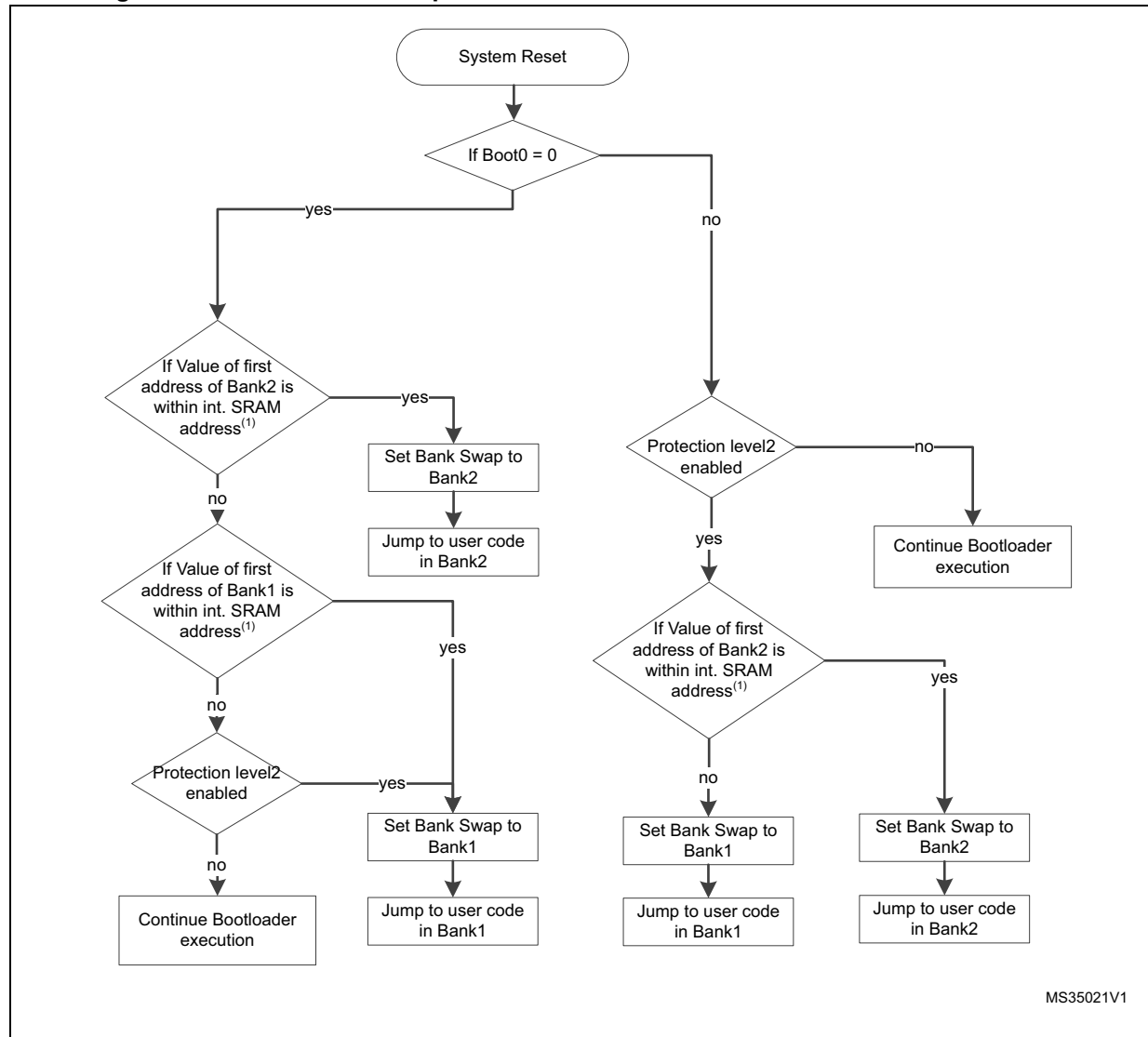
The system clock is derived from the embedded internal high-speed RC for USARTx and I2Cx bootloaders. This internal clock is also used for CAN and DFU (USB FS device), but only for the selection phase. An external clock multiple of 1 MHz (between 4 and 26 MHz) is required for CAN and DFU execution after the selection phase.

Note: *Due to HSI deviation and since HSI is used to detect HSE value, use low rather than high frequency HSE crystal values (low frequency values are better detected due to larger error margin). For example, it is better to use 8 MHz instead of 25 MHz.*

38.1.2 Bootloader selection

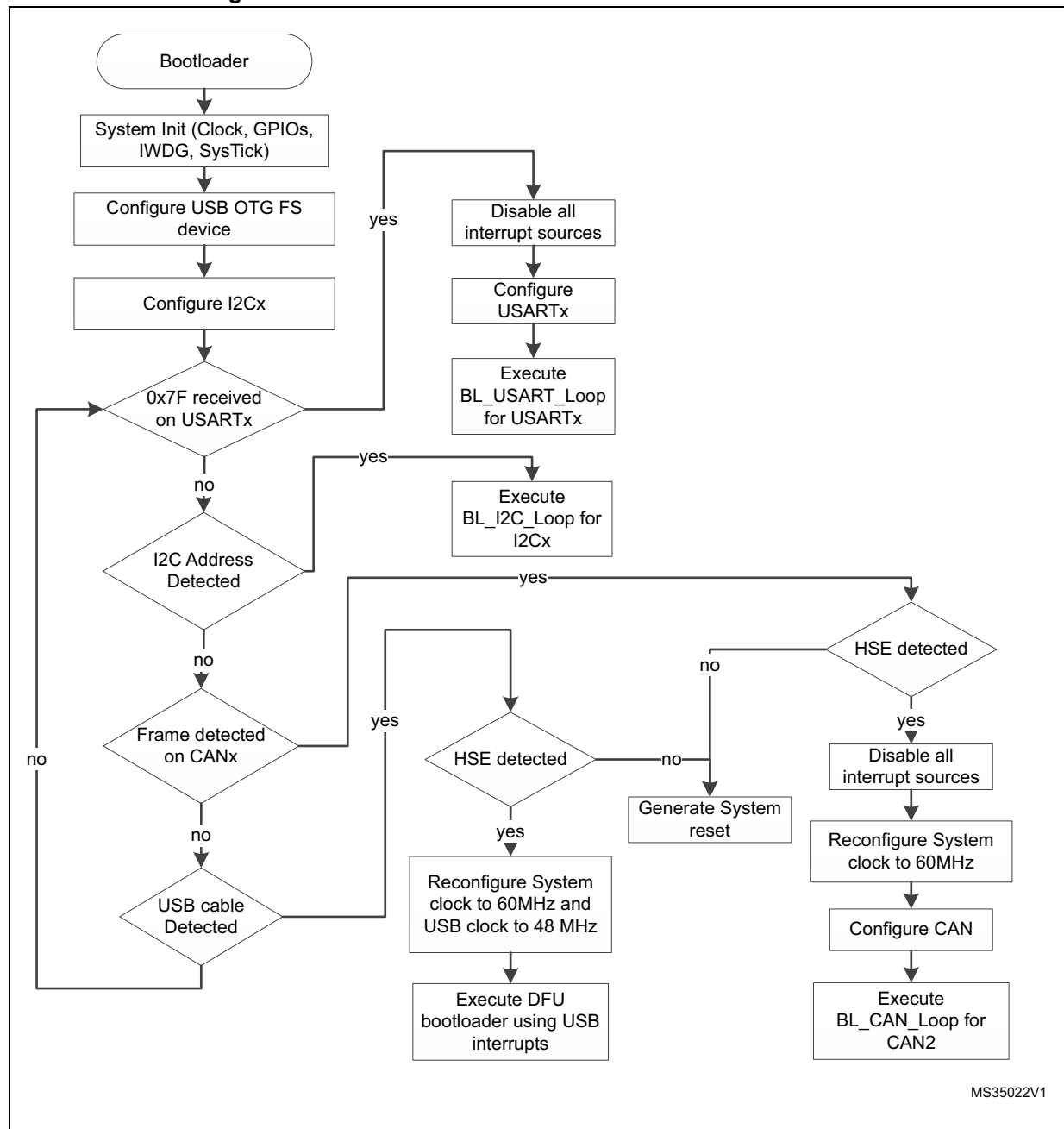
Figure 48 and Figure 49 show the bootloader selection mechanism.

Figure 48. Dual bank boot implementation for STM32F42xxx/43xxx Bootloader V7.x



1. CCM RAM is not considered valid as stack pointer address for the dual bank boot mechanism.

Figure 49. Bootloader V7.x selection for STM32F42xxx/43xxx



38.1.3 Bootloader version

The following table lists the STM32F42xxx/43xxx devices bootloader V7.x versions.

Table 81. STM32F42xxx/43xxx bootloader V7.x versions

Version number	Description	Known limitations
V7.0	Initial bootloader version	For the CAN interface, the Write Unprotect command is not functional. Use Write Memory command and write directly to the option bytes to disable the write protection. For the USB DFU interface, in Dual Bank mode, the Erase operation is not functional for the second bank. Return to Single Bank mode, erase desired sector(s) and then reactivate the Dual Bank mode. After executing Go command (jump to user code) the bootloader resets AHB1ENR value to 0x0000 0000 and thus CCM RAM, when present, is not active (must be re-enabled by user code at startup).

38.2 Bootloader V9.x

38.2.1 Bootloader configuration

The STM32F42xxx/43xxx bootloader is activated by applying Pattern 5 (described in [Table 2](#)). The following table shows the hardware resources used by this bootloader.

Table 82. STM32F42xxx/43xxx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 60 MHz using the PLL. The HSI clock source is used at startup (interface detection phase) and when USART or SPI or I2C interfaces are selected (once CAN or DFU is selected, the clock source is derived from the external crystal).
		HSE enabled	The system clock frequency is 60 MHz. The HSE clock source is used only when the CAN or the DFU (USB FS device) interfaces are selected. The external clock must provide a frequency multiple of 1 MHz and ranging from 4 MHz to 26 MHz.
		-	The CSS interrupt is enabled for the CAN and DFUs. Any failure (or removal) of the external clock generates system reset.
	RAM	-	12 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	29 Kbytes, starting from address 0x1FFF0000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	Power	-	Voltage range is set to [1.62 V, 2.1 V]. In this range internal flash write operations are allowed only in byte format (half-word, word, and double-word operations are not allowed). The voltage range can be configured in run time using bootloader commands.
	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input no pull mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in input no pull mode.

Table 82. STM32F42xxx/43xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
USART3 (on PB10/PB11)	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PB11 pin: USART3 in reception mode. Used in input pull-up mode.
	USART3_TX pin	Output	PB10 pin: USART3 in transmission mode. Used in input pull-up mode.
USART3 (on PC10/PC11)	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit.
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode.
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
CAN2	CAN2	Enabled	Once initialized the CAN2 configuration is: Baudrate 125 kbps, 11-bit identifier. Note: CAN1 is clocked during CAN2 bootloader execution because CAN1 manages the communication between CAN2 and SRAM.
	CAN2_RX pin	Input	PB5 pin: CAN2 in reception mode. Used in alternate push-pull, pull-up mode.
	CAN2_TX pin	Output	PB13 pin: CAN2 in transmission mode. Used in alternate push-pull, pull-up mode.
I2C1	I2C1	Enabled	The I2C1 configuration is: I2C speed: up to 400 kHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b0111000x (x = 0 for write and x = 1 for read).
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain mode.
	I2C1_SDA pin	Input/output	PB9 pin: data line is used in open-drain mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: I2C speed: up to 400 kHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b0111000x (x = 0 for write and x = 1 for read).
	I2C2_SCL pin	Input/output	PF1 pin: clock line is used in open-drain mode.
	I2C2_SDA pin	Input/output	PF0 pin: data line is used in open-drain mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: I2C speed: up to 400 kHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b0111000x (x = 0 for write and x = 1 for read).
	I2C3_SCL pin	Input/output	PA8 pin: clock line is used in open-drain mode.
	I2C3_SDA pin	Input/output	PC9 pin: data line is used in open-drain mode.

Table 82. STM32F42xxx/43xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI1	SPI1	Enabled	The SPI1 configuration is: Slave mode, Full Duplex, -bit MSB, Speed up to 8 MHz, Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode.
SPI2	SPI2	Enabled	The SPI2 configuration is: Slave mode, Full Duplex, 8-bit MSB, Speed up to 8 MHz, Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PI3 pin: slave data input line, used in push-pull, pull-down mode
	SPI2_MISO pin	Output	PI2 pin: slave data output line, used in push-pull, pull-down mode
	SPI2_SCK pin	Input	PI1 pin: slave clock line, used in push-pull, pull-down mode
	SPI2_NSS pin	Input	PI0 pin: slave chip select pin used in push-pull, pull-down mode.
SPI4	SPI4	Enabled	The SPI4 configuration is: Slave mode, Full Duplex, 8-bit MSB, Speed up to 8 MHz, Polarity: CPOL low, CPHA low, NSS hardware.
	SPI4_MOSI pin	Input	PE14 pin: slave data input line, used in push-pull, pull-down mode
	SPI4_MISO pin	Output	PE13 pin: slave data output line, used in push-pull, pull-down mode
	SP4_SCK pin	Input	PE12 pin: slave clock line, used in push-pull, pull-down mode
	SPI4_NSS pin	Input	PE11 pin: slave chip select pin used in push-pull, pull-down mode.
DFU	USB	Enabled	USB OTG FS configured in forced device mode
	USB_DM pin	Input/output	PA11: USB DM line. Used in alternate push-pull no pull mode.
	USB_DP pin		PA12: USB DP line. Used in alternate push-pull no pull mode. No external pull-up resistor is required
CAN2 and DFUs	TIM11	Enabled	This timer is used to determine the value of the HSE. Once the HSE frequency is determined, the system clock is configured to 60 MHz using PLL and HSE.

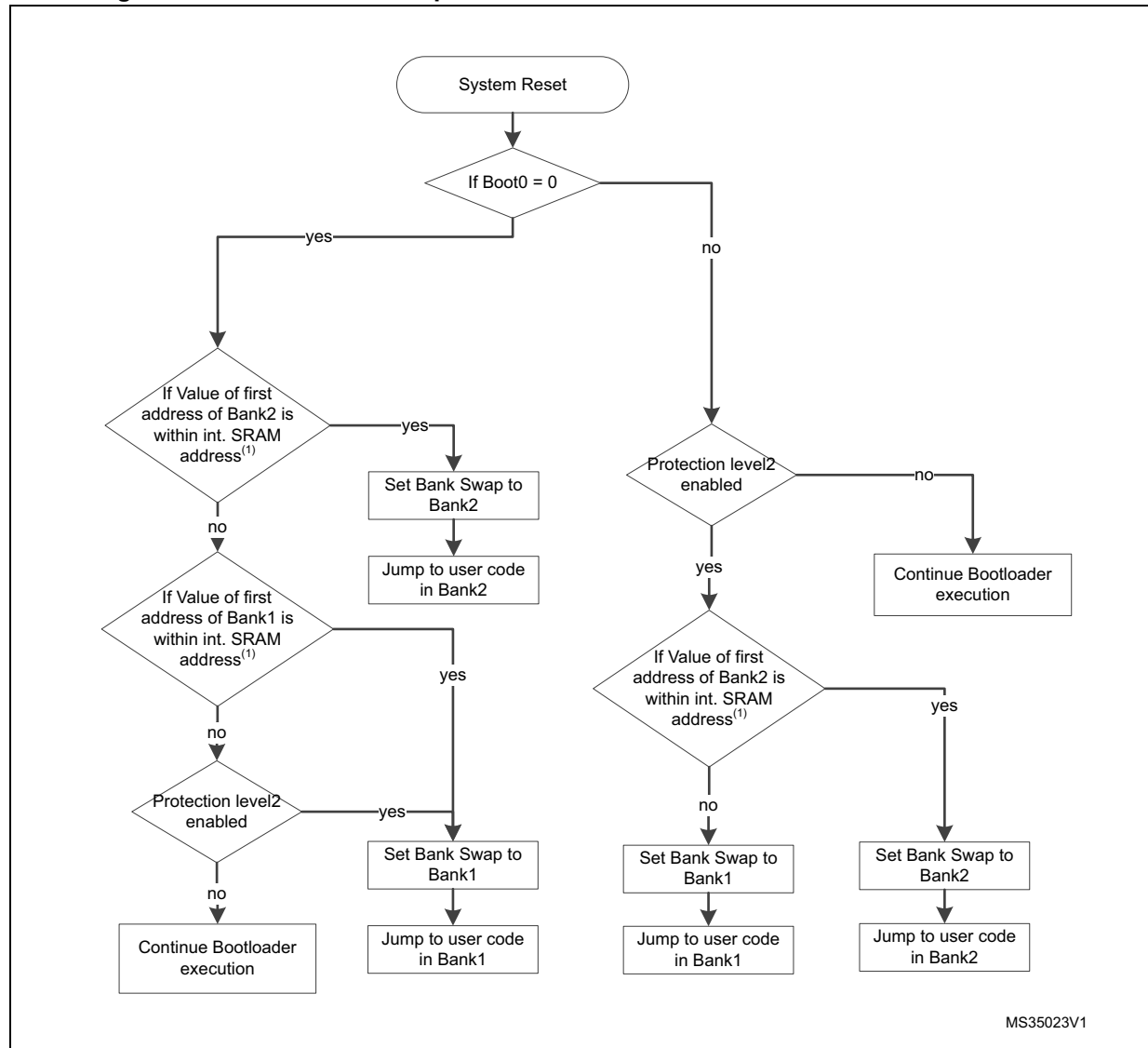
The system clock is derived from the embedded internal high-speed RC for USARTx, I2Cx, I2Cx, and SPIx bootloaders. This internal clock is also used for CAN and DFU (USB FS device), but only for the selection phase. An external clock multiple of 1 MHz (between 4 and 26 MHz) is required for CAN and DFU execution after the selection phase.

Note: Due to HSI deviation and since HSI is used to detect HSE value, use low rather than high frequency HSE crystal values (low frequency values are better detected due to larger error margin). For example, it is better to use 8 MHz instead of 25 MHz.

38.2.2 Bootloader selection

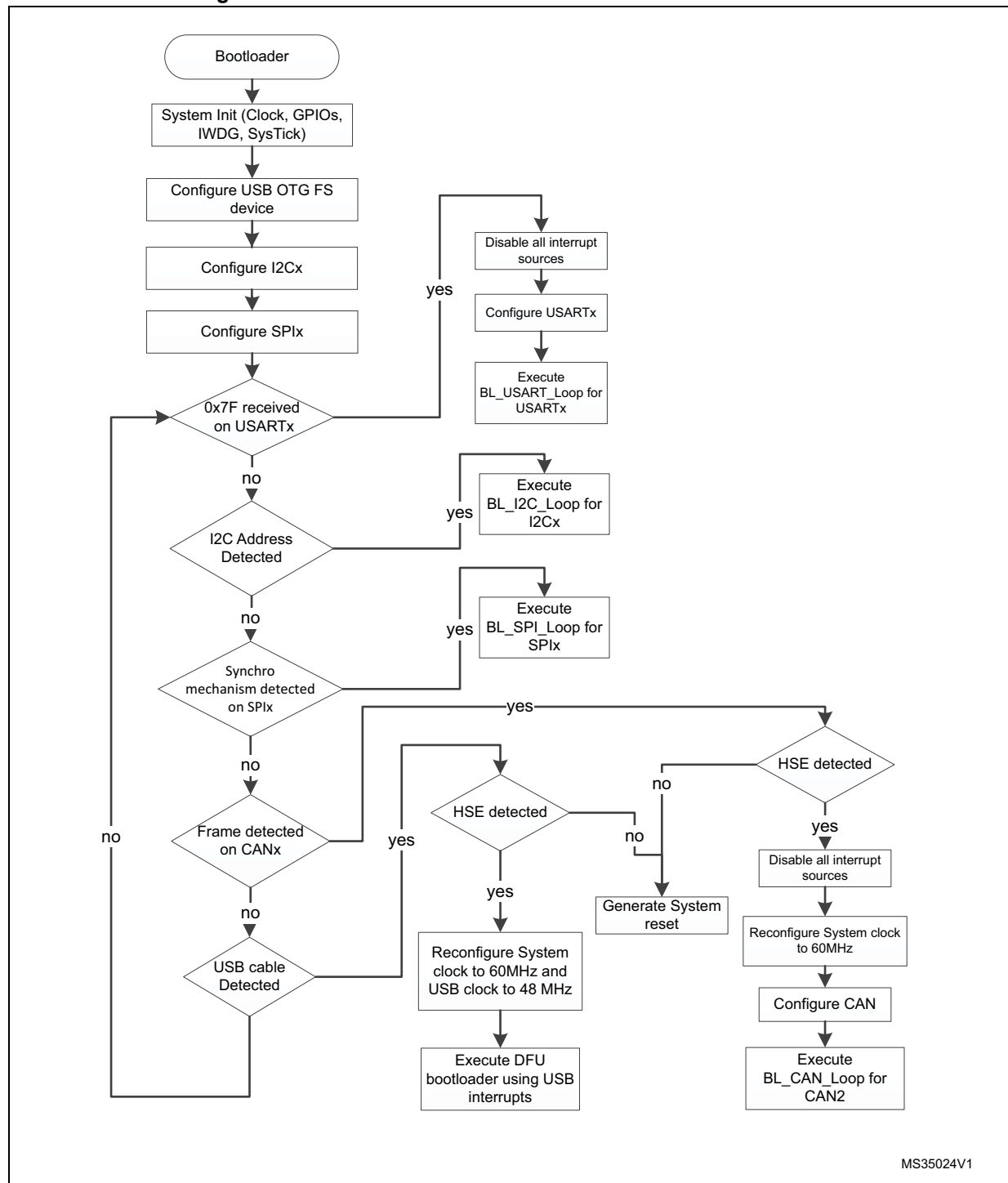
Figure 50 and Figure 51 show the bootloader selection mechanism.

Figure 50. Dual bank boot implementation for STM32F42xxx/43xxx bootloader V9.x



1. CCM RAM is not considered valid as stack pointer address for the dual bank boot mechanism.

Figure 51. Bootloader V9.x selection for STM32F42xxx/43xxx



38.2.3 Bootloader version

[Table 83](#) lists the STM32F42xxx/43xxx devices bootloader V9.x versions.

Table 83. STM32F42xxx/43xxx bootloader V9.x versions

Version number	Description	Known limitations
V9.0	This bootloader is an updated version of bootloader v7.0. This new version of bootloader supports I2C2, I2C3, SPI1, SPI2, and SPI4 interfaces. The RAM used by this bootloader is increased from 8 Kb to 12 Kb. The ID of this bootloader is 0x90. The connection time is increased.	For the USB DFU interface, in Dual Bank mode, the Erase operation is not functional for the second bank. Return to Single Bank mode, erase desired sector(s) and then reactivate the Dual Bank mode. After executing Go command (jump to user code) the bootloader resets AHB1ENR value to 0x0000 0000 and thus CCM RAM, when present, is not active (must be re-enabled by user code at startup)
V9.1	This bootloader is an updated version of bootloader v9.0. This new version implements the new I2C No-stretch commands (I2C protocol v1.1) and the capability of disabling PcROP when RDP1 is enabled with ReadOutUnprotect command for all protocols(USB, USART, CAN, I2C and SPI). The ID of this bootloader is 0x91	For the CAN interface, the Write Unprotect command is not functional. Use Write Memory command and write directly to the option bytes in order to disable the write protection. For the USB DFU interface, in Dual Bank mode, the Erase operation is not functional for the second bank. Return to Single Bank mode, erase desired sector(s) and then reactivate the Dual Bank mode. After executing Go command (jump to user code) the bootloader resets AHB1ENR value to 0x0000 0000 and thus CCM RAM, when present, is not active (must be re-enabled by user code at startup)

39 STM32F446xx devices

39.1 Bootloader configuration

The STM32F446xx bootloader is activated by applying Pattern 1 (described in [Table 2](#)). The following table shows the hardware resources used by this bootloader.

Table 84. STM32F446xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The HSI is used at startup as clock source for system clock configured to 60 MHz and for USART, I2C and SPI bootloader operation.
		HSE enabled	The HSE is used only when the CAN or the DFU (USB FS device) interfaces are selected. In this case the system clock configured to 60 MHz with HSE as clock source. The HSE frequency must be multiple of 1 MHz and ranging from 4 MHz to 26 MHz.
		-	The CSS interrupt is enabled for the CAN and DFUs. Any failure (or removal) of the external clock generates system reset.
	RAM	-	12 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	29 Kbytes, starting from address 0x1FFF0000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
	Power	-	The voltage range is [1.71 V, 3.6 V]. In this range: - Flash wait states: 3. - System Clock 60 MHz. - Prefetch disabled. - Flash write operation by byte (refer to section bootloader memory management for more information).

Table 84. STM32F446xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input no pull mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in input no pull mode.
USART3 (on PB10/PB11)	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PB11 pin: USART3 in reception mode. Used in input pull-up mode.
	USART3_TX pin	Output	PB10 pin: USART3 in transmission mode. Used in input pull-up mode.
USART3 (on PC10/PC11)	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. Used in input pull-up mode
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode. Used in input pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
CAN2	CAN2	Enabled	Once initialized the CAN2 configuration is: Baudrate 125 kbps, 11-bit identifier. Note: CAN1 is clocked during CAN2 bootloader execution because in CAN1 manages the communication between CAN2 and SRAM.
	CAN2_RX pin	Input	PB5 pin: CAN2 in reception mode. Used in alternate push-pull, pull-up mode.
	CAN2_TX pin	Output	PB13 pin: CAN2 in transmission mode. Used in alternate push-pull, pull-up mode.
I2C1	I2C1	Enabled	The I2C1 configuration is: I2C speed: up to 400 kHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b0111100x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin	Input/output	PB9 pin: data line is used in open-drain no pull mode.

Table 84. STM32F446xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I2C2	I2C2	Enabled	The I2C2 configuration is: I2C speed: up to 400 kHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b0111100x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PF1 pin: clock line is used in open-drain no pull mode.
	I2C2_SDA pin	Input/output	PF0 pin: data line is used in open-drain no pull mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: I2C speed: up to 400 kHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b0111100x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PA8 pin: clock line is used in open-drain no pull mode.
	I2C3_SDA pin	Input/output	PC9 pin: data line is used in open-drain no pull mode.
SPI1	SPI1	Enabled	The SPI1 configuration is: Slave mode, Full Duplex, 8-bit MSB, Speed up to 8 MHz, Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode.
SPI2	SPI2	Enabled	The SPI2 configuration is: Slave mode, Full Duplex, 8-bit MSB, Speed up to 8 MHz, Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PB15 pin: slave data input line, used in push-pull, pull-down mode
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, pull-down mode
	SPI2_SCK pin	Input	PC7 pin: slave clock line, used in push-pull, pull-down mode
	SPI2_NSS pin	Input	PB12 pin: slave chip select pin used in push-pull, pull-down mode.

Table 84. STM32F446xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI4	SPI4	Enabled	The SPI4 configuration is: Slave mode, Full Duplex, 8-bit MSB, Speed up to 8 MHz, Polarity: CPOL low, CPHA low, NSS hardware.
	SPI4_MOSI pin	Input	PE14 pin: slave data input line, used in push-pull, pull-down mode
	SPI4_MISO pin	Output	PE13 pin: slave data output line, used in push-pull, pull-down mode
	SPI4_SCK pin	Input	PE12 pin: slave clock line, used in push-pull, pull-down mode
	SPI4_NSS pin	Input	PE11 pin: slave chip select pin used in push-pull, pull-down mode.
DFU	USB	Enabled	USB OTG FS configured in forced device mode
	USB_DM pin	Input/output	PA11: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12: USB DP line. Used in alternate push-pull, no pull mode. No external pull-up resistor is required
CAN2 and DFUs	TIM17	Enabled	This timer is used to determine the value of the HSE. Once the HSE frequency is determined, the system clock is configured to 60 MHz using PLL and HSE.

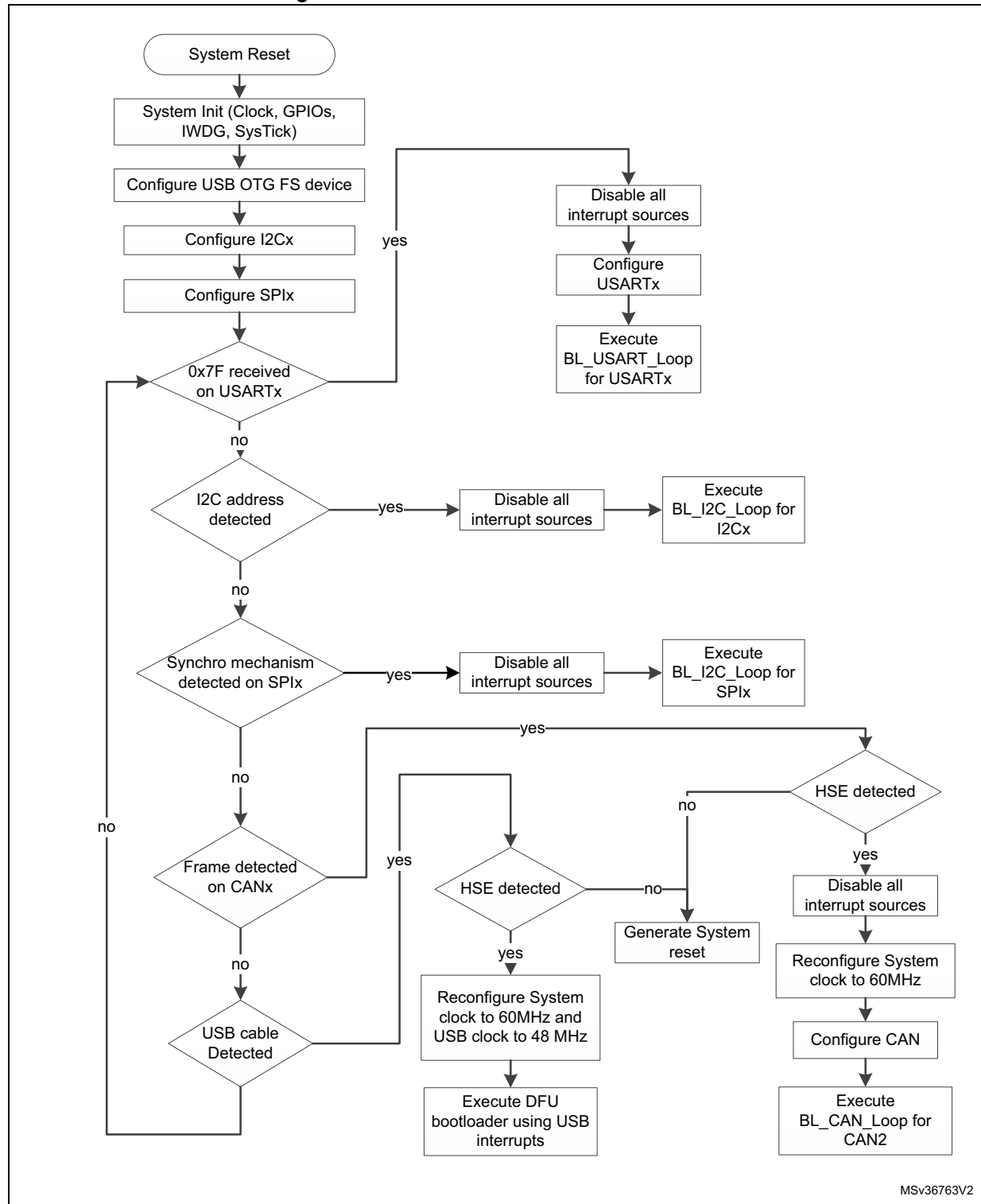
The system clock is derived from the embedded internal high-speed RC for USARTx and I2Cx bootloaders. This internal clock is also used for CAN and DFU (USB FS device), but only for the selection phase. An external clock multiple of 1 MHz (between 4 and 26 MHz) is required for CAN and DFU execution after the selection phase.

Note: *Due to HSI deviation and since HSI is used to detect HSE value, use low rather than high frequency HSE crystal values (low frequency values are better detected due to larger error margin). For example, it is better to use 8 MHz instead of 25 MHz.*

39.2 Bootloader selection

Figure 52 shows the bootloader selection mechanism.

Figure 52. Bootloader V9.x selection for STM32F446xx



MSv36763V2

39.3 Bootloader version

The following table lists the STM32F446xx devices bootloader V9.x versions:

Table 85. STM32F446xx bootloader V9.x versions

Version number	Description	Known limitations
V9.0	Initial bootloader version	After executing Go command (jump to user code) the bootloader resets AHB1ENR value to 0x0000 0000 and thus CCM RAM, when present, is not active (must be re-enabled by user code at startup)

40 STM32F469xx/479xx devices

40.1 Bootloader configuration

The STM32F469xx/479xx bootloader is activated by applying Pattern 5 (described in [Table 2](#)). [Table 86](#) shows the hardware resources used by this bootloader.

Table 86. STM32F469xx/479xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 60 MHz using the PLL. The HSI clock source is used at startup (interface detection phase) and when USART or SPI or I2C interfaces are selected (once CAN or DFU is selected, the clock source is derived from external crystal).
		HSE enabled	The system clock frequency is 60 MHz. The HSE clock source is used only when the CAN or the DFU (USB FS device) interfaces are selected. The external clock must provide a frequency multiple of 1 MHz and ranging from 4 MHz to 26 MHz.
		-	The CSS interrupt is enabled for the CAN and DFUs. Any failure (or removal) of the external clock generates system reset.
	RAM	-	12 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	29 Kbytes, starting from address 0x1FFF0000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
	Power	-	Voltage range is set to [1.62 V, 2.1 V]. In this range internal flash write operations are allowed only in byte format (half-word, word, and double-word operations are not allowed). The voltage range can be configured in run time using bootloader commands.

Table 86. STM32F469xx/479xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input no pull mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in input no pull mode.
USART3 (on PB10/PB11)	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PB11 pin: USART3 in reception mode.
	USART3_TX pin	Output	PB10 pin: USART3 in transmission mode. Used in input pull-up mode.
USART3 (on PC10/PC11)	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. Used in input pull-up mode.
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode. Used in input pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
CAN2	CAN2	Enabled	Once initialized the CAN2 configuration is: Baudrate 125 kbps, 11-bit identifier. Note: CAN1 is clocked during CAN2 bootloader execution because CAN1 manages the communication between CAN2 and SRAM.
	CAN2_RX pin	Input	PB05 pin: CAN2 in reception mode. Used in alternate push-pull, pull-up mode.
	CAN2_TX pin	Output	PB13 pin: CAN2 in transmission mode. Used in alternate push-pull, pull-up mode.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1000100x (x = 0 for write and x = 1 for read).
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin	Input/output	PB9 pin: data line is used in open-drain no pull mode.

Table 86. STM32F469xx/479xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I2C2	I2C2	Enabled	The I2C2 configuration is: I2C speed: up to 400 kHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b1000100x (x = 0 for write and x = 1 for read).
	I2C2_SCL pin	Input/output	PF0 pin: clock line is used in open-drain no pull mode.
	I2C2_SDA pin	Input/output	PF1 pin: data line is used in open-drain no pull mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: I2C speed: up to 400 kHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b1000100x (x = 0 for write and x = 1 for read).
	I2C3_SCL pin	Input/output	PA8 pin: clock line is used in open-drain no pull mode.
	I2C3_SDA pin	Input/output	PC9 pin: data line is used in open-drain no pull mode.
SPI1	SPI1	Enabled	The SPI1 configuration is: Slave mode, Full Duplex, 8-bit MSB, Speed up to 8 MHz, Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode.
SPI2	SPI2	Enabled	The SPI2 configuration is: Slave mode, Full Duplex, 8-bit MSB, Speed up to 8 MHz, Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PI3 pin: slave data input line, used in push-pull, pull-down mode
	SPI2_MISO pin	Output	PI2 pin: slave data output line, used in push-pull, pull-down mode
	SPI2_SCK pin	Input	PI1pin: slave clock line, used in push-pull, pull-down mode
	SPI2_NSS pin	Input	PI0 pin: slave chip select pin used in push-pull, pull-down mode.

Table 86. STM32F469xx/479xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI4	SPI4	Enabled	The SPI4 configuration is: Slave mode, Full Duplex, 8-bit MSB, Speed up to 8 MHz, Polarity: CPOL low, CPHA low, NSS hardware.
	SPI4_MOSI pin	Input	PE14 pin: slave data input line, used in push-pull, pull-down mode
	SPI4_MISO pin	Output	PE13 pin: slave data output line, used in push-pull, pull-down mode
	SPI4_SCK pin	Input	PE12 pin: slave clock line, used in push- pull, pull-down mode
	SPI4_NSS pin	Input	PE11 pin: slave chip select pin used in push-pull, pull-down mode.
DFU	USB	Enabled	USB OTG FS configured in forced device mode. USB_OTG_FS interrupt vector is enabled and used for USB DFU communications.
	USB_DM pin	Input/output	PA11 pin: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12 pin: USB DP line. Used in alternate push-pull, no pull mode. No external pull-up resistor is required.

The system clock is derived from the embedded internal high-speed RC for USARTx and I2Cx bootloaders. This internal clock is also used for CAN and DFU (USB FS device), but only for the selection phase. An external clock multiple of 1 MHz (between 4 and 48 MHz) is required for CAN and DFUs execution after the selection phase.

Note: *Due to HSI deviation and since HSI is used to detect HSE value, use low rather than high frequency HSE crystal values (low frequency values are better detected due to larger error margin). For example, it is better to use 8 MHz instead of 25 MHz.*

40.2 Bootloader selection

Figure 53 and Figure 54 show the bootloader selection mechanism.

Figure 53. Dual bank boot implementation for STM32F469xx/479xx Bootloader V9.x

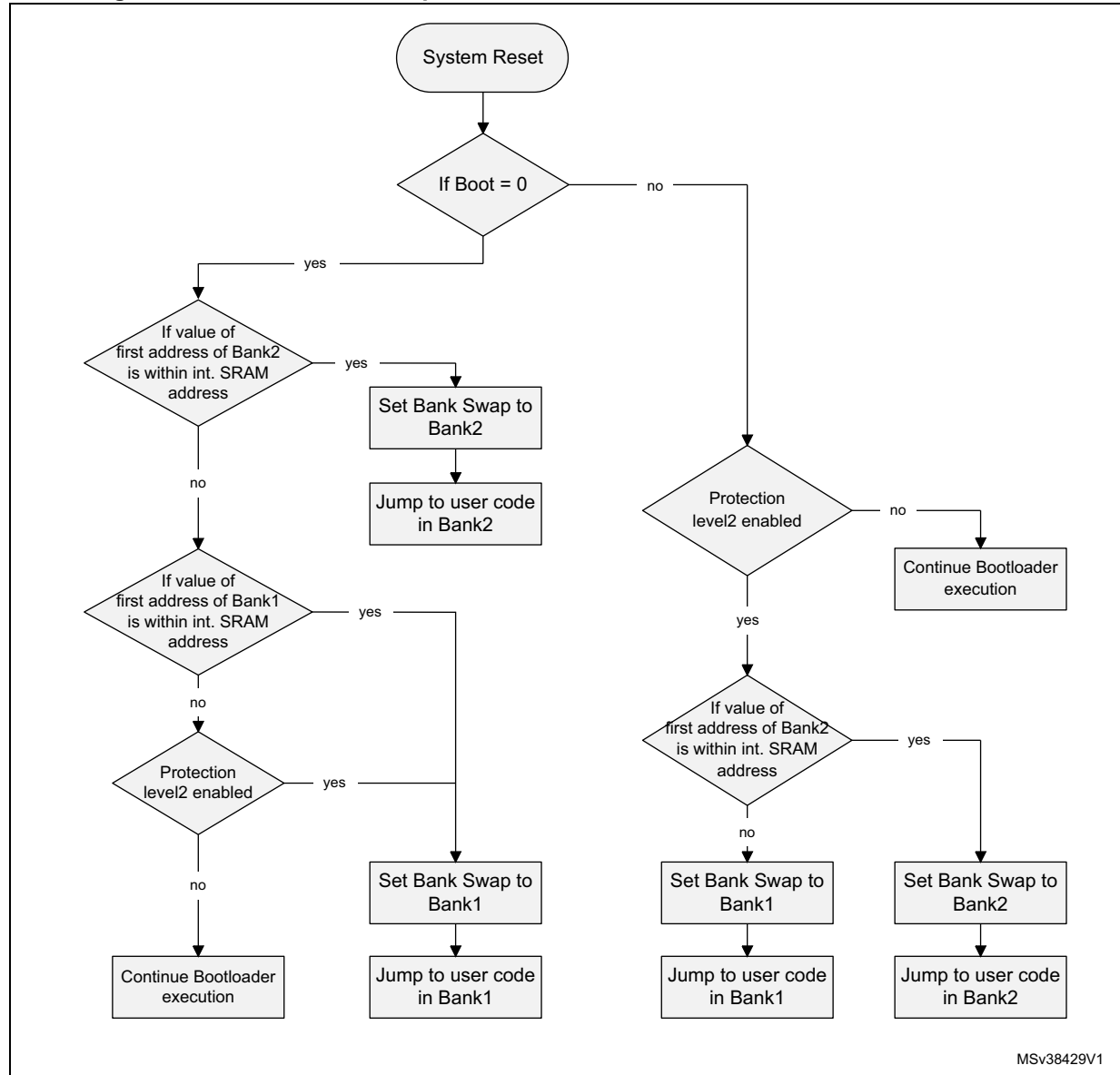
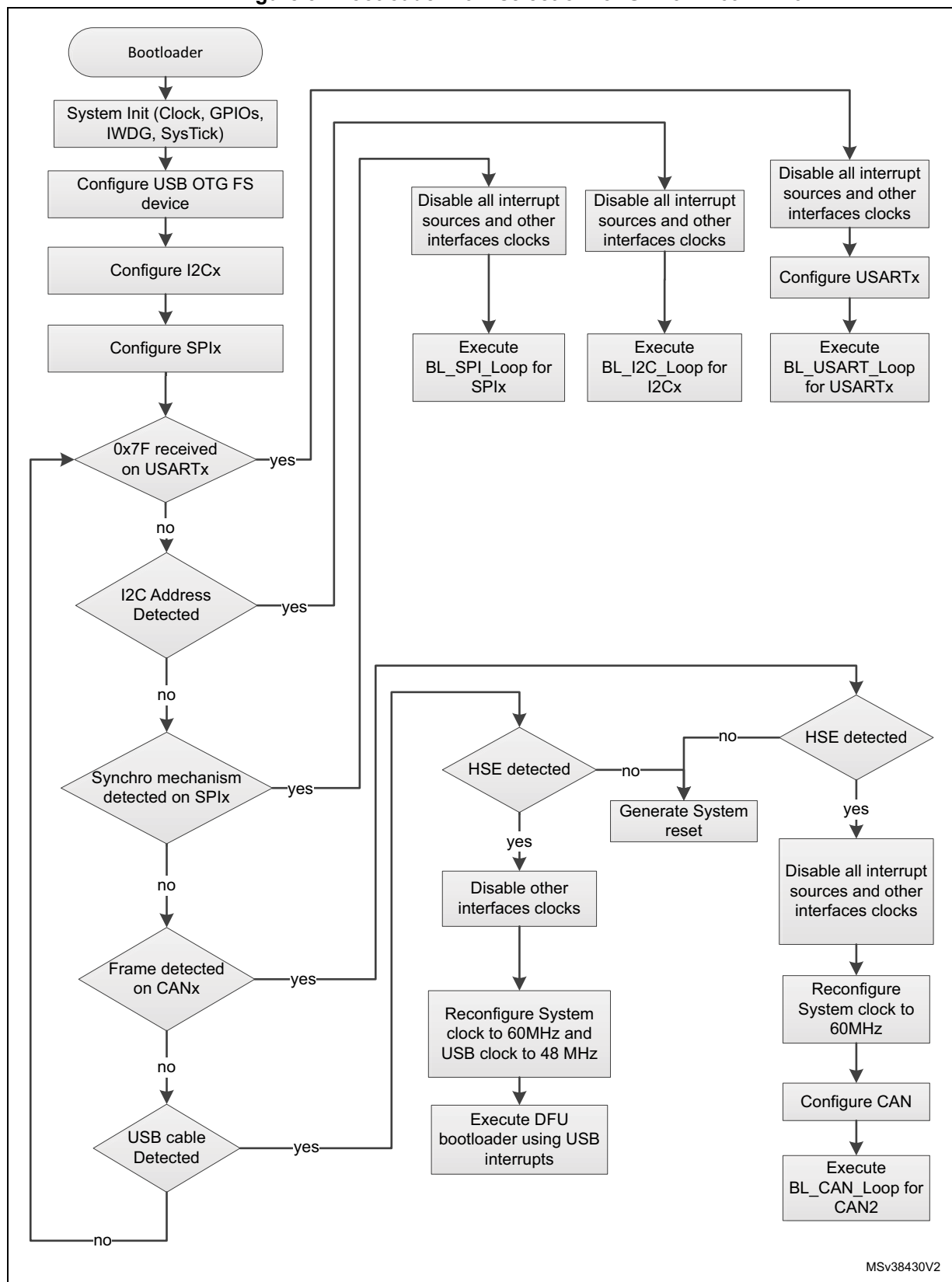


Figure 54. Bootloader V9.x selection for STM32F469xx/479xx



40.3 Bootloader version

[Table 87](#) lists the STM32F469xx/479xx devices V9.x bootloader versions:

Table 87. STM32F469xx/479xx bootloader V9.x versions

Version number	Description	Known limitations
V9.0	Initial bootloader version	After executing Go command (jump to user code) the bootloader resets AHB1ENR value to 0x0000 0000 and thus CCM RAM, when present, is not active (must be re-enabled by user code at startup).

41 STM32F72xxx/73xxx devices

41.1 Bootloader configuration

The STM32F72xxx/73xxx bootloader is activated by applying Pattern 8 (described in [Table 2](#)). [Table 88](#) shows the hardware resources used by this bootloader.

Table 88. STM32F72xxx/73xxx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The HSI is used at startup as clock source for system clock configured to 60 MHz and for USART and I2C bootloader operation.
		HSE enabled	The HSE is used only when the CAN or the DFU (USB FS device) interfaces are selected. In this case the system clock configured to 60 MHz with HSE as clock source. The HSE frequency must be a multiple of 1 MHz, ranging from 4 to 26 MHz.
		-	The CSS interrupt is enabled for the CAN and DFUs. Any failure (or removal) of the external clock generates system reset.
	RAM	-	16 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	59 Kbytes, starting from address 0x1FF00000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
	Power	-	The voltage range is [1.8V, 3.6V] In this range: - Flash wait states: 3. - System clock frequency 60 MHz. - ART Accelerator enabled. - Flash write operation by byte (refer to bootloader memory management section for more information).

Table 88. STM32F72xxx/73xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input no pull mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in input no pull mode.
USART3 (on PB11/PB10)	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PB11 pin: USART3 in reception mode. Used in input pull-up mode.
	USART3_TX pin	Output	PB10 pin: USART3 in transmission mode. Used in input pull-up mode.
USART3 (on PC11/PC10)	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. Used in input pull-up mode.
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode. Used in input pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
CAN1	CAN1	Enabled	Once initialized the CAN1 configuration is: Baudrate 125 kbps, 11-bit identifier.
	CAN1_RX pin	Input	PD0 pin: CAN1 in reception mode. Used in alternate push-pull, pull-up mode.
	CAN1_TX pin	Output	PD1 pin: CAN1 in transmission mode. Used in alternate push-pull, pull-up mode.
I2C1	I2C1	Enabled	The I2C1 configuration is: I2C speed: up to 400 kHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b1001001x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin	Input/output	PB9 pin: data line is used in open-drain no pull mode.

Table 88. STM32F72xxx/73xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I2C2	I2C2	Enabled	The I2C2 configuration is: I2C speed: up to 400 kHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b1001101x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PF1 pin: clock line is used in open-drain no pull mode.
	I2C2_SDA pin	Input/output	PF0 pin: data line is used in open-drain no pull mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: I2C speed: up to 400 kHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b1001001x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PA8 pin: clock line is used in open-drain no pull mode.
	I2C3_SDA pin	Input/output	PC9 pin: data line is used in open-drain no pull mode.
SPI1	SPI1	Enabled	The SPI1 configuration is: Slave mode, Full Duplex, 8-bit MSB, Speed up to 8 MHz, Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode.
SPI2	SPI2	Enabled	The SPI2 configuration is: Slave mode, Full Duplex, 8-bit MSB, Speed up to 8 MHz, Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PI3 pin: slave data input line, used in push-pull, pull-down mode
	SPI2_MISO pin	Output	PI2 pin: slave data output line, used in push-pull, pull-down mode
	SPI2_SCK pin	Input	PI1 pin: slave clock line, used in push-pull, pull-down mode
	SPI2_NSS pin	Input	PI0 pin: slave chip select pin used in push-pull, pull-down mode.

Table 88. STM32F72xxx/73xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI4	SPI4	Enabled	The SPI4 configuration is: Slave mode, Full Duplex, 8-bit MSB, Speed up to 8 MHz, Polarity: CPOL low, CPHA low, NSS hardware.
	SPI4_MOSI pin	Input	PE14 pin: slave data input line, used in push-pull, pull-down mode
	SPI4_MISO pin	Output	PE13 pin: slave data output line, used in push-pull, pull-down mode
	SP4_SCK pin	Input	PE12 pin: slave clock line, used in push-pull, pull-down mode
	SPI4_NSS pin	Input	PE11 pin: slave chip select pin used in push-pull, pull-down mode.
DFU	USB	Enabled	USB OTG FS configured in forced device mode
	USB_DM pin	Input/output	PA11 pin: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12 pin: USB DP line. Used in alternate push-pull, no pull mode. No external pull-up resistor is required.
CAN1 and DFUs	TIM11	Enabled	This timer is used to determine the value of the HSE. Once HSE frequency is determined, the system clock is configured to 60 MHz using PLL and HSE.

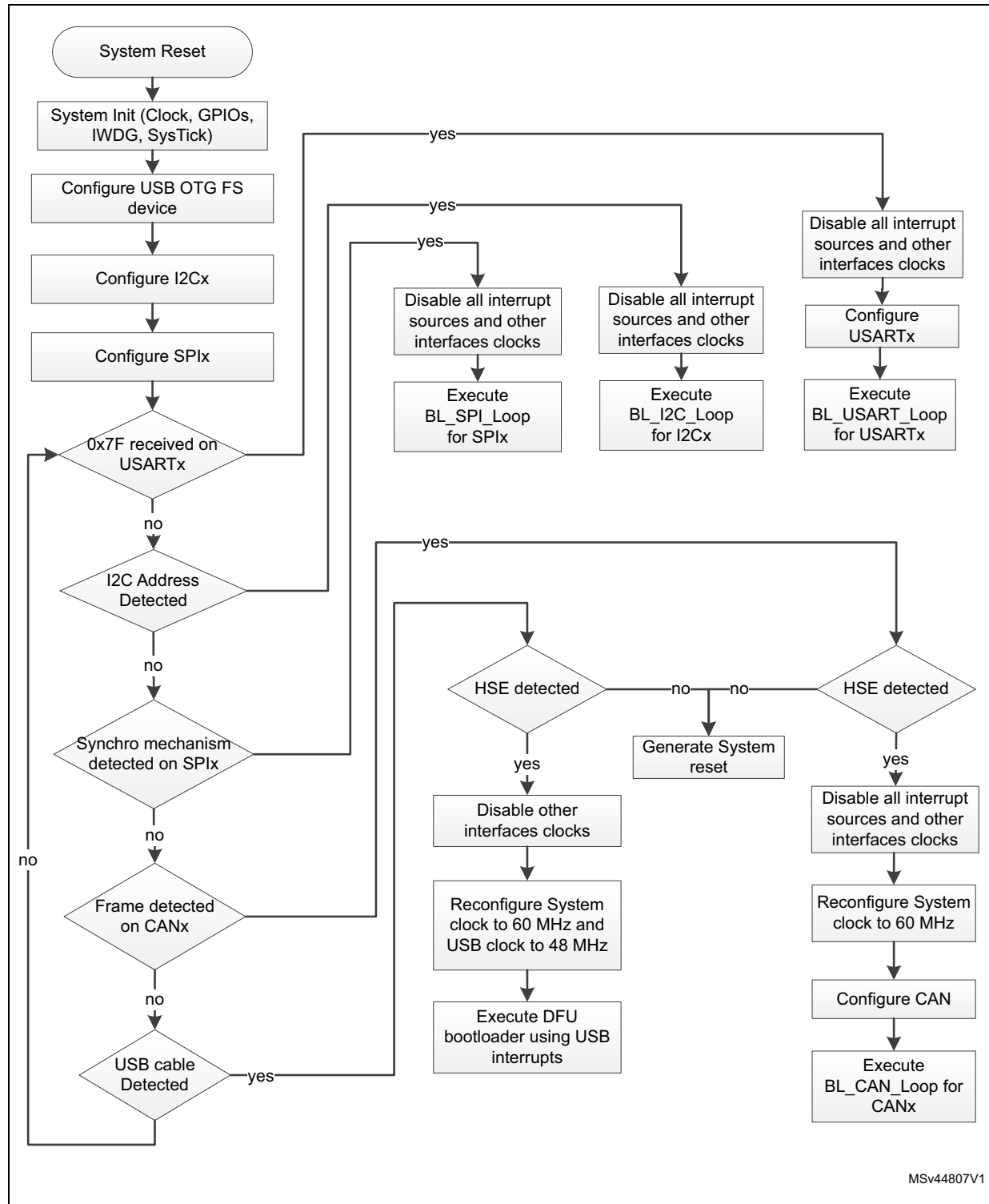
The system clock is derived from the embedded internal high-speed RC for USARTx and I2Cx bootloaders. This internal clock is also used for CAN and DFU (USB FS device), but only for the selection phase. An external clock multiple of 1 MHz (between 4 and 26 MHz) is required for CAN and DFU execution after the selection phase.

Note: *Due to HSI deviation and since HSI is used to detect HSE value, use low rather than high frequency HSE crystal values (low frequency values are better detected due to larger error margin). For example, it is better to use 8 MHz instead of 25 MHz.*

41.2 Bootloader selection

Figure 55 shows the bootloader selection mechanism.

Figure 55. Bootloader V9.x selection for STM32F72xxx/73xxx



MSv44807V1

41.3 Bootloader version

[Table 89](#) lists the STM32F72xxx/73xxx devices bootloader V9.x versions.

Table 89. STM32F72xxx/73xxx bootloader V9.x versions

Version number	Description	Known limitations
V9.0	Initial bootloader version	At high UART baudrates (115200 bps) connection may fail due to software jitter leading to wrong baudrate calculation. In that case bootloader may respond with a baudrate up to $\pm 5\%$ different from host baudrate. Workaround: use baudrates lower than 57600 bps if host tolerance to baudrate error is lower than $\pm 5\%$

42 STM32F74xxx/75xxx devices

Two bootloader versions are available:

- V7.x supporting USART1, USART3, CAN2, I2C1, I2C2, I2C3, and DFU (USB FS device). This version is embedded in STM32F74xxx/75xxx rev. A devices.
- V9.x supporting USART1, USART3, CAN2, I2C1, I2C2, I2C3, SPI1, SPI2, SPI4, and DFU (USB FS device). This version is embedded in STM32F74xxx/75xxx rev. Z and rev. 1 devices.

Note: When readout protection Level2 is activated, STM32F74xxx/75xxx devices can boot also on system memory and all commands are not accessible except Get, GetID, and GetVersion.

42.1 Bootloader V7.x

42.1.1 Bootloader configuration

The STM32F74xxx/75xxx bootloader is activated by applying Pattern 8 (described in [Table 2](#)). [Table 90](#) shows the hardware resources used by this bootloader.

Table 90. STM32F74xxx/75xxx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The HSI is used at startup as clock source for system clock configured to 60 MHz and for USART and I2C bootloader operation.
		HSE enabled	The HSE is used only when the CAN or the DFU (USB FS device) interfaces are selected. In this case the system clock configured to 60 MHz with HSE as clock source. The HSE frequency must be multiple of 1 MHz and ranging from 4 to 26 MHz.
		-	The CSS interrupt is enabled for the CAN and DFUs. Any failure (or removal) of the external clock generates system reset.
	RAM	-	16 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	60 Kbytes, starting from address 0x1FF00000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
	Power	-	The voltage range is [1.8V, 3.6V]. In this range: - Flash wait states: 3. - System clock frequency 60 MHz. - ART Accelerator enabled. - Flash write operation by byte (refer to bootloader memory management section for more information).

Table 90. STM32F74xxx/75xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input no pull mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in input no pull mode.
USART3 (on PB10/PB11)	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PB11 pin: USART3 in reception mode. Used in input pull-up mode.
	USART3_TX pin	Output	PB10 pin: USART3 in transmission mode. Used in input pull-up mode.
USART3 (on PC10/PC11)	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. Used in input pull-up mode.
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode. Used in input pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
CAN2	CAN2	Enabled	Once initialized the CAN2 configuration is: Baudrate 125 kbps, 11-bit identifier. Note: CAN1 is clocked during CAN2 bootloader execution because CAN1 manages the communication between CAN2 and SRAM.
	CAN2_RX pin	Input	PB5 pin: CAN2 in reception mode. Used in alternate push-pull, pull-up mode.
	CAN2_TX pin	Output	PB13 pin: CAN2 in transmission mode. Used in alternate push-pull, pull-up mode.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz – 7-bit address, Target mode – Analog filter ON – Target 7-bit address: 0b1000101x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin	Input/output	PB9 pin: data line is used in open-drain no pull mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 400 kHz – 7-bit address, Target mode – Analog filter ON – Target 7-bit address: 0b1000101x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PF1 pin: clock line is used in open-drain no pull mode.
	I2C2_SDA pin	Input/output	PF0 pin: data line is used in open-drain no pull mode.

Table 90. STM32F74xxx/75xxx configuration in system memory boot mode (continued)

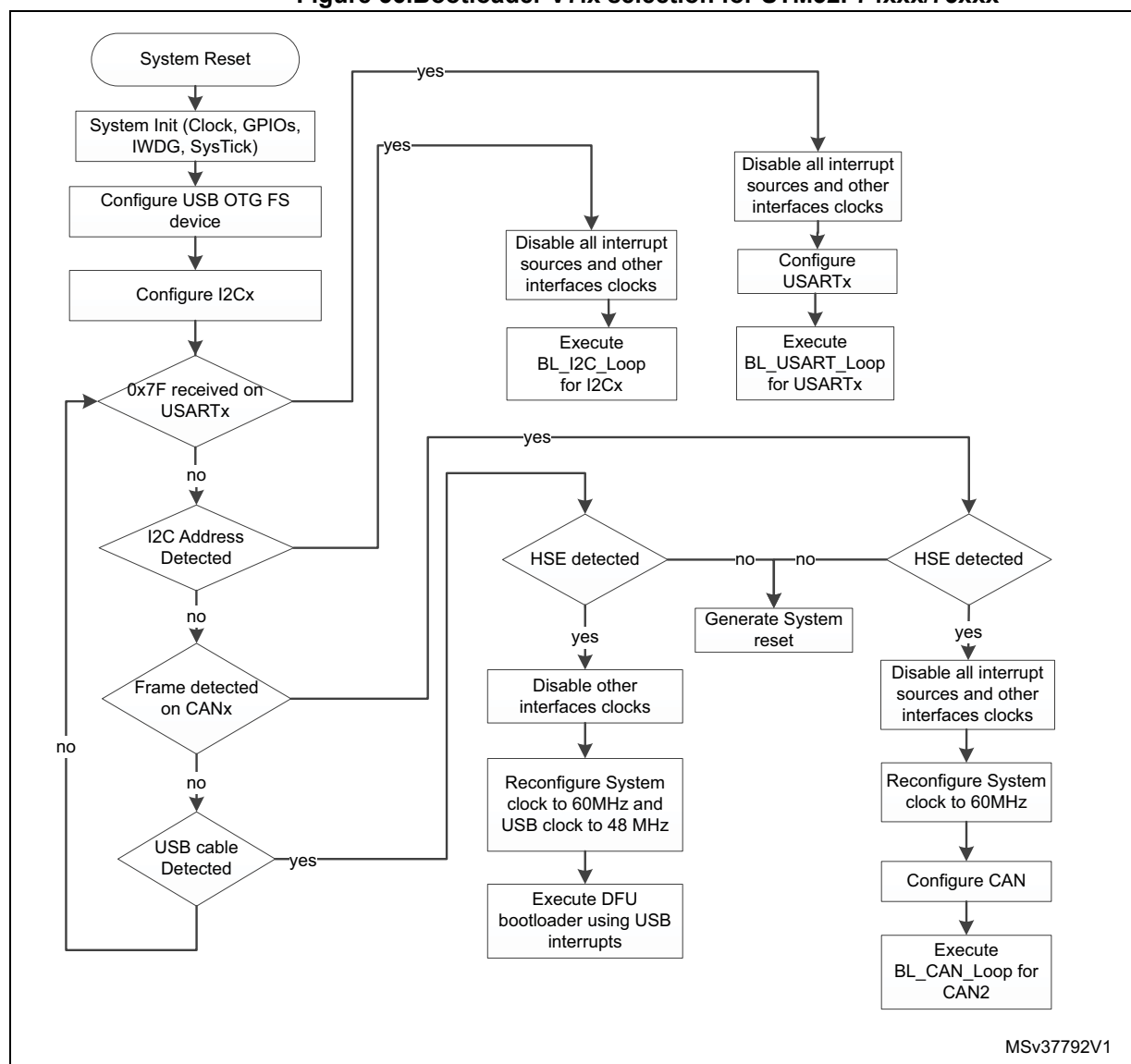
Bootloader	Feature/Peripheral	State	Comment
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode, Analog filter ON – Target 7-bit address: 0b1000101x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PA8 pin: clock line is used in open-drain no pull mode.
	I2C3_SDA pin	Input/output	PC9 pin: data line is used in open-drain no pull mode.
DFU	USB	Enabled	USB OTG FS configured in forced device mode.
	USB_DM pin	Input/output	PA11 pin: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12 pin: USB DP line. Used in alternate push-pull, no pull mode. No external pull-up resistor is required.
CAN2 and DFUs	TIM11	Enabled	This timer is used to determine the value of the HSE. Once HSE frequency is determined, the system clock is configured to 60 MHz using PLL and HSE.

The system clock is derived from the embedded internal high-speed RC for USARTx and I2Cx bootloaders. This internal clock is also used for CAN and DFU (USB FS device), but only for the selection phase. An external clock multiple of 1 MHz (between 4 and 26 MHz) is required for CAN and DFU execution after the selection phase.

Note: *Due to HSI deviation and since HSI is used to detect HSE value, use low rather than high frequency HSE crystal values (low frequency values are better detected due to larger error margin). For example, it is better to use 8 MHz instead of 25 MHz.*

42.1.2 Bootloader selection

Figure 56. Bootloader V7.x selection for STM32F74xxx/75xxx



42.1.3 Bootloader version

Table 91. STM32F74xxx/75xxx bootloader V7.x versions

Version number	Description	Known limitations
V7.0	Initial bootloader version	At high UART baudrates (115200 bps) connection may fail due to software jitter leading to wrong baudrate calculation. In that case bootloader may respond with a baudrate up to $\pm 5\%$ different from host baudrate. Workaround: use baudrates lower than 57600 bps if host tolerance to baudrate error is lower than $\pm 5\%$

42.2 Bootloader V9.x

42.2.1 Bootloader configuration

The STM32F74xxx/75xxx bootloader is activated by applying Pattern 8 (described in [Table 2](#)). The following table shows the hardware resources used by this bootloader.

Table 92. STM32F74xxx/75xxx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The HSI is used at startup as clock source for system clock configured to 60 MHz and for USART, I2C and SPI bootloader operation.
		HSE enabled	The HSE is used only when the CAN or the DFU (USB FS device) interfaces are selected. In this case the system clock configured to 60 MHz with HSE as clock source. The HSE frequency must be multiple of 1 MHz and ranging from 4 to 26 MHz.
		-	The CSS interrupt is enabled for the CAN and DFU bootloaders. Any failure (or removal) of the external clock generates system reset.
	RAM	-	16 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	60 Kbytes, starting from address 0x1FF00000, contain the bootloader firmware
USART1	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
	Power	-	The voltage range is 1.8 V, 3.6V. In this range: - Flash wait states: 3. - System clock frequency 60 MHz. - ART Accelerator enabled. - Flash write operation by byte (refer to bootloader memory management section for more information).
	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input no pull mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in input no pull mode.

Table 92. STM32F74xxx/75xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
USART3 (on PB10/PB11)	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PB11 pin: USART3 in reception mode. Used in input pull-up mode.
	USART3_TX pin	Output	PB10 pin: USART3 in transmission mode. Used in input pull-up mode.
USART3 (on PC10/PC11)	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. Used in input pull-up mode.
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode. Used in input pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
CAN2	CAN2	Enabled	Once initialized the CAN2 configuration is: Baudrate 125 kbps, 11-bit identifier. Note: CAN1 is clocked during CAN2 bootloader execution because CAN1 manages the communication between CAN2 and SRAM.
	CAN2_RX pin	Input	PB5 pin: CAN2 in reception mode. Used in alternate push-pull, pull-up mode.
	CAN2_TX pin	Output	PB13 pin: CAN2 in transmission mode. Used in alternate push-pull, pull-up mode.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1000101x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin	Input/output	PB9 pin: data line is used in open-drain no pull mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1000101x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PF1 pin: clock line is used in open-drain no pull mode.
	I2C2_SDA pin	Input/output	PF0 pin: data line is used in open-drain no pull mode.

Table 92. STM32F74xxx/75xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1000101x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PA8 pin: clock line is used in open-drain no pull mode.
	I2C3_SDA pin	Input/output	PC9 pin: data line is used in open-drain no pull mode.
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode.
SPI2	SPI2	Enabled	The SPI2 configuration is: Slave mode, Full Duplex, 8-bit MSB, Speed up to 8 MHz, Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PI3 pin: slave data input line, used in push-pull, pull-down mode
	SPI2_MISO pin	Output	PI2 pin: slave data output line, used in push-pull, pull-down mode
	SPI2_SCK pin	Input	PI1 pin: slave clock line, used in push-pull, pull-down mode
	SPI2_NSS pin	Input	PI0 pin: slave chip select pin used in push-pull, pull-down mode.

Table 92. STM32F74xxx/75xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI4	SPI4	Enabled	The SPI4 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI4_MOSI pin	Input	PE14 pin: slave data input line, used in push-pull, pull-down mode
	SPI4_MISO pin	Output	PE13 pin: slave data output line, used in push-pull, pull-down mode
	SP4_SCK pin	Input	PE12 pin: slave clock line, used in push-pull, pull-down mode
	SPI4_NSS pin	Input	PE11 pin: slave chip select pin used in push-pull, pull-down mode.
DFU	USB	Enabled	USB OTG FS configured in forced device mode.
	USB_DM pin	Input/output	PA11 pin: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12 pin: USB DP line. Used in alternate push-pull, no pull mode. No external pull-up resistor is required.
CAN2 and DFUs	TIM11	Enabled	This timer is used to determine the value of the HSE. Once HSE frequency is determined, the system clock is configured to 60 MHz using PLL and HSE.

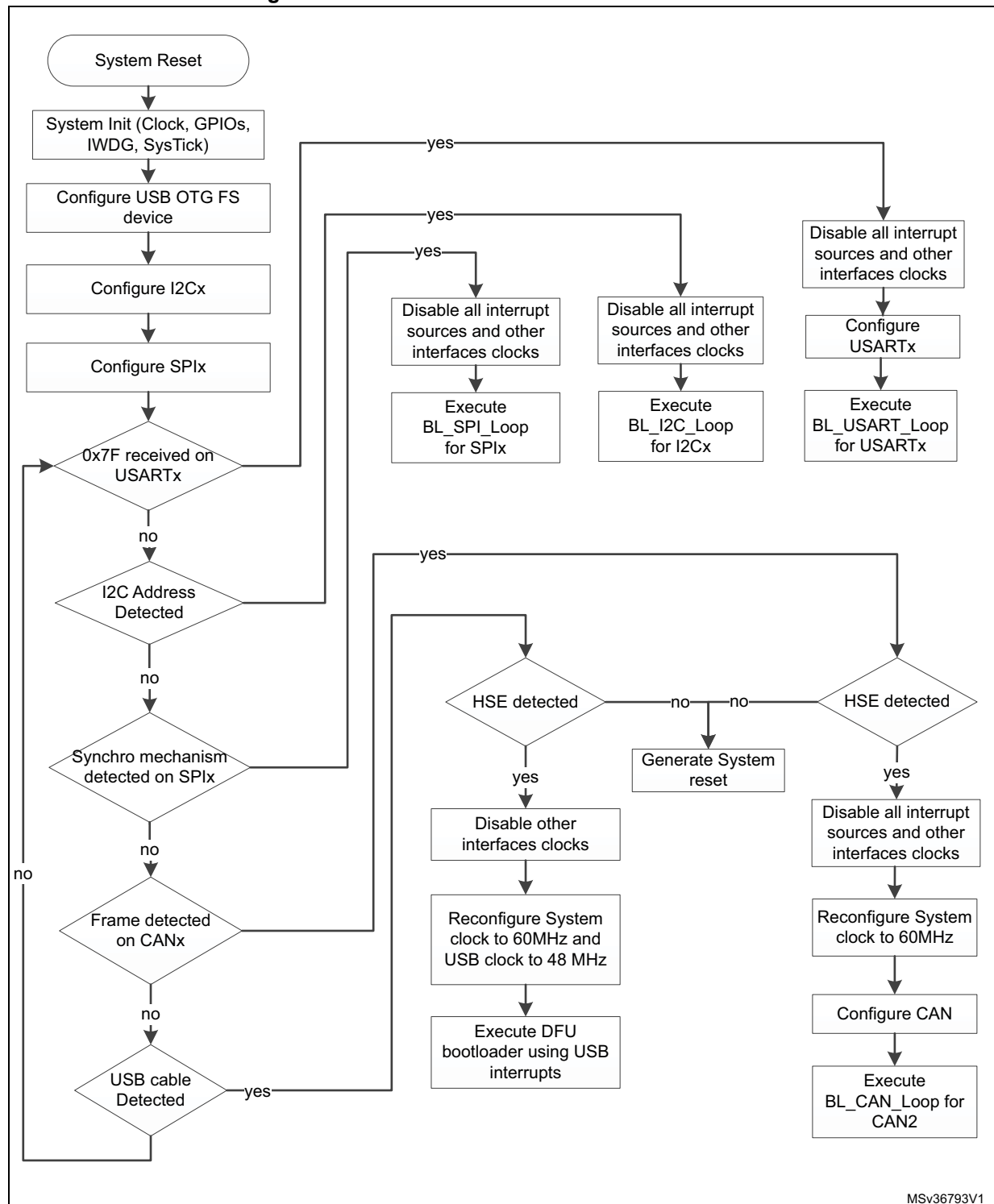
The system clock is derived from the embedded internal high-speed RC for USARTx, I2Cx, I2Cx, and SPIx bootloaders. This internal clock is also used for CAN and DFU (USB FS device), but only for the selection phase. An external clock multiple of 1 MHz (between 4 and 26 MHz) is required for CAN and DFU execution after the selection phase.

Note: *Due to HSI deviation and since HSI is used to detect HSE value, use low rather than high frequency HSE crystal values (low frequency values are better detected due to larger error margin). For example, it is better to use 8 MHz instead of 25 MHz.*

42.2.2 Bootloader selection

Figure 57 shows the bootloader selection mechanism.

Figure 57.Bootloader V9.x selection for STM32F74xxx/75xxx



MSv36793V1

42.2.3 Bootloader version

The following table lists the STM32F74xxx/75xxx bootloader V9.x versions:

Table 93. STM32F74xxx/75xxx bootloader V9.x versions

Version number	Description	Known limitations
V9.0	Initial bootloader version	At high UART baudrates (115200 bps) connection may fail due to software jitter leading to wrong baudrate calculation. In that case bootloader may respond with a baudrate up to $\pm 5\%$ different from host baudrate. Workaround: use baudrates lower than 57600 bps if host tolerance to baudrate error is lower than $\pm 5\%$

43 STM32F76xxx/77xxx devices

43.1 Bootloader configuration

The STM32F76xxx/77xxx bootloader is activated by applying Pattern 9 (described in [Table 2](#)). The following table shows the hardware resources used by this bootloader.

Table 94. STM32F76xxx/77xxx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The HSI is used at startup as clock source for system clock configured to 60 MHz and for USART and I2C bootloader operation.
		HSE enabled	The HSE is used only when the CAN or the DFU (USB FS device) interfaces are selected. In this case the system clock configured to 60 MHz with HSE as clock source. The HSE frequency must be a multiple of 1 MHz, ranging from 4 to 26 MHz.
		-	The CSS interrupt is enabled for the CAN and DFUs. Any failure (or removal) of the external clock generates system reset.
	RAM	-	16 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	59 Kbytes, starting from address 0x1FF00000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
	Power	-	The voltage range is [1.8 V, 3.6 V] In this range: - Flash wait states: 3. - System clock frequency 60 MHz. - ART Accelerator enabled. - Flash write operation by byte (refer to bootloader memory management section for more information).
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input no pull mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in input no pull mode.
USART3 (on PB11/PB10)	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PB11 pin: USART3 in reception mode. Used in input pull-up mode.
	USART3_TX pin	Output	PB10 pin: USART3 in transmission mode. Used in input pull-up mode.

Table 94. STM32F76xxx/77xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
USART3 (on PC11/PC10)	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. Used in input pull-up mode.
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode. Used in input pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
CAN2	CAN2	Enabled	Once initialized the CAN2 configuration is: Baudrate 125 kbps, 11-bit identifier. Note: CAN1 is clocked during CAN2 bootloader execution because CAN1 manages the communication between CAN2 and SRAM.
	CAN2_RX pin	Input	PB5 pin: CAN2 in reception mode. Used in alternate push-pull, pull-up mode.
	CAN2_TX pin	Output	PB13 pin: CAN2 in transmission mode. Used in alternate push-pull, pull-up mode.
I2C1	I2C1	Enabled	The I2C1 configuration is: I2C speed: up to 400 kHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b1001001x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin	Input/output	PB9 pin: data line is used in open-drain no pull mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: I2C speed: up to 400 kHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b1001001x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PF1 pin: clock line is used in open-drain no pull mode.
	I2C2_SDA pin	Input/output	PF0 pin: data line is used in open-drain no pull mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: I2C speed: up to 400 kHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b1001001x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PA8 pin: clock line is used in open-drain no pull mode.
	I2C3_SDA pin	Input/output	PC9 pin: data line is used in open-drain no pull mode.

Table 94. STM32F76xxx/77xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI1	SPI1	Enabled	The SPI1 configuration is: Slave mode, Full Duplex, 8-bit MSB, Speed up to 8 MHz, Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode.
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PI3 pin: slave data input line, used in push-pull, pull-down mode
	SPI2_MISO pin	Output	PI2 pin: slave data output line, used in push-pull, pull-down mode
	SPI2_SCK pin	Input	PI1 pin: slave clock line, used in push-pull, pull-down mode
	SPI2_NSS pin	Input	PI0 pin: slave chip select pin used in push-pull, pull-down mode.
SPI4	SPI4	Enabled	The SPI4 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI4_MOSI pin	Input	PE14 pin: slave data input line, used in push-pull, pull-down mode
	SPI4_MISO pin	Output	PE13 pin: slave data output line, used in push-pull, pull-down mode
	SPI4_SCK pin	Input	PE12 pin: slave clock line, used in push-pull, pull-down mode
	SPI4_NSS pin	Input	PE11 pin: slave chip select pin used in push-pull, pull-down mode.

Table 94. STM32F76xxx/77xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
DFU	USB	Enabled	USB OTG FS configured in forced device mode
	USB_DM pin	Input/output	PA11 pin: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12 pin: USB DP line. Used in alternate push-pull, no pull mode. No external pull-up resistor is required.
CAN2 and DFUs	TIM11	Enabled	This timer is used to determine the value of the HSE. Once HSE frequency is determined, the system clock is configured to 60 MHz using PLL and HSE.

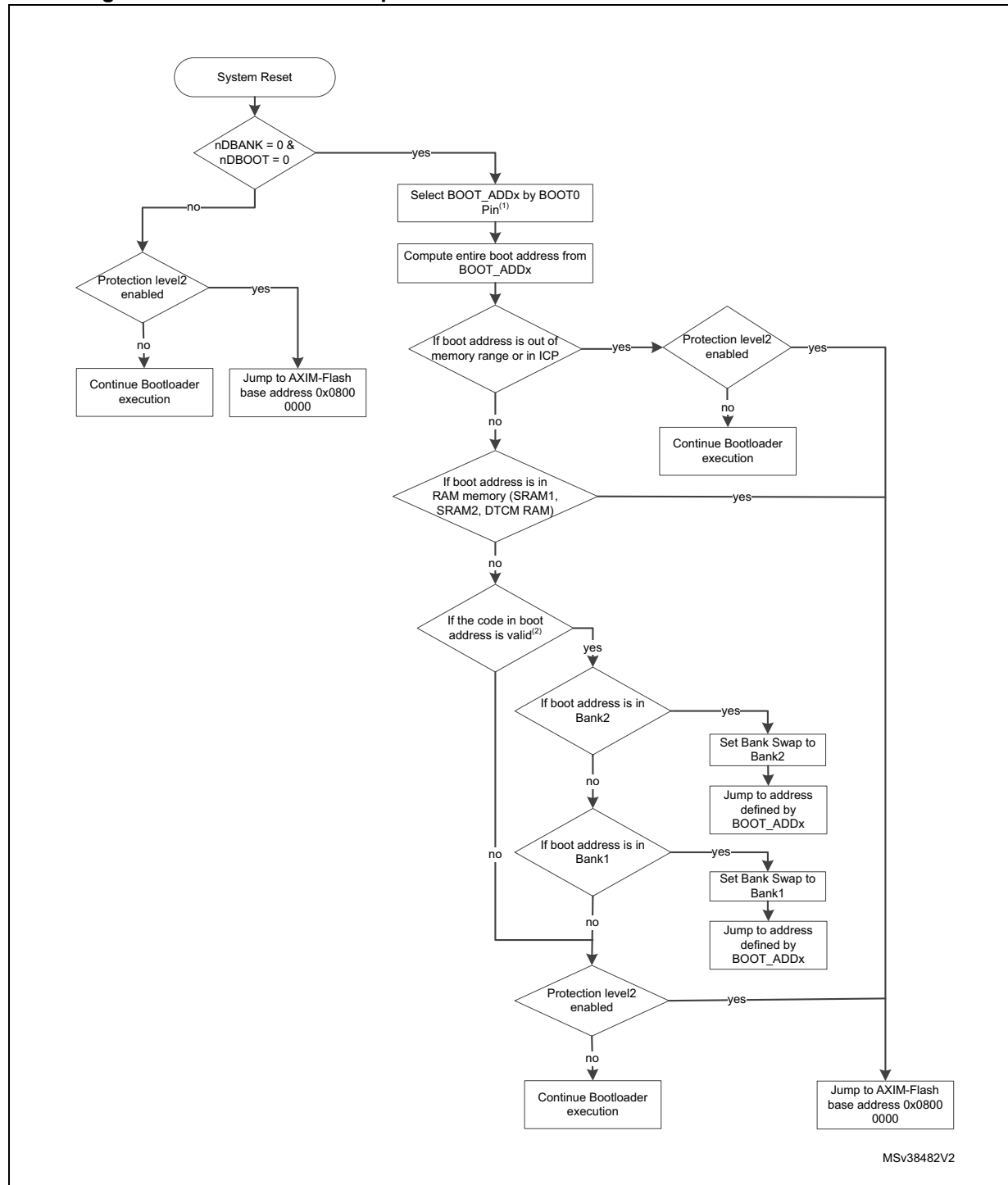
The system clock is derived from the embedded internal high-speed RC for USARTx and I2Cx bootloaders. This internal clock is also used for CAN and DFU (USB FS device), but only for the selection phase. An external clock multiple of 1 MHz (between 4 and 26 MHz) is required for CAN and DFU execution after the selection phase.

Note: *Due to HSI deviation and since HSI is used to detect HSE value, use low rather than high frequency HSE crystal values (low frequency values are better detected due to larger error margin). For example, it is better to use 8 MHz instead of 25 MHz.*

43.2 Bootloader selection

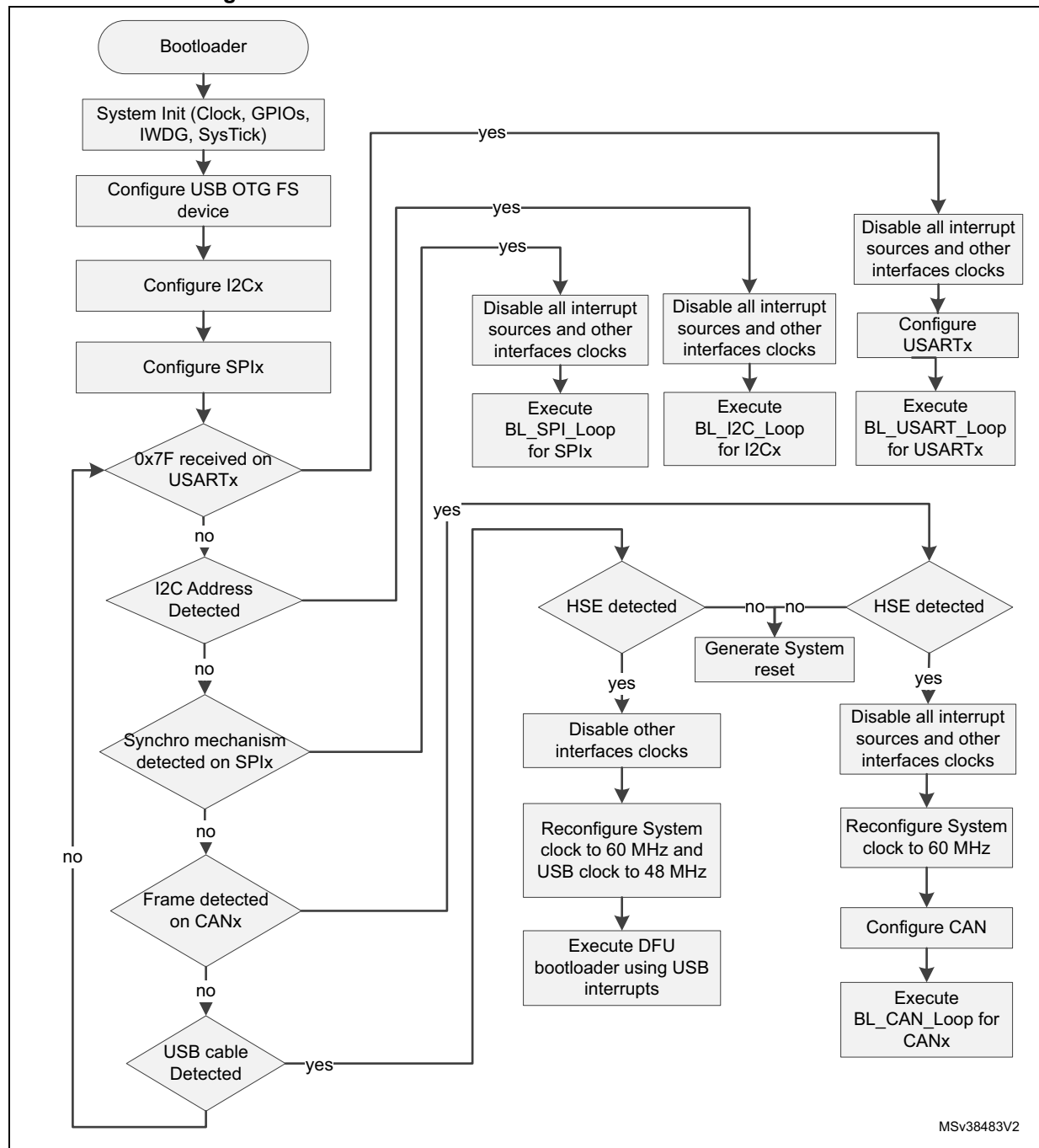
Figure 58 and Figure 59 show the bootloader selection mechanism.

Figure 58. Dual bank boot implementation for STM32F76xxx/77xxx Bootloader V9.x



1. Only BOOT_ADD0 value is considered whatever the BOOT0 pin state, as described in [Table 95](#).
2. ITCM RAM is not considered valid as stack pointer address for the dual bank boot mechanism.

Figure 59. Bootloader V9.x selection for STM32F76xxx/77xxx



43.3 Bootloader version

The following table lists the STM32F76xxx/77xxx devices bootloader V9.x versions.

Table 95. STM32F76xxx/77xxx bootloader V9.x versions

Version number	Description	Known limitations
V9.3	Initial bootloader version	When the flash memory is configured to the dual bank boot mode (nDBANK=nDBOOT=0), whatever the BOOT0 Pin state only BOOT_ADD0 value is considered (when BOOT0 Pin=1, BOOT_ADD0 value is considered not the BOOT_ADD1). Workaround: to manage dual bank boot with BOOT_ADD0 only, refer to AN4826 "<i>STM32F7 series flash memory dual bank mode</i>". At high UART baudrates (115200 bps) connection may fail due to software jitter leading to wrong baudrate calculation. In that case bootloader may respond with a baudrate up to $\pm 5\%$ different from host baudrate. Workaround: use baudrates lower than 57600 bps if host tolerance to baudrate error is lower than $\pm 5\%$. Bank2 sector erase issue when using USB interface. Erasing a sector from bank2 with index (i) leads to erase sector (i+4)

44 STM32G03xxx/STM32G04xxx devices

44.1 Bootloader configuration

The STM32G03xxx/G04xxx bootloader is activated by applying Pattern 11 (described in [Table 2](#)). The following table shows the hardware resources used by this bootloader. Note that STM32G030x do not have BOOT_LOCK(bit), so consider that when using Pattern 11.

Table 96. STM32G03xxx/G04xxx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 24 MHz (using PLL clocked by HSI).
	RAM	-	4 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	8 Kbytes, starting from address 0x1FFF0000
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
Securable memory area	-	-	The address to jump to for the securable memory area is 0x1FFF1D00
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USARTx bootloader	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
I2C1	I2C1	Enabled	The I2C1 configuration is: I2C speed: up to 1 MHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b1010110x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain pull-up mode.
	I2C1_SDA pin	Input/output	PB7 pin: data line is used in open-drain pull-up mode.

Table 96. STM32G03xxx/G04xxx configuration in system memory boot mode (continued)

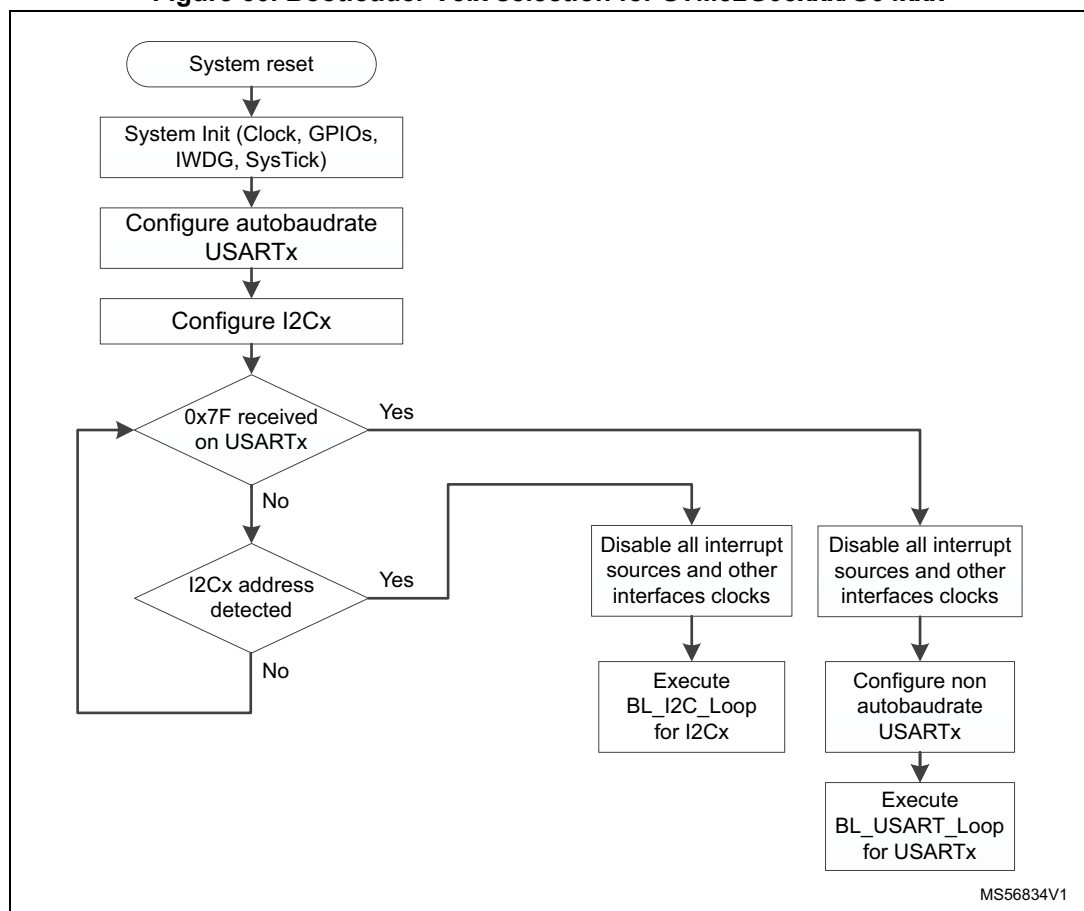
Bootloader	Feature/Peripheral	State	Comment
I2C2	I2C2	Enabled	The I2C2 configuration is: I2C speed: up to 1 MHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b1010110x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PB10 pin: clock line is used in open-drain pull-up mode.
	I2C2_SDA pin	Input/output	PB11 pin: data line is used in open-drain pull-up mode.

Note: On SO8, WLCSP18, TSSOP20, and UFQFN28 packages USART1 PA9/PA10 IOs are remapped on PA11/PA12.

44.2 Bootloader selection

Figure 60 shows the bootloader selection mechanism.

Figure 60. Bootloader V5.x selection for STM32G03xxx/G04xxx



44.3 Bootloader version

[Table 97](#) lists the STM32G03xxx/G04xxx devices bootloader versions.

Table 97. STM32G03xxx/04xxx bootloader versions

Version number	Description	Known limitations
V5.1	Initial bootloader version	– Supports only 48- and 32-pin packages – Issue is seen for both packages, if PA3 stays to low level, system is stuck in the USART2 detection sequence and no other interface is detected.
V5.2	Add support to small packages 8/20 and 28 pins	Issue is seen for all packages (except SO8, no PA3 pin), if PA3 stays to low level, system is stuck in the USART2 detection sequence and no other interface is detected.
V5.3	Fix V5.2 limitations	None

45 STM32G07xxx/08xxx device bootloader

45.1 Bootloader configuration

The STM32G07xxx/G08xxx bootloader is activated by applying Pattern 11 (described in [Table 2](#)). [Table 98](#) shows the hardware resources used by this bootloader.

When using Pattern 11, consider that STM32G070xx devices do not have BOOT_LOCK(bit).

Table 98. STM32G07xxx/8xxx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 24 MHz (using PLL clocked by HSI).
	RAM	-	12 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	28 Kbytes, starting from address 0x1FFF0000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
Securable memory area	-	-	The address to jump to for the securable memory area is 0x1FFF6800
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USART3	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. Used in alternate push-pull, pull-up mode.
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode. Used in alternate push-pull, pull-up mode.

Table 98. STM32G07xxx/8xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1010001x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain pull-up mode.
	I2C1_SDA pin	Input/output	PB7 pin: data line is used in open-drain pull-up mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1010001x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PB10 pin: clock line is used in open-drain pull-up mode.
	I2C2_SDA pin	Input/output	PB11 pin: data line is used in open-drain pull-up mode.
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode.
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode. ⁽¹⁾
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode.
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.

Table 98. STM32G07xxx/8xxx configuration in system memory boot mode (continued)

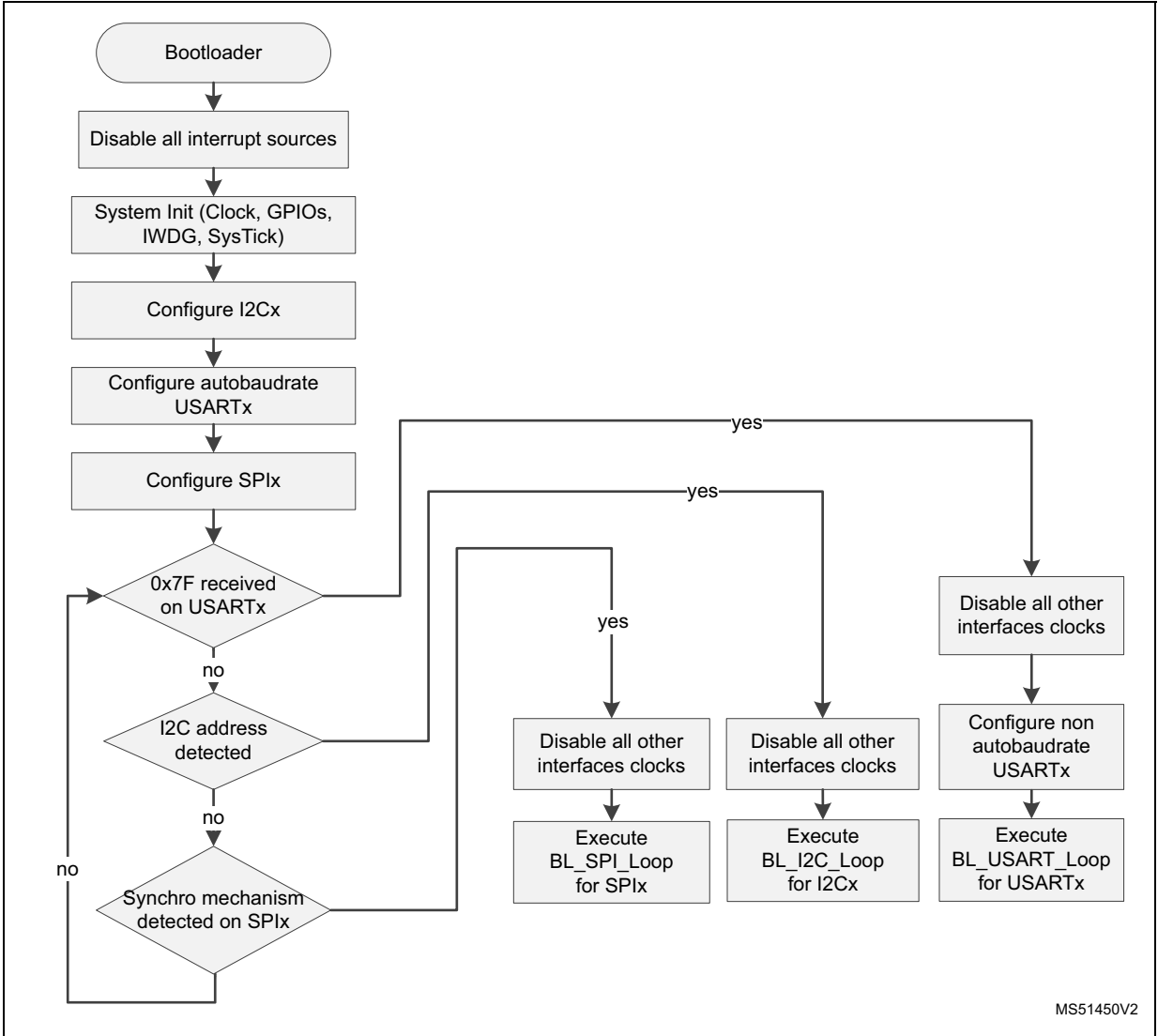
Bootloader	Feature/Peripheral	State	Comment
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PB15 pin: slave data input line, used in push-pull, pull-down mode.
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, pull-down mode. ⁽¹⁾
	SPI2_SCK pin	Input	PB13 pin: slave clock line, used in push-pull, pull-down mode.
	SPI2_NSS pin	Input	PB12 pin: slave chip select pin used in push-pull, pull-down. Note: This IO can be tied to GND if the SPI master does not use it.

1. SPI Tx (MISO) is handled by DMA. On the bootloader start-up after SPI initialization as soon as the bit DMATx enable on SPI CR2 register is set to 0x1, the MISO line is set to 3.3 V.

45.2 Bootloader selection

Figure 61 shows the bootloader selection mechanism.

Figure 61. Bootloader V11.0 selection for STM32G07xxx/G08xxx



45.3 Bootloader version

Table 99 lists the STM32G07xxx/8xxx devices bootloader versions.

Table 99. STM32G07xxx/08xxx bootloader versions

Version number	Description	Known limitations
V11.0	Initial bootloader version	Not supporting packages smaller than LQFP64
V11.1	Supporting all packages	None

Table 99. STM32G07xxx/08xxx bootloader versions (continued)

Version number	Description	Known limitations
V11.2	Added securable memory area feature	– Option byte launch missing when using USART protocol – RCC register RCC_ICSCR is not set to its default value when Go command is used. HSITRIM value is set to a value different from default. – RCC registers are not set to their default value when Go command is used (HSITRIM is not correctly reset). – Enabling SRAM parity check option byte causes bootloader crash if the SRAM is not initialized before enabling this feature.
V11.3	Fixed V11.2 limitations and added SW enhancements	– Empty check flag cleared by bootloader at boot – Compatibility break on boot sequence versus older versions⁽¹⁾ – Erase sectors not working as expected Root cause: Wrong FLITF BUSY bit check leading to not waiting for the erase operation termination. Workaround: Erase only by one sector and wait 40 ms for the erase termination before running the next operation. – I2C stretches the line on the connection causing issues with some HW hosts Root cause: USART3 detection method changed compared to the V11.2: a loop is added when a low edge is detected on the RX pin as the BL SW start baudrate calculations expecting it is the beginning of 0x7F byte. When the RX pin PC11 is tied to GND (manually on 64-pin packages or by production on low pin count packages), the USART3 detection loop is done on every Bootloader detection phase, causing a timeout wait when another peripheral is needed. Behavior: When connection to I2C is requested (host sends I2C address to the bootloader), the I2C HW detects the request ,but the BL SW is blocked on the USART3 SW loop, causing the I2C line stretching. Some HW hosts that do not support the stretching fail connecting with bootloader. Workaround: on 64-pin packages, do not put PC11 pin to GND. There is no workaround on low pin count packages, the only solution is to use a HW host supporting the clock stretching.
V11.4	Fixed V11.3 limitations	None

1. See [Section 45.3.1](#).

45.3.1 Compatibility break on boot sequence

Some enhancements introduced in V11.3 break the compatibility with the boot sequence of versions V11.2 and V11.1.

The major change is the addition of initialization of hardware peripherals (for the exact list refer to [Section 45.1](#)) used by the bootloader to their default values (as defined in RM0444).

The main impact is seen in two cases:

1. When jumping to the bootloader from user flash, the first operation is to reset the peripherals to their default values.
2. Usage of the “Empty check” boot mode
 - Empty check flag is raised by HW on a POR or option byte change when the user flash is empty.
 - This detection leads to boot on the bootloader.
 - As per the new boot sequence, the bootloader clears this flag.
 - Using a reset while the user flash is not yet programmed leads to a wrong boot on an empty user flash.

Avoid resets (except POR or option byte change) while the user flash is not programmed by the bootloader.

46 STM32G0B0xx device bootloader

46.1 Bootloader configuration

The STM32G0B0xx bootloader is activated by applying Pattern 11 (described in [Table 2](#)). The following table shows the hardware resources used by this bootloader. Note that STM32G0B0xx do not have BOOT_LOCK(bit), so consider that when using Pattern 11.

Table 100. STM32G0B0xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 60 MHz (using PLL clocked by HSI). If an external clock (HSE) is not present, the system is kept clocked from the HSI
		HSE enabled	The external clock can be used for all bootloader interfaces and must have one of the following values [48, 32, 16, 12, 8] MHz. The PLL is used to generate 48 MHz for USB and system clock.
	RAM	-	16 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	The bootloader firmware is shared on two banks: - 28 Kbytes, starting from address 0x1FFF0000 until 0x1FFF6FFF - Part of the 28 KB (0x1FFF8000 – 0x1FFFEFFF)
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
Securable memory area	-	-	The address to jump to for the exit securable memory area is 0x1FFF6800
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.

Table 100. STM32G0B0xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
USART3	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. Used in alternate push-pull, pull-up mode.
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode. Used in alternate push-pull, pull-up mode.
I2C1	I2C1	Enabled	The I2C1 configuration is: I2C speed: up to 1 MHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b1011101x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain pull-up mode.
	I2C1_SDA pin	Input/output	PB7 pin: data line is used in open-drain pull-up mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: I2C speed: up to 1 MHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b1011101x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PB10 pin: clock line is used in open-drain pull-up mode.
	I2C2_SDA pin	Input/output	PB11 pin: data line is used in open-drain pull-up mode.
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode.
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode. ⁽¹⁾
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode.
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.

Table 100. STM32G0B0xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PB15 pin: slave data input line, used in push-pull, pull-down mode.
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, pull-down mode. ⁽¹⁾
	SPI2_SCK pin	Input	PB13 pin: slave clock line, used in push-pull, pull-down mode.
	SPI2_NSS pin	Input	PB12 pin: slave chip select pin used in push-pull, pull-down. Note: This IO can be tied to GND if the SPI master does not use it.
DFU ⁽²⁾	USB	Enabled	USB FS configured in Forced Device mode. USB FS interrupt vector is enabled and used for USB DFU communications. Note: VDDUSB IO must be connected to 3.3 V as USB peripheral is used by the bootloader. ⁴
	USB_DM pin	Input/output	PA11: USB DM line. Used in input no pull mode.
	USB_DP pin		PA12: USB DP line. Used in input no pull mode. No external pull-up resistor is required

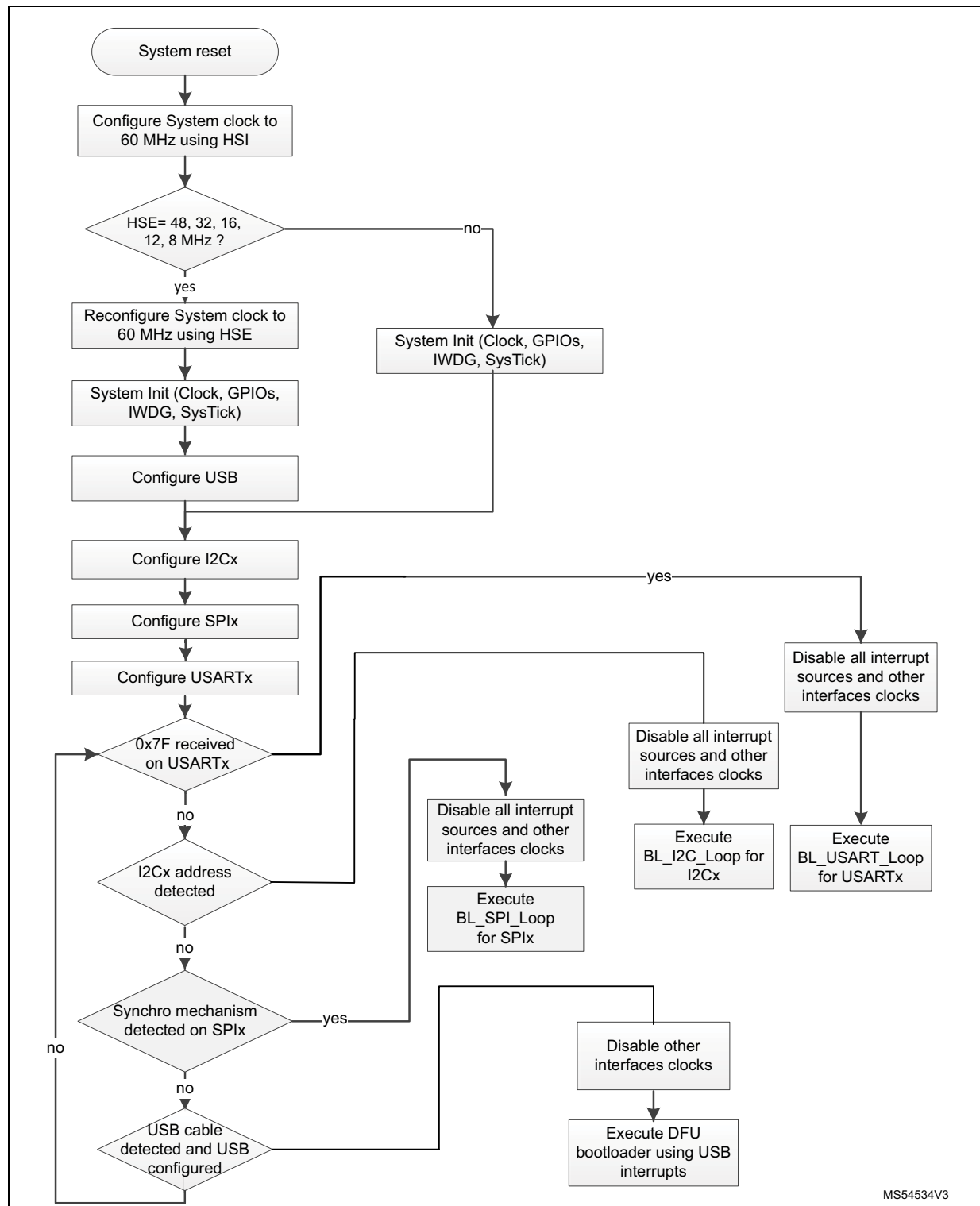
1. SPI Tx (MISO) is handled by DMA. On the bootloader start-up after SPI initialization as soon as the bit DMATx enable on SPI CR2 register is set to 0x1, the MISO line is set to 3.3 V.

2. USB DFU is not available on LQFP32 package.

46.2 Bootloader selection

Figure 62 shows the bootloader selection mechanism.

Figure 62. Bootloader selection for STM32G0B0xx



46.3 Bootloader version

[Table 101](#) lists the STM32G0B0xx devices bootloader versions.

Table 101. STM32G0B0xx bootloader versions

Version number	Description	Known limitations
V13.0	Initial bootloader version	Erase multiple sectors not working on Bank2 – Root cause: wrong BUSY bit check on Bank2 leads to generate FLITF error after one sector erase – Workaround: erase only by one sector when targeting Bank2 Empty check flag cleared by error on the bootloader startup phase – Root cause: on the startup phase the bootloader SW performs a system deinitialization, leading to write the default value on the FLASH_ACR register, which overrides the Empty check bit with 0 – Behavior: when Empty check boot mode is used and the flash memory is empty, the MCU boots on the bootloader but the flag is cleared by the SW. If a reset is triggered, the system tries to boot on the empty flash memory, and crashes. – Caution: Avoid using reset on this case. if the system crashes, an option byte change or POR is needed to reboot.

47 STM32G0B1xx/0C1xx device bootloader

47.1 Bootloader configuration

The STM32G0B1xx/0C1xx bootloader is activated by applying Pattern 11 (described in [Table 2](#)). The following table shows the hardware resources used by this bootloader.

Table 102. STM32G0B1xx/0C1xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 60 MHz (using PLL clocked by HSI).
		-	CRS is enabled for the DFU to allow USB to be clocked by HSI48 48 MHz.
		-	20 MHz derived from the PLLQ is used for FDCAN
	RAM	-	16 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	The bootloader firmware is shared on two banks: - 28 Kbytes, starting from address 0x1FFF0000 until 0x1FFF6FFF - Part of the 28 KB (0x1FFF8000 – 0x1FFFEFFF)
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
Securable memory area	-	-	The address to jump to for the securable memory area is 0x1FFF6800
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USART3	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. Used in alternate push-pull, pull-up mode.
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode. Used in alternate push-pull, pull-up mode.

Table 102. STM32G0B1xx/0C1xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 1 MHz, 7-bit address, Target mode, Analog filter ON – Target 7-bit address: 0b1011101x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain pull-up mode.
	I2C1_SDA pin	Input/output	PB7 pin: data line is used in open-drain pull-up mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 1 MHz, 7-bit address, Target mode, Analog filter ON – Target 7-bit address: 0b1011101x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PB10 pin: clock line is used in open-drain pull-up mode.
	I2C2_SDA pin	Input/output	PB11 pin: data line is used in open-drain pull-up mode.
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode.
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode. ⁽¹⁾
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode.
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.

Table 102. STM32G0B1xx/0C1xx configuration in system memory boot mode (continued)

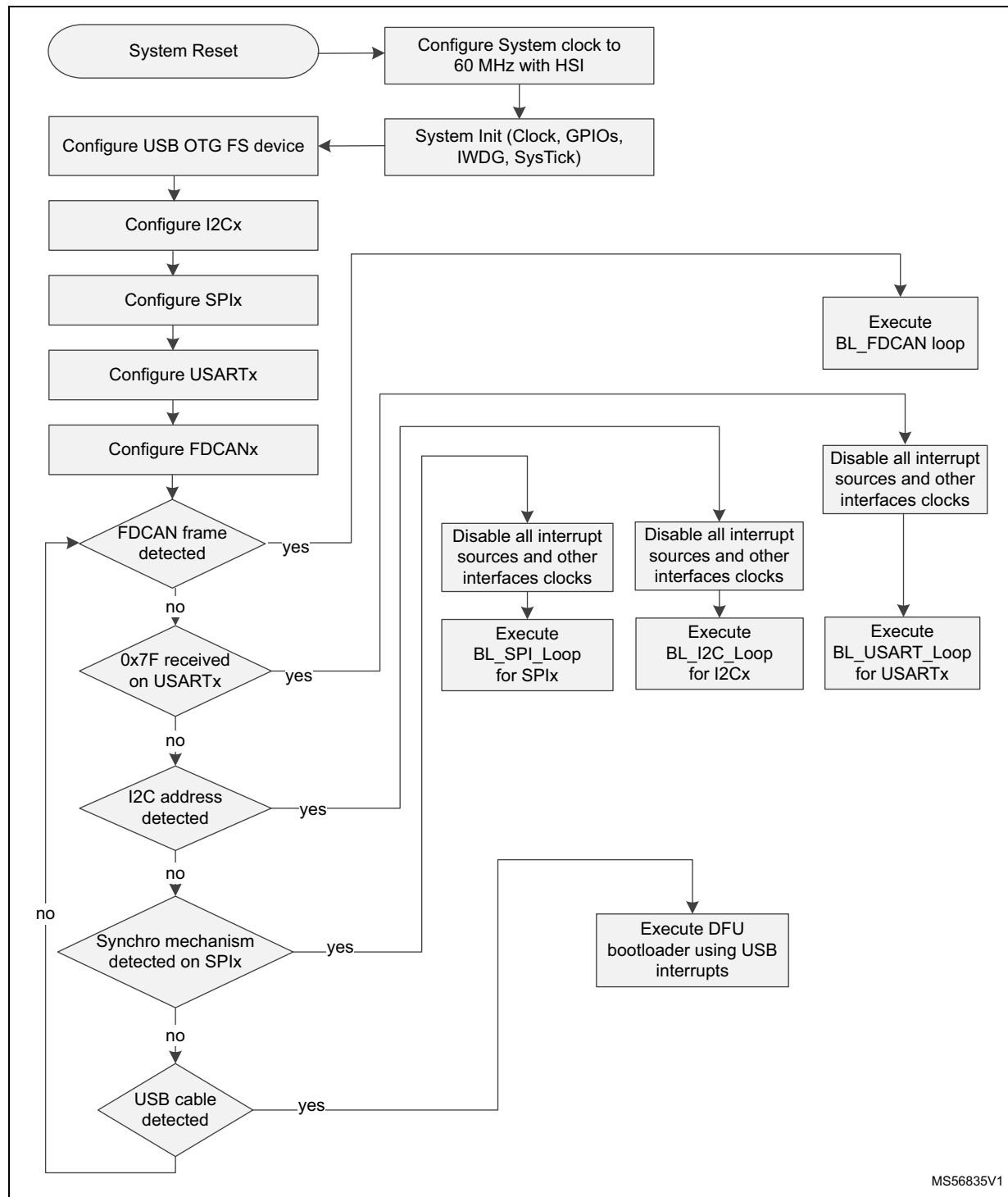
Bootloader	Feature/Peripheral	State	Comment
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PB15 pin: slave data input line, used in push-pull, pull-down mode.
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, pull-down mode. ⁽¹⁾
	SPI2_SCK pin	Input	PB13 pin: slave clock line, used in push-pull, pull-down mode.
	SPI2_NSS pin	Input	PB12 pin: slave chip select pin used in push-pull, pull-down. Note: This IO can be tied to GND if the SPI master does not use it.
DFU	USB	Enabled	USB FS configured in Forced device mode. USB FS interrupt vector is enabled and used for USB DFU communications. Note: VDDUSB IO must be connected to 3.3 V as USB peripheral is used by the bootloader. ¹
	USB_DM pin	Input/output	PA11: USB DM line. Used in no pull mode.
	USB_DP pin		PA12: USB DP line. Used in no pull mode. No external pull-up resistor is required
FDCAN	FDCAN1	Enabled	Once initialized the FDCAN1 configuration is: – Connection bit rate 250 kbit/s – Data bit rate 1000 kbit/s – FrameFormat = FDCAN_FRAME_FD_BRS – Mode = FDCAN_MODE_NORMAL – AutoRetransmission = ENABLE – TransmitPause = DISABLE – ProtocolException = ENABLE
	FDCAN1_Rx pin	Input	PD0 pin: FDCAN1 in reception mode. Used in alternate push-pull, pull-up mode.
	FDCAN1_Tx pin	Output	PD1 pin: FDCAN1 in transmission mode. Used in alternate push-pull, pull-up mode.

1. SPI Tx (MISO) is handled by DMA. On the bootloader start-up after SPI initialization as soon as the bit DMATx enable on SPI CR2 register is set to 0x1, the MISO line is set to 3.3 V.

47.2 Bootloader selection

Figure 63 shows the bootloader selection mechanism.

Figure 63. Bootloader selection for STM32G0B1xx/0C1xx



47.3 Bootloader version

[Table 103](#) lists the STM32G0B1xx/0C1xx devices bootloader versions.

Table 103. STM32G0B1xx/0C1xx bootloader versions

Version number	Description	Known limitations
V9.2	Initial bootloader version	When jumping to an application, the Go command disables the debug access port by writing a wrong value on the FLASH_ACR register (bit DBG_SWEN). Erase multiple sectors not working on Bank2 – Root cause: wrong BUSY bit check on Bank2 leads to generate FLITF error after one sector erase – Workaround: erase only by one sector when targeting Bank2 Empty check flag cleared by error on the bootloader startup phase – Root cause: on the startup phase the bootloader SW performs a system deinitialization, leading to write the default value on the FLASH_ACR register, which overrides the Empty check bit with 0 – Behavior: when Empty check boot mode is used and the flash memory is empty, the MCU boots on the bootloader but the flag is cleared by the SW. If a reset is triggered, the system tries to boot on the empty flash memory, and crashes. – Caution: Avoid using reset on this case. if the system crashes, an option byte change or POR is needed to reboot.

48 STM32G05xxx/061xx devices

48.1 Bootloader configuration

The STM32G05xxx/061xx bootloader is activated by applying Pattern 11 (described in [Table 2](#)). The following table shows the hardware resources used by this bootloader. Note that STM32G050x do not have BOOT_LOCK(bit), so consider that when using Pattern 11.

Table 104. STM32G05xxx/061xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 24 MHz (using PLL clocked by HSI).
	RAM	-	4 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	8 Kbytes, starting from address 0x1FFF0000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
Securable memory area	-	-	The address to jump to for the securable memory area is 0x1FFF6800
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USARTx bootloader	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USART2s.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1100010x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain pull-up mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain pull-up mode.

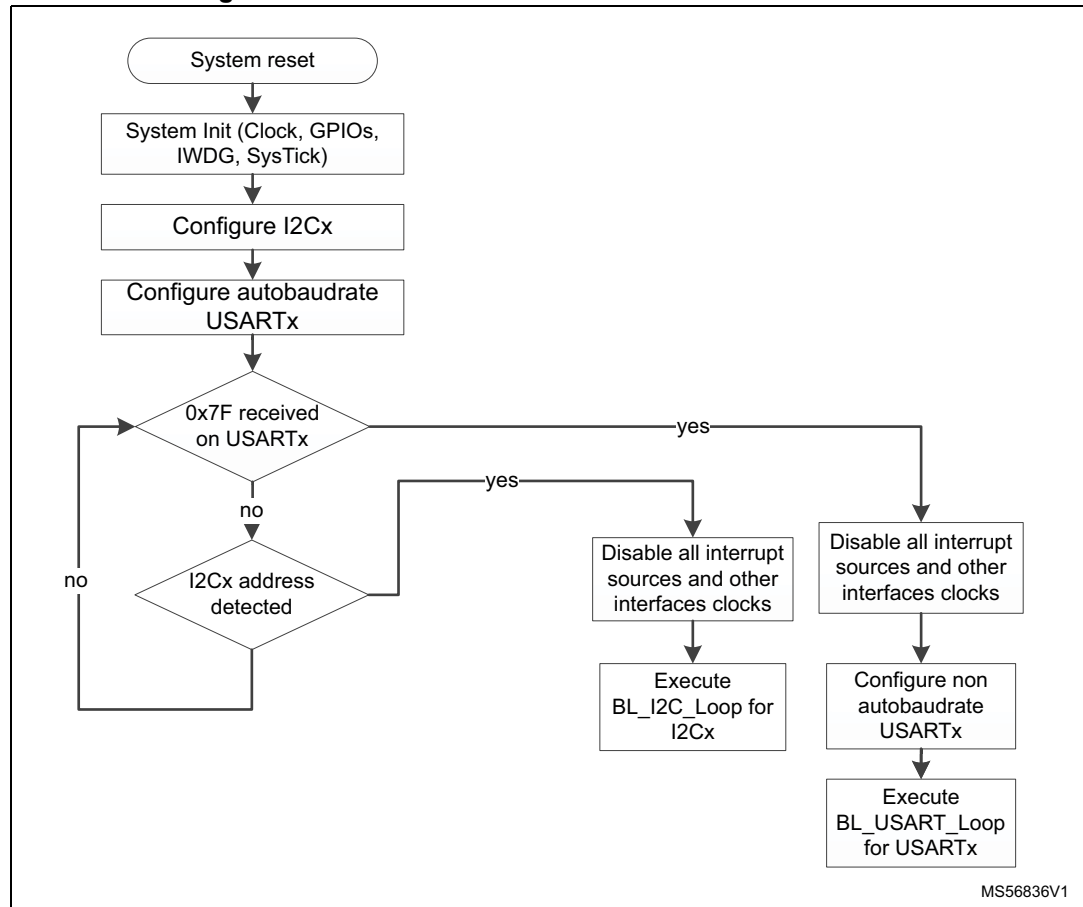
Table 104. STM32G05xxx/061xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1100010x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PB10 pin: clock line is used in open-drain pull-up mode.
	I2C2_SDA pin		PB11 pin: data line is used in open-drain pull-up mode.

48.2 Bootloader selection

Figure 64 shows the bootloader selection mechanism.

Figure 64. Bootloader selection for STM32G05xxx/061xx



48.3 Bootloader version

Table 105 lists the STM32G05xxx/061xx devices bootloader versions.

Table 105. STM32G05xxx/061xx bootloader versions

Version number	Description	Known limitations
V5.0	Initial bootloader version	USART2 SW jitter issue on detection phase
V5.1	Fix V5.0 limitation	Non-stretch command not working as expected and stretching the line – Root cause: wrong BUSY check leads to not entering BUSY byte generation while waiting for the Non stretch command to complete – Behavior: when running a non-stretch commands instead of receiving a BUSY byte (0x76) while command is running; the BL is stretching the line and no data is sent to the host.. This is noticed only on the non-stretch erase command, as it can take few ms, and can cause an issue if the host does not support the line stretching. – Workaround: patch in RAM to use the correct check

49 STM32G431xx/441xx devices

49.1 Bootloader configuration

The STM32G431xx/441xx bootloader is activated by applying Pattern 15 (described in [Table 2](#)). The following table shows the hardware resources used by this bootloader.

Table 106. STM32G431xx/441xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 72 MHz (using the PLL clocked by HSI)
		-	CRS is enabled for the DFU to allow USB to be clocked by HSI48 48 MHz
	RAM	-	16 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	28 Kbytes, starting from address 0x1FFF0000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
Securable memory area	-	-	The address to jump to the exit securable memory area @0x1FFF6800
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USART3	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. Used in alternate push-pull, pull-up mode.
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode. Used in alternate push-pull, pull-up mode.

Table 106. STM32G431xx/441xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1010100x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PC4 pin: clock line is used in open-drain pull-up mode.
	I2C2_SDA pin		PA8 pin: data line is used in open-drain pull-up mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1010100x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PC8 pin: clock line is used in open-drain pull-up mode.
	I2C3_SDA pin		PC9 pin: data line is used in open-drain pull-up mode.
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode.
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode. ⁽¹⁾
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode.
	SPI1_NSS pin		PA4 pin: slave chip select pin used in push-pull, pull-down mode.

Table 106. STM32G431xx/441xx configuration in system memory boot mode (continued)

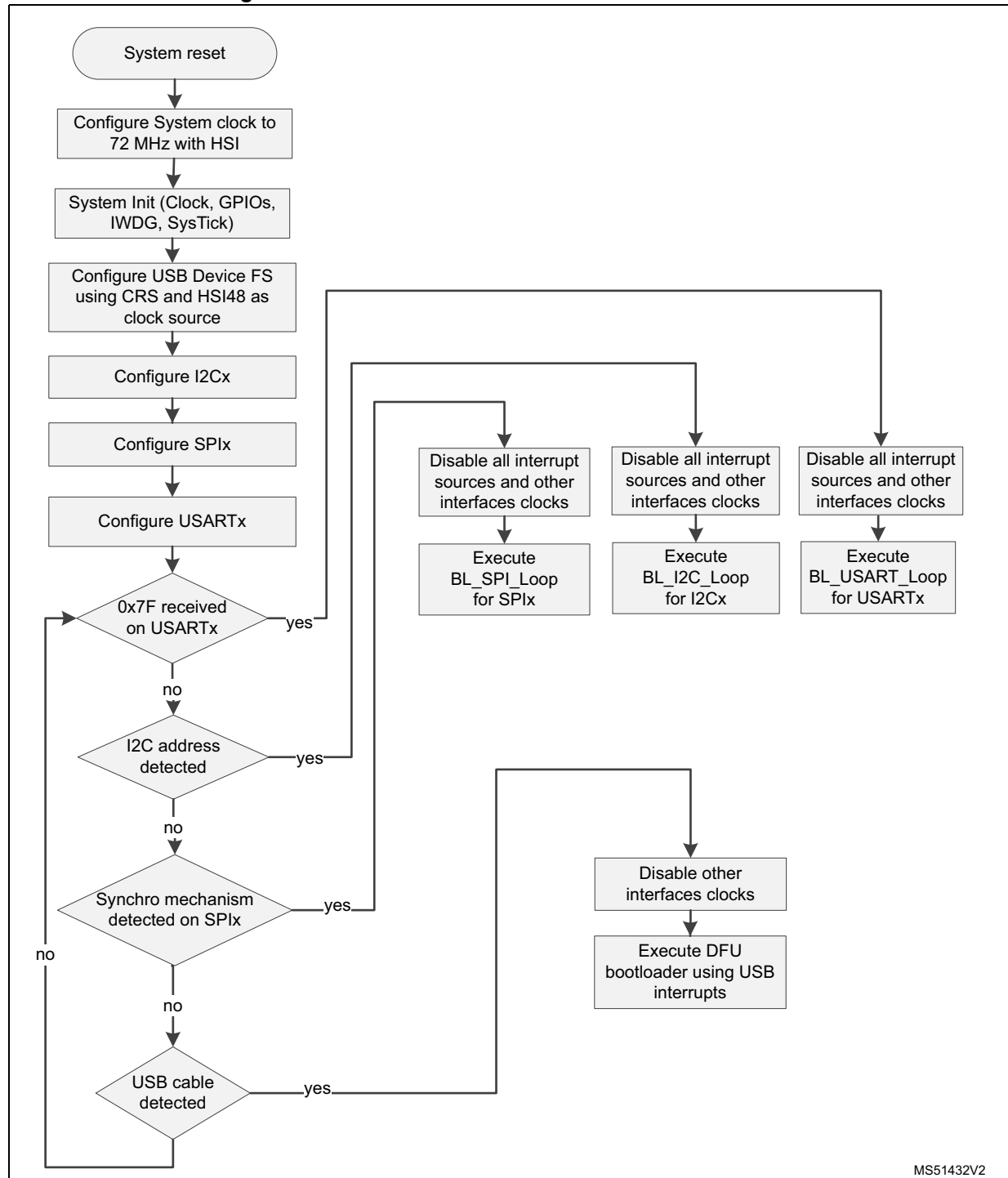
Bootloader	Feature/Peripheral	State	Comment
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PB15 pin: slave data input line, used in push-pull, pull-down mode.
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, pull-down mode. ⁽¹⁾
	SPI2_SCK pin	Input	PB13 pin: slave clock line, used in push-pull, pull-down mode.
	SPI2_NSS pin	Input	PB12 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.
DFU	USB	Enabled	USB FS configured in forced device mode. USB FS interrupt vector is enabled and used for USB DFU communications.
	USB_DM pin	Input/output	PA11: USB DM line. Used in input no pull mode.
	USB_DP pin		PA12: USB DP line. Used in input no pull mode. No external pull-up resistor is required

1. SPI Tx (MISO) is handled by DMA. On the bootloader start-up after SPI initialization as soon as the bit DMATx enable on SPI CR2 register is set to 0x1, the MISO line is set to 3.3 V.

49.2 Bootloader selection

The following figure shows the bootloader selection mechanism.

Figure 65. Bootloader selection for STM32G431xx/441xx



49.3 Bootloader version

Table 107. STM32G431xx/441xx bootloader version

Version number	Description	Known limitations
V13.3 (0xD3)	Initial bootloader version	CCSRAM not supported
V13.4 (0xD4)	Fix V13.3 limitations Add CCSRAM support	-

50 STM32G47xxx/48xxx devices

50.1 Bootloader configuration

The STM32G47xxx/48xxx bootloader is activated by applying Pattern 14 (described in [Table 2](#)). The following table shows the hardware resources used by this bootloader.

Table 108. STM32G47xxx/48xxx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 72 MHz (using the PLL clocked by HSI)
		-	CRS is enabled for the DFU to allow USB to be clocked by HSI48 48 MHz
	RAM	-	16 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	28 Kbytes, starting from address 0x1FFF0000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
		-	
Securable memory area	-	-	The address to jump to the exit securable memory area @0x1FFF6800
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USART3	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. Used in alternate push-pull, pull-up mode.
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode. Used in alternate push-pull, pull-up mode.

Table 108. STM32G47xxx/48xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I2C2	I2C2	Enabled	The I2C2 configuration is: I2C speed: up to 1 MHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b1010011x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PC4 pin: clock line is used in open-drain pull-up mode.
	I2C2_SDA pin	Input/output	PA8 pin: data line is used in open-drain pull-up mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: I2C speed: up to 1 MHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b1010011x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PC8 pin: clock line is used in open-drain pull-up mode.
	I2C3_SDA pin	Input/output	PC9 pin: data line is used in open-drain pull-up mode.
I2C4	I2C4	Enabled	The I2C4 configuration is: I2C speed: up to 1 MHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b1010011x (x = 0 for write and x = 1 for read)
	I2C4_SCL pin	Input/output	PC6 pin: clock line is used in open-drain pull-up mode.
	I2C4_SDA pin	Input/output	PC7 pin: data line is used in open-drain pull-up mode.
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode.
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode. ⁽¹⁾
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull no pull-up, pull-down mode.
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode.

Table 108. STM32G47xxx/48xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PB15 pin: slave data input line, used in push-pull, npull-down mode.
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, pull-down mode. ⁽¹⁾
	SPI2_SCK pin	Input	PB13 pin: slave clock line, used in push-pull, n pull-down mode.
	SPI2_NSS pin	Input	PB12 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.
DFU	USB	Enabled	USB FS configured in forced device mode. USB FS interrupt vector is enabled and used for USB DFU communications. Note: VDDUSB IO must be connected to 3.3 V as USB peripheral is used by the bootloader.
	USB_DM pin	Input/output	PA11: USB DM line. Used in input no pull mode.
	USB_DP pin		PA12: USB DP line. Used in input no pull mode. No external pull-up resistor is required

1. SPI Tx (MISO) is handled by DMA. On the bootloader start-up after SPI initialization as soon as the bit DMATx enable on SPI CR2 register is set to 0x1, the MISO line is set to 3.3 V.

50.2 Bootloader selection

The following figures show the bootloader selection mechanism.

Figure 66. Bootloader selection for STM32G47xxx/48xxx

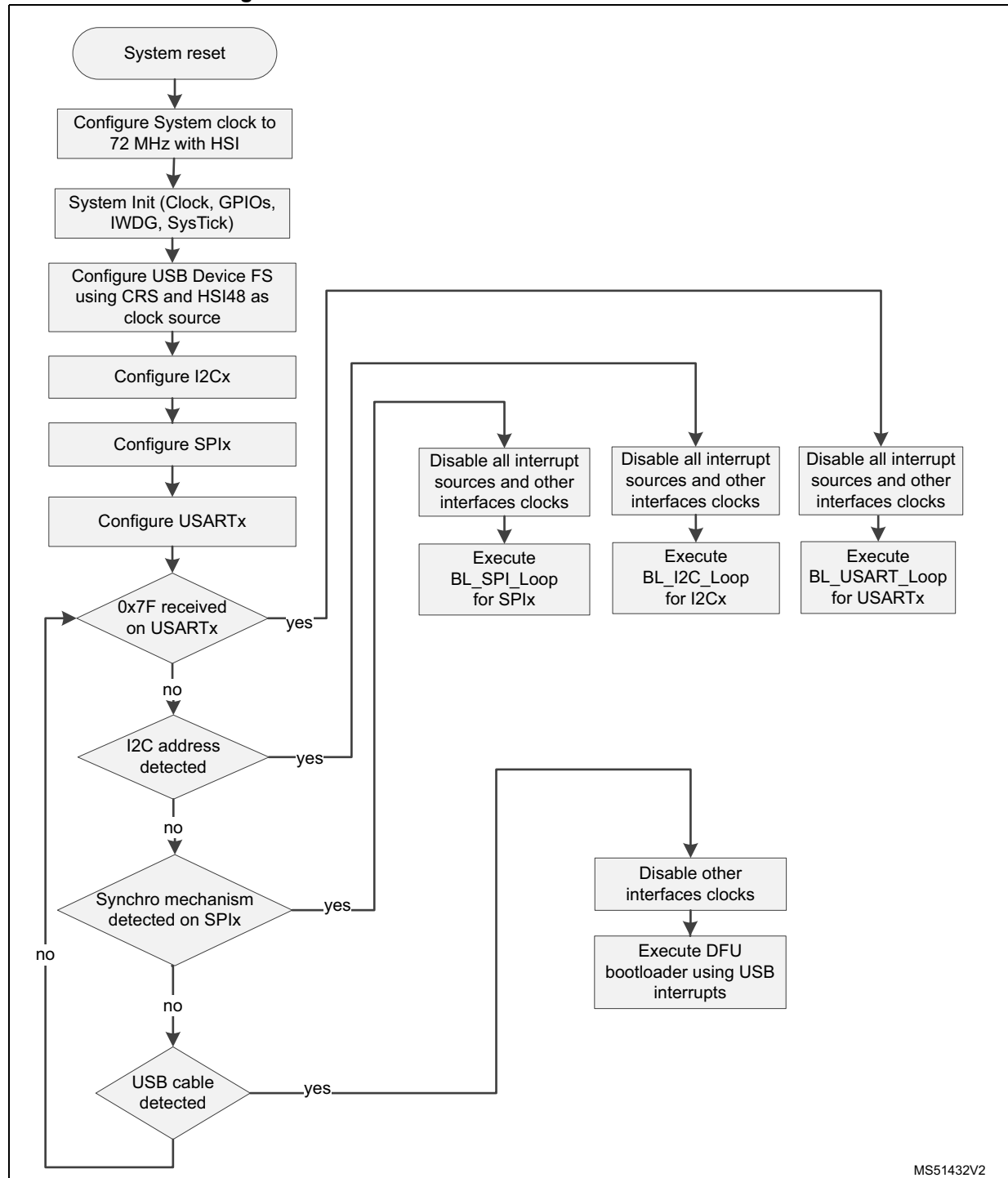
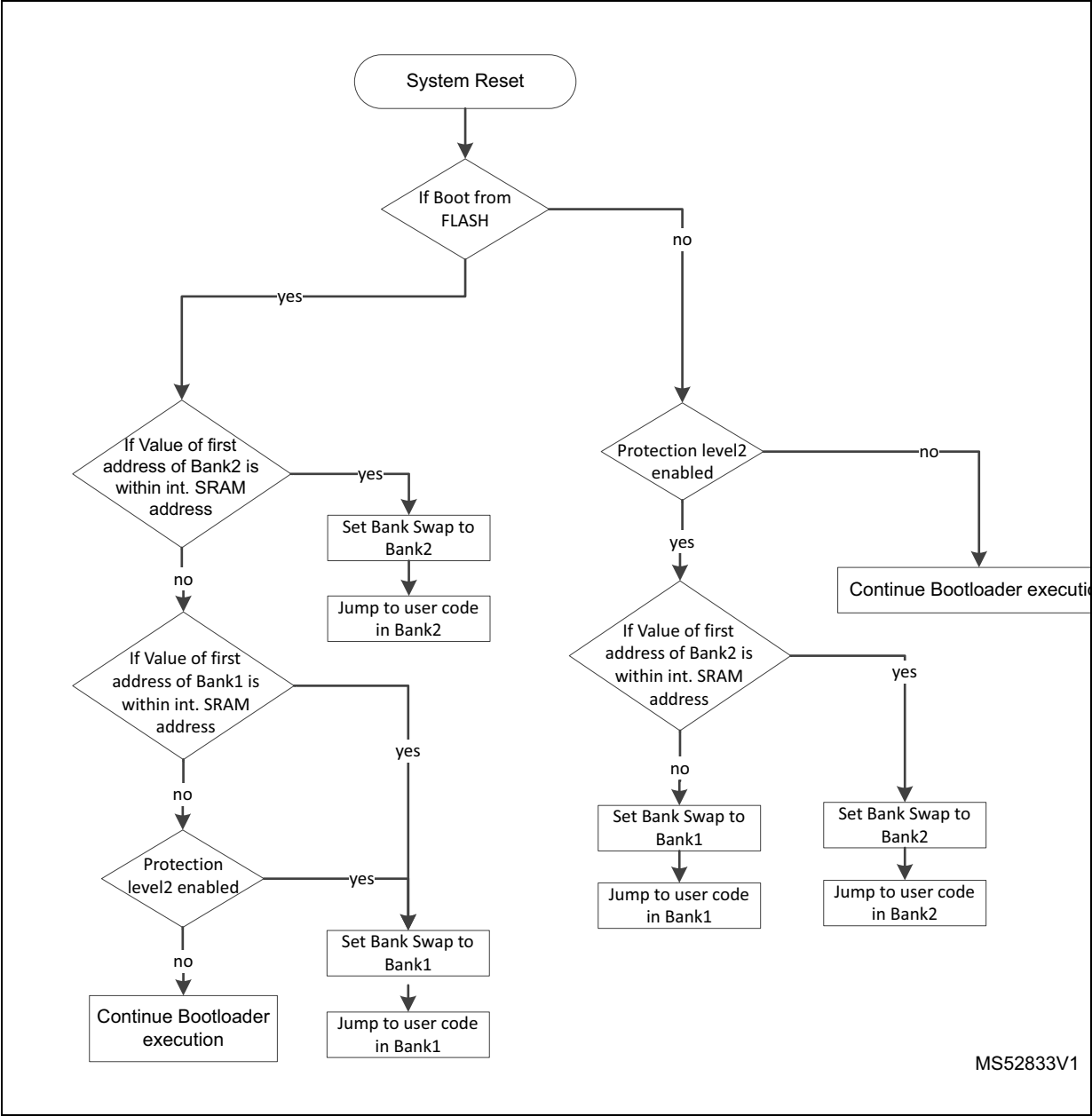


Figure 67. Dual bank boot implementation for STM32G47xxx/48xxx bootloader V13.x



50.3 Bootloader version

Table 109. STM32G47xxx/48xxx bootloader version

Version number	Description	Known limitations
V13.3 (0xD3)	Initial bootloader version	Boot from bank2 is not working

Table 109. STM32G47xxx/48xxx bootloader version (continued)

Version number	Description	Known limitations
V13.4 (0xD4)	Fix V13.3 limitations	CCSRAM/ENGI not supported
V13.5 (0xD5)	– Fix V13.4 limitations – Add CCSRAM/ENGI support	None

51 STM32G491xx/4A1xx devices

51.1 Bootloader configuration

The STM32G491xx/4A1xx bootloader is activated by applying Pattern 15 (described in [Table 2](#)). The following table shows the hardware resources used by this bootloader.

Table 110. STM32G491xx/4A1xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 72 MHz (using the PLL clocked by HSI)
		-	CRS is enabled for the DFU to allow USB to be clocked by HSI48 48 MHz
	RAM	-	16 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	28 Kbytes, starting from address 0x1FFF0000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
Securable memory area	-	-	The address to jump to the exit securable memory area @0x1FFF6800
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USART3	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. Used in alternate push-pull, pull-up mode.
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode. Used in alternate push-pull, pull-up mode.

Table 110. STM32G491xx/4A1xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I2C2	I2C2	Enabled	The I2C2 configuration is: I2C speed: up to 1 MHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b1011111x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PC4 pin: clock line is used in open-drain pull-up mode.
	I2C2_SDA pin	Input/output	PA8 pin: data line is used in open-drain pull-up mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: I2C speed: up to 1 MHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b1011111x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PC8 pin: clock line is used in open-drain pull-up mode.
	I2C3_SDA pin	Input/output	PC9 pin: data line is used in open-drain pull-up mode.
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode.
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode. ⁽¹⁾
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull no pull-up, pull-down mode.
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode.

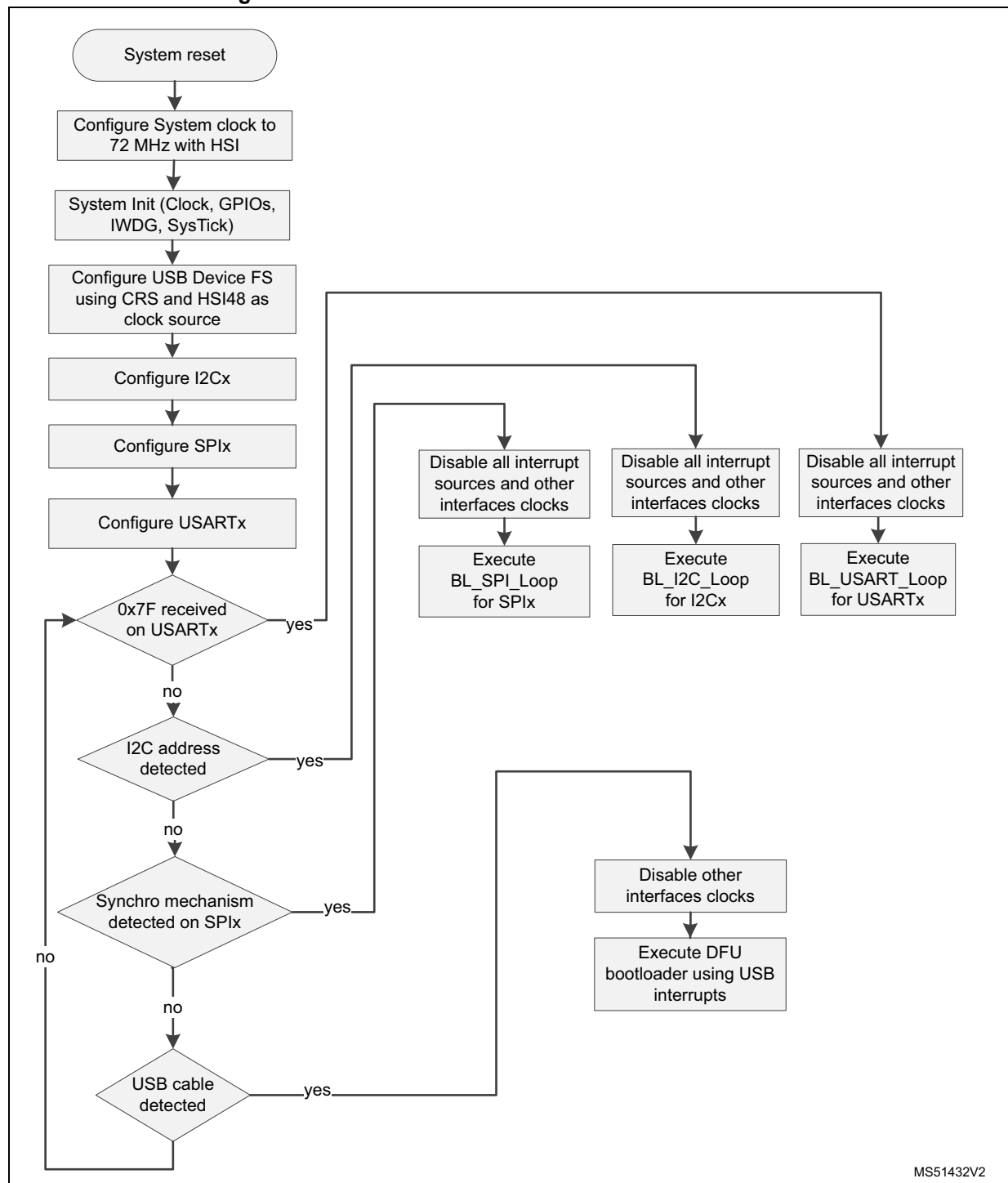
Table 110. STM32G491xx/4A1xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PB15 pin: slave data input line, used in push-pull, npull-down mode.
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, pull-down mode. ⁽¹⁾
	SPI2_SCK pin	Input	PB13 pin: slave clock line, used in push-pull, n pull-down mode.
	SPI2_NSS pin	Input	PB12 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.
DFU	USB	Enabled	USB FS configured in forced device mode. USB FS interrupt vector is enabled and used for USB DFU communications. Note: VDDUSB IO must be connected to 3.3 V as USB peripheral is used by the bootloader.
	USB_DM pin	Input/output	PA11: USB DM line. Used in input no pull mode.
	USB_DP pin		PA12: USB DP line. Used in input no pull mode. No external pull-up resistor is required

1. SPI Tx (MISO) is handled by DMA. On the bootloader start-up after SPI initialization, as soon as bit DMATx enable on SPI CR2 register is set to 0x1, the MISO line is set to 3.3 V.

51.2 Bootloader selection

Figure 68. Bootloader selection for STM32G491xx/4A1xx



51.3 Bootloader version

Table 111. STM32G491xx/4A1xx bootloader version

Version number	Description	Known limitations
V13.2	Initial bootloader version	None

52 STM32H503xx devices

52.1 Bootloader configuration

The STM32H503xx bootloader is activated by applying Pattern 17 (described in [Table 2](#)). [Table 112](#) shows the hardware resources used by this bootloader.

The bootloader follows boot model V3_2 (see [Section 4.10](#)), so it inherits all its constraints.

Table 112. STM32H503xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 200 MHz (using PLL clocked by the HSI)
		-	CRS is enabled for the DFU to allow USB to be clocked by HSI48 48 MHz
		-	20 MHz derived from the PLLQ is used for FDCAN
	RAM	-	16 Kbytes, starting from address 0x24000000, are used by the bootloader firmware
	System memory	-	35 Kbytes, starting from address 0x0BF87000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Set as input until USART1 is detected.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA15 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA5 pin: USART2 in transmission mode. Set as input until USART2 is detected.
USART3	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PA3 pin: USART3 in reception mode. Used in alternate push-pull, pull-down mode.
	USART3_TX pin	Output	PA4 pin: USART3 in transmission mode. Set as input until USART3 is detected.

Table 112. STM32H503xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I2C2	I2C2	Enabled	The I2C1 configuration is: – I2C speed: up to 1 MHz, 7-bit address, Target mode, Analog filter ON – Target 7-bit address: 0b1100111x, x = 0 for write and x = 1 for read
	I2C2_SCL pin	Input/output	PB3 pin: clock line is used in open-drain. pull up mode.
	I2C2_SDA pin	Input/output	PB4 pin: data line is used in open-drain, pull up mode.
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, no pull mode.
	SPI1_MISO pin	Output	PA0 pin: slave data output line, used in push-pull, no pull mode.
	SPI1_SCK pin	Input	PA8 pin: slave clock line, used in push-pull, no pull mode.
	SPI1_NSS pin	Input	PB8 pin: slave chip select pin used in push-pull, no pull mode.
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PB1 pin: slave data input line, used in push-pull no pull mode
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, no pull mode.
	SPI2_SCK pin	Input	PB10 pin: slave clock line, used in push-pull, no pull mode.
	SPI2_NSS pin	Input	PB12 pin: slave chip select pin used in push-pull, no pull mode.

Table 112. STM32H503xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI3	SPI3	Enabled	The SPI3 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz Polarity: CPOL low, CPHA low, NSS hardware.
	SPI3_MOSI pin	Input	PC12 pin: slave data input line, used in push-pull no pull mode
	SPI3_MISO pin	Output	PC11 pin: slave data output line, used in push-pull, no pull mode.
	SPI3_SCK pin	Input	PC10 pin: slave clock line, used in push-pull, no pull mode.
	SPI3_NSS pin	Input	PD2 pin: slave chip select pin used in push-pull, no pull mode.
DFU	USB	Enabled	USB FS configured in forced device mode. USB FS interrupt vector is enabled and used for USB DFU communications.
	USB_DM pin	Input/output	PA11: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12: USB DP line. Used in alternate push-pull, no pull mode. No external pull-up resistor is required
DFU	USB	Enabled	USB FS configured in forced device mode. USB FS interrupt vector is enabled and used for USB DFU communications. Note: VDDUSB IO must be connected to 3.3 V as USB peripheral is used by the bootloader.
	USB_DM pin	Input/output	PA11: USB DM line. Used in input no pull mode.
	USB_DP pin		PA12: USB DP line. Used in input no pull mode. No external pull-up resistor is required
FDCAN	FDCAN1	Enabled	Once initialized the FDCAN1 configuration is: – Connection bit rate 250 kbit/s – Data bit rate 1000 kbit/s – FrameFormat = FDCAN_FRAME_FD_BRS – Mode = FDCAN_MODE_NORMAL AutoRetransmission = ENABLE – TransmitPause = DISABLE – ProtocolException = ENABLE
	FDCAN1_Rx pin	Input	PB5 pin: FDCAN1 in reception mode. Used in alternate push-pull, no pull mode.
	FDCAN1_Tx pin	Output	PB15 pin: FDCAN1 in transmission mode. Used in alternate push-pull, no pull mode.

Table 112. STM32H503xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I3C	I3C	Enabled	– Mode: target mode – Aval timing: 0x4E – DMA Reg RX: disabled – DMA Req TX: disabled – Status FIFO: disabled – DMA Req status: disabled – DMA Req control: disabled – IBI: enabled – Additional data after IBI ack-ed: 1 byte – IBI configuration: Mandatory Data Byte (MDB) – All IT disabled except RXFNE (Receive FIFO Interrupt) The RXFNE interruption is disabled after SYNC byte detection by the bootloader.
	I3C1_SCL pin	Input/output	PB6 pin: I3C1 in transmission mode. Used in alternate push-pull, no pull mode.
	I3C1_SDA pin		PB7 pin: I3C1 in transmission mode. Used in alternate push-pull, no pull mode.

Table 113. STM32H503xx special commands

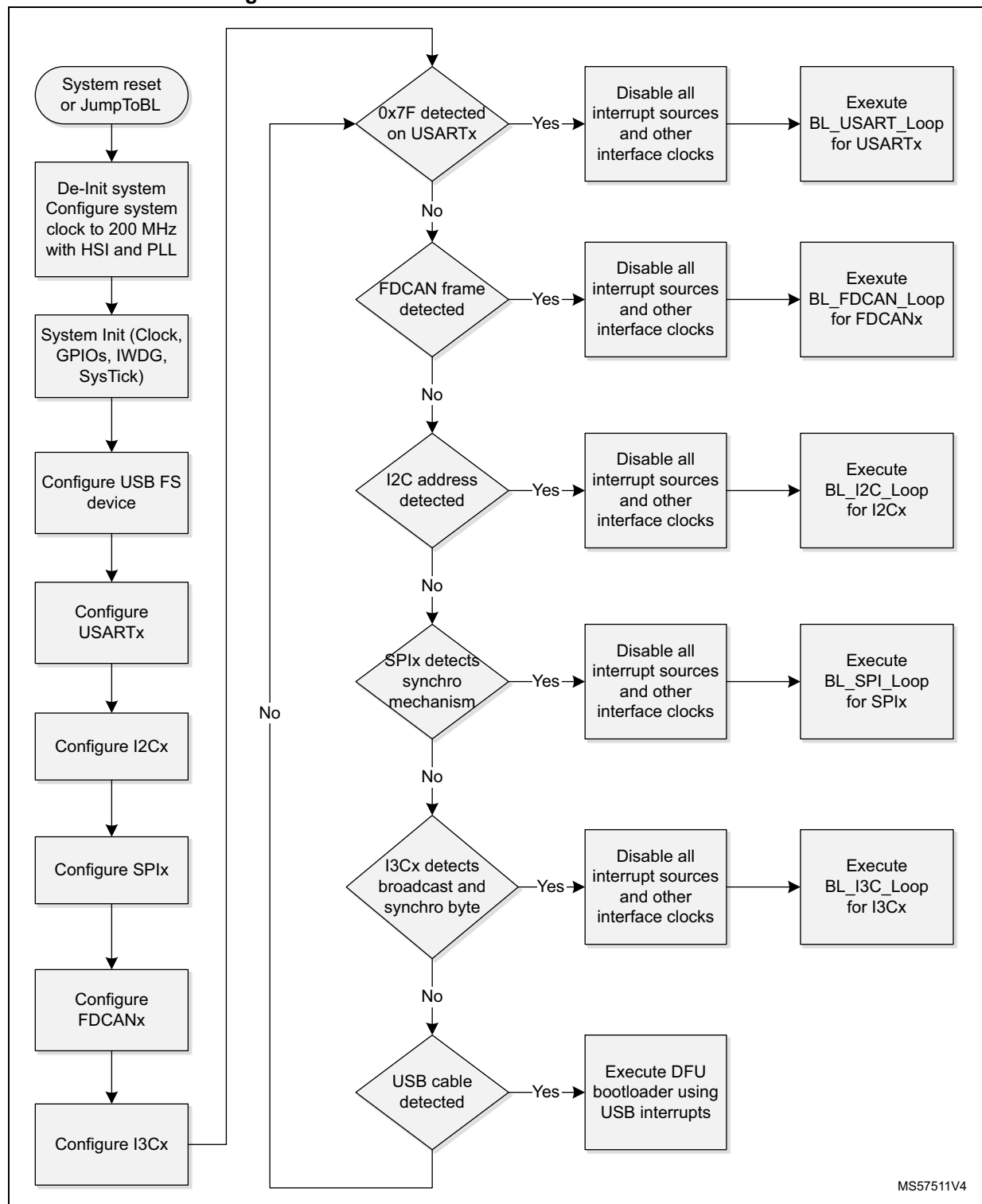
Special commands supported (USART/I2C/SPI/FDCAN/I3C) Opcode - 0x50							
Function	Sub-Opcode (2 bytes)	Number of data sent (2 bytes)	Data sent (MSB first)	Number of data received	Data received	Number of status data received (2 bytes)	Status data received
Change product state	0x01	0x4	Product state targeted Ex: 0x00000017	0x0	NA	0x1	0x0
Reset	0x02	0x4	0x0	0x0	NA	0x1	0x0

Note: USB special commands are slightly different from the other protocols as per the USB protocol specificities:

- No Opcode is used, Sub-Opcode is used directly
- Sub-Opcode is treated in a single byte and not two bytes
- Data are sent on USB frame by byte (LSB first). No need to add number of data to transmit
- Returned data and status are formatted on the USB native protocol

52.2 Bootloader selection

Figure 69. Bootloader V14 selection for STM32H503xx



52.3 Bootloader version

Table 114. STM32H503xx bootloader version

Version number	Description	Known limitations
V14.1	Initial bootloader version	Bootloader crash when jumping to it with (HiDe Protection Level = 3 + product state ≥ Provisioned)
V14.2	– Fix known limitations – Change BL system clock from 160 to 200 MHz	None

53 STM32H523xx/533xx devices

53.1 Bootloader configuration

The STM32H523xx/533xx bootloader is activated by applying Pattern 17 (described in [Table 2](#)). [Table 115](#) shows the hardware resources used by this bootloader.

The bootloader follows boot model V3_2 (see [Section 4.10](#)), so it inherits all its constraints.

Table 115. STM32H523xx/533xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 200 MHz (using PLL clocked by the HSI)
		-	CRS is enabled for the DFU to allow USB to be clocked by HSI48 48 MHz
		-	20 MHz derived from the PLLQ is used for FDCAN
	RAM	-	16 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	35 Kbytes, starting from address 0x0BF97000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Set as input until USART1 is detected.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Set as input until USART2 is detected.
USART3	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PD9 pin: USART3 in reception mode. Used in alternate push-pull, pull-down mode.
	USART3_TX pin	Output	PD8 pin: USART3 in transmission mode. Set as input until USART3 is detected.

Table 115. STM32H523xx/533xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1101110x, x = 0 for write and x = 1 for read
	I2C3_SCL pin	Input/output	PB8 pin: clock line is used in open-drain. pull up mode.
	I2C3_SDA pin		PB9 pin: data line is used in open-drain, pull up mode.
I2C4	I2C4	Enabled	The I2C4 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1101110x, x = 0 for write and x = 1 for read
	I2C4_SCL pin	Input/output	PA8 pin: clock line is used in open-drain. pull up mode.
	I2C4_SDA pin		PC9 pin: data line is used in open-drain, pull up mode.
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, no pull mode.
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, no pull mode.
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, no pull mode.
	SPI1_NSS pin		PA4 pin: slave chip select pin used in push-pull, no pull mode.

Table 115. STM32H523xx/533xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PC1 pin: slave data input line, used in push-pull, no pull mode in all packages except LQFP48, UFQFN48, and WLCSP39. PB15 pin: Slave data Input line, used in push-pull, no pull mode in LQFP48, UFQFN48, and WLCSP39 packages.
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, no pull mode.
	SPI2_SCK pin	Input	PB10 pin: slave clock line, used in push-pull, no pull mode.
	SPI2_NSS pin		PB12 pin: slave chip select pin used in push-pull, no pull mode.
SPI3	SPI3	Enabled	The SPI3 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz Polarity: CPOL low, CPHA low, NSS hardware.
	SPI3_MOSI pin	Input	PC12 pin: slave data input line, used in push-pull no pull mode
	SPI3_MISO pin	Output	PC11 pin: slave data output line, used in push-pull, no pull mode.
	SPI3_SCK pin	Input	PC10 pin: slave clock line, used in push-pull, no pull mode.
	SPI3_NSS pin		PA15 pin: slave chip select pin used in push-pull, no pull mode.
DFU	USB	Enabled	USB FS configured in forced device mode. USB FS interrupt vector is enabled and used for USB DFU communications.
	USB_DM pin	Input/output	PA11: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12: USB DP line. Used in alternate push-pull, no pull mode.- No external pull-up resistor is required

Table 115. STM32H523xx/533xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
FDCAN	FDCAN2	Enabled	Once initialized the FDCAN2 configuration is: – Connection bit rate 250 kbit/s – Data bit rate 1000 kbit/s – FrameFormat = FDCAN_FRAME_FD_BRS – Mode = FDCAN_MODE_NORMAL – AutoRetransmission = ENABLE – TransmitPause = DISABLE – ProtocolException = ENABLE
	FDCAN2_Rx pin	Input	PB5 pin: FDCAN2 in reception mode. Used in alternate push-pull, no pull mode.
	FDCAN2_Tx pin	Output	PB13 pin: FDCAN2 in transmission mode. Used in alternate push-pull, no pull mode.
I3C	I3C1	Enabled	– Mode: target mode – Aval timing: 0x4E – DMA Reg RX: disabled – DMA Reg TX: disabled – Status FIFO: disabled – DMA Req status: disabled – DMA Req control: disabled – IBI: enabled – Additional data after IBI ack-ed: 1 byte – IBI configuration: Mandatory Data Byte (MDB) – All IT disabled except RXFNE (Receive FIFO Interrupt). The RXFNE interruption is disabled after SYNC byte detection by the bootloader.
	I3C1_SCL pin	Input/output	PB6 pin: I3C1 in transmission mode. Used in alternate push-pull, no pull mode in all packages except LQFP48, UFQFN48, and WLCSP39. PB8 pin: I3C1 in transmission mode. Used in alternate push-pull, no pull mode in LQFP48, UFQFN48 and WLCSP39 packages..
	I3C1_SDA pin		PB7 pin: I3C1 in transmission mode. Used in alternate push-pull, no pull mode.

Table 116. STM32H523xx/533xx special commands

Special commands supported (USART/I2C/SPI/FDCAN/I3C) Opcode - 0x50							
Function	Sub-Opcode (2 bytes)	Number of data sent (2 bytes)	Data sent (MSB first)	Number of data received	Data received	Number of status data received (2 bytes)	Status data received
Change product state	0x01	0x4	Product state targeted Ex: 0x00000017	0x0	NA	0x1	0x0
Reset	0x02	0x4	0x0	0x0	NA	0x1	0x0
Data provisioning Only when BL is on HDPL = 1	0x83	0x4	RAM address where data to provision is written	– 0x0 if success – 0x1 if fail	– NA if success – Error code if fail	0x1	0x0

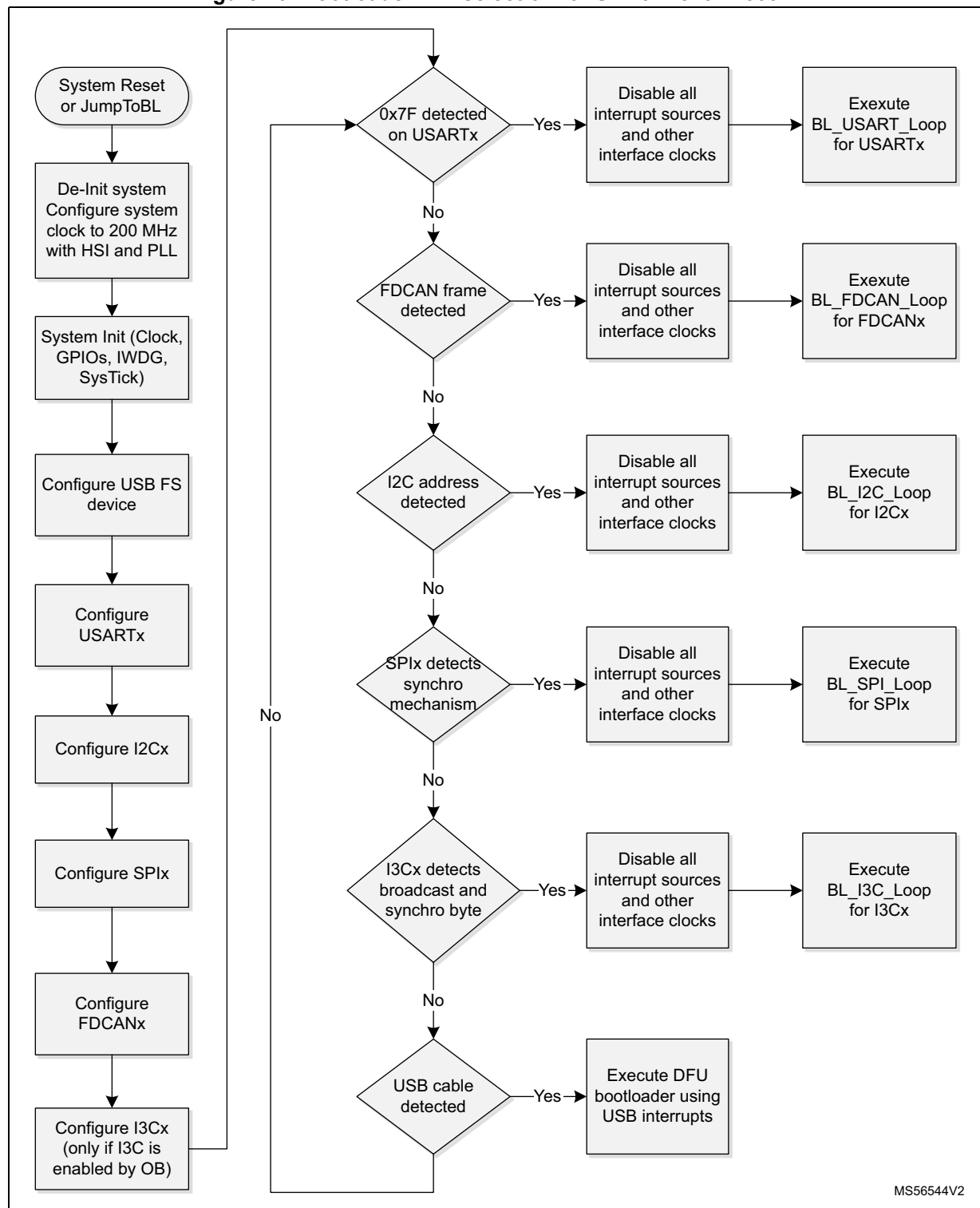
Note: USB special commands are slightly different from the other protocols as per the USB protocol specificities:

- No Opcode is used, Sub-Opcode is used directly
- Sub-Opcode is treated in a single byte and not two bytes
- Data are sent on USB frame by byte (LSB first). No need to add number of data to transmit
- Returned data and status is formatted on the USB native protocol

53.2 Bootloader selection

Figure 70 shows the bootloader selection mechanism.

Figure 70. Bootloader V14 selection for STM32H523xx/533xx



53.3 Bootloader version

Table 117. STM32H523xx/533xx bootloader version

Version number	Description	Known limitations
V14.0	Initial bootloader version	– PKG_ID wrongly detected when PKG_ID > 0xF – I2C/I3C not working on BL as wrong pinout applied
V14.2	Fix known limitations	None

54 STM32H562xx/563xx/573xx devices

54.1 Bootloader configuration

The STM32H562xx/563xx/573xx bootloader is activated by applying Pattern 17 (described in [Table 2](#)). [Table 118](#) shows the hardware resources used by this bootloader.

The bootloader follows boot model V3_2 (see [Section 4.10](#)), so it inherits all its constraints.

Table 118. STM32H562xx/563xx/573xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 200 MHz (using PLL clocked by the HSI)
		-	CRS is enabled for the DFU to allow USB to be clocked by HSI48 48 MHz
		-	20 MHz derived from the PLLQ is used for FDCAN
	RAM	-	16 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	35 Kbytes, starting from address 0x0BF97000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Set as input until USART1 is detected.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Set as input until USART2 is detected.
USART3	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PD9 pin: USART3 in reception mode. Used in alternate push-pull, pull-down mode.
	USART3_TX pin	Output	PD8 pin: USART3 in transmission mode. Set as input until USART3 is detected.

Table 118. STM32H562xx/563xx/573xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1100101x, x = 0 for write and x = 1 for read
	I2C3_SCL pin	Input/output	PA8 pin: clock line is used in open-drain, pull up mode.
	I2C3_SDA pin	Input/output	PC9 pin: data line is used in open-drain, pull up mode.
I2C4	I2C4	Enabled	The I2C4 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1100101x, x = 0 for write and x = 1 for read
	I2C4_SCL pin	Input/output	PD12 pin: clock line is used in open-drain, pull up mode.
	I2C4_SDA pin	Input/output	PD13 pin: data line is used in open-drain, pull up mode.
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, no pull mode.
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, no pull mode.
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, no pull mode.
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, no pull mode.
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PC1 pin: slave data input line, used in push, pull no pull mode
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, no pull mode.
	SPI2_SCK pin	Input	PB10 pin: slave clock line, used in push-pull, no pull mode.
	SPI2_NSS pin	Input	PB12 pin: slave chip select pin used in push-pull, no pull mode.

Table 118. STM32H562xx/563xx/573xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI3	SPI3	Enabled	The SPI3 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI3_MOSI pin	Input	PC12 pin: slave data input line, used in push-pull no pull mode
	SPI3_MISO pin	Output	PC11 pin: slave data output line, used in push-pull, no pull mode.
	SPI3_SCK pin	Input	PC10 pin: slave clock line, used in push-pull, no pull mode.
	SPI3_NSS pin	Input	PA15 pin: slave chip select pin used in push-pull, no pull mode.
DFU	USB	Enabled	USB FS configured in forced device mode. USB FS interrupt vector is enabled and used for USB DFU communications.
	USB_DM pin	Input/output	PA11: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12: USB DP line. Used in alternate push-pull, no pull mode. No external pull-up resistor is required
DFU	USB	Enabled	USB FS configured in forced device mode. USB FS interrupt vector is enabled and used for USB DFU communications. Note: VDDUSB IO must be connected to 3.3 V as USB peripheral is used by the bootloader.
	USB_DM pin	Input/output	PA11: USB DM line. Used in input no pull mode.
	USB_DP pin		PA12: USB DP line. Used in input no pull mode. No external pull-up resistor is required
FDCAN	FDCAN2	Enabled	Once initialized the FDCAN2 configuration is: – Connection bit rate 250 kbit/s – Data bit rate 1000 kbit/s – FrameFormat = FDCAN_FRAME_FD_BRS – Mode = FDCAN_MODE_NORMAL AutoRetransmission = ENABLE – TransmitPause = DISABLE – ProtocolException = ENABLE
	FDCAN2_Rx pin	Input	PB5 pin: FDCAN2 in reception mode. Used in alternate push-pull, no pull mode.
	FDCAN2_Tx pin	Output	PB13 pin: FDCAN2 in transmission mode. Used in alternate push-pull, no pull mode.

Table 118. STM32H562xx/563xx/573xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I3C	I3C	Enabled	– Mode: target mode – Aval timing: 0x4E – DMA Reg RX: disabled – DMA Reg TX: disabled – Status FIFO: disabled – DMA Req status: disabled – DMA Req control: disabled – IBI: enabled – Additional data after IBI ack-ed: 1 byte – IBI configuration: Mandatory Data Byte (MDB) – All IT disabled except RXFNE (Receive FIFO Interrupt) The RXFNE interruption is disabled after SYNC byte detection by the bootloader.
	I3C1_SCL pin	Input/output	PB6 pin: I3C1 in transmission mode. Used in alternate push-pull, no pull mode.
	I3C1_SDA pin		PB7 pin: I3C1 in transmission mode. Used in alternate push-pull, no pull mode.

Table 119. STM32H562xx/563xx/573xx special commands

Special commands supported (USART/I2C/SPI/FDCAN/I3C) Opcode - 0x50							
Function	Sub-Opcode (2 bytes)	Number of data sent (2 bytes)	Data sent (MSB first)	Number of data received	Data received	Number of status data received (2 bytes)	Status data received
Change product state	0x01	0x4	Product state targeted Ex: 0x00000017	0x0	NA	0x1	0x0
Reset	0x02	0x4	0x0	0x0	NA	0x1	0x0
Data provisioning Only when BL is on HDPL = 1	0x83	0x4	RAM address where data to provision is written	– 0x0 if success – 0x1 if fail	– NA if success – Error code if fail	0x1	0x0

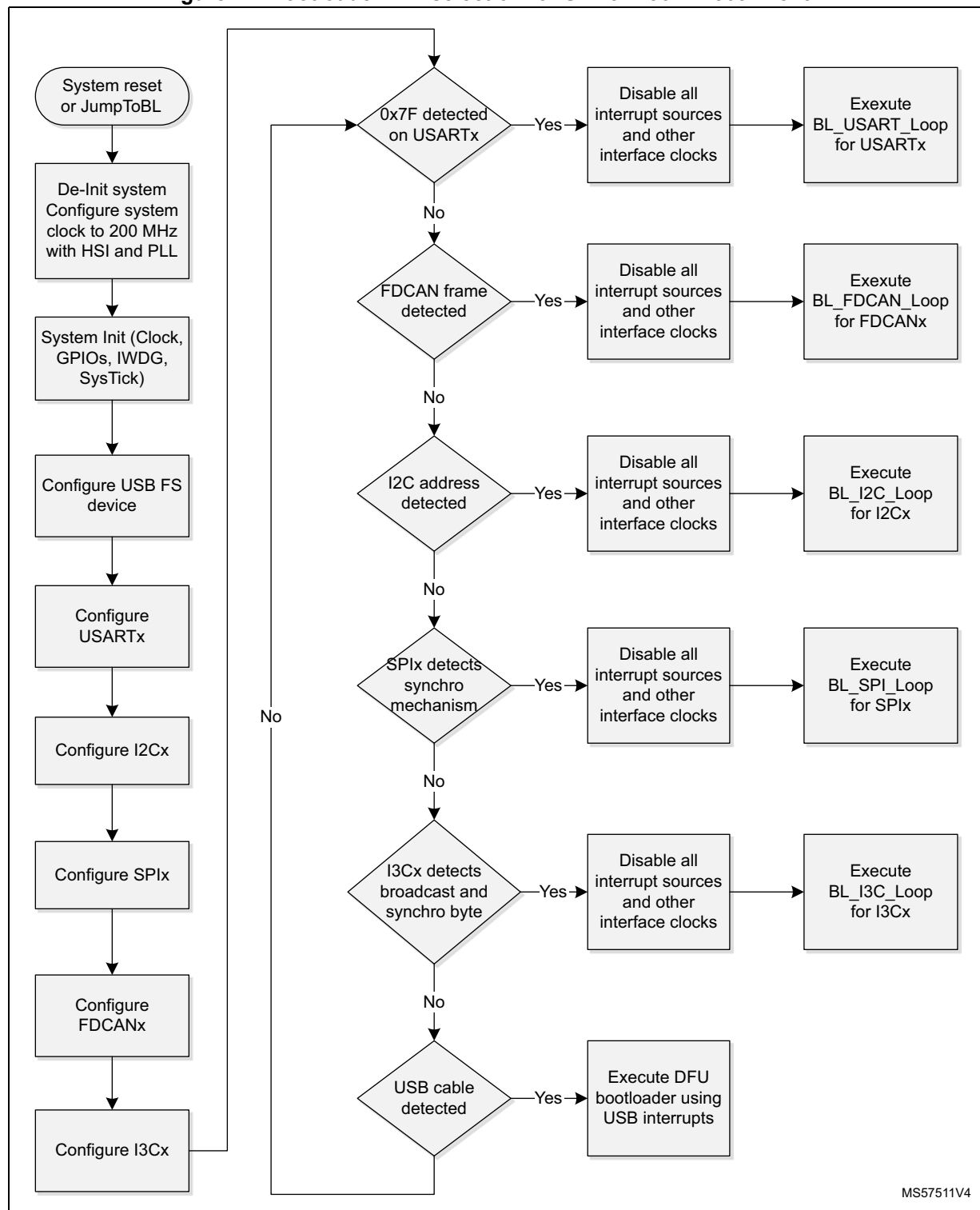
Note: USB special commands are slightly different from the other protocols as per the USB protocol specificities:

- No Opcode is used, Sub-Opcode is used directly
- Sub-Opcode is treated in a single byte and not two bytes
- Data are sent on USB frame by byte (LSB first). No need to add number of data to transmit
- Returned data and status is formatted on the USB native protocol

54.2 Bootloader selection

Figure 71 shows the bootloader selection mechanism.

Figure 71. Bootloader V14 selection for STM32H562xx/563xx/573xx



MS57511V4

54.3 Bootloader version

Table 120. STM32H562xx/563xx/573xx bootloader version

Version number	Description	Known limitations
V14.5	Fix known limitations ⁽¹⁾	None
V14.4	– Fix known limitations – Change BL system clock from 160 to 200 MHz	EEPROM sector erase not working on 1 Mbyte devices.
V14.3	Initial bootloader version	Bootloader crash when jumping to it with the following condition (TrustZone [®] enabled + HiDe Protection = 3 + Product state ≥ Provisioned)

1. Only on 1 Mbytes devices.

A standalone EraseEEPROM function is added on the system memory at address 0x0BF9 F500. When an erase sector is needed:

- Write at RAM address 0x2000 4000 (LSB to MSB)
 - Byte0: number of sectors to erase (N)
 - Byte1 to N (every byte contains the sector number, that is, 0 to 7 for Bank1, 8 to 15 for Bank2)
 - Example: to erase sector 3, 4, and 13, write 0x0303040D at address 0x20004000.
- After the erase, go back to the bootloader.
- To continue using the bootloader, a reconnect is needed.

55 STM32H72xxx/73xxx devices

55.1 Bootloader configuration

The STM32H72xxx/73xxx bootloader is activated by applying Pattern 10 (described in [Table 2](#)). [Table 121](#) shows the hardware resources used by this bootloader.

Table 121. STM32H72xxx/73xxx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 66 MHz (using PLL clocked by the HSI)
		-	CRS is enabled for the DFU to allow USB to be clocked by HSI48 48 MHz
		-	20 MHz derived from the PLLQ is used for FDCAN
	RAM	-	16 Kbytes, starting from address 0x24000000, are used by the bootloader firmware
	System memory	-	128 Kbytes, starting from address 0x1FF00000 contain the bootloader firmware. The bootloader start address is 0x1FF09800.
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
	Power	-	Voltage is set to Voltage Range 3. Bootloader SW is writing to the PWR_CR3 register using 4 bytes, locking this register. Only Power off/on unlocks it. This is fixed on the BL with 0x93 version.
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode. Set as input until USART1 is detected on the BL version 0x93.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode. Set as input until USART2 is detected on the BL version 0x93.

Table 121. STM32H72xxx/73xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
USART3 (on PB10/PB11)	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PB11 pin: USART3 in reception mode. Used in alternate push-pull, pull-down mode.
	USART3_TX pin	Output	PB10 pin: USART3 in transmission mode. Used in alternate push-pull, pull-down mode. Set as input until USART3 is detected on the BL version 0x93.
USART3 (on PD8/PD9)	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PD9 pin: USART3 in reception mode. Used in alternate push-pull, pull-down mode.
	USART3_TX pin	Output	PD8 pin: USART3 in transmission mode. Used in alternate push-pull, pull-down mode. Set as input until USART3 is detected on the BL version 0x93.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1011100x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin		PB9 pin: data line is used in open-drain no pull mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1011100x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PF1 pin: clock line is used in open-drain no pull mode.
	I2C2_SDA pin		PF0 pin: data line is used in open-drain no pull mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1011100x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PA8 pin: clock line is used in open-drain no pull mode.
	I2C3_SDA pin		PC9 pin: data line is used in open-drain no pull mode.

Table 121. STM32H72xxx/73xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, no pull mode.
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, no pull mode.
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, no pull mode.
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, no pull mode.
SPI3	SPI3	Enabled	The SPI3 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI3_MOSI pin	Input	PC12 pin: slave data input line, used in push-pull no pull mode
	SPI3_MISO pin	Output	PC11 pin: slave data output line, used in push-pull, no pull mode.
	SPI3_SCK pin	Input	PC10 pin: slave clock line, used in push-pull, no pull mode.
	SPI3_NSS pin	Input	PA15 pin: slave chip select pin used in push-pull, no pull mode.
SPI4	SPI4	Enabled	The SPI4 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI4_MOSI pin	Input	PE14 pin: slave data input line, used in push-pull, pull-down mode
	SPI4_MISO pin	Output	PE13 pin: slave data output line, used in push-pull, pull-down mode.
	SPI4_SCK pin	Input	PE12 pin: slave clock line, used in push-pull, pull-dpwn mode.
	SPI4_NSS pin	Input	PE11 pin: slave chip select pin used in push-pull, pull-up mode. Note: This IO can be tied to GND if the SPI master does not use it.

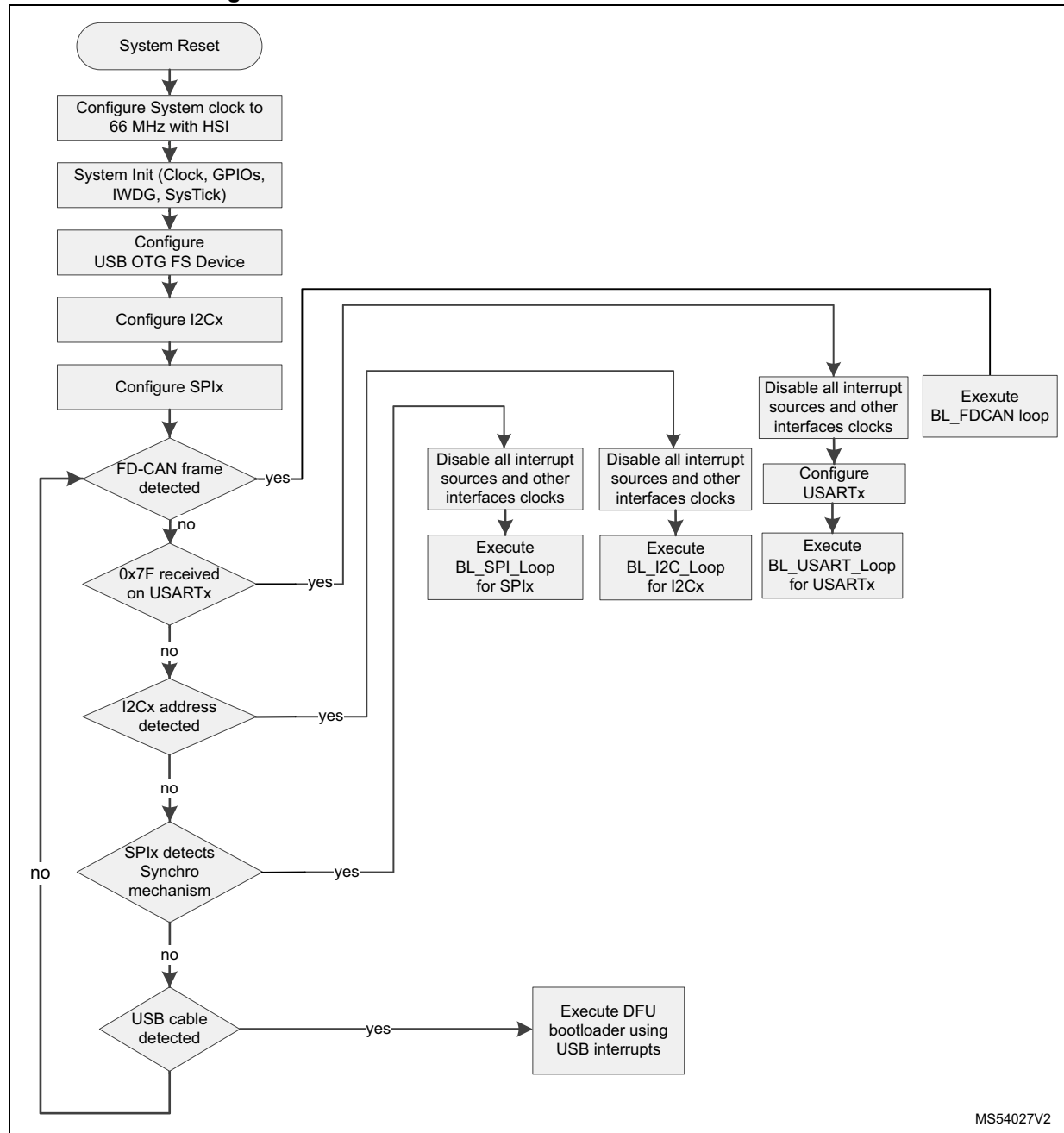
Table 121. STM32H72xxx/73xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
DFU	USB	Enabled	USB FS configured in forced device mode. USB FS interrupt vector is enabled and used for USB DFU communications.
	USB_DM pin	Input/output	PA11: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12: USB DP line. Used in alternate push-pull, no pull mode. No external pull-up resistor is required
FDCAN (on PH13/PH14)	FDCAN1	Enabled	Once initialized the FDCAN1 configuration is: Connection bit rate 250 kbit/s Data bit rate 1000 kbit/s FrameFormat = FDCAN_FRAME_FD_BRS Mode = FDCAN_MODE_NORMAL AutoRetransmission = ENABLE TransmitPause = DISABLE ProtocolException = ENABLE
	FDCAN1_Rx pin	Input	PH14 pin: FDCAN1 in reception mode. Used in alternate push-pull, pull-down mode.
	FDCAN1_Tx pin	Output	PH13 pin: FDCAN1 in transmission mode. Used in alternate push-pull, pull-down mode.
FDCAN (on PD1/PD0)	FDCAN1	Enabled	Once initialized the FDCAN1 configuration is: Connection bit rate 250 kbit/s Data bit rate 1000 kbit/s FrameFormat = FDCAN_FRAME_FD_BRS Mode = FDCAN_MODE_NORMAL AutoRetransmission = ENABLE TransmitPause = DISABLE ProtocolException = ENABLE
	FDCAN1_Rx pin	Input	PD0 pin: FDCAN1 in reception mode. Used in alternate push-pull, pull-down mode.
	FDCAN1_Tx pin	Output	PD1 pin: FDCAN1 in transmission mode. Used in alternate push-pull, pull-down mode.

55.2 Bootloader selection

Figure 72 shows the bootloader selection mechanism.

Figure 72. Bootloader V9.0 selection for STM32H72xxx/73xxx



55.3 Bootloader version

[Table 122](#) lists the STM32H72xxx/73xxx devices bootloader versions.

Table 122. STM32H72xxx/73xxx bootloader version

Version number	Description	Known limitations
V9.1	Initial bootloader version	– TCM_AXI OB cannot be modified using all BL interfaces – String returned describing the memory size when using USB is wrong
V9.2	Fix all issues of previous release	– Crash loop when booting on the BL, setting RDP to Level1, doing a reset or power on/off and the USB cable is plugged. – BL is not working in RDP Level1 when TCM_AXI_SHARED option byte is not “0”. Value of this OB must be set to “0” before going to RDP L1. – Bootloader SW is writing to the PWR_CR3 register using 4 bytes, which is locking this register. Only Power off/on will unlock it.
V9.3	– Fix all issues of previous release. – Modify USART TX from push pull mode in the previous versions to input.	None

56 STM32H74xxx/75xxx devices

56.1 Bootloader configuration

The STM32H74xxx/75xxx bootloader is activated by applying Pattern 10 (described in [Table 2](#)). [Table 123](#) shows the hardware resources used by this bootloader.

Table 123. STM32H74xxx/75xxx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 64 MHz using the HSI. The HSI clock source is used at startup (interface detection phase) and when USART or SPI or I2C interface is selected.
		-	CRS is enabled for the DFU to allow USB to be clocked by HSI48 48 MHz
		-	Clock used for the FDCAN is fixed to 20 MHz and is derived from PLLQ
	RAM	-	16 Kbytes, starting from address 0x20000000, and 208 Kbytes (reduced to 20 Kbytes in V9.1 version) starting from address 0x24000000, are used by the bootloader firmware
	System memory	-	122 Kbytes, starting from address 0x1FF00000 contain the bootloader firmware. The bootloader start address is 0x1FF09800.
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
	Power	-	Voltage is set to Range 3. Bootloader software writes to the PWR_CR3 register using 4 bytes, which locks this register. Only Power off/on unlocks it. This is fixed on the bootloader with 0x91 version.
USART1 (on PA9/PA10)	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode. Set as input until USART1 is detected on the bootloader version 0x91.
USART1 (on PB14/PB15)	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PB15 pin: USART1 in reception mode.Used in input pull-up mode.
	USART1_TX pin	Output	PB14 pin: USART1 in transmission mode. Used in alternate function push pull pull-up mode.

Table 123. STM32H74xxx/75xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate push-pull, no pull mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode. Set as input until USART3 is detected on the bootloader version 0x91.
USART3	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PB11 pin: USART3 in reception mode. Used in alternate push-pull, pull-up mode.
	USART3_TX pin	Output	PB10 pin: USART3 in transmission mode. Used in alternate push-pull, pull-up mode. Set as input until USART3 is detected on the bootloader version 0x91.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz, 7-bit address, Target mode, Analog filter ON – Target 7-bit address: 0b1001110x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin	Input/output	PB9 pin: data line is used in open-drain no pull mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 400 kHz, 7-bit address, Target mode, Analog filter ON – Target 7-bit address: 0b1001110x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PF1 pin: clock line is used in open-drain no pull mode.
	I2C2_SDA pin	Input/output	PF0 pin: data line is used in open-drain no pull mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 400 kHz, 7-bit address, Target mode, Analog filter ON – Target 7-bit address: 0b1001110x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PA8 pin: clock line is used in open-drain no pull mode.
	I2C3_SDA pin	Input/output	PC9 pin: data line is used in open-drain no pull mode.

Table 123. STM32H74xxx/75xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, no pull mode.
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, no pull mode.
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, no pull mode.
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, no pull mode.
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PI3 pin: slave data input line, used in push-pull, no pull mode.
	SPI2_MISO pin	Output	PI2 pin: slave data output line, used in push-pull, no pull mode.
	SPI2_SCK pin	Input	PI1 pin: slave clock line, used in push-pull, no pull mode.
	SPI2_NSS pin	Input	PI0 pin: slave chip select pin used in push-pull, no pull mode.
SPI3	SPI3	Enabled	The SPI3 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI3_MOSI pin	Input	PC12 pin: slave data input line, used in push-pull, no pull mode.
	SPI3_MISO pin	Output	PC11 pin: slave data output line, used in push-pull, no pull mode.
	SPI3_SCK pin	Input	PC10 pin: slave clock line, used in push-pull, no pull mode.
	SPI3_NSS pin	Input	PA15 pin: slave chip select pin used in push-pull, no pull mode.

Table 123. STM32H74xxx/75xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI4	SPI4	Enabled	The SPI4 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI4_MOSI pin	Input	PE14 pin: slave data input line, used in push-pull, no pull mode.
	SPI4_MISO pin	Output	PE13 pin: slave data output line, used in push-pull, no pull mode.
	SPI4_SCK pin	Input	PE12 pin: slave clock line, used in push-pull, no pull mode.
	SPI4_NSS pin	Input	PE11 pin: slave chip select pin used in push-pull, no pull mode.
DFU	USB	Enabled	USB FS configured in forced device mode. USB FS interrupt vector is enabled and used for USB DFU communications.
	USB_DM pin	Input/output	PA11: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12: USB DP line. Used in alternate push-pull, no pull mode. No external pull-up resistor is required
FDCAN	FDCAN1	Enabled	Once initialized the FDCAN1 configuration is: – Connection bit rate 250 kbit/s – Data bit rate 1000 kbit/s – FrameFormat = FDCAN_FRAME_FD_BRS – Mode = FDCAN_MODE_NORMAL AutoRetransmission = ENABLE – TransmitPause = DISABLE – ProtocolException = ENABLE
	FDCAN1_Rx pin	Input	PH14 pin: FDCAN1 in reception mode. Used in alternate push-pull, pull-up mode.
	FDCAN1_Tx pin	Output	PH13 pin: FDCAN1 in transmission mode. Used in alternate push-pull, pull-up mode.

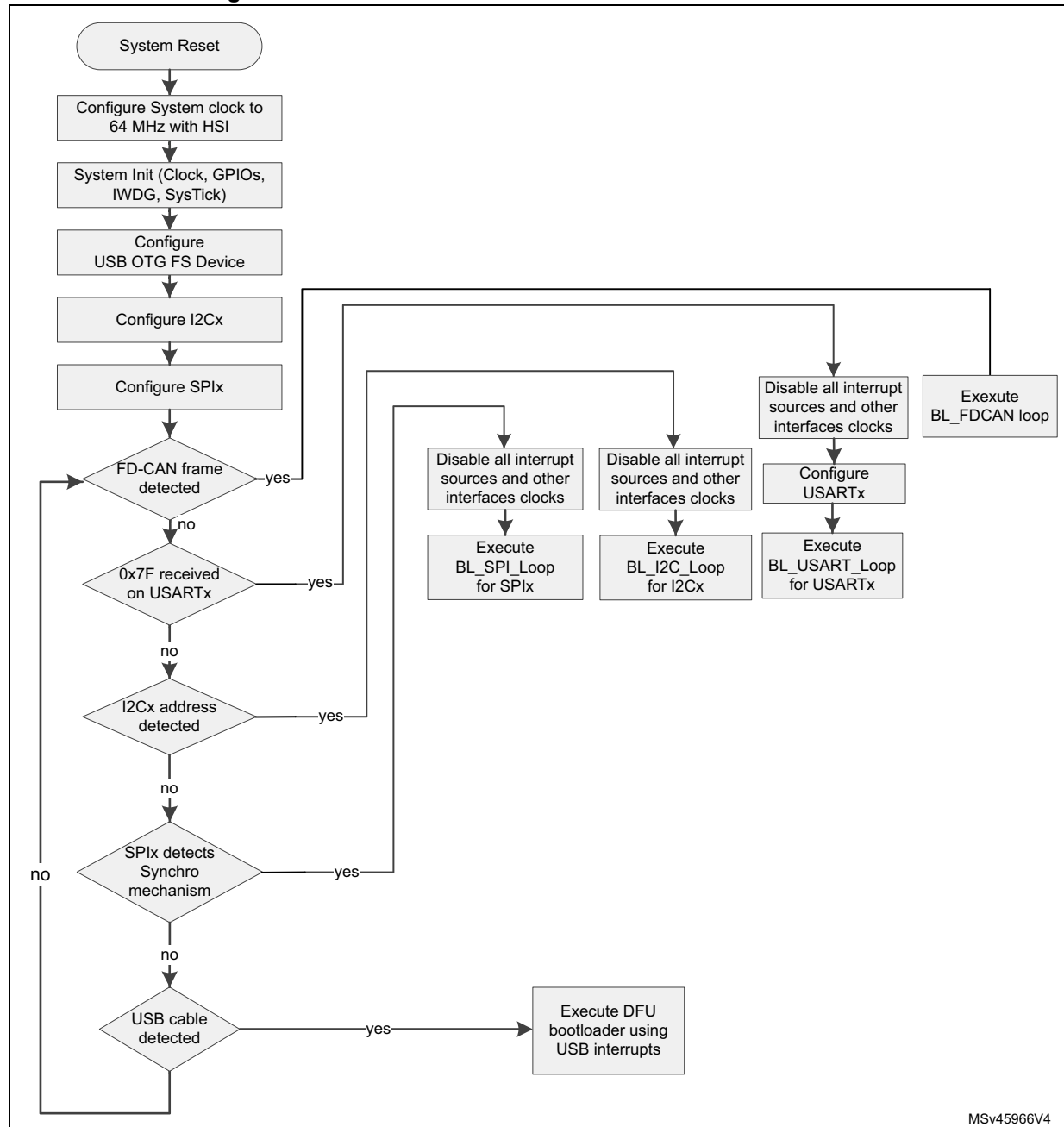
Note: To connect to the bootloader USART1 using PB14/PB15 pins, user must send two synchronization bytes. Baudrate is limited to 115200.

DFU mode does not support USBREGEN mode. If STM32 is powered by an 1.8 V source, it is not possible to use the BL DFU unless 3.3 V is provided

56.2 Bootloader selection

Figure 73 shows the bootloader selection mechanism.

Figure 73. Bootloader V9.x selection for STM32H74xxx/75xxx



56.3 Bootloader version

Table 124 lists the STM32H74xxx/75xxx devices bootloader versions.

Table 124. STM32H74xxx/75xxx bootloader version

Version number	Description	Known limitations
V13.2 (0xD2)	Initial bootloader version	– Go command is not working – USART2 connection is not working – SPI1 connection is not working – Mass erase does not work correctly on I2C (only Bank2 is erased in this command)
V13.3 (0xD3)	– Switch USB clock input from HSE to HSI48 with CRS – Fix known limitations on the V13.2	– Bank erase is not working on USART/SPI and I2C – DFU mass-erase not working
V9.0 (0x90)	– Add support of FDCAN interface – Fix V13.3 limitations – V9.0 is the latest version in production and replaces V13.2 and V13.3	– First ACK not received on Go command when using USART or SPI – On the FDCAN write memory, write of data with length > 63 bytes fails – If PB15 is set to GND, user cannot connect to BL interfaces. Only the USB is able to connect as it uses interrupt for detection. PB15 must not be pulled down if USART1 on PB14/PB15 is not used – Jump issue on some application. Application stack pointer must be lower than (RAM end @ - 16 bytes) to guarantee it is working – Additional reset needed after power off/on to enable connection to the BL interfaces. As a workaround user can add a pull up on PA11 pin.' – Cannot program the "CM4_BOOT_ADDx" option byte using BL in dual core case – FDCAN Get version command is giving a bad FDCAN protocol version (0x11). It must be 0 x10 (V1.0) – SRAM1/SRAM2/SRAM3 (0x30000000-0x30047FFF) and ITCM memories not accessible by the BL – Number of supported commands is wrong (13 instead of 11)

Table 124. STM32H74xxx/75xxx bootloader version (continued)

Version number	Description	Known limitations
V9.1 (0x91)	Fix V9.0 limitations – Fix the configuration of PWR control register CR3. Bootloader is no more blocking the change of PWR source – Adjust USB, I²C erase and program timings and fix them – Fix FDCAN version from V1.0 to V1.1 – Fix write issue when using FDCAN – Fix missing PCROP disable in RDP level1 regression – Update option byte support to handle all possible use cases	If PB15 is set to GND, user cannot connect to BL interfaces SRAM1/SRAM2/SRAM3 (0x30000000 to 0x30047FFF) and ITCM memories not accessible by the BL Jump issue on some applications. Application stack pointer must be lower than (RAM end address - 16 bytes) to guarantee it is working Erase on bank2 not working as expected – Root cause: check bad busy value while erasing on bank2. – Behavior: when erase on bank2 is requested, SW exits while the operation is ongoing. Sending a new command invoking the flash memory after the erase can hang the system. – Workaround: worst case erase timing from the product datasheet can be added after the erase command to allow the FLITF to complete the erase. A safer method is to use a RAM patch for the erase command. Same data are output on the USART1_TX PB14 when using USART1 on PA10/PA9. – Root cause: both USART1 TX pins PA9 and PB14 configured as alternate push pull-up and PB15/PB14 pins not disabled when PA10/PA9 set used by the customer. – Behavior: same data are output on USART1 TX pins PA9 and PB14. – Caution: take care to what PB14 is connected on your design to not damage it when using USART1 on PA10/PA9. Same data are output on the USART1_TX PA9 when using USART1 on PB15/PB14. – Root cause: both USART1 TX pins PB14 and PA9 configured as alternate push pull-up and PA10/PA9 pins not disabled when PB15/PB14 set used by the customer. – Behavior: same data are output on USART1 TX pins PA9 and PB14. – Caution: take care to what PA9 is connected on your design to not damage it when using USART1 on PB15/14.

Table 124. STM32H74xxx/75xxx bootloader version (continued)

Version number	Description	Known limitations
V9.2 (0x92)	Fix V9.1 limitations: – Same data on PA9 when USART1 is used on PB15/PB14. – Same data on PB14 when USART1 is used on PA10/PA9. – Erase on bank2 not working as expected. Enhancements: – Erase timeouts increased	– If PB15 is set to GND, user cannot connect to BL interfaces. – SRAM1/SRAM2/SRAM3 (0x30000000 to 0x30047FFF) and ITCM memories not accessible by the BL. – Jump issue on some applications. Application stack pointer must be lower than (RAM end address - 16 bytes) to guarantee it is working.

57 STM32H7A3xx/7B3xx/7B0xx devices

57.1 Bootloader configuration

The STM32H7A3xx/7B3xx/7B0xx bootloader is activated by applying Pattern 10 (described in [Table 2](#)). [Table 125](#) shows the hardware resources used by this bootloader.

Table 125. STM32H7A3xx/7B3xx/7B0xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 64 MHz using the HSI.
		-	CRS is enabled for the DFU to allow USB to be clocked by HSI48 48 MHz
		-	Clock used for the FDCAN is fixed to 20 MHz and is derived from PLLQ
	RAM	-	16 Kbytes, starting from address 0x24000000, are used by the bootloader firmware
	System memory	-	128 Kbytes, starting from address 0x1FF00000 contain the bootloader firmware. The bootloader start address is 0x1FF0A000
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
	Power	-	Bootloader software writes to PWR_CR3 register using four bytes, which locks it, only Power off/on unlocks it. Fixed on the bootloader with 0x92 versions.
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode. Set as input until USART1 is detected on the bootloader version 0x92.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode. Set as input until USART2 is detected on the bootloader version 0x92.

Table 125. STM32H7A3xx/7B3xx/7B0xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
USART3 on (PB10/PB11)	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PB11 pin: USART3 in reception mode. Used in alternate push-pull, pull-down mode.
	USART3_TX pin	Output	PB10 pin: USART3 in transmission mode. Used in alternate push-pull, pull-down mode. Set as input until USART3 is detected on the bootloader version 0x92.
USART3 on (PD8/PD9)	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PD9 pin: USART3 in reception mode. Used in alternate push-pull, pull-down mode.
	USART3_TX pin	Output	PD8 pin: USART3 in transmission mode. Used in alternate push-pull, pull-down mode. Set as input until USART3 is detected on the bootloader version 0x92.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz, 7-bit address, Target mode, Analog filter ON – Target 7-bit address: 0b10101111x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin	Input/output	PB9 pin: data line is used in open-drain no pull mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 400 kHz, 7-bit address, Target mode, Analog filter ON – Target 7-bit address: 0b10101111x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PF1 pin: clock line is used in open-drain no pull mode.
	I2C2_SDA pin	Input/output	PF0 pin: data line is used in open-drain no pull mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 400 kHz, 7-bit address, Target mode, Analog filter ON – Target 7-bit address: 0b10101111x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PA8 pin: clock line is used in open-drain no pull mode.
	I2C3_SDA pin	Input/output	PC9 pin: data line is used in open-drain no pull mode.

Table 125. STM32H7A3xx/7B3xx/7B0xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, no pull-up no pull-down mode.
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, no pull-up no pull-down mode.
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull no pull-up, no pull-up no pull-down mode.
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode.
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PI3 pin: slave data input line, used in push-pull, no pull mode.
	SPI2_MISO pin	Output	PI2 pin: slave data output line, used in push-pull, no pull mode.
	SPI2_SCK pin	Input	PI1 pin: slave clock line, used in push-pull, no pull mode.
	SPI2_NSS pin	Input	PI0 pin: slave chip select pin used in push-pull, no pull mode.
SPI3	SPI3	Enabled	The SPI3 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI3_MOSI pin	Input	PC12 pin: slave data input line, used in push-pull, no pull mode
	SPI3_MISO pin	Output	PC11 pin: slave data output line, used in push-pull, no pull mode.
	SPI3_SCK pin	Input	PC10 pin: slave clock line, used in push-pull, no pull mode.
	SPI3_NSS pin	Input	PA15 pin: slave chip select pin used in push-pull, no pull mode.

Table 125. STM32H7A3xx/7B3xx/7B0xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI4	SPI4	Enabled	The SPI4 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI4_MOSI pin	Input	PE14 pin: slave data input line, used in push-pull, no pull up, no pull down mode
	SPI4_MISO pin	Output	PE13 pin: slave data output line, used in push-pull, no pull up, no pull down mode.
	SPI4_SCK pin	Input	PE12 pin: slave clock line, used in push-pull, no pull up, no pull down mode.
	SPI4_NSS pin	Input	PE11 pin: slave chip select pin used in push-pull, no pull up, no pull down mode.
DFU	USB	Enabled	USB FS configured in forced device mode. USB FS interrupt vector is enabled and used for USB DFU communications.
	USB_DM pin	Input/output	PA11: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12: USB DP line. Used in alternate push-pull, no pull mode. No external pull-up resistor is required.
FDCAN on (PH13/PH14)	FDCAN1	Enabled	Once initialized the FDCAN1 configuration is: Connection bit rate 250 kbit/s Data bit rate 1000 kbit/s FrameFormat = FDCAN_FRAME_FD_BRS Mode = FDCAN_MODE_NORMAL AutoRetransmission = ENABLE TransmitPause = DISABLE ProtocolException = ENABLE
	FDCAN1_Rx pin	Input	PH14 pin: FDCAN1 in reception mode. Used in alternate push-pull, pull-down mode.
	FDCAN1_Tx pin	Output	PH13 pin: FDCAN1 in transmission mode. Used in alternate push-pull, pull-down mode.

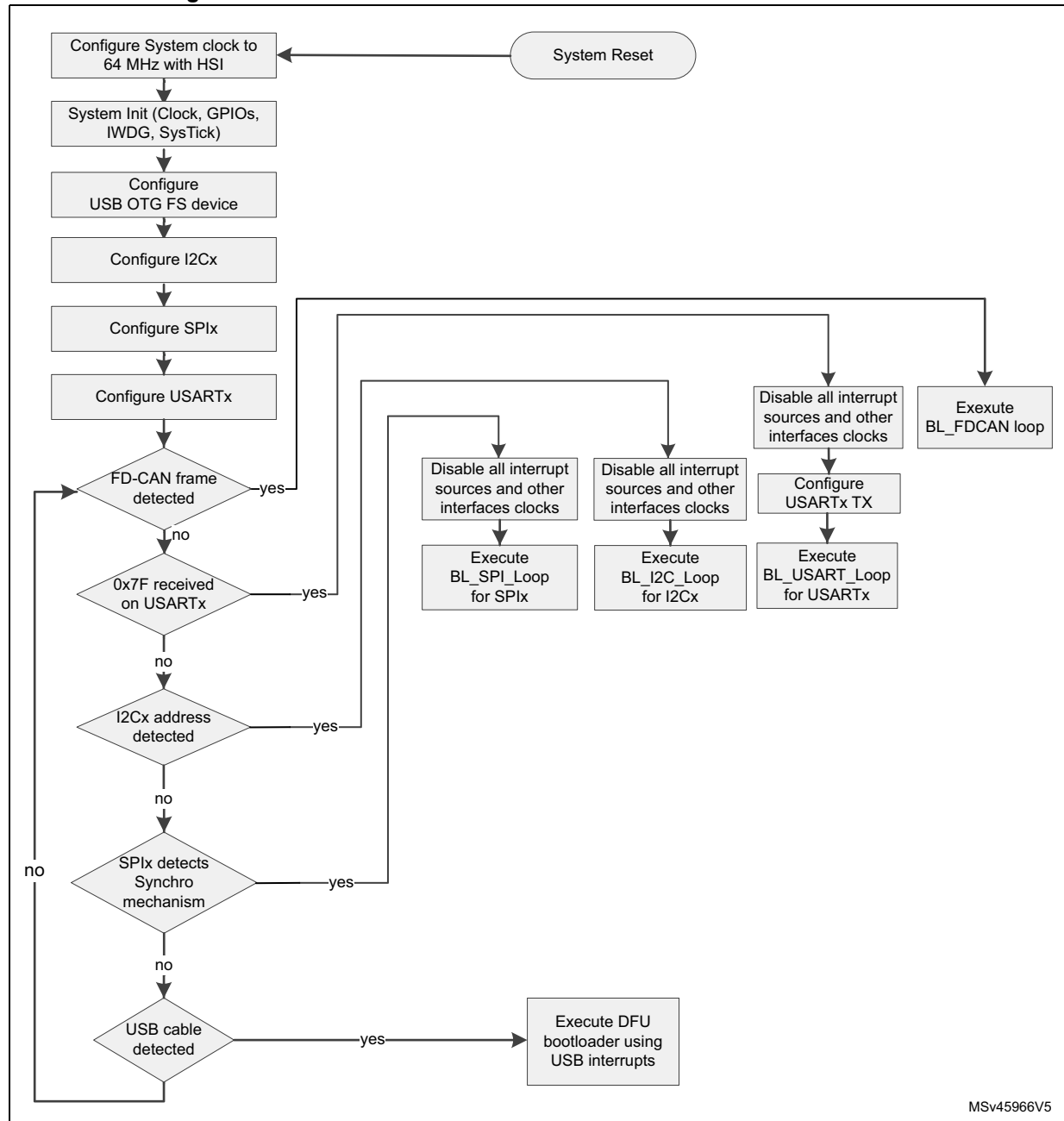
Table 125. STM32H7A3xx/7B3xx/7B0xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
FDCAN on (PD1/PD0)	FDCAN1	Enabled	Once initialized the FDCAN1 configuration is: Connection bit rate 250 kbit/s Data bit rate 1000 kbit/s FrameFormat = FDCAN_FRAME_FD_BRS Mode = FDCAN_MODE_NORMAL AutoRetransmission = ENABLE TransmitPause = DISABLE ProtocolException = ENABLE
	FDCAN1_Rx pin	Input	PD0 pin: FDCAN1 in reception mode. Used in alternate push-pull, pull-down mode.
	FDCAN1_Tx pin	Output	PD1 pin: FDCAN1 in transmission mode. Used in alternate push-pull, pull-down mode.

57.2 Bootloader selection

Figure 73 shows the bootloader selection mechanism.

Figure 74. Bootloader V9.x selection for STM32H7A3xx/7B3xx/7B0xx



57.3 Bootloader version

[Table 126](#) lists the STM32H7A3xx/7B3xx/7B0xx devices bootloader versions.

Table 126. STM32H7A3xx/7B3xx/7B0xx bootloader version

Version number	Description	Known limitations
V9.0	Initial bootloader version	– String returned describing the flash memory size when using USB is wrong (expected value 256 x 8 KB, but returns 256 x 2 KB) – OTP memory is not supported by the bootloader
V9.1	Fixes all issues of previous release.	– Crash loop when booting on the bootloader, setting RDP to Level1, doing a reset or power on/off and the USB cable is plugged. – Bootloader software is writing to the PWR_CR3 register using four bytes, which is locking this register. Only Power off/on unlocks it
V9.2	– Fix all issues of previous release – Modify USART TX from push pull mode in the previous versions to input.	None

58 STM32H7Rxxx/7Sxxx devices

58.1 Bootloader configuration

The STM32H7Rxxx/7Sxxx bootloader is activated by applying Pattern 17 (described in [Table 2](#)). [Table 127](#) shows the hardware resources used by this bootloader.

Table 127. STM32H7Rxxx/7Sxxx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 380 MHz using the PLL clocked by HSI
		-	20 MHz derived from the PLLQ is used for FDCAN.
		HSI48 enabled	CRS is enabled for the DFU to allow USB to be clocked by HSI48 48 MHz.
	RAM	-	16 Kbytes, starting from address 0x24020000, are used by the bootloader firmware
	System memory	-	32 Kbytes, starting from address 0x1FF18000 contain the bootloader firmware.
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode. Set as input until USART1 is detected on the bootloader version 0x91.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate push-pull, no pull mode. Used in alternate push-pull, no pull mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in alternate push-pull, no pull mode. Set as input until USART3 is detected on the bootloader version 0x91.
USART3	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PD9 pin: USART3 in reception mode. Used in alternate push-pull, no pull mode.
	USART3_TX pin	Output	PD8 pin: USART3 in transmission mode. Set as input until USART3.

Table 127. STM32H7Rxxx/7Sxxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
UART4	UART4	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	UART4_RX pin	Input	PD0 pin: UART4 in reception mode. Used in alternate push-pull, no pull mode.
	UART4_TX pin	Output	PD1 pin: UART4 in transmission mode. Set as input until UART4 is detected.
I2C1 ⁽¹⁾	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1100001x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB8 pin: clock line is used in open-drain pull-up mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain pull-up mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1100001x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PB10 pin: clock line is used in open-drain pull-up mode.
	I2C2_SDA pin		PB11 pin: data line is used in open-drain pull-up mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1100001x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PA8 pin: clock line is used in open-drain pull-up mode.
	I2C3_SDA pin		PC9 pin: data line is used in open-drain pull-up mode.

Table 127. STM32H7Rxxx/7Sxxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, no pull mode.
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, no pull mode.
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, no pull mode.
	SPI1_NSS pin		PA4 pin: slave chip select pin used in push-pull, no pull mode.
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PB15 pin: slave data input line, used in push-pull, no pull mode.
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, no pull mode.
	SPI2_SCK pin	Input	PB13 pin: slave clock line, used in push-pull, no pull mode.
	SPI2_NSS pin		PB12 pin: slave chip select pin used in push-pull, no pull mode.
SPI3	SPI3	Enabled	The SPI3 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI3_MOSI pin	Input	PC12 pin: slave data input line, used in push-pull, no pull mode.
	SPI3_MISO pin	Output	PB4 pin: slave data output line, used in push-pull, no pull mode.
	SPI3_SCK pin	Input	PB3 pin: slave clock line, used in push-pull, no pull mode.
	SPI3_NSS pin		PA15 pin: slave chip select pin used in push-pull, no pull mode.

Table 127. STM32H7Rxxx/7Sxxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
DFU	USB	Enabled	USB FS configured in forced device mode. USB FS interrupt vector is enabled and used for USB DFU communications.
	USB_DM pin	Input/output	PM12: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PM11: USB DP line. Used in alternate push-pull, no pull mode. No external pull-up resistor is required
FDCAN	FDCAN2	Enabled	Once initialized the FDCAN1 configuration is: – Connection bit rate 250 kbit/s – Data bit rate 1000 kbit/s – FrameFormat = FDCAN_FRAME_FD_BRS – Mode = FDCAN_MODE_NORMAL – AutoRetransmission = ENABLE – TransmitPause = DISABLE – ProtocolException = ENABLE
	FDCAN2_Rx pin	Input	PB5 pin: FDCAN2 in reception mode. Used in alternate push-pull, pull-up mode.
	FDCAN2_Tx pin	Output	PB1 pin: FDCAN2 in transmission mode. Used in alternate push-pull, pull-up mode.
I3C1 ⁽¹⁾	I3C1	Enabled	– Mode: target mode – Aval timing: 0x4E – DMA Reg RX: disabled – DMA Reg TX: disabled – Status FIFO: disabled – DMA Req status: disabled – DMA Req control: disabled – IBI: Enabled – Additional data after IBI ACK-ed: 1 byte – IBI configuration: Mandatory Data Byte (MDB) – All IT disabled except RXFNE (Receive FIFO Interrupt). The RXFNE interruption is disabled after SYNC byte detection by the bootloader.
	I3C1_SCL pin	Input/output	PB8 pin: I3C1 in transmission mode. Used in alternate push-pull, no pull mode.
	I3C1_SDA pin		PB7 pin: I3C1 in transmission mode. Used in alternate push-pull, no pull mode.

1. I2C1 and I3C1 are exclusive: only one of them can be used, depending on the option byte FLASH_OBW2SR_I2C_NI3C.

Table 128. STM32H7Rxxx/7Sxxx special commands

Special commands supported (USART/I2C/SPI/FDCAN/USB/I3C) Based on Go/Read commands on virtual addresses			
Function	Command	Virtual address	Data received
Get Security State	Read	0xFF010001	4 bytes security status
Get product state		0xFF010002	
Data Provisioning	Go	0xFF83xxxx (xxxx indicates LSB address of the wrapper on the SRAM: 0x2404xxxx)	N/A
Change Product State		0xFF0100xx (xx indicates new product state value: 17 for provisioning, 72 for closed, and 5C for locked)	
Change Secure OB		0xFF02xxxx (xxxx indicates LSB address of the wrapper on the SRAM: 0x2404xxxx) ⁽¹⁾	

1. According to data written at address 0x2404xxxx, you can select option byte:

- 0: secure option byte i-Rot Select
- 1: secure option byte Writeprotection
- 2: secure option byte writeunprotect
- 3: secure option byte HDP.

For secure option bytes i-Rot Select, Writeprotection or HDP start and HDP end, the register values will be at address (0x2404xxxx + 0x00000004).

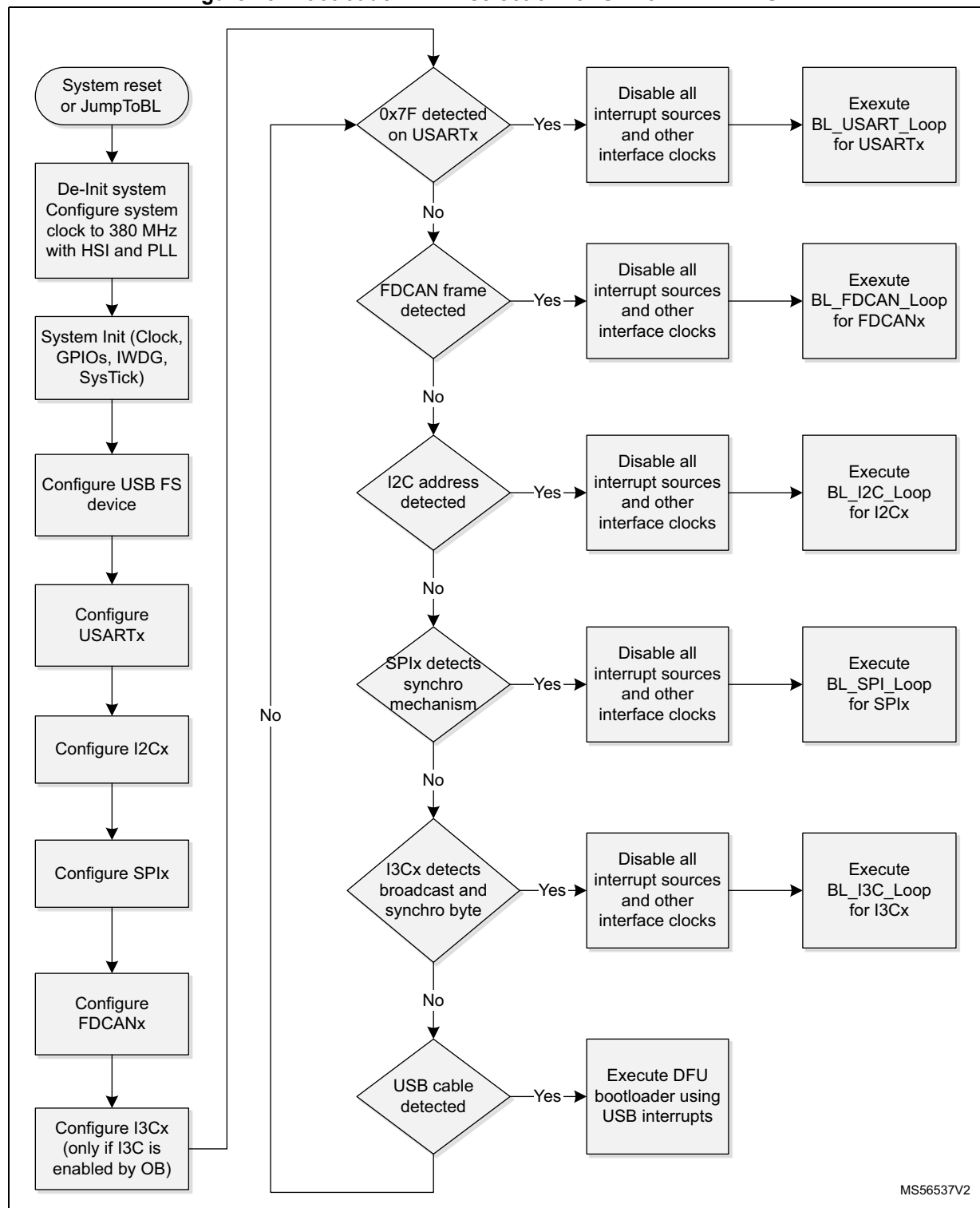
Example: HDP options bytes:

- write at 0x24040000 the value 0x00000003
- write at 0x24040004 the value 0x00EE0011 (EE: HDPend, 11:HDPstart)
- go at 0xFF020000.

58.2 Bootloader selection

Figure 73 shows the bootloader selection mechanism.

Figure 75. Bootloader V14.x selection for STM32H7Rxxx/7Sxxx



58.3 Bootloader version

[Table 129](#) lists the STM32H7Rxxx/7Sxxx devices bootloader versions.

Table 129. STM32H7Rxxx/7Sxxx bootloader version

Version number	Description	Known limitations
V14.3	Initial bootloader version	None

58.4 Jump to bootloader

If a user application jumping to the bootloader has the protection clock bits (xSPICKP or FMCKKP) enabled, the bootloader will be stuck/crash on the clock configuration.

The following fields cannot be modified when FMCKKP bit from RCC_CKPROT register is set to 1: PLL1ON, PLL2ON, PLL1QEN, PLL2REN, HSEON, HSION, CSION, FMCEN, FMCLPEN, and FMCRST.

The following fields cannot be modified when xSPICKP bit from RCC_CKPROTR register is set to 1: PLL2ON, PLL2SEN, PLL2TEN, HSEON, HSION, CSION, XSPiXEN, XSPiXLPEN, and XSPiXRST..

Refer to the section describing clock protection on RM0477.

At the startup; the bootloader deinitialises all the peripherals (including the clock), then reconfigures the clock based on PLL1 from HSI. In this case the bootloader is not be able to switch correctly to the needed default clock (crash or hang may be seen, depending on the application clock configuration)

Ensure that the protection clock bits (xSPICKP or FMCKKP) are disabled before jumping to the BL.

59 STM32L01xxx/02xxx devices

59.1 Bootloader configuration

The STM32L01xxx/02xxx bootloader is activated by applying Pattern 6 (described in [Table 2](#)). [Table 130](#) shows the hardware resources used by this bootloader.

Table 130. STM32L01xxx/02xxx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 32 MHz with HSI 16 MHz as clock source.
	RAM	-	2 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	4 Kbytes, starting from address 0x1FF00000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART2 (on PA9/PA10)	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA10 pin: USART2 in reception mode. Used in input pull-up mode.
	USART2_TX pin	Output	PA9 pin: USART2 in transmission mode. Used in input pull-up
USART2 (on PA2/PA3)	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode.Used in input pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in input pull-up mode.
USART2	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.

Table 130. STM32L01xxx/02xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI1 (for all device packages except TSSOP14)	SPI1	Enabled	The SPI1 configuration is: Slave mode, Full Duplex, 8-bit MSB, Speed up to 8 MHz, Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.
SPI1 (only for devices on TSSOP14 package)	SPI1	Enabled	The SPI1 configuration is: Slave mode, Full Duplex, 8-bit MSB, Speed up to 8 MHz, Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA14 pin: slave data output line, used in push-pull, pull-down mode. Note: This IO is also used as SWCLK for debug interface, as a consequence debugger cannot connect to the device in "on-the-fly" mode when the bootloader is running.
	SPI1_SCK pin	Input	PA13 pin: slave clock line, used in push-pull, pull-down mode
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode. Note: NSS pin synchronization is required on bootloader with SPI1 interface for devices on TSSOP14 package.

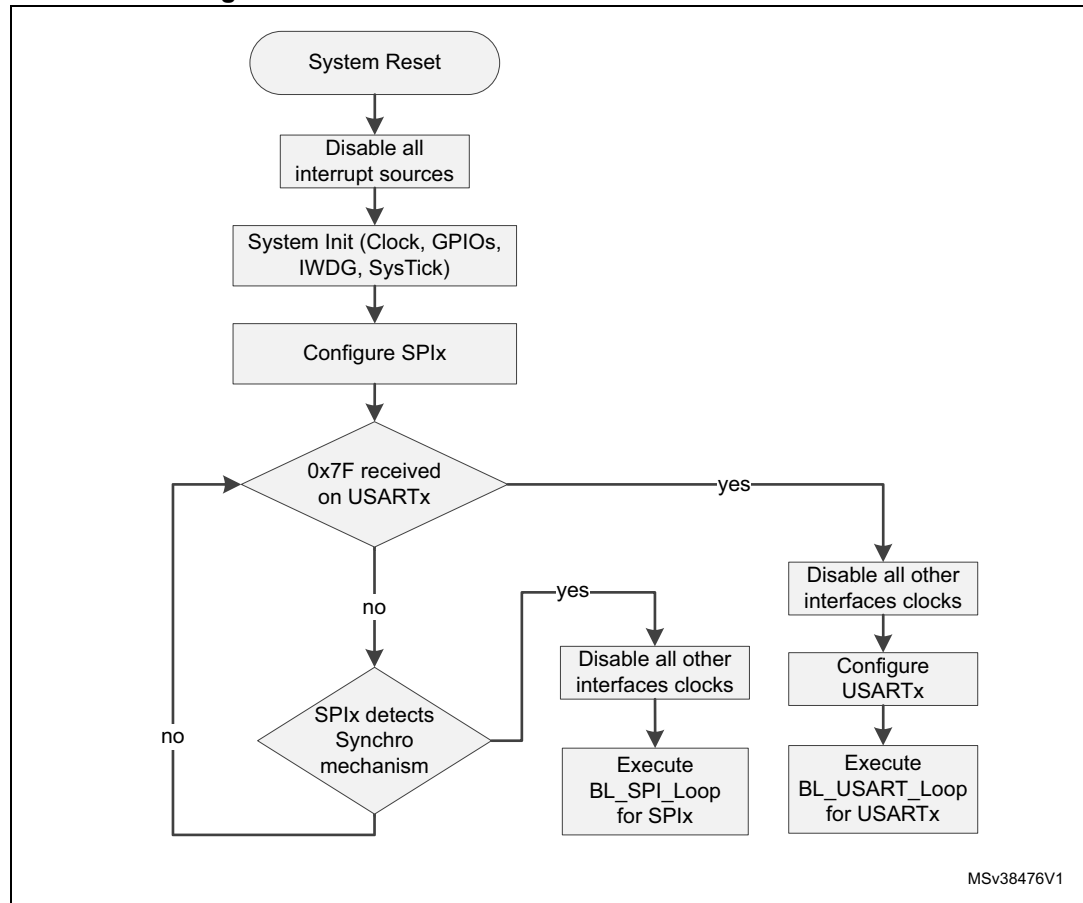
The system clock is derived from the embedded internal high-speed RC for all bootloader interfaces. No external quartz is required for bootloader operations.

Note: *Due to empty check mechanism present on this product, it is not possible to jump from user code to system bootloader. Such jump results in a jump back to user flash memory space. But if the first 4 bytes of user flash memory (at 0x0800 0000) are empty at the moment of the jump (i.e. erase first sector before jump or execute code from SRAM while flash is empty), then system bootloader is executed when jumped to.*

59.2 Bootloader selection

The [Table 76](#) shows the bootloader selection mechanism.

Figure 76. Bootloader selection for STM32L01xxx/02xxx



59.3 Bootloader version

The following table lists the STM32L01xxx/02xxx devices bootloader versions.

Table 131. STM32L01xxx/02xxx bootloader versions

Version number	Description	Known limitations
V12.2	Initial bootloader version	Bootloader not functional with SPI1 interface for devices on TSSOP14 package.
V12.3	Adds support of SPI interface for devices in TSSOP14 package.	For the SPI1 interface for devices in TSSOP14, a falling edge on NSS pin is required before starting communication, to properly synchronize the SPI interface. If the NSS pin is grounded (all time from device reset) the SPI communication is not synchronized and bootloader does not work properly with the SPI interface.

60 STM32L031xx/041xx devices

60.1 Bootloader configuration

The STM32L031xx/041xx bootloader is activated by applying Pattern 2 (described in [Table 2](#)). The following table shows the hardware resources used by this bootloader.

Table 132. STM32L031xx/041xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 32 MHz with HSI 16 MHz as clock source.
	RAM	-	4 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	4 Kbytes, starting from address 0x1FF00000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART2 (on PA9/PA10)	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA10 pin: USART2 in reception mode. Used in input pull-up mode.
	USART2_TX pin	Output	PA9 pin: USART2 in transmission mode. Used in input pull-up mode.
USART2 (on PA2/PA3)	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in input pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in input pull-up mode.
USART2	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.

Table 132. STM32L031xx/041xx configuration in system memory boot mode (continued)

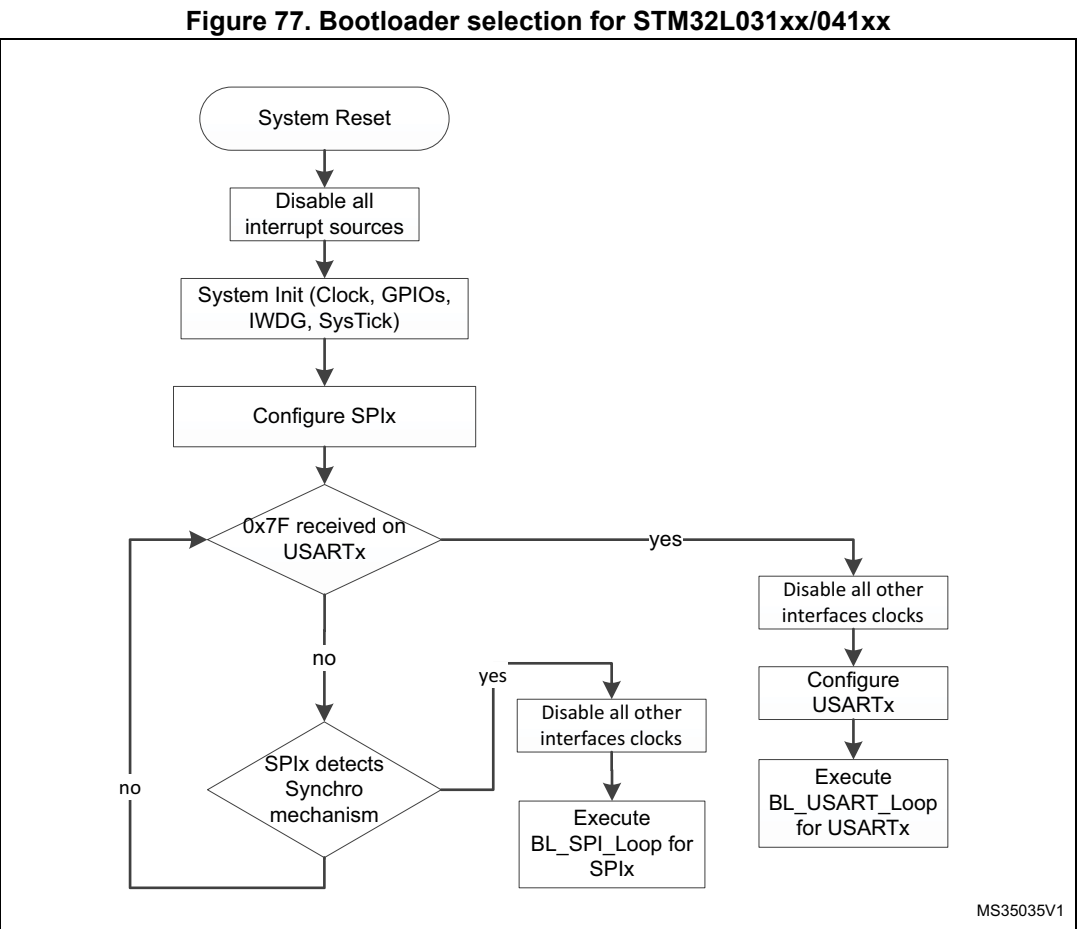
Bootloader	Feature/Peripheral	State	Comment
SPI1	SPI1	Enabled	The SPI1 configuration is: Slave mode, Full Duplex, 8-bit MSB, Speed up to 8 MHz, Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.

The system clock is derived from the embedded internal high-speed RC for all bootloader interfaces. No external quartz is required for bootloader operations.

The bootloader Read/Write commands do not support SRAM space for this product.

60.2 Bootloader selection

Figure 77 shows the bootloader selection mechanism.



60.3 Bootloader version

Table 133 lists the STM32L031xx/041xx devices bootloader versions.

Table 133. STM32L031xx/041xx bootloader versions

Version number	Description	Known limitations
V12.0	Initial bootloader version	None

61 STM32L05xxx/06xxx devices

61.1 Bootloader configuration

The STM32L05xxx/06xxx bootloader is activated by applying Pattern 1 (described in [Table 2](#)). [Table 134](#) shows the hardware resources used by this bootloader.

Table 134. STM32L05xxx/06xxx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 32 MHz with HSI 16 MHz as clock source.
	Power	-	Voltage range is set to Voltage Range 1.
	RAM	-	4 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	4 Kbytes, starting from address 0x1FF00000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.

Table 134. STM32L05xxx/06xxx configuration in system memory boot mode (continued)

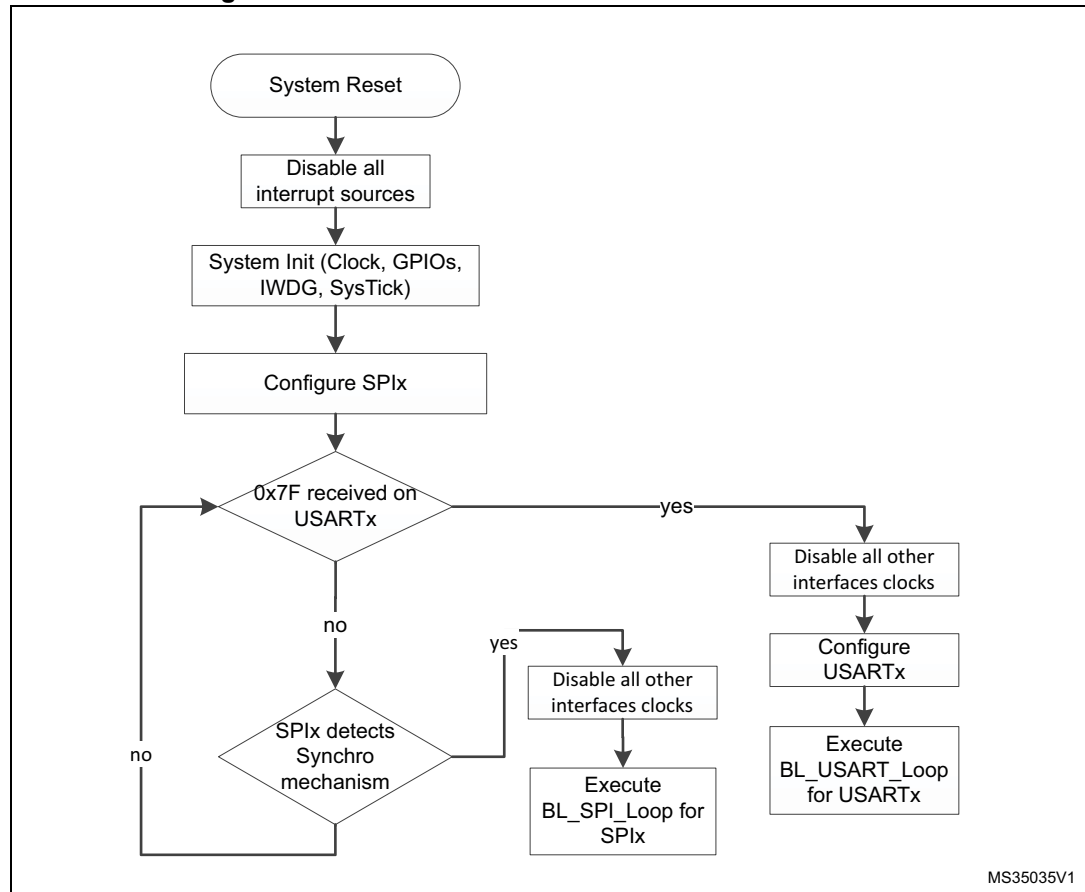
Bootloader	Feature/Peripheral	State	Comment
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode.
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PB15 pin: slave data input line, used in push-pull, pull-down mode
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, pull-down mode
	SPI2_SCK pin	Input	PB13 pin: slave clock line, used in push-pull, pull-down mode
	SPI2_NSS pin	Input	PB12 pin: slave chip select pin used in push-pull, pull-down mode.

The system clock is derived from the embedded internal high-speed RC for all bootloader interfaces. No external quartz is required for bootloader operations.

61.2 Bootloader selection

Figure 78 shows the bootloader selection mechanism.

Figure 78. Bootloader selection for STM32L05xxx/06xxx



61.3 Bootloader version

Table 135 lists the STM32L05xxx/06xxx devices bootloader versions:

Table 135. STM32L05xxx/06xxx bootloader versions

Version number	Description	Known limitations
V12.0	Initial bootloader version	PA13 set in alternate push-pull, pull-up mode and PA14 set in alternate pull-up pull-down mode even if not used.

62 STM32L07xxx/08xxx devices

Two bootloader versions are available on STM32L07xxx/08xxx devices:

- V4.x supporting USART1, USART2, and DFU (USB FS device). This version is embedded in STM32L072xx/73xx and STM32L082xx/83xx devices.
- V11.x supporting USART1, USART2, I2C1, I2C2, SPI1 and SPI2. This version is embedded in other STM32L071xx/081xx devices.

62.1 Bootloader V4.x

62.1.1 Bootloader configuration

The STM32L07xxx/08xxx bootloader is activated by applying Pattern 2 or Pattern 7 when dual bank boot feature is available (described in [Table 2](#)). [Table 136](#) shows the hardware resources used by this bootloader.

Table 136. STM32L07xxx/08xxx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 32 MHz with HSI 16 MHz as clock source.
	RAM	-	4 Kbytes, starting from address 0x20000000, are used by the bootloader firmware.
	System memory	-	8 Kbytes, starting from address 0x1FF00000, contain the bootloader firmware.
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.

Table 136. STM32L07xxx/08xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
DFU	USB	Enabled	USB FS configured in forced device mode. USB FS interrupt vector is enabled and used for USB DFU communications.
	USB_DM pin	Input/output	PA11 pin: USB FS DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12 pin: USB FS DP line. Used in alternate push-pull, no pull mode. No external pull-up resistor is required.

The system clock is derived from the embedded internal high-speed RC for all bootloader interfaces. No external quartz is required for bootloader operations.

62.1.2 Bootloader selection

Figure 79 and Figure 80 show the bootloader selection mechanism.

Figure 79. Dual bank boot implementation for STM32L07xxx/08xxx bootloader V4.x

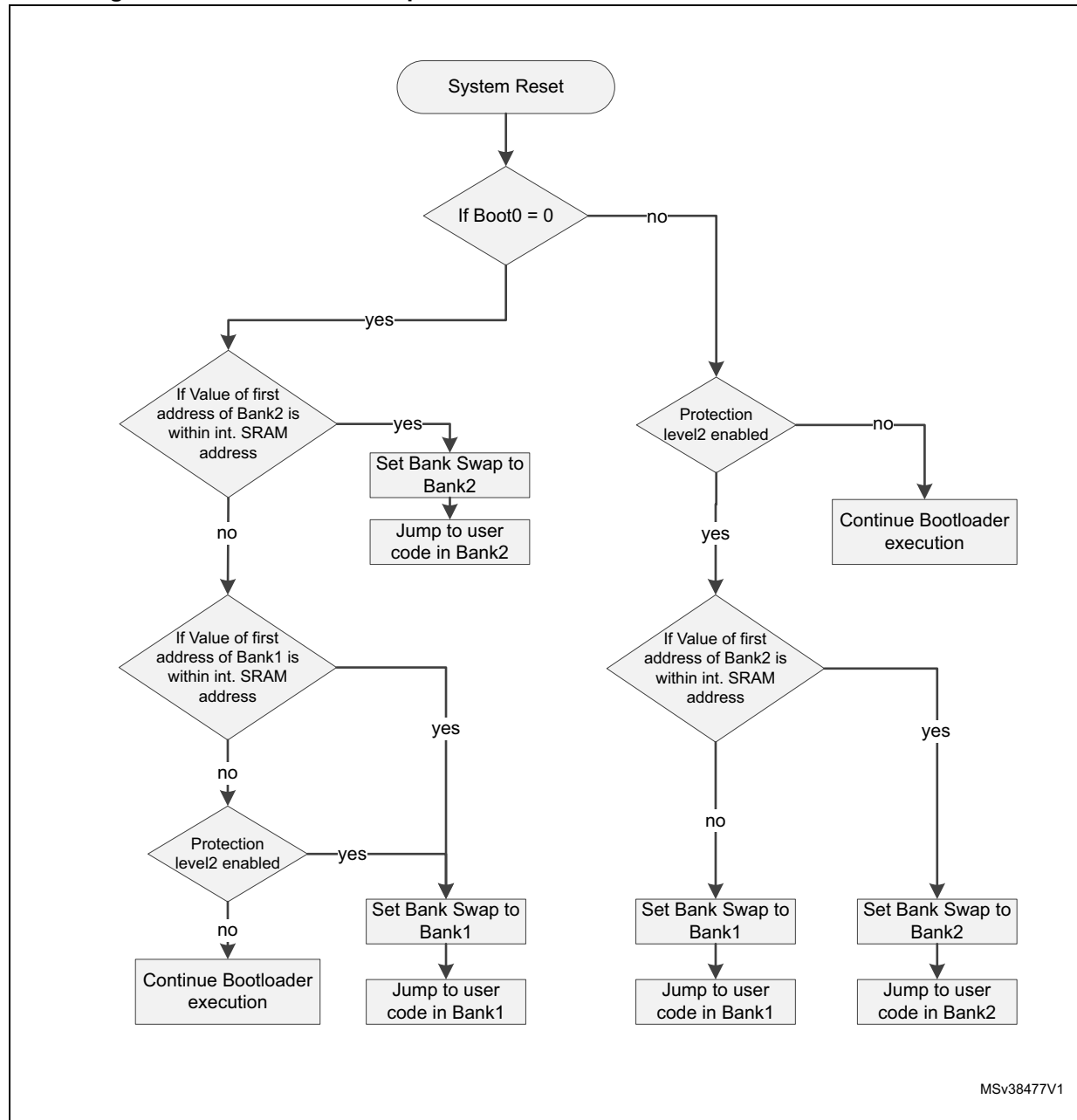
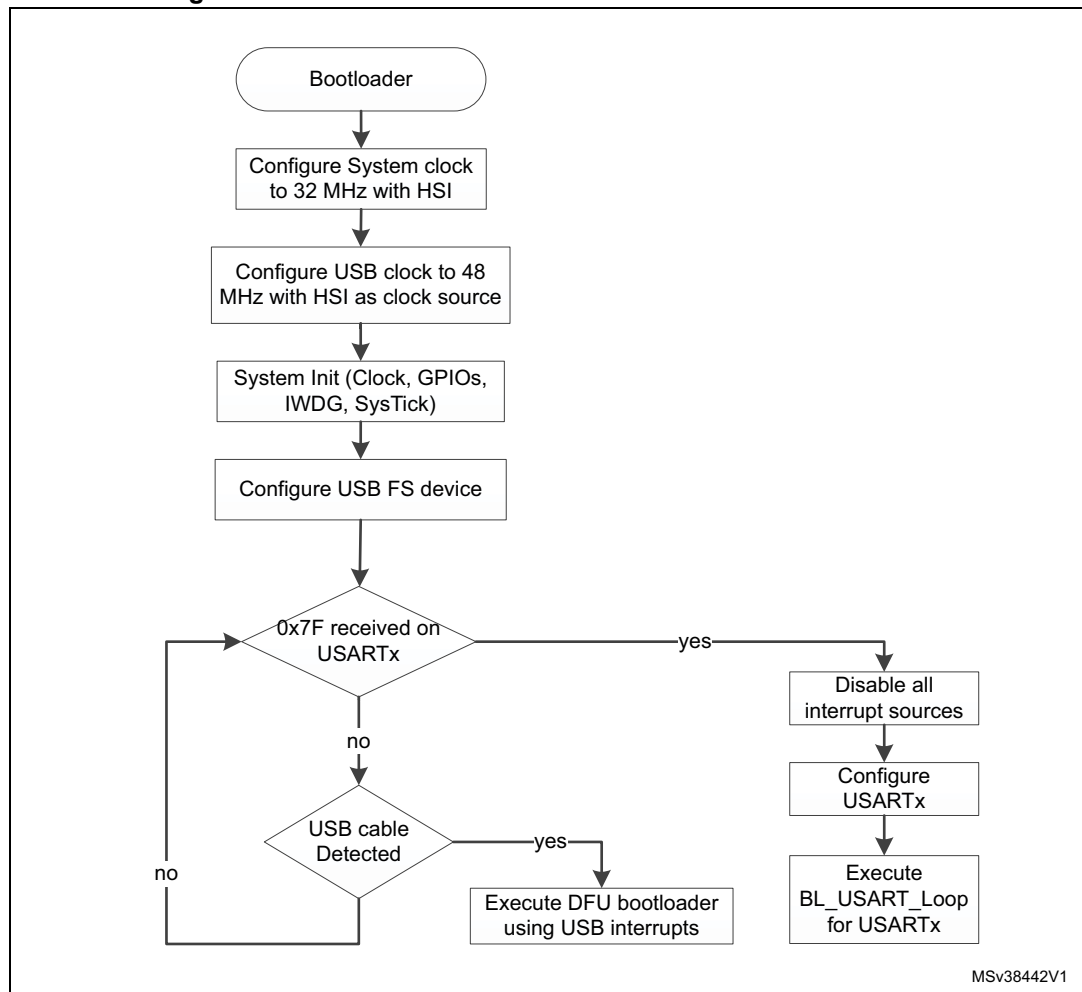


Figure 80. Bootloader V4.x selection for STM32L07xxx/08xxx

62.1.3 Bootloader version

Table 137 lists the STM32L07xxx/08xxx devices bootloader versions.

Table 137. STM32L07xxx/08xxx bootloader versions

Version number	Description	Known limitations
V4.0	Initial bootloader version	PA4, PA5, PA6 and PA7 IOs are configured in pull-down mode despite not used by bootloader
V4.1	This new version implements the Dual Bank Boot feature.	PA4, PA5, PA6 and PA7 IOs are configured in pull-down mode despite not used by bootloader

62.2 Bootloader V11.x

62.2.1 Bootloader configuration

The STM32L07xxx/08xxx bootloader is activated by applying Pattern 2 or Pattern 7 when dual bank boot feature is available (see in [Table 2](#)). [Table 138](#) shows the hardware resources used by this bootloader.

Table 138. STM32L07xxx/08xxx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 32 MHz with HSI 16 MHz as clock source.
	RAM	-	5 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	8 Kbytes, starting from address 0x1FF00000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz, 7-bit address, Target mode, Analog filter ON – Target 7-bit address: 0b1000010x – (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: I2C1 clock line is used in open-drain no pull mode.
	I2C1_SDA pin	Input/output	PB7 pin: I2C1 data line is used in open-drain no pull mode.

Table 138. STM32L07xxx/08xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 400 kHz, 7-bit address, Target mode, Analog filter ON – Target 7-bit address: 0b1000010x – (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PB10 pin: I2C2 clock line is used in open-drain no pull mode.
	I2C2_SDA pin	Input/output	PB11 pin: I2C2 data line is used in open-drain no pull mode.
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PB15 pin: slave data input line, used in push-pull, pull-down mode
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, pull-down mode
	SPI2_SCK pin	Input	PB13 pin: slave clock line, used in push-pull, pull-down mode
	SPI2_NSS pin	Input	PB12 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.

The system clock is derived from the embedded internal high-speed RC for all bootloader interfaces. No external quartz is required for bootloader operations.

62.2.2 Bootloader selection

Figure 81 and Figure 82 show the bootloader selection mechanism.

Figure 81. Dual bank boot implementation for STM32L07xxx/08xxx bootloader V11.x

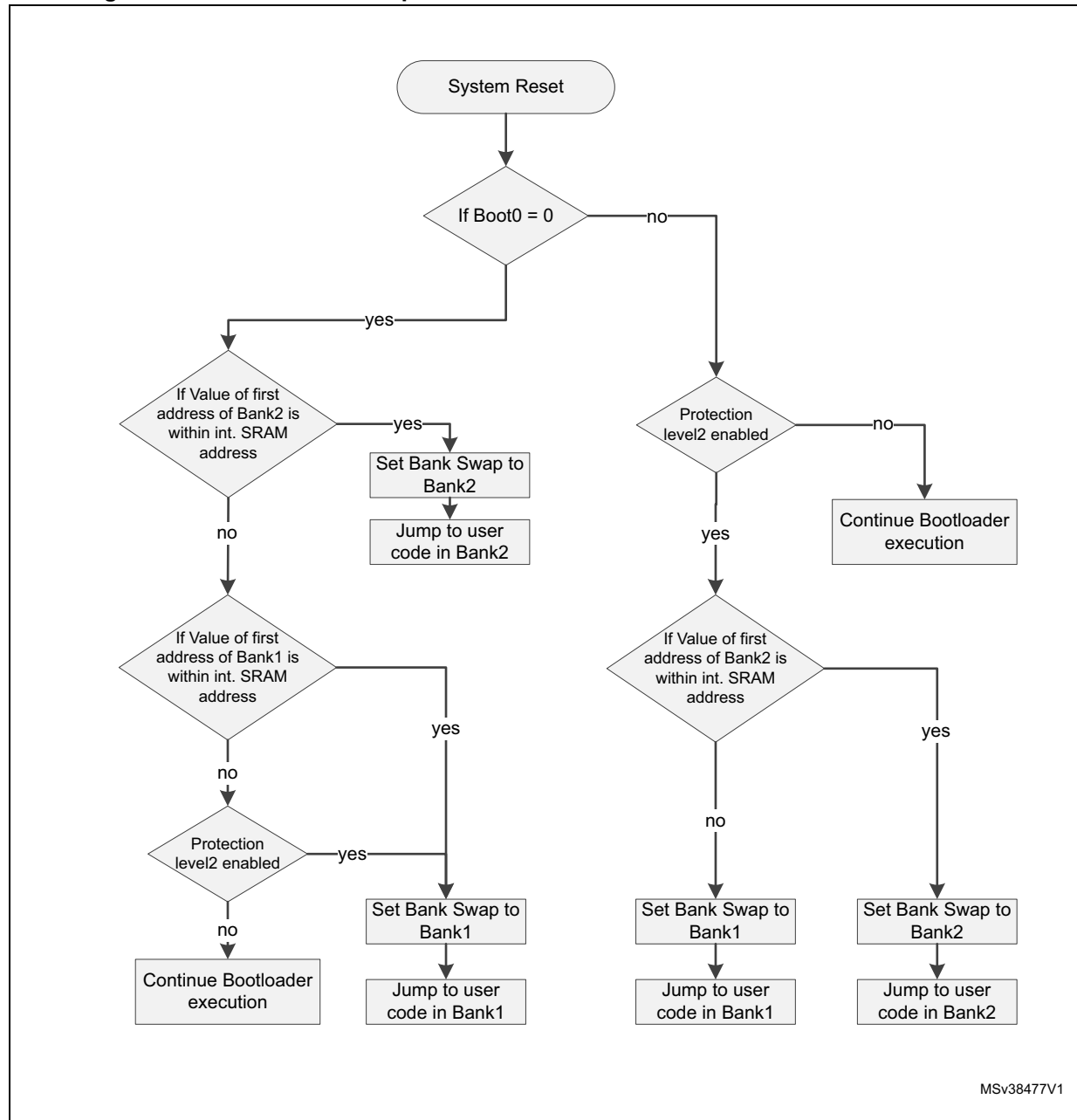
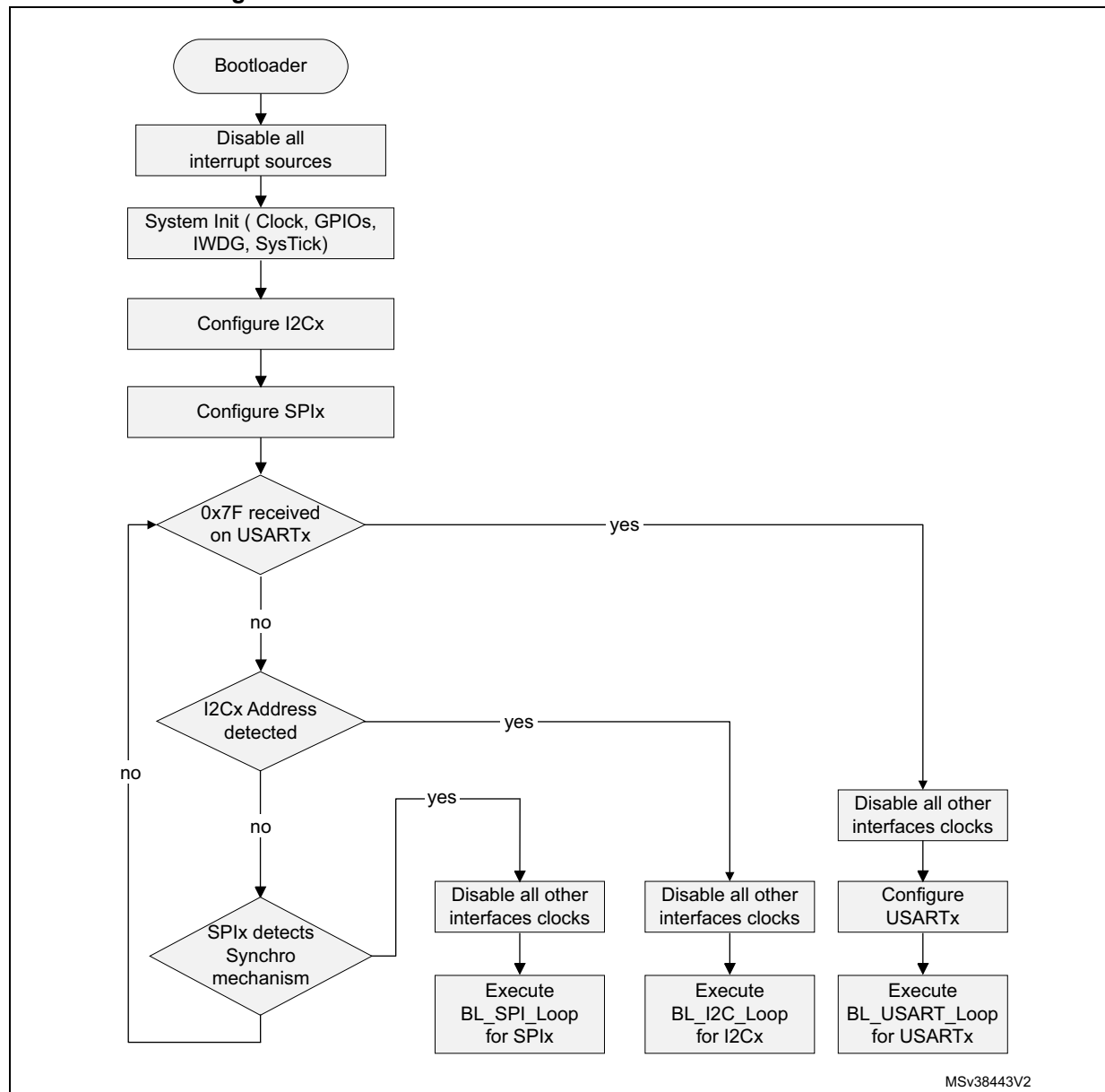


Figure 82. Bootloader V11.x selection for STM32L07xxx/08xxx



62.2.3 Bootloader version

Table 139 lists the STM32L07xxx/08xxx devices bootloader versions:

Table 139. STM32L07xxx/08xxx bootloader V11.x versions

Version number	Description	Known limitations
V11.1	Initial bootloader version	None
V11.2	This new version implements the Dual Bank Boot feature.	None

63 STM32L1xxx6(8/B)A devices

63.1 Bootloader configuration

The STM32L1xxx6(8/B)A bootloader is activated by applying Pattern 1 (described in [Table 2](#)). [Table 140](#) shows the hardware resources used by this bootloader.

Table 140. STM32L1xxx6(8/B)A configuration in system memory boot mode

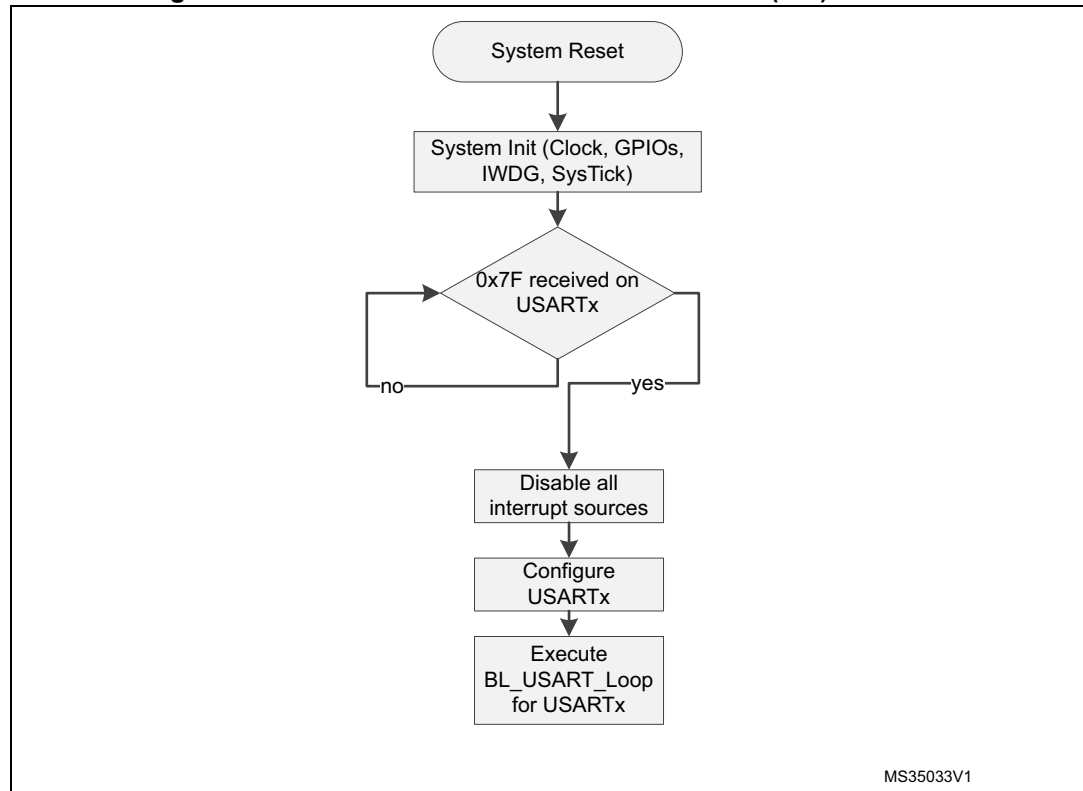
Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 16 MHz.
	RAM	-	2 Kbytes, starting from address 0x20000000, are used by the bootloader firmware.
	System memory	-	4 Kbytes, starting from address 0x1FF00000 contain the bootloader firmware.
	IWDG	-	The IWDG prescaler is configured to its maximum value and is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
	Power	-	Voltage is set to Voltage Range 1.
USART1	USART1	Enabled	Once initialized, the USART1 configuration is 8 bits, even parity, and one stop bit.
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
USART2	USART2	Enabled	Once initialized, the USART2 configuration is 8 bits, even parity, and one stop bit.
	USART2_RX pin	Input	PD6 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PD5 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host.

The system clock is derived from the embedded internal high-speed RC, no external . No external quartz is required for the bootloader execution.

63.2 Bootloader selection

The figure below shows the bootloader selection mechanism.

Figure 83. Bootloader selection for STM32L1xxx6(8/B)A devices



63.3 Bootloader version

The following table lists the STM32L1xxx6(8/B)A devices bootloader versions:

Table 141. STM32L1xxx6(8/B)A bootloader versions

Version number	Description	Known limitations
V2.0	Initial bootloader version	When a Read Memory or Write Memory command is issued with an unsupported memory address and a correct address checksum (i.e. address 0x6000 0000), the command is aborted by the bootloader device, but the NACK (0x1F) is not sent to the host. As a result, the next two bytes (the number of bytes to be read/written and its checksum) are considered as a new command and its checksum. ⁽¹⁾

1. If the "number of data - 1" (N-1) to be read/written is not equal to a valid command code, the limitation is not perceived from the host, as the command is NACK-ed anyway (as an unsupported new command).

64 STM32L1xxx6(8/B) devices

64.1 Bootloader configuration

The STM32L1xxx6(8/B) bootloader is activated by applying Pattern 1 (described in [Table 2](#)). The following table shows the hardware resources used by this bootloader.

Table 142. STM32L1xxx6(8/B) configuration in system memory boot mode

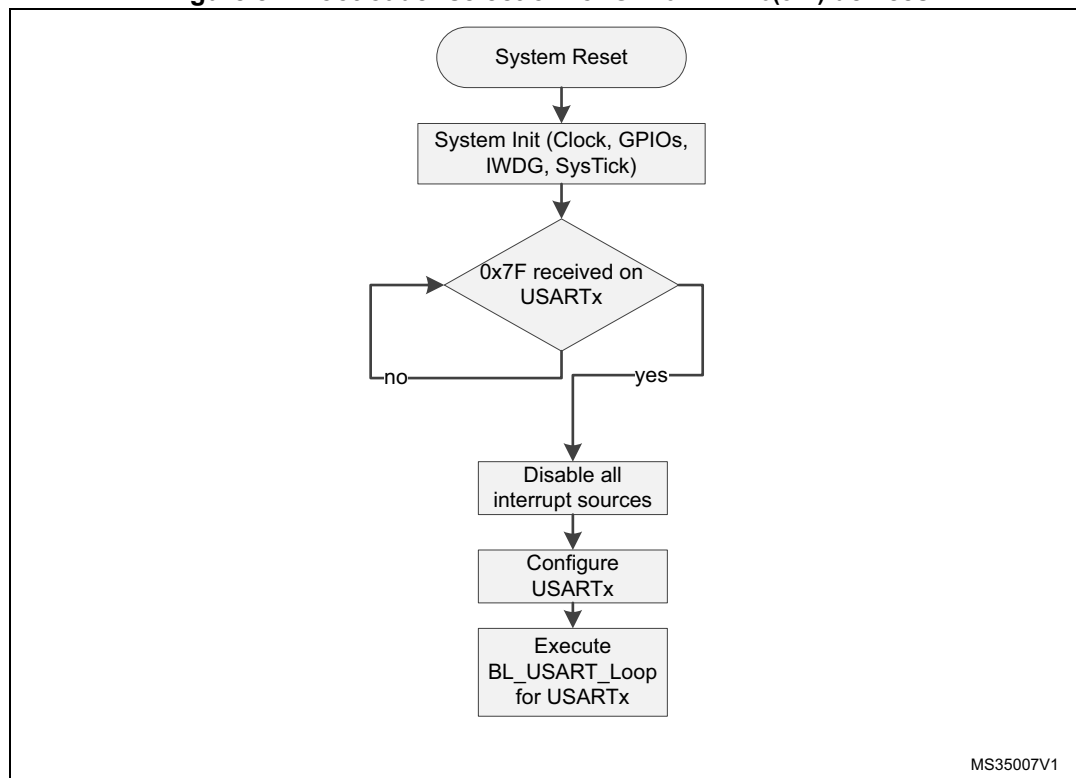
Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 16 MHz.
	RAM	-	2 Kbytes, starting from address 0x20000000, are used by the bootloader firmware.
	System memory	-	4 Kbytes, starting from address 0x1FF00000 contain the bootloader firmware.
	IWDG	-	The IWDG prescaler is configured to its maximum value and is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
	Power	-	Voltage is set to Voltage Range 1.
USART1	USART1	Enabled	Once initialized, the USART1 configuration is 8 bits, even parity, and one stop bit.
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
USART2	USART2	Enabled	Once initialized, the USART2 configuration is 8 bits, even parity, and one stop bit.
	USART2_RX pin	Input	PD6 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PD5 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host.

The system clock is derived from the embedded internal high-speed RC, no external . No external quartz is required for the bootloader execution.

64.2 Bootloader selection

The figure below shows the bootloader selection mechanism.

Figure 84. Bootloader selection for STM32L1xxx6(8/B) devices



64.3 Bootloader version

The following table lists the STM32L1xxx6(8/B) devices bootloader versions:

Table 143. STM32L1xxx6(8/B) bootloader versions

Version number	Description	Known limitations
V2.0	Initial bootloader version	- When a Read Memory or Write Memory command is issued with an unsupported memory address and a correct address checksum (i.e. address 0x6000 0000), the command is aborted by the bootloader device, but the NACK (0x1F) is not sent to the host. As a result, the next two bytes (the number of bytes to be read/written and its checksum) are considered as a new command and its checksum.⁽¹⁾ - PA13/14/15 is configured in alternate push-pull (PA14 in pull-down) even if not used.

1. If the "number of data - 1" (N-1) to be read/written is not equal to a valid command code, the limitation is not perceived from the host, as the command is NACK-ed anyway (as an unsupported new command).

65 STM32L1xxxC devices

65.1 Bootloader configuration

The STM32L1xxxC bootloader is activated by applying Pattern 1 (described in [Table 2](#)). The following table shows the hardware resources used by this bootloader.

Table 144. STM32L1xxxC configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 16 MHz using the HSI. This is used only for USARTx and during USB detection for DFU (once the DFU is selected, the clock source is derived from the external crystal).
		HSE enabled	The external clock is mandatory only for the DFU and must be in the following range: [24, 16, 12, 8, 6, 4, 3, 2] MHz. The PLL is used to generate the USB 48 MHz clock and the 32 MHz clock for the system clock.
		-	The CSS interrupt is enabled for the DFU. Any failure (or removal) of the external clock generates a system reset.
	RAM	-	4 Kbytes, starting from address 0x20000000, are used by the bootloader firmware.
	System memory	-	8 Kbytes, starting from address 0x1FF00000 contains the bootloader firmware.
	IWDG	-	The IWDG prescaler is configured to its maximum value and is periodically refreshed to prevent watchdog resets (if the hardware IWDG option was previously enabled by the user).
	Power	-	Voltage is set to Voltage Range 1.
USART1	USART1	Enabled	Once initialized, the USART1 configuration is 8 bits, even parity and 1 stop bit.
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.

Table 144. STM32L1xxxC configuration in system memory boot mode (continued)

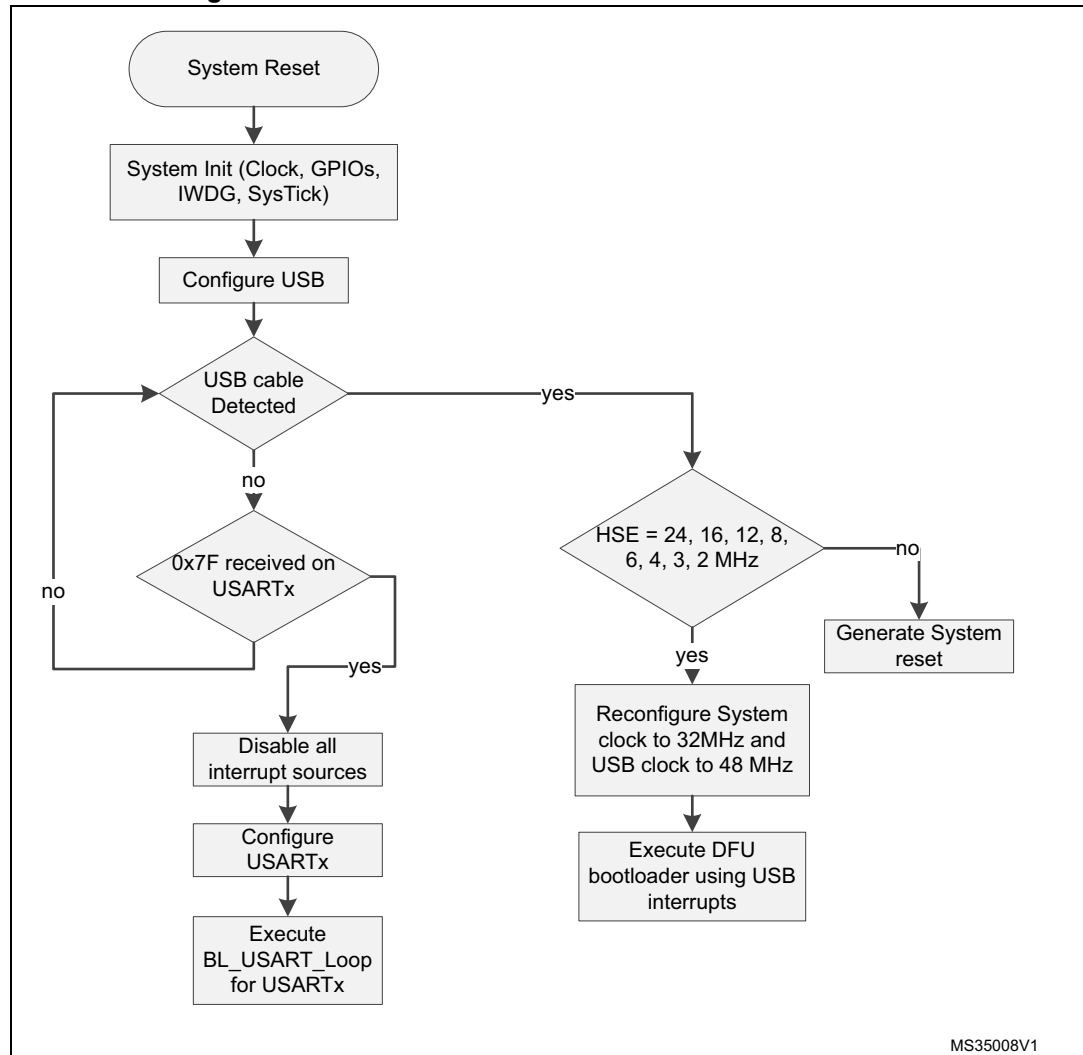
Bootloader	Feature/Peripheral	State	Comment
USART2	USART2	Enabled	Once initialized, the USART2 configuration is 8 bits, even parity and 1 stop bit. The USART2 uses its remapped pins.
	USART2_RX pin	Input	PD6 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PD5 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for the USARTx bootloader.
DFU	USB	Enabled	USB used in FS mode
	USB_DM pin	Input/output	PA11: USB DM line. Used in input no pull mode.
	USB_DP pin		PA12: USB DP line. Used in input no pull mode.

The system clock is derived from the embedded internal high-speed RC for the USARTx bootloader. This internal clock is also used for DFU, but only for the selection phase. An external clock in the range of [24, 16, 12, 8, 6, 4, 3, 2] MHz is required for the execution of the DFU after the selection phase.

65.2 Bootloader selection

The figure below shows the bootloader selection mechanism.

Figure 85. Bootloader selection for STM32L1xxxC devices



MS35008V1

65.3 Bootloader version

The following table lists the STM32L1xxxC devices bootloader versions.

Table 145. STM32L1xxxC bootloader versions

Version number	Description	Known limitations
V4.0	Initial bootloader version	For the USART interface, two consecutive NACKs instead of 1 NACK are sent when a Read Memory or Write Memory command is sent and the RDP level is active. PA13/14/15 configured in alternate push-pull, pull (PA14 in pull-down) even if not used

66 STM32L1xxxD devices

66.1 Bootloader configuration

The STM32L1xxxD bootloader is activated by applying Pattern 4 (described in [Table 2](#)). The following table shows the hardware resources used by this bootloader.

Table 146. STM32L1xxxD configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 16 MHz using the HSI. This is used only for USARTx and during USB detection for DFU (once the DFU is selected, the clock source is derived from the external crystal).
		HSE enabled	The external clock is mandatory only for DFU and it must be in the following range: [24, 16, 12, 8, 6, 4, 3, 2] MHz. The PLL is used to generate the USB 48 MHz clock and the 32 MHz clock for the system clock.
		-	The CSS interrupt is enabled for the DFU. Any failure (or removal) of the external clock generates system reset.
	RAM	-	4 Kbytes, starting from address 0x20000000, are used by the bootloader firmware.
	System memory	-	8 Kbytes, starting from address 0x1FF00000 contains the bootloader firmware.
	IWDG	-	The IWDG prescaler is configured to its maximum value and is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
	Power	-	Voltage is set to Voltage Range 1.
USART1	USART1	Enabled	Once initialized, the USART1 configuration is 8 bits, even parity, and one stop bit.
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.

Table 146. STM32L1xxxD configuration in system memory boot mode (continued)

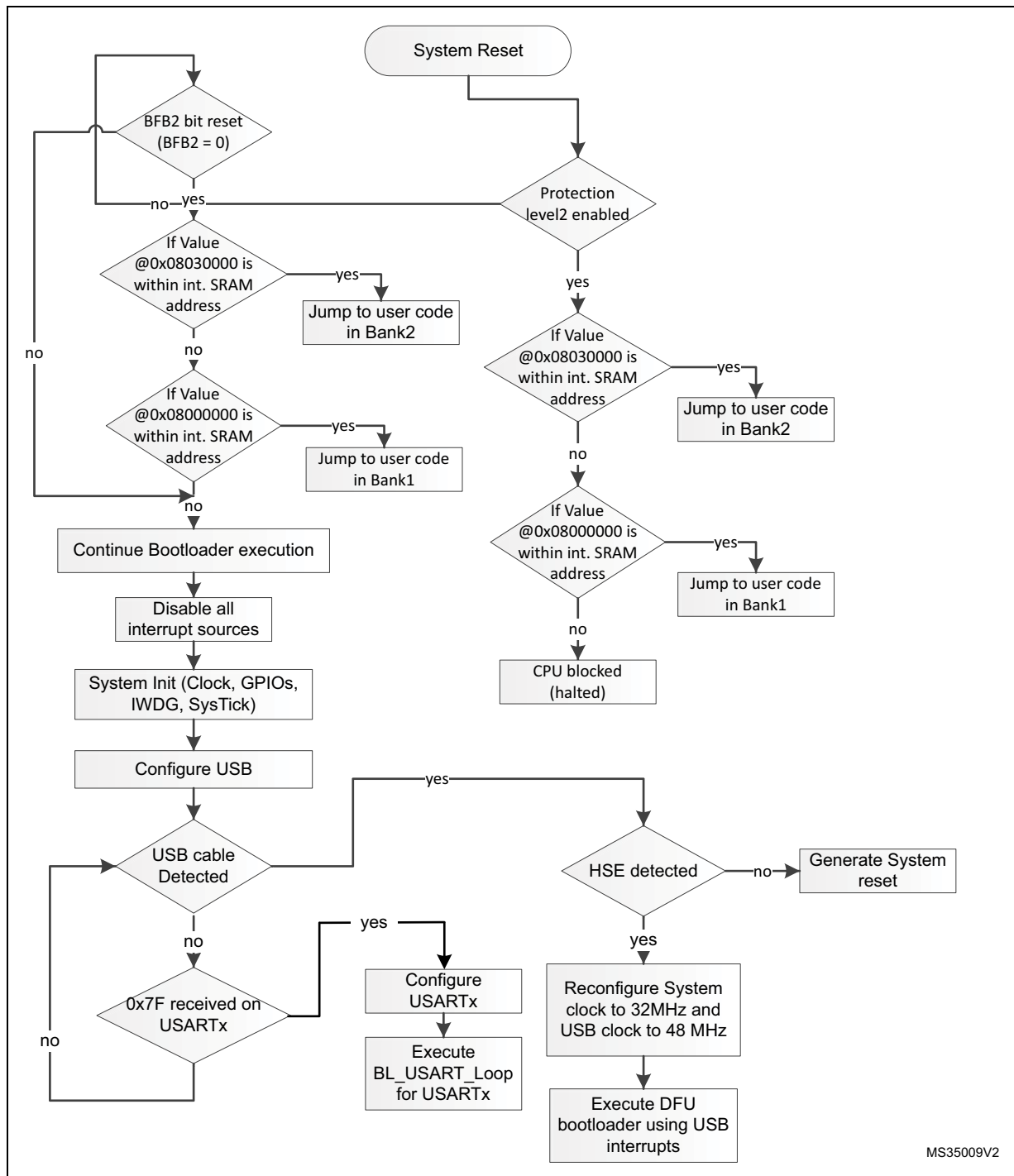
Bootloader	Feature/Peripheral	State	Comment
USART2	USART2	Enabled	Once initialized, the USART2 configuration is 8 bits, even parity, and one stop bit. The USART2 uses its remapped pins.
	USART2_RX pin	Input	PD6 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PD5 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx bootloader.
DFU	USB	Enabled	USB used in FS mode
	USB_DM pin	Input/output	PA11: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12: USB DP line. Used in alternate push-pull, no pull mode.

The system clock is derived from the embedded internal high-speed RC for USARTx bootloader. This internal clock is used also for DFU, but only for the selection phase. An external clock in the range of [24, 16, 12, 8, 6, 4, 3, 2] MHz is required for DFU execution after the selection phase.

66.2 Bootloader selection

The figure below shows the bootloader selection mechanism.

Figure 86. Bootloader selection for STM32L1xxxD devices



MS35009V2

66.3 Bootloader version

The following table lists the STM32L1xxxD devices bootloader versions:

Table 147. STM32L1xxxD bootloader versions

Version number	Description	Known limitations
V4.1	Initial bootloader version	– In the bootloader code the PA13 (JTMS/SWDIO) I/O output speed is configured to 400 kHz, as a consequence some debugger cannot connect to the device in Serial Wire mode when the bootloader is running. – When the DFU is selected, the RTC is reset and thus all RTC information (such as calendar, alarm) are lost including backup registers. Note: When the USART bootloader is selected there is no change on the RTC configuration (including backup registers).
V4.2	Fix V4.1 limitations (available on Rev.Z devices only)	– Stack overflow by 8 bytes when jumping to Bank1/Bank2 if BFB2=0 or when Read Protection level is set to 2. Workaround: the user code must force in the startup file the top of stack address before to jump to the main program. This can be done in the “Reset_Handler” routine. – When the Stack of the user code is placed outside the SRAM (i.e. @ 0x2000C000) the bootloader cannot jump to that user code which is considered invalid. This might happen when using compilers which place the stack at a non-physical address at the top of the SRAM (i.e. @ 0x2000C000). Workaround: place manually the stack at a physical address.
V4.5	Fix V4.2 limitations. DFU interface robustness enhancements (available on Rev.Y devices only).	– For the USART interface, two consecutive NACKs (instead of 1 NACK) are sent when a Read Memory or Write Memory command is sent and the RDP level is active.

67 STM32L1xxxE devices

67.1 Bootloader configuration

The STM32L1xxxE bootloader is activated by applying Pattern 4 (described in [Table 2](#)). The following table shows the hardware resources used by this bootloader.

Table 148. STM32L1xxxE configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 16 MHz using the HSI. This is used only for USARTx and during USB detection for DFU (once the DFU is selected, the clock source is derived from the external crystal).
		HSE enabled	The external clock is mandatory only for DFU and it must be in the following range: [24, 16, 12, 8, 6, 4, 3, 2] MHz. The PLL is used to generate the USB 48 MHz clock and the 32 MHz clock for the system clock.
		-	The CSS interrupt is enabled for the DFU. Any failure (or removal) of the external clock generates system reset.
	RAM	-	4 Kbytes, starting from address 0x20000000, are used by the bootloader firmware.
	System memory	-	8 Kbytes, starting from address 0x1FF00000 contains the bootloader firmware.
	IWDG	-	The IWDG prescaler is configured to its maximum value and is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
	Power	-	Voltage is set to Voltage Range 1.
USART1	USART1	Enabled	Once initialized, the USART1 configuration is 8 bits, even parity, and one stop bit.
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.

Table 148. STM32L1xxxE configuration in system memory boot mode (continued)

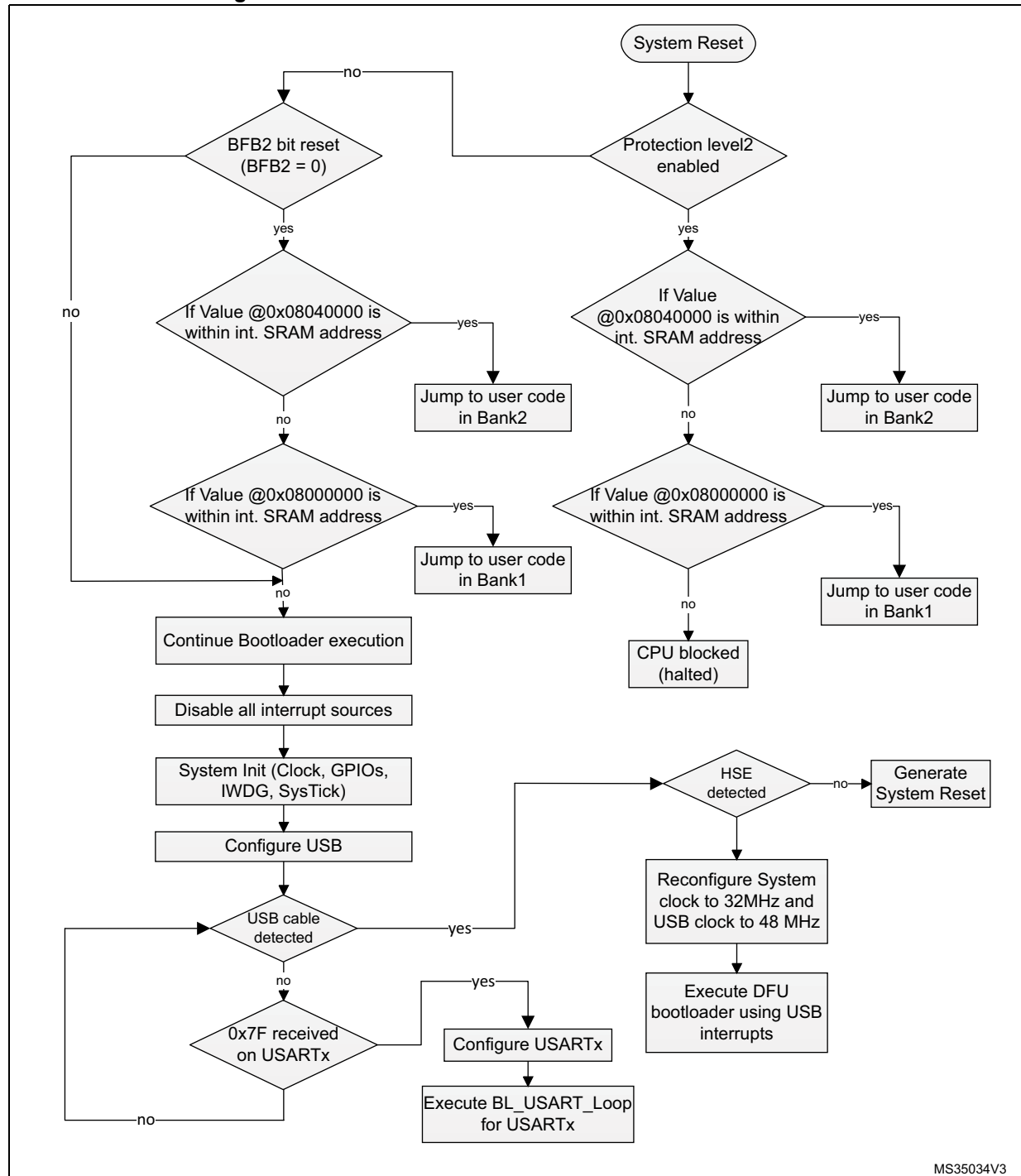
Bootloader	Feature/Peripheral	State	Comment
USART2	USART2	Enabled	Once initialized, the USART2 configuration is 8 bits, even parity, and one stop bit. The USART2 uses its remapped pins.
	USART2_RX pin	Input	PD6 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PD5 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx bootloader.
DFU	USB	Enabled	USB used in FS mode
	USB_DM pin	Input/output	PA11: USB DM line.
	USB_DP pin		PA12: USB DP line.

The system clock is derived from the embedded internal high-speed RC for USARTx bootloader. This internal clock is used also for DFU, but only for the selection phase. An external clock in the range of [24, 16, 12, 8, 6, 4, 3, 2] MHz is required for DFU execution after the selection phase.

67.2 Bootloader selection

The figure below shows the bootloader selection mechanism.

Figure 87. Bootloader selection for STM32L1xxxE devices



MS35034V3

67.3 Bootloader version

[Table 149](#) lists the STM32L1xxxE devices bootloader versions:

Table 149. STM32L1xxxE bootloader versions

Version number	Description	Known limitations
V4.0	Initial bootloader version	For the USART interface, two consecutive NACKs (instead of 1 NACK) are sent when a Read Memory or Write Memory command is sent and the RDP level is active. PA13/14/15 configured in alternate push-pull, pull (PA14 in pull-down) even if not used.

68 STM32L412xx/422xx devices

68.1 Bootloader configuration

The STM32L412xx/422xx bootloader is activated by applying Pattern 16 (described in [Table 2](#)). [Table 150](#) shows the hardware resources used by this bootloader.

Table 150. STM32L412xx/422xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The HSI is used at startup as clock source for system clock configured to 72 MHz and for USART, I2C, SPI and USB bootloader operation.
		-	CRS is enabled for the DFU to allow USB to be clocked by HSI48 48 MHz.
	RAM	-	12 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	28 Kbytes, starting from address 0x1FFF0000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
	Power	-	The DFU cannot be used to communicate with bootloader if the voltage scaling range 2 is selected. Bootloader firmware does not configure voltage scaling range value in PWR_CR1 register.
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input no pull mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in input no pull mode.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in input pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in input pull-up mode
USART3	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. Used in input pull-up mode.
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode. Used in input pull-up mode.

Table 150. STM32L412xx/422xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
I2C1	I2C1	Enabled	The I2C1 configuration is: I2C speed: up to 1 MHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b1010010x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin	Input/output	PB7 pin: data line is used in open-drain no pull mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: I2C speed: up to 1 MHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b1010010x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PB10 pin: clock line is used in open-drain no pull mode.
	I2C2_SDA pin	Input/output	PB11 pin: data line is used in open-drain no pull mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: I2C speed: up to 1 MHz, 7-bit address, Target mode, Analog filter ON Target 7-bit address: 0b1010010x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PC0 pin: clock line is used in open-drain no pull mode.
	I2C3_SDA pin	Input/output	PC1 pin: data line is used in open-drain no pull mode.
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode. ⁽¹⁾
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.

Table 150. STM32L412xx/422xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz Polarity: CPOL low, CPHA low, NSS hardware
	SPI2_MOSI pin	Input	PB15 pin: slave data input line, used in push-pull, pull-down mode
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, pull-down mode. ⁽¹⁾
	SPI2_SCK pin	Input	PB13 pin: slave clock line, used in push-pull, pull-down mode
	SPI2_NSS pin	Input	PB12 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.
DFU	USB	Enabled	USB FS configured in forced device mode. USB FS interrupt vector is enabled and used for USB DFU communications. Note: VDDUSB IO must be connected to 3.3 V as USB peripheral is used by the bootloader.
	USB_DM pin	Input/output	PA11: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12: USB DP line. Used in alternate push-pull, no pull mode. No external pull-up resistor is required

1. SPI Tx (MISO) is handled by DMA. On the bootloader startup after SPI initialisation as soon as the bit DMATx enable on SPI CR2 register is set to 0x1, the MISO line will be set to 3.3 V.

Note: If VDDUSB pin is not connected to V_{DD} , SPI flash memory write operations may be corrupted due to voltage issue. For more details, refer to product's datasheet and errata sheet.

68.2 Bootloader selection

The following figures show the bootloader selection mechanism.

Figure 88. Dual bank boot Implementation for STM32L412xx/422xx bootloader V9.x

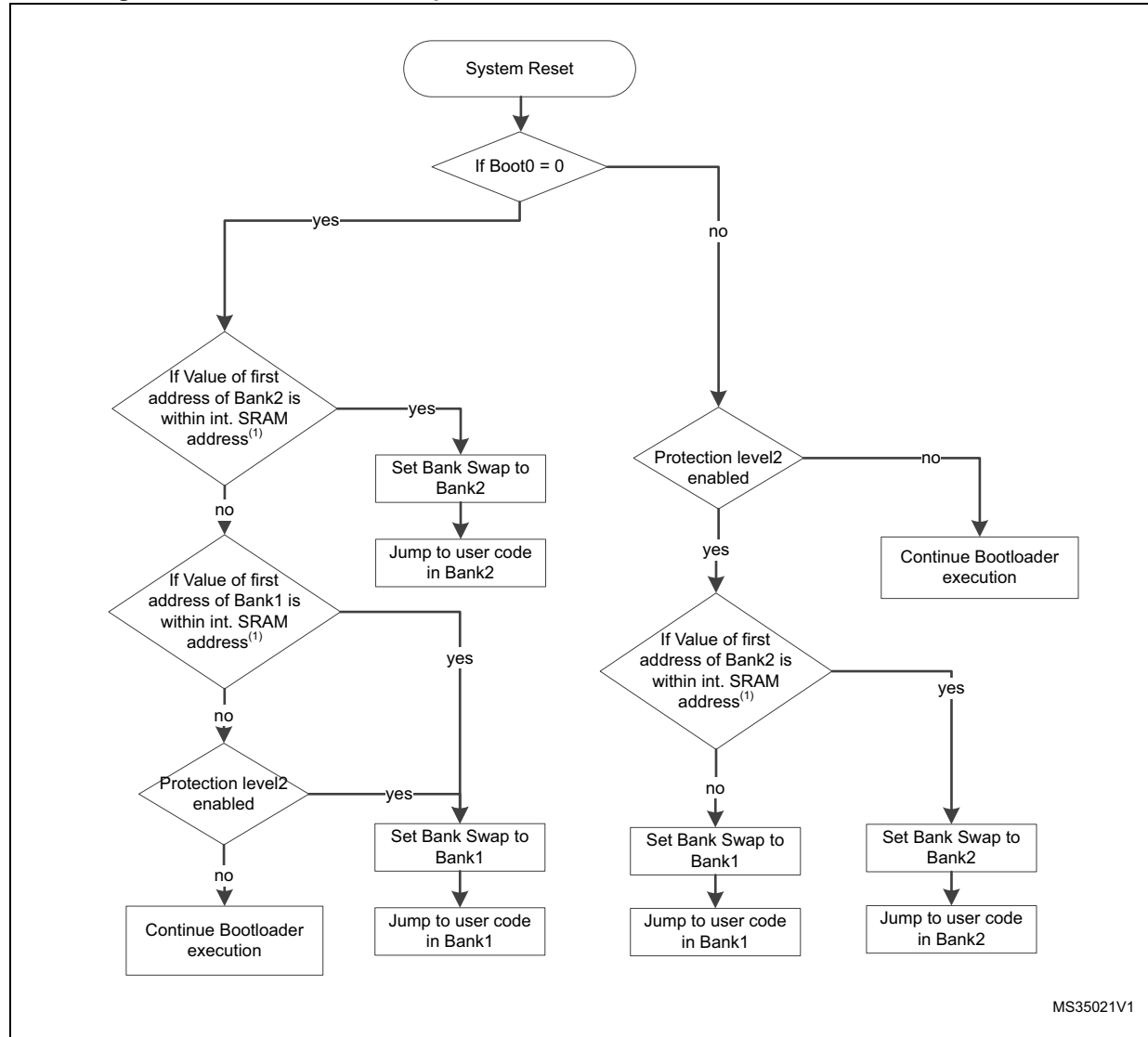
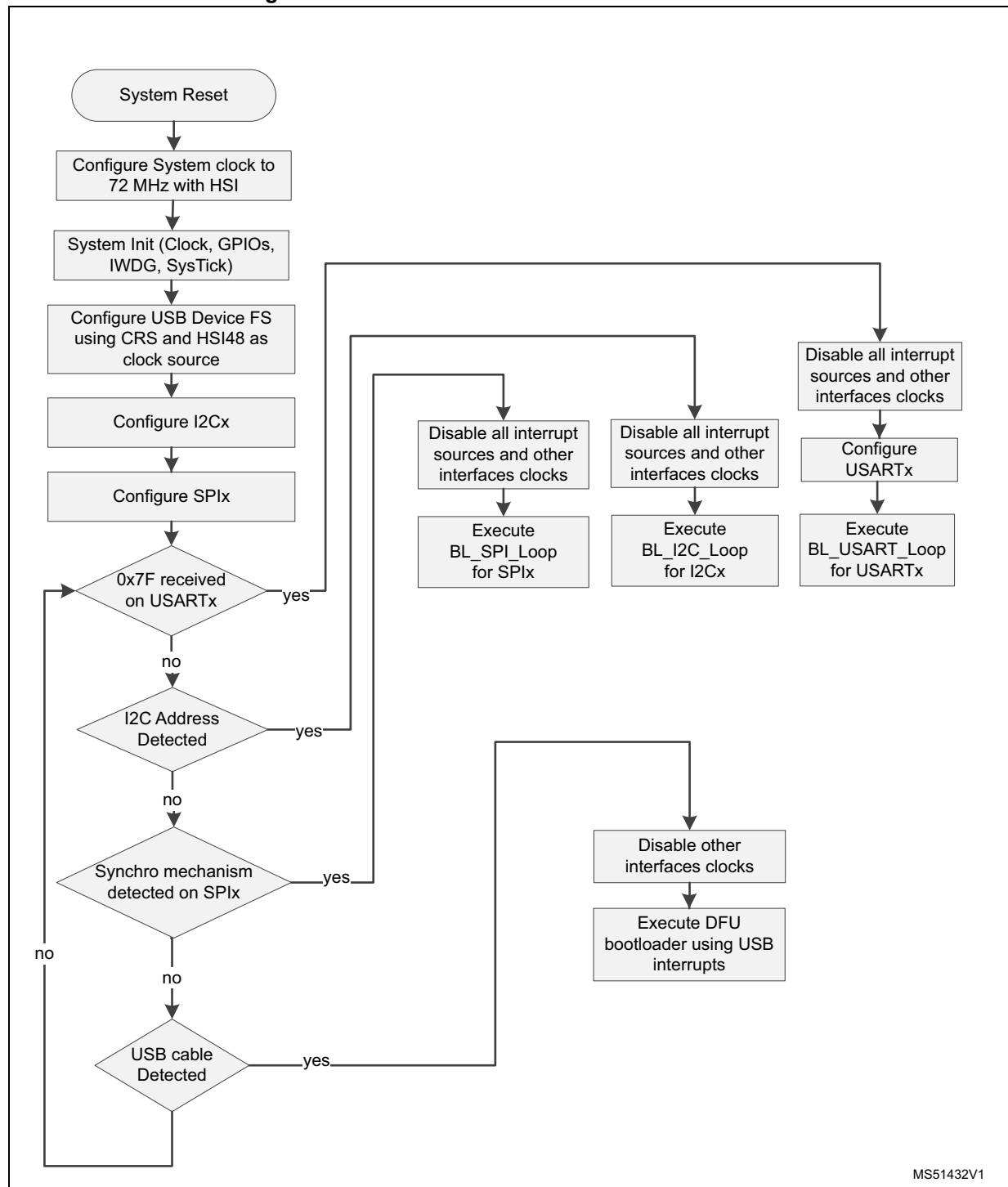


Figure 89. Bootloader V13.x selection for STM32L412xx/422xx



68.3 Bootloader version

[Table 151](#) lists the STM32L412xx/422xx devices bootloader version.

Table 151. STM32L412xx/422xx bootloader versions

Version number	Description	Known limitations
V13.1	Initial bootloader version	– On connection phase, USART responds with two ACK bytes (0x79) instead of one. – PcROP option bytes cannot be written as Bootloader uses Byte access while PcROP must be accessed using half-word access. Workaround: load a code snippet in SRAM using Bootloader interface, then jump to it, and that code writes PcROP value.

69 STM32L43xxx/44xxx devices

69.1 Bootloader configuration

The bootloader V9.1 version is updated to fix known limitations relative to USB-DFU interface, and is implemented on devices with version information ID equal to 0x10 (refer to [Table 153](#) for more details).

The STM32L43xxx/44xxx bootloader is activated by applying Pattern 16 (described in [Table 2](#)). [Table 152](#) shows the hardware resources used by this bootloader.

Table 152. STM32L43xxx/44xxx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The HSI is used at startup as clock source for system clock configured to 60 MHz and for USART, I2C, SPI and USB bootloader operation.
		-	CRS is enabled for the DFU to allow USB to be clocked by HSI48 48 MHz.
		HSE enabled	The HSE is used only when the CAN interface is selected. The HSE must have one of the following values [24,20,18,16,12,9,8,6,4] MHz.
		-	The CSS interrupt is enabled when HSE is enabled. Any failure (or removal) of the external clock generates system reset
	RAM	-	12 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	28 Kbytes, starting from address 0x1FFF0000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
	Power	-	The DFU cannot be used to communicate with bootloader if the voltage scaling range 2 is selected. Bootloader firmware does not configure voltage scaling range value in PWR_CR1 register.
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input no pull mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in input no pull mode.

Table 152. STM32L43xxx/44xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in input pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in input pull-up mode.
USART3	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. Used in input pull-up mode.
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 1 MHz, 7-bit address, Target mode, Analog filter ON – Target 7-bit address: 0b1001000x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin	Input/output	PB7 pin: data line is used in open-drain no pull mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 1 MHz, 7-bit address, Target mode, Analog filter ON – Target 7-bit address: 0b1001000x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PB10 pin: clock line is used in open-drain no pull mode.
	I2C2_SDA pin	Input/output	PB11 pin: data line is used in open-drain no pull mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 1 MHz, 7-bit address, Target mode, Analog filter ON – Target 7-bit address: 0b1001000x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PC0 pin: clock line is used in open-drain no pull mode.
	I2C3_SDA pin	Input/output	PC1 pin: data line is used in open-drain no pull mode.

Table 152. STM32L43xxx/44xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware
	SPI2_MOSI pin	Input	PB15 pin: slave data input line, used in push-pull, pull-down mode
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, pull-down mode
	SPI2_SCK pin	Input	PB13 pin: slave clock line, used in push-pull, pull-down mode
	SPI2_NSS pin	Input	PB12 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.
CAN1	CAN1	Enabled	Once initialized the CAN1 configuration is: Baudrate 125 kbps, 11 -bit identifier.
	CAN1_RX pin	Input	PB8 pin: CAN1 in reception mode. Used in alternate push-pull, pull-up mode.
	CAN1_TX pin	Output	PB9 pin: CAN1 in transmission mode. Used in alternate push-pull, pull-up mode.
	TIM16	Enabled	This timer is used to determine the value of the HSE. Once the HSE frequency is determined, the system clock is configured to 60 MHz using PLL and HSE.

Table 152. STM32L43xxx/44xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
DFU	USB	Enabled	USB FS configured in forced device mode. USB FS interrupt vector is enabled and used for USB DFU communications. Note: VDDUSB IO must be connected to 3.3 V as USB peripheral is used by the bootloader.
	USB_DM pin	Input/output	PA11: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12: USB DP line. Used in alternate push-pull, no pull mode. No external pull-up resistor is required

Note: If VDDUSB pin is not connected to V_{DD} , SPI flash memory write operations may be corrupted due to voltage issue. For more details, refer to product's datasheet and errata sheet.

SPI Tx (MISO) is handled by DMA. On the bootloader start-up after SPI initialization, as soon as the bit DMATx enable on SPI CR2 register is set to 0x1, the MISO line is set to 3.3 V.

69.2 Bootloader selection

The following figures show the bootloader selection mechanism.

Figure 90. Dual bank boot Implementation for STM32L3x2xx/44xxx bootloader V9.x

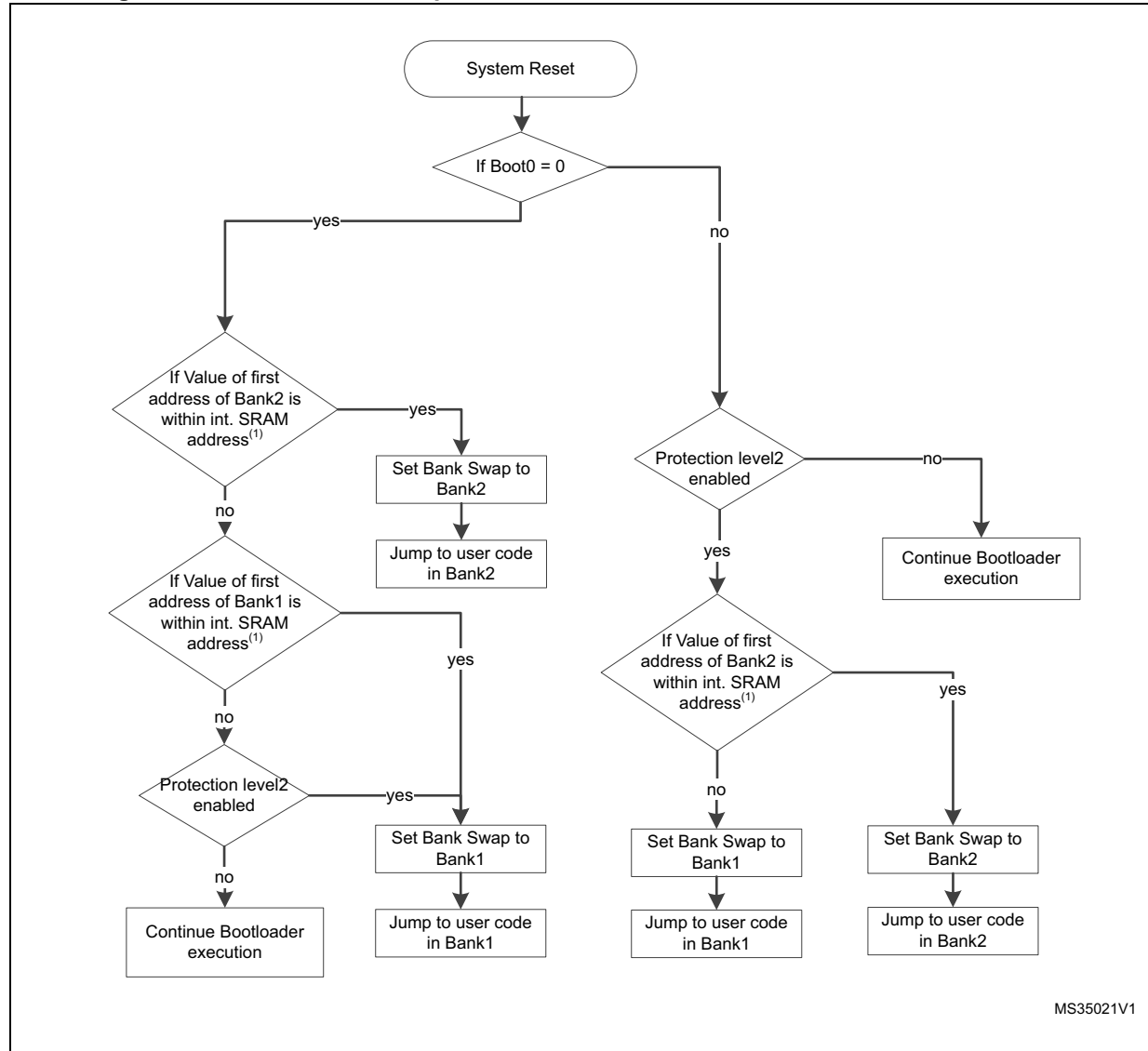
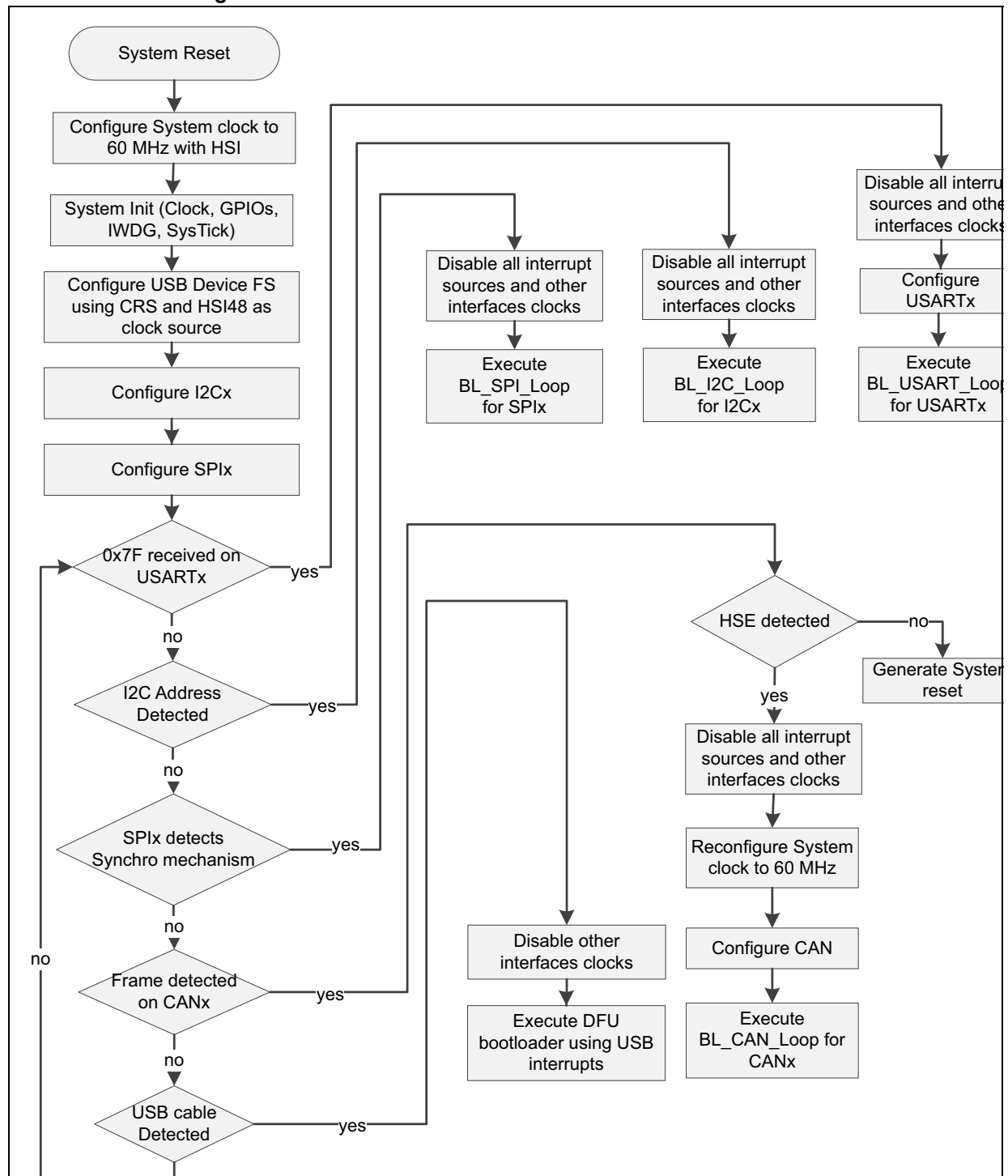


Figure 91. Bootloader V9.x selection for STM32L43xxx/44xxx



69.3 Bootloader version

[Table 153](#) lists the STM32L43xxx/44xxx devices bootloader versions.

Table 153. STM32L43xxx/44xxx bootloader versions

Version number	Description	Known limitations
V9.1	Initial bootloader version	Check the Version Information ID of your STM32L43xxx/44xxx device, which can be read at 0x1FFF6FF2 address. Version Information ID equal to 0xFF: - For memory write operations using DFU interface: If the buffer size is larger than 256 bytes and not multiple of 8 bytes, the write memory operation result is corrupted. Workaround: if the file size is larger than 256 bytes, add byte padding to align it on 8-bytes multiple size. - For the USB-DFU interface, the CRS (clock recovery system) is not correctly configured and this may lead to random USB communication errors (depending on temperature and voltage). In most case communication error will manifest by a "Stall" response to setup packets. - On the Go command, system bootloader de-init clears the RTCAPBEN bit in the RCC_APB1ENR register Workaround: manually call <code>__HAL_RCC_RTC_CLK_ENABLE()</code> in the software which sets the RTCAPBEN bit. Version Information ID equal to 0x10: None - PcROP option bytes cannot be written as Bootloader uses Byte access while PcROP must be accessed using half-word access. Workaround: load a code snippet in SRAM using Bootloader interface then jump to it, and that code writes the PcROP value.

Table 153. STM32L43xxx/44xxx bootloader versions (continued)

Version number	Description	Known limitations
V9.1 (continued)	Initial bootloader version (continued)	– SPI write operation fail Limitation: a. During bootloader SPI write flash memory operation, some random 64-bits (2 double-words) may be left blank at 0xFF. Root cause: a. Bootloader uses 64-bits cast write operation which is interrupted by SPI DMA and it leads to double access on same flash memory address and the 64-bits are not written. Workarounds: a. WA1: add a delay between sending write command and its ACK request. Its duration must be the duration of the 256-Bytes flash memory write time. b. WA2: read back after write and in case of error start write again. c. WA3: Patch in RAM to write in flash memory that implements write memory without 64-bits cast. WA1 and WA3 are more efficient than WA2 in terms of total programming time. How critical is the limitation: a. The limitation leads to a modification in customer SPI host software by adding 3-4 ms delay to each write operation. b. The delay is not waste because it is anyway the flash memory write period of time that host has to wait anyway (so instead of waiting by sending ACK requests, host will wait by delay). c. Limitation has been seen only on SPI and cannot impact USART/I2C/CAN – If the RTC is used by application prior to booting (through a system reset) on system bootloader, it is possible that CAN interface does not work correctly (cannot establish connection) unless a power cycle is performed or RTC is reset by application before booting on System Bootloader

70 STM32L45xxx/46xxx devices

70.1 Bootloader configuration

The STM32L45xxx/46xxx bootloader is activated by applying Pattern 16 (described in [Table 2](#)). [Table 154](#) shows the hardware resources used by this bootloader.

Table 154. STM32L45xxx/46xxx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The HSI is used at startup as clock source for system clock configured to 72 MHz and for USART, I2C, SPI and USB bootloader operation.
		-	CRS is enabled for the DFU to allow USB to be clocked by HSI48 48 MHz.
		HSE enabled	The system clock frequency is 60 MHz. The HSE is used only when the CAN interface is selected. The HSE must have one of the following values [24,20,18,16,12,9,8,6,4] MHz.
		-	The CSS interrupt is enabled when HSE is enabled. Any failure (or removal) of the external clock generates system reset
	RAM	-	12 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	28 Kbytes, starting from address 0x1FFF0000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	Power	-	The DFU cannot be used to communicate with bootloader if the voltage scaling range 2 is selected. Bootloader firmware does not configure voltage scaling range value in PWR_CR1 register.
	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input no pull mode.
USART2	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in input no pull mode.
	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in input pull-up mode.
USART2	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in input pull-up mode.

Table 154. STM32L45xxx/46xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
USART3	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. Used in input pull-up mode.
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 1 MHz, 7-bit address, Target mode, Analog filter ON – Target 7-bit address: 0b1001010x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin	Input/output	PB7 pin: data line is used in open-drain no pull mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 1 MHz, 7-bit address, Target mode, Analog filter ON – Target 7-bit address: 0b1001010x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PB10 pin: clock line is used in open-drain no pull mode.
	I2C2_SDA pin	Input/output	PB11 pin: data line is used in open-drain no pull mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 1 MHz, 7-bit address, Target mode, Analog filter ON – Target 7-bit address: 0b1001010x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PC0 pin: clock line is used in open-drain no pull mode.
	I2C3_SDA pin	Input/output	PC1 pin: data line is used in open-drain no pull mode.

Table 154. STM32L45xxx/46xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PB15 pin: slave data input line, used in push-pull, pull-down mode
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, pull-down mode
	SPI2_SCK pin	Input	PB13 pin: slave clock line, used in push-pull, pull-down mode
	SPI2_NSS pin	Input	PB12 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.
CAN1	CAN1	Enabled	Once initialized the CAN1 configuration is: Baudrate 125 kbps, 11 -bit identifier.
	CAN1_RX pin	Input	PB8 pin: CAN1 in reception mode. Used in alternate push-pull, pull-up mode.
	CAN1_TX pin	Output	PB9 pin: CAN1 in transmission mode. Used in alternate push-pull, pull-up mode.
	TIM16	Enabled	This timer is used to determine the value of the HSE. Once the HSE frequency is determined, the system clock is configured to 60 MHz using PLL and HSE.

Table 154. STM32L45xxx/46xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
DFU	USB	Enabled	USB FS configured in forced device mode. USB FS interrupt vector is enabled and used for USB DFU communications. Note: VDDUSB IO must be connected to 3.3 V as USB peripheral is used by the bootloader.
	USB_DM pin	Input/output	PA11: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12: USB DP line. Used in alternate push-pull, no pull mode. No external pull-up resistor is required

Note: If VDDUSB pin is not connected to V_{DD} , SPI flash memory write operations may be corrupted due to voltage issue. For more details, refer to product's datasheet and errata sheet.

70.2 Bootloader selection

The following figures show the bootloader selection mechanism.

Figure 92. Dual bank boot implementation for STM32L45xxx/46xxx bootloader V9.x

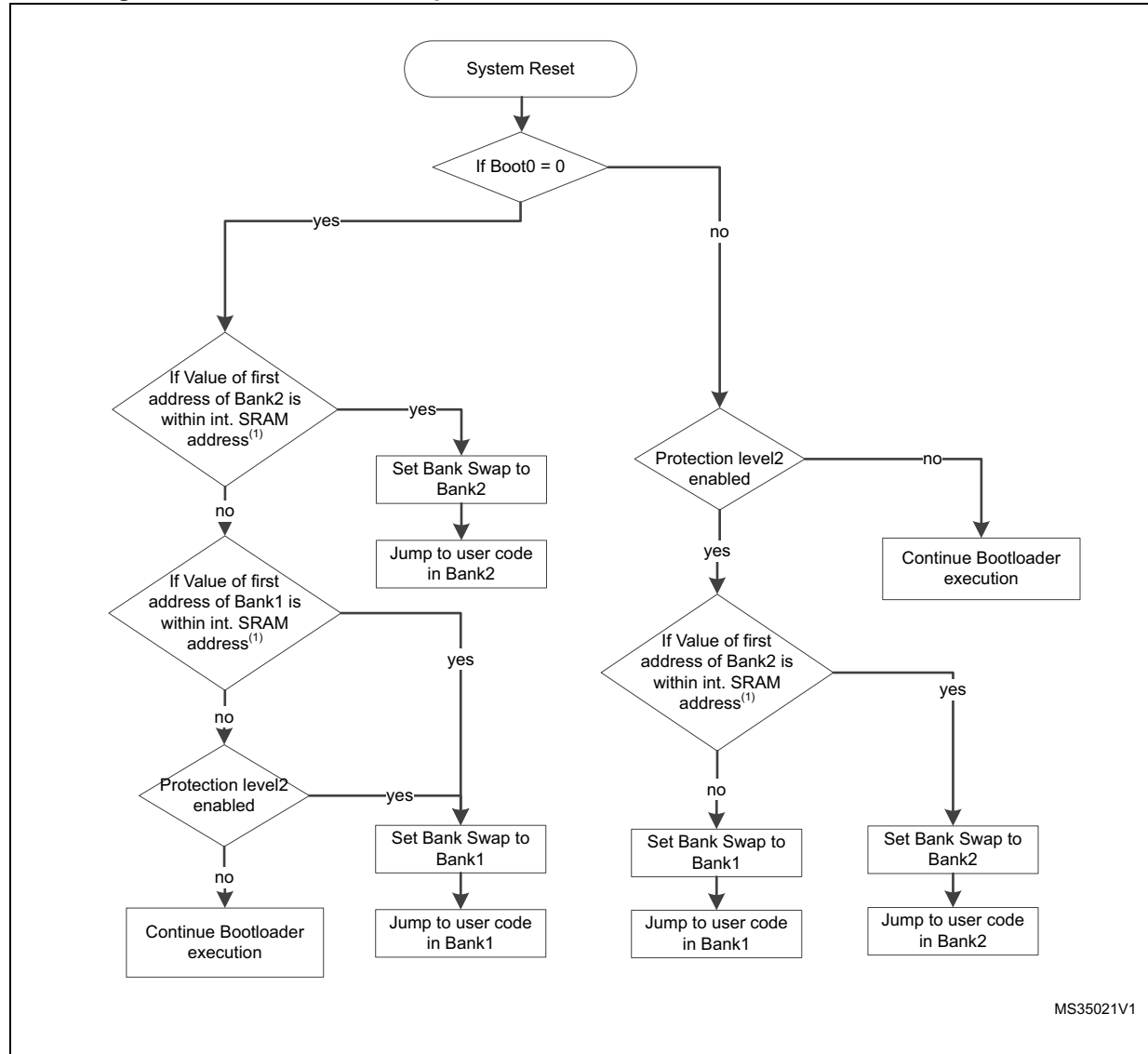
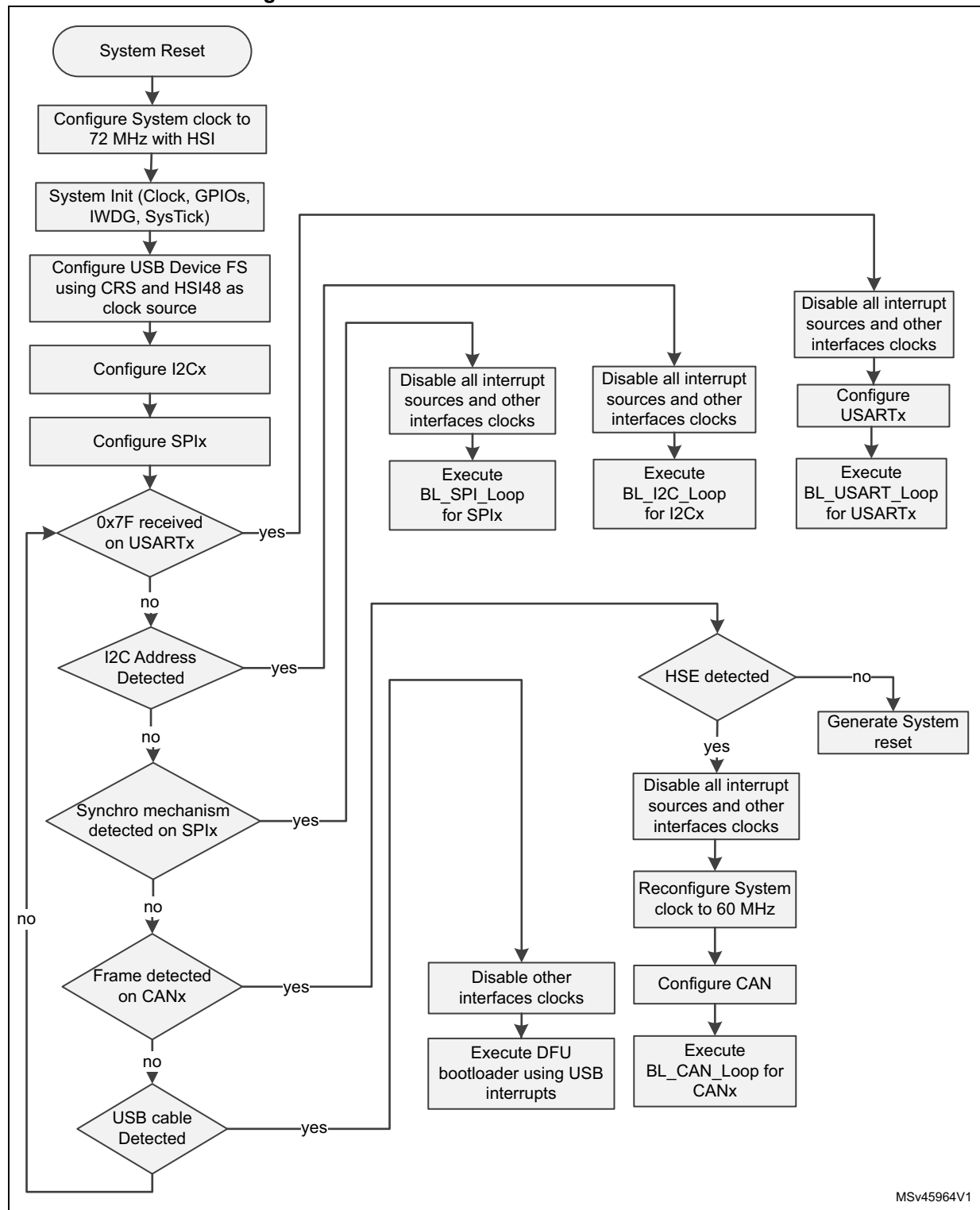


Figure 93. Bootloader V9.x selection for STM32L45xxx/46xxx



MSv45964V1

70.3 Bootloader version

Table 155 lists the STM32L45xxx/46xxx devices bootloader versions.

Table 155. STM32L45xxx/46xxx bootloader versions

Version number	Description	Known limitations
V9.2	Initial bootloader version	– PcROP option bytes cannot be written as Bootloader uses Byte access while PcROP must be accessed using half-word access. Workaround: load a code snippet in SRAM using Bootloader interface then jump to it, and that code writes the PcROP value. – SPI write operation fail limitation: a. During Bootloader SPI write flash memory operation, some random 64-bits (2 double-words) may be left blank at 0xFF. Root cause: a. Bootloader uses 64-bits cast write operation which is interrupted by SPI DMA and it leads to double access on same flash memory address and the 64-bits are not written Workarounds: a. WA1: add a delay between sending write command and its ACK request. Its duration must be the duration of the 256-Bytes flash memory write time. b. WA2: read back after write and in case of error start write again. c. WA3: Patch in RAM to write in flash memory that implements write memory without 64-bits cast. WA1 and WA3 are more efficient than WA2 in terms of total programming time How critical is the limitation: a. The limitation leads to a modification in customer SPI host software by adding 3-4 ms delay to each write operation. b. The delay is not waste because it is anyway the flash memory write period of time that host has to wait anyway (so instead of waiting by sending ACK requests, host will wait by delay). c. Limitation has been seen only on SPI and cannot impact USART/I2C/CAN.

71 STM32L47xxx/48xxx devices

Two bootloader versions are available:

- V10.x supporting USART, I2C and DFU (USB FS device).
This version is embedded in STM32L47xxx/48xxx rev. 2 and rev. 3 devices.
- V9.x supporting USART, I2C, SPI, CAN and DFU (USB FS device).
This version is embedded in STM32L47xxx/48xxx rev. 4 devices.

71.1 Bootloader V10.x

71.1.1 Bootloader configuration

The STM32L47xxx/48xxx bootloader is activated by applying Pattern 7 (described in [Table 2](#)). [Table 156](#) shows the hardware resources used by this bootloader.

Table 156. STM32L47xxx/48xxx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The HSI is used at startup as clock source for system clock configured to 24 MHz and for USART and I2C bootloader operation.
		HSE enabled	The HSE is used only when the USB interface is selected and the LSE is not present. The HSE must have one of the following values: 24, 20, 18, 16, 12, 9, 8, 6, and 4 MHz.
		LSE enabled	The LSE is used to trim the MSI which is configured to 48 MHz as USB clock source. The LSE must be equal to 32.768 kHz. If the LSE is not detected, the HSE is used instead if USB is connected.
		MSI enabled	The MSI is configured to 48 MHz and is used as USB clock source. The MSI is used only if LSE is detected, otherwise, HSE is used if USB is connected.
		-	The CSS interrupt is enabled when LSE or HSE is enabled. Any failure (or removal) of the external clock generates system reset.
	RAM	-	12 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	28 Kbytes, starting from address 0x1FFF0000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
	Power	-	The DFU cannot be used to communicate with bootloader if the voltage scaling range 2 is selected. Bootloader firmware does not configure voltage scaling range value in PWR_CR1 register.

Table 156. STM32L47xxx/48xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input no pull mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in input no pull mode.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in input pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in input pull-up mode.
USART3	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. Used i input pull-up mode.
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1000011x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain no pull mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1000011x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PB10 pin: clock line is used in open-drain no pull mode.
	I2C2_SDA pin		PB11 pin: data line is used in open-drain no pull mode.

Table 156. STM32L47xxx/48xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address is 0b1000011x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PC0 pin: clock line is used in open-drain no pull mode.
	I2C3_SDA pin		PC1 pin: data line is used in open-drain no pull mode.
DFU	USB	Enabled	USB OTG FS configured in forced device mode
	USB_DM pin	Input/output	PA11: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12: USB DP line. Used in alternate push-pull, no pull mode. No external pull-up resistor is required
	TIM17	Enabled	This timer is used to determine the value of the HSE. Once the HSE frequency is determined, the system clock is configured to 24 MHz using PLL and HSE.

For USARTx and I2Cx bootloaders no external clock is required.

USB bootloader (DFU) requires either an LSE (low-speed external clock) or a HSE (high-speed external clock):

- If the LSE is present regardless of the HSE presence, the MSI is configured and trimmed by the LSE to provide an accurate clock equal to 48 MHz, which is the clock source of the USB. The system clock is kept clocked to 24 MHz by the HSI.
- If the HSE is present, the system clock and USB clock are configured, respectively, to 24 MHz and 48 MHz with HSE as clock source.

71.1.2 Bootloader selection

Figure 94 and Figure 95 show the bootloader selection mechanism.

Figure 94. Dual bank boot implementation for STM32L47xxx/48xxx bootloader V10.x

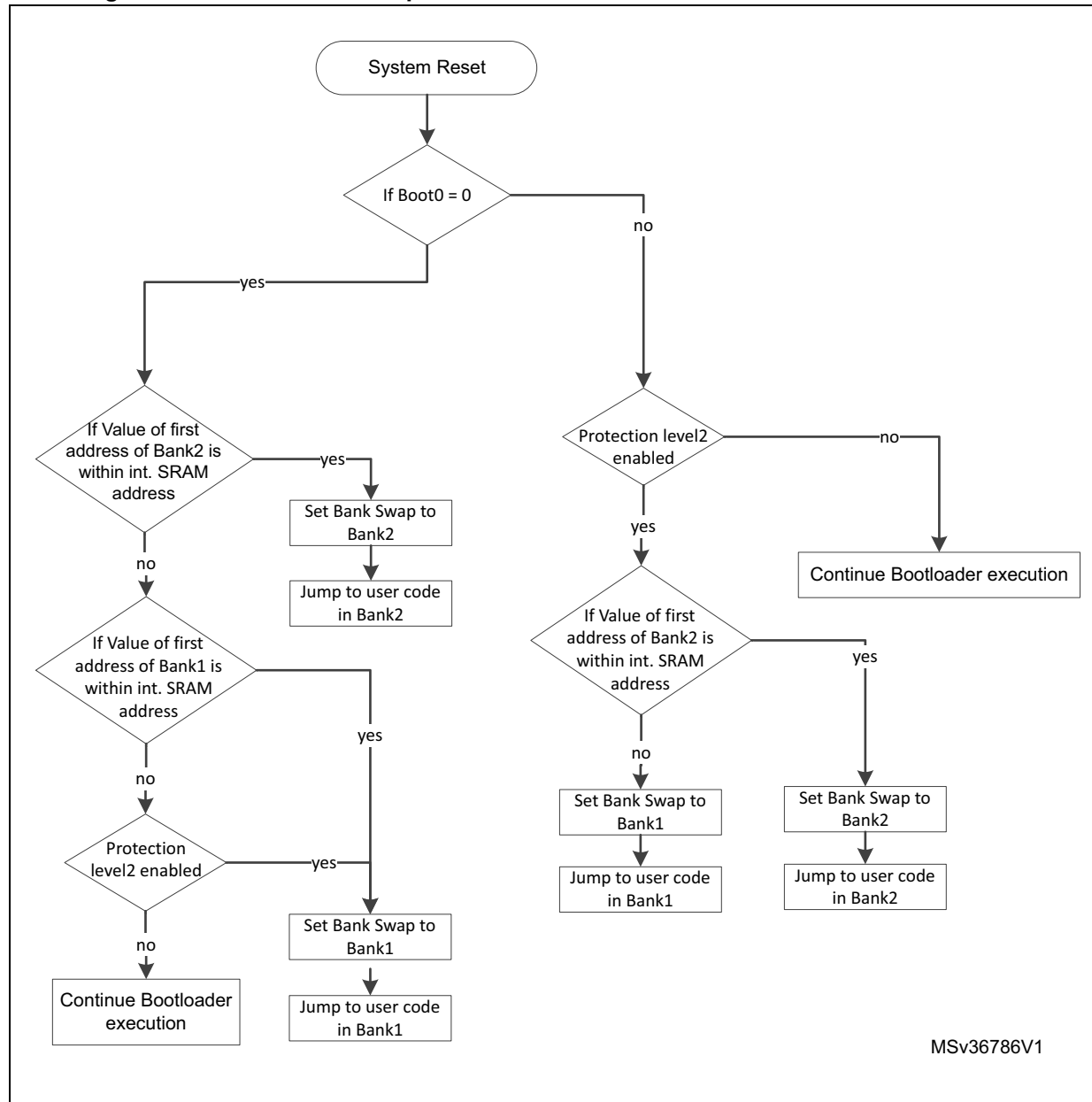
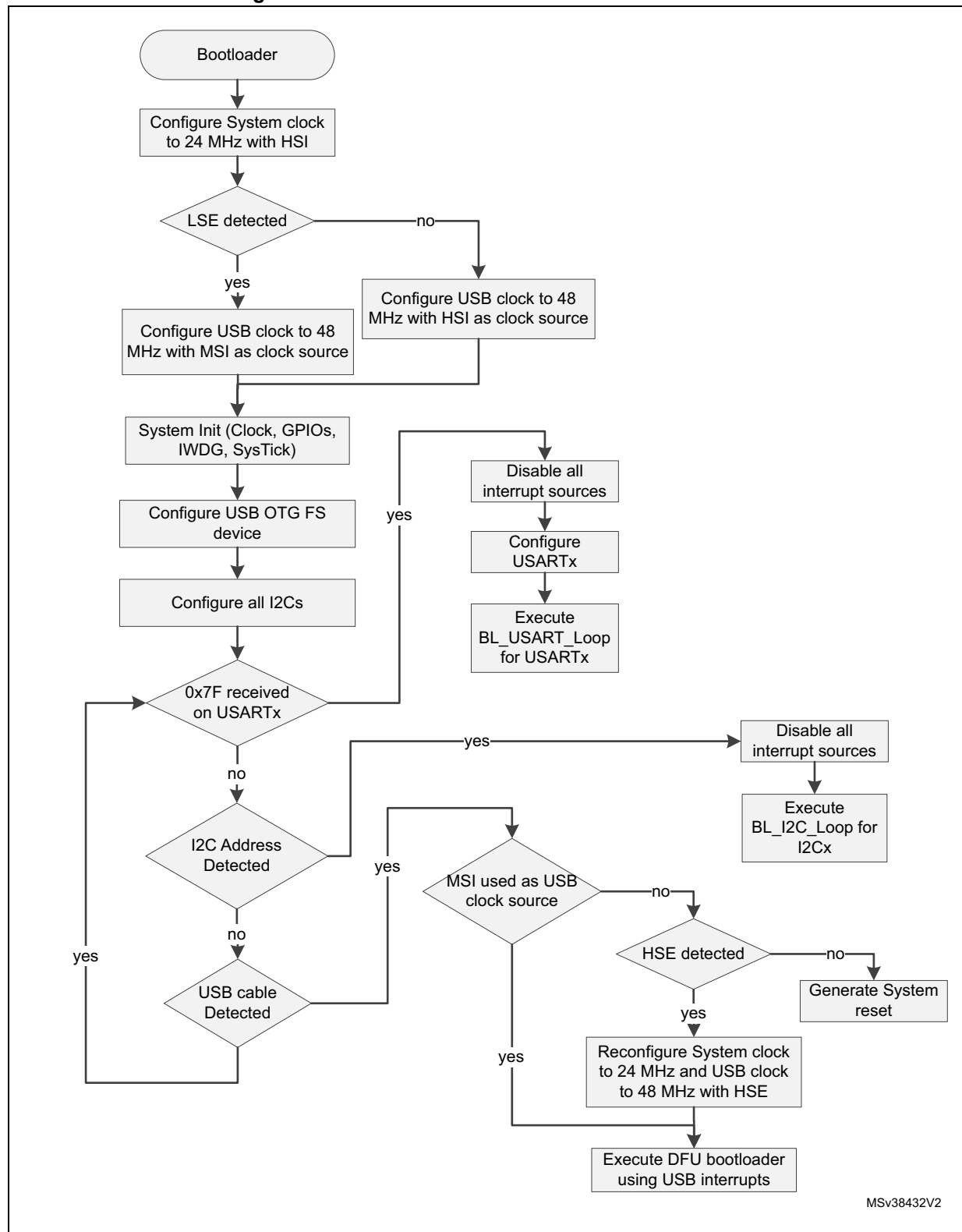


Figure 95. Bootloader V10.x selection for STM32L47xxx/48xxx



71.1.3 Bootloader version

Table 157 lists the STM32L47xxx/48xxx devices bootloader V10.x versions:

Table 157. STM32L47xxx/48xxx bootloader V10.x versions

Version number	Description	Known limitations
V10.1	Initial bootloader version	For memory write operations using DFU interface: If the buffer size is larger than 256 bytes and not multiple of 8 bytes, the write memory operation result is corrupted. Workaround: if the file size is larger than 256 bytes, add byte padding to align it on 8-bytes multiple size. Write in SRAM is corrupted.
V10.2	Fix write in SRAM issue	For memory write operations using DFU interface: If the buffer size is larger than 256 bytes and not multiple of 8 bytes, the write memory operation result is corrupted. Workaround: if the file size is larger than 256 bytes, add byte padding to align it on 8-bytes multiple size.
V10.3	Add support of MSI as USB clock source (MSI is trimmed by LSE). Update dual bank boot feature to support the case when user stack is mapped in SRAM2.	- For memory write operations using DFU interface: If the buffer size is larger than 256 bytes and not multiple of 8 bytes, the write memory operation result is corrupted. Workaround: if the file size is larger than 256 bytes, add byte padding to align it on 8-bytes multiple size. - PcROP option bytes cannot be written as Bootloader uses Byte access while PcROP must be accessed using half-word access. Workaround: load a code snippet in SRAM using Bootloader interface then jump to it, and that code writes the PcROP value.

71.2 Bootloader V9.x

71.2.1 Bootloader configuration

The STM32L47xxx/48xxx bootloader is activated by applying Pattern 7 (described in [Table 2](#)). [Table 158](#) shows the hardware resources used by this bootloader.

Table 158. STM32L47xxx/48xxx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The HSI is used at startup as clock source for system clock configured to 72 MHz and for USART and I2C bootloader operation.
		HSE enabled	The HSE is used only when the USB interface is selected and the LSE is not present. The HSE must have one of the following values: 24, 20, 18, 16, 12, 8, 6, 4 MHz. System is clocked at 72 MHz if USB is used or 60 MHz if CAN is used.
		LSE enabled	The LSE is used to trim the MSI which is configured to 48 MHz as USB clock source. The LSE must be equal to 32.768 kHz. If the LSE is not detected, the HSE is used instead if USB is connected.
		MSI enabled	The MSI is configured to 48 MHz and is used as USB clock source. The MSI is used only if LSE is detected, otherwise, HSE is used if USB is connected.
		CSS	The CSS interrupt is enabled when LSE or HSE is enabled. Any failure (or removal) of the external clock generates system reset.
	RAM	-	13 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	28 Kbytes, starting from address 0x1FFF0000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	Power	-	The DFU cannot be used to communicate with bootloader if the voltage scaling range 2 is selected. Bootloader firmware does not configure voltage scaling range value in PWR_CR1 register.
	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input no pull mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in input no pull mode.

Table 158. STM32L47xxx/48xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in input pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in input pull-up mode.
USART3	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. Used in input pull-up mode.
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode. Used in input pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1000011x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain no pull mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1000011x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PB10 pin: clock line is used in open-drain no pull mode.
	I2C2_SDA pin		PB11 pin: data line is used in open-drain no pull mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1000011x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PC0 pin: clock line is used in open-drain no pull mode.
	I2C3_SDA pin		PC1 pin: data line is used in open-drain no pull mode.

Table 158. STM32L47xxx/48xxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode ⁽¹⁾
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode
	SPI1_NSS pin		PA4 pin: slave chip select pin used in push-pull, pull-down mode.
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware
	SPI2_MOSI pin	Input	PB15 pin: slave data input line, used in push-pull, pull-down mode
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, pull-down mode ⁽¹⁾
	SPI2_SCK pin	Input	PB13 pin: slave clock line, used in push-pull, pull-down mode
	SPI2_NSS pin		PB12 pin: slave chip select pin used in push-pull, pull-down mode.
CAN1	CAN1	Enabled	Once initialized the CAN1 configuration is: Baudrate 125 kbps, 11-bit identifier.
	CAN1_RX pin	Input	PB8 pin: CAN1 in reception mode. Used in alternate push-pull, pull-up mode.
	CAN1_TX pin	Output	PB9 pin: CAN1 in transmission mode. Used in alternate push-pull, pull-up mode.
DFU	USB	Enabled	USB FS configured in forced device mode. USB FS interrupt vector is enabled and used for USB DFU communications. Note: VDDUSB IO must be connected to 3.3 V as USB peripheral is used by the bootloader.
	USB_DM pin	Input/output	PA11 pin: USB FS DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12 pin: USB FS DP line. Used in alternate push-pull, no pull mode. No external pull-up resistor is required.

1. SPI Tx (MISO) is handled by DMA. On the bootloader start-up after SPI initialization, as soon as the bit DMATx enable on SPI CR2 register is set to 0x1, the MISO line is set to 3.3 V.

If the HSE is present, the system clock and USB clock are configured, respectively, to 72 and 48 MHz with PLL (clocked by HSE) as a clock source.

Note: If VDDUSB pin is not connected to V_{DD} , SPI flash memory write operations may be corrupted due to voltage issue. For more details, refer to product's datasheet and errata sheet.

71.2.2 Bootloader selection

Figure 96 and Figure 97 show the bootloader selection mechanism.

Figure 96. Dual bank boot implementation for STM32L47xxx/48xxx bootloader V9.x

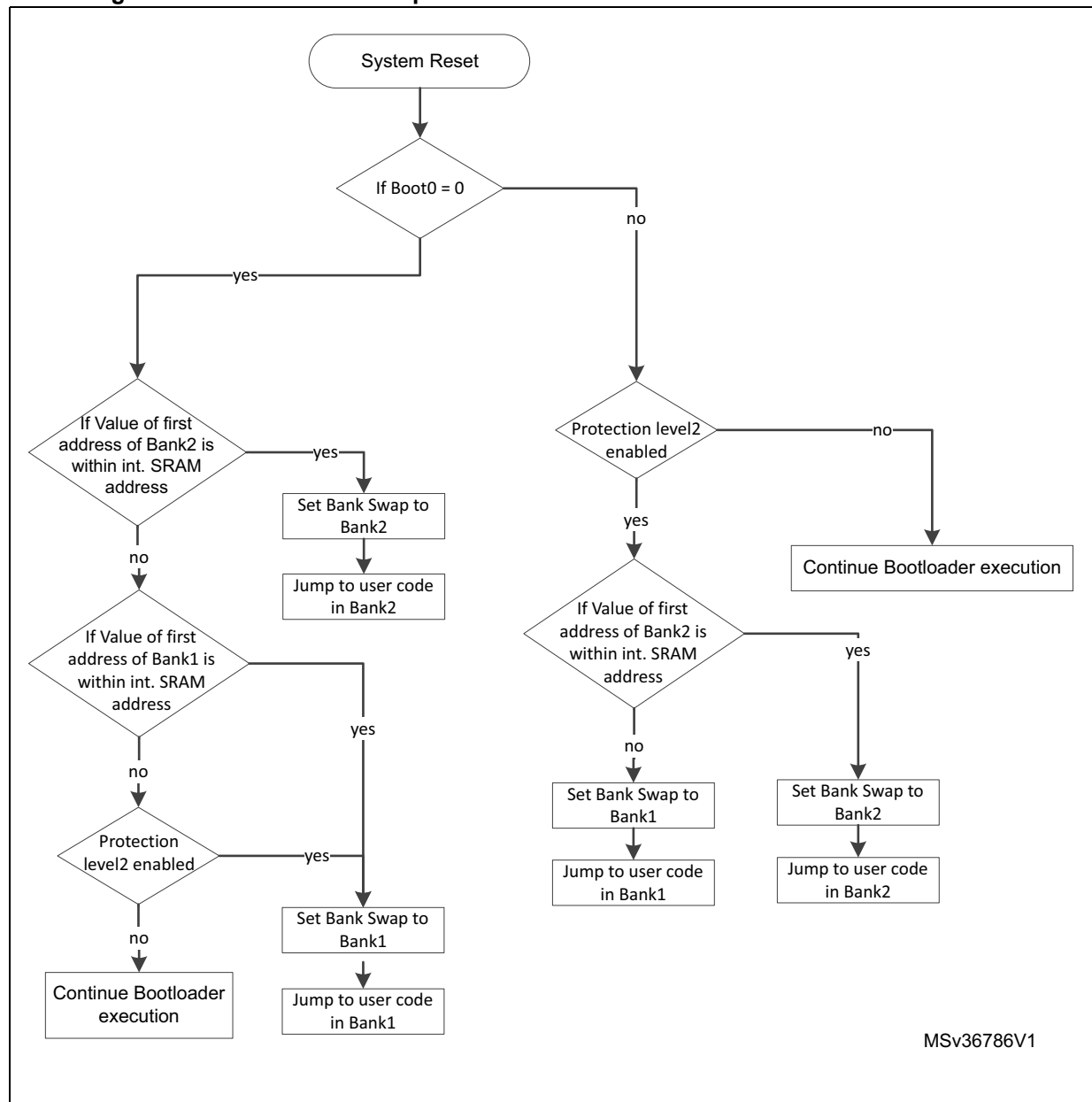
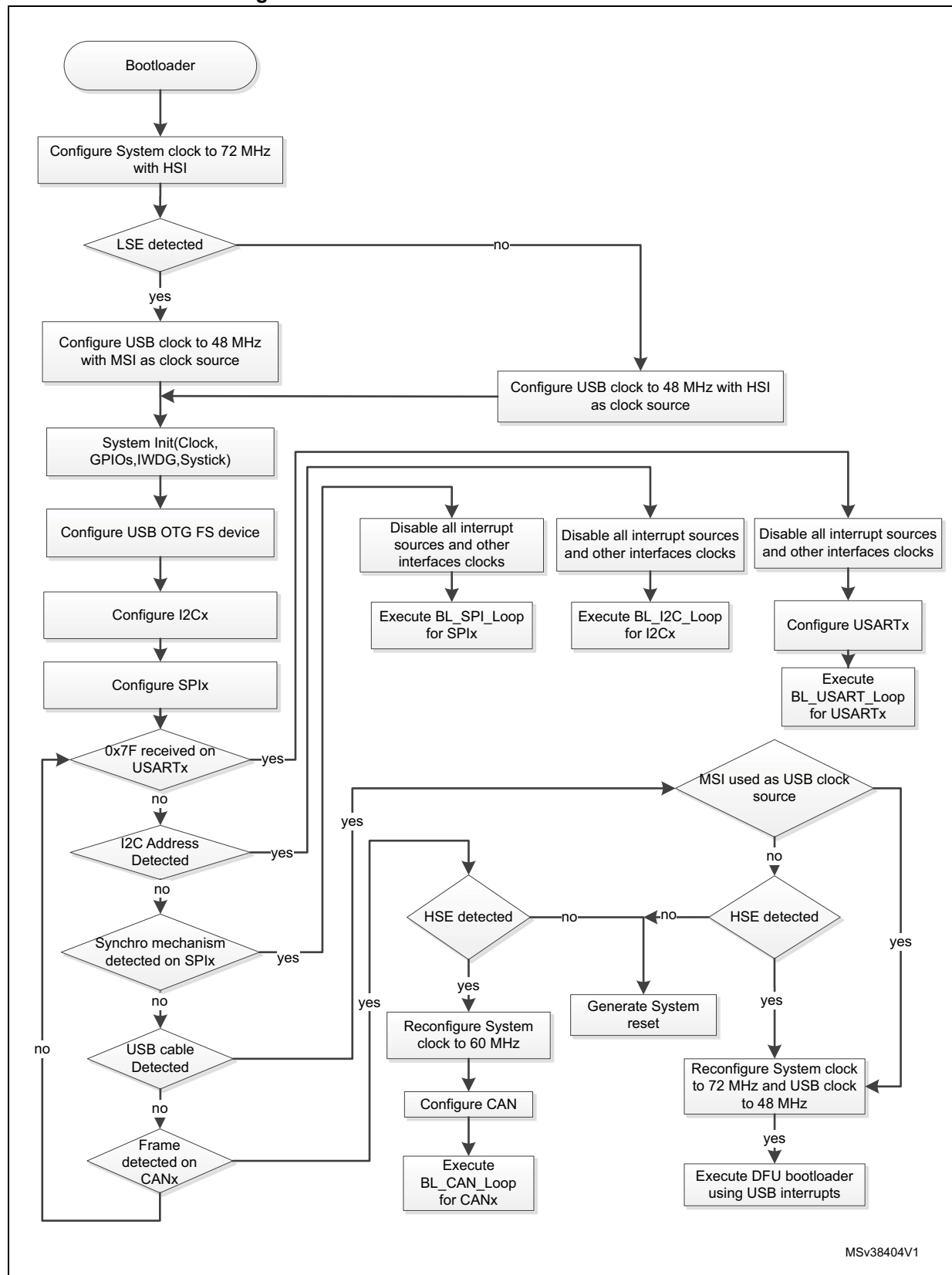


Figure 97. Bootloader V9.x selection for STM32L47xxx/48xxx



71.2.3 Bootloader version

Table 159 lists the STM32L47xxx/48xxx devices bootloader V9.x versions:

Table 159. STM32L47xxx/48xxx bootloader V9.x versions

Version number	Description	Known limitations
V9.0	Initial bootloader version	For memory write operations using DFU interface: If the buffer size is larger than 256 bytes and not multiple of 8 bytes, the write memory operation result is corrupted. Workaround: if the file size is larger than 256 bytes, add byte padding to align it on 8-bytes multiple size. Write in SRAM is corrupted
V9.1	Deprecated version (not used)	None
V9.2	Fix write in SRAM issue	For memory write operations using DFU interface: If the buffer size is larger than 256 bytes and not multiple of 8 bytes, the write memory operation result is corrupted. Workaround: if the file size is larger than 256 bytes, add byte padding to align it on 8-bytes multiple size. PcROP option bytes cannot be written as the bootloader uses byte access while PcROP must be accessed using half-word access. Workaround: load a code snippet in SRAM using Bootloader interface then jump to it, and that code writes the PcROP value. During bootloader SPI write flash memory operation, some random 64 bits (2 double-words) may be left blank at 0xFF. Root cause: the bootloader uses 64-bit cast write operation, interrupted by SPI DMA. This leads to double access on same flash memory address, and the 64 bits are not written. Workarounds: – WA1: add a delay between sending write command and its ACK request. Its duration must be the duration of the 256-byte flash memory write time. – WA2: read back after write, and, in case of error, start write again. – WA3: patch in RAM to write in flash memory that implements write without 64-bit cast. WA1 and WA3 are more efficient than WA2 in terms of total programming time. The limitation leads to a modification in customer SPI host software by adding 3-4 ms delay to each write operation. This time is not lost, because it is anyway the flash memory write time, the host must wait anyway (so instead of waiting by sending ACK requests, host waits by delay). Limitation has been seen only on SPI and cannot impact USART/I2C/CAN/USB.

72 STM32L496xx/4A6xx devices

72.1 Bootloader configuration

The STM32L496xx/4A6xx bootloader is activated by applying Pattern 6 (described in [Table 2](#)). [Table 160](#) shows the hardware resources used by this bootloader.

Table 160. STM32L496xx/4A6xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The HSI is used at startup as clock source for system clock configured to 72 MHz and for USART, I2C and SPI bootloader operation.
		-	CRS is enabled for the DFU to allow USB to be clocked by HSI 48 MHz.
		HSE enabled	The HSE is used only when the CAN interface is selected. The HSE must have one of the following values: 24,20,18,16,12,9,8,6,4 MHz.
		-	The CSS interrupt is enabled when HSE is enabled. Any failure (or removal) of the external clock generates system reset
	RAM	-	12 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	28 Kbytes, starting from address 0x1FFF0000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	Power	-	The DFU cannot be used to communicate with bootloader if voltage scaling range 2 is selected. Bootloader firmware does not configure voltage scaling range value in PWR_CR1 register.
	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input no pull mode.
USART2	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in input no pull mode.
	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in input no pull mode.
USART2	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in input no pull mode.

Table 160. STM32L496xx/4A6xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
USART3	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. Used in input no pull mode.
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode. Used in input no pull mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1001100x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain no pull mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1001100x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PB10 pin: clock line is used in open-drain no pull mode.
	I2C2_SDA pin		PB11 pin: data line is used in open-drain no pull mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1001100x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PC0 pin: clock line is used in open-drain no pull mode.
	I2C3_SDA pin		PC1 pin: data line is used in open-drain no pull mode.

Table 160. STM32L496xx/4A6xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PB15 pin: slave data input line, used in push-pull, pull-down mode
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, pull-down mode
	SPI2_SCK pin	Input	PB13 pin: slave clock line, used in push-pull, pull-down mode
	SPI2_NSS pin	Input	PB12 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.
CAN1	CAN1	Enabled	Once initialized the CAN1 configuration is: Baudrate 125 kbps, 11-bit identifier.
	CAN1_RX pin	Input	PB8 pin: CAN1 in reception mode. Used in alternate push-pull, pull-up mode.
	CAN1_TX pin	Output	PB9 pin: CAN1 in transmission mode. Used in alternate push-pull, pull-up mode.
	TIM16	Enabled	This timer is used to determine the value of the HSE. Once the HSE frequency is determined, the system clock is configured to 60 MHz using PLL and HSE.

Table 160. STM32L496xx/4A6xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
DFU	USB	Enabled	USB OTG FS configured in forced device mode. USB OTG FS interrupt vector is enabled and used for USB DFU communications. Note: VDDUSB IO must be connected to 3.3 V as USB peripheral is used by the bootloader.
	USB_DM pin	Input/output	PA11: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12: USB DP line. Used in alternate push-pull, no pull mode. No external pull-up resistor is required

Note: *If VDDUSB pin is not connected to V_{DD} , SPI flash memory write operations may be corrupted due to voltage issue. For more details, refer to product's datasheet and errata sheet.*

72.2 Bootloader selection

The figures below show the bootloader selection mechanism.

Figure 98. Dual bank boot Implementation for STM32L496xx/4A6xx bootloader V9.x

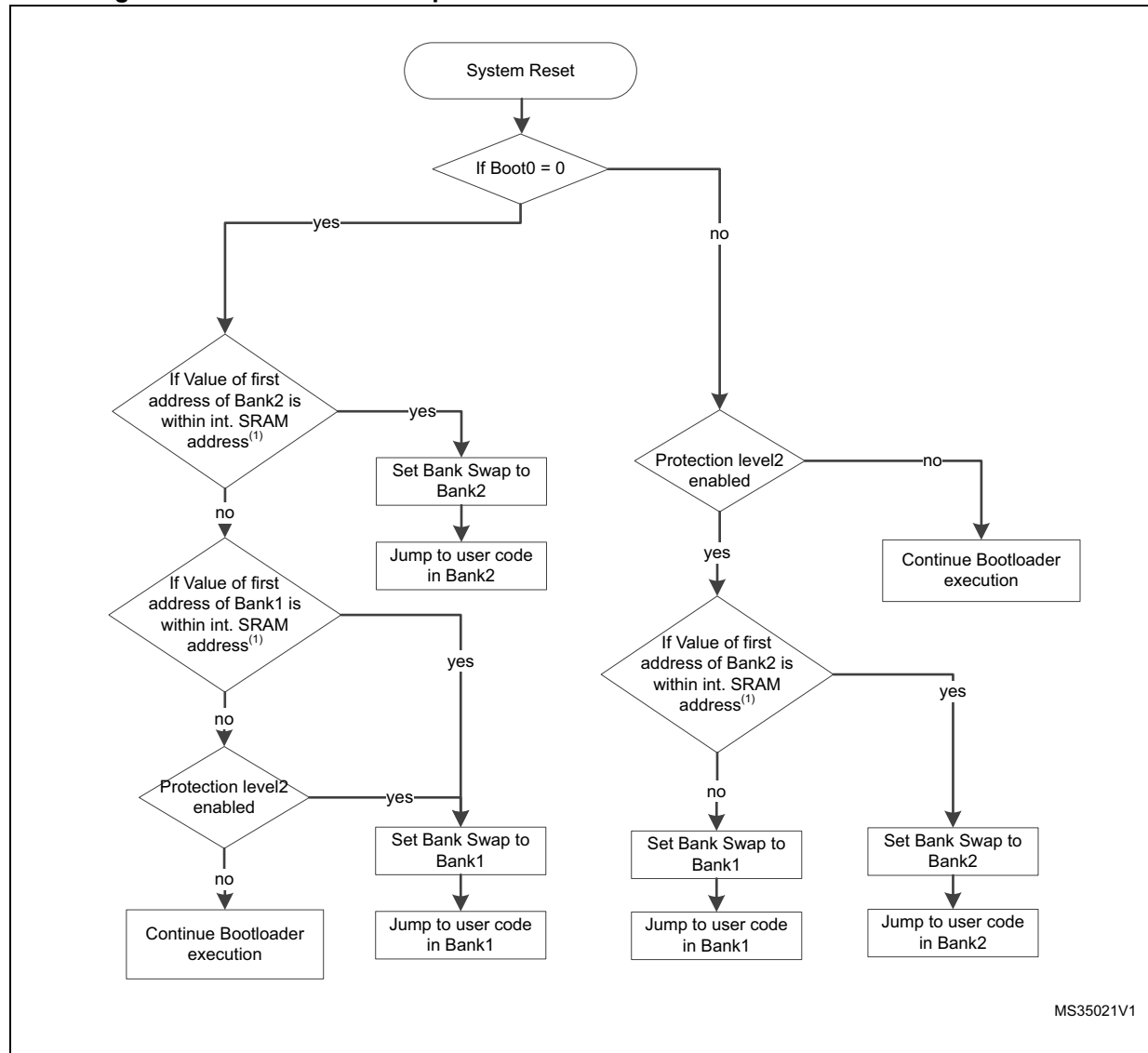
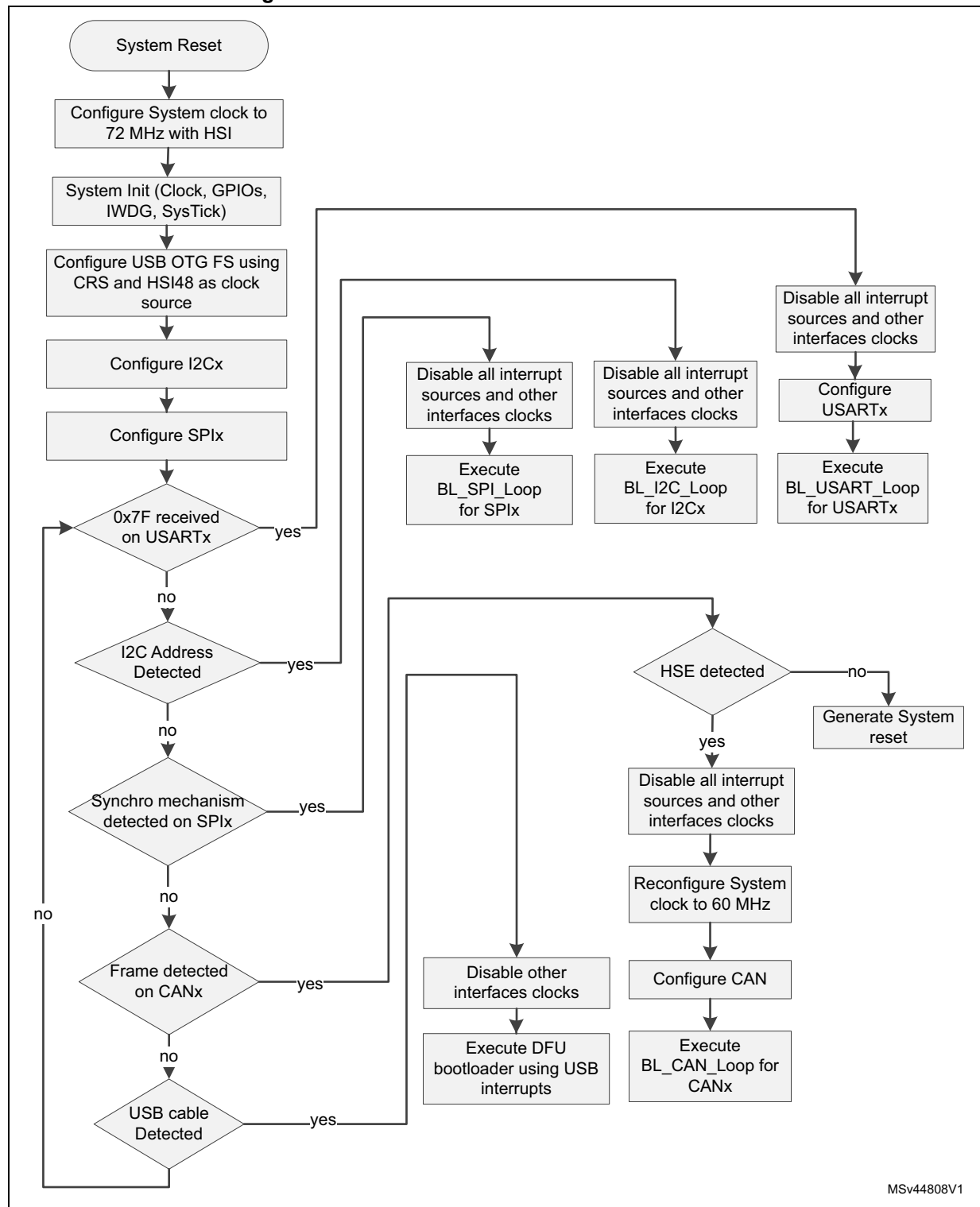


Figure 99. Bootloader V9.x selection for STM32L496xx/4A6xx



72.3 Bootloader version

[Table 161](#) lists the STM32L496xx/4A6xx devices bootloader versions.

Table 161. STM32L496xx/4A6xx bootloader version

Version number	Description	Known limitations
V9.3	Initial bootloader version	– The Bank Erase command is aborted by the bootloader device, and the NACK (0x1F) is sent to the host. Workaround: Perform Bank erase operation through page erase using the Erase command (0x44). – SPI write operation fail Limitation: a. During Bootloader SPI write flash memory operation, some random 64-bits (2 double-words) may be left blank at 0xFF. Root cause: a. Bootloader uses 64-bits cast write operation which is interrupted by SPI DMA and it leads to double access on same flash memory address and the 64-bits are not written Workarounds: a. WA1: add a delay between sending write command and its ACK request. Its duration must be the duration of the 256-Bytes flash memory write time. b. WA2: read back after write and in case of error start write again. c. WA3: Patch in RAM to write in flash memory that implements write memory without 64-bits cast. WA1 and WA3 are more efficient than WA2 in terms of total programming time How critical is the limitation: a. The limitation leads to a modification in customer SPI host software by adding 3-4 ms delay to each write operation. b. The delay is not waste because it is anyway the flash memory write period of time that host has to wait anyway (so instead of waiting by sending ACK requests, host will wait by delay). c. Limitation has been seen only on SPI and cannot impact USART/I2C/CAN. – PcROP option bytes cannot be written as Bootloader uses Byte access while PcROP must be accessed using half-word access. Workaround: load a code snippet in SRAM using Bootloader interface then jump to it, and that code writes the PcROP value.

73 STM32L4P5xx/4Q5xx devices

73.1 Bootloader configuration

The STM32L4P5xx/4Q5xx bootloader is activated by applying Pattern 7 (described in [Table 2](#)). [Table 164](#) shows the hardware resources used by this bootloader.

Table 162. STM32L4P5xx/4Q5xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The HSI is used at startup as clock source for system clock configured to 60 MHz and for USART, I2C, SPI and USB bootloader operation.
		-	CRS is enabled for the DFU to allow USB to be clocked by HSI 48 MHz.
		HSE enabled	The HSE is used only when the CAN interface is selected. The HSE must have one of the following values 24, 20, 18, 16, 12, 9, 8, 6, 4 MHz.
		-	The CSS interrupt is enabled when HSE is enabled. Any failure (or removal) of the external clock generates system reset
	RAM	-	16 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	28 Kbytes, starting from address 0x1FFF0000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
	Power	-	The DFU cannot be used to communicate with bootloader if the voltage scaling range 2 is selected. Bootloader firmware does not configure voltage scaling range value in PWR_CR1 register.
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input no pull mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in input no pull mode.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in input pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode

Table 162. STM32L4P5xx/4Q5xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
USART3	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. Used in input pull-up mode.
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode. Used in input pull-up mode.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1011011x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain no pull mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1011011x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PB10 pin: clock line is used in open-drain no pull mode.
	I2C2_SDA pin		PB11 pin: data line is used in open-drain no pull mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1011011x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PC0 pin: clock line is used in open-drain no pull mode.
	I2C3_SDA pin		PC1 pin: data line is used in open-drain no pull mode.

Table 162. STM32L4P5xx/4Q5xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PB15 pin: slave data input line, used in push-pull, pull-down mode
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, pull-down mode
	SPI2_SCK pin	Input	PB13 pin: slave clock line, used in push-pull, pull-down mode
	SPI2_NSS pin	Input	PB12 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.
CAN1	CAN1	Enabled	Once initialized the CAN1 configuration is: Baudrate 125 kbps, 11 -bit identifier.
	CAN1_RX pin	Input	PB8 pin: CAN1 in reception mode. Used in alternate push-pull, pull-up mode.
	CAN1_TX pin	Output	PB9 pin: CAN1 in transmission mode. Used in alternate push-pull, pull-up mode.

Table 162. STM32L4P5xx/4Q5xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
DFU	USB	Enabled	USB FS configured in forced device mode. USB FS interrupt vector is enabled and used for USB DFU communications. Note: VDDUSB IO must be connected to 3.3 V as USB peripheral is used by the bootloader.
	USB_DM pin	Input/output	PA11: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PA12: USB DP line. Used in alternate push-pull, no pull mode. No external pull-up resistor is required

73.2 Bootloader selection

Figure 102 and Figure 103 show the bootloader selection mechanisms.

Figure 100. Dual bank boot implementation for STM32L4P5xx/4Q5xx bootloader V9.x

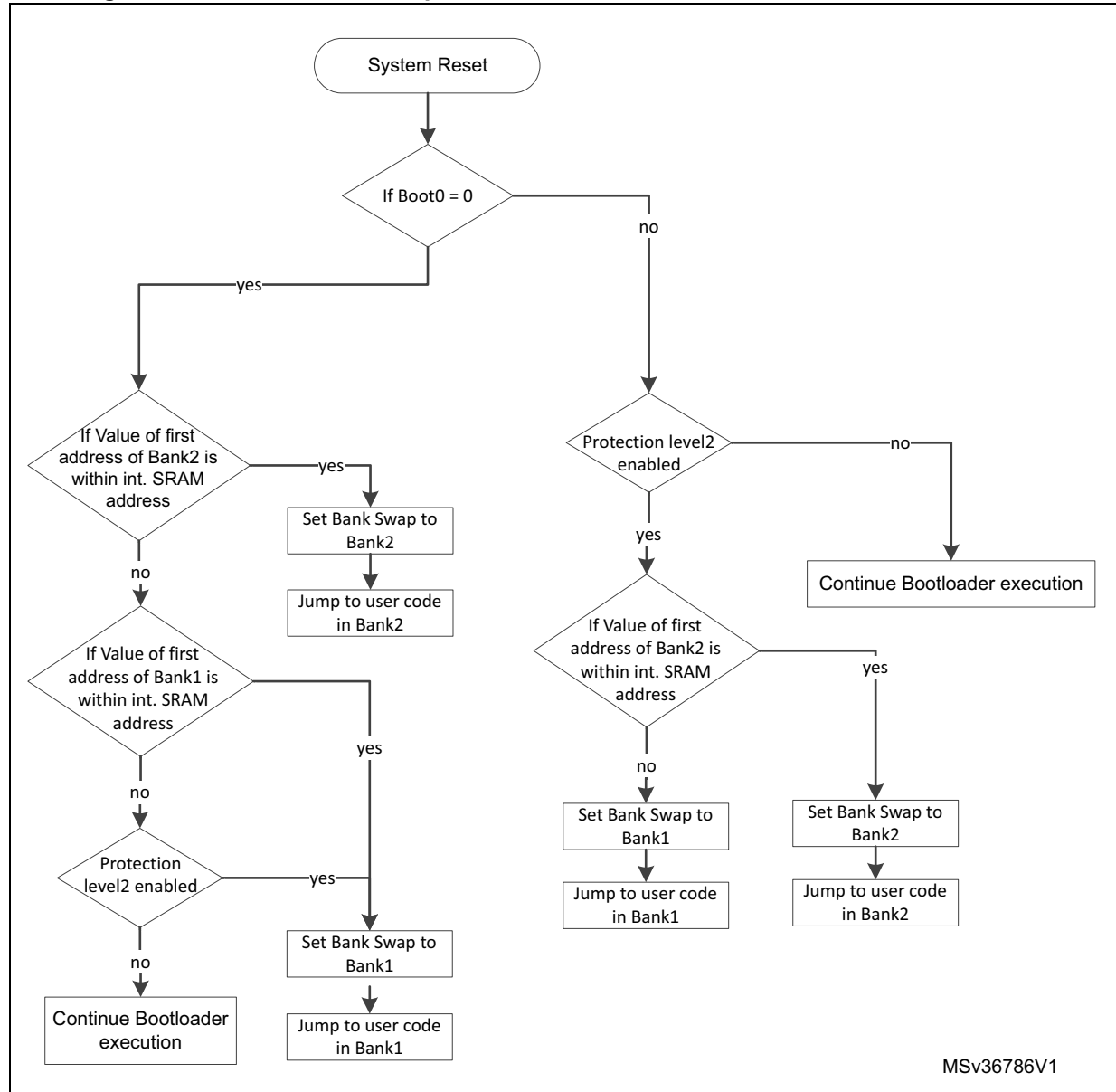
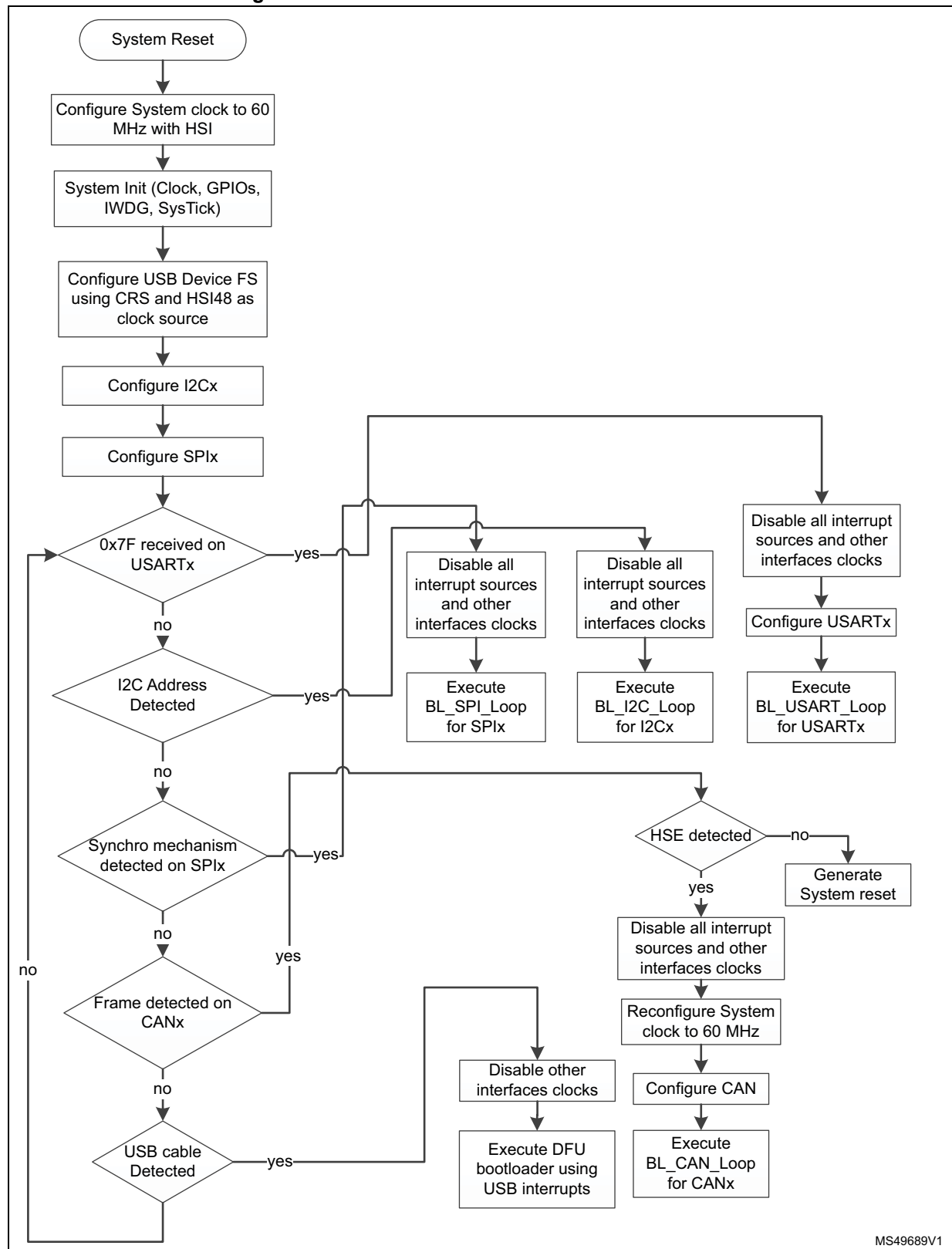


Figure 101. Bootloader V9.x selection for STM32L4P5xx/4Q5xx



73.3 Bootloader version

[Table 163](#) lists the STM32L4P5xx/4Q5xx devices bootloader versions.

Table 163. STM32L4P5xx/4Q5xx bootloader versions

Version number	Description	Known limitations
V9.0	Initial bootloader version on cut 1.0 samples	– PcROP option bytes cannot be written as bootloader uses byte access while PcROP must be accessed using half-word access. Workaround: load a code snippet in SRAM using bootloader interface then jump to it, and that code writes PcROP value.

74 STM32L4Rxxx/4Sxxx devices

74.1 Bootloader configuration

The STM32L4Rxx/4Sxx bootloader is activated by applying Pattern 6 (described in [Table 2](#)). [Table 164](#) shows the hardware resources used by this bootloader.

Table 164. STM32L4Rxxx/4Sxxx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The HSI is used at startup as clock source for system clock configured to 60 MHz and for USART, I2C, SPI and USB bootloader operation.
		-	CRS is enabled for the DFU to allow USB to be clocked by HSI 48 MHz.
		HSE enabled	The HSE is used only when the CAN interface is selected. The HSE must have one of the following values: 24, 20, 18, 16, 12, 9, 8, 6, 4 MHz.
		-	The CSS interrupt is enabled when HSE is enabled. Any failure (or removal) of the external clock generates system reset.
	RAM	-	12 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	28672 bytes starting from address 0x1FFF0000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	Power	-	The DFU cannot be used to communicate with bootloader if the voltage scaling range 2 is selected. Bootloader firmware does not configure voltage scaling range value in PWR_CR1 register.
	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input no pull mode.
USART2	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in input no pull mode.
	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in input pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in input pull-up mode.

Table 164. STM32L4Rxxx/4Sxxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
USART3	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. Used in input pull-up mode.
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode. Used in input pull-up mode.
USARTx	SysTick timer	Enabled	Used to automatically detect the serial baud rate from the host for USARTx.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1010000x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin	Input/output	PB7 pin: data line is used in open-drain no pull mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1010000x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PB10 pin: clock line is used in open-drain no pull mode.
	I2C2_SDA pin	Input/output	PB11 pin: data line is used in open-drain no pull mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1010000x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PC0 pin: clock line is used in open-drain no pull mode.
	I2C3_SDA pin	Input/output	PC1 pin: data line is used in open-drain no pull mode.

Table 164. STM32L4Rxxx/4Sxxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PB15 pin: slave data input line, used in push-pull, pull-down mode
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, pull-down mode
	SPI2_SCK pin	Input	PB13 pin: slave clock line, used in push-pull, pull-down mode
	SPI2_NSS pin	Input	PB12 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.
CAN1	CAN1	Enabled	Once initialized the CAN1 configuration is: Baudrate 125 kbps, 11 -bit identifier.
	CAN1_RX pin	Input	PB8 pin: CAN1 in reception mode. Used in alternate push-pull, pull-up mode.
	CAN1_TX pin	Output	PB9 pin: CAN1 in transmission mode. Used in alternate push-pull, pull-up mode.
	TIM16	Enabled	This timer is used to determine the value of the HSE. Once the HSE frequency is determined, the system clock is configured to 60 MHz using PLL and HSE.

Table 164. STM32L4Rxxx/4Sxxx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
DFU	USB	Enabled	USB FS configured in forced device mode. USB FS interrupt vector is enabled and used for USB DFU communications. Note: VDDUSB IO must be connected to 3.3 V as USB peripheral is used by the bootloader.
	USB_DM pin	Input/output	PA11: USB DM line.
	USB_DP pin		PA12: USB DP line No external pull-up resistor is required

74.2 Bootloader selection

Figure 102 and Figure 103 show the bootloader selection mechanisms.

Figure 102. Dual bank boot implementation for STM32L4Rxxx/STM32L4Sxxx bootloader V9.x

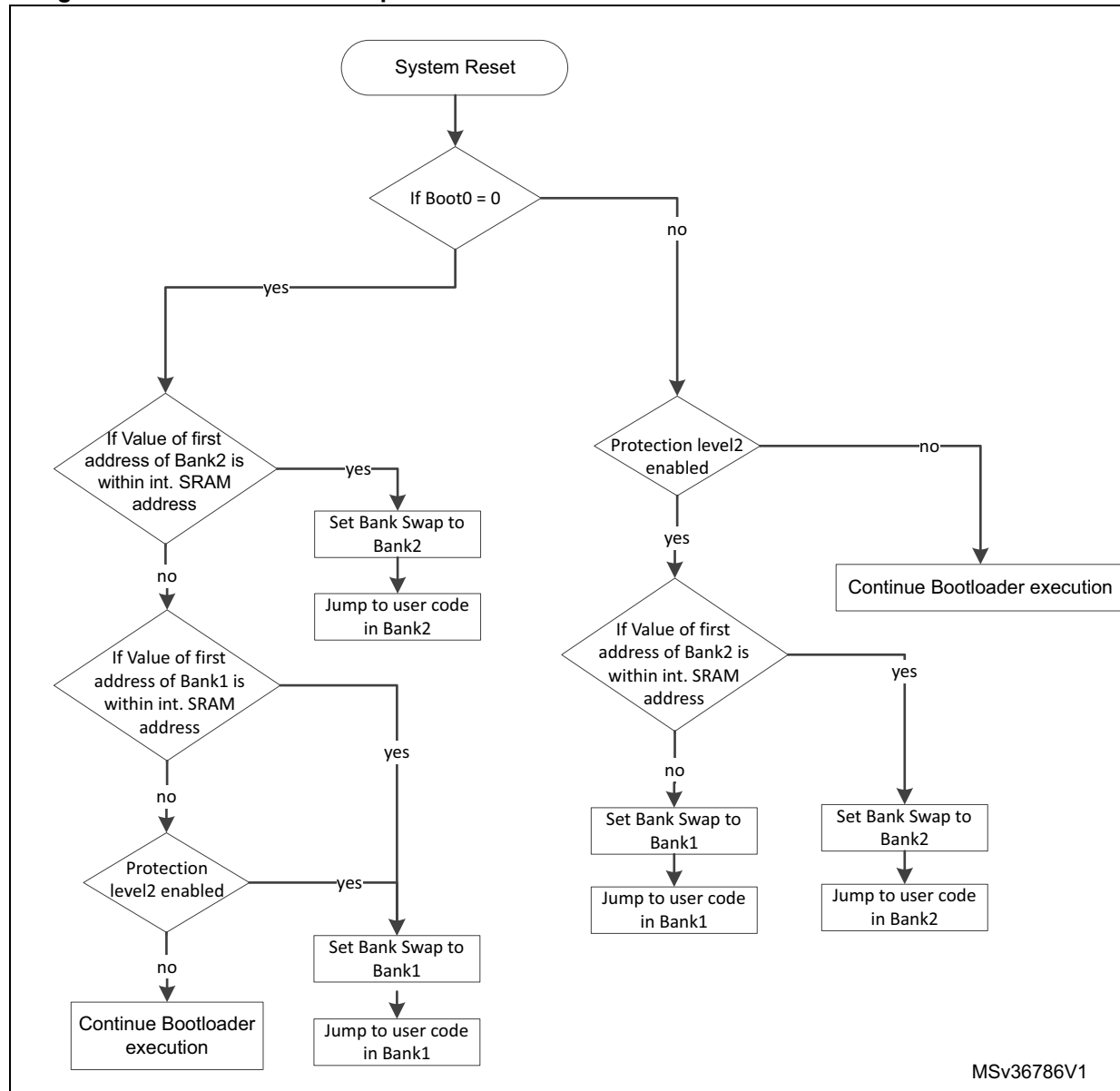
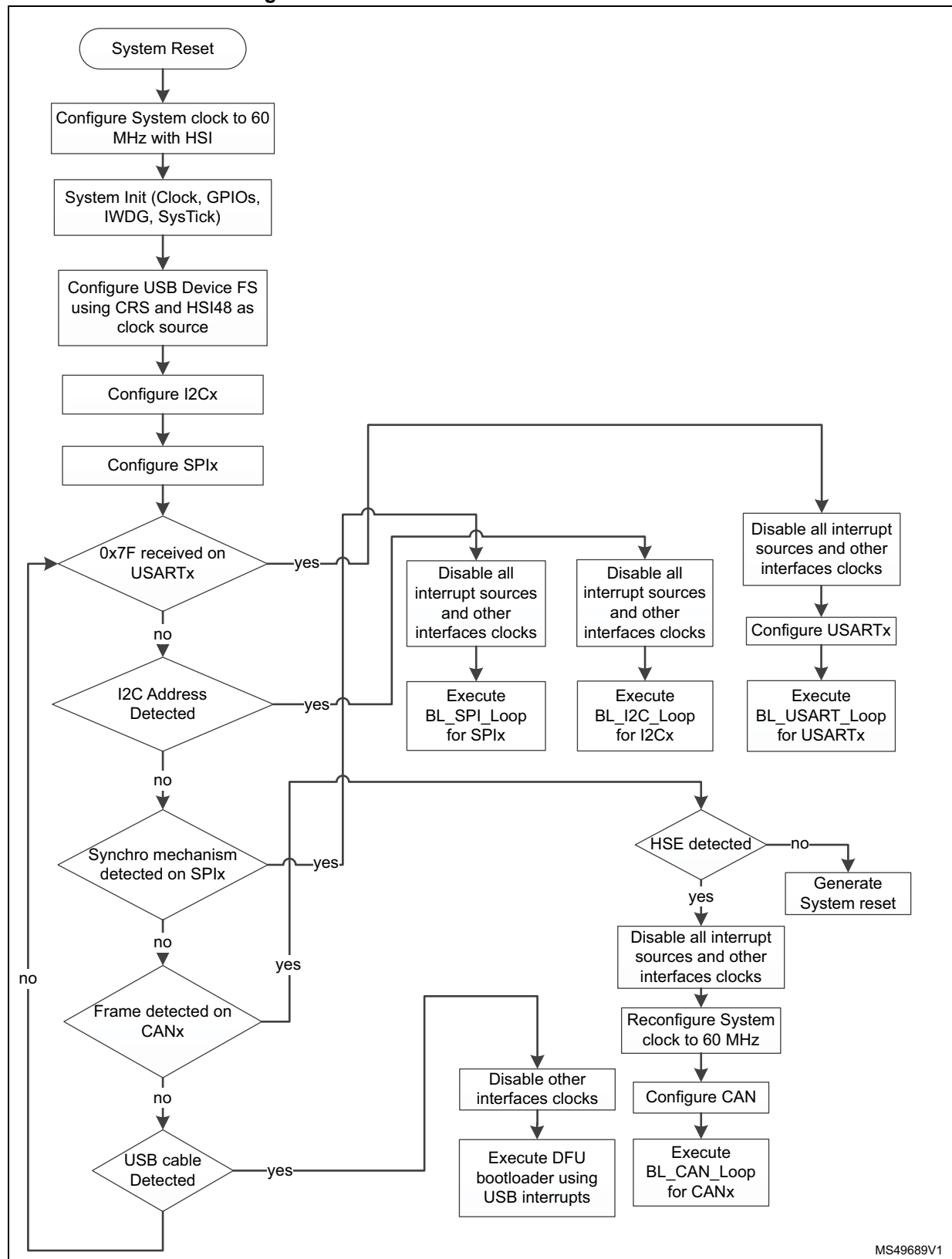


Figure 103. Bootloader V9.x selection for STM32L4Rxx/4Sxx



74.3 Bootloader version

[Table 165](#) lists the STM32L4Rxx/4Sxx devices bootloader versions.

Table 165. STM32L4Rxx/4Sxx bootloader versions

Version number	Description	Known limitations
V9.0	Initial bootloader version on cut 1.0 samples	None

75 STM32L552xx/62xx devices

75.1 Bootloader configuration

The STM32L552xx/62xx bootloader is activated by applying Pattern 12 (described in [Table 2](#)). [Table 166](#) shows the hardware resources used by this bootloader.

The bootloader follows boot model V3_1 (see [Section 4.10](#)), so it inherits all its constraints.

Table 166. STM32L552xx/62xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 60 MHz (using PLL clocked by HSI).
		-	CRS is enabled for the DFU to allow USB to be clocked by HSI 48 MHz.
		-	20 MHz derived from the PLLQ is used for FDCAN
	RAM	-	16 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	32 Kbytes, starting from address 0x0BF90000.
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USART3	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. Used in alternate push-pull, pull-up mode.
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode. Used in alternate push-pull, pull-up mode.

Table 166. STM32L552xx/62xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b0101100x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain pull-up mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain pull-up mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b0101100x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PB10 pin: clock line is used in open-drain pull-up mode.
	I2C2_SDA pin		PB11 pin: data line is used in open-drain pull-up mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b0101100x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PC0 pin: clock line is used in open-drain pull-up mode.
	I2C3_SDA pin		PC1 pin: data line is used in open-drain pull-up mode.
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.

Table 166. STM32L552xx/62xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PB15 pin: slave data input line, used in push-pull, pull-down mode
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, pull-down mode
	SPI2_SCK pin	Input	PB13 pin: slave clock line, used in push-pull, pull-down mode
	SPI2_NSS pin	Input	PB12 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.
SPI3	SPI3	Enabled	The SPI configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI3_MOSI pin	Input	PB5 pin: slave data input line, used in push-pull, pull-down mode
	SPI3_MISO pin	Output	PG10 pin: slave data output line, used in push-pull, pull-down mode
	SPI3_SCK pin	Input	PG9 pin: slave clock line, used in push-pull, pull-down mode
	SPI3_NSS pin	Input	PG12 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.
FDCAN	FDCAN1	Enabled	Once initialized the FDCAN1 configuration is: – Connection bit rate 250 kbit/s – Data bit rate 1000 kbit/s – FrameFormat = FDCAN_FRAME_FD_BRS – Mode = FDCAN_MODE_NORMAL – AutoRetransmission = ENABLE – TransmitPause = DISABLE – ProtocolException = ENABLE
	FDCAN1_Rx pin	Input/	PB9 pin: FDCAN1 in reception mode. Used in alternate push-pull, pull-up mode.
	FDCAN1_Tx pin	Output	PB8 pin: FDCAN1 in transmission mode. Used in alternate push-pull, pull-up mode.

Table 166. STM32L552xx/62xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
DFU	USB	Enabled	USB FS configured in forced device mode. USB FS interrupt vector is enabled and used for USB DFU communications. Note: VDDUSB IO must be connected to 3.3 V as USB peripheral is used by the bootloader.
	USB_DM pin	Input/output	PA11: USB DM line. Used in input no pull mode.
	USB_DP pin		PA12: USB DP line. Used in input no pull mode. No external pull-up resistor is required

Table 167. STM32L552xx/62xx special commands

Special commands supported (USART/I2C/SPI/FDCAN) Opcode - 0x50							
Function	Sub-Opcode (2 bytes)	Number of data sent (2 bytes)	Data sent	Number of data received	Data received	Number of status data received (2 bytes)	Status data received
TrustZone disable Must be run when TZEN = 1 and RDP = 1	0x82	0x4	0x0	0x0	NA	0x1	0x0
Regression from RDP L1 to RDP 0.5 Must be run when TZEN = 1 and RDP = 1	0x82	0x4	0x1	0x0	NA	0x1	0x0

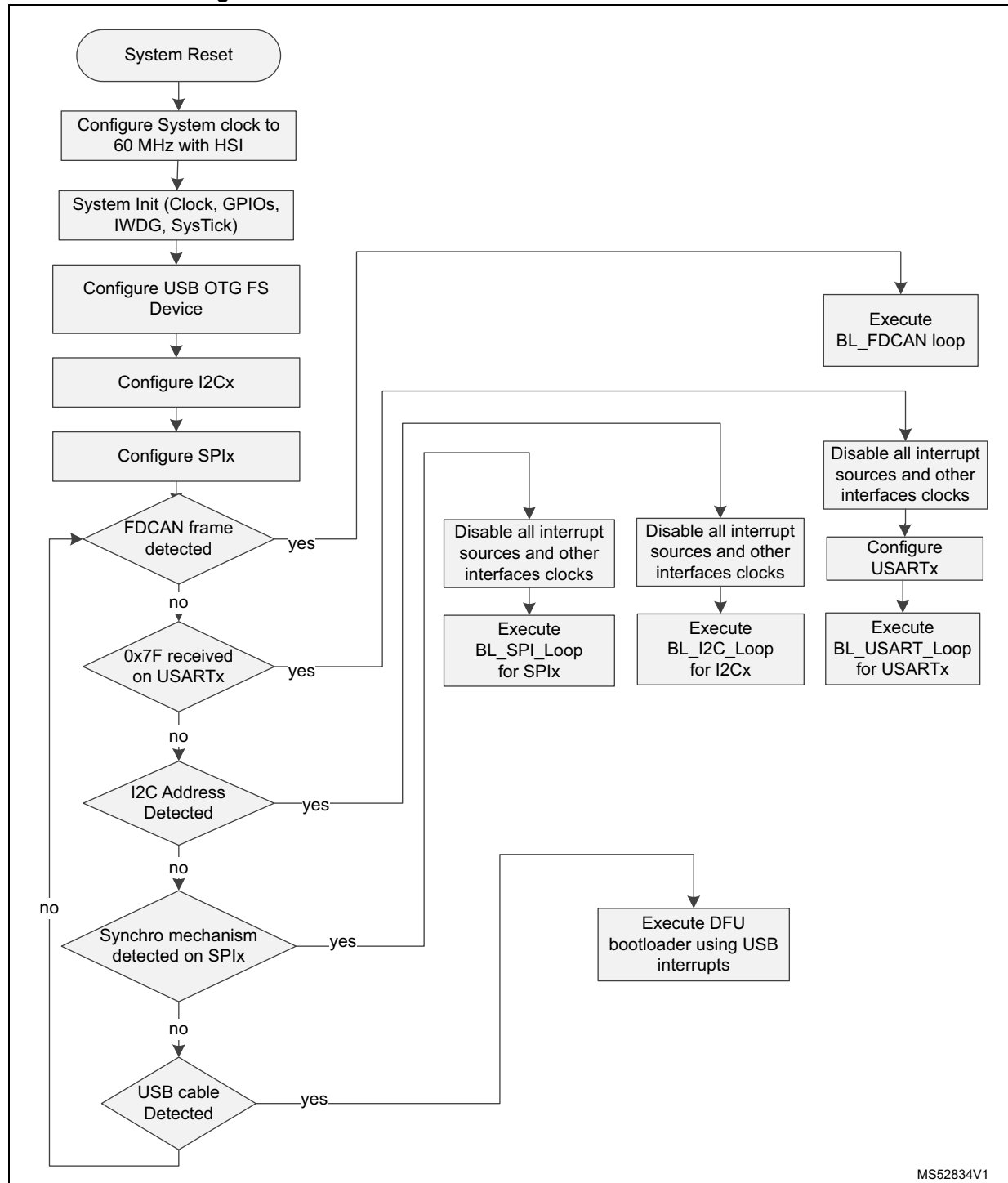
Note: USB special commands are slightly different from the other protocols as per the USB protocol specificities:

- No Opcode is used, Sub-Opcode is used directly
- Sub-Opcode is treated in a single byte and not two bytes
- Data is sent on USB frame byte per byte. No need to add number of data to be transmitted
- Returned data and status is formatted on the USB native protocol

75.2 Bootloader selection

Figure 104 shows the bootloader selection mechanism.

Figure 104. Bootloader V9.x selection for STM32L552xx/62xx



75.3 Bootloader version

Table 168 lists the STM32L552xx/62xx devices bootloader versions.

Table 168. STM32L552xx/62xx bootloader versions

Version number	Description	Known limitations
V13.0	Initial bootloader version on cut1.0 samples	– USART3 not working – SPI3 not working – OB launch not working on USB-DFU – No read/write SRAM2 in all protocols – Read Secure Option bytes only implemented on USART/I2C – Regression from TZEN = 1 to TZEN = 0 is done automatically on RDP regression
V9.0	Release supported only in cut2.0 – Fix all issues on previous release – Add FDCAN support – New command added for TZEN disable – Support of sales type 256 KB	– Unable to set TZEN to 1 option byte using all interfaces of the BL No workarounds available – Cannot set RDP level 0.5 nor option bytes in RDP level 0.5 using BL interfaces No workarounds available – Multiple reset seen when enabling HW IWDG option byte in TZEN = 1 No workarounds available – Unable to set secure option bytes setting when TZEN = 1 and RDP level is 0 No workarounds available – Go command on USB is not working – FDCAN erase not working as page number endianness is not aligned with the protocol (device waits for LSB first but host sends MSB first) – WA - Send data MSB first to the BL
V9.1	– Fix all known limitations of previous release – Add enable BOOT_LOCK BL command – Add support of RDP L1 to 0.5 regression	– Option byte programming is not working properly when using FDCAN interface. This makes the change of the Option byte not effective until a power off/power on. – FDCAN erase not working as page number endianness is not aligned with the protocol (device waits for LSB first but host sends MSB first) – WA - Send data MSB first to the BL
V9.2	– Fix all known limitations of previous release – Version for silicon revision Z	FDCAN Readout unprotect command does not send the command ID to the host

Note: When jumping to the BL the cache must be disabled.

76 STM32WB10xx/15xx devices

76.1 Bootloader configuration

The STM32WB10xx/15xx bootloader is activated by applying Pattern 6 (described in [Table 2](#)). [Table 171](#) shows the hardware resources used by this bootloader.

Table 169. STM32WB10xx/15xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	MSI enabled	The system clock frequency is 64 MHz (using PLL clocked by MSI).
	RAM	-	16 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	28 Kbytes, starting from address 0x1FFF0000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1001111x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain no pull mode.

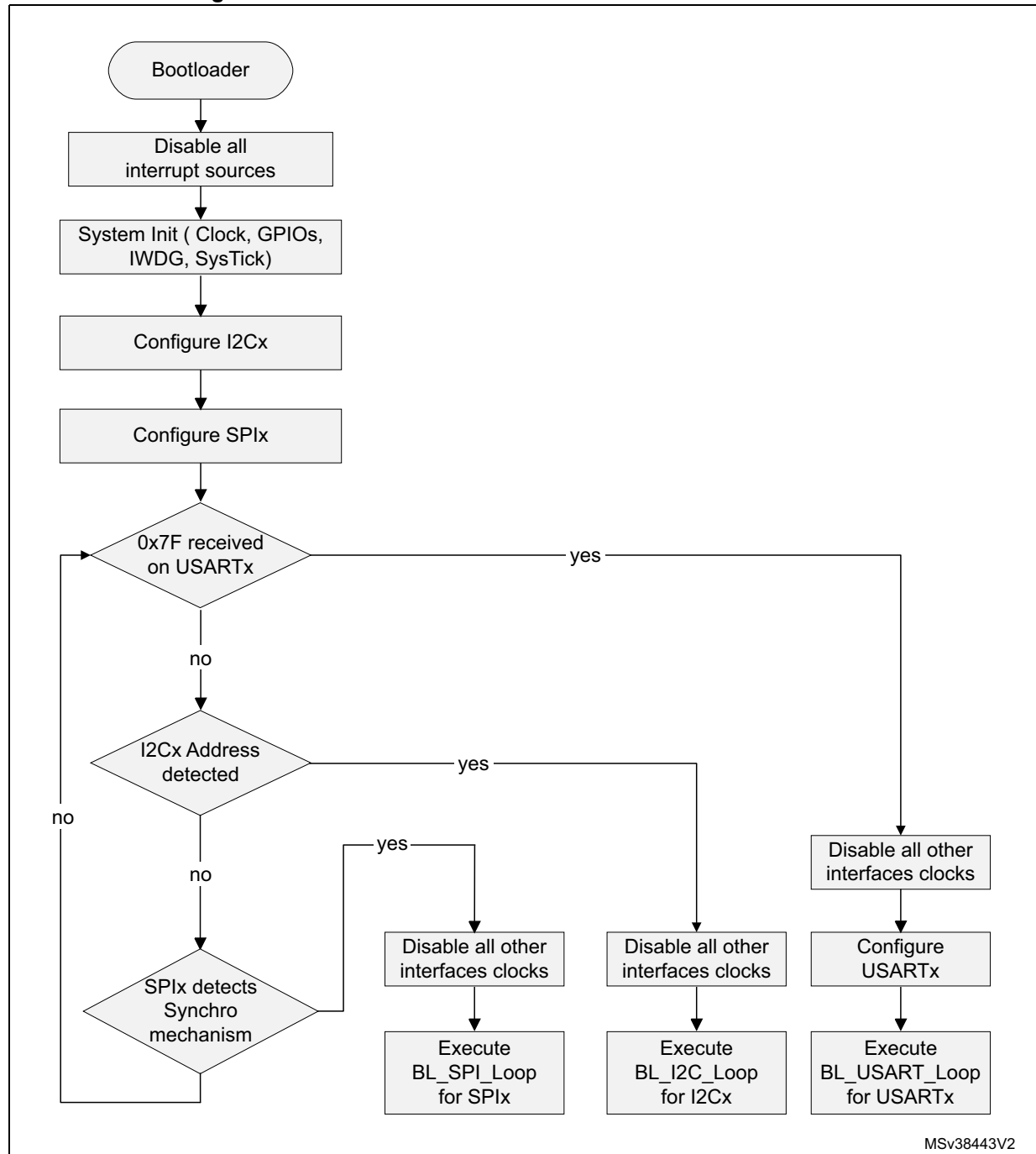
Table 169. STM32WB10xx/15xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.

76.2 Bootloader selection

Figure 105 shows the bootloader selection mechanism.

Figure 105. Bootloader V11.x selection for STM32WB10xx/15xx



76.3 Bootloader version

Table 170. STM32WB10xx/15xx bootloader versions

Version number	Description	Known limitations
V11.1	Initial bootloader version	– I2C Write Protect command (0x73) performs a Read Unprotect instead of disabling write protection. – Workaround: Use No-Stretch Write Unprotect command (0x74) that is performing correctly the write unprotect operation

77 STM32WB30xx/35xx/50xx/55xx devices

77.1 Bootloader configuration

The STM32WB30xx/35xx/50xx/55xx bootloader is activated by applying Pattern 16 (described in [Table 2](#)). [Table 171](#) shows the hardware resources used by this bootloader.

Table 171. STM32WB30xx/35xx/50xx/55xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	MSI enabled	The system clock frequency is 64 MHz (using PLL clocked by MSI).
		-	CRS is enabled for the DFU to allow USB to be clocked by HSI 48 MHz.
	RAM	-	20 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	28 Kbytes, starting from address 0x1FFF0000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1001111x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin	Input/output	PB7 pin: data line is used in open-drain no pull mode.

Table 171. STM32WB30xx/35xx/50xx/55xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1001111x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PC0 pin: clock line is used in open-drain no pull mode.
	I2C3_SDA pin	Input/output	PC1 pin: data line is used in open-drain no pull mode.
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.

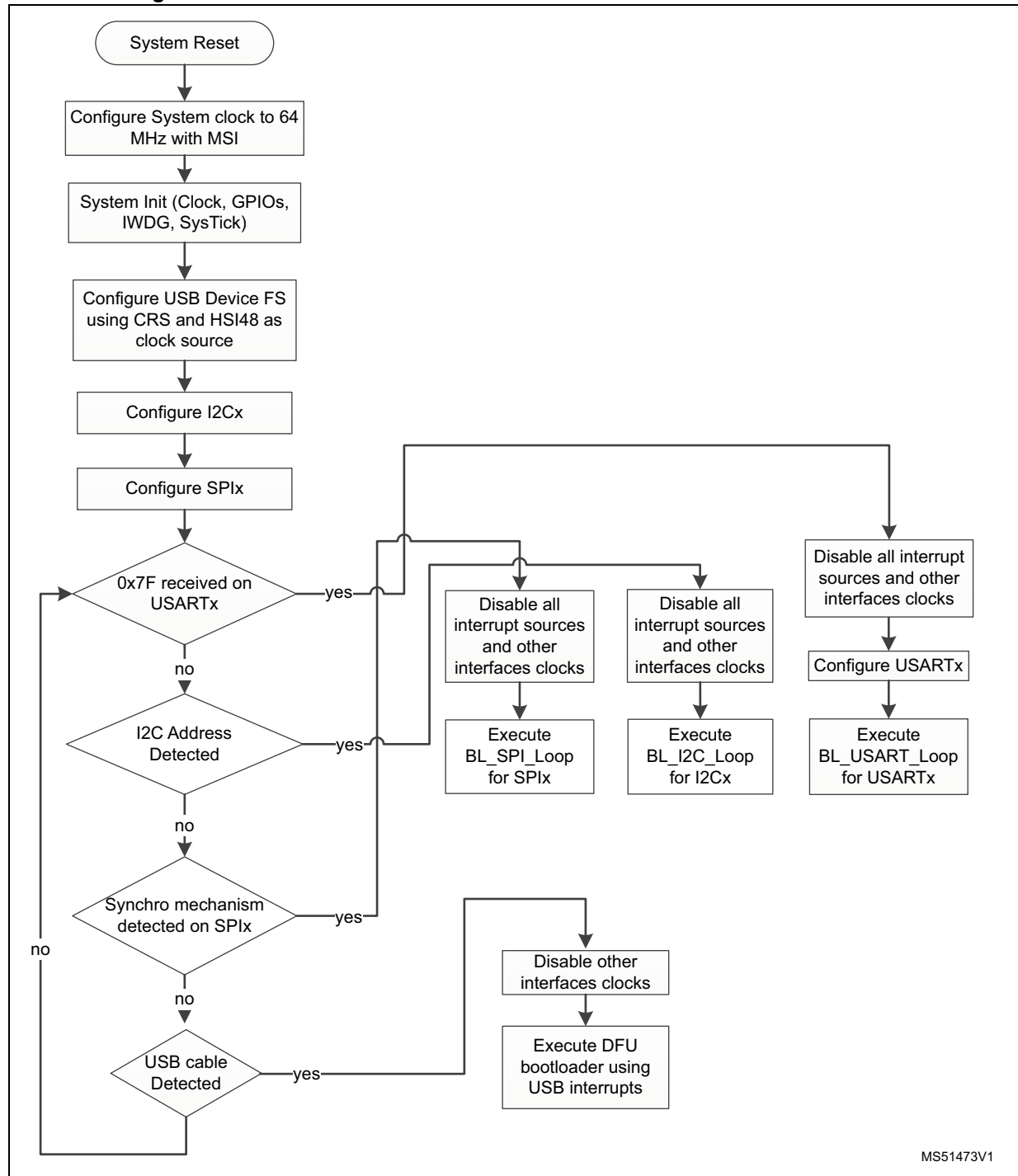
Table 171. STM32WB30xx/35xx/50xx/55xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PB15 pin: slave data input line, used in push-pull, pull-down mode
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, pull-down mode
	SPI2_SCK pin	Input	PB13 pin: slave clock line, used in push-pull, pull-down mode
	SPI2_NSS pin	Input	PB12 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.
DFU	USB	Enabled	USB FS configured in forced device mode. USB FS interrupt vector is enabled and used for USB DFU communications. Note: VDDUSB IO must be connected to 3.3 V as USB peripheral is used by the bootloader.
	USB_DM pin	Input/output	PA11: USB DM line. Used in input no pull mode.
	USB_DP pin		PA12: USB DP line. Used in input no pull mode. No external pull-up resistor is required

77.2 Bootloader selection

Figure 106 shows the bootloader selection mechanism.

Figure 106. Bootloader V13.0 selection for STM32WB30xx/35xx/50xx/55xx



77.3 Bootloader version

Table 172. STM32WB30xx/35xx/50xx/55xx bootloader versions

Version number	Description	Known limitations
V13.5	Initial bootloader version	– Readout Unprotect Command is not working properly as at the end of the command an NVIC_SystemReset is done instead of a flash option bytes reload. This makes the change of the RDP level not effective until a power off/on. – I2C Write Protect command (0x73) performs a Read Unprotect instead of disabling write protection. Workaround: Uses No-Stretch Write Unprotect command (0x74) that is performing correctly the write unprotect operation

Note: Instability when performing multiple resets during operations ongoing causing Overrun or FrameError errors on USART Bootloader and not recoverable unless Hardware Reset is performed. Fixed by workaround in FUS V1.0.1 and V1.0.2.

78 STM32WBA52xx devices

78.1 Bootloader configuration

The STM32WBA52xx bootloader is activated by applying Pattern12 (described in [Table 2](#)). [Table 173](#) shows the hardware resources used by this bootloader.

The bootloader follows boot model V3_1 (see [Section 4.10](#)), so it inherits all its constraints.

Table 173. STM32WBA52xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 60 MHz (using PLL clocked by HSI).
	RAM	-	16 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	32 Kbytes, starting from address 0x0BF88000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA8 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PB12 pin: USART1 in transmission mode. Not set until USART1 is detected.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA11 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA12 pin: USART2 in transmission mode. Not set until USART2 is detected.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1100110x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB2 pin: clock line is used in open-drain pull up mode.
	I2C1_SDA pin		PB1 pin: data line is used in open-drain pull up mode.

Table 173. STM32WBA52xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1100110x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PA6 pin: clock line is used in open-drain pull up mode.
	I2C3_SDA pin		PA7 pin: data line is used in open-drain pull up mode.
SPI3	SPI3	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI3_MOSI pin	Input	PB8 pin: slave data input line, used in push-pull, no pull mode
	SPI3_MISO pin	Output	PB9 pin: slave data output line, used in push-pull, no pull mode
	SPI3_SCK pin	Input	PA0 pin: slave clock line, used in push-pull, no pull mode
	SPI3_NSS pin	Input	PA5 pin: slave chip select pin used in push-pull, no pull mode.

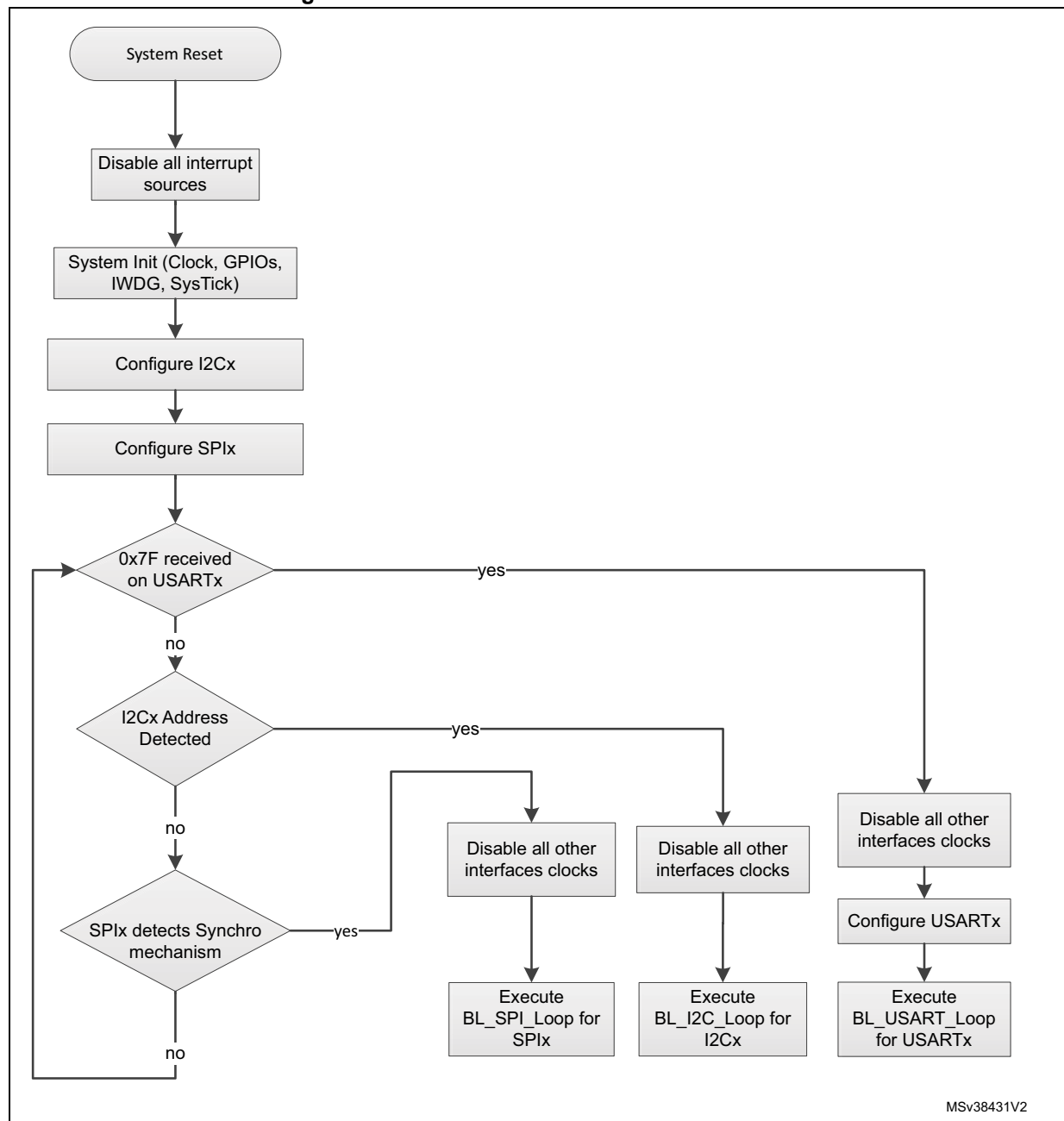
Table 174. STM32WBA52xx special commands

Special commands supported (USART/I2C/SPI) Opcode - 0x50							
Function	Sub-Opcode (2 bytes)	Number of data sent (2 bytes)	Data sent	Number of data received	Data received	Number of status data received (2 bytes)	Status data received
TrustZone disable Must be run when TZEN = 1 and RDP = 1	0x82	0x4	0x0	0x0	NA	0x1	0x0
Regression from RDP L1 to RDP 0.5 Must be run when TZEN = 1 and RDP = 1	0x82	0x4	0x1	0x0	NA	0x1	0x0
Unlock write protection Must be run when RDP = 1	0x82	0x4	0xYY02 ⁽¹⁾	0x0	NA	0x1	0x0

1. 0xYY can have 3 values (0: WRP area, 1: WRP1A, 2: WRP2A)

78.2 Bootloader selection

Figure 107. Bootloader V11.x selection for STM32WBA52xx



78.3 Bootloader version

Table 175. STM32WBA52xx bootloader versions

Version number	Description	Known limitations
V11.0	Initial bootloader version	None

79 STM32WBA54xx/55xx devices

79.1 Bootloader configuration

The STM32WBA54xx/55xx bootloader is activated by applying Pattern12 (described in [Table 2](#)). [Table 176](#) shows the hardware resources used by this bootloader.

The bootloader follows boot model V3_1 (see [Section 4.10](#)), so it inherits all its constraints.

Table 176. STM32WBA54xx/55xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 60 MHz (using PLL clocked by HSI).
	RAM	-	6 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	28 Kbytes, starting from address 0x0BF88000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA8 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PB12 pin: USART1 in transmission mode. Not set until USART1 is detected.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA11 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA12 pin: USART2 in transmission mode. Not set until USART2 is detected.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1100110x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB2 pin: clock line is used in open-drain pull up mode.
	I2C1_SDA pin		PB1 pin: data line is used in open-drain pull up mode.

Table 176. STM32WBA54xx/55xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1100110x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PA6 pin: clock line is used in open-drain pull up mode.
	I2C3_SDA pin		PA7 pin: data line is used in open-drain pull up mode.
SPI3	SPI3	Enabled	The SPI3 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI3_MOSI pin	Input	PB8 pin: slave data input line, used in push-pull, no pull mode
	SPI3_MISO pin	Output	PB9 pin: slave data output line, used in push-pull, no pull mode
	SPI3_SCK pin	Input	PA0 pin: slave clock line, used in push-pull, no pull mode
	SPI3_NSS pin		PA5 pin: slave chip select pin used in push-pull, no pull mode.
DFU	USB	Enabled	USB FS configured in forced device mode. USB FS interrupt vector is enabled and used for USB DFU communications.
	USB_DM pin	Input/output	PM12: USB DM line. Used in alternate push-pull, no pull mode.
	USB_DP pin		PM11: USB DP line. Used in alternate push-pull, no pull mode. No external pull-up resistor is required.

Table 177. STM32WBA54xx/55xx special commands

Special commands supported (USART/I2C/SPI) Opcode - 0x50							
Function	Sub-Opcode (2 bytes)	Number of data sent (2 bytes)	Data sent	Number of data received	Data received	Number of status data received (2 bytes)	Status data received
TrustZone disable Must be run when TZEN = 1 and RDP = 1	0x82	0x4	0x0	0x0	NA	0x1	0x0

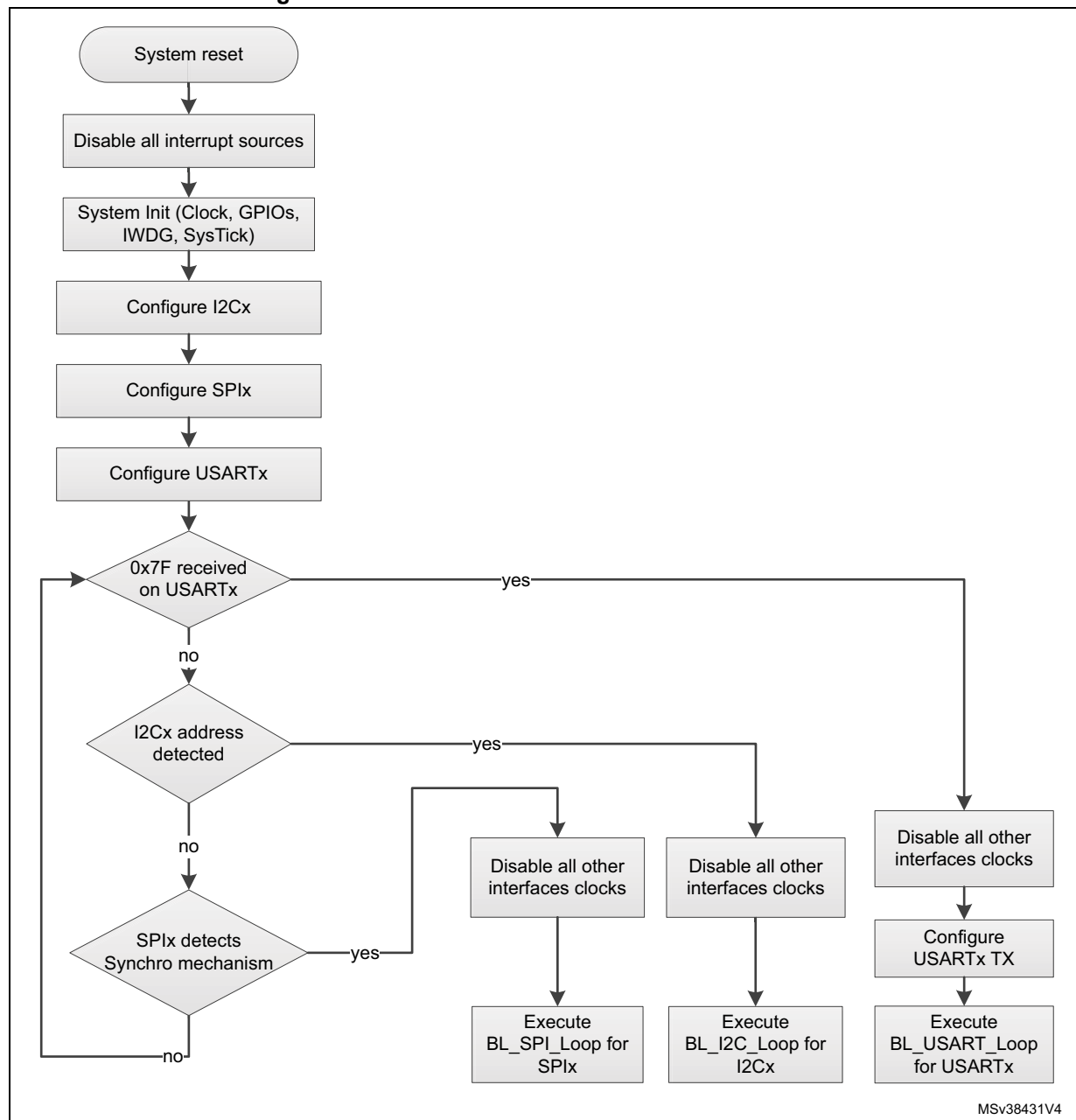
Table 177. STM32WBA54xx/55xx special commands (continued)

Special commands supported (USART/I2C/SPI) Opcode - 0x50							
Function	Sub- Opcode (2 bytes)	Number of data sent (2 bytes)	Data sent	Number of data received	Data received	Number of status data received (2 bytes)	Status data received
Regression from RDP L1 to RDP 0.5 Must be run when TZEN = 1 and RDP = 1	0x82	0x4	0x1	0x0	NA	0x1	0x0
Unlock write protection Must be run when RDP = 1	0x82	0x4	0xYY02 ⁽¹⁾	0x0	NA	0x1	0x0

1. 0xYY can have three values (0: WRP area, 1: WRP1A, 2: WRP2A)

79.2 Bootloader selection

Figure 108. Bootloader V11.x selection for STM32WBA54xx/55xx



79.3 Bootloader version

Table 178. STM32WBA54xx/55xx bootloader versions

Version number	Description	Known limitations
V11.1	Initial bootloader version	None

80 STM32WBA62xx/63xx/64xx/65xx devices

80.1 Bootloader configuration

The STM32WBA62xx/63xx/64xx/65xx bootloader is activated by applying Pattern12 (described in [Table 2](#)). [Table 179](#) shows the hardware resources used by this bootloader.

The bootloader follows boot model V3_1 (see [Section 4.10](#)), so it inherits all its constraints.

Table 179. STM32WBA62xx/63xx/64xx/65xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 60 MHz (using HSI through PLL1 source).
		HSE enabled	Enabled only when USB cable detected. HSE must be 32 MHz to have the USB working.
	RAM	-	12 Kbytes, starting from address 0x2000 0000 are used by the bootloader firmware
	System memory	-	64 Kbytes, starting from address 0x0BF9 0000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA8 pin: USART1 in reception mode. Used in alternate function, pull-up mode.
	USART1_TX pin	Output	PB12 pin: USART1 in transmission mode. Kept in reset configuration until 0x7F detected on USART_RX.
USART3	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PA11 pin: USART3 in reception mode. Used in alternate push-pull, pull-up mode.
	USART3_TX pin	Output	PA12 pin: USART2 in transmission mode. Kept in reset configuration until 0x7F detected on USART_RX.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1101111x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB2 pin: clock line is used in open-drain no pull mode.
	I2C1_SDA pin		PB1 pin: data line is used in open-drain no pull mode.

Table 179. STM32WBA62xx/63xx/64xx/65xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1101111x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PA6 pin: clock line is used in open-drain no pull mode.
	I2C3_SDA pin		PA7 pin: data line is used in open-drain no pull mode.
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PB14 pin: Slave data Input line, used in alternate function with no pull.
	SPI2_MISO pin ⁽¹⁾	Output	PB0 pin: Slave data output line used in alternate function with no pull.
	SPI2_SCK pin	Input	PA9 pin: Slave clock line, used in alternate function with no pull.
	SPI2_NSS pin		PA10 pin: slave chip select pin used in alternate function with no pull.
SPI3	SPI3	Enabled	The SPI3 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI3_MOSI pin	Input	PB8 pin: Slave data Input line, used in alternate function with pull down.
	SPI3_MISO pin	Output	PB9 pin: Slave data output line used in alternate function with no pull.
	SPI3_SCK pin	Input	PA0 pin: Slave clock line, used in alternate function with no pull.
	SPI3_NSS pin		PA5 pin: slave chip select pin used in alternate function with no pull.
DFU	USB	Enabled	USB configured in device mode. USB interrupt vector is enabled and used for DFU communications.
	USB_DM pin	Input/output	PD7: USB DM line. Used in alternate function, no pull mode.
	USB_DP pin		PD6: USB DP line. Used in alternate function, no pull mode. No external pull-up resistor is required.

1. SPI Tx (MISO) is handled by DMA. On the bootloader start-up after SPI initialization, as soon as the bit DMATx enable on SPI CR2 register is set to 0x1, the MISO line is set to 3.3 V.

Table 180. STM32WBA62xx/63xx/64xx/65xx special commands

Special commands supported (USART/I2C/SPI) Opcode - 0x50							
Function	Sub-Opcode (2 bytes)	Number of data sent (2 bytes)	Data sent	Number of data received	Data received	Number of status data received (2 bytes)	Status data received
TrustZone disable Must be run when TZEN = 1 and RDP = 1	0x82	0x4	0x0	0x0	NA	0x1	0x0
Regression from RDP L1 to RDP 0.5 Must be run when TZEN = 1 and RDP = 1	0x82	0x4	0x1	0x0	NA	0x1	0x0
Unlock write protection Must be run when RDP = 1	0x82	0x4	0xYY02 ⁽¹⁾	0x0	NA	0x1	0x0

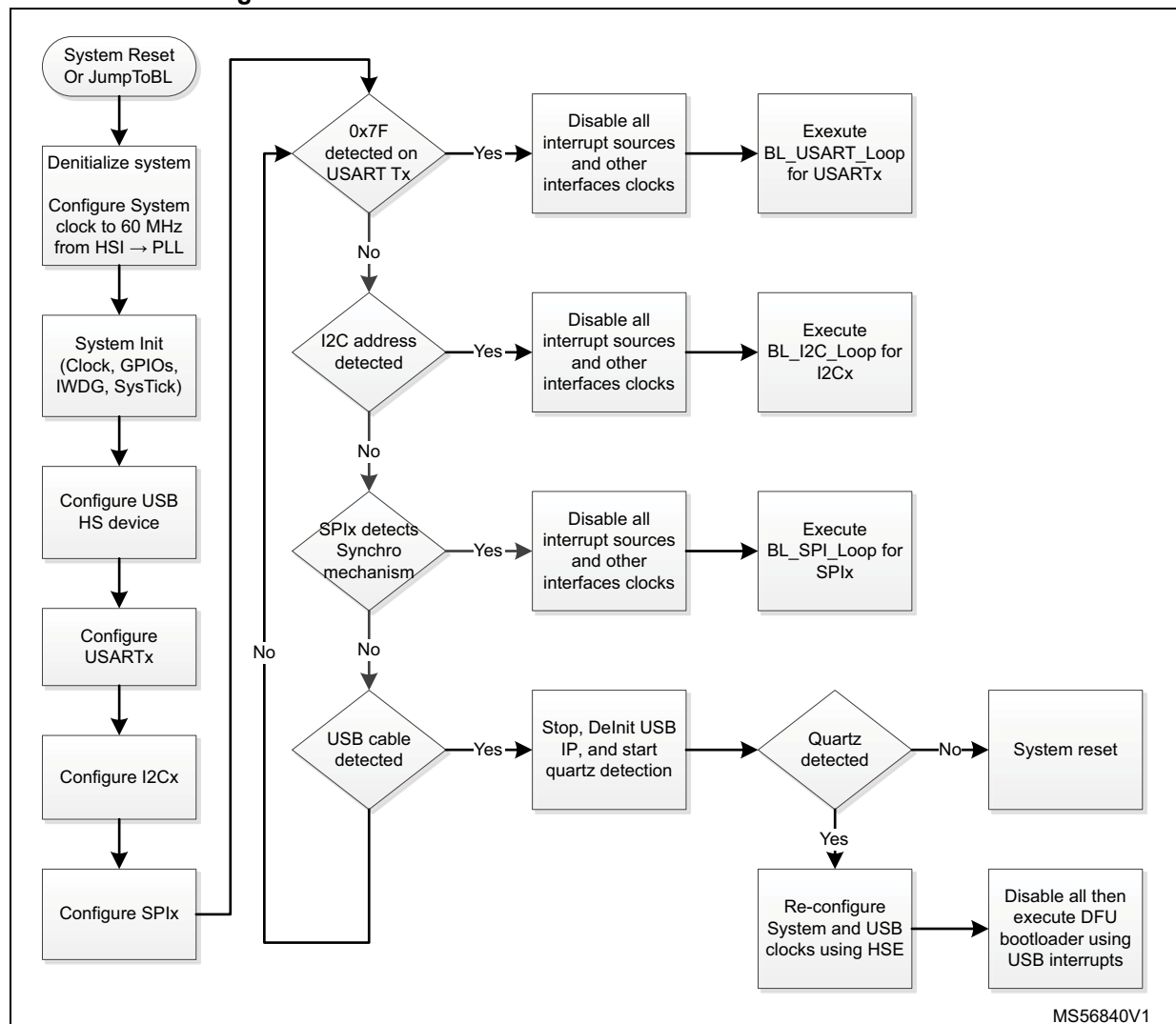
1. 0xYY can have three values (0: WRP area, 1: WRP1A, 2: WRP2A)

Note: *USB special commands are slightly different from the other protocols as per the USB protocol specificities:*

- *No Opcode is used, Sub-Opcode is used directly*
- *Sub-Opcode is treated in a single byte and not in two bytes*
- *Data is sent on USB frame byte per byte. No need to add the number of data to be transmitted*
- *Returned data and status is formatted on the USB native protocol*

80.2 Bootloader selection

Figure 109. Bootloader V13.2 selection for STM32WBA62xx/63xx/64xx/65xx



80.3 Bootloader version

Table 181. STM32WBA62xx/63xx/64xx/65xx bootloader versions

Version number	Description	Known limitations
V13.2	Initial bootloader version	None

81 STM32WB05xx devices

81.1 Bootloader configuration

The STM32WB05xx bootloader is activated by applying Pattern 18 (described in [Table 2](#)). [Table 182](#) shows the hardware resources used by this bootloader.

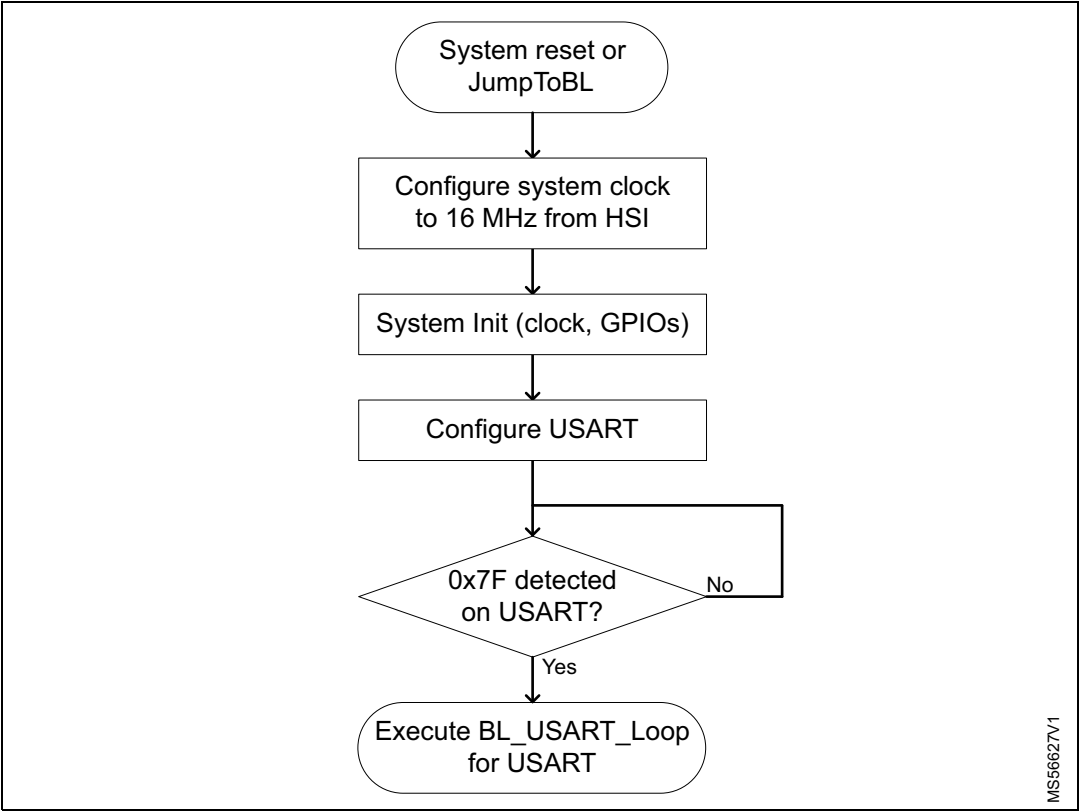
Table 182. STM32WB05xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 16 MHz using HSI
	RAM	-	The last 3 Kbytes of the last bank
	System memory	-	6 Kbytes, starting from address 0x10000000, contain the bootloader firmware
USART	USART	Enabled	Once initialized, the configuration is 8-bit, no parity
	USART_RX pin	Input	PB0 pin: USART in reception mode. Used in alternate push-pull, pull-up mode.
	USART_TX pin	Output	PA1 pin: USART in transmission mode. Not set until USART is detected.

81.2 Bootloader selection

Figure 110 shows the bootloader selection mechanism.

Figure 110. Bootloader V2.x selection for STM32WB05xx



81.3 Bootloader version

Table 183. STM32WB05xx bootloader versions

Version number	Description	Known limitations
V2.0	Final bootloader version	None

82 STM32WB06xx/07xx devices

82.1 Bootloader configuration

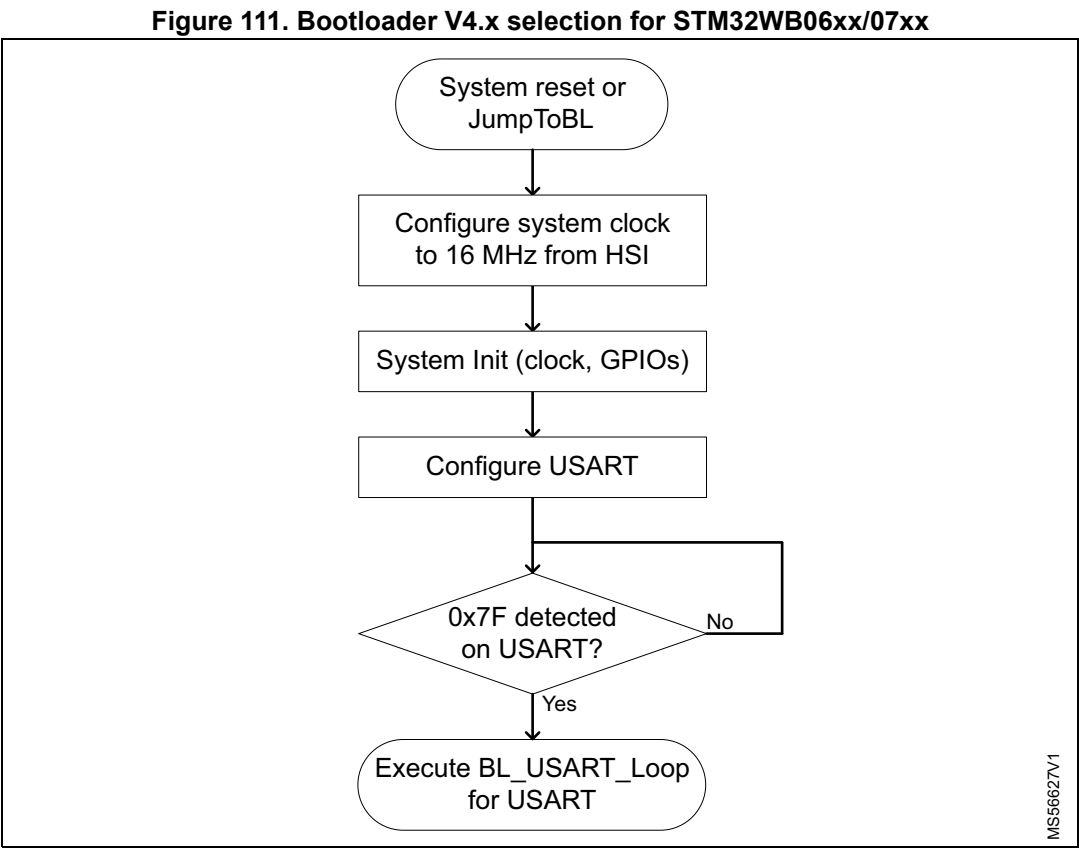
The STM32WB06xx/07xx bootloader is activated by applying Pattern 18 (described in [Table 2](#)). [Table 184](#) shows the hardware resources used by this bootloader.

Table 184. STM32WB06xx/07xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 16 MHz using HSI
	RAM	-	The last 3 Kbytes of the last bank
	System memory	-	6 Kbytes, starting from address 0x10000000, contain the bootloader firmware
USART	USART	Enabled	Once initialized, the configuration is 8-bit, no parity
	USART_RX pin	Input	PA8 pin: USART in reception mode. Used in alternate push-pull, pull-up mode.
	USART_TX pin	Output	PA9 pin: USART in transmission mode. Not set until USART is detected.

82.2 Bootloader selection

Figure 111 shows the bootloader selection mechanism.



82.3 Bootloader version

Table 185. STM32WB06xx/07xx bootloader versions

Version number	Description	Known limitations
V4.0	Final bootloader version	None

83 STM32WB09xx devices

83.1 Bootloader configuration

The STM32WB09xx bootloader is activated by applying Pattern 18 (described in [Table 2](#)). [Table 186](#) shows the hardware resources used by this bootloader.

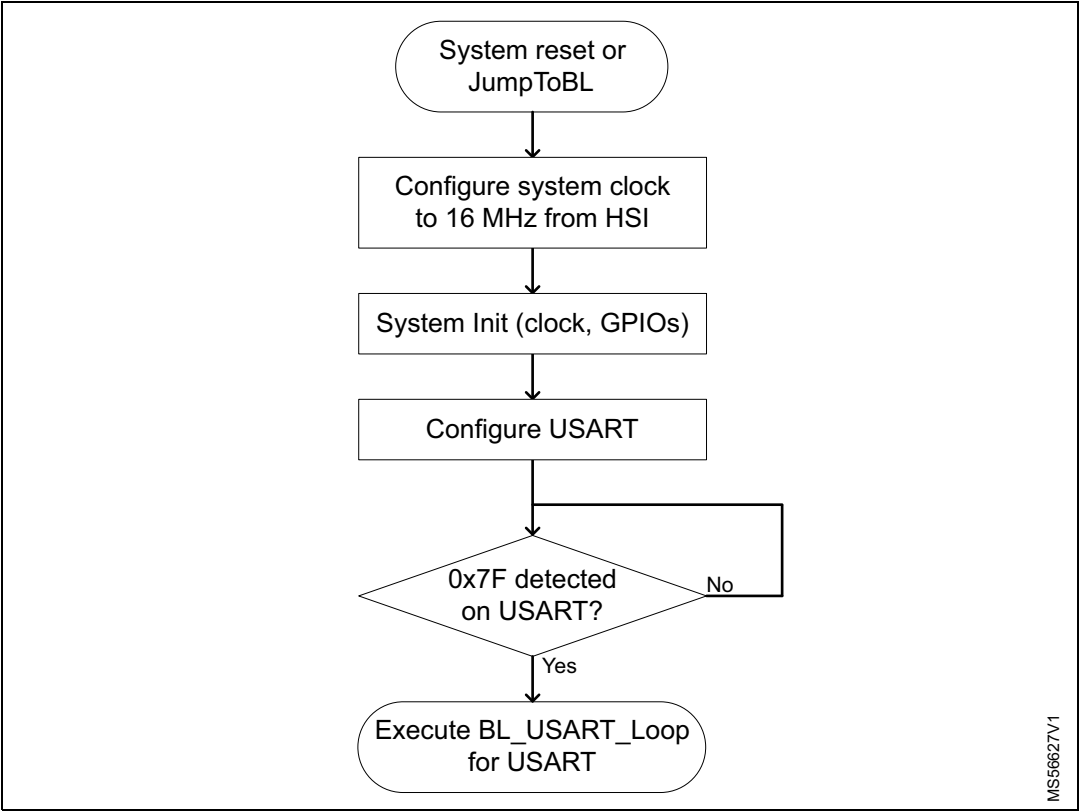
Table 186. STM32WB09xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 16 MHz using HSI
	RAM	-	The last 3 Kbytes of the last bank
	System memory	-	6 Kbytes, starting from address 0x10000000, contain the bootloader firmware
USART	USART	Enabled	Once initialized, the configuration is 8-bit, no parity
	USART_RX pin	Input	PB0 pin: USART in reception mode. Used in alternate push-pull, pull-up mode.
	USART_TX pin	Output	PA1 pin: USART in transmission mode. Not set until USART is detected.

83.2 Bootloader selection

Figure 113 shows the bootloader selection mechanism.

Figure 112. Bootloader V1.x selection for STM32WB09xx



83.3 Bootloader version

Table 187. STM32WB09xx bootloader versions

Version number	Description	Known limitations
V1.0	Final bootloader version	None

84 STM32WLE5xx/55xx devices

84.1 Bootloader configuration

The STM32WLE5xx/55xx bootloader is activated by applying Pattern 13 (described in [Table 2](#)). [Table 188](#) shows the hardware resources used by this bootloader.

Table 188. STM32WLE5xx/55xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 48 MHz (using PLL clocked by HSI).
	RAM	-	8 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	16 Kbytes, starting from address 0x1FFF0000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.

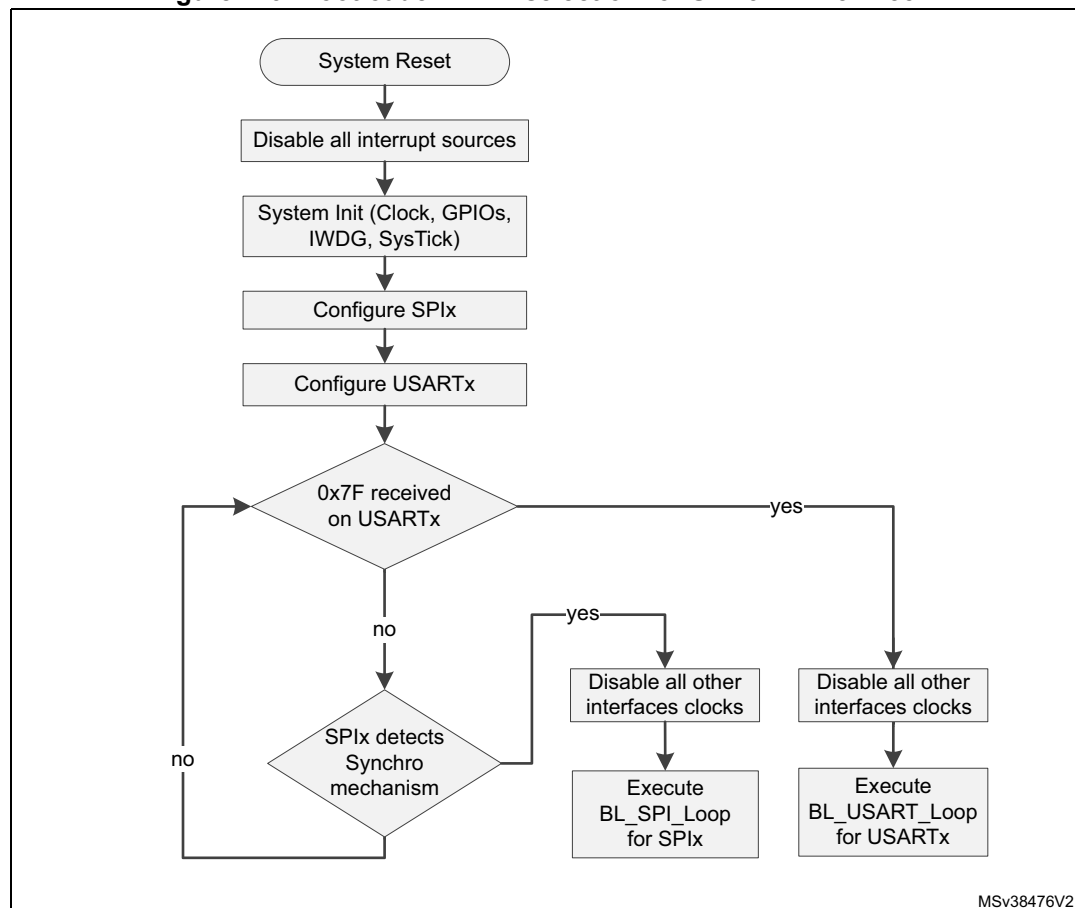
Table 188. STM32WLE5xx/55xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware
	SPI2_MOSI pin	Input	PB15 pin: slave data input line, used in push-pull, pull-down mode
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, pull-down mode
	SPI2_SCK pin	Input	PB13 pin: slave clock line, used in push-pull, pull-down mode
	SPI2_NSS pin	Input	PB12 pin: slave chip select pin used in push-pull, pull-down mode. Note: This IO can be tied to GND if the SPI master does not use it.

84.2 Bootloader selection

Figure 113 shows the bootloader selection mechanism.

Figure 113. Bootloader V12.x selection for STM32WLE5xx/55xx



84.3 Bootloader version

Table 189. STM32WLE5xx/55xx bootloader versions

Version number	Description	Known limitations
V12.2	Initial bootloader version on rev. Z samples	BL cannot write/read the following option bytes: – FLASH_SFR (Offset - 0x80) – FLASH_SRRVR (Offset - 0x84)
V12.3	Final bootloader version on rev. Z samples	BL cannot write/read the following option bytes: – FLASH_SFR (Offset - 0x80) – FLASH_SRRVR (Offset - 0x84)
V12.4	Final bootloader version on rev. Y samples	BL cannot write/read the following option bytes: – FLASH_SFR (Offset - 0x80) – FLASH_SRRVR (Offset - 0x84)

85 STM32U031xx devices

85.1 Bootloader configuration

The STM32U031xx bootloader is activated by applying Pattern 11 (described in [Table 2](#)). [Table 190](#) shows the hardware resources used by this bootloader.

Table 190. STM32U031xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI16 enabled	The system clock frequency is 24 MHz (using PLL clocked by HSI16).
	RAM	-	5 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	14 Kbytes, starting from address 0x1FFF0000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
Securable memory area	-	-	The address to jump to for the securable memory area is 0x1FFF3500
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART in reception mode. Used in alternate function input, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART in transmission mode. Kept in reset configuration until 0x7F is detected on USART_RX. PA11 is remapped to PA9 on the TSSOP20 package.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART in reception mode. Used in alternate function input, pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART in transmission mode. Kept in reset configuration until 0x7F is detected on USART_RX.
USART3	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART in reception mode. Used in alternate function input, pull-up mode.
	USART3_TX pin	Output	PC10 pin: USART in transmission mode. Kept in reset configuration until 0x7F is detected on USART_RX.
	EXTI line 11	Input	Used on detection on USART and its IT for baudrate calculation

Table 190. STM32U031xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1101011x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain pull-up mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain pull-up mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1101011x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PB10 pin: clock line is used in open-drain pull-up mode.
	I2C2_SDA pin		PB11 pin: data line is used in open-drain pull-up mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1101011x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PB3 pin: clock line is used in open-drain pull-up mode.
	I2C3_SDA pin		PB4 pin: data line is used in open-drain pull-up mode.
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in alternate function, pull-down mode
	SPI1_MISO pin ⁽¹⁾	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in alternate function, pull-down mode. PA1 pin is used instead of PA5 on TSSOP20 package. Used on push-pull, pull-up mode
	SPI1_NSS pin		PA4 pin: slave chip select pin used in alternate function, pull-down mode.

Table 190. STM32U031xx configuration in system memory boot mode (continued)

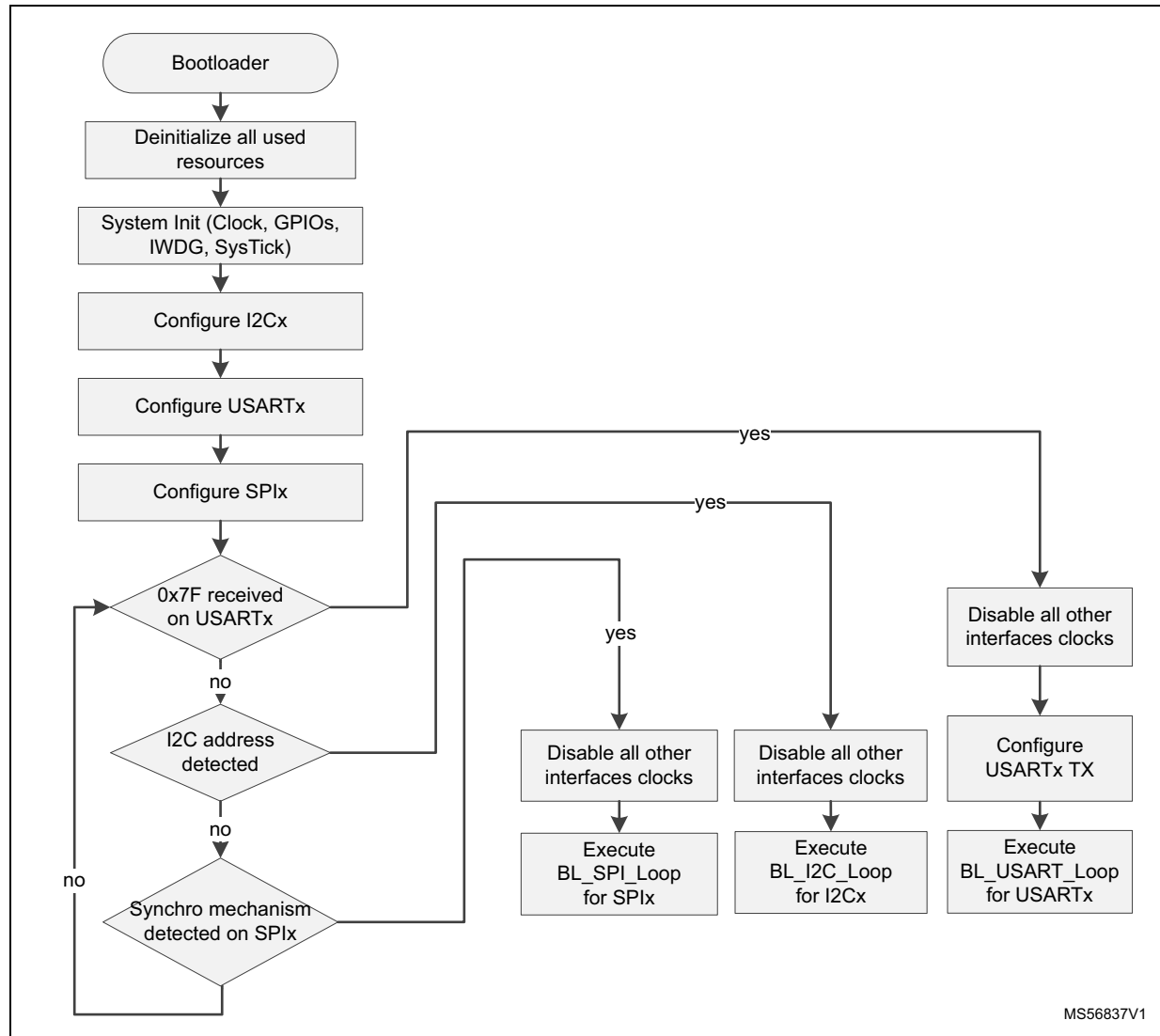
Bootloader	Feature/Peripheral	State	Comment
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PB15 pin: slave data input line, used in alternate function, no pull mode
	SPI2_MISO pin ⁽¹⁾	Output	PB14 pin: slave data output line, used in push-pull, no pull mode
	SPI2_SCK pin	Input	PB13 pin: slave clock line, used in alternate function, no pull mode
	SPI2_NSS pin	Input	PB12 pin: slave chip select pin used in alternate function, no pull mode.

1. SPI Tx (MISO) is handled by DMA. On the bootloader start-up after SPI initialization, as soon as the bit DMATx enable on SPI CR2 register is set to 1, the MISO line is set to 3.3 V.

85.2 Bootloader selection

Figure 114 shows the bootloader selection mechanism.

Figure 114. Bootloader V11.x selection for STM32U031xx



85.3 Bootloader version

Table 191. STM32U031xx bootloader versions

Version number	Description	Known limitations
V11.1	Initial bootloader version	Empty check flag cleared by error on the bootloader startup phase – Root cause: on the startup phase the bootloader SW performs a system deinitialization, leading to write the default value on the FLASH_ACR register, which overrides the Empty check bit with 0 – Behavior: when Empty check boot mode is used and the flash memory is empty, the MCU boots on the bootloader but the flag is cleared by the SW. If a reset is triggered, the system tries to boot on the empty flash memory, and crashes. – Caution: Avoid using reset on this case. if the system crashes, an option byte change or POR is needed to reboot.

86 STM32U073xx/83xx devices

86.1 Bootloader configuration

The STM32U073xx/83xx bootloader is activated by applying Pattern 11 (described in [Table 2](#)). [Table 192](#) shows the hardware resources used by this bootloader.

Table 192. STM32U073xx/83xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 24 MHz (using PLL clocked by HSI).
		HSI48 enabled	The clock recovery system (CRS) is enabled for the DFU bootloader to allow USB to be clocked by HSI48 48 MHz
	RAM	-	8.5 Kbytes, starting from address 0x2000000, are used by the bootloader firmware
	System memory	-	26 Kbytes, starting from address 1FFF0000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
Securable memory area	-	-	The address to jump to for the securable memory area is 0x1FFF6000
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART in reception mode. Used in alternate function, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART in transmission mode. Kept in reset configuration until 0x7F is detected on USART_RX.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART in reception mode. Used in alternate function, pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART in transmission mode. Kept in reset configuration until 0x7F is detected on USART_RX.
USART3	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART in reception mode. Used in alternate function, pull-up mode.
	USART3_TX pin	Output	PC10 pin: USART in transmission mode. Kept in reset configuration until 0x7F is detected on USART_RX.
	EXTI line 11	Input	Used on detection on USART and its IT for baudrate calculation

Table 192. STM32U073xx/83xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1101010x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain pull-up mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain pull-up mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1101010x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PB10 pin: clock line is used in open-drain pull-up mode.
	I2C2_SDA pin		PB11 pin: data line is used in open-drain pull-up mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1101010x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PB3 pin: clock line is used in open-drain pull-up mode.
	I2C3_SDA pin		PB4 pin: data line is used in open-drain pull-up mode.
SPI1	SPI	Enabled	The SPI configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in alternate function, pull-down mode
	SPI1_MISO pin ⁽¹⁾	Output	PA6 pin: slave data output line, used in push pull, pull-down mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in alternate function, pull-down mode
	SPI1_NSS pin		PA4 pin: slave chip select pin used in alternate function, pull-down mode.

Table 192. STM32U073xx/83xx configuration in system memory boot mode (continued)

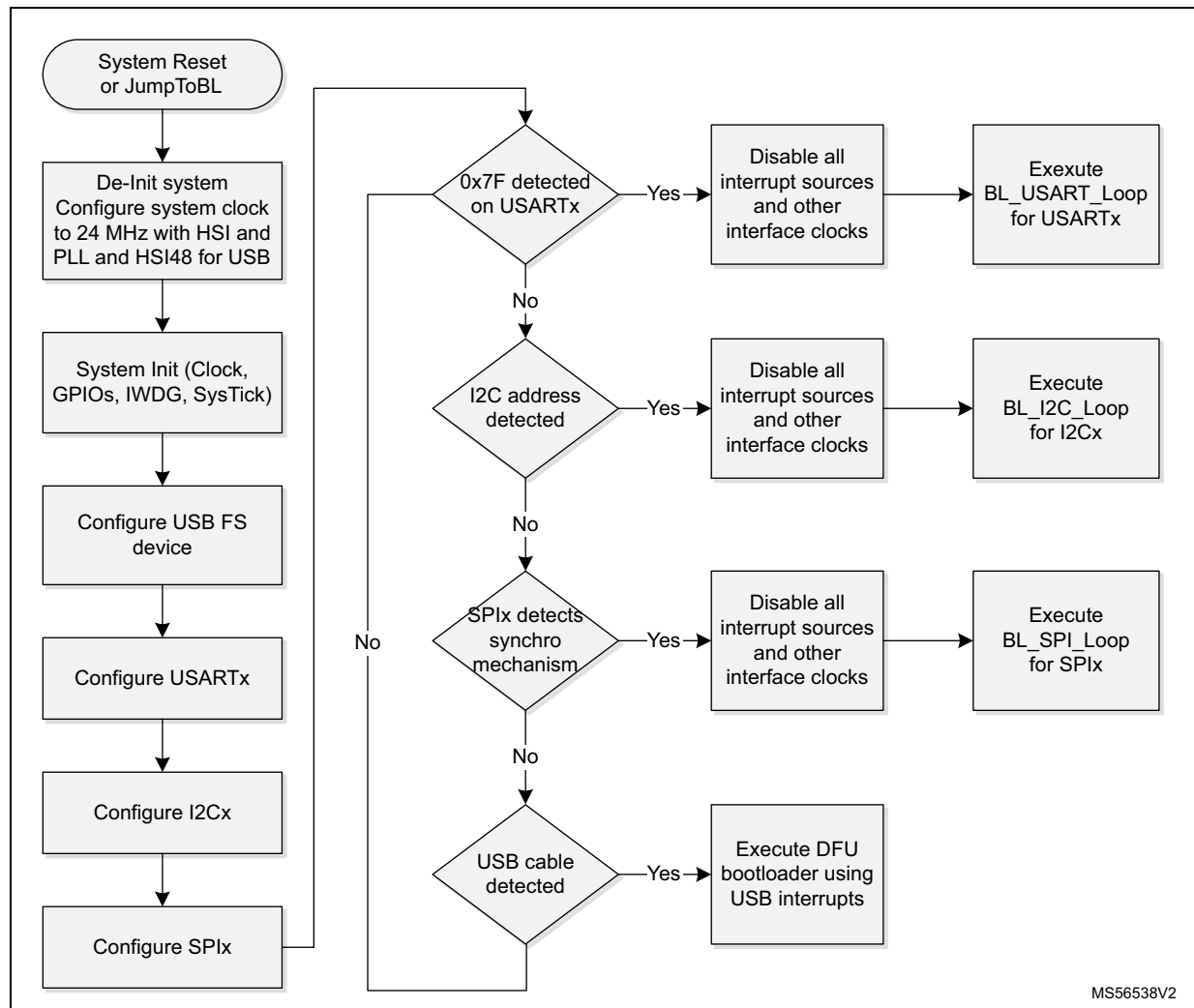
Bootloader	Feature/Peripheral	State	Comment
SPI2	SPI	Enabled	The SPI configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PB15 pin: slave data input line, used in alternate function, pull-down mode
	SPI2_MISO pin ⁽¹⁾	Output	PB14 pin: slave data output line, used in push-pull, pull mode
	SPI2_SCK pin	Input	PB13 pin: slave clock line, used in alternate function, pull-down mode
	SPI2_NSS pin		PB12 pin: slave chip select pin used in alternate function, pull-down mode.
DFU	USB	Enabled	USB configured in device mode. USB interrupt vector is enabled and used for USB DFU communications.
	USB_DM pin	Input/output	PA11: USB DM line. Used in additional function mode, behaving as input until communication starts.
	USB_DP pin		PA12: USB DP line. Used in additional function mode, behaving as input until communication starts.

1. SPI Tx (MISO) is handled by DMA. On the bootloader start-up after SPI initialization, as soon as the bit DMATx enable on SPI CR2 register is set to 1, the MISO line is set to 3.3 V.

86.2 Bootloader selection

Figure 115 shows the bootloader selection mechanism.

Figure 115. Bootloader V13.x selection for STM32U073xx/83xx



86.3 Bootloader version

Table 193. STM32U073xx/83xx bootloader versions

Version number	Description	Known limitations
V13.0	Initial bootloader version	Empty check flag cleared by error on the bootloader startup phase – Root cause: on the startup phase the bootloader SW performs a system deinitialization, leading to write the default value on the FLASH_ACR register, which overrides the Empty check bit with 0 – Behavior: when Empty check boot mode is used and the flash memory is empty, the MCU boots on the bootloader but the flag is cleared by the SW. If a reset is triggered, the system tries to boot on the empty flash memory, and crashes. – Caution: Avoid using reset on this case. if the system crashes, an option byte change or POR is needed to reboot.

87 STM32U375xx/85xx devices

87.1 Bootloader configuration

The STM32U375xx/85xx bootloader is activated by applying Pattern 12 (described in [Table 2](#)). [Table 194](#) shows the hardware resources used by this bootloader.

The bootloader follows boot model V3_1 (see [Section 4.10](#)), so it inherits all its constraints.

Table 194. STM32U375xx/85xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	MSI enabled	The system clock frequency is 48 MHz (using MSIS source, that is, MSI divided by 2)).
		-	48 MHz derived from the MSIK (MSI divided by 2) is used for FDCAN
		HSI48	CRS is enabled for the DFU so that USB can be clocked by HSI 48 MHz
	RAM	-	16 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	40 Kbytes, starting from address 0xBF8 F000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate function, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Kept in reset configuration until 0x7F detected on USART_RX.
USART3	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. Used in alternate function, pull-up mode.
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode. Kept in reset configuration until 0x7F detected on USART_RX.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 400 kHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1101100x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain pull-up mode.
	I2C1_SDA pin	Input/output	PB7 pin: data line is used in open-drain pull-up mode.

Table 194. STM32U375xx/85xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1011010x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PB10 pin: clock line is used in open-drain pull-up mode.
	I2C2_SDA pin	Input/output	PB11 pin: data line is used in open-drain pull-up mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1011010x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PC0 pin: clock line is used in open-drain pull-up mode.
	I2C3_SDA pin	Input/output	PC1 pin: data line is used in open-drain pull-up mode.
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: Slave data Input line, used in alternate function with pull down. On release v14.2, PG4 pin is used instead of PA7 on WLCSP68-G package. Used on alternate function with pull down.
	SPI1_MISO pin ⁽¹⁾	Output	PA6 pin: Slave data output line used in alternate function with pull down. On release v14.2, PG3 pin is used instead of PA6 on WLCSP68-G package. Used on alternate function with pull down.
	SPI1_SCK pin	Input	PA5 pin: Slave clock line, used in alternate function with pull down. On release v14.2, PG2 pin is used instead of PA5 on WLCSP68-G package. Used on alternate function with pull down.
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in alternate function with pull down. On release v14.2, PG5 pin is used instead of PA4 on WLCSP68-G package. Used on alternate function with pull down.

Table 194. STM32U375xx/85xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PD4 pin: Slave data Input line, used in alternate function with pull down.
	SPI2_MISO pin ⁽¹⁾	Output	PD3 pin: Slave data output line used in alternate function with pull down.
	SPI2_SCK pin	Input	PD1 pin: Slave clock line, used in alternate function with pull down.
	SPI2_NSS pin	Input	PD0 pin: slave chip select pin used in alternate function with pull down.
SPI3	SPI3	Enabled	The SPI3 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI3_MOSI pin	Input	PB5 pin: Slave data Input line, used in alternate function with pull down.
	SPI3_MISO pin ⁽¹⁾	Output	PB4 pin: Slave data output line used in alternate function with pull down.
	SPI3_SCK pin	Input	PB3 pin: Slave clock line, used in alternate function with pull down.
	SPI3_NSS pin	Input	PA15 pin: slave chip select pin used in alternate function with pull down.
FDCAN1	FDCAN1	Enabled	Once initialized the configuration is: – Connection bit rate 600 kbit/s – Data bit rate 2400 kbit/s – FrameFormat = FDCAN_FRAME_FD_BRS – Mode = FDCAN_MODE_NORMAL – AutoRetransmission = ENABLE – TransmitPause = DISABLE – ProtocolException = ENABLE
	FDCAN1_Rx pin	Input/	PB8 pin: FDCAN1 in reception mode. Used in alternate function, pull-up mode.
	FDCAN1_Tx pin	Output	PB9 pin: FDCAN1 in transmission mode. Used in alternate function, pull-up mode.

Table 194. STM32U375xx/85xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
DFU	USB	Enabled	USB configured in device mode. USB interrupt vector is enabled and used for DFU communications.
	USB_DM pin	Input/output	PA11: USB DM line. Used in alternate function with no pull mode.
	USB_DP pin		PA12: USB DP line. Used in alternate function with no pull mode. No external pull-up resistor is required.
I3C1	I3C	Enabled	Mode: target mode Aval timing: 0x4E DMA Reg RX: disabled DMA Reg TX: disabled Status FIFO: disabled DMA Req status: disabled DMA Req control: disabled IBI: enabled Additional data after IBI ack-ed: 1 byte IBI configuration: Mandatory Data Byte (MDB) All IT disabled except RXFNE (Receive FIFO Interrupt) The RXFNE interruption is disabled after SYNC byte detection by the bootloader.
	I3C_SCL pin	Input/Output	PB13 pin: clock line is used in open-drain pull up mode.
	I3C_SDA pin	Input/Output	PA1 pin: clock line is used in open-drain pull up mode.

1. SPI Tx (MISO) is handled by DMA. On the bootloader start-up after SPI initialization as soon as the bit DMATx enable on SPI CR2 register is set to 0x1, the MISO line is set to 3.3 V.

87.2 SPI1 pinout on WLCSP68-G

The SPI1 pinout on the WLCSP68-G package is different from all the other packages: it uses pins PG5/PG2/PG3/PG4 instead of pins PA4/PA5/PA6/PA7. This makes it possible to use an independent voltage for the SPI1 pins through VDDIO2, to provide them a voltage different from the global MCU voltage (3.3 V).

When using 1.2 V for these pins, the HSLV feature must be enabled on PortG. To protect this feature, which can damage the pins in case of bad usage, some SW option bytes are reserved. Bits 22 and 23 from Flash OPTR registers must be written to 0b11 while enabling IOHSLV feature (refer to RM0487 for OPTR register description).

87.3 Boot model

The bootloader follows boot model V3_1 (see [Section 4.10](#)), so it inherits all its constraints.

Table 195. STM32U375xx/385xx special commands

Special commands supported (USART/I2C/SPI/FDCAN) Opcode - 0x50							
Function	Sub-Opcode (2 bytes)	Number of data sent (2 bytes)	Data sent	Number of data received	Data received	Number of status data received (2 bytes)	Status data received
TrustZone disable Must be run when TZEN = 1 and RDP = 1	0x82	0x4	0x0	0x0	NA	0x1	0x0
Regression from RDP L1 to RDP 0.5 Must be run when TZEN = 1 and RDP = 1	0x82	0x4	0x1	0x0	NA	0x1	0x0
Unlock write protection Must be run when RDP = 1	0x82	0x4	0xYY02 ⁽¹⁾	0x0	NA	0x1	0x0

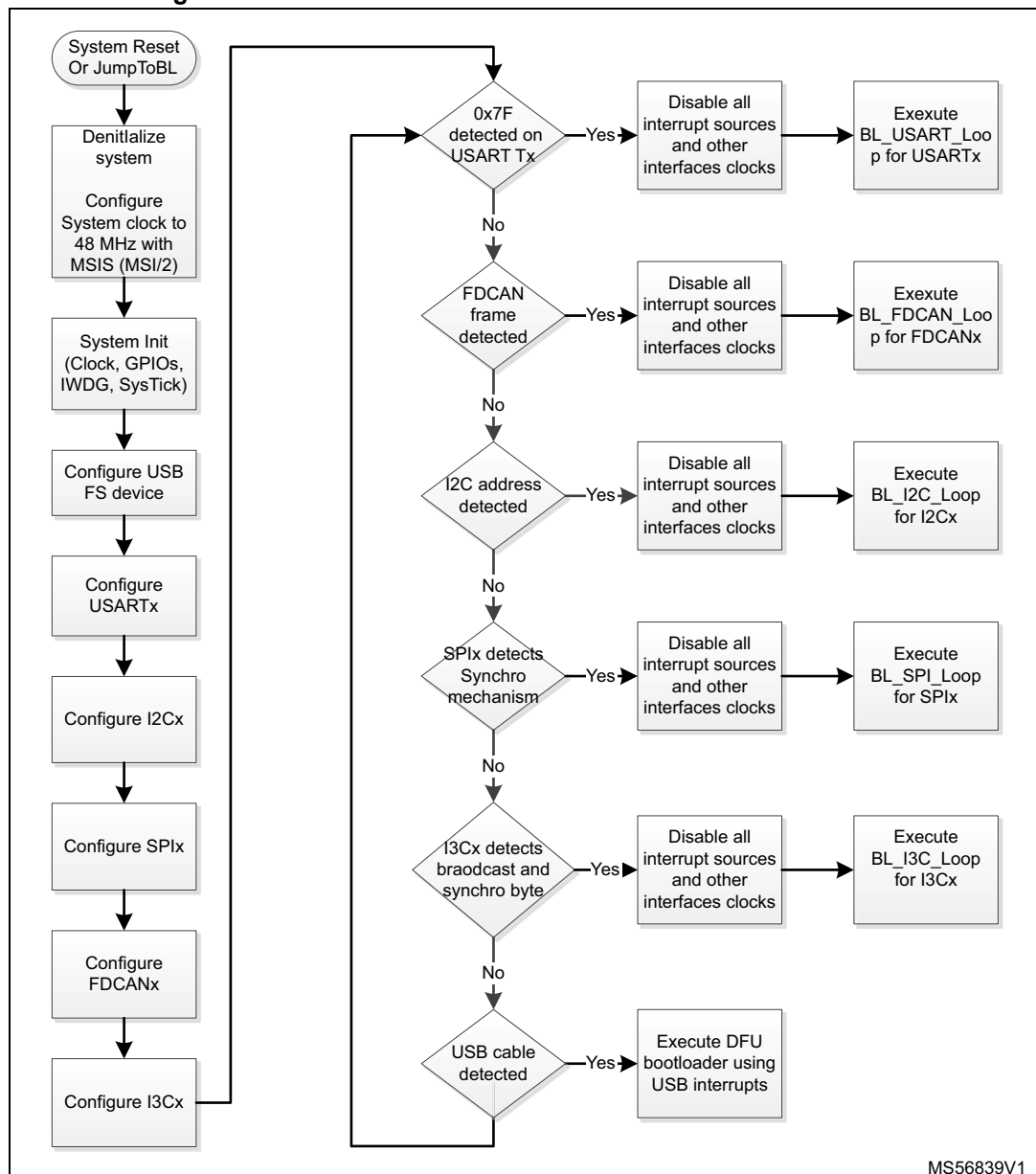
1. 0xYY can have four values (0: WRP area, 1: WRP1A, 2: WRP2A, 3: WRP1B, 4: WRP2B)

Note: *USB special commands are slightly different from the other protocols as per the USB protocol specificities:*

- *No Opcode is used, Sub-Opcode is used directly*
- *Sub-Opcode is treated in a single byte and not two bytes*
- *Data is sent on USB frame byte per byte. No need to add number of data to be transmitted*
- *Returned data and status is formatted on the USB native protocol*

87.4 Bootloader selection

Figure 116. Bootloader V14.2 selection for STM32U375xx/85xx



87.5 Bootloader version

Table 196. STM32U375xx/85xx bootloader versions

Version number	Description	Known limitations
V14.1	Initial bootloader version	– Erasing multiple flash memory pages at the same time is not working, only the first page is erased – FDCAN: cannot erase more than 32 flash memory sectors on the same time
V14.2	– Correct known limitations – Switch SPI1 support on PortG when using WLCSP68-G package instead of PortA – Add SW OB support to Enable HSLV for PortG SPI1 pins	– FDCAN: cannot erase more than 32 flash memory sectors on the same time

88 STM32U535xx/545xx devices

88.1 Bootloader configuration

The STM32U535xx/545xx bootloader is activated by applying Pattern 12 (described in [Table 2](#)). [Table 197](#) shows the hardware resources used by this bootloader.

The bootloader follows boot model V3_1 (see [Section 4.10](#)), so it inherits all its constraints.

Table 197. STM32U535xx/545xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 60 MHz (using PLL clocked by HSI).
		HSI48 enabled	CRS is enabled for the DFU so that USB can be clocked by HSI 48 MHz.
		-	20 MHz derived from the PLLQ is used for FDCAN
	RAM	-	16 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	64 Kbytes, starting from address 0x0BF90000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Set as input until USART1 is detected.
USART3	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. Used in alternate push-pull, pull-up mode.
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode. Set as input until USART1 is detected.
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1011010x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain pull-up mode.
	I2C1_SDA pin	Input/output	PB7 pin: data line is used in open-drain pull-up mode.

Table 197. STM32U535xx/545xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1011010x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PB10 pin: clock line is used in open-drain pull-up mode.
	I2C2_SDA pin	Input/output	PB11 pin: data line is used in open-drain pull-up mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1011010x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PC0 pin: clock line is used in open-drain pull-up mode.
	I2C3_SDA pin	Input/output	PC1 pin: data line is used in open-drain pull-up mode.
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, no pull mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, no pull mode
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, no pull mode.

Table 197. STM32U535xx/545xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PB15 pin: slave data input line, used in push-pull, no pull mode
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, no pull mode
	SPI2_SCK pin	Input	PB13 pin: slave clock line, used in push-pull, no pull mode
	SPI2_NSS pin	Input	PB12 pin: slave chip select pin used in push-pull, no pull mode.
SPI3	SPI3	Enabled	The SPI3 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI3_MOSI pin	Input	PB5 pin: slave data input line, used in push-pull, no pull mode
	SPI3_MISO pin	Output	PG10 pin: slave data input line, used in push-pull, no pull mode
	SPI3_SCK pin	Input	PG9 pin: slave data output line, used in push-pull, no pull mode
	SPI3_NSS pin	Input	PG12 pin: slave chip select pin used in push-pull, no pull mode.
FDCAN	FDCAN1	Enabled	Once initialized the configuration is: – Connection bit rate 250 kbit/s – Data bit rate 1000 kbit/s – FrameFormat = FDCAN_FRAME_FD_BRS – Mode = FDCAN_MODE_NORMAL – AutoRetransmission = ENABLE – TransmitPause = DISABLE – ProtocolException = ENABLE
	FDCAN1_Rx pin	Input/	PB8 pin: FDCAN1 in reception mode. Used in alternate push-pull, no pull mode.
	FDCAN1_Tx pin	Output	PB9 pin: FDCAN1 in transmission mode. Used in alternate push-pull, no pull mode.

Table 197. STM32U535xx/545xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
DFU	USB	Enabled	USB FS configured in forced device mode. USB FS interrupt vector is enabled and used for USB DFU communications.
	USB_DM pin	Input/output	PA11: USB DM line. Used in input no pull mode.
	USB_DP pin		PA12: USB DP line. Used in input no pull mode. No external pull-up resistor is required

Table 198. STM32U535xx/545xx special commands

Special commands supported (USART/I2C/SPI/FDCAN) Opcode - 0x50							
Function	Sub-Opcode (2 bytes)	Number of data sent (2 bytes)	Data sent	Number of data received	Data received	Number of status data received (2 bytes)	Status data received
TrustZone disable Must be run when TZEN = 1 and RDP = 1	0x82	0x4	0x0	0x0	NA	0x1	0x0
Regression from RDP L1 to RDP 0.5 Must be run when TZEN = 1 and RDP = 1	0x82	0x4	0x1	0x0	NA	0x1	0x0
Unlock write protection Must be run when RDP = 1	0x82	0x4	0xYY02 ⁽¹⁾	0x0	NA	0x1	0x0

1. 0xYY can have four values (0: WRP area, 1: WRP1A, 2: WRP2A, 3: WRP1B, 4: WRP2B)

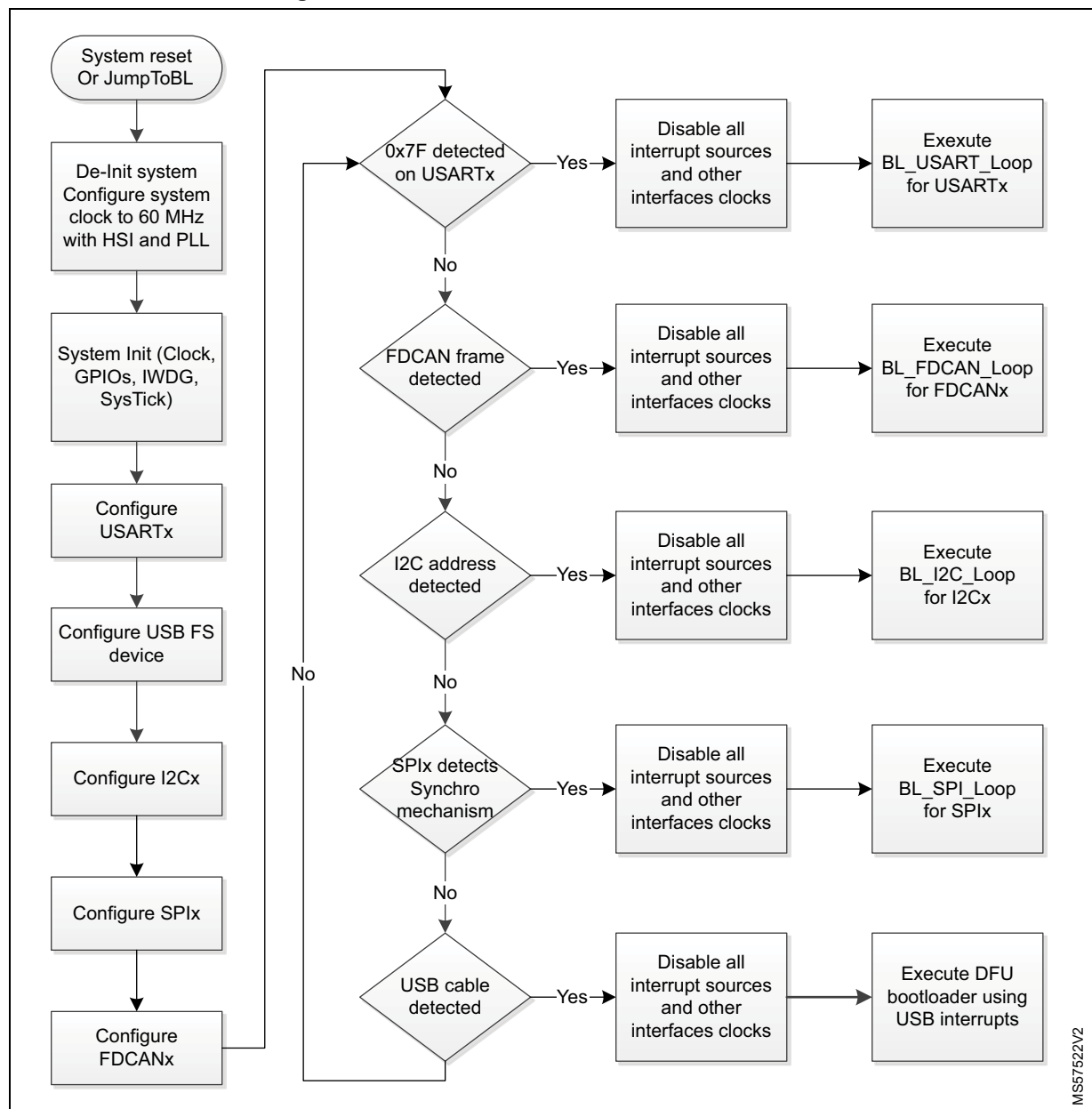
Note: USB special commands are slightly different from the other protocols as per the USB protocol specificities:

- No Opcode is used, Sub-Opcode is used directly
- Sub-Opcode is treated in a single byte and not two bytes
- Data is sent on USB frame byte per byte. No need to add number of data to be transmitted
- Returned data and status is formatted on the USB native protocol

88.2 Bootloader selection

Figure 117 shows the bootloader selection mechanism.

Figure 117. Bootloader V9.x selection for STM32U535xx/545xx



88.3 Bootloader version

Table 199. STM32U535xx/545xx bootloader versions

Version number	Description	Known limitations
V9.1	Initial bootloader version	FDCAN Readout unprotect command does not send the command ID to the host

89 STM32U575xx/85xx devices

89.1 Bootloader configuration

The STM32U575xx/85xx bootloader is activated by applying Pattern 12 (described in [Table 2](#)). [Table 200](#) shows the hardware resources used by this bootloader.

The bootloader follows boot model V3_1 (see [Section 4.10](#)), so it inherits all its constraints.

Table 200. STM32U575xx/85xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 60 MHz (using PLL clocked by HSI).
		HSI48 enabled	CRS is enabled for the DFU so that USB can be clocked by HSI 48 MHz.
		-	20 MHz derived from the PLLQ is used for FDCAN
	RAM	-	16 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	64 Kbytes, starting from address 0x0BF90000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in alternate push-pull, pull-up mode.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Used in alternate push-pull, pull-up mode.
USART3	USART3	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. Used in alternate push-pull, pull-up mode.
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode. Used in alternate push-pull, pull-up mode.

Table 200. STM32U575xx/85xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1011010x (x = 0 for write and x = 1 for read)
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain pull-up mode.
	I2C1_SDA pin	Input/output	PB7 pin: data line is used in open-drain pull-up mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1011010x (x = 0 for write and x = 1 for read)
	I2C2_SCL pin	Input/output	PB10 pin: clock line is used in open-drain pull-up mode.
	I2C2_SDA pin	Input/output	PB11 pin: data line is used in open-drain pull-up mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1011010x (x = 0 for write and x = 1 for read)
	I2C3_SCL pin	Input/output	PC0 pin: clock line is used in open-drain pull-up mode.
	I2C3_SDA pin	Input/output	PC1 pin: data line is used in open-drain pull-up mode.
SPI1	SPI1	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, pull-down mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, pull-down mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, pull-down mode
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, pull-down mode.

Table 200. STM32U575xx/85xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
SPI2	SPI2	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PB15 pin: slave data input line, used in push-pull, pull-down mode
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, pull-down mode
	SPI2_SCK pin	Input	PB13 pin: slave clock line, used in push-pull, pull-down mode
	SPI2_NSS pin	Input	PB12 pin: slave chip select pin used in push-pull, pull-down mode.
SPI3	SPI3	Enabled	The SPI3 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz – Polarity: CPOL low, CPHA low, NSS hardware.
	SPI3_MOSI pin	Input	PB5 pin: slave data input line, used in push-pull, pull-down mode
	SPI3_MISO pin	Output	PG10 pin: slave data input line, used in push-pull, pull-down mode
	SPI3_SCK pin	Input	PG9 pin: slave data output line, used in push-pull, pull-down mode
	SPI3_NSS pin	Input	PG12 pin: slave chip select pin used in push-pull, pull-down mode.
FDCAN	FDCAN1	Enabled	Once initialized the configuration is: – Connection bit rate 250 kbit/s – Data bit rate 1000 kbit/s – FrameFormat = FDCAN_FRAME_FD_BRS – Mode = FDCAN_MODE_NORMAL – AutoRetransmission = ENABLE – TransmitPause = DISABLE – ProtocolException = ENABLE
	FDCAN1_Rx pin	Input/	PB8 pin: FDCAN1 in reception mode. Used in alternate push-pull, pull-up mode.
	FDCAN1_Tx pin	Output	PB9 pin: FDCAN1 in transmission mode. Used in alternate push-pull, pull-up mode.

Table 200. STM32U575xx/85xx configuration in system memory boot mode (continued)

Bootloader	Feature/Peripheral	State	Comment
DFU	USB	Enabled	USB FS configured in forced device mode. USB FS interrupt vector is enabled and used for USB DFU communications. Note: VDDUSB IO must be connected to 3.3 V as USB peripheral is used by the bootloader.
	USB_DM pin	Input/output	PA11: USB DM line. Used in input no pull mode.
	USB_DP pin		PA12: USB DP line. Used in input no pull mode. No external pull-up resistor is required

Table 201. STM32U575xx/585xx special commands

Special commands supported (USART/I2C/SPI/FDCAN) Opcode - 0x50							
Function	Sub-Opcode (2 bytes)	Number of data sent (2 bytes)	Data sent	Number of data received	Data received	Number of status data received (2 bytes)	Status data received
TrustZone disable Must be run when TZEN = 1 and RDP = 1	0x82	0x4	0x0	0x0	NA	0x1	0x0
Regression from RDP L1 to RDP 0.5 Must be run when TZEN = 1 and RDP = 1	0x82	0x4	0x1	0x0	NA	0x1	0x0
Unlock write protection Must be run when RDP = 1	0x82	0x4	0xYY02 ⁽¹⁾	0x0	NA	0x1	0x0

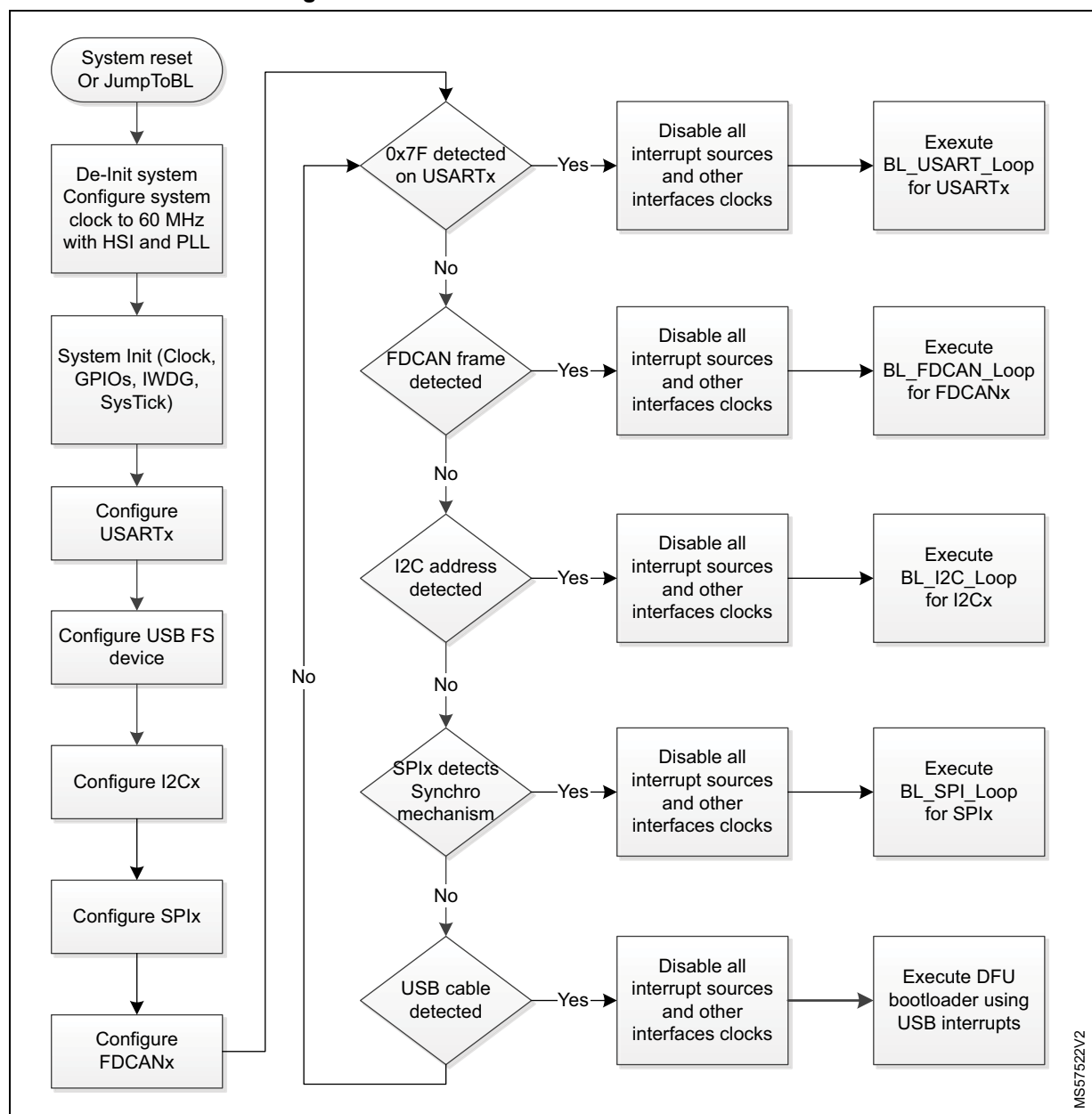
1. 0xYY can have four values (0: WRP area, 1: WRP1A, 2: WRP2A, 3: WRP1B, 4: WRP2B)

Note: USB special commands are slightly different from the other protocols as per the USB protocol specificities:

- No Opcode is used, Sub-Opcode is used directly
- Sub-Opcode is treated in a single byte and not two bytes
- Data is sent on USB frame byte per byte. No need to add number of data to be transmitted
- Returned data and status is formatted on the USB native protocol

89.2 Bootloader selection

Figure 118. Bootloader V9.x selection for STM32U575xx/85xx



89.3 Bootloader version

Table 202. STM32U575xx/85xx bootloader versions

Version number	Description	Known limitations
V9.2	Initial bootloader version	FDCAN Readout unprotect command does not send the command ID to the host

90 STM32U595xx/99xx/A5xx/A9xx devices

90.1 Bootloader configuration

The STM32U595xx/99xx/A5xx/A9xx bootloader is activated by applying Pattern 12 (described in [Table 2](#)). [Table 203](#) shows the hardware resources used by this bootloader.

The bootloader follows boot model V3_1 (see [Section 4.10](#)), so it inherits all its constraints.

Table 203. STM32U595xx/99xx/A5xx/A9xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 60 MHz (using PLL clocked by HSI).
		-	20 MHz derived from the PLLQ is used for FDCAN
		HSE enabled	When USB cable is detected, SW tries to detect if a quartz is plugged in the board to configure the USBPHY clock. Supported quartz: 8, 12, 16, 20, 24, 26, and 32 MHz If no quartz is detected a system reset is triggered.
	RAM	-	16 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	64 Kbytes, starting from address 0x0BF90000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Set as input until USART1 is detected.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Set as input until USART1 is detected.
USART3	USART3	Enabled	Once initialized the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. Used in alternate push-pull, pull-up mode.
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode. Set as input until USART1 is detected.

Table 203. STM32U595xx/99xx/A5xx/A9xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1100000x (x = 0 for write and x = 1 for read).
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain pull-up mode.
	I2C1_SDA pin	Input/output	PB7 pin: data line is used in open-drain pull-up mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1100000x (x = 0 for write and x = 1 for read).
	I2C2_SCL pin	Input/output	PB10 pin: clock line is used in open-drain pull-up mode.
	I2C2_SDA pin	Input/output	PB11 pin: data line is used in open-drain pull-up mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1100000x (x = 0 for write and x = 1 for read).
	I2C3_SCL pin	Input/output	PC0 pin: clock line is used in open-drain pull-up mode.
	I2C3_SDA pin	Input/output	PC1 pin: data line is used in open-drain pull-up mode.
SPI1	SPI	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, no pull mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, no pull mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, no pull mode
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, no pull mode.

Table 203. STM32U595xx/99xx/A5xx/A9xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
SPI2	SPI	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PB15 pin: slave data input line, used in push-pull, no pull mode
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, no pull mode
	SPI2_SCK pin	Input	PB13 pin: slave clock line, used in push-pull, no pull mode
	SPI2_NSS pin	Input	PB12 pin: slave chip select pin used in push-pull, no pull mode.
SPI3	SPI	Enabled	The SPI3 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz Polarity: CPOL low, CPHA low, NSS hardware.
	SPI3_MOSI pin	Input	PB5 pin: slave data input line, used in push-pull, no pull mode
	SPI3_MISO pin	Output	PG10 pin: slave data input line, used in push-pull, no pull mode
	SPI3_SCK pin	Input	PG9 pin: slave data output line, used in push-pull, no pull mode
	SPI3_NSS pin	Input	PG12 pin: slave chip select pin used in push-pull, no pull mode.
DFU	USB	Enabled	USB HS configured in forced device mode. USB HS interrupt vector is enabled and used for USB DFU communications.
	USB_DM pin	Input/output	PA11: USB DM line. Used in input no pull mode.
	USB_DP pin		PA12: USB DP line. Used in input no pull mode. No external pull-up resistor is required

Table 203. STM32U595xx/99xx/A5xx/A9xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
FDCAN	FDCAN1	Enabled	Once initialized the configuration is: – Connection bit rate 250 kbit/s – Data bit rate 1000 kbit/s – FrameFormat = FDCAN_FRAME_FD_BRS – Mode = FDCAN_MODE_NORMAL – AutoRetransmission = ENABLE – TransmitPause = DISABLE – ProtocolException = ENABLE
	FDCAN1_Rx pin	Input/	PB8 pin: FDCAN1 in reception mode. Used in alternate push-pull, no pull mode.
	FDCAN1_Tx pin	Output	PB9 pin: FDCAN1 in transmission mode. Used in alternate push-pull, no pull mode.

Table 204. STM32U595xx/99xx/A5xx/A9xx special commands

Special commands supported (USART/I2C/SPI/FDCAN) Opcode - 0x50							
Function	Sub-Opcode (2 bytes)	Number of data sent (2 bytes)	Data sent	Number of data received	Data received	Number of status data received (2 bytes)	Status data received
TrustZone disable Must be run when TZEN = 1 and RDP = 1	0x82	0x4	0x0	0x0	NA	0x1	0x0
Regression from RDP L1 to RDP 0.5 Must be run when TZEN = 1 and RDP = 1	0x82	0x4	0x1	0x0	NA	0x1	0x0
Unlock write protection Must be run when RDP = 1	0x82	0x4	0xYY02 ⁽¹⁾	0x0	NA	0x1	0x0

1. 0xYY can have four values (0: WRP area, 1: WRP1A, 2: WRP2A, 3: WRP1B, 4: WRP2B)

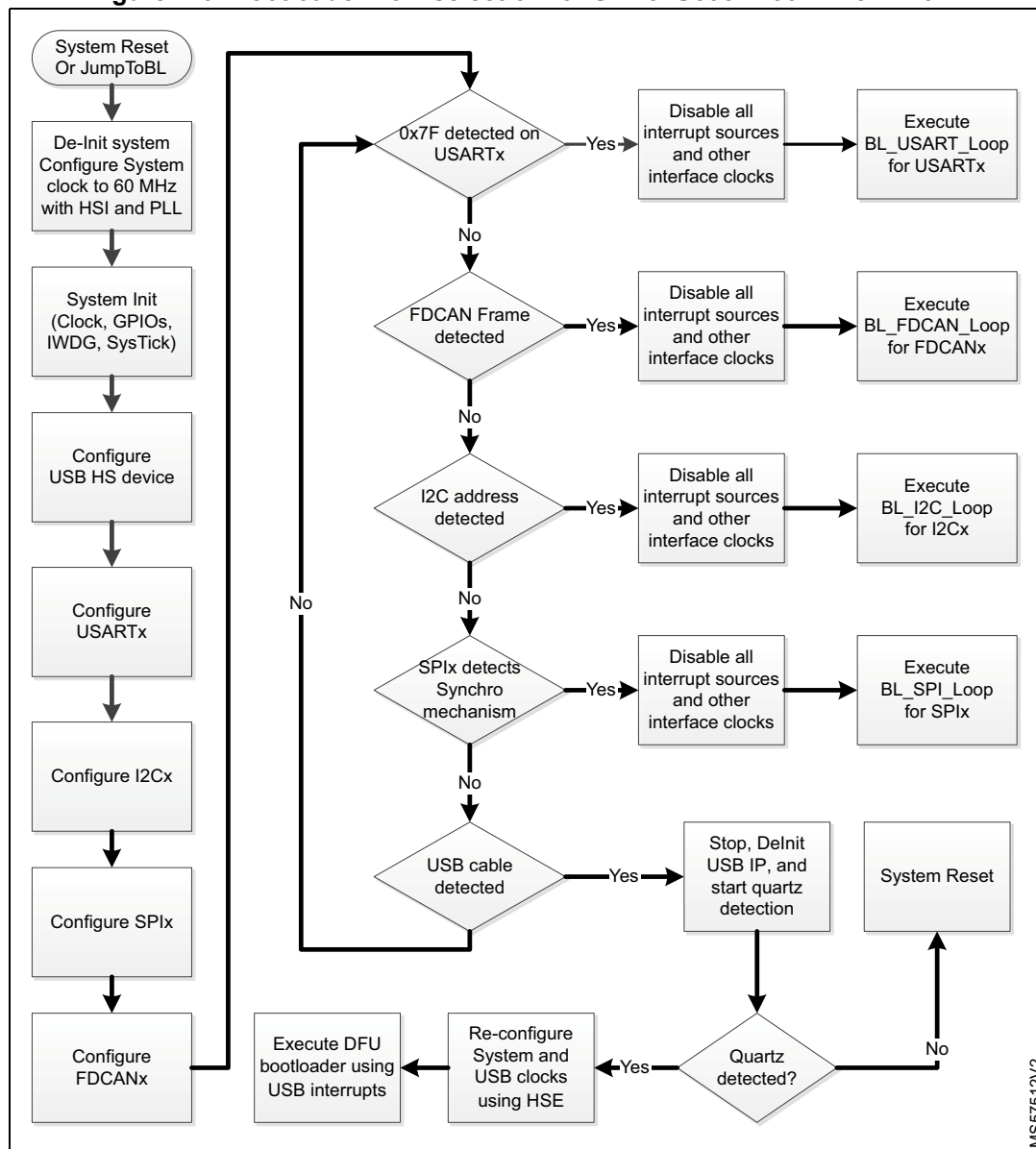
Note: *USB special commands are slightly different from the other protocols as per the USB protocol specificities:*

- *No Opcode is used, Sub-Opcode is used directly*
- *Sub-Opcode is treated in a single byte and not two bytes*
- *Data is sent on USB frame byte per byte. No need to add number of data to be transmitted*
- *Returned data and status is formatted on the USB native protocol*

90.2 Bootloader selection

Figure 119 shows the bootloader selection mechanism.

Figure 119. Bootloader V9.x selection for STM32U595xx/99xx/A5xx/A9xx



90.3 Bootloader version

Table 205. STM32U595xx/99xx/A5xx/A9xx bootloader versions

Version number	Description	Known limitations
V9.2	Initial bootloader version	FDCAN Readout unprotect command does not send the command ID to the host

91 STM32U5F7xx/F9xx/G7xx/G9xx devices

91.1 Bootloader configuration

The STM32U5F7xx/F9xx/G7xx/G9xx bootloader is activated by applying Pattern 12 (described in [Table 2](#)). [Table 206](#) shows the hardware resources used by this bootloader.

Table 206. STM32U5F7xx/F9xx/G7xx/G9xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
Common to all	RCC	HSI enabled	The system clock frequency is 60 MHz (using PLL clocked by HSI).
		-	20 MHz derived from the PLLQ is used for FDCAN
		HSE enabled	When USB cable is detected, SW tries to detect if a quartz is plugged in the board to configure the USBPHY clock. Supported quartz: 8, 12, 16, 20, 24, 26, and 32 MHz. If no quartz is detected, a system reset is triggered.
	RAM	-	16 Kbytes, starting from address 0x20000000, are used by the bootloader firmware
	System memory	-	64 Kbytes, starting from address 0x0BF90000, contain the bootloader firmware
	IWDG	-	The IWDG prescaler is configured to its maximum value. It is periodically refreshed to prevent watchdog reset (if the hardware IWDG option was previously enabled by the user).
USART1	USART1	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in alternate push-pull, pull-up mode.
	USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Set as input until USART1 is detected.
USART2	USART2	Enabled	Once initialized, the configuration is 8-bit, even parity, and one stop bit
	USART2_RX pin	Input	PA3 pin: USART2 in reception mode. Used in alternate push-pull, pull-up mode.
	USART2_TX pin	Output	PA2 pin: USART2 in transmission mode. Set as input until USART1 is detected.
USART3	USART3	Enabled	Once initialized the configuration is 8-bit, even parity, and one stop bit
	USART3_RX pin	Input	PC11 pin: USART3 in reception mode. Used in alternate push-pull, pull-up mode.
	USART3_TX pin	Output	PC10 pin: USART3 in transmission mode. Set as input until USART1 is detected.

Table 206. STM32U5F7xx/F9xx/G7xx/G9xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
I2C1	I2C1	Enabled	The I2C1 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1101001x (x = 0 for write and x = 1 for read).
	I2C1_SCL pin	Input/output	PB6 pin: clock line is used in open-drain pull-up mode.
	I2C1_SDA pin		PB7 pin: data line is used in open-drain pull-up mode.
I2C2	I2C2	Enabled	The I2C2 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1101001x (x = 0 for write and x = 1 for read).
	I2C2_SCL pin	Input/output	PB10 pin: clock line is used in open-drain pull-up mode.
	I2C2_SDA pin		PB11 pin: data line is used in open-drain pull-up mode.
I2C3	I2C3	Enabled	The I2C3 configuration is: – I2C speed: up to 1 MHz – 7-bit address – Target mode – Analog filter ON – Target 7-bit address: 0b1101001x (x = 0 for write and x = 1 for read).
	I2C3_SCL pin	Input/output	PC0 pin: clock line is used in open-drain pull-up mode.
	I2C3_SDA pin		PC1 pin: data line is used in open-drain pull-up mode.
SPI1	SPI	Enabled	The SPI1 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz Polarity: CPOL low, CPHA low, NSS hardware.
	SPI1_MOSI pin	Input	PA7 pin: slave data input line, used in push-pull, no pull mode
	SPI1_MISO pin	Output	PA6 pin: slave data output line, used in push-pull, no pull mode
	SPI1_SCK pin	Input	PA5 pin: slave clock line, used in push-pull, no pull mode
	SPI1_NSS pin	Input	PA4 pin: slave chip select pin used in push-pull, no pull mode.

Table 206. STM32U5F7xx/F9xx/G7xx/G9xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
SPI2	SPI	Enabled	The SPI2 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz Polarity: CPOL low, CPHA low, NSS hardware.
	SPI2_MOSI pin	Input	PB15 pin: slave data input line, used in push-pull, no pull mode
	SPI2_MISO pin	Output	PB14 pin: slave data output line, used in push-pull, no pull mode
	SPI2_SCK pin	Input	PB13 pin: slave clock line, used in push-pull, no pull mode
	SPI2_NSS pin	Input	PB12 pin: slave chip select pin used in push-pull, no pull mode.
SPI3	SPI	Enabled	The SPI3 configuration is: – Slave mode – Full Duplex – 8-bit MSB – Speed up to 8 MHz Polarity: CPOL low, CPHA low, NSS hardware.
	SPI3_MOSI pin	Input	PB5 pin: slave data input line, used in push-pull, no pull mode
	SPI3_MISO pin	Output	PG10 pin: slave data input line, used in push-pull, no pull mode
	SPI3_SCK pin	Input	PG9 pin: slave data output line, used in push-pull, no pull mode
	SPI3_NSS pin	Input	PG12 pin: slave chip select pin used in push-pull, no pull mode.
DFU	USB	Enabled	USB HS configured in forced device mode. USB HS interrupt vector is enabled and used for USB DFU communications.
	USB_DM pin	Input/output	PA11: USB DM line. Used in input no pull mode.
	USB_DP pin		PA12: USB DP line. Used in input no pull mode. No external pull-up resistor is required

Table 206. STM32U5F7xx/F9xx/G7xx/G9xx configuration in system memory boot mode

Bootloader	Feature/Peripheral	State	Comment
FDCAN	FDCAN1	Enabled	Once initialized the configuration is: – Connection bit rate 250 kbit/s – Data bit rate 1000 kbit/s – FrameFormat = FDCAN_FRAME_FD_BRS – Mode = FDCAN_MODE_NORMAL – AutoRetransmission = ENABLE – TransmitPause = DISABLE – ProtocolException = ENABLE
	FDCAN1_Rx pin	Input/	PB8 pin: FDCAN1 in reception mode. Used in alternate push-pull, no pull mode.
	FDCAN1_Tx pin	Output	PB9 pin: FDCAN1 in transmission mode. Used in alternate push-pull, no pull mode.

Table 207. STM32U5F7xx/F9xx/G7xx/G9xx special commands

Special commands supported (USART/I2C/SPI/FDCAN) Opcode - 0x50							
Function	Sub-Opcode (2 bytes)	Number of data sent (2 bytes)	Data sent	Number of data received	Data received	Number of status data received (2 bytes)	Status data received
TrustZone disable Must be run when TZEN = 1 and RDP = 1	0x82	0x4	0x0	0x0	NA	0x1	0x0
Regression from RDP L1 to RDP 0.5 Must be run when TZEN = 1 and RDP = 1	0x82	0x4	0x1	0x0	NA	0x1	0x0
Unlock write protection Must be run when RDP = 1	0x82	0x4	0xYY02 ⁽¹⁾	0x0	NA	0x1	0x0

1. 0xYY can have four values (0: WRP area, 1: WRP1A, 2: WRP2A, 3: WRP1B, 4: WRP2B)

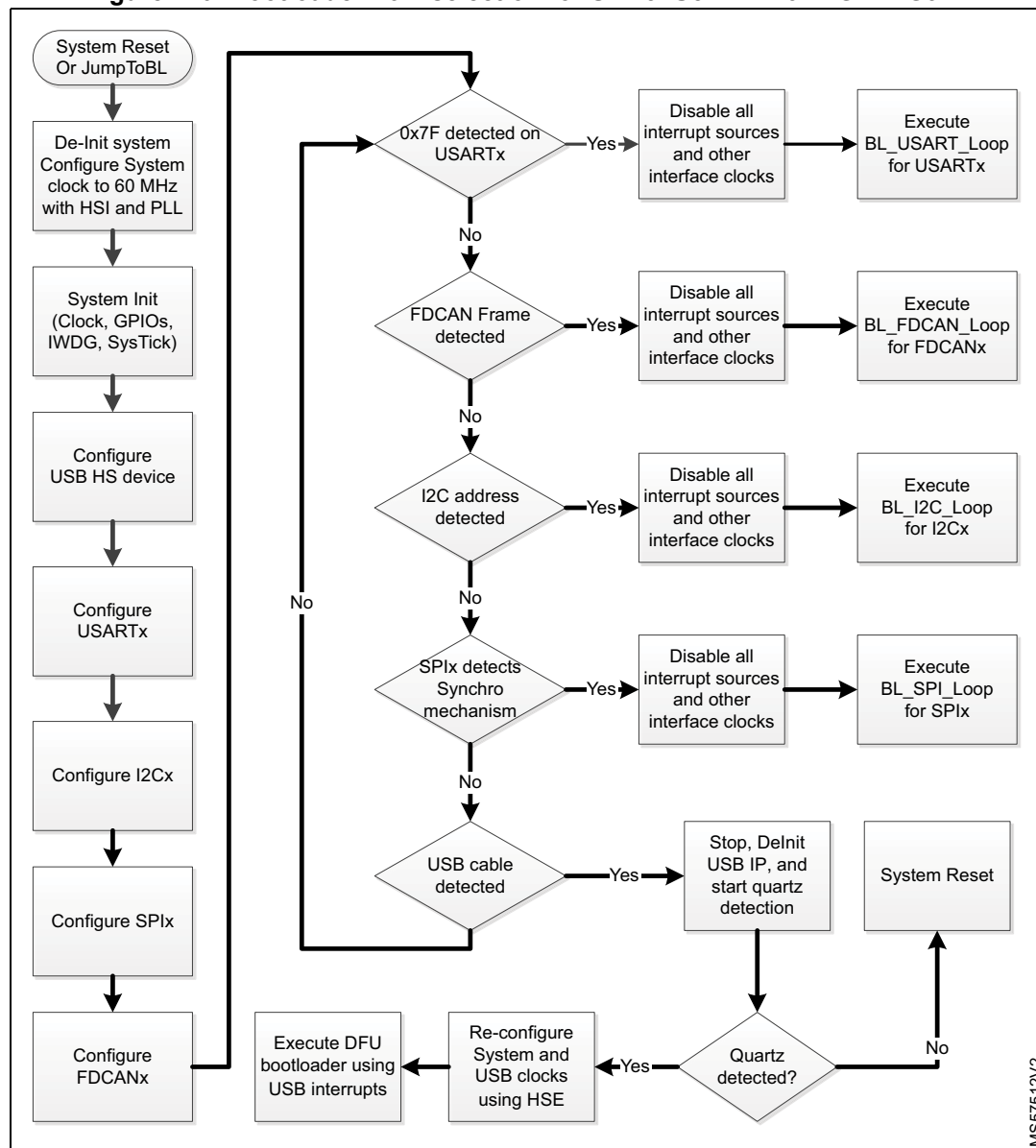
Note: USB special commands are slightly different from the other protocols as per the USB protocol specificities:

- No Opcode is used, Sub-Opcode is used directly
- Sub-Opcode is treated in a single byte and not two bytes
- Data is sent on USB frame byte per byte. No need to add number of data to be transmitted
- Returned data and status is formatted on the USB native protocol

91.2 Bootloader selection

Figure 119 shows the bootloader selection mechanism.

Figure 120. Bootloader V9.x selection for STM32U5F7xx/F9xx/G7xx/G9xx



91.3 Bootloader version

Table 208. STM32U5F7xx/F9xx/G7xx/G9xx bootloader versions

Version number	Description	Known limitations
V9.0	Initial bootloader version	FDCAN Readout unprotect command does not send the command ID to the host

92 Device-dependent bootloader parameters

The bootloader protocol command set and sequences for each serial peripheral are the same for all STM32 devices. Some parameters depend on device and bootloader version:

- PID (Product ID)
- Valid RAM addresses (RAM area used during bootloader execution is not accessible) accepted by the bootloader when the Read Memory, Go and Write Memory commands are requested.
- System memory area.

[Table 209](#) shows the values of these parameters for each STM32 device.

Table 209. Bootloader device-dependent parameters

STM32 series	Device	PID	BL ID	RAM	System memory
C0	STM32C011xx	0x443	0x51	0x20000000 - 0x20002FFF	0x1FFF0000 - 0x1FFF17FF
	STM32C031xx	0x453	0x52	0x20002000 - 0x200017FF	
	STM32C051xx	0x44C	0xB0	0x20001400 - 0x20002FFF	0x1FFF0000 - 0x1FFF2FFF
	STM32C071xx	0x493	0xD1	0x20002300 - 0x20005FFF	0x1FFF0000 - 0x1FFF6FFF
	STM32C091xx/92xx	0x44D	0x121	0x20002400 - 0x200077FF	0x1FFF0000 - 0x1FFF3FFF
F0	STM32F05xxx and STM32F030x8	0x440	0x21	0x20000800 - 0x20001FFF	0x1FFFE000 - 0x1FFFF7FF
	STM32F03xx4/6	0x444	0x10	0x20000800 - 0x20000FFF	
	STM32F030xC	0x442	0x52	0x20001800 - 0x20007FFF	0x1FFFD800 - 0x1FFFF7FF
	STM32F04xxx	0x445	0xA1	NA	0x1FFFC400 - 0x1FFFF7FF
	STM32F070x6	0x445	0xA2	NA	0x1FFFC400 - 0x1FFFF7FF
	STM32F070xB	0x448	0xA2	NA	0x1FFFC800 - 0x1FFFF7FF
	STM32F071xx/072xx	0x448	0xA1	0x20001800 - 0x20003FFF	0x1FFFC800 - 0x1FFFF7FF
	STM32F09xxx	0x442	0x50	NA	0x1FFFD800 - 0x1FFFF7FF

Table 209. Bootloader device-dependent parameters (continued)

STM32 series	Device		PID	BL ID	RAM	System memory
F1	STM32F10xxx	Low-density	0x412	NA	0x20000200 - 0x200027FF	0x1FFFF000 - 0x1FFFF7FF
		Medium-density	0x410	NA	0x20000200 - 0x20004FFF	
		High-density	0x414	NA	0x20000200 - 0x2000FFFF	
		Medium-density value line	0x420	0x10	0x20000200 - 0x20001FFF	
		High-density value line	0x428	0x10	0x20000200 - 0x20007FFF	
	STM32F105xx/107xx		0x418	NA	0x20001000 - 0x2000FFFF	0x1FFFB000 - 0x1FFFF7FF
	STM32F10xxx XL-density		0x430	0x21	0x20000800 - 0x20017FFF	0x1FFFE000 - 0x1FFFF7FF
F2	STM32F2xxxx		0x411	0x20	0x20002000 - 0x2001FFFF	0x1FFF0000 - 0x1FFF77FF
				0x33		
F3	STM32F373xx		0x432	0x41	0x20001400 - 0x20007FFF	0x1FFFD800 - 0x1FFFF7FF
	STM32F378xx			0x50	0x20001000 - 0x20007FFF	
	STM32F302xB(C)/303xB(C)		0x422	0x41	0x20001400 - 0x20009FFF	
	STM32F358xx			0x50		
	STM32F301xx/302x4(6/8)		0x439	0x40	0x20001800 - 0x20003FFF	
	STM32F318xx			0x50		
	STM32F303x4(6/8)/334xx/328xx		0x438	0x50	0x20001800 - 0x20002FFF	
	STM32F302xD(E)/303xD(E)		0x446	0x40	0x20001800 - 0x2000FFFF	
	STM32F398xx		0x446	0x50	0x20001800 - 0x2000FFFF	

Table 209. Bootloader device-dependent parameters (continued)

STM32 series	Device	PID	BL ID	RAM	System memory
F4	STM32F40xxx/41xxx	0x413	0x31	0x20002000 - 0x2001FFFF	0x1FFF0000 - 0x1FFF77FF
			0x91	0x20003000 - 0x2001FFFF	
	STM32F42xxx/43xxx	0x419	0x70	0x20003000 - 0x2002FFFF	
			0x91	0x2002FFFF	
	STM32F401xB(C)	0x423	0xD1	0x20003000 - 0x2000FFFF	
	STM32F401xD(E)	0x433	0xD1	0x20003000 - 0x20017FFF	
	STM32F410xx	0x458	0xB1	0x20003000 - 0x20007FFF	
	STM32F411xx	0x431	0xD0	0x20003000 - 0x2001FFFF	
	STM32F412xx	0x441	0x90	0x20003000 - 0x2003FFFF	
	STM32F446xx	0x421	0x90	0x20003000 - 0x2001FFFF	
	STM32F469xx/479xx	0x434	0x90	0x20003000 - 0x2005FFFF	
	STM32F413xx/423xx	0x463	0x90	0x20003000 - 0x2004FFFF	
F7	STM32F72xxx/73xxx	0x452	0x90	0x20004000 - 0x2003FFFF	0x1FF00000 - 0x1FF0EDBF
	STM32F74xxx/75xxx	0x449	0x70	0x20004000 - 0x2004FFFF	0x1FF00000 - 0x1FF0EDBF
			0x90	0x20004000 - 0x2004FFFF	0x1FF00000 - 0x1FF0EDBF
	STM32F76xxx/77xxx	0x451	0x93	0x20004000 - 0x2007FFFF	0x1FF00000 - 0x1FF0EDBF

Table 209. Bootloader device-dependent parameters (continued)

STM32 series	Device	PID	BL ID	RAM	System memory
G0	STM32G03xxx/04xxx	0x466	0x52	0x20001000 - 0x20001FFF	0x1FFF0000 - 0x1FFF1FFF
	STM32G05xxx/061xx	0x456	0x51	0x20001000 - 0x20002000	0x1FFF0000 - 0x1FFF1FFF
	STM32G07xxx/08xxx	0x460	0xB4	0x20002700 - 0x20009000	0x1FFF0000 - 0x1FFF6FFF
	STM32G0B0xx	0x467	0xD0	0x20004000 - 0x20020000	0x1FFF0000 - 0x1FFF6FFF 0x1FFF8000 - 0x1FFFEFFF
	STM32G0B1xx/0C1xx	0x467	0x92	0x20004000 - 0x20020000	0x1FFF0000 - 0x1FFF6FFF 0x1FFF8000 - 0x1FFFEFFF
G4	STM32G431xx/441xx	0x468	0xD4	0x20004000 - 0x20005800	0x1FFF0000 - 0x1FFF7000
	STM32G47xxx/48xxx	0x469	0xD5	0x20004000 - 0x20018000	0x1FFF0000 - 0x1FFF7000
	STM32G491xx/A1xx	0x479	0xD2	0x20004000 - 0x2001C000	0x1FFF0000 - 0x1FFF7000
H5	STM32H503xx	0x474	0xE1	0x20004000 - 0x20007FFF	0x0BF87000 - 0x0BF8FFFF
	STM32H562xx/563xx/573xx	0x484	0xE3	0x20000000 - 0x2009FFFF	0x0BF97000 - 0x0BF9FFFF
	STM32H523xx/533xx	0x478	0xE2	0x20004000 - 0x20043FFF	0x0BF97000 - 0x0BF9FFFF

Table 209. Bootloader device-dependent parameters (continued)

STM32 series	Device	PID	BL ID	RAM	System memory
H7	STM32H72xxx/73xxx	0x483	0x93	0x20004100 - 0x2001FFFF 0x24004000 - 0x2404FFFF	0x1FF00000 - 0x1FF1E7FF
	STM32H74xxx/75xxx	0x450	0x92	0x20004100 - 0x2001FFFF 0x24005000 - 0x2407FFFF	0x1FF00000 - 0x1FF1E7FF
	STM32H7A3xx/7B3xx/7B0xx	0x480	0x92	0x20004100 - 0x2001FFFF 0x24034000 - 0x2407FFFF	0x1FF00000 - 0x1FF13FFF
	STM32H7Rxxx/7Sxxx	0x485	0xE3	0x24000000 - 0x2401FFFF 0x24024000 - 0x24071FFF 0x20000000 - 0x2002FFFF 0x00000000 - 0x0002FFFF ⁽¹⁾	0x1FF18000 - 0x1FF1FFFF
L0	STM32L01xxx/02xxx	0x457	0xC3	NA	0x1FF00000 - 0x1FF00FFF
	STM32L031xx/041xx	0x425	0xC0	0x20001000 - 0x20001FFF	0x1FF00000 - 0x1FF00FFF
	STM32L05xxx/06xxx	0x417	0xC0	0x20001000 - 0x20001FFF	0x1FF00000 - 0x1FF00FFF
	STM32L07xxx/08xxx	0x447	0x41	0x20001000 - 0x20004FFF	0x1FF00000 - 0x1FF01FFF
0xB2			0x20001400 - 0x20004FFF		
L1	STM32L1xxx6(8/B)	0x416	0x20	0x20000800 - 0x20003FFF	0x1FF00000 - 0x1FF01FFF
	STM32L1xxx6(8/B)A	0x429	0x20	0x20001000 - 0x20007FFF	
	STM32L1xxxC	0x427	0x40		
	STM32L1xxxD	0x436	0x45	0x20001000 - 0x2000BFFF	
	STM32L1xxxE	0x437	0x40	0x20001000 - 0x20013FFF	

Table 209. Bootloader device-dependent parameters (continued)

STM32 series	Device	PID	BL ID	RAM	System memory
L4	STM32L412xx/422xx	0x464	0xD1	0x20002100 - 0x20008000	0x1FFF 0000 - 0x1FFF6FFF
	STM32L43xxx/44xxx	0x435	0x91	0x20003100 - 0x2000BFFF	0x1FFF 0000 - 0x1FFF6FFF
	STM32L45xxx/46xxx	0x462	0x92	0x20003100 - 0x2001FFFF	0x1FFF 0000 - 0x1FFF6FFF
	STM32L47xxx/48xxx	0x415	0xA3	0x20003000 - 0x20017FFF	0x1FFF 0000 - 0x1FFF6FFF
			0x92	0x20003100 - 0x20017FFF	
	STM32L496xx/4A6xx	0x461	0x93	0x20003100 - 0x2003FFFF	0x1FFF 0000 - 0x1FFF6FFF
	STM32L4Rxx/4Sxx	0x470	0x95	0x20003200 - 0x2009FFFF	0x1FFF 0000 - 0x1FFF6FFF
	STM32L4P5xx/Q5xx	0x471	0x90	0x20004000 - 0x2004FFFF	0x1FFF 0000 - 0x1FFF6FFF
L5	STM32L552xx/562xx	0x472	0x92	0x20004000 - 0x2003FFFF	0x0BF9 0000 - 0x0BF9 7FFF
U0	STM32U031xx	0x459	0xB0	0x20001500 - 0x20002FFF	0x1FFF 0000 - 0x1FFF 37FF
	STM32U073xx/83xx	0x489	0xD0	0x20002170 - 0x20009FFF	0x1FFF 0000 - 0x1FFF 67FF
U3	STM32U375xx/ STM32U385xx	0x454	0xE2	0x20004000 - 0x2003FFFF	0x0BF8 F000 - 0x0BF9 FFFF
U5	STM32U535xx/545xx	0x455	0x91	0x20004000 - 0x2023FFFF	0x0BF90000 - 0x0BF9FFFF
	STM32U575xx/ STM32U585xx	0x482	0x92	0x20004000 - 0x200BFFFF	0x0BF90000 - 0x0BF9FFFF
	STM32U595xx/599xx/5A9xx	0x481	0x92	0x20004000 - 0x2026FFFF	0x0BF90000 - 0x0BF9FFFF
	STM32U5F7xx/5F9xx/5G7xx/5G9xx	0x476	0x90	0x20004000 - 0x202EFFFF	0x0BF90000 - 0x0BF9FFFF
WB	STM32WB10xx/15xx	0x494	0xB1	0x20005000 - 0x20040000	0x1FFF 0000 - 0x1FFF7000
	STM32WB30xx/35xx/50xx/WB55xx	0x495	0xD5	0x20004000 - 0x2000BFFF	0x1FFF 0000 - 0x1FFF7000

Table 209. Bootloader device-dependent parameters (continued)

STM32 series	Device	PID	BL ID	RAM	System memory
WBA	STM32WBA52xx	0x492	0xB0	0x2000 0000 - 0x2000 1FFF	0x0BF8 8000 - 0x0BF8 FFFF
	STM32WBA54xx/55xx	0x492	0xB1	0x2000 1800 - 0x2000 1FFF	0x0BF8 8000 - 0x0BF8 FFFF
	STM32WBA62xx/63xx/64xx/65xx	0x4B0	0xD2	0x2000 3000 - 0x2007 FFFF	0x0BF9 0000 - 0x0BF9 FFFF
WL	STM32WLE5xx/WL55xx	0x497	0xC4	0x2000 2000 - 0x2000 FFFF	0x1FFF 0000 - 0x1FFF3FFF

1. Addresses are listed with the max values, but depending on option bytes, the end values can change.

93 Bootloader timings

This section details the timings of the bootloader firmware to use for correct synchronization between the host and the STM32 device.

Two types of timings are described, namely STM32 device bootloader resources initialization duration, and communication interface selection duration.

After these timings the bootloader is ready to receive and execute host commands.

93.1 Bootloader startup timing

After bootloader reset, the host must wait until the STM32 bootloader is ready to start detection phase with a specific interface communication. This time corresponds to bootloader startup timing, during which resources used by bootloader are initialized.

Figure 121. Bootloader startup timing description

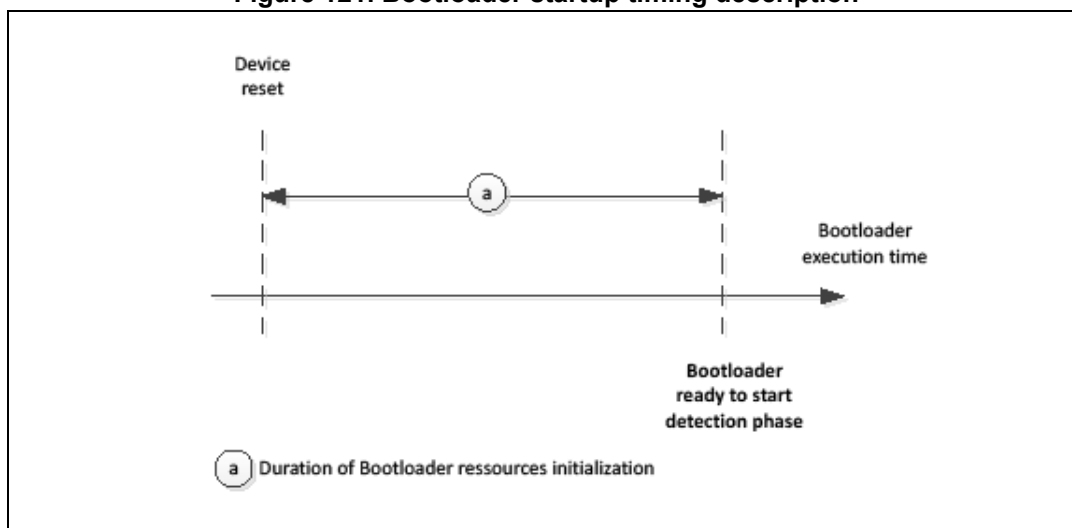


Table 210. Bootloader startup timings (ms)

Device	Minimum startup	HSE timeout
STM32C011xx	2.1	NA
STM32C031xx	2.1	NA
STM32C051xx	2.262	NA
STM32C071xx	2.358	NA
STM32C091xx/92xx	1.527	NA
STM32F03xx4/6	1.612	NA
STM32F05xxx and STM32F030x8 devices	1.612	NA
STM32F04xxx	0.058	NA
STM32F071xx/072xx	0.058	NA

Table 210. Bootloader startup timings (ms) (continued)

Device		Minimum startup	HSE timeout
STM32F070x6	HSE connected	3	200
	HSE not connected	230	
STM32F070xB	HSE connected	6	200
	HSE not connected	230	
STM32F09xxx		2	NA
STM32F030xC		2	NA
STM32F10xxx		1.227	NA
STM32F105xx/107xx	PA9 pin low	1.396	NA
	PA9 pin high	524.376	
STM32F10xxx XL-density		1.227	NA
STM32F2xxxx	V2.x	134	NA
	V3.x	84.59	0.790
STM32F301xx/302x4(6/8)	HSE connected	45	560.5
	HSE not connected	560.8	
STM32F302xB(C)/303xB(C)	HSE connected	43.4	2.236
	HSE not connected	2.36	
STM32F302xD(E)/303xD	HSE connected	7.53	NA
	HSE not connected	146.71	NA
STM32F303x4(6/8)/334xx/328xx		0.155	NA
STM32F318xx		0.182	NA
STM32F358xx		1.542	NA
STM32F373xx	HSE connected	43.4	2.236
	HSE not connected	2.36	
STM32F378xx		1.542	NA
STM32F398xx		1.72	NA
STM32F40xxx/41xxx	V3.x	84.59	0.790
	V9.x	74	96
STM32F401xB(C)		74.5	85
STM32F401xD(E)		74.5	85
STM32F410xx		0.614	NA
STM32F411xx		74.5	85
STM32F412xx		0.614	180
STM32F413xx/423xx		0.642	165
STM32F429xx/439xx	V7.x	82	97
	V9.x	74	97

Table 210. Bootloader startup timings (ms) (continued)

Device			Minimum startup	HSE timeout
STM32F446xx			73.61	96
STM32F469xx/479xx			73.68	230
STM32F72xxx/73xxx			17.93	50
STM32F74xxx/75xxx			16.63	50
STM32G03xxx/04xxx			0.390	NA
STM32G07xxx/08xxx			0.390	NA
STM32G0Bxxx/Cxxx			0.390	NA
STM32G05xxx/061xx			0.390	NA
STM32G4xxxx			0.390	NA
STM32H503xx			0,238	NA
STM32H562xx/63xx/73xx			0.292	NA
STM32H72xxx/73xxx			53.975	NA
STM32H74xxx/75xxx			53.975	2
STM32H7A3xx/7B3xx/7B0xx			545	NA
STM32L01xxx/02xxx			0.63	NA
STM32L031xx/041xx			0.62	NA
STM32L05xxx/06xxx			0.22	NA
STM32L07xxx/08xxx	V4.x		0.61	NA
	V11.x		0.71	NA
STM32L1xxx6(8/B)A			0.542	NA
STM32L1xxx6(8/B)			0.542	NA
STM32L1xxxC			0.708	80
STM32L1xxxD			0.708	80
STM32L1xxxE			0.708	200
STM32L43xxx/44xxx			0.86	100
STM32L45xxx/46xxx			0.86	NA
STM32L47xxx/48xxx	V10.x	LSE connected	55	100
		LSE not connected	2560	
	V9.x	LSE connected	55.40	100
		LSE not connected	2560.5	
STM32L412xx/422xx			0.86	NA
STM32L496xx/4A6xx			76.93	100
STM32L4P5xx /Q5xx			9.891	NA
STM32L4Rxx/4Sxx			10.12	NA
STM32L552xx/562xx			0.390	NA

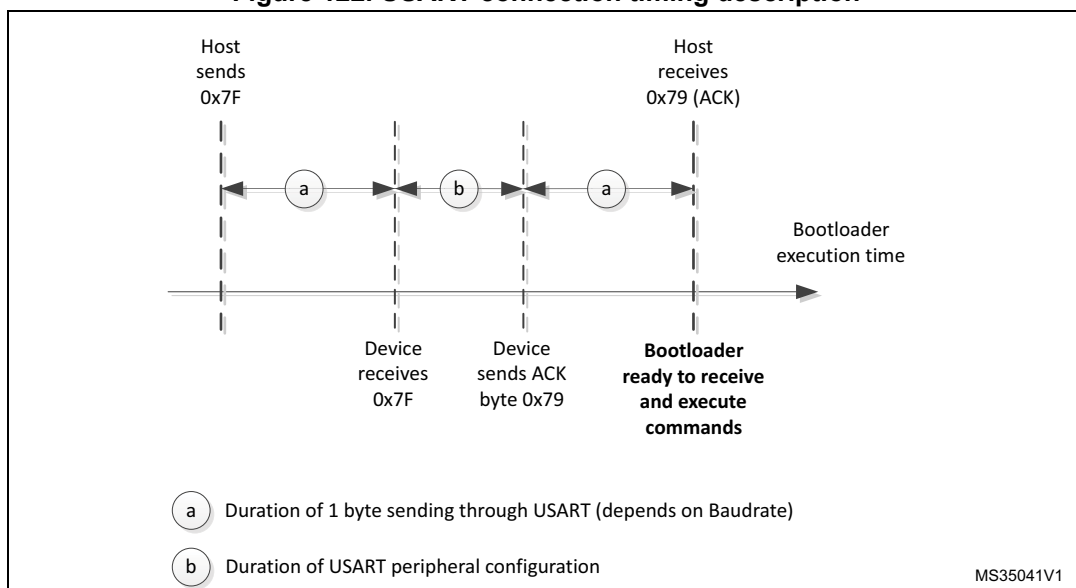
Table 210. Bootloader startup timings (ms) (continued)

Device	Minimum startup	HSE timeout
STM32U031xx	4.534	NA
STM32U073xx/ STM32U083xx	5.626	NA
STM32U375xx/385xx	4.418	NA
STM32U535xx/545xx	0.390	NA
STM32U575xx/85xx	0.390	NA
STM32U595xx/599xx/5A5xx/5A9xx	0.390	NA
STM32U5F7xx/5F9xx/5G7xx/5G9xx	0.390	NA
STM32WB10xx/15xx/30xx/35xx/50xx/55xx	0.390	NA
STM32WBA52xx	0.390	NA
STM32WBA62xx/63xx/64xx/65xx	2,73	NA
STM32WLE5xx/WL55xx	0.390	NA

93.2 USART connection timing

USART connection timing is the time that the host must wait for between sending the synchronization data (0x7F) and receiving the first acknowledge response (0x79).

Figure 122. USART connection timing description



1. Receiving characters different from 0x7F (or line glitches) causes bootloader to start communication using a wrong baudrate. Bootloader measures the signal length between rising edge of the first bit to the falling edge of the last bit to deduce the baudrate value
2. Bootloader does not realign the calculated baudrate to standard baudrate values (i.e. 1200, 9600, 115200).

Note:

The PA9 pin (USB_VBUS) on STM32F105xx/107xx devices is used to detect the USB host connection. The initialization of USB peripheral is performed only if PA9 is high at detection

phase, which means that a host is connected to the port and delivering 5 V on the USB bus. When PA9 level is high at detection phase, more time is required to initialize and shutdown the USB peripheral. To minimize bootloader detection time when PA9 pin is not used, keep PA9 low during USART detection phase, from the moment the device is reset, until a device ACK is sent.

Table 211. USART bootloader minimum timings (ms)

Device		One USART byte sending	USART configuration	USART connection
STM32C011xx		0.138	0.043	0.182
STM32C031xx		0.112	0.028	0.168
STM32C051xx		0.1	0.033	0.23
STM32C071xx		0.1	0.104	0.182
STM32C091xx/92xx		0.1	0.102	0.383
STM32F03xx4/6		0.078125	0.0064	0.16265
STM32F05xxx and STM32F030x8 devices		0.078125	0.0095	0.16575
STM32F04xxx		0.078125	0.007	0.16325
STM32F071xx/072xx		0.078125	0.007	0.16325
STM32F070x6		0.078125	0.014	0.17
STM32F070xB		0.078125	0.08	0.23
STM32F09xxx		0.078125	0.07	0.22
STM32F030xC		0.078125	0.07	0.22
STM32F10xxx		0.078125	0.002	0.15825
STM32F105xx/107xx	PA9 pin low	0.078125	0.007	0.16325
	PA9 pin High		105	105.15625
STM32F10xxx XL-density		0.078125	0.006	0.16225
STM32F2xxxx	V2.x	0.078125	0.009	0.16525
	V3.x			
STM32F301xx/302x4(6/8)	HSE connected	0.078125	0.002	0.15825
	HSE not connected			
STM32F302xB(C)/303xB(C)	HSE connected	0.078125	0.002	0.15825
	HSE not connected			
STM32F302xD(E)/303xD		0.078125	0.002	0.15885
STM32F303x4(6/8)/334xx/328xx		0.078125	0.002	0.15825
STM32F318xx		0.078125	0.002	0.15825
STM32F358xx		0.15625	0.001	0.3135
STM32F373xx	HSE connected	0.078125	0.002	0.15825
	HSE not connected			
STM32F378xx		0.15625	0.001	0.3135

Table 211. USART bootloader minimum timings (ms) (continued)

Device		One USART byte sending	USART configuration	USART connection
STM32F398xx		0.078125	0.002	0.15885
STM32F40xxx/41xxx	V3.x	0.078125	0.009	0.16525
	V9.x		0.0035	0.15975
STM32F401xB(C)		0.078125	0.00326	0.15951
STM32F401xD(E)		0.078125	0.00326	0.15951
STM32F410xx		0.078125	0.002	0.158
STM32F411xx		0.078125	0.00326	0.15951
STM32F412xx		0.078125	0.002	0.158
STM32F413xx/423xx		0.078125	0.002	0.158
STM32F429xx/439xx	V7.x	0.078125	0.007	0.16325
	V9.x		0.00326	0.15951
STM32F446xx		0.078125	0.004	0.16
STM32F469xx/479xx		0.078125	0.003	0.159
STM32F72xxx/73xxx		0.078125	0.070	0.22
STM32F74xxx/75xxx		0.078125	0.065	0.22
STM32G03xxx/04xxx		0.078125	0.01	0.11
STM32G07xxx/08xxx		0.078125	0.01	0.11
STM32G0Bxxx/Cxxx		0.078125	0.01	0.11
STM32G05xxx/061xx		0.078125	0.01	0.11
STM32G4xxxx		0.078125	0.003	0.159
STM32H503xx		0.048	0.05	0.101
STM32H562xx/63xx/73xx		0.047	0.06	0.100
STM32H72xxx/73xxx		0.078125	0.072	0.22825
STM32H74xxx/75xxx		0.078125	0.072	0.22825
STM32H7A3xx/7B3xx/7B0xx		0.078125	0.072	0.22825
STM32L01xxx/02xxx		0.078125	0.016	0.17
STM32L031xx/041xx		0.078125	0.018	0.174
STM32L05xxx/06xxx		0.078125	0.018	0.17425
STM32L07xxx/08xxx	V4.x	0.078125	0.017	0.173
	V11.x	0.078125	0.017	0.158
STM32L1xxx6(8/B)A		0.078125	0.008	0.16425
STM32L1xxx6(8/B)		0.078125	0.008	0.16425
STM32L1xxxC		0.078125	0.008	0.16425
STM32L1xxxD		0.078125	0.008	0.16425

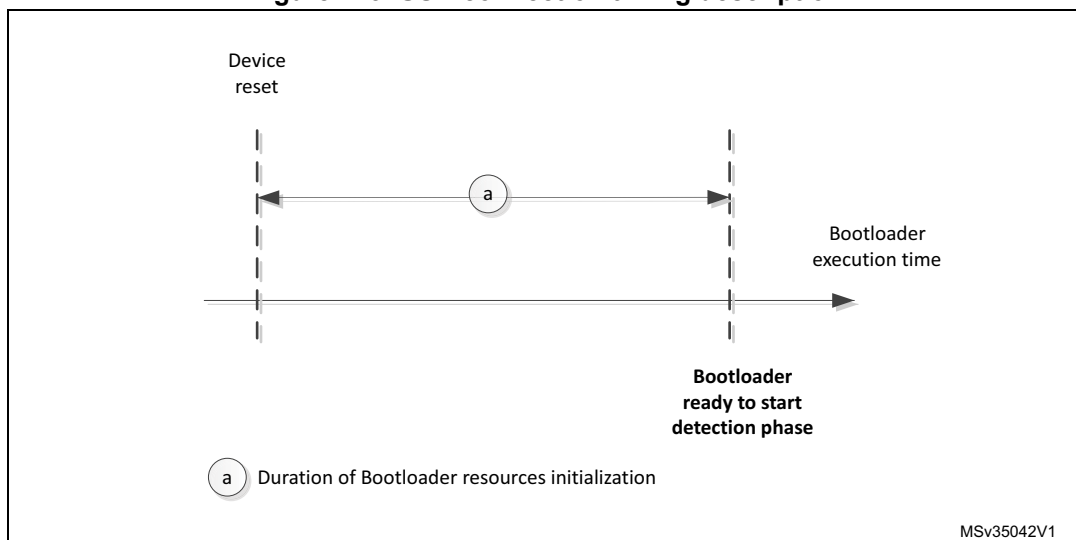
Table 211. USART bootloader minimum timings (ms) (continued)

Device		One USART byte sending	USART configuration	USART connection
STM32L1xxxE		0.078125	0.008	0.16425
STM32L412xx/422xx		0.078125	0.005	0.2
STM32L43xxx/44xxx		0.078125	0.003	0.159
STM32L45xxx/46xxx		0.078125	0.07	0.22
STM32L47xxx/48xxx	V10.x	0.078125	0.003	0.159
	V9.x	0.078125	0.003	0.159
STM32L496xx/4A6xx		0.078125	0.003	0.159
STM32L4Rxx/4Sxx		0.0062	0.0235	0.0307
STM32L4P5xx/4Q5xx		0.0062	0.0235	0.0307
STM32L552xx/562xx		0.078125	0.01	0.11
STM32U031xx		0.012	0.050	0.062
STM32U073xx/STM32U083xx		0.014	0.049	0.077
STM32U031xx		0.012	0.050	0.062
STM32U535xx/545xx		0.078125	0.001	NA
STM32U575xx/85xx		0.078125	0.001	NA
STM32U595xx/599xx/5A5xx/5A9xx		0.078125	0.001	NA
STM32U5F7xx/5F9xx/5G7xx/5G9xx		0.078125	0.001	NA
STM32WB10xx/15xx/30xx/35xx/50xx/55xx		0.078125	0.003	0.159
STM32WBA52xx		0.078125	0.001	NA
STM32WBA62xx/63xx/64xx/65xx		0.102	0.004	0,106
STM32WLE5xx/WL55xx		0.078125	0.001	0.110

93.3 USB connection timing

This is the time that the host must wait for between plugging the USB cable and establishing a correct connection with the device. It includes enumeration and DFU components configuration. The USB connection depends upon the host.

Figure 123. USB connection timing description



Note:

For STM32F105xx/107xx devices, if the external HSE crystal frequency is different from 25 MHz (14.7456 or 8 MHz), the device performs several unsuccessful enumerations (with connect/disconnect sequences) before establishing a correct connection with the host. This is due to the HSE detection mechanism based on Start Of Frame (SOF) detection.

Table 212. USB bootloader minimum timings (ms)

Device		USB connection
STM32C011xx		NA
STM32C031xx		NA
STM32C051xx		NA
STM32C071xx		552
STM32C091xx/92xx		NA
STM32F04xxx		350
STM32F070x6		TBD
STM32F070xB		320
STM32F105xx/107xx	HSE = 25 MHz	460
	HSE = 14.7465 MHz	4500
	HSE = 8 MHz	13700
STM32F2xxxx		270
STM32F301xx/302x4(6/8)		300
STM32F302xB(C)/303xB(C)		300
STM32F302xD(E)/303xD		100
STM32F373xx		300
STM32F40xxx/41xxx	V3.x	270
	V9.x	250

Table 212. USB bootloader minimum timings (ms) (continued)

Device		USB connection
STM32F401xB(C)		250
STM32F401xD(E)		250
STM32F411xx		250
STM32F412xx		380
STM32F413xx/423xx		350
STM32F429xx/439xx	V7.x	250
	V9.x	
STM32F446xx		200
STM32F469xx/479xx		270
STM32F72xxx/73xxx		320
STM32F74xxx/75xxx		230
STM32G0B1xx/C1xx		300
STM32G4xxxx		300
STM32H503xx		251
STM32H562xx/63xx/73xx		245
STM32H72xxx/73xxx		53.9764
STM32H74xxx/75xxx		53.9764
STM327A3xx/7B3xx/7B0xx		53.9764
STM32L07xxx/08xxx		140
STM32L1xxxC		849
STM32L1xxxD		849
STM32L412xx/422xx		820
STM32L43xxx/44xxx		820
STM32L45xxx/46xxx		330
STM32L47xxx/48xxx	V10.x	300
	V9.x	
STM32L496xx/4A6xx		430
STM32L4P5xx/4Q5xx		322
STM32L4Rxx/4Sxx		322
STM32L552xx/L562xx		300
STM32U031xx		NA
STM32U073xx/ STM32U083xx		241
STM32U375xx/385xx		210
STM32U535xx/545xx		300
STM32U575xx/85xx		300

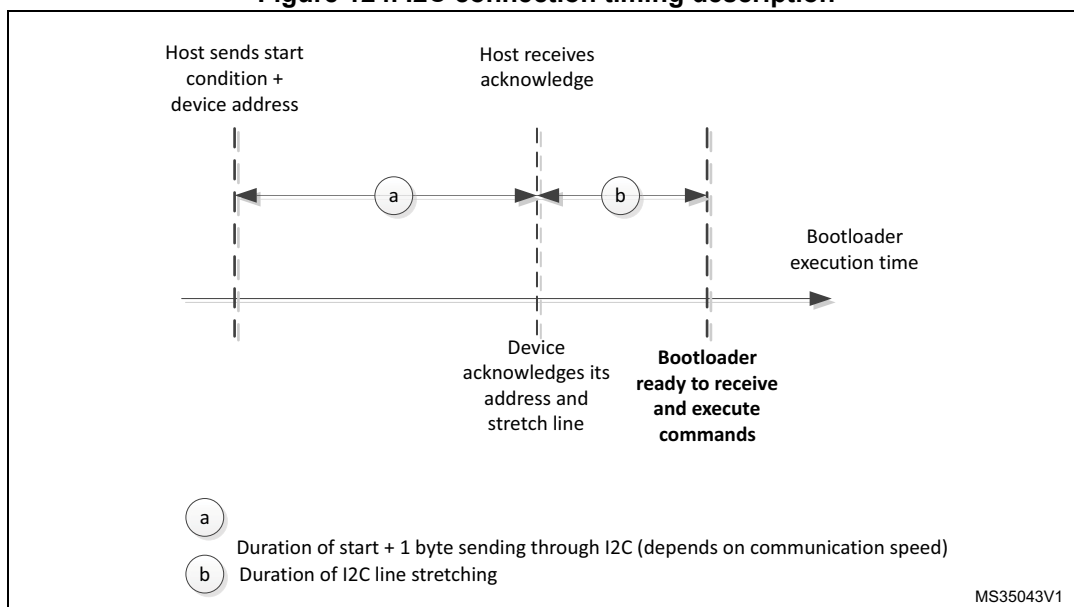
Table 212. USB bootloader minimum timings (ms) (continued)

Device	USB connection
STM32U595xx/599xx/5A5xx/5A9xx	300
STM32U5F7xx/5F9xx/5G7xx/5G9xx	300
STM32WB30xx/35xx/50xx/55xx	300
STM32WBA62xx/63xx/64xx/65xx	69

93.4 I2C connection timing

I2C connection timing is the time that the host must wait for between sending I2C device address and sending command code. This timing includes I2C line stretching duration.

Figure 124. I2C connection timing description



Note: For I2C communication, a timeout mechanism is implemented and must be respected to execute bootloader commands correctly. This timeout is implemented between two I2C frames in the same command (example: for Write memory command, a timeout is inserted between command sending frame and address memory sending frame). The same timeout period is inserted between two successive data receptions or transmissions in the same I2C frame. If the timeout period elapses, a system reset is generated to avoid bootloader crash.

In Erase memory and Read-out unprotect commands, consider the duration of the operation when implementing the host side. After sending the code of pages to erase, the host must wait until the bootloader device performs page erasing to complete the remaining steps of erase command.

Table 213. I2C bootloader minimum timings (ms)

Device		Start condition + one I2C byte sending	I2C line stretching	I2C connection	I2C timeout
STM32C011xx		0.254	0.004	0.060	2000
STM32C031xx		0.028	0.003	0.031	2000
STM32C051xx		0.031	0.019	0.050	2200
STM32C071xx		0.028	0.017	0.046	2000
STM32C091xx/92xx		0.031	0.018	0.049	2000
STM32F030xC		0.0225	0.0025	0.0250	1000
STM32F04xxx		0.0225	0.0025	0.0250	1000
STM32F070x6		0.0225	0.0025	0.0245	1000
STM32F070xB		0.0225	0.0025	0.0245	1000
STM32F071xx/072xx		0.0225	0.0025	0.0250	1000
STM32F09xxx		0.0225	0.0025	0.0245	1000
STM32F303x4(6/8)/334xx/328xx		0.0225	0.0027	0.0252	1000
STM32F318xx		0.0225	0.0027	0.0252	1000
STM32F358xx		0.0225	0.0055	0.0280	10
STM32F378xx		0.0225	0.0055	0.0280	10
STM32F398xx		0.0225	0.0020	0.0245	1500
STM32F40xxx/41xxx		0.0225	0.0022	0.0247	1000
STM32F401xB(C)		0.0225	0.0022	0.0247	1000
STM32F401xD(E)		0.0225	0.0022	0.0247	1000
STM32F410xx		0.0225	0.0020	0.0245	1000
STM32F411xx		0.0225	0.0022	0.0247	1000
STM32F412xx		0.0225	0.0020	0.0245	1000
STM32F413xx/423xx		0.0225	0.0020	0.0245	1000
STM32F42xxx/43xxx	V7.x	0.0225	0.0033	0.0258	1000
	V9.x	0.0225	0.0022	0.0247	1000
STM32F446xx		0.0225	0.0020	0.0245	1000
STM32F469xx/479xx		0.0225	0.0020	0.0245	1000
STM32F72xxx/73xxx		0.0225	0.0020	0.0245	1000
STM32F74xxx/75xxx		0.0225	0.0020	0.0245	500
STM32G03xxx/04xxx		0.0225	0.0020	0.0245	1000
STM32G07xxx/08xxx		0.0225	0.0020	0.0245	1000
STM32G0Bxx/Cxx		0.0225	0.0020	0.0245	1000
STM32G05xxx/061xx		0.0225	0.0020	0.0245	1000

Table 213. I2C bootloader minimum timings (ms) (continued)

Device		Start condition + one I2C byte sending	I2C line stretching	I2C connection	I2C timeout
STM32G4xxx		0.0225	0.0020	0.0245	1000
STM32H503xx		0.038	0.03	0.041	1000
STM32H562xx/63xx/73xx		0.039	0.02	0.041	1000
STM32H72xxx/73xxx		0.0225	0.05	0.0745	1000
STM32H74xxx/75xxx		0.0225	0.05	0.0725	1000
STM32H7A3xx/7B3xx/7B0xx		0.0225	0.05	0.0745	1000
STM32L07xxx/08xxx		0.0225	0.0020	0.0245	1000
STM32L412xx/422xx		0.0225	0.0020	0.0245	1000
STM32L43xxx/44xxx		0.0225	0.0020	0.0245	1000
STM32L45xxx/46xxx		0.0225	0.0020	0.0245	1000
STM32L47xxx/48xxx	V10.x	0.0225	0.0020	0.0245	1000
	V9.x	0.0225	0.0020	0.0245	1000
STM32L496xx/4A6xx		0.0225	0.0020	0.0245	1000
STM32L4P5xx/4Q5xx		0.0109	0.0020	0.0642	1000
STM32L4Rxx/4Sxx		0.0109	0.0020	0.0642	1000
STM32L552xx/L562xx		0.0225	0.0020	0.0245	1000
STM32U031xx		0.035	0.019	0.106	1000
STM32U073xx/ STM32U083xx		0.016	0.019	0.024	1000
STM32U375xx/385xx		0.021	0.018	0.092	1000
STM32U535xx/545xx		0.0225	0.0020	0.0245	1000
STM32U575xx/85xx		0.0225	0.0020	0.0245	1000
STM32U595xx/599xx/5A5xx/5A9xx		0.0225	0.0020	0.0245	1000
STM32U5F7xx/5F9xx/5G7xx/5G9xx		0.0225	0.0020	0.0245	1000
STM32WB10xx/15xx/30xx/35xx/50xx/55xx		0.0225	0.0020	0.0245	1000
STM32WBA52xx		0.0225	0.0020	0.0245	1000
STM32WBA62xx/63xx/64xx/65xx		0.003	0.019	0.539	1000

93.5 SPI connection timing

SPI connection timing is the time that the host must wait for between sending the synchronization data (0xA5) and receiving the first acknowledge response (0x79).

Figure 125. SPI connection timing description

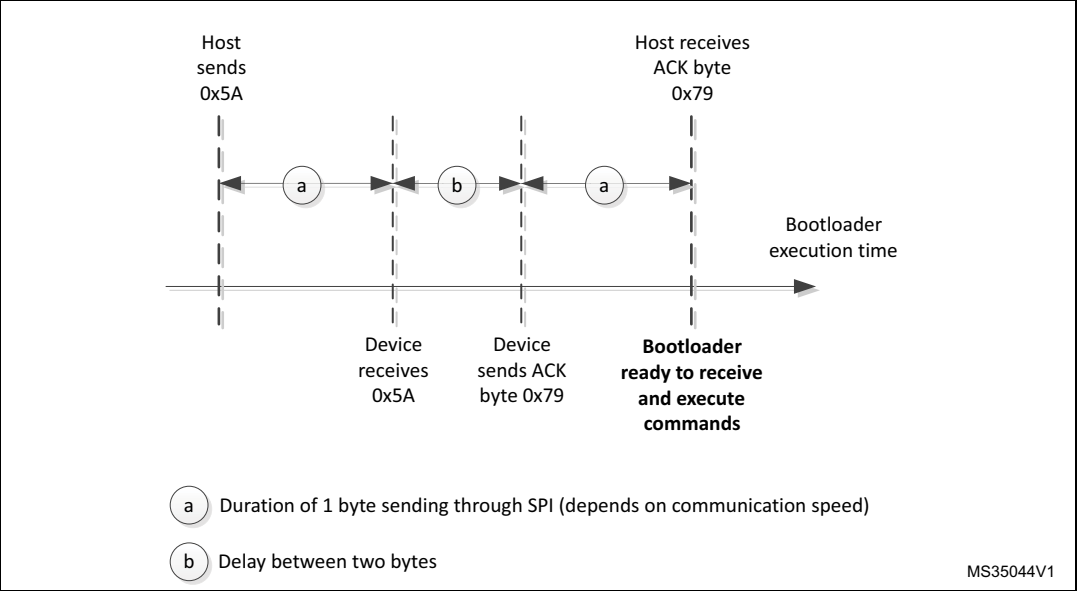


Table 214. SPI bootloader minimum timings (ms) for STM32 devices

Device	One SPI byte sending	Delay between two bytes	SPI connection
All products	0.001	0.008	0.01

Appendix A Example of *ExitSecureMemory* v1.0 function

```

/**
*****
****
* @file    main.c
*****
****
*/

/* Includes -----
---*/

#include "main.h"

/* Private function prototypes -----
---*/

static void ConfigClock(void);

void JUMP_WITHOUT_PARAM(uint32_t jump_address);
void JUMP_WITH_PARAM(uint32_t jump_address, uint32_t magic, uint32_t
applicationVectorAddress);

/* Private functions -----
---*/

/**
* @brief   Main program
* @param   None
* @retval  None
*/
int main(void)
{
    ConfigClock();

    uint32_t application_address          = 0x08000800;
    uint32_t exit_secure_memory_address  = 0x1FFF1E00;
    uint32_t magic_number                 = 0x08192A3C;
    uint32_t exit_with_magic_number      = 0x0;

    if (exit_with_magic_number)
    {
        JUMP_WITH_PARAM(exit_secure_memory_address, magic_number,
application_address);
    }
    else
    {

```

```

        JUMP_WITHOUT_PARAM(exit_secure_memory_address);
    }
}

/**
 * @brief ConfigClock
 * @param None
 * @retval None
 */
static void ConfigClock(void)
{
    /* Will be developped as per the template of the needed project */
}

/**
 * @brief JUMP_WITHOUT_PARAM
 * @param jump_address
 * @retval None
 */
void JUMP_WITHOUT_PARAM(uint32_t jump_address)
{
    asm ("LDR R1, [R0]");      // jump_address
    asm ("LDR R2, [R0,#4]");
    asm ("MOV SP, R1");
    asm ("BX R2");
}

/**
 * @brief JUMP_WITH_PARAM
 * @param jump_address, magic, applicationVectorAddress
 * @retval None
 */
void JUMP_WITH_PARAM(uint32_t jump_address, uint32_t magic, uint32_t
applicationVectorAddress)
{
    asm ("MOV R3, R0");      // jump_address
    asm ("LDR R0, [R3]");
    asm ("MOV SP, R0");
    asm ("LDR R0, [R3,#4]");
    asm ("BX R0");
}

/***** (C) COPYRIGHT STMicroelectronics *****/
FILE****/

```


Appendix B Example of *ExitSecureMemory* v1.1 function

```

/**
*****
****
* @file    main.c
*****
****
*/

/* Includes -----
---*/
#include "main.h"

/* Private function prototypes -----
---*/
static void ConfigClock(void);
static void Jump_Without_Param(uint32_t exit_secure_address, uint32_t
dummy1, uint32_t dummy2, uint8_t mpu_region_number);
static void Jump_With_Param(uint32_t exit_secure_address, uint32_t magic,
uint32_t application_address, uint8_t mpu_region_number);

/* Private functions -----
---*/

/**
* @brief   Main program.
* @retval  None.
*/
int main(void)
{
    ConfigClock();

    uint32_t application_address = 0x08008000;
    uint32_t exit_secure_address = 0x1FFF6000;
    uint8_t mpu_region_number    = 0x04;
    uint32_t magic_number        = 0x08192A3C;
    uint32_t with_magic_number   = 0x0;

    if (with_magic_number)
    {
        Jump_With_Param(exit_sticky_address, magic_number, application_address,
mpu_region_number);
    }
}

```

```

    else
    {
        Jump_Without_Param(exit_sticky_address, 0x0, 0x0, mpu_region_number);
    }
}

/**
 * @brief  Configure system clocks.
 * @retval None.
 */
static void ConfigClock(void)
{
    /* Will be developped as per the template of the needed project */
}

/**
 * @brief  Jump to secure exit memory without magic number and user
address.
 * @param  exit_secure_address Address of exit secure memory.
 * @param  dummy1 Not used.
 * @param  dummy2 Not used.
 * @param  mpu_region_number MPU region to enable.
 * @retval None.
 */
static void Jump_Without_Param(uint32_t exit_secure_address, uint32_t
dummy1, uint32_t dummy2, uint8_t mpu_region_number)
{
    /**
     * R0 = exit_secure_address  --> Exit secure memory stack pointer
     * R1 = dummy1               --> Dummy data
     * R2 = dummy2               --> Dummy data
     * R3 = mpu_region_number    --> MPU region number
     *
     * NOTE: Assume R1 and R2 registers are useless so can be modified, but R3
is used for MPU region number and must not be changed
     */
    asm ("LDR R1, [R0]");      // Load stack pointer (content of address
pointed by R0) in R1
    asm ("LDR R2, [R0,#4]");   // Load jump address (content of address pointed
by R4 + 4) in R2
    asm ("MOV SP, R1");        // Change stack pointer register with value from
R1
    asm ("BX  R2");            // Jump to address pointed by R2
}

/**

```

```

* @brief Jump to secure exit memory with parameters.
* @param exit_secure_address Address of exit secure memory.
* @param magic Magic number value.
* @param application_address Application address.
* @param mpu_region_number MPU region to enable.
* @retval None
*/
static void Jump_With_Param(uint32_t exit_secure_address, uint32_t magic,
uint32_t application_address, uint8_t mpu_region_number)
{
    /**
    * R0 = exit_secure_address --> Exit secure memory stack pointer
    * R1 = magic --> Magic number
    * R2 = application_address --> User application start address
    * R3 = mpu_region_number --> MPU region number
    *
    * NOTE: R1, R2 and R3 registers must not be changed in below code
    */
    asm ("MOV R4, R0"); // Backup R0 in R4
    asm ("LDR R0, [R4]"); // Load stack pointer (content of address
pointed by R4) in R0
    asm ("MOV SP, R0"); // Change stack pointer register with value from
R0
    asm ("LDR R0, [R4,#4]"); // Load jump address (content of address pointed
by R4 + 4) in R0
    asm ("BX R0"); // Jump to address pointed by R0

```

94 Revision history

Table 215. Document revision history

Date	Revision	Changes
21-Feb-2019	36	Updated Table 1: Applicable products , Section 3: Glossary , Table 3: Embedded bootloaders , Table 209: Bootloader device-dependent parameters , Table 210: Bootloader startup timings (ms) , Table 211: USART bootloader minimum timings (ms) , Table 212: USB bootloader minimum timings (ms) , Table 213: I2C bootloader minimum timings (ms) . Added Section 77: STM32WB30xx/35xx/50xx/55xx devices
06-May-2019	37	Updated Table 1: Applicable products , Section 3: Glossary , Table 209: Bootloader device-dependent parameters , Table 210: Bootloader startup timings (ms) , Table 211: USART bootloader minimum timings (ms) , Table 212: USB bootloader minimum timings (ms) , Table 213: I2C bootloader minimum timings (ms) . Added Section 49: STM32G431xx/441xx devices , Section 50: STM32G47xxx/48xxx devices .
08-Jul-2019	38	Updated: – Table 1: Applicable products , Table 2: Bootloader activation patterns , Table 3: Embedded bootloaders , Table 78: STM32F413xx/423xx configuration in system memory boot mode , Table 123: STM32H74xxx/75xxx configuration in system memory boot mode , Table 124: STM32H74xxx/75xxx bootloader version , Table 132: STM32L031xx/041xx configuration in system memory boot mode , Table 153: STM32L43xxx/44xxx bootloader versions , Table 154: STM32L45xxx/46xxx configuration in system memory boot mode , Table 161: STM32L496xx/4A6xx bootloader version , Table 172: STM32WB30xx/35xx/50xx/55xx bootloader versions , Table 209: Bootloader device-dependent parameters , Table 210: Bootloader startup timings (ms) , Table 211: USART bootloader minimum timings (ms) , Table 212: USB bootloader minimum timings (ms) , Table 213: I2C bootloader minimum timings (ms) – Section 3: Glossary , Section 4.1: Bootloader activation , Section 44.1: Bootloader configuration , Section 49.1: Bootloader configuration – Figure 73: Bootloader V9.x selection for STM32H74xxx/75xxx , Figure 102: Dual bank boot implementation for STM32L4Rxxx/STM32L4Sxxx bootloader V9.x Added Note: in Section 4.2 , Note: in Section 18.3 , Note: in Section 56.1 , Note: in Section 59.1 , Section 44: STM32G03xxx/STM32G04xxx devices
16-Sep-2019	39	Updated: – Table 1: Applicable products , Table 2: Bootloader activation patterns , Table 3: Embedded bootloaders , Table 97: STM32G03xxx/04xxx bootloader versions , Table 151: STM32L412xx/422xx bootloader versions , Table 153: STM32L43xxx/44xxx bootloader versions , Table 155: STM32L45xxx/46xxx bootloader versions , Table 157: STM32L47xxx/48xxx bootloader V10.x versions , Table 159: STM32L47xxx/48xxx bootloader V9.x versions , Table 161: STM32L496xx/4A6xx bootloader version , Table 163: STM32L4P5xx/4Q5xx bootloader versions , Table 209: Bootloader device-dependent parameters , Table 210: Bootloader startup timings (ms) , Table 211: USART bootloader minimum timings (ms) , Table 212: USB bootloader minimum timings (ms) , Table 213: I2C bootloader minimum timings (ms) – Section 3: Glossary , Section 4.2: Bootloader identification Added Figure 67: Dual bank boot implementation for STM32G47xxx/48xxx bootloader V13.x , Section 75: STM32L552xx/62xx devices , note in Section 77.3: Bootloader version

Table 215. Document revision history (continued)

Date	Revision	Changes
03-Oct-2019	40	Updated Table 3: Embedded bootloaders , Table 168: STM32L552xx/62xx bootloader versions , Table 172: STM32WB30xx/35xx/50xx/55xx bootloader versions
25-Oct-2019	41	Updated: – Table 89: STM32F72xxx/73xxx bootloader V9.x versions, Table 91: STM32F74xxx/75xxx bootloader V7.x versions, Table 93: STM32F74xxx/75xxx bootloader V9.x versions, Table 95: STM32F76xxx/77xxx bootloader V9.x versions, Table 96: STM32G03xxx/G04xxx configuration in system memory boot mode, Table 124: STM32H74xxx/75xxx bootloader version, Table 163: STM32L4P5xx/4Q5xx bootloader versions, Table 166: STM32L552xx/62xx configuration in system memory boot mode, Table 210: Bootloader startup timings (ms), Table 211: USART bootloader minimum timings (ms), Table 213: I2C bootloader minimum timings (ms) – Section 21: STM32F2xxxx devices
05-Dec-2019	42	Updated: – Table 1: Applicable products, Table 2: Bootloader activation patterns, Table 3: Embedded bootloaders, Table 209: Bootloader device-dependent parameters, Table 210: Bootloader startup timings (ms), Table 211: USART bootloader minimum timings (ms), Table 212: USB bootloader minimum timings (ms), Table 213: I2C bootloader minimum timings (ms) – Section 3: Glossary Added: Section 57: STM32H7A3xx/7B3xx/7B0xx devices , Section 73: STM32L4P5xx/4Q5xx devices , Section 84: STM32WLE5xx/55xx devices

Table 215. Document revision history (continued)

Date	Revision	Changes
04-Jun-2020	43	Updated: - Table 1: Applicable products, Table 2: Bootloader activation patterns, Table 3: Embedded bootloaders, Table 106: STM32G431xx/441xx configuration in system memory boot mode, Table 108: STM32G47xxx/48xxx configuration in system memory boot mode, Table 109: STM32G47xxx/48xxx bootloader version, Table 124: STM32H74xxx/75xxx bootloader version, Table 126: STM32H7A3xx/7B3xx/7B0xx bootloader version, Table 163: STM32L4P5xx/4Q5xx bootloader versions, Table 166: STM32L552xx/62xx configuration in system memory boot mode, Table 168: STM32L552xx/62xx bootloader versions, Table 171: STM32WB30xx/35xx/50xx/55xx configuration in system memory boot mode, Table 209: Bootloader device-dependent parameters - Section 3: Glossary, Section 42: STM32F74xxx/75xxx devices, Section 44.1: Bootloader configuration, Section 45.1: Bootloader configuration, Section 49.1: Bootloader configuration, Section 50.1: Bootloader configuration, Section 56.1: Bootloader configuration Added: - Section 4.5: Bootloader UART baudrate detection, Section 4.6: Programming constraints, Section 4.7: ExitSecureMemory feature - Note in: Section 31.1.1: Bootloader configuration, Section 31.2.1: Bootloader configuration, Section 32.1: Bootloader configuration, Section 33.1: Bootloader configuration, Section 35.1: Bootloader configuration, Section 36.1: Bootloader configuration, Section 37.1: Bootloader configuration, Section 38.1.1: Bootloader configuration, Section 38.2.1: Bootloader configuration, Section 39.1: Bootloader configuration, Section 40.1: Bootloader configuration, Section 41.1: Bootloader configuration, Section 42.1.1: Bootloader configuration, Section 42.2.1: Bootloader configuration, Section 43.1: Bootloader configuration - Figure 90: Dual bank boot Implementation for STM32L3x2xx/44xxx bootloader V9.x, Figure 92: Dual bank boot implementation for STM32L45xxx/46xxx bootloader V9.x, Figure 98: Dual bank boot Implementation for STM32L496xx/4A6xx bootloader V9.x - Appendix A: Example of ExitSecureMemory v1.0 function Deleted Figure 48. Access to securable memory area from the bootloader for STM32G03xxx/G04xxx, Figure 50. Access to securable memory area from the bootloader for STM32G07xxx/G08xxx, Figure 52. Access to securable memory area, Figure 54. Access to securable memory area
29-Jul-2020	44	Introduced STM32H72xxx/73xxx devices, hence added Section 55: STM32H72xxx/73xxx devices and its subsections. Updated Section 3: Glossary, note in Section 44.1: Bootloader configuration and Section 77.1: Bootloader configuration. Updated Table 1: Applicable products, Table 2: Bootloader activation patterns, Table 3: Embedded bootloaders, Table 8: ExitSecureMemory entry address, Table 108: STM32G47xxx/48xxx configuration in system memory boot mode, Table 126: STM32H7A3xx/7B3xx/7B0xx bootloader version, Table 144: STM32L1xxxC configuration in system memory boot mode, Table 146: STM32L1xxxD configuration in system memory boot mode, Table 148: STM32L1xxxE configuration in system memory boot mode, Table 166: STM32L552xx/62xx configuration in system memory boot mode, Table 209: Bootloader device-dependent parameters, Table 210: Bootloader startup timings (ms), Table 211: USART bootloader minimum timings (ms), Table 212: USB bootloader minimum timings (ms) and Table 213: I2C bootloader minimum timings (ms). Updated Figure 73: Bootloader V9.x selection for STM32H74xxx/75xxx. Minor text edits across the whole document.

Table 215. Document revision history (continued)

Date	Revision	Changes
06-Nov-2020	45	Introduced STM32WB30xx, STM32WB35xx, STM32WI55xx in Table 1: Applicable products, Table 3: Embedded bootloaders and in Section 3: Glossary Updated: – Table 72: STM32F410xx configuration in system memory boot mode, Table 78: STM32F413xx/423xx configuration in system memory boot mode, Table 84: STM32F446xx configuration in system memory boot mode, Table 86: STM32F469xx/479xx configuration in system memory boot mode, Table 88: STM32F72xxx/73xxx configuration in system memory boot mode, Table 92: STM32F74xxx/75xxx configuration in system memory boot mode, Table 94: STM32F76xxx/77xxx configuration in system memory boot mode, Table 98: STM32G07xxx/8xxx configuration in system memory boot mode, Table 106: STM32G431xx/441xx configuration in system memory boot mode, Table 107: STM32G431xx/441xx bootloader version, Table 108: STM32G47xxx/48xxx configuration in system memory boot mode, Table 123: STM32H74xxx/75xxx configuration in system memory boot mode, Table 124: STM32H74xxx/75xxx bootloader version, Table 125: STM32H7A3xx/7B3xx/7B0xx configuration in system memory boot mode, Table 126: STM32H7A3xx/7B3xx/7B0xx bootloader version, Table 130: STM32L01xxx/02xxx configuration in system memory boot mode, Table 132: STM32L031xx/041xx configuration in system memory boot mode, Table 137: STM32L07xxx/08xxx bootloader versions, Table 138: STM32L07xxx/08xxx configuration in system memory boot mode, Table 150: STM32L412xx/422xx configuration in system memory boot mode, Table 162: STM32L4P5xx/4Q5xx configuration in system memory boot mode, Table 164: STM32L4Rxxx/4Sxxx configuration in system memory boot mode, Table 166: STM32L552xx/62xx configuration in system memory boot mode, Table 171: STM32WB30xx/35xx/50xx/55xx configuration in system memory boot mode, Table 188: STM32WLE5xx/55xx configuration in system memory boot mode, Table 189: STM32WLE5xx/55xx bootloader versions, Table 209: Bootloader device-dependent parameters, Table 210: Bootloader startup timings (ms), Table 211: USART bootloader minimum timings (ms) – title of Table 77: STM32WB30xx/35xx/50xx/55xx devices, Table 171: STM32WB30xx/35xx/50xx/55xx configuration in system memory boot mode, Table 84: STM32WLE5xx/55xx devices, Table 188: STM32WLE5xx/55xx configuration in system memory boot mode, Table 113: Bootloader V12.x selection for STM32WLE5xx/55xx, Table 189: STM32WLE5xx/55xx bootloader versions
02-Dec-2020	46	Updated: – Table 3: Embedded bootloaders, Table 96: STM32G03xxx/G04xxx configuration in system memory boot mode, Table 108: STM32G47xxx/48xxx configuration in system memory boot mode, Table 153: STM32L43xxx/44xxx bootloader versions, Table 155: STM32L45xxx/46xxx bootloader versions Added following notes: – Note: on page 350, Note: on page 357, Note: on page 365, Note: on page 378, Note: on page 384

Table 215. Document revision history (continued)

Date	Revision	Changes
16-Feb-2021	47	Updated: – Table 1: Applicable products, Table 3: Embedded bootloaders, Table 8: ExitSecureMemory entry address, Table 95: STM32F76xxx/77xxx bootloader V9.x versions, Table 108: STM32G47xxx/48xxx configuration in system memory boot mode, Table 150: STM32L412xx/422xx configuration in system memory boot mode, Table 152: STM32L43xxx/44xxx configuration in system memory boot mode, Table 154: STM32L45xxx/46xxx configuration in system memory boot mode, Table 158: STM32L47xxx/48xxx configuration in system memory boot mode, Table 160: STM32L496xx/4A6xx configuration in system memory boot mode, Table 162: STM32L4P5xx/4Q5xx configuration in system memory boot mode, Table 164: STM32L4Rxxx/4Sxxx configuration in system memory boot mode, Table 166: STM32L552xx/62xx configuration in system memory boot mode, Table 171: STM32WB30xx/35xx/50xx/55xx configuration in system memory boot mode, Table 209: Bootloader device-dependent parameters, Table 210: Bootloader startup timings (ms), Table 211: USART bootloader minimum timings (ms), Table 212: USB bootloader minimum timings (ms), Table 213: I2C bootloader minimum timings (ms) – Section 3: Glossary Added Section 46: STM32G0B0xx device bootloader and Section 47: STM32G0B1xx/0C1xx device bootloader
01-Apr-2021	48	Updated: – Table 1: Applicable products, Table 3: Embedded bootloaders, Table 8: ExitSecureMemory entry address, Table 209: Bootloader device-dependent parameters, Table 210: Bootloader startup timings (ms), Table 211: USART bootloader minimum timings (ms), Table 213: I2C bootloader minimum timings (ms) Added Section 48: STM32G05xxx/061xx devices and Section 51: STM32G491xx/4A1xx devices
06-Jul-2021	49	Updated: – Section 3: Glossary, Section 31.2.1: Bootloader configuration – Table 3, Table 20, from Table 22 to Table 27, from Table 30 to Table 35, Table 36, Table 38, Table 40, Table 44, Table 46, Table 48, Table 50, Table 52, Table 54, Table 56, Table 58, Table 60, Table 62, Table 64, Table 66, Table 67, Table 68, Table 70, Table 72, Table 74, Table 76, Table 78, Table 80, Table 82, Table 84, Table 86, Table 88, Table 90, Table 92, Table 94, Table 96, Table 98, Table 100, Table 102, Table 104, Table 106, Table 108, Table 109, Table 110, Table 121, Table 123, Table 125, Table 125, Table 130, Table 132, Table 134, Table 135, Table 136, Table 138, Table 140, Table 142, Table 143, Table 144, Table 145, Table 146, Table 148, Table 149, Table 150, Table 152, Table 154, Table 156, Table 158, Table 160, Table 162, Table 164, Table 166, Table 171, Table 188, Table 209 Added Table 167: STM32L552xx/62xx special commands and Section 76: STM32WB10xx/15xx devices
23-Sep-2021	50	Updated: – Section 3: Glossary, Section 46.1: Bootloader configuration, Section 47.1: Bootloader configuration, – Table 1, Table 2, Table 3, Table 99, Table 122, Table 124, Table 126, Table 153, Table 170, Table 172, Table 209, Table 210, Table 211, Table 212, Table 213 Added Section 89: STM32U575xx/85xx devices

Table 215. Document revision history (continued)

Date	Revision	Changes
20-Oct-2021	51	Updated: – Table 3 , Table 67 , Table 99 , Table 122 , Table 209 – Section 31.2.1: Bootloader configuration
04-Feb-2022	52	Updated: – Section 3: Glossary , Section 4.1: Bootloader activation , – Table 1 , Table 2 , Table 3 , Table 7 , Table 99 , Table 124 , Table 209 – Figure 62 Added Section 5: STM32C011xx devices and Section 6: STM32C031xx devices
01-Mar-2022	53	Updated: – Table 3 , Table 121 , Table 122 , Table 209 . – Section 4.1: Bootloader activation , Section 44.1: Bootloader configuration , Section 45.1: Bootloader configuration , Section 46.1: Bootloader configuration , Section 48.1: Bootloader configuration
20-Apr-2022	54	Updated: – Table 3 , Table 121 , Table 125 , Table 126 , Table 209
22-Jun-2022	55	Updated: – Table 3 , Table 123 , Table 122 , Table 209
14-Dec-2022	56	Added Section 4.8: IWDG usage , Section 78: STM32WBA52xx devices Updated: – Table 1 , Table 2 , Table 3 , Table 7 , Table 124 , Table 125 , Table 189 , Table 209 , Table 210 , Table 211 , Table 213 – Section 3: Glossary , Section 4.1: Bootloader activation – added note 1 in Table 98 , Table 100 , Table 102 , Table 106 , Table 108 , Table 110 , Table 150
21-Feb-2023	57	Updated: – Table 1: Applicable products , Table 2: Bootloader activation patterns , Table 3: Embedded bootloaders , Table 7: Flash memory alignment constraints , Table 209: Bootloader device-dependent parameters , Table 210: Bootloader startup timings (ms) , Table 211: USART bootloader minimum timings (ms) , Table 212: USB bootloader minimum timings (ms) , Table 213: I2C bootloader minimum timings (ms) – Section 3: Glossary , Section 4.2: Bootloader identification Added Section 52: STM32H503xx devices , Section 54: STM32H562xx/563xx/573xx devices , Section 90: STM32U595xx/99xx/A5xx/A9xx devices
04-Apr-2023	58	Updated: – Table 1: Applicable products , Table 3: Embedded bootloaders , Table 209: Bootloader device-dependent parameters , Table 210: Bootloader startup timings (ms) , Table 211: USART bootloader minimum timings (ms) , Table 212: USB bootloader minimum timings (ms) , Table 213: I2C bootloader minimum timings (ms) – Section 3: Glossary Added Section 88: STM32U535xx/545xx devices
21-Jun-2023	59	Updated Table 3: Embedded bootloaders , Table 99: STM32G07xxx/08xxx bootloader versions , and Table 209: Bootloader device-dependent parameters . Minor text edits across the whole document.

Table 215. Document revision history (continued)

Date	Revision	Changes
25-Oct-2023	60	Updated Section 2: Related documents, note in Section 4.1: Bootloader activation, Section 54.3: Bootloader version, and Section 68.1: Bootloader configuration. Updated Table 2: Bootloader activation patterns, Table 3: Embedded bootloaders, Table 69: STM32F401xB(C) bootloader versions, Table 102: STM32G0B1xx/0C1xx configuration in system memory boot mode, Table 112: STM32H503xx configuration in system memory boot mode, Table 118: STM32H562xx/563xx/573xx configuration in system memory boot mode, Table 120: STM32H562xx/563xx/573xx bootloader version, Table 121: STM32H72xxx/73xxx configuration in system memory boot mode, Table 123: STM32H74xxx/75xxx configuration in system memory boot mode, Table 125: STM32H7A3xx/7B3xx/7B0xx configuration in system memory boot mode, Table 166: STM32L552xx/62xx configuration in system memory boot mode, Table 168: STM32L552xx/62xx bootloader versions, Table 197: STM32U535xx/545xx configuration in system memory boot mode, Table 197: STM32U535xx/545xx configuration in system memory boot mode, Table 199: STM32U535xx/545xx bootloader versions, Table 202: STM32U575xx/85xx bootloader versions, Table 203: STM32U595xx/99xx/A5xx/A9xx configuration in system memory boot mode, Table 205: STM32U595xx/99xx/A5xx/A9xx bootloader versions, and Table 210: Bootloader startup timings (ms). Updated Figure 69: Bootloader V14 selection for STM32H503xx, Figure 71: Bootloader V14 selection for STM32H562xx/563xx/573xx, Figure 117: Bootloader V9.x selection for STM32U535xx/545xx, Figure 118: Bootloader V9.x selection for STM32U575xx/85xx, and Figure 119: Bootloader V9.x selection for STM32U595xx/99xx/A5xx/A9xx. Minor text edits across the whole document.
11-Jan-2024	61	Added STM32U5F7xx, STM32U5F9xx, STM32U5G7xx, and STM32U5G9xx devices. Updated Table 1: Applicable products, Table 3: Embedded bootloaders, and tables 209 to 213. Updated Section 3: Glossary. Added Section 91: STM32U5F7xx/F9xx/G7xx/G9xx devices and its subsections. Minor text edits across the whole document.

Table 215. Document revision history (continued)

Date	Revision	Changes
06-Mar-2024	62	Updated Table 1: Applicable products, Table 3: Embedded bootloaders, Table 7: Flash memory alignment constraints, Table 8: ExitSecureMemory entry address, Table 36: STM32F10xxx configuration in system memory boot mode, Table 102: STM32G0B1xx/0C1xx configuration in system memory boot mode, Table 112: STM32H503xx configuration in system memory boot mode, Table 114: STM32H503xx bootloader version, Table 118: STM32H562xx/563xx/573xx configuration in system memory boot mode, Table 121: STM32H72xxx/73xxx configuration in system memory boot mode, Table 123: STM32H74xxx/75xxx configuration in system memory boot mode, Table 124: STM32H74xxx/75xxx bootloader version, Table 125: STM32H7A3xx/7B3xx/7B0xx configuration in system memory boot mode, Table 158: STM32L47xxx/48xxx configuration in system memory boot mode, and Table 209: Bootloader device-dependent parameters. Updated Section 3: Glossary, Section 4.1: Bootloader activation, Section 69.1: Bootloader configuration, and Section 70.1: Bootloader configuration. Added Section 4.7.1: ExitSecureMemory v1.0, Section 4.7.2: ExitSecureMemory v1.1, Section 53: STM32H523xx/533xx devices, Section 58: STM32H7Rxxx/7Sxxx devices, Section 79: STM32WBA54xx/55xx devices, Section 85: STM32U031xx devices, Section 86: STM32U073xx/83xx devices, and their subsections. Added Appendix B: Example of ExitSecureMemory v1.1 function. Updated Figure 6: ExitSecureMemory function usage, Figure 7: Access to securable memory area from the bootloader, Figure 69: Bootloader V14 selection for STM32H503xx, and Figure 71: Bootloader V14 selection for STM32H562xx/563xx/573xx. Minor text edits across the whole document.
13-May-2024	63	Added STM32H562xx devices and STM32WB0 series, hence updated Table 1: Applicable products, Table 3: Embedded bootloaders, Table 7: Flash memory alignment constraints, and tables 210 to 214. Updated Section 3: Glossary and note in Section 69.1: Bootloader configuration. Added Section 81: STM32WB05xx devices, Section 82: STM32WB06xx/07xx devices, Section 83: STM32WB09xx devices, and their subsections. Updated Figure 52: Bootloader V9.x selection for STM32F446xx and Figure 108: Bootloader V11.x selection for STM32WBA54xx/55xx. Updated Table 2: Bootloader activation patterns, Table 103: STM32G0B1xx/0C1xx bootloader versions, Table 121: STM32H72xxx/73xxx configuration in system memory boot mode, Table 123: STM32H74xxx/75xxx configuration in system memory boot mode, Table 197: STM32U535xx/545xx configuration in system memory boot mode, and Table 200: STM32U575xx/85xx configuration in system memory boot mode. Minor text edits across the whole document.

Table 215. Document revision history (continued)

Date	Revision	Changes
18-Sep-2024	64	Added STM32C071xx and STM32H7B0xx devices, hence updated Table 1: Applicable products, Table 3: Embedded bootloaders, and tables 209 to 214. Updated Section 3: Glossary, Section 4.2: Bootloader identification, Section 52.1: Bootloader configuration, Section 53.1: Bootloader configuration, Section 54.1: Bootloader configuration, Section 57: STM32H7A3xx/7B3xx/7B0xx devices, Section 75.1: Bootloader configuration, Section 78.1: Bootloader configuration, Section 79.1: Bootloader configuration, Section 88.1: Bootloader configuration, Section 89.1: Bootloader configuration, and Section 90.1: Bootloader configuration. Added Section 4.9: Bootloader models, Section 4.10: Boot constraints on BL, Section 7: STM32C051xx devices, and Section 45.3.1: Compatibility break on boot sequence. Updated Table 8: ExitSecureMemory entry address, Table 99: STM32G07xxx/08xxx bootloader versions, Table 157: STM32L47xxx/48xxx bootloader V10.x versions, and Table 158: STM32L47xxx/48xxx configuration in system memory boot mode.
14-Feb-2025	65	Added STM32C051/91/92xx, STM32U375/385xx, and STM32WBA62/63/64/65xx devices. Updated Table 1: Applicable products, Table 3: Embedded bootloaders, tables 6 to 10, Table 14: STM32C051xx configuration in system memory boot mode, Table 101: STM32G0B0xx bootloader versions, Table 103: STM32G0B1xx/0C1xx bootloader versions, Table 105: STM32G05xxx/061xx bootloader versions, Table 107: STM32G431xx/441xx bootloader version, Table 115: STM32H523xx/533xx configuration in system memory boot mode, Table 118: STM32H562xx/563xx/573xx configuration in system memory boot mode, Table 124: STM32H74xxx/75xxx bootloader version, Table 127: STM32H7Rxxx/7Sxxx configuration in system memory boot mode, Table 159: STM32L47xxx/48xxx bootloader V9.x versions, Table 176: STM32WBA54xx/55xx configuration in system memory boot mode, tables 190 to 194, Table 203: STM32U595xx/99xx/A5xx/A9xx configuration in system memory boot mode, and tables 209 to 214. Added footnote to Table 100: STM32G0B0xx configuration in system memory boot mode. Updated Section 3: Glossary, Section 4.2: Bootloader identification, Section 4.3: Hardware connection requirements, Section 85.1: Bootloader configuration, and Section 86.1: Bootloader configuration. Replaced master/slave with controller target when referring to I2C. Added Section 7: STM32C051xx devices, Section 9: STM32C091xx/92xx devices, Section 58.4: Jump to bootloader, Section 80: STM32WBA62xx/63xx/64xx/65xx devices, Section 87: STM32U375xx/85xx devices, and their subsections. Updated Figure 4: SPI connection, figures 13 to 15, 60 to 66, 68 to 71, 74 and 75, Figure 108: Bootloader V11.x selection for STM32WBA54xx/55xx, and figures 113 to 115. Minor text edits across the whole document.
10-Apr-2025	66	Updated Table 3: Embedded bootloaders, Table 99: STM32G07xxx/08xxx bootloader versions, Table 124: STM32H74xxx/75xxx bootloader version, and Table 209: Bootloader device-dependent parameters.

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